

PROGRAMME AND ABSTRACTS

9th International Conference on Econometrics and Statistics (EcoSta 2026)

<https://cmstatistics.org/EcoSta2026>

Ryukoku University, Kyoto, Japan

8-10 August 2026



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Co-chairs:

Masayuki Hirukawa, Ansgar Steland, Kin Wai Chan, Joan Hu, Marianthi Markatou and Lihua Lei.

EcoSta Editors:

Ana Colubi, George Kapetanios and Erricos J. Kontoghiorghes.

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Organizers:

Ryukoku University, EcoSta, CMStatistics and CFEnetwork.

Chair of the Local Organizing Committee:

Masayuki Hirukawa, Ryukoku University, Kyoto, Japan.

Dear Colleagues,

It is a great pleasure to welcome you to the 9th International Conference on Econometrics and Statistics (EcoSta 2026). The conference is held in a hybrid format, accommodating both in-person and virtual attendance, ensuring flexibility for participants based on their circumstances and local restrictions.

The conference program has been thoughtfully designed to support the effective presentation of research and facilitate networking. EcoSta 2026 comprises about 210 sessions, four keynote talks, six invited sessions, and over 810 presentations. These figures reflect the strong support of our research communities and highlight the significance of this initiative. We trust that the EcoSta conference will continue to serve as an excellent platform for disseminating high-quality research in econometrics and statistics while fostering valuable connections among participants.

The conference is jointly organized by the Computational and Methodological Statistics Working Group (CMStatistics), the Computational and Financial Econometrics Network (CFEnetwork), the journal *Econometrics and Statistics* (EcoSta), and Ryukoku University. Building upon the achievements of previous editions, our aim is for this conference to remain a leading event in the fields of econometrics, statistics, and their applications.

The co-chairs express their gratitude to the scientific program committee, session organizers, and local organizing committee for their collective efforts in delivering a comprehensive program that covers various areas of econometrics and statistics. Ryukoku University, as the local host, along with the dedicated assistants and staff of the International Foundation of Methodological Statistics & Econometrics (IFMSE), have played an instrumental role in ensuring the smooth organization of the conference. We extend our heartfelt thanks to all of them for their invaluable support.

The Elsevier journals *Econometrics and Statistics* (EcoSta) and *Computational Statistics & Data Analysis* (CSDA) are associated with CFEnetwork, CMStatistics, and the EcoSta 2026 conference. Participants are encouraged to join the networks and submit their papers to special or regular peer-reviewed issues of EcoSta and *Annals of Statistical Data Science* (SDS).

The Journal Impact Factor of *Econometrics and Statistics* (EcoSta) for 2025, announced in June 2026, increased to 3. EcoSta is ranked Q1 in both Economics and in Statistics and Probability.

Finally, we are pleased to announce that the 10th International Conference on Econometrics and Statistics (EcoSta 2027) will take place at ShanghaiTech University, Shanghai, China, 6-9 July 2027. The city and the local host will provide an ideal setting for productive discussions, with impeccable infrastructure and a rich cultural environment. We warmly invite and encourage your active participation in EcoSta 2026.

Ana Colubi, Erricos J. Kontoghiorghes and Masayuki Hirukawa
on behalf of the Co-Chairs, Organizers and EcoSta Editors

**CMStatistics network on
COMPUTATIONAL AND METHODOLOGICAL STATISTICS**

<http://www.cmstatistics.org>

The working group (WG) CMStatistics comprises a number of specialized teams in various research areas of computational and methodological statistics. The teams act autonomously within the framework of the WG in order to promote their own research agenda. Their activities are endorsed by the WG. They submit research proposals, organize sessions, tracks and tutorials during the annual WG meetings and edit journal special issues. The Econometrics and Statistics (EcoSta) and Computational Statistics & Data Analysis (CSDA) are the official journals of the CMStatistics.

Specialized teams

Currently, the CMStatistic network has about 1950 members and the following specialized teams

BIO: Biostatistics	NPS: Non-Parametric Statistics
BS: Bayesian Statistics	RS: Robust Statistics
DMC: Dependence Models and Copulas	SA: Survival Analysis
DOE: Design Of Experiments	SAE: Small Area Estimation
FDA: Functional Data Analysis	SDS: Statistical Data Science: Methods and Computations
HDS: High-Dimensional Statistics	SEA: Statistics of Extremes and Applications
IS: Imprecision in Statistics	SL: Statistical Learning
LVSEM: Latent Variable and Structural Equation Models	TSMC: Time Series
MM: Mixture Models	

You are encouraged to become a member of the WG. For further information, please contact the Chairs of the specialized groups (see the WG's website), or by email at info@cmstatistics.org.

**CFEnetwork
COMPUTATIONAL AND FINANCIAL ECONOMETRICS**

<http://www.CFEnetwork.org>

The Computational and Financial Econometrics (CFEnetwork) comprises a number of specialized teams in various research areas of theoretical and applied econometrics, financial econometrics and computation, and empirical finance. The teams contribute to the network's activities by organizing sessions, tracks and tutorials during the annual CFEnetwork meetings and submitting research proposals. Furthermore, the teams edit special issues currently published under the Annals of CFE. The Econometrics and Statistics (EcoSta) is the official journal of the CFEnetwork. Now, the CFEnetwork has over 1100 members.

You are encouraged to become a member of the CFEnetwork. For further information, please see the website or contact by email at info@cfenetwork.org.

SCHEDULE (Japan time, UTC+9)

2026-08-08	2026-08-09	2026-08-10
Opening , 08:50 - 09:05	F EcoSta2026 08:15 - 09:30	K EcoSta2026 08:15 - 09:55
A - Keynote EcoSta2026 09:05 - 09:55	Coffee break 09:30 - 10:00	Coffee break 09:55 - 10:25
Coffee break 09:55 - 10:25	G EcoSta2026 10:00 - 12:05	L EcoSta2026 10:25 - 12:05
B EcoSta2026 10:25 - 12:05	Lunch break 12:05 - 13:35	Lunch break 12:05 - 13:35
Lunch break 12:05 - 13:35	H EcoSta2026 13:35 - 14:50	M - Keynote EcoSta2026 13:35 - 14:25
C EcoSta2026 13:35 - 15:15	I - Keynote EcoSta2026 15:00 - 15:45	N EcoSta2026 14:35 - 16:15
Coffee break 15:15 - 15:45	Coffee break 15:45 - 16:15	Coffee break 16:15 - 16:45
D - Keynote EcoSta2026 15:45 - 16:35	J EcoSta2026 16:15 - 17:55	O EcoSta2026 16:45 - 18:25
E EcoSta2026 16:45 - 18:50		
Welcome reception 19:00 - 20:30	Conference dinner 19:00 - 21:00	

TUTORIAL, MEETINGS, SOCIAL EVENTS AND ACCESS TO THE CONFERENCE

TUTORIAL

The tutorial “Accurate inference in time series and network data: Learning from small samples” will take place on Friday, the 7th of August 2026, 14:00-18:30 (UTC+9) at the Fukakusa Campus of Ryukoku University, 67 Tsukamoto-cho, Fukakusa, Fushimi-ku, Kyoto 612-8577, Japan (Building Tokokan, Room 104). It will be delivered by Prof. Elvezio Ronchetti, University of Geneva, Switzerland. Only participants who have subscribed for the tutorial can attend. Registered participants will be able to access the tutorial either in person or virtually.

SPECIAL MEETINGS by invitation to group members

The EcoSta (Econometrics and Statistics) Editorial Board meeting will take place on Friday, the 7th of August 2026, at 19:00. Instructions to attend the Editorial Board meeting will be sent to the associate editors participating in the conference in due course.

ACCESS TO THE CONFERENCE

- Participants can attend virtually or in person according to what they selected while registering.
- The in-person venue is the Fukakusa campus of Ryukoku University, 67 Tsukamoto-cho, Fukakusa, Fushimi-ku, Kyoto 612-8577, Japan, Building Tokokan. Please note that the university opens at 8:00, so you will not be able to enter before that time.
- The **registration** will be located at the entrance hall (1F) of Tokokan from 8:00 to 17:00 from Saturday to Monday. Please note that no food or drink is allowed in the entrance hall.
- The **coffee breaks** will take place in the basement (B1) of Building 4 (opposite Tokokan).
- The **welcome reception** will take place in the second basement (B2) of Building 22.
- The **lunches** for those who had booked will take place in the second basement (B2) of Building 22.
- The **conference dinner** for those who had booked will take place at the restaurant Funatsuru Kyoto Kamogawa Resort (180 Minoyacho, Shimogyo Ward, Kyoto).
- For environmental sustainability reasons, the conference endeavors to minimize paper usage and overall consumption. While there will be a limited number of printed Books of Abstracts, bags, pens, and pads available for those who requested them in their registration, we strongly encourage all participants to opt for digital materials by downloading them onto their personal devices. QR codes will be displayed in the registration area. These codes will enable participants to quickly access essential information, further reducing the need for printed materials and promoting a paperless conference experience.
- The full conference will be livestreamed through Zoom, and it will not be recorded. The conference programme time is set at UTC+9.
- In order to access the virtual conference, you must first log in to the registration tool, get the daily password there and leave the session open. Then you should open another tab and go to the interactive programme (schedule). Click on the slot you wish to attend and then on the session. You will be redirected to Zoom, where you will need to use the daily password.
- Please note that for security reasons, the Zoom links will not be sent to the speakers, and they can only be found on the online interactive programme (schedule).
- Detailed instructions for virtual and in-person attendance, hybrid sessions, speakers, chairs, posters, networking, test sessions, as well as FAQs, can be found on the webpage.

Presentation instructions

The paper presentations must be shared through Zoom. The in-person rooms will be visible in Zoom as the corresponding hybrid session. Virtual speakers should have a stable internet connection, and make sure their video and audio are working. They will share their slides when the chair requires it, present their talk, and be ready to answer the questions after the presentation. Detailed instructions for speakers can be found on the website. Standard talks are generally 20 minutes long, followed by 3-4 minutes for discussion. Strict timing must be observed.

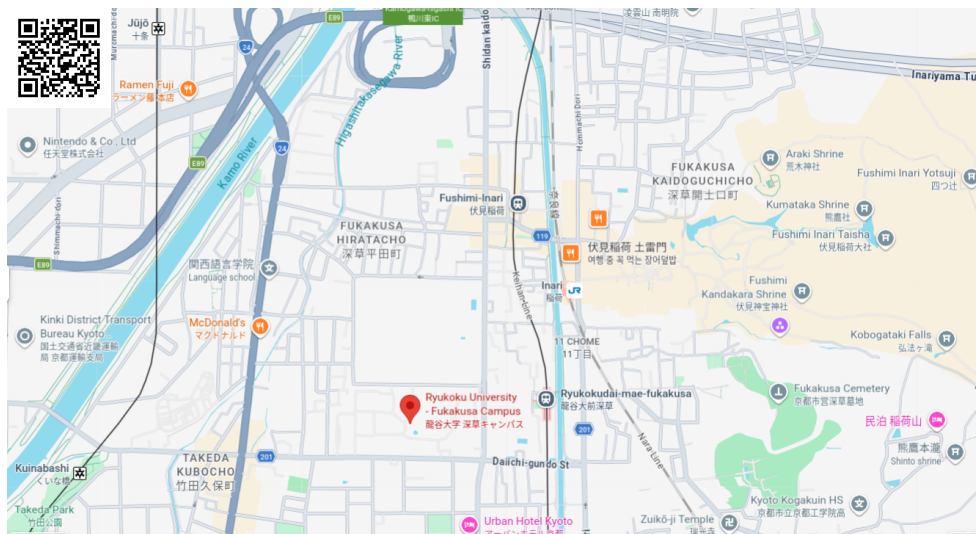
Posters

Posters will be displayed in a common Zoom session and do not need to be uploaded or sent in advance. In-person participants can join the poster session from the room designated in the programme with their devices. The Chair will give time to the presenters to show their posters, which will be rotated sequentially to be seen by the session audience. Presenters may use their time to briefly talk about their posters or simply show them and wait for questions or comments from the audience. Detailed instructions for the poster presentations can be found on the website.

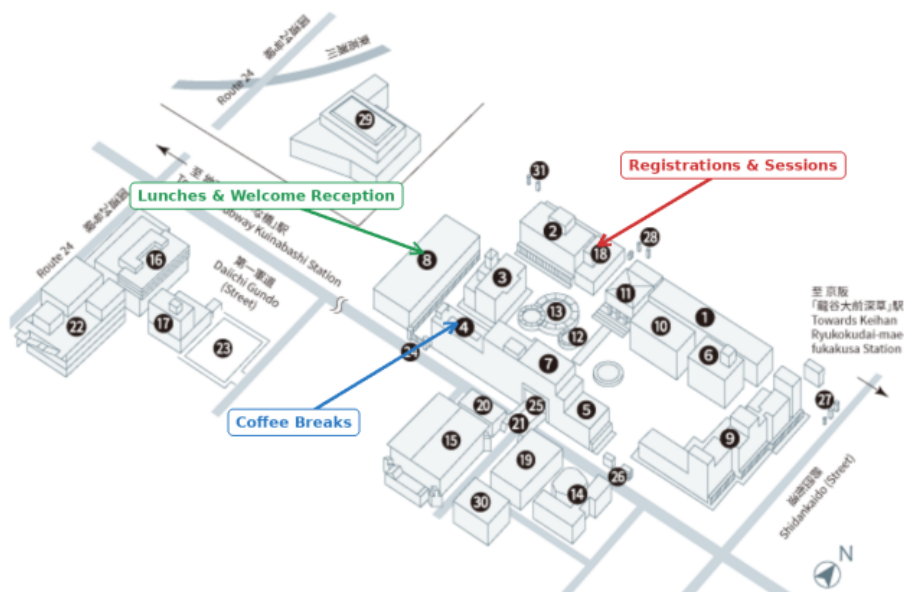
Session chairs

Session chairs are responsible for introducing the session, the speakers, making sure that virtual and in-person participants can listen and see the talks and coordinating the discussion time. A member of the conference staff, identified by the name *Angel* followed by a number, will assist virtually. If any speaker is missing or has a technical problem, the chair can move on the next speaker and come back later to resume if possible. Detailed instructions for the (virtual or in-person) session chairs can be found on the website.

Map of the venue area



Map of the campus



CONFERENCE VENUE GUIDE · Fukakusa Campus, Ryukoku University

Conference Venues

- 18 Tokokan**
Registrations & Sessions
- 4 Building 4**
Coffee Breaks
- 22 Building 22**
Lunches & Welcome Reception

All Campus Buildings

- | | |
|---|---------------------------------------|
| ① Wagenkan | ①⑦ Shishinkan |
| ② Building 2 | ①⑩ Tokokan — Registrations & Sessions |
| ③ Building 3 | ①⑪ Ekokan |
| ④ Building 4 — Coffee Breaks | ①⑫ Monshikan |
| ⑤ Building 5 | ①⑬ Yui renkyo |
| ⑥ Building 8 | ①⑭ Hoyukan |
| ⑦ Building 21 | ①⑮ Tennis Court |
| ⑧ Building 22 — Lunches & Welcome Reception | ①⑯ West Gate |
| ⑨ Shieikan | ①⑰ Main Gate |
| ⑩ Library | ①⑱ Side Gate |
| ⑪ Kenshinkan | ①⑲ East Gate |
| ⑫ Cafe "Jurin" | ①⑳ North Gate |
| ⑬ Stage | ②① Senshokan |
| ⑭ Jojukan | ②② Mitrakan |
| ⑮ Gymnasium | ②③ Northwest Gate |
| ⑯ Shikokan | |

Route for the conference dinner

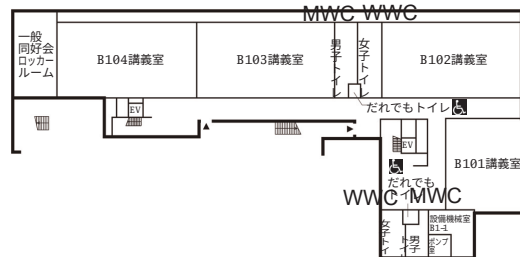


Floor maps

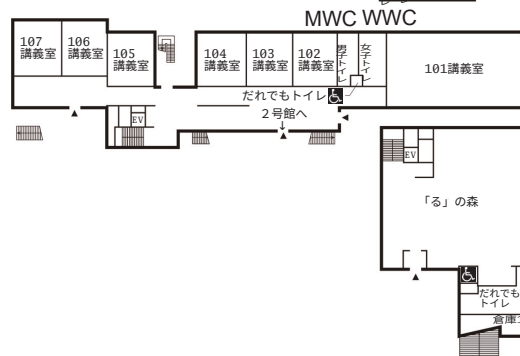
Tokokan



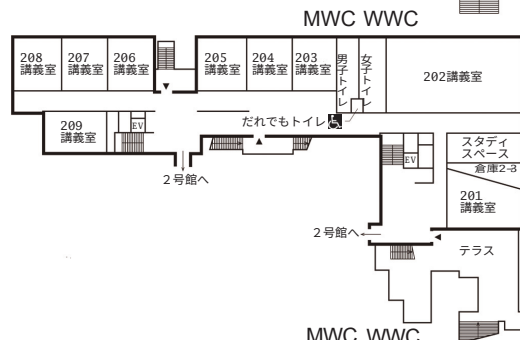
B1F



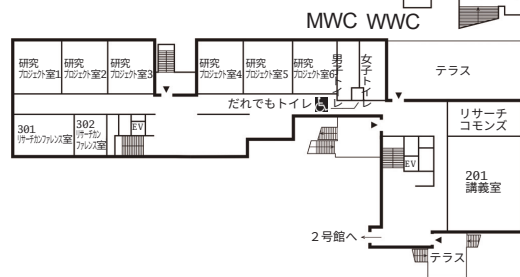
1F



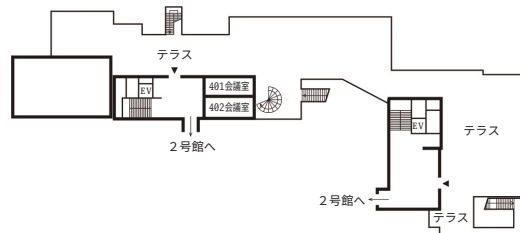
2F



3F



4F



RECEPTION DESK
located here
**NO FOOD OR DRINK
ALLOWED**

Room 201 stretches to
2nd and 3rd floors.

PUBLICATION OUTLETS

The Elsevier journal *Econometrics and Statistics* (EcoSta) is the official journal of the conference. The CMStatistics and CFE networks, co-organizers of the conference, also publish the *Annals of Computational and Financial Econometrics* and *Annals of Statistical Data Science*.

Econometrics and Statistics (EcoSta)

<http://www.elsevier.com/locate/ecosta>

Econometrics and Statistics is the official journal of the networks *Computational and Financial Econometrics* and *Computational and Methodological Statistics* published by Elsevier (<http://www.journals.elsevier.com/econometrics-and-statistics/>). It publishes research papers in all aspects of econometrics and statistics and comprises of two sections:

- **Part A: Econometrics.** Emphasis will be given to methodological and theoretical papers containing substantial econometrics derivations or showing a potential of a significant impact in the broad area of econometrics. Topics of interest include the estimation of econometric models and associated inference, model selection, panel data, measurement error, Bayesian methods, and time series analyses. Simulations are to be considered when they involve an original methodology. Innovative papers in financial econometrics and its applications will be considered. The topics to be covered include portfolio allocation, option pricing, quantitative risk management, systemic risk and market microstructure. Interest will be focused as well on well-founded applied econometric studies that demonstrate the practicality of new procedures and models. Such studies should involve the rigorous application of statistical techniques, including estimation, inference and forecasting. Topics will include volatility and risk, credit risk, pricing models, portfolio management, and emerging markets. Innovative contributions in empirical finance and financial data analysis that use advanced statistical methods are encouraged. The results of the submissions should be replicable. Applications consisting only of routine calculations will not be of interest to the journal.
- **Part B: Statistics.** Papers providing important original contributions to methodological statistics inspired in applications will be considered for this section. Papers dealing, directly or indirectly, with computational and technical elements will be particularly encouraged. These cover developments concerning issues of high-dimensionality, re-sampling, dependence, robustness, filtering, and, in general, the interaction of mathematical methods, numerical implementations and the extra burden of analysing large and/or complex datasets with such methods in different areas such as medicine, epidemiology, biology, psychology, climatology and communication. Innovative algorithmic developments are also of interest, as are the computer programs and the computational environments that implement them as a complement.

The journal consists, preponderantly, of original research. Occasionally, review and short papers from experts are published, which may be accompanied by discussions. Special issues and sections within important areas of research are occasionally published.

EcoSta Annals of Computational and Financial Econometrics (CFE)

<http://www.elsevier.com/locate/ecosta>

The CFEnetwork is currently publishing the *Annals of Computational and Financial Econometrics* as a supplement to the journal of *Econometrics and Statistics*. The *Annals of CFE* serve as an outlet for distinguished research papers in computational econometrics and financial econometrics. The high standards of the *Annals* will make it a valuable resource for econometric research. Each issue will be edited by several Guest Editors who will be responsible, together with the EcoSta Co-Editors, for the selection of the papers. Authors submitting a paper to EcoSta may request that it be considered for inclusion in the *Annals*. All submissions must contain original unpublished work that is not being considered for publication elsewhere. Please submit your paper electronically using the Editorial Manager system (choose Article Type: Research paper, and then select “*Annals of Computational and Financial Econometrics*”).

EcoSta Annals of Statistical Data Science (SDS)

<http://www.elsevier.com/locate/ecosta>

CMStatistics is inviting submissions for the *EcoSta Annals of Statistical Data Science*. The *Annals of Statistical Data Science* are published as a supplement to the journal of *Econometrics and Statistics*. They will serve as an outlet for distinguished research papers using advanced computational and/or statistical methods for tackling challenging data analytic problems. The *Annals* will become a valuable resource for well-founded theoretical and applied data-driven research. Authors submitting a paper to EcoSta may request that it be considered for inclusion in the *Annals*. Each issue will be assigned to several Guest Associate Editors who will be responsible, together with the EcoSta Co-Editors, for the selection of papers. Submissions for the *Annals* should contain a significant computational or statistical methodological component for data analytics. In particular, the *Annals* welcome contributions at the interface of computing and statistics addressing problems involving large and/or complex data. Emphasis will be given to comprehensive and reproducible research, including data-driven methodology, algorithms, and software. All submissions must contain original unpublished work that is not being considered for publication elsewhere. Please submit your paper electronically using the Editorial Manager system (choose Article Type: Research paper, and then select “*Annals of Statistical Data Science*”).

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Saturday 08.08.2026 09:05 - 09:55

Room: 202 Chair: Erricos Kontoghiorghes

Keynote talk I

A bridge between statistical approximations, optimal transportation, and information theorySpeaker: **Elvezio Ronchetti, University of Geneva, Switzerland**

Davide La Vecchia, Andrej Ilievski

Unexplored connections between saddlepoint approximations, measure transportation, and key topics in information theory are presented. A selective review of fundamental results establishes links between Esscher's tilting, a result rooted in information theory and central to saddlepoint approximations, and the solution of the dual Kantorovich problem, a cornerstone of measure transportation theory, via the Legendre transform of the cumulant generating function, also known as the Bahadur half slope. These links are investigated in the framework of M-estimators and quantile regression. The unveiled connections enable saddlepoint approximations to be viewed from different angles, highlighting links to convex analysis through the notion of duality and to differential geometry through the notion of geodesic. Finally, robustness issues of optimal transportation and their connections to penalization methods are addressed.

Saturday 08.08.2026 15:45 - 16:35

Room: 202 Chair: Ana Colubi

Keynote talk II

Modeling and forecasting financial tail risk: Methods and applicationsSpeaker: **Cathy W-S Chen, Feng Chia University, Taiwan**

A unified Bayesian framework is presented for modeling and forecasting financial tail risk, with a focus on value-at-risk (VaR) and expected shortfall (ES). The first part develops nonlinear threshold extensions of ESCAViaR-type models, allowing for regime-dependent tail behavior and improved characterization of extreme risks in cryptocurrency markets. The second part introduces a semi-parametric approach that incorporates geopolitical narratives and range-based volatility measures to capture forward-looking uncertainty and high-frequency market dynamics. The third part considers a flexible distributional framework based on skewed and heavy-tailed specifications within a GARCH-type structure, enabling coherent joint modeling of VaR and ES while accounting for asymmetry and tail thickness. Across all three parts, the analysis employs Bayesian inference to accommodate parameter uncertainty and enhance predictive performance. Empirical studies on major financial assets, including cryptocurrencies and crude oil markets, demonstrate that incorporating nonlinear dynamics, distributional flexibility, and external information leads to substantial improvements in tail risk forecasting. The discussion includes evaluation using scoring rules and backtesting procedures, with emphasis on joint assessment of accuracy and calibration.

Sunday 09.08.2026 15:00 - 15:45

Room: 202 Chair: Masayuki Hirukawa

Keynote talk III

Robust and efficient copulas: From moment redundancy to Neyman orthogonalitySpeaker: **Artem Prokhorov, University of Sydney, Australia**

Consider a general multivariate model where univariate marginal distributions are known up to a parameter vector and we are interested in estimating that parameter vector without specifying the joint distribution, except for the marginals. If we assume independence between the marginals and maximize the resulting quasi-likelihood, we obtain a consistent but inefficient QMLE. If we assume a parametric copula (other than independence) we obtain a full MLE, which is efficient but only under a correct copula specification and is biased otherwise. The Holy Grail of dependence modelling is to characterize a copula function that improves efficiency without sacrificing robustness. The talk is about a few parametric and semiparametric results obtained in search of the holy grail in recent years.

Monday 10.08.2026 13:35 - 14:25

Room: 202 Chair: Takeshi Emura

Keynote talk IV

Prediction for several infinitely divisible distribution modelsSpeaker: **Fumiyasu Komaki, The University of Tokyo, Japan**

In high-dimensional settings, predictive procedures based on shrinkage priors dominating noninformative priors such as the Jeffreys prior are effective in various models including normal, Poisson, and normal regression models. In the predictive theory of these models, the relationship between prediction and estimation based on the infinite divisibility of the underlying models and distributions plays an important role. The aim is to survey these results and discuss prediction problems for subordinators such as gamma distribution models, where shrinkage estimation remains meaningful even in one-dimensional settings. Additionally, how such exceptional behavior can be understood from the viewpoint of Bayesian prediction is discussed. In contrast to the normal and Poisson cases, estimation of Levy measures plays an essential role in these models.

Saturday 08.08.2026

10:25 - 12:05

Parallel Session B – EcoSta2026

EI004 Room 202 RECENT ADVANCES IN TIME SERIES ANALYSIS**Chair: Kin Wai Chan****E1275: Forecasting under structural breaks in nonparametric time series regression***Presenter:* **Sze Him Isaac Leung**, The Chinese University of Hong Kong, Hong Kong*Co-authors:* Chun Yip Yau, Yuanbo Li

In this presentation, the challenge of forecasting in non-parametric time series regression models that experience a structural break is addressed. A novel weighted kernel estimator designed to estimate the post-break regression function and predict future observations is introduced. Unlike traditional methods, this approach assigns weights that are both time and location dependent, enabling the model to make effective use of pre-break data. The findings reveal that incorporating pre-break observations can significantly enhance the estimation of the post-break regression function by reducing the mean squared forecast error. This improvement arises from managing the bias-variance trade-off associated with using pre-break data. The optimal weight allocation for both conditional and unconditional forecasting scenarios is also explored, providing theoretical insights into the asymptotic properties that guide this process. Simulation studies demonstrate the superior performance of the weighted kernel estimator compared to standard post-break methods, showing notable reductions in MSFE. To showcase the practical applicability of the method, a case study on forecasting the volatility of the Nasdaq 100 index is provided.

E1271: High dimensional changing region detection through distributional test*Presenter:* **Ting Tin Ma**, The Chinese University of Hong Kong, Hong Kong*Co-authors:* Kin Wai Chan

Different components of high-dimensional time series may respond to structural changes at varying rates, leading to change points clustering within an interval rather than occurring precisely at isolated points. A test for change intervals, or more broadly, change regions, is studied. The discrepancy from the null hypothesis is measured by the asymptotic proportion of coordinates that experience a change point over an unspecified change region. The test is constructed by comparing the empirical distribution of estimated change points across coordinates to the null distribution. This form of testing is highly scalable for high-dimensional time series and is tuning-free. The proposed framework covers the traditional change point test as a special case and avoids long-run variance estimation and self-normalization employed in mainstream testing approaches. The test statistic is proven to be pivotal asymptotically, and Monte Carlo simulations confirm good size control and high power even in finite samples.

E1315: Difference-based inference and estimation for long-run variance function*Presenter:* **Kin Wai Chan**, The Chinese University of Hong Kong, Hong Kong*Co-authors:* Yi Ho Ngan

A difference-based locally smoothed estimator is proposed for the long-run variance function of locally stationary time series in the presence of a general trend. In contrast to conventional long-run variance function estimators, the new estimator admits a closed-form expression for the mean integrated squared error, revealing that the optimal bandwidths for lag truncation and local smoothing depend on two key quantities: the dependence-to-nonstationarity ratio and the dependence-and-nonstationarity strength. Based on a normal approximation, the asymptotic distribution of the maximal deviation of the proposed estimator is derived, which serves as a unified tool for constructing simultaneous confidence bands and for testing homogeneity and parametric forms of the long-run variance function in the presence of a trend. Additionally, it is shown that local and global mean variability can be quantified by the difference between difference-based and non-difference-based estimators, and new maximum-type and integrated-type tests for mean constancy and parametric trends are developed. The importance of addressing local stationarity is justified by identifying and analyzing the anti-conservative behavior of stationary quadratic-form tests when applied to nonstationary time series.

E1335: Changepoint detection as model selection: A general framework and L0 approximation*Presenter:* **Xueheng Shi**, University of Nebraska-Lincoln, United States

Changepoint detection is reframed as a model selection problem and introduces a unified framework that fundamentally broadens its scope. The framework accommodates a wide range of model structures including seasonal patterns, linear and quadratic trends, and autoregressive dependence, which allows changepoint analysis to move beyond restrictive, simplified settings. At the core of this framework is the iteratively reweighted fused lasso (IRFL), an efficient and scalable approximation to the L0 optimization problem. By adaptively reweighting penalties, IRFL directly targets sparsity in changepoint configurations, substantially improving support recovery and information criterion optimization compared to existing L1-based methods. Extensive simulations demonstrate that IRFL delivers reliable and accurate changepoint detection even in challenging settings with strong nuisance components, including trends, seasonality, and serial correlation. The framework naturally extends beyond time series to image data, enabling edge-preserving denoising and segmentation. Applications to real-world datasets highlight the practical impact of the approach. In particular, analysis of the Mauna Loa CO2 series reveals scientifically meaningful changepoints associated with volcanic eruptions and ENSO events, while providing a sharper and more interpretable trend decomposition than conventional methods.

EO159 Room B101 EVALUATING AND CHOOSING POLICIES**Chair: Liyang Sun****E1548: Segment fixed effects in spatial regression discontinuity designs***Presenter:* **Soonwoo Kwon**, Brown University, United States*Co-authors:* Tobias Grossboelting, Michael Pollmann

When a policy varies discontinuously across geographic space, researchers often identify its effects by comparing outcomes on either side of the policy border in a spatial regression discontinuity design. For estimation, most regression specifications control for location along the border using border-segment fixed effects, and for location away from the border using a bandwidth and sometimes linear adjustments in distance. The conditions under which segment fixed effects are needed are clarified, and a modeling technique is proposed to characterize the benefits and costs of fixed effects in finite samples. Theoretical and practical guidance is provided: Many prior studies chose border segments that are likely longer than optimal, and a common local linear adjustment for distance to the border can yield distant comparisons. The proposed auxiliary analyses and robustness checks are illustrated by reanalyzing empirical studies.

E1650: Optimal estimation for regression discontinuity design with binary outcomes*Presenter:* **Masayuki Sawada**, Hitotsubashi University, Japan*Co-authors:* Takuya Ishihara, Kohei Yata

A finite-sample optimal estimator is developed for regression discontinuity design when outcomes are bounded, including binary outcomes as the leading case. The estimator achieves minimax mean squared error among linear shrinkage estimators with nonnegative weights when the regression function lies in a Lipschitz class. Although the original minimax problem involves an iterative nonconvex optimization problem, the estimator is obtained by solving a convex optimization problem. A key advantage of the proposed estimator is that the Lipschitz constant is its only tuning parameter. A uniformly valid inference procedure is also proposed without a large-sample approximation. In a simulation exercise for small samples, the estimator exhibits smaller mean squared errors and shorter confidence intervals than those of conventional large-sample techniques.

In an empirical multi-cutoff design in which the sample size for each cutoff is small, the method yields informative confidence intervals, in contrast to the leading large-sample approach.

E1689: **Compound selection decisions: An almost SURE approach**

Presenter: **Liyang Sun**, UCL and CEMFI, Spain

Co-authors: Kevin Chen, Lihua Lei, Timothy Sudijono, Tian Xie

Methods are proposed for compound selection decisions in a Gaussian sequence model. Given unknown, fixed parameters $\mu_{1:n}$, known $\sigma_{1:n}$, observations $Y_i \sim \text{Norm}(\mu_i, \sigma_i^2)$, and known costs k_i , the decision maker chooses thresholds δ_i to maximize $\frac{1}{n} \sum_{i=1}^n \mathbb{1}(Y_i > \delta_i)(\mu_i - k_i)$. Inspired by Stein's unbiased risk estimate (SURE), a family of estimators termed ASSURE is introduced for the expected utility of a set of proposed thresholds. ASSURE enables practitioners to select thresholds within a pre-specified class by maximizing an estimated objective, borrowing strength across noisy estimates in compound settings. A leading member of this family, ASSURE*, is nearly unbiased for the true expected utility and converges at near-parametric rates. ASSURE* has favorable regret properties. Applications include the selection of Census tracts for economic opportunity, the identification of discriminating firms, and the analysis of p -value decision procedures in A/B testing.

E1832: **A LATE-targeting experimental design under multivalued treatment**

Presenter: **Tetsuya Kaji**, University of Chicago, United States

Experiments with multivalued treatments are appealing when experimentation is costly. Yet, under imperfect compliance, the identification and even the definition of compliers is unclear. First, the impossibility of identifying margin-specific local average treatment effects (LATEs) in a standard randomized experiment, even under a strong generalization of one-sided compliance, is established. Then, a novel experimental design is proposed that achieves identification of margin-specific LATEs under a mild generalization of one-sided compliance.

EO091 Room B102 INNOVATIONS IN EVENT HISTORY DATA: PRIVACY ENHANCEMENT AND CAUSAL DISCOVERY

Chair: Yi Xiong

E1559: **Privacy-aware data analytics via deconvolution**

Presenter: **Farhad Farokhi**, The University of Melbourne, Australia

Stochastic programs provide a framework for modelling optimization problems that involve uncertainty, such as machine learning, parameter estimation, and data-driven optimization. Most often the distribution of the uncertainty is not known, and samples must be used to approximate the distribution. In many situations, the distribution must be inferred from noisy data samples, such as privacy-preserving or corrupt samples. This can be done via density deconvolution or distributionally robust optimization. The development of data analytics algorithms that work efficiently under differential privacy is discussed through these approaches.

E1610: **Optimal adaptive treatment strategies to maximize quality-adjusted lifetime**

Presenter: **Brent Johnson**, University of Rochester, United States

Amyotrophic lateral sclerosis (ALS) is a fatal, incurable neurodegenerative disorder in which patients eventually lose the ability to swallow as the disease progresses. Percutaneous endoscopic gastrostomy (PEG) provides access for medications and nutrition, potentially extending survival. However, PEG tube placement carries surgical risks and adversely impacts quality of life, creating a trade-off that has complicated guidelines on when to insert PEG. To address this challenge, methods for constructing optimal adaptive palliative care strategies (APCS) that maximize quality-adjusted lifetime, a composite outcome of survival time and quality of life, are proposed. Specifically, methods of optimal dynamic treatment regimes are extended to the palliative care setting for a quality-adjusted lifetime outcome. Using data from the Ceftriaxone Trial, an optimal APCS for PEG insertion is constructed using two key intermediate outcomes, forced vital capacity and body mass index, for the quality-adjusted lifetime outcome, where quality of life is measured by the ALS Functional Rating Scale instrument. The optimal APCS is compared to other proposed palliative care strategies in the literature and shown to be superior. Results demonstrate that quality-adjusted lifetime can be significantly improved using the proposed methods, offering a data-driven framework to inform PEG tube placement decisions for palliative care in ALS.

E1672: **Accelerated failure time models with error-prone response and nonlinear covariates**

Presenter: **Li-Pang Chen**, National Chengchi University, Taiwan

Survival analysis of patient survival time for a specific Cancer using gene expressions as covariates is a main interest in medical studies. In the framework of survival analysis, the accelerated failure time model in parametric form is a common approach. However, gene expressions are possibly nonlinear and the survival time as well as censoring status are subject to measurement error. To address these complex features simultaneously, measurement error in survival time and censoring status is first corrected, and these corrections are used to develop a corrected Buckley-James estimator. Subsequently, the boosting algorithm with cubic spline estimation is used to iteratively recover the nonlinear relationship between covariates and survival time. Theoretically, the validity of measurement error correction and the estimation procedure are justified. Numerical studies show that the proposed method improves the performance of estimation and is able to capture informative covariates. The methodology is applied to analyze breast Cancer data provided by the Netherlands Cancer Institute for research.

E1671: **A longitudinal item response theory approach to create an integrative of child neurodevelopment**

Presenter: **Shelley Liu**, Icahn School of Medicine at Mount Sinai, United States

A common challenge in environmental epidemiology cohort studies examining longitudinal neurodevelopment is the use of different assessment tasks across follow-up periods. Due to various factors such as cohort design, reducing participant burden, and age-appropriateness of measures, tasks administered at different time points may differ in format and content. This makes it difficult to study the longitudinal trajectory of a neurodevelopmental domain. Consequently, studies often limit analyses to the common tasks that are used across ages when studying longitudinal development. However, using only the common tasks across ages could result in studying a shorter span of development than desired, and valuable information from age-specific tasks may be missed. An integrative modeling approach using longitudinal item response theory (IRT) was developed to integrate multiple executive functioning assessments. Using working memory, an executive functioning domain, as an example, multiple assessments (DMTS, IRA, NIH Toolbox, and CANTAB) at ages 6, 8, and 10 in the PROGRESS cohort were integrated. Three approaches -direct estimation, latent growth curve IRT, and parameter linking- were compared to place scores from tasks onto a common scale and generate longitudinal latent working memory scores. The harmonized framework enabled examination of the impacts of early childhood lead exposure on longitudinal working memory trajectories with greater precision.

EO282 Room B103 APPLIED MACROECONOMICS**Chair: Michael Owyang****E1273: Analyzing regional variation in the response to policy uncertainty shocks****Presenter:** Laura Jackson Young, Bentley University, United States**Co-authors:** Michael Owyang

A factor-augmented VAR (FAVAR) is estimated using a panel of monthly state- and national-level data to examine how policy uncertainty influences state-level economic conditions. The model includes three types of factors: a national factor, regional factors which model the correlation within a particular BEA region, and state-level factors which model the correlation across variables within a particular state. Measures of economic policy uncertainty are included, and the impulse responses of state-level employment, unemployment, manufacturing hours, and labor force participation are considered. Manufacturing employment share, export dependencies, and capital share are found to be important for explaining variation in state-level responses.

E1317: Model-average impulse responses for nonlinear VARMA**Presenter:** Daniel Soques, University of North Carolina Wilmington, United States**Co-authors:** Miguel Faria-e-Castro, Neville Francis, Michael Owyang

In linear settings, local projections (LPs) can be more robust to misspecification than VAR-based methods when the true data-generating process has a VARMA representation. However, regime-dependent LPs can be biased in nonlinear settings, particularly when shocks affect the future regime. Monte Carlo experiments based on Threshold VARs that vary along two dimensions are used: (i) the degree of moving-average behavior in the data generating process and (ii) the strength of nonlinear regime dependence. These simulations allow comparison of the relative performance of generalized impulse response functions (GIRFs) and regime-dependent LPs across a range of misspecifications. A model-average impulse response that combines GIRFs and LPs using weights based on relative mean squared error is proposed, thereby allowing the estimator to adjust to the underlying source of error rather than relying on a single method. The usefulness of this approach is then assessed in an application based on a regime-dependent DSGE model, showing how model averaging can improve impulse response estimation when both nonlinearities and misspecification arising from moving-average components are present.

E1319: Screening and monitoring in consumer credit: Evidence from credit unions**Presenter:** Olivia Wilkinson, University of Wisconsin-Madison, United States

Banking consolidation and technological advancements have led to a decline in in-person banking relationships. Credit unions are financial institutions that continue to emphasize consumer relationships and community ties. Using an anonymous random panel of consumer credit reports from Equifax, a matched sample of credit union and non-credit union credit card borrowers is constructed to compare delinquency outcomes across lenders. Credit union borrowers exhibit lower delinquency rates on average, though there is heterogeneity across borrower quality. Two mechanisms are examined empirically and structurally: screening and monitoring. The screening channel is evaluated by testing whether credit limits contain differential predictive information of future delinquency across lenders. Credit limits issued by credit unions are more predictive of subsequent delinquency, consistent with stronger screening of borrower risk. Evidence for monitoring is more difficult to identify. Some evidence of slower transitions into delinquency appears with zip code-score-time fixed effects, but not with consumer-time fixed effects since the set of borrowers with both lenders is not representative of the full sample. A structural life-cycle model with hidden information and endogenous lender choice is estimated to quantify the relative importance of these two mechanisms and to evaluate welfare implications of changes to credit union market share.

E1332: Re-evaluating inflation forecasts**Presenter:** Michael Owyang, Federal Reserve Bank of St Louis, United States**Co-authors:** Julie Bennett, Brooke Hathhorn

Inflation forecasting is essential to macroeconomic modeling and policy. A few stylized facts about inflation forecasting were established in a foundational paper. Their evaluation sample, however, covered a period of mostly low and stable inflation. High and volatile inflation during the recent COVID-era has renewed interest in the dynamics of inflation forecasting. The conclusions are revisited, comparing the accuracy of judgment-based and model-based inflation forecasts from 2004-2024. The findings are that (i) judgment-based forecasts outperform model-based ones, (ii) simpler models outperform more complicated models, especially at shorter horizons, and (iii) model performance can change over time and, especially, with the level of inflation.

EO145 Room B104 FACTOR MODELS IN ECONOMETRICS/FINANCIAL ECONOMETRICS**Chair: Sung Hoon Choi****E1286: Low-rank structured nonparametric prediction of instantaneous volatility****Presenter:** Sung Hoon Choi, University of Connecticut, United States**Co-authors:** Donggyu Kim

Based on Ito semimartingale models, several studies have proposed methods for forecasting intraday volatility using high-frequency financial data. These approaches typically rely on restrictive parametric assumptions and are often vulnerable to model misspecification. To address this issue, a novel nonparametric prediction method for the future intraday instantaneous volatility process during trading hours is introduced, which leverages both previous days' data and the current day's observed intraday data. The approach imposes an interday-by-intraday matrix representation of the instantaneous volatility, which is decomposed into a low-rank conditional expectation component and a noise matrix. To predict the future conditional expected volatility vector, this low-rank structure is exploited and the structural intraday-volatility prediction (SIP) procedure is proposed. The asymptotic properties of the SIP estimator are established and its effectiveness is demonstrated through an out-of-sample prediction study using real high-frequency trading data.

E1372: Connected trade flows via trade costs: A spatial autoregressive framework**Presenter:** Hanbat Jeong, Macquarie University, Australia**Co-authors:** Jieun Lee

A new gravity specification grounded in countries' network-leveraging features is proposed. Building on spatial autoregressive models, trade costs are endogenized as a function of a network of country proximities. The resulting specification induces interdependence among trade flows and pair-specific heterogeneity, and nests the conventional iceberg cost as a special case when countries do not leverage network connections. For estimation, PPML is used and robust inference accommodating heteroskedasticity and unknown error correlations is developed. A fast algorithm for the large-dimensional network multiplier transformation is also proposed. Empirically, evidence of endogenous trade costs and substantially improved fit relative to the conventional gravity equation (up to 30%) is found, implying pair-specific heterogeneity missed by existing models. Counterfactual analysis shows that a bilateral increase in trade costs generates sizable reallocations of import shares across many third countries once network interactions are accounted for.

E1701: Matrix completion and causal inference*Presenter:* **Jungjun Choi**, University of Rhode Island, United States*Co-authors:* Ming Yuan

Matrix completion concerns the imputation of missing entries in a partially observed matrix. While its rapid development was originally motivated by applications in recommendation systems, it has also opened new possibilities in causal inference, where missing counterfactual outcomes can be imputed to estimate individual treatment effects. In causal inference problems, the missingness pattern in potential outcomes is induced by the treatment adoption process. Although treatment assignment may satisfy the missing-at-random (MAR) assumption in randomized experiments and certain quasi-experimental designs, this assumption is often violated in observational studies. For instance, when a program is introduced at a particular time for a subset of units, the potential outcomes under control become unobserved for treated units after treatment adoption, creating a block missingness structure. An inferential framework for matrix completion under missing-not-at-random (MNAR) mechanisms is presented and applied to treatment effect estimation. In particular, fixed effects are incorporated into the model, which is more appropriate for economic data, and the inclusion of these fixed effects is shown to improve estimation quality.

E1474: AI-driven residual learning with robust factor models for stock return prediction*Presenter:* **Zhuolan Liu**, University of Macau, FBA, China*Co-authors:* Xinyi Ren, Lianjie Shu

Accurate stock return prediction remains a central challenge in financial econometrics, particularly under structural instability and nonlinear market dynamics. A hybrid framework is proposed that integrates robust factor modeling with machine learning to improve predictive performance. Specifically, Huber Principal Component Analysis (HPCA) is employed to extract stable latent factors from high-dimensional financial data, followed by vector autoregression (VAR) to capture temporal dynamics. A residual neural network is then applied to model nonlinear components unexplained by traditional factor structures. Using U.S. equity data from 1973 to 2019, results indicate that the proposed approach improves out-of-sample predictive accuracy compared to standard benchmarks such as CAPM and Fama-French models, as well as standalone machine learning methods. The findings highlight the value of combining structured econometric models with AI-driven techniques, providing a robust and flexible framework for financial forecasting.

EO211 Room 101 PATTERNS IN MOTION: TIME SERIES, CHANGE POINTS, AND NETWORKS (VIRTUAL)**Chair: Abolfazl Safikhani****E1705: Online change point detection for multivariate inhomogeneous Poisson processes time series***Presenter:* **Carlos Misael Madrid Padilla**, Washington University in St Louis, United States*Co-authors:* Xiaokai Luo, Haotian Xu, Oscar Hernan Madrid Padilla

Change point detection in streaming data is a fundamental problem in statistics and machine learning, with important applications in areas such as seismology, climate monitoring, and epidemic surveillance. The problem of online change point detection for multivariate inhomogeneous Poisson point process time series is studied, a setting that captures event-based data with complex temporal and spatial structure. A novel method is introduced that leverages low-rank Matrix representations of multivariate intensity functions, enabling flexible nonparametric modeling while maintaining computational scalability. The proposed procedure operates in a single-pass online fashion, requiring only constant computational cost per incoming observation, independent of the length of the time series. On the theoretical side, guarantees are established for controlling the false alarm probability and the detection delay is characterized under temporal dependence. A key technical contribution is a new Matrix Bernstein inequality for temporally dependent Poisson point processes, which may be of independent interest. Numerical experiments demonstrate the statistical robustness and computational efficiency of the proposed approach in realistic scenarios.

E1803: Multiple change point detection in time series with non-stationary dynamics*Presenter:* **Yuhan Tian**, Technische Universität München, Germany*Co-authors:* Abolfazl Safikhani

Change point detection (CPD) is fundamental for analyzing time series data, with applications in finance, climate science, and quality control. However, most existing methods rely on piecewise stationarity, an assumption often violated in practice due to continuous non-stationary dynamics. Such dynamics can mask abrupt changes or lead to false detections. A model-based CPD framework is proposed that explicitly accounts for non-stationarity and cross-correlations. The method decomposes the time series into a random walk component capturing non-stationary behavior and a vector autoregressive component modeling dependencies. Change points are identified by evaluating error reduction within moving windows when structural changes are introduced. The proposed approach is flexible and supports a broad class of component selection procedures, while admitting theoretical guarantees on consistent estimation of both the number and locations of change points. Although derived under a model-based framework, the method demonstrates strong empirical robustness. Simulation studies show that it performs well across a wide range of non-stationary settings, including seasonality and long-memory dependence. Experiments on simulated and real data confirm the effectiveness of the method. In particular, it successfully identifies surface defects in steel rolling images despite gradual background variations, highlighting its robustness in complex non-stationary environments.

E2070: A generalized Bayesian approach to multiple changepoint analysis*Presenter:* **Michael Jauch**, Florida State University, United States*Co-authors:* Yuhui Wang, Joshua Loyal, Andrew Thomas

A generalized Bayesian method for multiple changepoint analysis is introduced, employing a loss function inspired by multinomial logistic regression. The method does not require specification of the data-generating process and avoids restrictive assumptions on the nature of changepoints. From the joint posterior distribution, simultaneous inference can be made on the locations of changepoints and the coefficients of a multinomial logistic regression model for distinguishing data across homogeneous segments. The multinomial logistic regression coefficients provide a familiar means of interpreting potentially complex changes. To select the number of changepoints, posterior summaries are leveraged that measure whether the multinomial logistic classifier can distinguish data from either side of a potential changepoint. To simulate from the generalized posterior distribution, a Gibbs sampler based on Polya-Gamma data augmentation is presented. The accuracy and flexibility of the method are assessed through simulation studies featuring different types of changes, and its interpretability is demonstrated through applications to financial network data and topological data derived from nanoparticle videos.

E2077: Nonparametric inference under Gaussian subordinated process*Presenter:* **Haihan Yu**, University of Rhode Island, United States

Long-range dependent time series $\{X_t\}$ can often be embedded in Gaussian subordinated processes, where they are regarded as an unknown transformation $G(\cdot)$ of a long-memory Gaussian process $\{Z_t\}$ (i.e., $X_t = G(Z_t)$). Despite substantial methodological developments for long-memory time series over the past 50 years, inference for such possibly nonlinear processes remains largely an open problem. This difficulty stems from the inaccessibility of key parameters, the Hermite rank m of G and the latent memory exponent α of $\{Z_t\}$, as well as the non-Gaussian limiting distributions that commonly arise under long memory. A formal method is introduced for jointly estimating (m, α) and its consistency is established under mild conditions. Building on these estimators, a broadly applicable bootstrap-based inference framework is proposed, which is exemplified for the mean, approximately linear statistics, and the empirical distribution. The performance of the proposed methods is assessed via simulation

studies and their practical utility is illustrated through applications to the Dow Jones Index.

EO054 Room 102 STATISTICAL INFERENCE AND APPLICATIONS FOR NETWORK DATA
Chair: Youngseok Song
E1293: A sequential two-stage test for large random graphs of unequal size

Presenter: **Xin Jin**, The University of Tampa, United States

Two-sample hypothesis testing for large graphs is increasingly important in cognitive science, probabilistic machine learning, and artificial intelligence. While numerous methods have been proposed, less attention has been given to settings involving unequal graph sizes or situations with only one or a few graph samples. An efficient sequential two-stage procedure tailored for such scenarios is proposed. In Stage 1, spectral clustering is applied to each graph to estimate the number of communities, serving as a screening step for structural compatibility. In Stage 2, bootstrapped adjacency matrices are generated from estimated community-wise edge probability matrices, forming the basis of a Frobenius test statistic. The asymptotic distribution of the statistic is established and it is demonstrated, via extensive simulations, that the proposed procedure is stable, efficient, and effective in detecting subtle structural differences, while offering rigorous theoretical guarantees. Finally, its utility is demonstrated on fMRI data, successfully distinguishing brain activity patterns when subjects are exposed to sentences, pictures, and control stimuli.

E1534: Multivariate Hawkes process modeling of dynamic functional connectivity network in MEG for epilepsy

Presenter: **Jaehye Kim**, Duksung Womens University, Korea, South

Epilepsy is increasingly recognized as a disorder of large scale brain networks, in which seizures and interictal dysfunction arise from abnormal interactions among distributed regions on fast temporal scales. Conventional functional connectivity approaches, including static and dynamic methods, rely on temporal averaging or sliding windows, limiting sensitivity to transient, directed, and history dependent neural interactions central to epileptic dynamics. Magnetoencephalography provides millisecond temporal resolution and direct sensitivity to neuronal activity. However, most connectivity analyses rely on correlation or coherence measures that assume continuous signals and symmetric interactions, while dynamic approaches impose temporal smoothness and stationarity, restricting detection of brief and asymmetric excitation. An event driven framework using the multivariate Hawkes process is proposed to model directed and history dependent functional connectivity. This approach captures self and cross excitation, enabling inference of causal and temporally asymmetric interactions without arbitrary windowing. Applied to resting state MEG data from 44 patients and 46 controls, the model revealed increased recurrent excitation and lateralized propagation in patients, whereas controls showed weaker and more balanced dynamics.

E1570: Network autoregression for binary responses

Presenter: **Lijia Wang**, City University of Hong Kong, Hong Kong

Studying the propagation of binary responses on nodes in a large-scale social network is critical for understanding how individual behaviors and decisions are shaped by social structures and for predicting collective outcomes. A network autoregressive model for binary-valued responses is proposed, in which the probability of response at each node is influenced by its neighbors' past decisions, its own past decision, and node-specific covariates, through a logistic link function. The model accounts for network noise and community structure by assuming the underlying network is generated from a block model, with autoregressive parameters that are community-specific. Conditions are established under which the long-term behavior of the high-dimensional binary vector converges to a community-specific distribution and the associated convergence rate, illustrating when individuals in the same community or across the whole network reach a consensus regardless of their initial positions. Given an observed network and response vectors, asymptotic consistency and normality of the maximum likelihood estimators are established. The efficiency and validity of the inference procedure are demonstrated through simulated and real data. In particular, the model is shown to be applicable to studying the dynamics of strike occurrences in China and highlights the impact of online social networks in facilitating collective actions.

E1763: Configuration recovery for distance-based latent space models

Presenter: **Haoran Zhang**, Southern University of Science and Technology, China

Co-authors: Yinghang Chen, Yunxiao Chen

Multidimensional Unfolding (MDU) is a fundamental tool for visualizing joint embeddings of distinct object sets, such as respondents and items, based on their similarities. However, existing MDU methods often lack rigorous statistical guarantees and are also computationally intensive when applied to high-dimensional, noisy data. A Generalized Multidimensional Unfolding (GMDU) framework is proposed that handles continuous, binary, and count data within a unified statistical model. A three-stage estimation procedure is further developed, which includes matrix denoising, singular space refinement, and configuration retrieval. The matrix denoising step employs a truncated singular value thresholding algorithm to recover the underlying drifted distance matrix with average Frobenius-norm consistency guarantee, followed by a refinement step that upgrades the initial estimator to achieve entry-wise uniform consistency. The final step consists of a low-dimensional optimization to recover the configuration by resolving rotation and translation ambiguities. Non-asymptotic error bounds are established for the estimators under mild conditions. Extensive simulations and an application to the 2020 Cooperative Election Study (CES) reveal a primary ideological spectrum and an orthogonal institutional dimension, providing clear visualization of geopolitical and ideological structures.

EO160 Room 103 RECENT ADVANCES IN TEXT PATTERN RECOGNITION AND SEQUENCE ANALYSIS
Chair: Yaofang Hu
E1600: Model-based clustering of music pieces

Presenter: **Xuwen Zhu**, The University of Alabama, United States

Co-authors: Yingying Zhang, Hung Tong, Volodymyr Melnykov

Grouping songs with similar music compositions plays an important role in streaming services as group characteristics can be used to provide personalized song recommendations and enhance user engagement. In essence, a song is characterized by a categorical sequence consisting of notes from different octaves and the sojourn time of every note. A cluster-weighted model is proposed to cluster songs based on their compositions, with marginal distributions as multivariate first-order Markov models and conditional distributions as mixtures of Gammas. The proposal enjoys great flexibility in capturing potentially multimodal behavior of sojourn time. Application of the model to cluster piano songs achieves meaningful results.

E1214: A state-restricted hidden Markov model for authorship attribution of the deutero-Pauline and pastoral epistles

Presenter: **Josiah Leinbach**, Bowling Green State University, United States

Co-authors: Shuchismita Sarkar, Xuwen Zhu

The New Testament contains thirteen epistles attributed to the Apostle Paul, all of which were traditionally accepted as authentically Pauline by early Christian theologians. Since the 19th century, however, many scholars have questioned Paul's authorship of certain epistles due to differences in vocabulary and writing style compared to the undisputed Pauline epistles. In particular, two clusters of epistles, known as the Deutero-Pauline Epistles (Ephesians, Colossians, and 2 Thessalonians) and the Pastoral Epistles (1 Timothy, 2 Timothy, and Titus) have been subject to the most doubt. A novel state-restricted hidden Markov model is presented that constructs a constrained state space for the undisputed Pauline epistles and an unrestricted state space for other epistles. The model jointly analyzes all thirteen epistles and some additional biblical texts, employing a novel first-order Markov emission for transitions between parts of speech to classify sentences based on Pauline and non-Pauline style detection.

Competing models are analyzed using Bayesian model averaging, which also quantifies model uncertainty. Then, informed by New Testament scholarship, the result of the model is interpreted and the possibility of Pauline authorship for the Deutero-Pauline and Pastoral Epistles has been examined.

E1236: A variational Bayesian approach for multimodal multi-instance classification

Presenter: **Yaofang Hu**, The University of Alabama, United States

Co-authors: Yichen Cheng, Yusen Xia, Xinlei Wang

In multiple instance learning (MIL), objects are represented by bags of instances. Each instance shares the same feature set but has unique feature values. MIL aims to train models that predict bag-level outcomes based on features of these instances, making it a weakly supervised approach due to the lack of instance-level labels. While MIL methods focusing on binary classification are abundant, they often cannot identify which specific instances drive bag labels and have limited or little interpretability. MICProB addresses these issues as a Bayesian multiple instance classification (MIC) method. However, MICProB is computationally intensive and best suited for unimodal instances. To overcome these limitations, vM-MIC is proposed, a novel variational Bayesian approach for multimodal MIC. vMMIC handles diverse instance types and significantly improves computational efficiency by leveraging Bayesian variational inference, coupled with the data augmentation technique. vMMIC is benchmarked against MICProB and many other commonly used MIC approaches on both simulated and real-world data. Results demonstrate vMMIC's superior performance, faster computation, and improved interpretability.

E1586: On regime changes in text data using hidden Markov model of contaminated vMF distribution

Presenter: **Yingying Zhang**, Western Michigan University, United States

Co-authors: Shuchismita Sarkar, Yuanyuan Chen, Xuwen Zhu

A novel methodology for analyzing temporal directional data with scatter and heavy tails is presented. A hidden Markov model with contaminated von Mises-Fisher emission distribution is developed. The model is implemented using forward and backward selection approach that provides additional flexibility for contaminated as well as non-contaminated data. The utility of the method for finding homogeneous time blocks (regimes) is demonstrated on several experimental settings and two real-life text datasets containing presidential addresses and corporate financial statements respectively.

EO107 Room 105 STATISTICAL METHODS FOR COMPLEX AND STRUCTURED DATA

Chair: Jingru Mu

E1391: Least-squares estimator-based inference for survival and reliability data

Presenter: **Bowen Liu**, University of Missouri - Kansas City, United States

Co-authors: Malwane Ananda, Sam Weerahandi

Statistical inference for survival and reliability measures is essential in engineering, biomedicine, and risk analysis. An inferential framework based on least-squares estimators (LSEs) for survival probabilities, reliability functions, and related quantities in location-scale families is proposed. Using these estimators, statistical inferential procedures are developed and bootstrap methods for interval estimation are investigated. Results show that LSE-based confidence intervals perform particularly well compared with likelihood-based methods. In simulation studies, they achieve strong coverage probabilities and favorable interval efficiency, often outperforming likelihood-based methods, especially in small samples where asymptotic approximations may be unreliable. The proposed methodology applies to several widely used lifetime models, including the Weibull, log-logistic, and Gumbel distributions. Real-data examples further illustrate the practical effectiveness of the proposed methods in survival and reliability analysis.

E1612: Bayesian inference for geometrically anisotropic spatial random fields on regular lattices

Presenter: **Fan Dai**, North Dakota State University, United States

Co-authors: Somak Dutta

Geometric anisotropy arises when the dependence structure of a spatial random field varies with direction. A Bayesian inference framework is developed for a class of geometrically anisotropic random fields defined on regular lattices. The models are constructed through fractional Laplacian differencing on the lattice, which provides a flexible way to capture directional dependence and spatial smoothness. It is further shown that, as the lattice spacing decreases, this class of lattice-based models converges to a continuum family of anisotropic Matern-type random fields, thereby establishing a theoretical link between discrete and continuous spatial models. The framework enables principled uncertainty quantification for anisotropy and other dependence parameters. The methodology is illustrated through an analysis of ocean chlorophyll concentration data from NASA's MODIS-Aqua project.

E1738: Robust sparse Bayesian semiparametric modeling for longitudinal genetic studies

Presenter: **Jie Ren**, Indiana University School of Medicine, United States

Co-authors: Kun Fan, Cen Wu, Shuangge Ma

In high-dimensional longitudinal interaction studies, performing variable selection and statistical inference is challenging when the assumed interaction model is misspecified under heterogeneous disease phenotypes. Partially motivated by an analysis of the Childhood Asthma Management Program (CAMP) data with high-dimensional single nucleotide polymorphism (SNP) measurements, a novel sparse robust Bayesian mixed model is developed to capture nonlinear longitudinal interactions. The proposed mixed model is robust to outliers and heavy-tailed distributions in the response variable, as well as to the misspecification of nonlinear interaction effects, in terms of both parameter estimation and statistical inference. A Gibbs sampler incorporating spike-and-slab priors is developed to promote exact sparsity in the identification of appropriate forms of main and interaction effects. Simulation results reveal superior performance of the proposed fully robust Bayesian analysis in identification, estimation, and statistical inference compared with alternative methods. In particular, the proposed analysis enables exact statistical inference in misspecified interaction models for the first time by yielding Bayesian credible intervals with nominal coverage probabilities in finite samples. Application of the proposed and alternative methods to the CAMP study sheds novel insight into Asthma etiology.

E1255: Nonlinearity of spending effects on public education: Evidence of spatial dependence

Presenter: **Shengwu Shang**, Kansas State University, United States

Co-authors: John Anderson, Juan Du

The effects of fiscal expenditures on educational outcomes in U.S. public education are studied, with particular attention to spatial dependence and nonlinearities. Using data from the 2010 Michigan educational assessment program (MEAP) on fourth-grade mathematics pass rates, the analysis replicates earlier findings on spending effects following Michigan's proposal a reforms and extends them by incorporating spatial dependence through the nonparametric covariance estimator. While prior research reports little difference between linear and nonlinear models in terms of average partial effects, it is shown that linear specifications fail to capture spatial interactions. Accounting for spatial dependence reveals meaningful nonlinear spending effects, and estimates from linear models using level pass rates overstate these effects by approximately 4-6%.

EO116 Room 106 GRAPHICAL AND CAUSAL LEARNING FOR COMPLEX STOCHASTIC PROCESSES**Chair: Kuang-Yao Lee****E2049: Robust causal effect estimation in high dimensional survival analysis via nonparametric learning****Presenter:** Shanshan Ding, University of Delaware, United States**Co-authors:** Zhezhen Jin

Causal inference is an important tool for making inferences about causal relationships between variables. A robust nonparametric framework for estimating causal treatment effects in high-dimensional survival analysis is developed. The proposed framework is flexible and alleviates assumptions in existing causal survival analysis. It allows the covariate dimension to grow exponentially fast with the sample size without imposing stringent model or distributional assumptions for estimating the causal effect. The effectiveness of the methods is demonstrated through both theoretical and numerical studies.

E1960: Mapping risk contagion through graphical models and early warning signals in Taiwan's stock market**Presenter:** Huei-Wen Teng, National Yang Ming Chiao Tung University, Taiwan**Co-authors:** Nok Hang Chan, Po-Yao Hsu

Graphical models are leveraged to investigate financial risk in Taiwan's stock market, focusing on the dynamic interactions among the 30 largest market-capitalization firms from January 1985 to April 2025. Granger causality networks are constructed to capture directional predictability among stock returns and track how the resulting network structure evolves over four decades of market history. Analysis reveals a consistent pattern in the dynamics of network centrality over time, aligning with the widely held view that technology firms dominate the Taiwanese market. By measuring network connectivity through degree of causality, number of connections, closeness, and eigenvector centrality, an early warning mechanism is proposed for identifying periods of heightened systemic risk. Empirical results show that sharp increases in aggregate connectivity tend to precede major market downturns, providing a forward-looking signal of contagion risk. The findings offer practical implications for institutional risk management and individual investment strategies, and demonstrate the value of graphical models as a tool for financial risk monitoring in an emerging-market setting.

E1927: Graphical modeling of multivariate Cox process**Presenter:** Kuang-Yao Lee, Temple University, United States**Co-authors:** Jiehuan Sun, Tianxi Cai, Lexin Li

A central question in the analysis of multivariate temporal point process data that are simultaneously recorded for multiple subjects is to model the inter-dependency among different point processes. A novel hidden point process graphical model is proposed to address this question. Built on a log Gaussian multivariate Cox process, the model hinges on random intensity functions to wholly determine the point process distribution, and the conditional dependency among the intensity functions characterizes the network's edges. The approach distinguishes itself from most existing vectorial or functional graphical models by dealing with entirely unobserved and non-Hilbertian random intensity functions. A precision operator is introduced, and a one-to-one correspondence between the zero element of this operator and the conditional independence of the intensity function is established. A hard thresholding-based estimation procedure is developed, and its asymptotic consistency is established. The empirical performance of the method is investigated through both simulations and the analysis of electronic health records (EHR) data, which infers relatedness of different EHR encounters and reconstructs the healthcare knowledge network for patients with type 2 diabetes.

E1972: Joint accelerated rate and hazard model for recurrent events and informative terminals**Presenter:** Chuan-Fa Tang, University of Texas at Dallas, United States**Co-authors:** Sy Han Chiou, Weixi Zhu

Recurrent and terminal events are commonly observed and often correlated in longitudinal studies. A joint accelerated rate and hazard modeling framework is proposed that simultaneously accommodates recurrent events and an informative terminal event via subject-specific frailty. A rank-based estimation procedure is employed. A perturbation-based approach is refined for estimating the variance-covariance matrix, and Wald and score tests are developed for statistical inference. The proposed methods are supported through numerical studies and illustrated using real-world colorectal cancer datasets.

EO050 Room 107 STATISTICAL LEARNING FOR COMPLEX DATA**Chair: Wenbo Wu****E1225: A partial envelope approach for modeling multivariate spatial-temporal data****Presenter:** Reisa Widjaja, University of Wisconsin - La Crosse, United States

In the new era of big data, modeling multivariate spatial-temporal data is a challenging task due to both the high dimensionality of the features and complex associations among the responses across different locations and time-points. To improve the estimation efficiency, a spatial-temporal partial envelope model is proposed, which is parsimonious and effective in modeling high-dimensional spatial-temporal data. The partial envelope model is proposed under a linear coregionalization model framework which allows heterogeneous covariance structures for different variables of the response vector. The asymptotic behavior of the estimator is studied and a thorough simulation study is conducted to demonstrate the soundness and effectiveness of the proposed method. The proposed model is also applied to analyze the crowdsourcing weather data collected from personal weather stations in the city of Syracuse, New York, United States.

E1532: Sufficient dimension reduction via explained kernel embedding variation**Presenter:** Chenlu Ke, Virginia Commonwealth University, United States

A new framework for sufficient dimension reduction based on explained kernel embedding variation is proposed. The main idea is to view dimension reduction through a model-free ANOVA decomposition in reproducing kernel Hilbert spaces and to estimate low-dimensional linear combinations of predictors that maximize kernel regression sum of squares. This formulation unifies several sufficient dimension reduction targets: with characteristic kernels it targets the central subspace, while with linear kernels it reduces to central mean subspace estimation. On the computational side, a sequential procedure that builds an ordered set of directions is combined with a joint refinement step over the Grassmann manifold. A validation-based backward test is also used to determine the structural dimension. The framework is further extended to right-censored outcomes and an IPCW version of kernel regression sum of squares for estimating sufficient reductions for the event time is developed. Numerical studies show that the proposed method can recover the relevant dimension and directions in both fully observed and censored settings.

E1917: Deep ranking with heterogeneous effects**Presenter:** Shuxing Fang, The Hong Kong Polytechnic University, Hong Kong

Classical latent-score ranking models often fail to distinguish objects' intrinsic scores from contextual effects, which are typically nonlinear and can dominate the observed outcomes. A semiparametric ranking framework is introduced in which the log-score of each object is modeled as the sum of a utility parameter and nonparametric covariate effects. Within this framework, model identifiability is established under mild regularity and connectivity conditions. For estimation, the covariate effects are approximated using a neural network and the parameters are estimated via maximum likelihood. Under random design assumptions, it is proven that the resulting estimator exists with high probability and non-asymptotic

error bounds are derived that achieve minimax optimality for both the parametric and nonparametric components. Numerical experiments on both synthetic data and an ATP tennis dataset are conducted to support these findings.

E1920: **Learning guarantee of reward modeling using deep neural networks**

Presenter: **Yuanhang Luo**, Hong Kong Polytechnic University, Hong Kong

Co-authors: Ruijian Han, Guohao Shen, Yeheng Ge

The Learning theory of reward modeling with pairwise comparison data using deep neural networks is studied. A novel non-asymptotic regret bound for deep reward estimators in a non-parametric setting is established, which depends explicitly on the network architecture. Furthermore, to underscore the critical importance of clear Human beliefs, a margin-type condition is introduced that assumes the conditional winning probability of the optimal action in pairwise comparisons is significantly distanced from 1/2. This condition enables a sharper regret bound, which substantiates the empirical efficiency of Reinforcement Learning from Human Feedback and highlights clear Human beliefs in its success. Notably, this improvement stems from high-quality pairwise comparison data implied by the margin-type condition, is independent of the specific estimators used, and thus applies to various Learning algorithms and models.

EO110 Room 201 RECENT ADVANCES IN INDUSTRIAL AND APPLIED STATISTICS

Chair: Chang-Yun Lin

E1245: **Orthogonalized moment aberration for multi-stratum factorial designs**

Presenter: **Ming-Chung Chang**, Academia Sinica, Taiwan

Multi-stratum factorial designs, such as block designs and row-column designs, are widely used for screening treatment factors in experiments with complex experimental-unit structures arising from multiple sources of variability. A unified, model-free approach termed orthogonalized moment aberration is introduced for comparing similarities among level combinations of treatment factors assigned to heterogeneous experimental units. The proposed approach evaluates the rows of design matrices through kernel functions, rather than the columns, enabling the assessment of a broad class of mixed-level regular and nonregular factorial designs under heterogeneous experimental-unit structures known as partially relaxed orthogonal block structures. This framework is highly flexible: different choices of kernel functions allow adaptation to various experimental scenarios, with certain choices recovering well-known minimum aberration criteria from the literature. Although model-free in nature, the proposed method admits rigorous justification via linear mixed-effects models and Gaussian process models. Theoretical results and numerical examples demonstrate that the approach can generate multi-stratum factorial designs with high Bayesian D-efficiencies.

E1263: **Robust estimation of small area poverty measures**

Presenter: **Sanjoy Sinha**, Carleton University, Canada

Small area estimators of poverty indicators have been extensively studied in the literature in recent years. Existing model-based estimators are efficient under standard normality assumptions on the assumed nested error models, but are generally sensitive to outliers or other deviations from the underlying assumptions. Some robust methods for estimating poverty measures will be presented and explored by extending the robust method for small area means. These methods may be used to bound the influence of potential outliers when predicting the small area poverty measures. Empirical properties of the proposed estimators will be investigated using Monte Carlo simulations.

E1553: **Bridging minimax design and robust active learning in nonlinear regression**

Presenter: **Xiaojuan Xu**, Brock University, Canada

Active learning is a widely used framework for improving data efficiency by enabling learners to strategically select training samples to enhance estimation and prediction accuracy. The robust design strategies for active learning under potential model misspecification are investigated. Model departures within an L_2 -type neighbourhood are considered and analytical expressions for optimal and robust sampling densities are derived using a minimax approach. These designs explicitly balance variance minimization and bias reduction, ensuring reliable performance even when the assumed model is misspecified. The effectiveness of the proposed designs is evaluated using the integrated mean squared error criterion, demonstrating their robustness and efficiency in general nonlinear regression settings.

E1852: **Reliable Shannon entropy estimation from incomplete species inventories**

Presenter: **Tsung-Jen Shen**, National Chung Hsing University, Taiwan

Shannon's entropy is widely used to quantify species diversity because it captures the overall complexity of community composition while balancing the contributions of rare and abundant species. In ecological surveys, however, many species remain undetected, especially when sampling effort is limited. Such incompleteness can lead to biased and unstable entropy estimates. A new estimator for Shannon's entropy that explicitly accounts for the missing portion of a species inventory is presented. The proposed method combines information from observed species and species frequency counts to improve estimation under incomplete sampling and provides uncertainty assessment via bootstrap resampling. Its performance is evaluated through simulation studies based on several empirical communities, including butterflies, beetles, small mammals, corals, and insects. Compared with eight existing estimators, the proposed estimator generally yields smaller bias and more stable performance, particularly when sample sizes are limited or moderate. These results suggest that the method offers a practical and reliable tool for biodiversity assessment when complete inventories are difficult to obtain.

EO095 Room 203 RECENT ADVANCES AND DEVELOPMENT IN STATISTICAL MODELING

Chair: Li-Hsien Sun

E1224: **Bayesian modeling of degradation data with dependent increments using the t copula**

Presenter: **I-Tang Yu**, Tunghai University, Taiwan

Co-authors: Raf Loreto

Degradation measures serve as surrogates of lifetime, that is, they allow to perform inference about the lifetime of a product unit based on the increments of its degradation path. To preserve mathematical tractability, most existing stochastic-process-based degradation models assume that increments from non-overlapping time intervals are independent, which limits their ability to capture temporal correlation commonly observed in real data. A Bayesian modeling framework is introduced that explicitly accounts for dependence among degradation log-increments by employing the t copula. A key advantage of this approach is its flexibility in accommodating arbitrary marginal distributions for the log-increments, which makes the proposed model applicable across diverse degradation processes. Computations are carried out using a Markov chain Monte Carlo algorithm, and the model is validated through an application to a real-world degradation dataset. Additionally, a simulation study is conducted to evaluate the model's goodness-of-fit, robustness, and general applicability. The simulation study confirms the effectiveness of the proposed approach in capturing complex degradation dynamics.

E1478: **Detecting structural shifts in interval-based time series**

Presenter: **Li-Hsien Sun**, National Central University, Taiwan

Co-authors: Chi-Yang Chiu

Multiple change-points analysis for interval-valued time series data is addressed through a proposed model containing daily opening (O), up (U), low (L), and closing (C) values, based on the log-transformed geometric Brownian motion model rather than traditional opening and closing values

models. The joint distribution for these interval-valued observations is obtained using the reflection principle and Girsanov's theorem. Change-points are identified by the MLE method and PELT algorithm. Performance of the model is investigated through extensive simulations and real data analysis using S&P500 returns during the 2020 COVID-19 pandemic. Results demonstrate that the proposed OULC model consistently outperforms the traditional OC model in offering more accurate and reliable change-point detection.

E1489: **An extension of the dynamic seasonal grey model with covariates**

Presenter: **Jenyu Lee**, Fengchia University, Taiwan

The aim is to extend the dynamic seasonal grey model by generalizing the univariate DSGM(1,1) to a covariate-inclusive DSGM(1,p+1), aiming to enhance forecasting performance for data exhibiting both exogenous influences and dynamic seasonality. In the DSGM(1,p+1) framework, the dynamic seasonal adjustment factors (DSAF) for the forecasting period are estimated using several methods, including the naive approach, Holt-Winters, SARIMA, and Fourier expansion. These estimates are then integrated with the predicted values of the underlying trend component to obtain improved forecasting results. Furthermore, various simulation scenarios are designed, considering patterns such as stable seasonality, varying seasonal amplitude, and unstable seasonal structures, along with different types of covariates, to compare the performance of the proposed models. Finally, empirical validation is conducted using real-world data with pronounced seasonal patterns and available covariate information. It is expected that the approach will broaden the theoretical and practical scope of dynamic seasonal grey models and provide an integrated forecasting approach that simultaneously accounts for covariate effects and the estimation of seasonal factors during the forecasting period.

E1758: **Outlier-sensitive forecast evaluation of value-at-risk and expected shortfall**

Presenter: **Takaaki Koike**, Hitotsubashi University, Japan

Co-authors: Cathy W-S Chen

Backtesting via calibration tests is standard for financial risk measures such as Value-at-Risk and Expected Shortfall. However, for highly volatile assets, their risk forecasts can themselves become extremely large in absolute value, and existing calibration tests may not adequately penalize such outlier forecasts. An outlier-sensitive extension of VaRES calibration tests is proposed that reweights the identification functions so that extreme forecasts receive stronger penalties. This improves the ability of the tests to detect models that generate overly volatile or excessively conservative tail risk forecasts. At the same time, stronger penalization creates a trade-off since the validity of asymptotic inference becomes more demanding in terms of moment conditions. To address this issue, an admissible region is estimated in which these conditions are likely to remain satisfied and test statistics are developed that aggregate calibration evidence over that region. The proposed framework provides a practical diagnostic for assessing robustness and calibration under tail stress. An empirical application to cryptocurrency return forecasts shows that the method can distinguish models with stable tail risk forecasts from those whose apparent performance deteriorates once extreme forecasts are appropriately penalized.

EO113 Room 204 RECENT ADVANCES IN MULTIVARIATE DEPENDENCE MODELLING

Chair: Pavel Krupskiy

E1441: **Proxy methods for factor copula-based clustered data with random covariates**

Presenter: **Bruno Remillard**, HEC Montreal, Canada

Co-authors: Pavel Krupskiy, Bouchra Nasri

A proxy method is proposed for the estimation of parameters and latent factors for clustered data based on factor copulas, under the assumption that the number of observations per cluster is very large. The model can also incorporate random covariates, and the conditional distribution of the response variable, given covariates, is a parametric family of continuous or discrete distributions, while the effect of a common latent variable pertaining to a cluster is modelled with a factor copula. As a by-product, a goodness-of-fit test is also proposed. This extends previous work on similar models where the number of observations per cluster was assumed to be small.

E1446: **Computationally tractable factor copula models**

Presenter: **Bahareh Ghanbari**, RMIT University, Australia

Co-authors: Pavel Krupskiy, Laleh Tafakori, Yan Wang

Gaussian factor models are extensively employed to characterise multivariate dependence structures. Nevertheless, their applicability in high-dimensional settings is limited by restrictive tail behaviour and strong distributional assumptions. In particular, they fail to capture asymmetric and tail dependence structures commonly observed in real data. To overcome these limitations, a novel class of factor copula models is proposed that combines the tractability of Gaussian factor models with the flexibility of copula-based dependence modelling. The proposed framework represents each observed variable as a mixture of a latent Gaussian factor and a residual component, where the mixing weights are random loadings. Dependence among the residuals is introduced through an exchangeable one-factor copula driven by an additional latent variable, allowing for flexible tail dependence and asymmetric structures. Both continuous and discrete specifications of the random loadings are considered, leading to analytically tractable expressions for the correlation matrix. Theoretical properties of the proposed model are established and it is shown that the latent factor can be consistently recovered as both the sample size and dimension increase. Simulation studies demonstrate the accuracy and robustness of the estimation procedure in high-dimensional settings. An empirical application illustrates the model's ability to capture complex dependence patterns, particularly in the tails.

E1453: **Spatial clustering of big time series by Gibbs sampling and empirical BIC with an application to Australian rainfall**

Presenter: **Guoqi Qian**, The University of Melbourne, Australia

Co-authors: Yiyang Chen

Spatially distributed time series data arise ubiquitously in environmental monitoring, economics, and biomedical systems, where data from each spatial location evolves dynamically over time. Identifying spatial grouping structures among such series is fundamental which enables understanding of underlying patterns, regional similarities, and common temporal dynamics. However, spatially clustering time series remains a challenging problem due to the strong temporal dependence and high spatial dimensionality involved. A two-stage framework is developed to tackle this challenge. In the first stage, temporal dynamics are explicitly modelled under a seasonally adjusted autoregression (SAAR) framework. In the second stage, partitioning spatial locations is converted into partitioning the collection of the estimated parameter vectors in SAAR models corresponding to all spatial locations. Then partitioning the estimated parameter vectors is equivalent to generating random samples from a computationally intractable probability distribution defined on the sample space of the estimated parameter vectors and derived by empirical BIC. Generating such random samples can be implemented by a computationally feasible Gibbs sampling algorithm. It is shown that Gibbs sampling can find the optimal spatial partition with probability one. In the end, the developed method is evaluated through extensive simulation studies and further applied to a fused 22-year monthly Australian rainfall dataset.

E1466: **Tests of independence and randomness for arbitrary data using copula-based covariances**

Presenter: **Bouchra Nasri**, University of Montreal, Canada

Co-authors: Bruno Remillard

Tests of independence for data with arbitrary distributions in both non-serial and serial settings are discussed. These tests are constructed using copula-based covariances and their multivariate extensions via Mobius transforms. Their asymptotic properties are examined, results from numerical experiments are presented, and the methodology is illustrated with a data-driven example.

EO154 Room 205 RECENT DEVELOPMENTS ON COMPLEX DATA**Chair: Jingnan Zhang****E1699: Diffusion structure inference under heterogeneous network cascade****Presenter:** Yubai Yuan, Penn State University, United States

A cascade over a network refers to the diffusion process where behavior changes occurring in one part of an interconnected population lead to a series of sequential changes throughout the entire population. In recent years, there has been a surge in interest and efforts to understand and model cascade mechanisms, as they motivate many significant research topics across different disciplines. Inferring diffusion networks enables interventions in the cascading process to maximize information propagation and provides insights into the Granger causality of interaction mechanisms among individuals. A novel double mixture graphical model is proposed for inferring latent diffusion networks in the presence of strong cascade heterogeneity. The model represents cascade pathways as a distributional mixture over diffusion networks, where these networks capture different cascading patterns at the population level. A data-driven optimization method is developed to infer diffusion networks using only visible temporal cascade records, without requiring the modeling of complex and heterogeneous individual states. Both statistical and computational guarantees for the proposed method are established. The double mixture cascade model is applied to analyze research topic cascades in social science across U.S. universities and uncover the latent research topic diffusion networks among the top U.S. social science programs.

E1483: Generalized tensor completion with non-random missingness**Presenter:** Biao Cai, City University of Hong Kong, United States

Tensor completion plays a crucial role in applications such as recommender systems and medical imaging, where data are often highly incomplete. While extensive prior work has addressed tensor completion with data missingness, most assume that each entry of the tensor is available independently with probability. However, real-world tensor data often exhibit missing-not-at-random (MNAR) patterns, where the probability of missingness depends on the underlying tensor values. A generalized tensor completion framework is introduced for noisy data with MNAR, where the observation probability is modeled as a function of underlying tensor values. The flexible framework accommodates various tensor data types, such as continuous, binary and count data. For model estimation, an alternating maximization algorithm is developed and non-asymptotic error bounds are derived for the estimator at each iteration, under considerably relaxed conditions on the observation probabilities. Additionally, a statistical inference procedure is proposed to test whether observation probabilities depend on underlying tensor values, offering a formal assessment of the missingness assumption within the modeling framework. The utility and efficacy of the approach are demonstrated through comparative simulation studies and analyses of two real-world datasets.

E1394: Community detection in heterogeneous signed networks**Presenter:** Jingnan Zhang, University of Science and Technology of China, China

Network data has attracted growing interest across scientific domains, prompting the development of various network models. Existing network analysis methods mainly focus on unsigned networks, whereas signed networks, consisting of both positive and negative edges, have been frequently encountered in practice but much less investigated. A signed block beta-model is formally defined along with strong and weak balance in signed networks, which is capable of modeling strong- and weak-balanced signed networks simultaneously. The identifiability of the proposed model is established by leveraging properties of bipartite graphs, and an efficient alternating updating algorithm is developed to optimize the resulting log-likelihood function. More importantly, the asymptotic consistencies of the proposed model are established in terms of both probability estimation and community detection. Its advantages are also demonstrated through extensive numerical experiments and the application to a real-world international relationship network.

E1692: Constrained contextual bandits under systemic fluctuations**Presenter:** Yi Chen, Hong Kong University of Science and Technology, Hong Kong

In online decision-making, unobserved systemic fluctuations often confound perceived outcomes, leading to a profit illusion where transient environmental surges trigger the premature exhaustion of scarce resources on suboptimal actions. Standard methods typically fail by conflating these global shocks with intrinsic treatment effects. To address this, a constrained contextual bandit model is formulated incorporating unobservable, time-varying intercepts to capture arbitrary exogenous volatility. An online framework is proposed that achieves statistical consistency by centralizing real-time data, a simple yet effective algebraic approach that decouples systemic noise from true signals without requiring evolution forecasting. By overcoming the complex temporal dependencies inherent in adaptive processes, sublinear regret bounds are rigorously established, demonstrating the policy's robustness. The method proves its efficacy in operational settings such as e-commerce coupon assignment and personalized medicine by preventing resource misallocation. Ultimately, this research establishes a critical design principle: explicitly modeling and isolating systemic fluctuations is essential for achieving operational resilience and long-term efficiency in volatile, budget-constrained environments.

EO238 Room 208 MODERN STATISTICS FOR TIMING, TRENDS, AND TRANSITIONS IN PUBLIC HEALTH DATA**Chair: Maricela Cruz****E1253: An investigation of the choice of time scale for studies with time-to-dementia-related outcomes****Presenter:** Kyle Conniff, University of Wisconsin Madison, United States**Co-authors:** Daniel Gillen

Time to event analyses using Cox proportional hazards models require specifying a time scale, which determines the risk sets used to estimate covariate effects. Although age and time on study are the two most common choices, prior simulation studies have evaluated them only under correct specification of the underlying time process, producing mixed guidance for applied researchers. This literature is extended by examining the performance of the Cox model when the analysts chosen time scale is misspecified relative to the true event generating mechanism. Scenarios are constructed in which the hazard function changes at an unobserved time T_0 that is unrelated to study entry. Under an age based time scale, the hazard is low before age T_0 and increases afterward; under a time on study time scale, the hazard is zero until the individual reaches T_0 , regardless of entry age. Both time scales are evaluated across settings where the predictor of interest is either correlated or uncorrelated with age at entry. When the predictor and entry age are uncorrelated, both time scales yield similar performance even under misspecification. However, when they are correlated and the true time process aligns more closely with time on study, the age time scale model has much more bias than the time-on-study model. Overall, the results indicate that time on study is more robust to misspecification of the true underlying time scale.

E1276: Interrupted time series and seasonality: An investigation of the ups and downs**Presenter:** Simon Turner, Monash University, Australia

Interrupted time series (ITS) are frequently used to evaluate population-level interventions or exposures in public health research. Seasonality may be present for some outcomes, yet its impact in estimating the effects of the interruption have received little attention. Continuous daily data from a segmented linear regression model with a single interruption across scenarios (e.g., different level and slope changes, varying series length, varying magnitude of seasonality) are simulated. The data are analysed using segmented linear regression, applying different estimation methods with and without seasonal adjustment, and bias in the level- and slope-change parameters is examined. Not adjusting for seasonality produced biased level- and slope-change estimates and standard errors. Bias depended on the magnitude of seasonality as well as the relative timing of the seasonal curve to the beginning, interruption and end points of the series. Adjusting for seasonality removes the biases in all parameters, but can result in large

standard errors, particularly for short time series. If seasonality is anticipated, researchers should seek to collect data for longer series (>24 data points) and adjust for seasonality in the model. For shorter series, the appropriate approach is less clear because of the trade-off between bias and imprecision.

E1213: **A doubly hierarchical changepoint model for multi-unit interrupted time series**

Presenter: **Maricela Cruz**, Kaiser Permanente Washington Health Research Institute, United States

Co-authors: Andrew Holbrook

Interrupted time series (ITS) designs are aptly situated for studying the impacts of large-scale public health policies, as they borrow from case-crossover designs and can retrospectively assess the impact of an intervention. Recent advances in ITS methods include formal tests for the existence of a change point, procedures to estimate a change point when appropriate, models that capture post-intervention changes in higher-order moments, and models that estimate marginal effects. However, no ITS methods that estimate a change point also quantify the uncertainty around those estimates. DHITS is proposed, a Bayesian doubly hierarchical change point model that identifies unit-specific change points and quantifies their uncertainty, while sharing information across units. The model also estimates a global change point across all units (along with its variance) and accounts for changes in temporal dependence following the intervention. The methodology is demonstrated by analyzing multi-unit patient centered data from a hospital that implemented a new care delivery model.

E1280: **The growth curve model and multivariate bilinear regression models useful in the analysis of longitudinal data**

Presenter: **Jemila Hamid**, University of Ottawa, Canada

Inference involving the growth curve model (GCM) will be discussed, which is a generalized multivariate analysis of variance (GMANOVA) model that has been demonstrated to be useful in the analysis of longitudinal data, growth curves, and other response curves. Unlike traditional regression models, the GCM assumes a structured mean (eg. a polynomial mean trajectory over time), which is incorporated through two design matrices: the within and between individual design matrices. Inference involving the mean parameters of the model involve bilinear projections, consequently, the gcm is also referred to as a bilinear regression model. A trace test that was developed as an alternative to the likelihood ratio (LR) test will be presented and its null distribution discussed, including finite sample and asymptotic (large sample) approximations that are useful in practical applications. Results from extensive simulations demonstrating the test's performance will be presented, along with applications to real data sets.

EO083 Room 209 SURVIVAL ANALYSIS, BIOSTATISTICS AND LONGITUDINAL DATA ANALYSIS

Chair: Sheng-Yueh Chang

E1488: **A Bayesian adaptive randomization design for many-to-one comparison in multi-arm clinical trials**

Presenter: **Yu-Mei Chang**, Tunghai University, Taiwan

In multi-arm phase II oncology trials, survival endpoints require prolonged follow-up, limiting the practicality of outcome-adaptive randomization (OAR). A novel Bayesian adaptive randomization (BAR) design is proposed for many-to-one comparisons that leverages short-term surrogate responses to guide allocation while retaining progression-free survival (PFS) time as the primary endpoint. Building on previous work, the key contribution is a unified framework that links multinomial response categories to arm-specific Weibull survival distributions via a Bayesian mixture model. The proposed BAR design enables continually posterior updating of mean survival and interim futility monitoring such that final selection can be made based on the posterior probability that an experimental arm exceeds control by a prespecified clinically meaningful margin. Two adaptive allocation strategies are examined: one driven by transformed posterior superiority probabilities and another that targets desired allocations while correcting observed imbalances. Simulation studies across realistic scenarios demonstrate nominal 5% Type I error rate control, high power under heterogeneous treatment effects, and robust performance under survival-model misspecification and alternative prior specifications.

E1651: **Flexible spatial modeling with variable selection: A higher-order nonparametric spatial autoregressive framework**

Presenter: **Sheng-Yueh Chang**, National Cheng Kung University, Taiwan

Co-authors: Kuo-Jung Lee

Deep neural networks have attracted increasing attention because of their strong predictive performance and flexibility in modeling complex nonlinear relationships. A higher-order nonparametric spatial autoregressive framework with variable selection for spatial data analysis is proposed. Unlike conventional spatial autoregressive models that mainly rely on prespecified spatial weight matrices, the proposed method uses neural networks to learn multiple spatial weighting mechanisms from the data and to model nonparametric endogenous spatial effects. To perform variable selection while preserving nonlinear modeling flexibility, an L1 penalty and a hierarchical sparsity constraint are imposed on the main-effect branch. A regularization-path training procedure with a hierarchical proximal update is developed to estimate the model and identify important variables. Simulation studies show that the proposed method can recover relevant variables and achieve competitive predictive performance under nonlinear and multiple spatial dependence structures. An empirical study using the California Housing Prices data further demonstrates its ability to improve prediction and identify meaningful explanatory variables. Overall, the proposed framework integrates flexible spatial mechanism learning, nonparametric endogenous spatial effects, and interpretable variable selection in a unified deep learning model.

E1741: **A flexible joint modeling approach for recurrent events, death, and cure fraction with application to breast cancer data**

Presenter: **Pingyu Lu**, National Yang Ming Chiao Tung University, Taiwan

In clinical research, patients are typically followed from baseline until the occurrence of a failure event or censoring. However, individuals may experience recurrent events repeatedly, and such events are often associated with, and ultimately terminated by, the failure event. Moreover, with advances in medical treatment, a subgroup of patients may remain free of both recurrent events and the terminal event over a sufficiently long follow-up period, suggesting the presence of a cure fraction. Cured individuals are assumed to be not susceptible to either recurrent events or the terminal event, whereas uncured individuals may experience recurrent events until the occurrence of the terminal event. To appropriately account for the cessation of recurrent events following the terminal event, a conditional additive mean function defined among survivors is proposed. A three-stage estimation procedure is developed: Stages 1, 2, and 3 estimate the cure fraction, the terminal-event model, and the recurrent-event process, respectively. Without specifying the association among these processes, estimating equations for all model parameters are derived. Large-sample properties of the estimators are established and their performance is evaluated through extensive simulation studies. Finally, the proposed method is illustrated using breast cancer data.

E1753: **Generalized method of moments for semiparametric transformation models of recurrent events with informative censoring**

Presenter: **Chang-Yu Tsai**, National University of Kaohsiung, Taiwan

Co-authors: Yu-Jen Cheng

Semiparametric transformation models for recurrent events with a shared frailty variable are considered, allowing for dependence between the event process and censoring. Unlike standard shared frailty proportional rate models, this framework relaxes the proportionality assumption and allows the rate functions to vary flexibly across covariates over time. Motivated by the decomposition of the rate function into shape and size components, an inverse-rate weighting approach and a generalized method of moments framework are developed that combine information from different components of the rate function to improve efficiency. Large-sample properties are established, finite-sample performance is evaluated through simulations, and practical utility is demonstrated using a real dataset.

E1162: Functional marked ETAS models for the 2009 L'Aquila earthquake sequence*Presenter:* **Giada Adelfio**, Università degli Studi di Palermo, Italy

The Epidemic Type Aftershock Sequence (ETAS) model is extended by incorporating functional marks derived from waveform data into the triggering component. Each waveform associated with an event is represented through its functional principal component analysis (FPCA) scores, which are then included as covariates in the triggering term. The model is estimated using the forward likelihood for prediction (FLP) approach and applied to the 2009 L'Aquila earthquake sequence. A local random labelling test for functional marks reveals spatial dependence in the waveform-related features, particularly during the first day of activity. Comparative diagnostics between a standard ETAS model and the proposed functional ETAS model indicate that including waveform information improves the overall goodness-of-fit and the description of the early aftershock clustering.

E1185: On optimal prediction of missing functional data with memory*Presenter:* **Tommi Sottinen**, University of Vaasa, Finland*Co-authors:* Pauliina Ilmonen, Nourhan Shafik, Germain Van Bever, Lauri Viitasaari

The problem of reconstructing missing parts of functions based on their observed segments is considered. It provides, for Gaussian processes and arbitrary bijective transformations thereof, theoretical expressions for the L2-optimal reconstruction of the missing parts. These functions are obtained as solutions of explicit integral equations. In the discrete case, approximations of the solutions provide consistent expressions of all missing values of the processes. Rates of convergence of these approximations, under extra assumptions on the transformation function, are provided. In the case of Gaussian processes with a parametric covariance structure, the estimation can be conducted separately for each function, and yields nonlinear solutions in presence of memory. Simulated examples show that the proposed reconstruction indeed fares better than the conventional interpolation methods in various situations.

E1943: Graph-guided group detection and sparse estimation for functional regression with correlated functional covariates*Presenter:* **Xinyue Li**, City University of Hong Kong, Hong Kong

With the development of wearable sensor technologies, it has become common to collect multiple densely sampled signals from a single subject. These Functional covariates are rarely independent, as spatially adjacent sensors, physiologically linked channels, or biomechanically synergistic body parts produce highly correlated signals. Such structured dynamic dependence induces high collinearity that complicates coefficient estimation and interpretation, and conventional methods are not well-suited to this situation. A new analytic framework based on the Functional Adaptive Graph-Guided Sparsity estimator is introduced. The method constructs a graph to capture the time-varying correlation structure among Functional covariates, applies community detection to reveal densely connected subgroups, and incorporates this structure into a penalty that encourages similarity among coefficient functions within clusters while preserving heterogeneity across clusters. An Adaptive LASSO penalty further induces local Sparsity, enabling simultaneous recovery of clustered and sparse Functional effects. Theoretical properties are discussed, and simulations demonstrate superior performance and interpretability over existing approaches. Application to a Kinect sensor study uncovers biologically coherent joint-movement groups predictive of balance-related outcomes, highlighting the framework's utility for Kinect sensor and multi-channel sensing applications with correlated signals.

E2009: Functional modeling of sparse longitudinal trajectories with varying observation weights*Presenter:* **Maryam Al Alawi**, Sultan Qaboos University, Oman

A Functional data framework is proposed for modeling sparse longitudinal trajectories observed through unequally weighted assessments with varying temporal importance. Unlike conventional longitudinal approaches that rely on aggregate summaries or treat observations equally over time, the proposed methodology incorporates assessment-specific weights directly into the trajectory construction process, allowing highly influential observations to contribute more strongly to the Estimation of the underlying latent stochastic dynamics. Trajectory reconstruction is performed within the Principal Analysis by Conditional Estimation (PACE) framework, followed by Functional Principal Component Analysis (FPCA) to capture the dominant modes of temporal variation. To investigate the effect of explanatory variables on trajectory evolution, a function-on-scalar regression model with time-varying coefficient functions is employed. The proposed approach provides a flexible framework for analyzing sparse longitudinal processes in which observations contribute unequally over time, while offering a richer characterization of latent temporal dynamics than conventional pointwise methods. Although motivated by weighted educational assessment data, the methodology is sufficiently general to extend to broader longitudinal settings involving heterogeneous observation structures.

Saturday 08.08.2026

13:35 - 15:15

Parallel Session C – EcoSta2026

EO150 Room B101 OPTIMAL TRANSPORT IN ECONOMETRICS**Chair: Toru Kitagawa****E1525: Statistical deconvolution via entropic iterative proportional fitting***Presenter:* **Antoine Jacquet**, Sciences Po, France*Co-authors:* Alfred Galichon, Guillaume Pouliot

A linear programming approach to deconvolution is proposed for recovering the latent distribution of X from observations of $Y = X + \epsilon$, when the noise distribution is known or can be simulated. From an entropically regularized formulation, primal and dual iterative proportional fitting algorithms with closed-form updates are derived, in the spirit of Sinkhorn's algorithm for optimal transport. Numerical experiments suggest that the proposed approach is tractable and compares favorably with existing procedures.

E1602: Dynamic decision-making under model misspecification: A stochastic stability approach*Presenter:* **Xinyu Dai**, Brown University, United States*Co-authors:* Daniel Chen, Yian Qian

Dynamic decision-making under model uncertainty is central to many economic environments, yet existing bandit and reinforcement learning algorithms rely on the assumption of correct model specification. The behavior and performance of Thompson Sampling (TS), one of the most commonly used Bayesian reinforcement learning algorithms, is studied when the model class is misspecified. A complete dynamic classification of posterior evolution in a misspecified two-armed Gaussian bandit is provided, identifying distinct regimes: correct model concentration, incorrect model concentration, and persistent belief mixing, characterized by the direction of statistical evidence and the model-action mapping. These regimes yield sharp predictions for limiting beliefs, action frequencies, and asymptotic regret. The analysis is then extended to a general finite model class and a unified stochastic stability framework is developed that represents posterior evolution as a Markov process on the belief simplex. This approach characterizes two sufficient conditions to classify the ergodic and transient behaviors and provides inductive dimensional reductions of the posterior dynamics. The results offer the first qualitative and geometric classification of TS under misspecification, bridging Bayesian learning with evolutionary dynamics, and build the foundations of robust decision-making in structured bandits.

E1596: Who with whom: Learning optimal matching policies*Presenter:* **Toru Kitagawa**, Brown University, United States*Co-authors:* Yagan Hazard

Many economic contexts exist where the productivity and welfare performance of institutions and policies depend on who matches with whom. Examples include caseworkers and job seekers in job search assistance programs, medical doctors and patients, teachers and students, attorneys and defendants, and tax auditors and taxpayers, among others. Although reallocating individuals through a change in matching policy can be less costly than training personnel or introducing a new program, methods for learning optimal matching policies and their statistical performance are less studied than methods for other policy interventions. A method is developed to learn welfare optimal matching policies for two-sided matching problems in which a planner matches individuals based on a rich set of observable characteristics of the two sides. The learning problem is formulated as an empirical optimal transport problem with a match cost function estimated from training data, and an optimal matching policy is estimated by maximizing the entropy regularized empirical welfare criterion. A welfare regret bound for the estimated policy is derived and its convergence is characterized. The proposal is applied to the problem of matching caseworkers and job seekers in a job search assistance program, and its welfare performance is assessed in a simulation study calibrated with French administrative data.

E1712: Optimal treatment assignment rules under capacity constraints*Presenter:* **Keita Sunada**, University of Rochester, United States*Co-authors:* Kohei Izumi

Treatment assignment under supply constraints is studied when a planner aims to maximize social welfare by assigning treatments based on observable covariates. Such constraints are common when treatments are scarce and costly, but they complicate the analysis of optimal assignment rules because assignment probabilities must be coordinated across the entire covariate distribution. A new approach reformulates the planner's problem as an optimal transport problem, which makes the constraints analytically tractable. Using a limits of experiments framework, local asymptotic optimality results are established for two canonical decision rules: the plug-in rule and the Bayesian rule. The latter rule can dominate the former rule, with simulations demonstrating sizable risk reductions. An empirical illustration using school voucher program data demonstrates how the two rules differ in practice.

EO088 Room B102 QUANTITATIVE METHODS IN ECONOMICS AND FINANCE: THEORY AND PRACTICE**Chair: Hsin-Chieh Wong****E1406: A country of two theories: Coexistence of collusion and efficiency hypotheses***Presenter:* **Wen-Jen Tsay**, Soochow University, Taiwan

The relationship between market concentration, labor productivity, and real output is examined, evaluating the efficiency and collusion hypotheses. An oligopolistic model is developed showing that, under technological progress, a positive concentration-output relationship arises in large and competitive markets, such as internationally exposed sectors, implying that the efficiency and collusion hypotheses can coexist within an economy. Using Taiwanese census data, concentration is found to reduce output in locally-oriented service sectors despite rising productivity, while it increases output in manufacturing sectors. These findings reveal sectoral heterogeneity and suggest that uniform antitrust enforcement may lead to over-deterrence risk in Taiwan's manufacturing sector.

E1490: Distributional reinforcement learning for tail risk control in weekly options sell-side strategies*Presenter:* **ChiFang Chao**, National Chengchi University, Taiwan*Co-authors:* Ming-Hua Hsieh, Mu-En Wu

Options sell-side stop-loss is conventionally treated as a prediction problem, where fixed price thresholds trigger position exit. The decision is reframed as one of tail risk control and an IQN-CVaR framework is proposed that learns the full return distribution via Implicit Quantile Networks and selects exit actions by minimizing Conditional Value-at-Risk. A single parameter alpha controls risk sensitivity at inference without retraining. The framework is evaluated on 8,923 out-of-the-money TXO weekly option episodes spanning 2019 to 2025, using six-fold walk-forward validation and a 2025 hold-out test. At alpha = 0.50, CVaR at the 25% level improves by 41% relative to a hold-to-stop baseline in walk-forward validation, and by 71% on a 2025 hold-out test. A standard DQN with the same training budget achieves only 13% improvement in walk-forward and 2% in hold-out. The distributional agent exhibits risk-sensitive behavior that adapts to the asymmetric tail risk structure of call versus put positions. Exit timing varies smoothly with alpha, enabling practitioners to calibrate the trade-off between tail risk reduction and premium income retention.

E1429: Simulation-based american option pricing: Primal-dual approaches and machine learning*Presenter:* **Sheng-Feng Luo**, Chung Yuan Christian University, Taiwan*Co-authors:* Yu-Ying Tzeng

An integrated simulation-based framework is proposed for pricing American and Bermudan options, combining a primal-dual bounding approach with machine learning to improve accuracy and computational efficiency. The early-exercise boundary is parameterized using a double-exponential functional form and calibrated via K-fold cross-validation on simulated paths, yielding a lower bound on the option price. This boundary is then incorporated into a martingale duality framework with weighted Laguerre polynomial basis functions to derive an upper bound, forming a confidence interval for the true price. To address the suboptimality inherent in parametric stopping rules, logistic regression is applied by treating the exercise decision as a binary classification problem trained on labeled stopping data pooled across multiple initial asset prices. Numerical experiments benchmarked against finite-difference results demonstrate that the primal-dual midpoint estimator achieves low variance but carries a modest upward bias, while the logistic regression-based estimator substantially reduces bias at the cost of higher variance, illustrating a characteristic bias-variance trade-off. Strong performance is achieved without large training datasets, highlighting the practical advantage of combining parametric structure with data-driven refinement. The framework is computationally tractable and naturally extensible to multi-dimensional settings.

E1420: Model averaging predictor in an infinite-order autoregressive process*Presenter:* **Hsin-Chieh Wong**, National Taipei University, Taiwan

Least-squares forecast combination is investigated for a stationary infinite-order autoregressive process under same-realization prediction. One-step-ahead predictors fitted from a growing collection of finite-order AR models are averaged, allowing both the candidate orders and the size of the model family to increase with the sample size. An explicit asymptotic expression for the mean-squared prediction error (MSPE) of the least-squares averaging predictor in the increasing-order setting is obtained. The main technical advance is a uniform convergence theorem establishing that the MSPE approximation holds uniformly across the candidate models, providing a theoretical basis for validating model averaging criteria.

EO118 Room B103 APPLIED MICROECONOMETRICS FOR SOCIAL AND ECONOMIC OUTCOMES**Chair: Stanley Iat-Meng Ko****E1424: Toxic tradeoffs: Ambient air quality and mental health in Africa***Presenter:* **Chon-Kit Ao**, National Cheng Kung University, Taiwan*Co-authors:* Naveen Sunder

Over 90% of Africa's population is exposed to unsafe levels of airborne particulate matter, yet the mental health implications of this exposure remain largely unexplored. Using spatially linked survey and satellite data from 16 African countries, the short-run relationship between ambient PM_{2.5} pollution and mental well-being is examined. Contrary to findings from higher-income settings, increased PM_{2.5} concentrations are found to be associated with modest reductions in depression- and anxiety-related symptoms. Higher pollution may signal rising industrial activity and income in these low-income contexts, with the mental health benefits of increased economic opportunity outweighing the direct harms of pollution exposure. The results highlight the need for context-specific approaches to reconcile economic growth with environmental and mental health, emphasizing the importance of transferring cleaner production technologies to low-income settings.

E1561: Effects of gender imbalance within and across generations: Men's marrying up and family dynamics*Presenter:* **Yanjun Li**, Tohoku University, Japan*Co-authors:* Midori Wakabayashi, Yu Bai

The aim is to examine the short-run and intergenerational consequences of male scarcity induced by World War II on marriage market sorting, household outcomes, and children's development. Using prefecture-level variation in wartime male mortality to identify shifts in local gender ratios, three sets of findings emerge. First, male scarcity shifted assortative matching toward hypergamy, with men increasingly marrying women of higher social status and education. Second, despite improving men's position in the marriage market, this demographic shock increased men's bargaining power within households, constraining wives' post-marriage labor force participation while raising fertility and divorce rates. Third, children born into these mismatched households experienced lower educational attainment and weaker labor market outcomes. These intergenerational effects are concentrated among boys and are more pronounced in resource-constrained regions, consistent with limited institutional capacity to absorb demographic shocks. Together, these findings demonstrate how a temporary demographic disruption can become embedded in persistent shifts in household power structures and intergenerational mobility.

E1875: Assessing the relationship between immigration and crime in Taiwan*Presenter:* **Yi-Ju Hung**, National Chung Cheng University, Taiwan

Patterns and trends of immigration and crime rates in Taiwan are examined using administrative data to construct a county panel dataset that discusses the relationship between the immigration population and local crime rates. Leveraging the TWFE model and instrumental variables, the analysis shows that there is no significant association between the share of the population that is immigrant and local crime rates.

E1625: Log-OLS and mean elasticities: A specification test*Presenter:* **Stanley Iat-Meng Ko**, Tohoku University, Japan

Log-linear regressions are widely used to estimate elasticities and proportional effects in applied economics. However, coefficients from log-OLS generally recover derivatives of the conditional expectation of the log outcome, whereas many empirical interpretations concern elasticities of the conditional mean. These two objects need not coincide. The aim is to study when log-OLS admits an elasticity interpretation for the conditional mean. It is shown that, under a multiplicative conditional mean benchmark, this interpretation requires an orthogonality restriction between regressors and the log regression error. A specification test of this restriction is developed using residuals implied by multiplicative mean estimators, and the relative asymptotic efficiency of log-OLS and standard multiplicative alternatives is characterized. Monte Carlo simulations illustrate the finite-sample behavior of the test, and applications to trade gravity equations, health expenditure models, and firm-level investment outcomes show that the distinction is empirically consequential. The results provide a practical diagnostic for assessing when log-linear regressions recover mean elasticities and when multiplicative estimators should be preferred.

EO153 Room B104 FINANCIAL ECONOMETRICS AND FINANCIAL BIG DATA**Chair: Yi Ding****E1295: Intraday correlation patterns***Presenter:* **Zhiyuan Zhang**, Shanghai University of Finance and Economics, China*Co-authors:* Torben G Andersen, Viktor Todorov, Yingwen Tan

A functional central limit theorem is established for estimating the diurnal pattern of instantaneous correlations between two financial assets, using high frequency data over a long span of time. This is the first functional central limit theory ever built for such high frequency characteristics of asset return correlations. An estimator of the asymptotic covariance function is proposed, rendering the limit theorem feasible in a two-sample test for the equivalence of diurnal patterns of two non-overlapping time periods. Simulation evidence supports the theoretical findings, while empirical results on the E-mini S&P 500 and US Treasury bond futures show that correlations exhibit pronounced intraday periodicity, with the diurnal patterns varying significantly over time.

E1345: A general score-difference framework for independence testing*Presenter:* **Wenhao Cui**, Beihang University, China

A general framework is developed for testing unconditional and conditional independence based on comparisons of predictive distributions using strictly proper scoring rules. The central idea is to characterize independence through the invariance of predictive performance, whereby under the null, augmenting the information set does not improve expected scores, whereas violations induce systematic score differences. This perspective yields a flexible class of distributional tests that accommodates discrete, continuous, and mixed data, and is naturally compatible with modern machine learning methods for estimating conditional distributions via cross-fitting. To address degeneracy in score-based criteria, a randomized construction is introduced that delivers asymptotically pivotal inference under weak temporal dependence. In addition, localized tests are developed that detect dependence in specific regions of the outcome distribution, capturing heterogeneous and tail-specific relationships. Simulation studies and empirical applications demonstrate the effectiveness of the proposed approach.

E1384: A fine lens on common trading flows*Presenter:* **Songze Shi**, University of Macau, China*Co-authors:* Yi Ding

A continuous-time, high-dimensional factor model for trading volume is proposed. The model features a common trading intensity (CTI) factor that captures aggregated trading information flow and reflects common systematic risk. A statistical framework for spot estimation of the CTI process using trading volume aggregated over short intervals from a large cross-section of stocks is developed, and a new statistical test for deciding its roughness property is proposed. Using high frequency trading data of U.S. stocks, strong evidence is found that the CTI displays rough dynamics.

E1471: Modeling and forecasting realized volatility with multivariate fractional Brownian motion*Presenter:* **Chen Zhang**, Sun Yat-sen University, China*Co-authors:* Jun Yu, Markus Bibinger

A multivariate fractional Brownian motion (mfBm) with component-wise Hurst exponents is used to model and forecast realized volatility. The interplay between correlation coefficients and Hurst exponents is investigated and a novel estimation method for model parameters is proposed, establishing consistency and asymptotic normality of the estimators. Additionally, a time-reversibility test is developed, which is typically not rejected by real volatility data. When the data generating process is a time-reversible mfBm, optimal forecasting formulae are derived and their properties analyzed. A key insight is that an mfBm with different Hurst exponents and non-zero correlations can reduce forecasting errors compared to a one-dimensional model. Consistent with this theory, out-of-sample forecasts using the time-reversible mfBm show improvements over univariate fBm, particularly when the estimated Hurst exponents differ significantly. Empirical results demonstrate that mfBm-based forecasts outperform the scalar HAR, vector HAR and vector HAR with a common factor.

EO190 Room 101 RECENT ADVANCES IN TIME SERIES ANALYSIS**Chair: Giovanni Motta****E1260: Nonstationarity and cyclicity extended Whittle estimation***Presenter:* **Edward Hill**, Queen Mary University of London, United Kingdom

The estimation of a time series exhibiting a persistent cycle is considered. Such dynamics are common in economic, epidemiological and geophysical data and encompass stationary cyclical long memory, nonstationary cyclical dependence and cyclical unit-root behaviour. The generalised Gegenbauer autoregressive moving average (GARMA) process provides a flexible framework for modelling this wide class of oscillatory phenomena. Using a modified Whittle likelihood function based on a corrected periodogram, the parameter estimates of the GARMA model are shown to be consistent and asymptotically normal in both stationary and nonstationary settings. In addition, the asymptotic properties of the estimator are unaffected by preliminary estimation of the cycle frequency. Monte Carlo simulations confirm the good finite-sample performance of the procedure and an application to United Kingdom influenza infections data illustrates its practical usefulness for modelling persistent seasonal dynamics.

E1564: Semi-parametric estimation of non-stationary autoregressive models*Presenter:* **Giovanni Motta**, Columbia University, United States*Co-authors:* Qiyuan Wang

A novel semi-parametric approach is developed for accurately estimating time-varying mean and variance in autoregressive (AR) models. By combining B-splines with (i) generalized least squares (GLS) estimation to account for serial correlation, and (ii) weighted least squares (WLS) for smooth parametrization, the approach addresses the challenges posed by time-varying dynamics in time series data. The covariance matrix in the GLS estimation of the spline coefficients is iteratively updated through the WLS estimation of the AR coefficients in a band-limited manner. A new autoregressive model is proposed that incorporates time-varying variance with a finite bounded envelope function, and a novel method is introduced to estimate it through splines. Additionally, the order of the AR model is determined through a generalized Bayesian information criterion (GBICp) that incorporates prior information. The effectiveness of the methodology is demonstrated through extensive simulations and applications to Federal Reserve Economic Data. Asymptotic theory is derived for the approach and rates of convergence are compared with those achieved by existing methods.

E1726: Linear hypothesis test for dynamic factor model*Presenter:* **Rikako Senoo**, Waseda University, Japan*Co-authors:* Giovanni Motta, Yan Liu

A hypothesis testing framework is proposed for the linear structure of latent factors in an approximate factor model. Despite the widespread use of dynamic factor models and dimensional reduction via PCA, formal hypothesis testing for latent factor dynamics remains unexplored. To address this gap, a linear hypothesis testing problem for factor dynamics is considered, where the latent factors follow a VAR(p) process within a general approximate factor model. Estimation proceeds in two steps: latent factors are first estimated via PCA, followed by OLS estimation of the VAR coefficient matrices. Asymptotic normality of the estimated VAR coefficient matrices is derived under two regimes, $\sqrt{N}/T \rightarrow 0$ and $\sqrt{N}/T$ does not vanish as $N, T \rightarrow \infty$, and a Wald-type statistic for general linear hypotheses is proposed. The statistic is constructed using a rotated OLS estimator and is shown to be asymptotically chi-square distributed. The rotated OLS estimator converges directly to the true VAR coefficient matrices, enabling valid inference on the underlying linear structure and overcoming the rotational indeterminacy inherent in the PCA-based approach. Simulation results demonstrate strong finite-sample performance and are consistent with the theoretical findings. Empirical applications of the proposed methods are also presented.

E1736: Statistical inference for multinomial-valued time series*Presenter:* **Yan Liu**, Waseda University, Japan*Co-authors:* Wangzhengtian Su, Yan Liu

Statistical inference for multinomial-valued time series, which commonly arise in economics, epidemiology, and social sciences, is studied. Classical time series models usually fail to account for discreteness, non-negativity, and dependence across multiple categories. A multinomial count time series model is considered and a quasi-maximum likelihood estimation framework tailored to this setting is developed. The proposed estimator is a multinomial quasi-maximum likelihood estimator constructed from the multinomial quasi-likelihood function. Consistency and asymptotic

normality of the estimator are established under general regularity conditions, and its asymptotic distribution is derived, including non-standard limits when parameters lie on the boundary of the parameter space. A local asymptotic normality representation of the multinomial quasi-likelihood is obtained, which enables a precise characterization of these boundary asymptotic behaviors. Monte Carlo simulations demonstrate that the finite sample performance of the estimator is consistent with the theoretical asymptotic results. The simulation results show accurate parameter estimation and support the validity of the derived asymptotic distributions. A real data application illustrates the practical usefulness of the proposed method.

EO097 Room 103 ADVANCED STATISTICS AND MACHINE LEARNING FOR COMPLEX DATA ANALYSIS
Chair: Henghsiu Tsai
E1510: An automatic approach to explainable AI with applications to medical image classification

Presenter: **Tso-Jung Yen**, Academia Sinica, Taiwan

Recent advances in explainable AI have inspired researchers to develop methods that can effectively detect input impact on the output of a predictive model. However, most of these methods require multiple stages to process the input. Some of the stages involve re-sampling the input data. When there are large amounts of input data, such a re-sampling procedure may take time to proceed. A method is proposed that can automatically detect input impact on the output of the predictive model. The method relies on the idea that turns the impact detection problem to a prediction problem. Simulation experiments show that the method can correctly carry out the input impact detection without spending too much time on input processing.

E1497: The kriging-correlation score for feature importance in high dimensional clustering

Presenter: **Cheng-Han Chua**, Tunghai University, Taiwan

Co-authors: Meihui Guo, Shih-Feng Huang

A kriging-correlation (KC) score is proposed to identify the most critical clustering features from high-dimensional datasets. The KC score ranks feature importance through a three-stage procedure. In the first stage, an appropriate distance metric is learned from the data, and the original data is projected into a two-dimensional space using the t-distributed stochastic neighbor embedding (t-SNE) method. The resulting 2D coordinates are treated as essential clustering variables in the reduced space. In the second stage, each original feature is reconstructed from the reduced space using the AutoFRK method. In the final stage, feature importance is assessed by measuring the dependence between each original feature and its reconstructed version; the number of significant features is determined using MANOVA. The KC score is applied to four single-cell datasets and its performance compared with the Laplacian score, a widely used unsupervised feature ranking method. The results demonstrate that the KC score can identify a smaller set of features while achieving equal or better classification accuracy compared to the Laplacian score.

E1512: Statistical analysis of the genetic correlations between quantitative traits

Presenter: **Chen-Hung Kao**, Institute of Statistical Science, Taiwan

It is well known that phenotypic correlation between two traits is an outcome of interplays among genetic correlation, residual correlation, and heritability. Also, genetic correlations between different quantitative traits are caused by the closely linked and pleiotropic QTL. It shows by example that, for a pair of traits, high (low) phenotypic correlation does not imply high (low) genetic correlation, and vice versa, as the heritability and residual correlation also play roles in affecting the strength of phenotypic correlation. The phenotypic and genetic correlations seem to have an arbitrary relationship, making accurate prediction of each other difficult. Statistical analysis is presented to dissect the genetic correlations between traits by showing that, depending on the effect sizes and linkage parameters, multigenic traits controlled by tightly linked and pleiotropic QTL do not necessarily show significant genetic correlations with each other, because linkages between different QTL may make different levels of (positive or negative) contribution and may combine to produce no genetic correlation, or very weak to very strong genetic correlations. Analytical and numerical examples are presented.

E1416: A doubly-threshold diffusion model: One threshold in drift, one in diffusion

Presenter: **Henghsiu Tsai**, Academia Sinica, Taiwan

Co-authors: Edward Meng-Hua Lin

A novel threshold diffusion model is proposed in which the drift and diffusion components undergo regime shifts at distinct threshold values. Specifically, the drift term switches regimes at one threshold, while the diffusion term transitions at a potentially different level. This doubly threshold specification offers greater flexibility for capturing asymmetric dynamics in both the conditional mean and volatility. To estimate the model, an approximate maximum likelihood estimation (AMLE) procedure is developed that remains computationally tractable despite the model's structural complexity. Monte Carlo simulations demonstrate that the proposed estimator is both consistent and efficient, and that standard information criteria (AIC, BIC, HQIC) can effectively distinguish between models with shared versus separate thresholds. Empirical applications to U.S. Treasury interest rate data-including the 3-Month Treasury Bill and the 10-Year/3-Month yield spread-further support the proposed framework. The results reveal structural asymmetries that are better captured by allowing distinct threshold mechanisms in the drift and diffusion terms.

EO180 Room 104 ADVANCES IN CAUSAL INFERENCE AND FUNCTIONAL DATA ANALYSIS
Chair: Won-Ki Seo
E1259: Vector-to-density cointegration for climate damage assessment

Presenter: **Kyungsik Nam**, Hankuk University of Foreign Studies, Korea, South

Co-authors: Won-Ki Seo

Global mean temperature is the standard summary of climate change, but economically relevant climate risk is distributional. A vector-to-density cointegration framework is developed that treats the temperature anomaly distribution as a density-valued functional outcome and relates its persistent evolution to a finite-dimensional vector of heterogeneous radiative forcing processes. A formal test for this long-run relation is derived and, conditional on its empirical support, estimation and simulation-based inference for forcing-specific distributional responses are developed. Empirically, stock-like long-lived greenhouse forcing is dominated by warming translation with dispersion and upper-tail expansion; chemically coupled atmospheric forcing combines moderate warming with central compression; and localized non-greenhouse forcing generates a cooling-oriented central response with outward redistribution. These findings suggest that scalar climate sensitivity may be an incomplete guide to climate-risk assessment when forcing processes differ in how they reshape temperature risk.

E1575: Functional regression with nonstationarity and error contamination: Application to the economic impact of climate change

Presenter: **Won-Ki Seo**, University of Sydney, Australia

A regression model with functional dependent and explanatory variables exhibiting nonstationary dynamics is studied. The model assumes that the nonstationary stochastic trends of the dependent variable are explained by those of the explanatory variable, and hence that a stable long-run relationship exists between the two variables despite their nonstationary behavior. Functional observations may be error-contaminated. Novel autocovariance-based estimation and inference methods are developed for this model. The methodology is broadly applicable to economic and statistical functional time series with nonstationary dynamics. Application to evaluating the global economic impact of climate change demonstrates the methodology's usefulness and addresses an issue of intrinsic importance.

E1658: Counterfactual concentration curves in difference-in-differences settings: Identification and inference*Presenter:* **Mohamad Khaled**, University of Queensland, Australia*Co-authors:* Paul Makdissi, Myra Yazbeck

A framework for identifying concentration curves in a difference-in-differences setting is introduced. Building on the changes-in-changes and distributional difference-in-differences frameworks for identifying quantile treatment effects, the identification strategy is generalized to cases where the outcome variable is multidimensional, thereby allowing for nonparametric identification of bivariate objects such as concentration curves. Nonparametric estimators of counterfactual concentration curves are derived, and their asymptotic properties are investigated.

E1725: Dynamic difference-in-discontinuities*Presenter:* **Dakyung Seong**, University of Sydney, Australia*Co-authors:* Shu Shen

The difference-in-discontinuities (diff-in-disc) design is a widely used empirical framework to address identification failures in the traditional regression discontinuity (RD) design due to recurring confounding treatments that utilize the same policy eligibility cutoff as the new policy of interest. The repeated treatment nature of the diff-in-disc design is formalized within a general potential outcome framework that accounts for both treatment effect heterogeneity and dynamic treatment effects, including carryover effects and path dependency in contemporaneous effects. Both standard sharp and fuzzy diff-in-disc setups are considered. New identification and estimation strategies for different scenarios are proposed, and the small sample performance of the proposed estimators is studied using Monte Carlo simulations. The proposed method is applied to a seminal study on the impact of relaxing fiscal rules, illustrating how the proposed methods complement existing approaches.

EO082 Room 105 NEW METHODS FOR SPATIAL AND NETWORK DATA**Chair: Chao Yang****E1454: Differentially private estimation and inference for spatial autoregressive models***Presenter:* **Danyang Huang**, Renmin University of China, China

Privacy-preserving data analysis has attracted increasing attention in modern statistics. However, how to simultaneously protect both the network structure and the nodal information remains a challenging problem. Differentially private estimation and inference for spatial autoregressive (SAR) models is investigated, aiming to protect the privacy of network structure and individual nodal information. The minimax lower bounds for differentially private estimators under the SAR model are first derived, providing a theoretical benchmark for optimal algorithm design. Building on these insights, a differentially private algorithm for SAR parameter estimation is proposed and a comprehensive analysis of its convergence properties is presented, including examinations of representative cases in specific networks. Additionally, differentially private inference procedures for the SAR model are developed. Empirical validation through simulations and real-world data analysis confirms the reliable finite-sample performance of the proposed methodology.

E1730: Consistent generalized method of moments estimation of spatial autoregressive models based on non-Gaussian distributions*Presenter:* **Fei Jin**, School of Economics, Fudan University, China

A generalized method of moments (GMM) estimation approach for the spatial autoregressive (SAR) model is proposed using generalized linear and quadratic moments based on non-Gaussian distributions. Both homoskedastic and heteroskedastic innovations are investigated. The proposed GMM estimator of parameters of interest is consistent and asymptotically normal under regularity conditions. It is computationally simple and can yield significant efficiency improvements over popular estimators of the SAR model. In the homoskedastic case, conditions for the existence of best moments that generate a GMM estimator with minimum asymptotic variance are provided. Simulation experiments demonstrate that the considered GMM estimator performs well in finite samples.

E1743: Heteroskedasticity-robust GMM for hierarchical network models with random group effects*Presenter:* **Jiankun Chen**, University of International Business and Economics, China*Co-authors:* Yanli Lin, Yang Yang

Identification and estimation are studied in a cross-sectional hierarchical network model in which individuals interact within groups while groups interact across groups. The model accommodates random group effects and unknown heteroskedasticity at the individual level. Although the stacked system can be written in a SAR-like form, the hierarchical structure creates a distinct econometric problem: after eliminating group effects by within-group demeaning, the transformed error covariance is no longer diagonal, so the conventional zero-diagonal quadratic-moment construction used in spatial GMM is not robust to heteroskedasticity. A corrected class of quadratic moments for the within-transformed equation is derived and paired with a second class of between-group quadratic moments based on group averages. The two classes of moments exploit different sources of variation and separately identify the within-group and inter-group network parameters. Consistency and asymptotic normality are established for robust and optimally weighted GMM estimators, and a feasible estimator for the covariance matrix of the linear and between-group quadratic moments is provided under unknown heteroskedasticity. Monte Carlo results show that the proposed procedure improves substantially on linear-moment estimators and remains robust under heteroskedasticity.

E1756: Multidimensional asset networks and momentum spillover effects*Presenter:* **Chao Yang**, Shanghai University of Finance and Economics, China

Owing to misspecification bias, it is challenging for traditional asset pricing models to capture the multidimensional network correlations of stock returns in the A-share market. Meanwhile, existing research on momentum spillover effects predominantly focuses on a single dimension of association, resulting in incomplete analysis. This study examines Chinese Shanghai and Shenzhen A-share listed companies from 2006 to 2024, introducing model averaging methods into momentum spillover effect analysis. Five types of spatial weight matrices are constructed: analyst co-coverage, geographic proximity, industry classification, technological similarity, and statistical similarity. Optimal weights are determined by minimizing the Mallows criterion and integrating multi-matrix information to form a comprehensive asset network structure. The return performance and pricing effectiveness of long-short portfolios under single-matrix versus model-averaged matrix specifications are then compared. Findings reveal significant heterogeneity in how different spatial association dimensions explain stock return differentials. Single-dimension portfolios generally exhibit weak alpha significance and time-series stability. In contrast, portfolios constructed using the model averaging method demonstrate clear quantile monotonicity characteristics, with outstanding performance in raw excess returns and cumulative excess returns.

EO187 Room 106 STOCHASTIC PROCESSES AND PROBABILISTIC INFERENCE WITH DEPENDENT DATA**Chair: Geoffrey Wolfer****E1217: Empirical PAC-Bayes bounds for Markov chains****Presenter:** Pierre Alquier, ESSEC Business School, Singapore**Co-authors:** Vahe Karagulyan

The core of generalization theory was developed for independent observations. Some PAC and PAC-Bayes bounds are available for data that exhibit a temporal dependence. However, there are constants in these bounds that depend on properties of the data-generating process: mixing coefficients, mixing time, spectral gap... Such constants are unknown in practice. A new PAC-Bayes bound for Markov chains is proven. This bound depends on a quantity called the pseudo-spectral gap. The main novelty is that an empirical bound on the pseudo-spectral gap can be provided when the state space is finite. Thus, the first fully empirical PAC-Bayes bound for Markov chains is obtained. This extends beyond the finite case, although this requires additional assumptions. On simulated experiments, the empirical version of the bound is essentially as tight as the non-empirical one.

E1451: Sample complexity for covariance estimation via the unadjusted Langevin algorithm**Presenter:** Shogo Nakakita, The University of Osaka, Japan

Estimating the covariance matrix of a Gibbs distribution with an unknown normalizing constant is a ubiquitous problem in statistics and machine learning. In particular, the high dimensionality of target distributions is frequently encountered, making the sample complexity and query complexity required to achieve a prescribed error tolerance with high probability an important concern of the community. The unadjusted Langevin algorithm (ULA) is studied, for which the sample complexity matches the query complexity, and its sample efficiency is analyzed. Furthermore, two settings are compared: (i) running a single long trajectory and estimating the covariance from the resulting dependent sequence, and (ii) implementing an embarrassingly parallel strategy that generates independent outputs and estimates the covariance from independent samples. It is shown that the sample complexity of the single-path approach is better than that of the parallel approach by a logarithmic factor.

E1615: Recent progress of minimum information dependence modeling**Presenter:** Keisuke Yano, The Institute of Statistical Mathematics, Japan

The Minimum Information Dependency Model is a statistical Model designed to handle higher-order dependencies in mixed-domain data. Applications of the Model are discussed, along with its theoretical properties including its relationship to multimarginal entropic optimal transport. Estimation methods and optimization techniques are presented, including acceleration, stabilization, and uncertainty quantification by combining Adam with the Ruppert-Polyak-Juditsky method.

E1223: Group-averaged Markov chains II: Tuning of group action in finite state space**Presenter:** Michael Choi, National University of Singapore, Singapore

Group-averaged Markov chains obtained by augmenting a π -stationary transition kernel P with a group action on the state space via orbit kernels are studied. Given a group G with orbits $(O_i)_{i=1}^k$, three canonical orbit kernels are analysed: namely the Gibbs (G), Metropolis-Hastings (M), and Barker (B) kernels, as well as their multiplicative sandwiches QPQ and the additive mixtures $\frac{1}{2}(P+Q)$ where $Q \in \{G, M, B\}$. It is shown that orbit averaging never deteriorates the absolute spectral gap or asymptotic variance when P is reversible. A direct and simple proof of pythagorean identity under the Kullback-Leibler (KL) divergence is given, showing that GPG arises naturally as an information projection of P onto the set of G -invariant transition matrices. For a given P , the optimal choice of G with a fixed number of orbits that minimises the one-step KL divergence to stationarity is characterised. Analogously, for a given G , the optimal choice of P is characterised and sufficient conditions under which $GPG = \Pi$ are given. It is further shown that alternating projections over multiple group actions converge at a rate governed by the singular values of an overlap matrix, and that in structured cases, this yields exact sampling where the number of group actions grows logarithmically with the size of the state space.

EO284 Room 107 RECENT DEVELOPMENTS IN CAUSAL INFERENCE AND SURVIVAL ANALYSIS**Chair: Liming Xiang****E1681: Multilevel joint model for correlated survival and binary outcome data: Application to a multicentre trial****Presenter:** Richard Tawiah, Monash University, Australia

In many individually randomised trials, randomisation is stratified by centre; appropriate methods of analysis require centre to be accounted for in the analysis of outcomes. This is usually done by including random intercepts for centres in outcome regression models. Moreover, in most situations, multiple outcomes that are not commensurate in scale, such as time-to-event (e.g., survival) and binary or continuous endpoints (e.g., biomarker), are measured on the same subjects, thus additionally introducing intra-subject correlation between the outcomes. Joint models remain the centrepiece for simultaneous modelling of outcomes to account for their correlation. However, joint models allowing centre-specific random effects have rarely been considered, possibly due to estimation and computational complexities. A multilevel joint model is described, presenting a unified framework that simultaneously models survival and binary outcomes, utilising bivariate nested random intercepts to characterise intra-subject correlation and centre effect. Estimation is facilitated by a semiparametric restricted maximum likelihood (REML) method. Estimation difficulties are discussed, showcasing how the method simplifies calculations. Using simulation studies and the allogeneic bone marrow transplantation trial, an individually randomised multicentre trial, the method is shown to improve statistical inference as opposed to methods that provide separate analysis for the outcomes.

E1266: Estimating conditional complier quantile treatment effect via stratified quantile regression**Presenter:** Mengjiao Peng, East China Normal University, China, China

Understanding the causal effect of a treatment in randomized experiments with non-compliance is of fundamental interest in many domains. Within the instrumental variable (IV) framework, the causal treatment effect can only be reliably assessed for compliers, as they are the only subpopulation whose treatment assignment is influenced by the instrument. The conditional complier quantile treatment effect based on individual characteristics is studied through stratified quantile regression models for compliers with and without treatment, which are flexible in capturing the interaction between treatment and covariates and include the past unified model as a special case. A tuning parameter-free method is introduced that directly utilizes the mixture structure in the complier problem, departing from past approaches that relied on minimizing a weighted check function with nonparametric method-estimated weights. A novel iterated algorithm is proposed to solve discontinuous equations that involve unknown parameters in a complicated way. The consistency and asymptotic normality of the proposed estimators are established. Numerical results, including extensive simulation studies and real data analysis of the Oregon health insurance experiment and a job training study, show the practical utility of the proposed approach.

E1329: Robust estimation of population attributable fractions in the presence of multiple ordered mediators**Presenter:** Han Chi Peng, Institute of Statistics, National Yang Ming Chiao Tung University, Taiwan

Population attributable fraction (PAF) is a key epidemiological measure used to quantify the contribution of risk factors to the overall disease burden. However, when an exposure affects an outcome through multiple ordered mediators, traditional PAF estimation methods face challenges in accurately identifying the impact of each mediating pathway. These challenges arise from mediator-outcome relationships, interactions among mediators, and the presence of potential confounders. New measures, termed mPAFs, are proposed to quantify the fraction of disease attributable

to a specific mediation pathway. The proposed framework incorporates a multiply robust estimator yielding consistent mPAF estimates if at least two of the three types of models are correctly specified: exposure, mediator, or outcome models. The asymptotic properties of the estimator are formally established, and simulations demonstrate its robustness against misspecification. In a TCGA lung cancer cohort application, smoking's effect on mortality mediated through TTK and MAD2L1 is analyzed. In lung adenocarcinoma, total PAF was 4.45%, with a direct effect of 1.82% and pathway-specific contributions of -1.95% (TTK) and 0.68% (MAD2L1). In contrast, lung squamous cell carcinoma showed a 10.43% total PAF, with most of the effect attributable to the direct pathway (10.22%), suggesting minimal mediation via the selected genes.

E1265: **Beyond hazard-based illness-death models for semi-competing risks data**

Presenter: **Liming Xiang**, Nanyang Technological University, Singapore

Co-authors: Rui Huang

The illness-death model has been developed as a common approach for regression analysis of semicompeting risks data by modeling transitions between states using the proportional hazards or accelerated failure time model. An illness-death model is proposed that is formulated based on the mean residual life function, offering a more straightforward and interpretable framework for modeling state transitions in the semicompeting risks setting. Estimation is facilitated through novel estimating equations derived from a penalized quasi-likelihood approach incorporating inverse probability weighting. The method requires no distributional assumption on the latent frailty term, thereby reducing the risk of model misspecification. Simulation studies demonstrate its promising performance across a range of realistic scenarios and an application to a myeloma progression study illustrates its practical utility.

EO014 Room 201 ADVANCES AND FRONTIERS IN STATISTICAL MODELING AND COMPUTING

Chair: Tsung-I Lin

E1165: **Finite mixtures of multivariate contaminated normal censored regression models**

Presenter: **Wan-Lun Wang**, National Cheng Kung University, Taiwan

Co-authors: Tsung-I Lin

The complexity of model-based clustering grows as outliers become more prevalent, compounded by restrictions imposed by the detection of quantification. A finite mixture of multivariate contaminated normal censored regression (FM-MCNCR) model tailored for handling censored data in a linear regression scenario is introduced. For the estimation of model parameters, a computationally analytical alternating expectation conditional maximization (AECM) algorithm is devised. Additionally, an information matrix-based formula to approximate the asymptotic standard errors of parameter estimates is presented. Importantly, the AECM algorithm serves a dual role by not only facilitating parameter estimation but also providing methods to recover censored measurements and detect outlier data points as a by-product when it converges. The efficacy and advantages of the proposed methodology are illustrated through a simulation study and an example related to U.S. women's labor force participation data.

E1166: **Modeling overdispersed spatial data by using mixture random fields**

Presenter: **Luis Mauricio Castro Cepero**, Pontificia Universidad Catolica de Chile, Chile

Co-authors: Diego Morales, Moreno Bevilacqua

Modeling and estimating overdispersed count data present significant challenges, particularly when data are continuously indexed in space. A spatial random field model with negative binomial marginal distributions is introduced, called the spatial PoissonErlang mixture random field. A Poisson random field is first constructed via a renewal-process representation with exponential inter-arrival times. Its random mean is then modeled using an Erlang random field. The resulting PoissonErlang mixture random field is overdispersed, has negative binomial marginals, and enjoys desirable second-order properties, including mean-square continuity. For the proposed Poisson-Erlang spatial random field, analytic expressions for the covariance function and bivariate distribution are provided. These features facilitate likelihood-based inferences using (potentially misspecified) weighted pairwise likelihood methods. An extensive simulation study was conducted to investigate the performance of the weighted pairwise likelihood approach for estimating the parameters of the Poisson-Erlang random field. The model was applied to analyze weed count data from the Bjertop Farm in southwest Sweden.

E1168: **Differential gene expression via local false discovery rate with nonparametric maximum likelihood**

Presenter: **Sangkon Oh**, Pukyong National University, Korea, South

Co-authors: Geoffrey McLachlan

Two-component mixture models are particularly useful for identifying differentially expressed genes, but their performance can deteriorate markedly when the alternative distribution departs from parametric assumptions or symmetry. A semiparametric mixture model is proposed in which the null component follows a standard normal distribution, while the alternative is modeled as a skew-normal scale mixture with an unspecified scale-mixing distribution. This formulation accommodates skewness and heavy-tailed behavior, providing a flexible and computationally tractable framework for differential gene expression analysis under a nonparametric maximum likelihood paradigm, without imposing restrictive distributional assumptions. Identifiability and consistency of the proposed model are established and an efficient estimation algorithm is developed that incorporates nonparametric maximum likelihood estimation of the scale distribution. Numerical studies demonstrate notable improvements over existing parametric and nonparametric approaches for modeling the alternative distribution, and applications to colon cancer and leukemia datasets illustrate reduced false discovery and false negative rates.

E1208: **Stein-rule shrinkage for stochastic gradient estimation in high dimensions**

Presenter: **Mohammad Arashi**, Ferdowsi University of Mashhad, Iran

Co-authors: Mahmood Amintoosi

Stochastic gradient computation is formulated as a high-dimensional estimation problem and a decision-theoretic framework based on Stein-rule shrinkage is introduced. A shrinkage gradient estimator is constructed that adaptively contracts noisy mini-batch gradients toward a stable restricted estimator derived from historical momentum. The shrinkage intensity is determined data-drivenly using an online estimate of the gradient noise variance, leveraging second-moment statistics commonly maintained by adaptive optimization methods. Under a Gaussian noise model, it is shown that the proposed estimator uniformly dominates the standard stochastic gradient under squared error loss and is minimax-optimal in the classical decision-theoretic sense. It is further demonstrated how this estimator can be incorporated into the Adam optimizer, yielding a practical algorithm with negligible additional computational cost. Empirical evaluations show consistent improvements over Adam in the large-batch regime. The results illustrate that classical shrinkage principles provide a principled and effective approach to improving stochastic gradient estimation in modern deep learning.

EO216 Room 202 RECENT ADVANCES IN ROBUST STATISTICS**Chair: Masao Ueki****E1240: New robust estimation methods for diagnostic classification models****Presenter:** Elena Castilla, Universidad Rey Juan Carlos, Spain

A new estimation framework is presented for Cognitive Diagnosis Models (CDMs), focusing on the Loglinear CDM (LCDM). CDMs aim to provide detailed information about individuals' mastery of multiple skills, making them especially useful in educational assessment. While joint maximum likelihood estimation (JMLE) has recently been shown to be statistically consistent for these models, it may be sensitive to model misspecification and data contamination. To address this limitation, a class of joint minimum divergence estimators based on the Cressie Read family of divergences is introduced. This approach generalizes JMLE and allows for a flexible trade-off between efficiency and robustness. The theoretical properties of the proposed estimator are established and its performance is evaluated through simulation studies and real data applications. Results suggest that the new method preserves the desirable asymptotic properties of JMLE while offering improved robustness in finite samples.

E1337: Robust determination of the number of factors via divergence information criteria**Presenter:** Subhrajyoti Roy, Washington University in St. Louis, United States**Co-authors:** Abhik Ghosh, Ayanendranath Basu

Estimating the true rank of a noisy data matrix is a fundamental problem in multivariate statistics and econometrics, including the determination of the number of latent factors in panel data. Existing rank estimation criteria are categorized into two broad types: information-based penalization methods and cross-validation techniques. Existing information criteria are often inaccurate in the presence of outliers, even when combined with robust factorization methods. On the other hand, cross-validation approaches quickly become computationally intractable when combined with robust estimators. A new criterion is proposed, the divergence information criterion for matrix rank (DICMR), that achieves both robustness and computational simplicity. Derived from the density power divergence framework, DICMR inherits strong robustness properties while avoiding the heavy computational burdens of resampling methods. Asymptotic bounds on its overestimation and underestimation probabilities are derived and the first-order B-robustness of the criteria is demonstrated. Extensive simulations show that DICMR delivers accuracy comparable to robustified cross-validation methods but with far lower computational costs. The practical utility of DICMR is demonstrated in real-world applications, e.g., extracting latent factors from cross-country panel data, and modelling recommender systems.

E1574: Consistency as a testable property: Statistical methods to evaluate AI agent reliability**Presenter:** Subhabrata Majumdar, Indian Institute of Management Bangalore, India**Co-authors:** Cheryl Flynn, Aritra Guha, Niranjana Krishna, Harsh Raj, Suvrorup Mukherjee

A statistical framework is introduced for measuring and testing consistency in AI agents along two dimensions: observable reliability (output-level consistency) and internal reliability (trajectory-level stability). Treating an agent's execution trajectory as a stochastic process, both dimensions are formalized as testable hypotheses using U-statistic theory. For observable reliability, a distinction is made between output randomness (intrinsic sampling stochasticity) and input sensitivity (robustness to semantic perturbations), developing asymptotic hypothesis tests with provable size and power guarantees. For internal reliability, this method is extended using Maximum Mean Discrepancy over execution traces, with kernels based on action histograms and sequential alignment. The framework is evaluated on three diverse agentic benchmarks under prompt-level and task-level perturbations. Results reveal significant consistency gaps: introducing a decoy MCP server drops accuracy from 65% to 9% and sequence consistency to 33%, exposing extreme tool-selection fragility even when core capability is preserved.

E1688: A robust persistent homology trimming approach**Presenter:** Subhra Sankar Dhar, IIT Kanpur, India**Co-authors:** Tuhin Subhra Mahato

A robust version of persistent homology based on trimming methodology is studied to capture geometric features through support of the data in the presence of outliers. The proposed methodology operates when outliers lie outside the main data cloud as well as inside the data cloud. Theoretical analysis establishes that the Bottleneck distance between the proposed robust version of persistent homology and its population analogue can be made arbitrarily small at a certain rate for sufficiently large sample size. The practicality of the methodology is demonstrated on various simulated data and benchmark real data associated with cellular biology.

EO139 Room 203 EXPERIMENTAL DESIGNS: AN INTERSECTION BETWEEN STATISTICS AND ALGEBRA Chair: Frederick Kin Hing Phoa**E1250: D-optimality of multi-level circulant almost orthogonal arrays****Presenter:** Xiao-Nan Lu, Gifu University, Japan

Circulant almost orthogonal arrays (CAOAs) are a type of circulant matrix used in the statistical design of fMRI experiments. D-optimality, which maximizes the determinant of the information matrix of a design, plays a key role in ensuring experimental efficiency. The presentation focuses on strength-2 CAOAs with at least three levels, emphasizing the characterization of their D-optimality. If time permits, related topics on combinatorial designs with favorable algebraic and combinatorial properties are also presented.

E1243: Construction and analysis of (near-)CCOP design for mixed-level order-of-addition experiments**Presenter:** Frederick Kin Hing Phoa, Academia Sinica, Taiwan**Co-authors:** Ching-Yen Chuang

Mixed-level order-of-addition (OofA) experiments arise in settings where responses depend jointly on the sequence in which experimental components are added and on the levels at which these components are set, with different components potentially having different numbers of levels. In many applications, some components contribute only through their order, whereas others possess additional levels that further influence the response. Identifying optimal order level combinations is therefore a fundamental yet challenging task, as the number of possible configurations grows combinatorially and experimental trials are often expensive. To address this challenge, an extension of complete consecutive order-pairing (CCOP) designs to mixed-level OofA experiments is proposed. The proposed designs ensure complete coverage of all consecutive component pairs across levels while achieving minimal or near-minimal run sizes. Explicit construction methods are provided, structural properties of the designs are established, and their optimality with respect to run size under broad conditions is proved. For analysis, a procedure based on the distance-based linear model (DBLM) is developed, which accommodates both ordering effects and level effects within a unified framework. Numerical studies and a real drug-combination experiment demonstrate that the proposed design analysis strategy reliably identifies superior order level settings using substantially fewer runs than competing methods.

E1249: Design and analysis of a new class of three-level factorial design for screening experiments**Presenter:** Cinyu Shih, National Taiwan University, Taiwan**Co-authors:** Frederick Kin Hing Phoa

Three-level screening designs aim to estimate main effects and second-order terms under limited experimental runs while minimizing aliasing and capturing nonlinear behavior. However, widely used designs such as definitive screening designs (DSDs) and orthogonal-minimal-aliasing response surface (OMARS) designs rely fully or partially on search-based or optimization-driven constructions. This reliance increases computational costs,

limits scalability, and restricts flexibility. Moreover, changing factor sets often necessitates a full reconstruction. To address these limitations, a new class of designs is proposed, termed OMAHLD, constructed via a closed-form, search-free procedure. The proposed design accommodates both odd and even numbers of factors, scales naturally to high-dimensional settings, and allows sequential expansion by adding columns with minimal increases in run size. OMAHLD preserves the minimal-aliasing principle, and any residual aliasing can be efficiently resolved using simple matrix operations. To further mitigate main-effect aliasing, a two-stage Dantzig selector tailored to OMAHLD designs is developed. Simulation studies demonstrate that this strategy achieves true-positive and false-discovery rates comparable to those of DSDs, while substantially outperforming conventional one-stage methods. These results establish OMAHLD as a flexible and high-performance alternative for large-scale three-level screening experiments.

E2018: Number-theoretic design theory

Presenter: Masanori Sawa, Kobe University, Japan

A new branch of Design Theory, called Number-Theoretic Design Theory, is introduced. Specifically, a systematic treatment of the r -dimensional PTE problem in additive number Theory (abbreviated as PET_r) is provided through its connection with combinatorial designs such as orthogonal arrays and t -designs, incorporating some new results.

EO162 Room 204 RECENT DEVELOPMENTS IN STATISTICAL METHODOLOGY AND APPLICATIONS

Chair: HaoYun Huang

E1248: Extreme value theory for individuals control charts: A semiparametric approach to ensuring in-control performance

Presenter: Hong-Ji Yang, National Dong Hwa University, Taiwan

A semiparametric method for designing individuals control charts in phase I statistical process monitoring is introduced, aimed at ensuring guaranteed in-control performance. Motivated by the limitations of parametric and nonparametric methods under small to moderate sample sizes, the proposed approach combines extreme value theory with the exceedance probability criterion to construct control limits that adjust for estimation variability. The methodology uses Pickands and moment estimators to model extreme tail behavior, offering a more reliable solution than purely data-driven nonparametric methods. Simulation results show that a sample size of approximately $n = 1112$ is sufficient to achieve a nominal coverage probability of $P_n = 0.90$. The method demonstrates strong and consistent performance across a variety of distributions, avoids the need for computationally intensive bootstrap procedures, and provides an R package for implementation to support reproducibility and real-world applications, such as semiconductor manufacturing.

E1528: Spatial ZIEGP models with varying coefficient parameters

Presenter: Leng-Yi Ku, National Central University, Taiwan

Co-authors: Chun-Shu Chen

Understanding and modeling spatial heterogeneity is critical for climate and environmental studies, particularly in rainfall analysis where both the frequency of zero precipitation and the tail behavior of extreme events exhibit substantial geographical variation. The zero-inflated extended generalized Pareto (ZIEGP) model offers a unified probabilistic framework that simultaneously describes dry days and the full spectrum of rainfall intensities, including low, moderate, and extreme values, without relying on subjective threshold selection. However, existing applications of the ZIEGP model often assume spatially constant parameters, which restricts their ability to reflect local climatological characteristics. To overcome this limitation, a spatial ZIEGP model is developed in which all parameters vary smoothly across space through basis functions. This formulation enables flexible location-specific inference while remaining computationally feasible for large spatial datasets. Simulation studies further confirm the effectiveness of the approach in accurately recovering spatially varying parameters under different data-generating scenarios. The methodology is applied to high-resolution rainfall data from Taiwan, confirming its practical utility and effectiveness for climate risk assessment.

E1828: Accounting for correlated detectability in time-to-detection wildlife surveys

Presenter: Wen-Han Hwang, National Tsing Hua University, Taiwan

Time-to-detection (TTD) data are increasingly used in multi-visit wildlife surveys to estimate species occupancy and abundance under imperfect detection. At each site, observers conduct repeated visits and record the time to first detection (or a censored time if no detection occurs). While efficient in the field and more informative than binary detection/non-detection data, detectability across visits within a site is often correlated due to persistent habitat features, local abundance, or unmeasured conditions. Standard TTD models typically assume independence across visits, which can lead to biased inference and underestimated uncertainty. A framework is proposed that allows detectability to be correlated across visits while incorporating habitat and survey covariates. A shared detectability component captures persistent site-level differences and links commonly used TTD occupancy and abundance models. Building on this framework, a mixed gamma exponential TTD model is developed that balances flexibility and interpretability. Simulation studies show improved estimation accuracy compared to independent-detection models, and an application to bird survey data from South Africa demonstrates clearer and more reliable ecological inference.

E1906: Spatial adapter: Structured spatial decomposition and closed-form covariance for frozen predictors

Presenter: HaoYun Huang, National Dong Hwa University, Taiwan

The Spatial Adapter is a parameter-efficient post-hoc layer that equips any frozen first-stage predictor with a structured Spatial representation of its residual field and an induced closed-form Spatial covariance. The Adapter operates as a cascade second stage on residuals, jointly learning a spatially regularized orthonormal basis and per-sample scores via a tractable mini-batch ADMM procedure. Without retraining the frozen backbone, its role is to supply a compressed distributional summary of the residual field. Smoothness, sparsity, and orthogonality together turn a generic low-rank factorization into an identifiable Spatial representation whose induced residual covariance admits a closed-form low-rank-plus-noise estimator; the effective rank is determined data-adaptively by spectral thresholding, while the nominal rank K is an optimization-side upper bound. This covariance enables kriging-style Spatial prediction at unobserved locations, yielding plug-in uncertainty quantification. Across synthetic data, Weather2K (Spatial-holdout prediction), and GWHD patch grids (basis-transferability diagnostic), the Adapter recovers residual Spatial structure when paired with frozen first stages from linear models to deep spatiotemporal and vision backbones; the added representation uses fewer than $K(N+T)$ parameters alongside a compact residual-trend network.

EO070 Room 205 STATISTICAL INFERENCE AND LEARNING

Chair: Seonghun Cho

E1432: Online testing with dependent streaming data

Presenter: Seohwa Hwang, Seoul National University, Korea, South

The problem of converting offline multiple testing procedures into valid online testing rules under sequential data acquisition is studied. While classical offline methods assume access to the full dataset and control error rates such as the false discovery rate (FDR) in a batch setting, directly applying them in an online environment leads to invalid inference. To address this gap, a principled offline-to-online conversion framework based on sliding-window inference is proposed. At each time step, the procedure constructs local statistics using a finite window of recent observations and applies a calibrated decision rule that mimics its offline counterpart while respecting the online filtration. Sufficient conditions are established under which the resulting procedure provably controls the FDR asymptotically, even in the presence of dependence.

E1426: Conformalized method for empirical bayes normal mean inference problem with heteroscedastic variance*Presenter:* Kwangok Seo, Inha University, Korea, South

The normal mean inference problem is studied, which involves simultaneous testing of the means of many normal distributions. This problem has been extensively studied within the empirical Bayes (EB) framework. However, the reliability of most EB methods heavily depends on two key conditions: (i) the prior distribution is correctly specified, and (ii) it can be accurately estimated. In practice, both conditions are difficult to satisfy, and it is often unclear whether they hold in a given application. To overcome these limitations, a new algorithm is proposed, called COIN (conformal inference for normal mean inference problem). Unlike traditional empirical Bayes approaches, COIN produces decision rules whose validity does not depend on the correct specification or accurate estimation of the prior. It is theoretically proven that COIN asymptotically controls the false discovery rate at the nominal level, even in the presence of prior misspecification or estimation errors. Since the COIN algorithm requires an external training dataset to estimate the prior distribution and conformity score function, two data-splitting strategies are introduced—sample-splitting and feature-splitting—for the case where such external data are unavailable. Theoretical guarantees are provided for the data-splitting strategies and their effectiveness is demonstrated through extensive numerical studies and three real data examples.

E1404: Agnostic model-assisted estimation in finite population sampling*Presenter:* Mehdi Dagdou, McGill University, Canada*Co-authors:* Ziming An, David Haziza

The aim of survey sampling is to estimate finite population parameters. In many settings, auxiliary information is available at the population level and can be incorporated through modeling within the model-assisted framework. Establishing the asymptotic properties of such estimators has traditionally been carried out on a method-by-method basis and has remained an active area of research over the past 25 years. Although these estimators often share similar properties, the techniques used to analyze them are typically tailored to the specific prediction method, and a general theoretical framework applicable to arbitrary prediction methods is lacking. A novel form of conditional cross-fitting adapted to the survey sampling setting is introduced to address this gap. Leveraging this approach, a unified theory for the asymptotic behavior of model-assisted estimators based on general prediction methods is developed. In particular, the asymptotic normality of the proposed estimators and the consistency of a corresponding cross-fitting variance estimator are established, thereby enabling the construction of asymptotically valid confidence intervals. Simulation studies demonstrating the favorable finite-sample performance of the proposed methodology will also be presented.

E1855: Hypothesis testing for penalized estimating equations with cross-fitted covariance calibration*Presenter:* Jing Zhou, University of Manchester, United Kingdom*Co-authors:* Zhe Zhang

Hypothesis testing for penalized estimators is studied in settings where the full marginal distribution of a multivariate response is difficult to specify, such as longitudinal data with correlated measurements or high-dimensional heteroscedastic regression. Assuming that the conditional mean model is correctly specified, it is established that the penalized estimating equations admit a $\sqrt{n}$ -consistent solution, even when the working covariance structure is misspecified. The inferential target is a low-dimensional subvector of parameters associated with the mean model. The resulting test statistic is shown to converge to a χ^2 distribution, and its asymptotic power depends on the nuisance covariance function. To mitigate this dependence, a procedure for estimating the covariance function via cross-fitting is proposed, which provides a calibrated and robust approach for inference.

EC292 Room 102 VOLATILITY MODELS**Chair:** Toshiaki Watanabe**E1836: A novel Markov-switching GAS jump-in-mean model: State-dependent dynamics and risk heterogeneity in U.S. stock indices***Presenter:* Kuang-Liang Chang, National Sun Yat-sen University, Taiwan

A novel Markov-switching GAS Jump-in-Mean model is proposed that integrates a Markov-switching Generalized Autoregressive Score (GAS) process into both the conditional variance of non-jump innovations and the conditional jump intensity. The framework is employed to investigate the state-dependent effects of non-jump risk and jump risk on excess stock returns for the S&P 500, DJIA, and Nasdaq indices. Furthermore, the analysis examines how macroeconomic and financial factors—including economic policy uncertainty (EPU), the VIX, and the U.S. dollar index—influence conditional volatility across different volatility states. The empirical results indicate that: (i) excess stock returns display pronounced Markov-switching and jump dynamics; (ii) the effects of non-jump risk and jump risk are state-dependent and heterogeneous across stock indices; (iii) the contribution of non-jump innovations to conditional variance is stronger in the high-volatility state for all indices, while the contribution of jump innovations to conditional variance is greater in the low-volatility state for the S&P 500 and Nasdaq; and (iv) stock market volatility responds differently to macroeconomic and financial variables across indices, depending on volatility states and the underlying sources of innovation.

E1865: Stochastic coefficient GARCH-X models: Evidence from the US and ASEAN-5 equity markets*Presenter:* Tsubasa Furuichi, Soka University, Japan*Co-authors:* Manabu Asai

The Stochastic Coefficient GARCH-X (SC-GARCH-X) model extends the standard GARCH-X framework by allowing conditional variance parameters to vary over time as a Stochastic process driven by exogenous variables. Stationarity conditions and asymptotic properties of the Quasi-Maximum Likelihood (QML) estimator are discussed. Monte Carlo simulations examine the finite sample properties of the QML estimator to ensure its reliability. An empirical application to US and ASEAN-5 equity markets, incorporating range-based volatility as the exogenous variable, reveals that the SC-GARCH-X model provides superior in-sample fit and out-of-sample forecasts for developed markets by capturing dynamic volatility structures more effectively. However, certain emerging markets are better characterized by a simple asymmetric GARCH model, suggesting that the complexity of volatility dynamics varies across different stages of market development.

E1952: Joint tail risk forecasting for high-volatility assets based on copula semiparametric framework*Presenter:* Aaron E-P Lin, Feng Chia University, Taiwan*Co-authors:* Cathy W-S Chen, Takaaki Koike

A copula-based bivariate ES-CAViaR framework is proposed to jointly forecast value-at-risk (VaR) and expected shortfall (ES), while investigating downside dependence between financial assets. Four ES-CAViaR-type marginal specifications are considered, including the baseline ES-CAViaR model, ES-CAViaR-st with a scaled exceedance ratio, ES-CAViaR-v incorporating Rogers-Satchell volatility, and ES-CAViaR-R incorporating Garman-Klass volatility. To model cross-asset dependence, the Clayton and BB1 copulas are employed based on pseudo-observations. The Clayton copula is designed to capture lower-tail dependence and joint downside risk, whereas the BB1 copula accommodates both lower- and upper-tail dependence, allowing for greater flexibility in modeling asymmetric tail behavior. Model parameters are estimated using a Bayesian adaptive Markov chain Monte Carlo (MCMC) framework that combines random-walk Metropolis and independent-kernel Metropolis-Hastings sampling. An empirical application to Bitcoin and Ethereum is conducted using rolling-window one-step-ahead forecasts at the 1% risk level over a five-year evaluation period. Forecast performance is evaluated using violation-based backtests and scoring criteria. The empirical results indicate that the scaled exceedance ratio specification provides more robust joint VaR and ES forecasts, especially during periods of elevated downside risk.

E1299: State-dependent copula particle filtering for stochastic volatility models*Presenter:* **Tingfu Chen**, National Central University, Taiwan

A state-dependent copula-particle filter (SD-CoPF) is developed for the joint estimation of parameters and latent states in multivariate stochastic volatility models. By treating model parameters as time-varying states and incorporating kernel smoothing, the framework ensures convergence of posterior distributions and mitigates particle degeneracy. The methodology integrates copula functions to capture complex cross-market dependency structures, distinguishing between particle-correlated and state-correlated filtering architectures. Theoretical properties, including parameter convergence and the validity of the auxiliary-like importance weighting scheme, are mathematically established. Numerical simulations across Gaussian, Student-t, and Archimedean copula families demonstrate that SD-CoPF achieves high estimation accuracy. The proposed algorithm effectively tracks hidden volatility dynamics and adapts to various non-linear dependence structures in high-dimensional financial time series.

EC294 Room 209 BAYESIAN COMPUTATION**Chair: Shih-Feng Huang****E1321: Matrix-exponential variational inference***Presenter:* **Martin Magris**, ITAM - Instituto Tecnológico Autónomo de México, Mexico

Variational Inference (VI) has become an important tool for approximating complex posterior distributions in modern Bayesian applications, particularly in machine learning, where models are often non-conjugate and high-dimensional. A widely used approach is Gaussian fixed-form VI. While this choice enables scalable optimization with gradient-based methods, enforcing the positive definiteness of the covariance matrix Σ remains challenging. A novel and analytically tractable parameterization of the variational covariance matrix is proposed. Specifically, the variational covariance matrix Σ is represented as $\Sigma = MDM^T$, where $M = \exp(\Omega)$ is the matrix exponential of a skew-symmetric matrix and D is a diagonal matrix. This construction guarantees positive-definiteness while preserving numerical stability and computational efficiency. Under this parameterization, closed-form expressions for both the standard and natural gradients of the Evidence Lower Bound are derived and a close connection to Riemann optimization is established. Efficient algorithms for computing derivatives of the matrix exponential are developed and the method is demonstrated on logistic regression, time-series models with GARCH volatility, and Gaussian mixture models. The proposed approach yields stable posterior approximations comparable to those of existing VI and MCMC methods while enabling fast and scalable optimization.

E1334: Moment constrained cutting feedback for modular Bayesian models*Presenter:* **Pushkar Kale**, National University of Singapore, Singapore*Co-authors:* David Nott, Ajay Jasra, Xin Tong

Statistical models are often constructed from multiple linked submodels (modules), each informed by different data sources and domain expertise. Misspecification in any module can distort posterior inference and propagate across the entire model. Cutting feedback methods address this by modifying the joint posterior so unreliable modules cannot distort inference in trusted components. The standard formulation finds the Kullback-Leibler closest distribution to the full posterior whose marginal for the shared parameters matches the posterior from the trusted module alone. However, sampling from this posterior is challenging because it requires evaluating an intractable marginal density at each step. Consequently, naive nested MCMC is typically used, leading to high computational cost. An alternative formulation, called the moment-constrained cut (MCC) posterior, replaces the full distributional constraint with a finite set of marginal mean and variance constraints on the shared parameters. The resulting distribution admits an exponential tilting representation, with the tilting coefficients estimated via stochastic gradient descent coupled with MCMC updates, whose convergence properties are established theoretically. This eliminates the need to run a separate Markov chain for each sample. Three benchmark examples demonstrate that the MCC posterior produces results similar to the conventional cut posterior while being much easier to compute.

E1354: Scalable stochastic gradient variational inference framework for microbiome differential abundance analysis*Presenter:* **Saurabh Panchasara**, York University, Canada*Co-authors:* Hanna Jankowski, Kevin McGregor

Differential abundance analysis (DAA) attempts to answer a core question in microbiome studies: which microbial features change reliably between conditions, or with an exposure? DAA is challenging because microbiome data are high-dimensional, compositional, zero-inflated, and skewed in additive log-ratio transformation. The proposed Bayesian zero inflated factor analysis logistic skew normal multinomial (ZIFA-LSNM) model, with skew-normal priors on latent factors, addresses these issues effectively and can be considered as an effective tool box to perform DAA. However, standard inference via coordinate ascent variational inference (CAVI) or fixed form variational Bayes (FFVB) suffers major computational and analytical challenges in high-dimensional settings. To overcome this, a scalable inference framework based on stochastic gradient variational inference (SGVI) is developed that avoids rigid closed-form analytical updates. Across available models such as ZIPPICA-LPNM and ZIFA-LSNM, SGVI achieves comparable optimized fits while substantially reducing runtime. This efficiency enables fast, scalable estimation of latent microbial structures, providing an accessible tool for large-scale DAA.

E2033: Fast Bayesian analysis of mixture of Gaussian structural equation models*Presenter:* **Luca Maestrini**, The Australian National University, Australia*Co-authors:* Khue-Dung Dang, Francis Hui

Structural equation models (SEMs) are widely used in social and behavioral sciences to study the structural relationship between measured outcomes and latent constructs. Recently, Bayesian fitting procedures for SEMs have gained increasing popularity due to their potential to facilitate more flexible model structures. Variational Bayes approximations, in particular, have been shown to provide fast and accurate inference for Bayesian analysis of SEMs, although to date their application has been limited to simple forms involving no covariates and assuming normality for all measured outcomes. Motivated by a study assessing the impact of prenatal alcohol exposure on child cognitive function, mean field variational Bayes for a SEM formulation is developed where measured outcomes exhibit non-Gaussian features such as skewness and multimodality. The proposed SEMMIX model utilizes finite mixtures of Gaussian distributions to accommodate such features; covariates are incorporated into the modeling of the latent traits, and an automated approach is developed to account for missing at random outcomes. Two variational information criteria for model selection are also proposed that are straightforward to compute in the variational inference framework. Simulations and application to the motivating child cognitive function dataset demonstrate that SEMMIX fitted using fast variational Bayes methods captures key non-Gaussian features of the data.

EV303 Room Virtual R01 ESTIMATION AND DENSITY METHODS**Chair: Degui Li****E1886: Probability density estimators for tail-weighted mean integrated squared error***Presenter:* **Yusuke Omura**, Yokohama City University, Japan*Co-authors:* Taku Moriyama

Large-scale risks often arise from heavy-tailed distributions, making accurate density estimation in the tail crucial for appropriate risk management decisions. A loss function for parameter selection in density estimation when large-scale risk management is the objective is proposed. A tail-weighted mean integrated squared error (WMISE) is introduced, given by an integral of the tail-weighted mean squared error. The kernel density estimator (KDE) is a common density estimator whose accuracy depends strongly on bandwidth selection. Cross-validation and plug-in criteria based on WMISE are developed to select the bandwidth. Additionally, the Hall-type estimator is considered as an existing tail density estimator. For this estimator, the optimal parameter of a tail-weighted maximum log-likelihood estimator (WMLE) is introduced in terms of WMISE under correct model specification. The behavior of the proposed bandwidth and WMISE is investigated analytically and numerically. The performances of the KDE and the Hall-type estimator with proposed and existing parameter selection methods are compared through both analytical and numerical results.

E1979: Bayes linear estimator in the general linear model*Presenter:* **Hirai Mukasa**, Kyushu University, Japan

The Bayes linear estimator is derived by minimizing the Bayes risk with respect to squared loss in the general linear model. When certain prior information is available, it is often natural to consider Bayes linear estimators. For example, generalized least squares, ridge, and shrinkage estimators can be represented as Bayes linear estimators or limit points thereof. Statistical properties and equivalence of Bayes linear estimators are discussed. First, properties of Bayes linear estimators such as linear sufficiency and linear completeness are explored. These properties are important concepts that characterize whether a linear estimator retains sufficient and nonredundant information to construct the best linear unbiased estimator for a linear predictor. Second, necessary and sufficient conditions under which two Bayes linear estimators coincide are derived. These conditions clarify when different choices of covariance structures or prior information lead to the same estimator. As an illustration, linear mixed-effects models are considered and results are shown to provide a more efficient estimation procedure.

E2037: Bayesian semiparametric hierarchical copula density estimation for grouped data*Presenter:* **Jichan Park**, Korea University, Korea, South*Co-authors:* Taeryon Choi

Copula models provide a flexible framework for modeling dependence in multivariate data, but copula density estimation for grouped bivariate data remains challenging when dependence structures vary across related groups and exhibit latent heterogeneity within groups. A Bayesian hierarchical copula density estimator for grouped bivariate data is proposed. The model uses a parametric copula baseline and accounts for family uncertainty through reversible jump selection. It further introduces a hierarchical Dirichlet process mixture of latent copula atoms together with a group level smooth nonparametric adjustment. The hierarchical Dirichlet process shares copula atoms across groups while allowing multiple latent dependence components within each group. Residual departures from the parametric baseline are represented by a two dimensional hierarchical spectral expansion with tensor product basis functions and group level smoothing priors. Posterior inference is carried out by Markov chain Monte Carlo with reversible jump updates, blocked Gibbs sampling for the hierarchical Dirichlet process component, and updates for the smooth adjustment. Simulation studies and an empirical application illustrate the proposed framework in terms of copula family selection, dependence estimation, cross group sharing, and within group heterogeneity.

Saturday 08.08.2026

16:45 - 18:50

Parallel Session E – EcoSta2026

EO102 Room B101 THEORY AND APPLICATIONS IN ECONOMETRICS: SOME RECENT DEVELOPMENTS**Chair: Yu-Chang Chen****E1296: Learning preference from observed rankings****Presenter:** Yu-Chang Chen, National Taiwan University, Taiwan**Co-authors:** Chen Chian Fuh, Shang En Tsai

Estimating consumer preferences is central to many problems in economics and marketing. A flexible framework is developed for learning individual preferences from partial ranking data by interpreting observed rankings as collections of pairwise comparisons with logistic choice probabilities. Latent utility is modeled as the sum of product attributes, item fixed effects, and a low rank user item factor structure, allowing both interpretability and information sharing across consumers and products. Selection in which comparisons are observed is also addressed. A comparison is recorded only if both items enter the consumer consideration set, which creates exposure bias toward frequently encountered items. Pair observability is modeled as the product of item level propensities and these propensities are estimated using a logistic model for the probability that an item is observable. Preference parameters are then estimated using an inverse probability weighted and ridge regularized log likelihood that reweights observed comparisons toward a target comparison population. To scale computation, a stochastic gradient descent algorithm based on inverse probability resampling is developed. In an application to transaction data from an online wine retailer, the method improves out of sample recommendation performance relative to a popularity benchmark, especially for predicting purchases of previously unconsumed products.

E1297: Hourly versus monthly minimum wages and employment adjustment: Evidence from Taiwan**Presenter:** Po-Chun Huang, National Chengchi University, Taiwan

Minimum wage increases in Taiwan are examined using two-year matched panels from the 2012-2020 Manpower Utilization Survey, where separate monthly and hourly wage floors coexist and the hourly minimum has risen faster since 2016. By comparing directly affected workers with higher-wage comparison workers before and after larger minimum wage hikes, the analysis finds that minimum wage increases raise wages for affected workers, with little evidence of reduced continued employment. Relative to control workers, affected workers become more likely to be paid monthly, less likely to work part-time, and more likely to work longer hours, consistent with substitution away from hourly-paid and part-time arrangements toward monthly-paid jobs. Complementary county-level exposure estimates show similar patterns: more exposed counties experience stronger wage growth at the lower tail, with little evidence of aggregate employment loss, and low-wage workers exhibit similar contract adjustments.

E1304: Structural changes, copula asymmetry, and tail dependence in Japan's equity and FX markets**Presenter:** Tsung-Chih Lai, National Chung Hsing University, Taiwan**Co-authors:** Yi-Chi Chen

Time-varying dependence between Japan's equity and foreign exchange markets is examined using a nonparametric copula framework that accommodates asymmetry and tail comovement. Using daily returns over the past five decades, annual empirical copulas are estimated and randomization-based tests for copula symmetry and equality across adjacent years are applied. Three main findings are documented: (i) stock-currency dependence exhibits discrete regime shifts over time; (ii) extreme equity declines are strongly associated with yen appreciations, indicating pronounced downside tail dependence consistent with the yen's safe-haven role; and (iii) violations of exchangeability and radial symmetry imply that correlation-based hedge ratios may substantially mismeasure risk during stress episodes. Conditioning on macrofinancial variables reveals systematic variation in dependence across monetary and risk regimes, underscoring the importance of tail dependence for hedging and asset allocation involving Japanese assets.

E1314: Endogeneity, heterogeneity, and corporate default risk: A correlated random effects panel approach**Presenter:** Chien-Ho Wang, National Taipei University, Taiwan**Co-authors:** Yi-Chi Chen

A parametric panel framework is developed for predicting corporate probabilities of default that explicitly addresses endogeneity arising from persistent firm-level heterogeneity. In corporate finance, financial policies such as leverage are systematically related to underlying firm risk. When these persistent features are unobserved, conventional logit, probit, and hazard models may confound cross-sectional risk differences with the effects of financial variables. The focus is therefore on correlated heterogeneity, and a correlated random effects design is adopted rather than an instrumental variable method. The specification embeds a control function type adjustment within a discrete-time forward hazard model. A longer time dimension strengthens the approximation, as time averages more precisely summarize persistent components of firm behavior, thereby improving the quality and stability of the correction. Estimation proceeds by random effects probit maximum likelihood, and economic magnitudes are reported through average partial effects. The resulting framework provides a theoretically coherent and empirically tractable approach to PD estimation, enhancing the credibility and interpretability of default risk assessment for financial institutions, investors, and regulators.

E1729: Macroeconomic consequences of sustained warming: A bias-corrected dynamic heterogeneous panel approach**Presenter:** Jui-Chung Yang, National Taiwan University, Taiwan

The aim is to study how increasing temperatures have affected the economies of 195 countries between 1960 and 2022, focusing on income losses caused by gradual shifts to new climate conditions. A dynamic heterogeneous panel model is developed to distinguish between long-term and short-term effects, account for adaptation through rolling climate norms, and address key econometric challenges including nonstationarity, cross-country heterogeneity, and unobserved global factors. Findings reveal that a sustained 0.01C annual increase in temperatures above historical climate norms reduces global GDP per capita growth by 0.05 percentage points per year, with income losses accumulating as long as temperatures keep increasing. This effect is 70% larger than what would be estimated under a homogeneous panel specification. Contrary to much of the existing literature, no country appears immune to the impacts of rising temperatures: middle- and high-income nations, as well as those in temperate or cold and hot climate zones, all exhibit persistent (though not permanent) growth slowdowns, with income losses linked to how quickly they adapt.

EO016 Room B102 MODELING HIGH-DIMENSIONAL ECONOMETRICS**Chair: Degui Li****E1383: Taming high dimensional cointegrated regressors****Presenter:** Ziwei Mei, University of Macau, China

LASSO has been the state-of-the-art methodology for high dimensional linear regressions. When cointegration is present in a high dimensional predictive regression, LASSO is inconsistent due to the excessive penalty on cointegrated regressors. Nonconvex penalties, including the widely used SCAD and MCP, are proposed to fix the over-penalization and restore consistent estimation of the high dimensional predictive regressions. The consistency of estimation and variable selection by SCAD and MCP is established. In the numerical studies, substantial improvements by the nonconvex penalties relative to LASSO in terms of parameter estimation and out-of-sample prediction are observed.

E1965: Moduli of continuity of Brownian motion and related processes over an expanding time interval*Presenter:* **Shin Kanaya**, Kyoto University and University of Essex, Japan*Co-authors:* Jihyun Kim, Joon Park

Sharp results on the uniform path continuity properties of Brownian motion and related continuous-time processes, such as Brownian semimartingales and Gaussian processes, are established over an expanding time interval $[0, T]$, where T tends to infinity. While such continuity properties are well known over a fixed time span, they have not been investigated in full generality in the long span setting where T grows. The theoretical implications of these results for the estimation of continuous-time processes from discretely observed samples and for volatility regressions are further discussed.

E1837: Estimation of large dynamic precision matrices with a latent semiparametric structure*Presenter:* **Jia Chen**, University of Macau, China*Co-authors:* Yuning Li, Oliver Linton

The estimation of dynamic precision matrices with multiple conditioning variables is studied for high-dimensional time series. The high-dimensional time series is assumed to have an approximate factor structure plus an idiosyncratic error term, which allows the time series to have a non-sparse dynamic precision matrix and enhances the applicability of the method. Exploiting the Sherman-Morrison-Woodbury formula, the estimation of the dynamic precision matrix for the time series reduces to the estimation of a low-rank factor structure and the precision matrix of the idiosyncratic error term. For the latter, a semiparametric method is introduced to estimate the entries of the corresponding dynamic covariance matrix via the Model Averaging MArginal Regression (MAMAR) before applying the constrained l_1 minimisation for inverse matrix estimation (CLIME) method to obtain the dynamic precision matrix. Under some regularity conditions, uniform consistency for the proposed estimators is derived. A simulation study illustrates the finite-sample performance of the developed methodology, and an application in the construction of minimum-variance portfolios using daily returns of S&P 500 constituents from 2000 to 2024 is provided.

E1895: Estimating the number of significant components in high-dimensional principal component analysis*Presenter:* **Zhixiang Zhang**, University of Macau, China

A new penalized approach for estimating the number of significant components in high-dimensional principal component analysis is proposed, using the explained variance ratio and the rigidity of the nonspiked sample eigenvalues of sample covariance matrices of variables. Compared with methods in the existing literature, consistency of the proposed estimator holds not only for independent data but also for some time series data when the dimension and the sample size both tend to infinity. Even for independent data, the estimator works under weaker conditions than existing approaches such as AIC and BIC, including allowing heterogeneity in the bulk of the population eigenvalues. Simulation studies are conducted to illustrate the performance of the proposed estimator.

E1492: Model averaging of multi-layer time-varying network vector autoregressions*Presenter:* **Degui Li**, University of Macau, China

A time-varying multi-layer network vector autoregression (VAR) model framework is introduced for large-scale time series, allowing agents in dynamic systems to interact through multiple channels and incorporating multiple adjacency matrices to capture network spillover effects. A penalized model averaging method is proposed to determine a time-varying optimal combination of multi-layer network VAR candidate models whose number is allowed to be divergent. Under regularity conditions, asymptotic properties such as asymptotic optimality and convergence rates of the proposed time-varying weight estimation are derived in the contexts of both in-sample fitting and out-of-sample prediction. Additionally, conformal prediction is introduced to construct interval forecasts. Monte-Carlo simulation and empirical studies illustrate reliable finite-sample estimation and predictive performance of the developed methodology.

EO013 Room B104 EMPIRICAL ANALYSIS OF MACROECONOMIC AND FINANCIAL DATA**Chair: Kaiji Motegi****E1505: Testing the hysteresis hypothesis of unemployment in Japan: A cyclical time trend approach***Presenter:* **Yumeri Ninomiya**, Kobe University, Japan*Co-authors:* Kaiji Motegi

The hysteresis hypothesis of unemployment is known as a potential factor for secular stagnation. Whether the unemployment hysteresis is present or absent in reality can be transformed to a testable question whether the unemployment rate follows a non-stationary or stationary process. The unit root hypothesis of unemployment rates is tested with a cyclical trend regression model, where the cyclical time trend is specified by trigonometric functions. The irrelevance hypothesis of the cyclical trend is pre-tested by a data-driven approach which does not require a prior knowledge of the non-stationarity versus stationarity of the unemployment rate. When the cyclical trend is deemed necessary, the unit root hypothesis of the unemployment rate is tested via the Lagrange Multiplier test associated with the cyclical trend model. The proposed methodology is applied to quarterly unemployment data of Japan, finding that cyclical trends are necessary for the whole country and multiple regions. Further, the presence of unemployment hysteresis is observed for most regions. These empirical results suggest that Japan's secular stagnation, which is sometimes called the lost three decades, could be attributed to the unemployment hysteresis.

E1572: Testing and quantifying economic resilience*Presenter:* **Naoko Hara**, Seikei University, Japan*Co-authors:* Yohei Yamamoto

A formal testing procedure is proposed to examine economic resilience that applies even when a cross-section of a control group is unavailable and circumvents bias in time-series regressions using data with structural breaks. Novel measures of shock absorption and cumulative recovery are provided. Empirical analysis shows that most advanced economies were not resilient to the Global Financial Crisis, while many were resilient during the COVID-19 pandemic. These results challenge the view that a single set of determinants governs economic resilience, showing instead that factors such as Financial leverage and labor market regulation are associated with lower resilience, while other determinants matter in a Crisis-dependent manner.

E1484: Why do firms prefer after-hours disclosure? Evidence from Japan*Presenter:* **Huixing Jin**, Tokyo Metropolitan University, Japan*Co-authors:* Konari Uchida

The Tokyo Stock Exchange's extension of its closing time from 3:00 p.m. to 3:30 p.m. in November 2024 provides an opportunity to examine whether firms deliberately announce earnings after market close. The extension reduces the proportion of after-hours announcements by approximately 15%, indicating that operational constraints significantly influence announcement timing. Importantly, about 40% of firms announce earnings immediately after market close in both the pre- and post-extension periods, implying that many firms deliberately choose after-hours disclosure. Firms with more volatile Stock returns consistently make after-hours announcements around the trading hours extension. These consistent after-hours announcements are also associated with lower ownership by relationship-based shareholders. Moreover, these factors are related to the likelihood that firms choose after-hours announcements for all earnings releases during the sample period. Finally, the market-to-book ratio, particularly its misvaluation component, is positively associated with the likelihood of making after-hours announcements.

E1566: The impact of consumption stimulus design on the MPC: Evidence from a survey experiment*Presenter:* **Kento Tango**, Yokohama City University, Japan*Co-authors:* Yoshiyuki Nakazono, Jun Takahashi

While policymakers increasingly use non-cash consumption stimulus policies, there is little evidence comparing their effects on the marginal propensity to consume (MPC). An online survey in Japan examines how payment formats affect the MPC by eliciting self-reported spending responses to hypothetical windfall gains delivered in different formats: cash, time-limited vouchers, and cashless points. Three main findings emerge. First, consumption responses are broadly similar across payment formats. While more than half of respondents report no change in spending, those who do respond exhibit substantial increases in spending. Second, the average MPC is below 0.2 across all payment formats, with no substantial differences across formats. Third, MPC heterogeneity depends on the payment format even among individuals with identical observed characteristics. This suggests that households respond differently to identical payment benefits depending on the consumption stimulus design.

E1498: Conditional threshold autoregression (CoTAR)*Presenter:* **Kaiji Motegi**, Kobe University, Japan*Co-authors:* John Dennis, Shigeyuki Hamori

A new time series model called conditional threshold autoregression (CoTAR) is proposed, in which the threshold is specified as an empirical quantile of recent observations of a threshold variable. The conditional threshold is expected to trace economic and financial variables well, as a cut-off level of low and high regimes likely changes over time. The presence versus absence of conditional threshold effects can be tested via the wild bootstrap, and the out-of-sample predictive ability of CoTAR can be evaluated via the Diebold-Mariano test. It is shown that CoTAR satisfies desired statistical properties in both large and small samples. The proposed model is applied to the CBOE Volatility Indices of S&P 500 and major U.S. shares, obtaining desired in-sample and out-of-sample results.

EO206 Room 101 BAYESIAN TIME SERIES ANALYSIS OF CRYPTOCURRENCY MARKETS**Chair: Teruo Nakatsuma****E1480: Minimum variance versus market cap portfolios in cryptocurrency market***Presenter:* **Sakae Oya**, Keio University, Japan*Co-authors:* Teruo Nakatsuma

Portfolio construction in cryptocurrency markets has primarily relied on market capitalization-weighted benchmarks, while alternative allocation strategies remain less explored. The risk-return performance of minimum variance portfolios relative to market cap portfolios within the cryptocurrency asset class is examined. Portfolios are constructed from a limited set of major cryptocurrencies, reflecting basic liquidity considerations. Covariance matrices are estimated using regularization-based methods to ensure stable estimation. Using historical return data, out-of-sample portfolio performance is evaluated over multiple periods. The main objective is to investigate whether empirical findings established in equity markets, where minimum variance portfolios often outperform market cap benchmarks in terms of risk-return efficiency, also hold in cryptocurrency markets. The analysis provides empirical evidence on the applicability of portfolio optimization techniques developed for traditional asset classes to this emerging market.

E1775: LLM-Orchestrated Bayesian Nonparametric Risk framework for Cryptocurrency*Presenter:* **Yuhong Chen**, Keio University, Japan

Cryptocurrency markets present a persistent challenge for institutional risk management, as return distributions defy parametric assumptions, regimes shift abruptly, and tail events cluster in ways that conventional models systematically underestimate. At the same time, the rapid maturation of large language models and Bayesian nonparametric theory opens a new design space in which probabilistic AI systems can reason about uncertainty in a principled, data-adaptive manner. Research at the intersection of these two developments explores how LLM-orchestrated Bayesian inference represents a promising direction for next-generation crypto risk intelligence, drawing on recent advances in heavy-tailed nonparametric priors, the theoretical grounding of in-context learning as posterior inference, and AI-driven quantitative research workflows. Rather than prescribing a fixed model, this research outlines a research agenda connecting distributionally robust risk measures, sequential uncertainty quantification, and hybrid local-cloud deployment.

E1500: Forecasting Bitcoin volatility with non-Gaussian realized stochastic volatility models*Presenter:* **Yosuke Mori**, Keio University, Japan*Co-authors:* Makoto Nakakita, Teruo Nakatsuma

An empirical comparison of Gaussian and non-Gaussian realized stochastic volatility models for Bitcoin is presented. Using daily Bitcoin returns and one-minute realized volatility from Binance (2022-2024), the analysis compares seven return-error specifications within a common realized stochastic volatility architecture: Gaussian, variance gamma, skew variance gamma, Student t, skew Student t, Laplace, and asymmetric Laplace. The non-Gaussian models are estimated through a unified generalized inverse Gaussian latent-scale representation, so the comparison isolates the empirical role of heavy tails and asymmetry while keeping the state and the realized volatility measurement equation fixed. Forecast evaluation uses a 730day rolling design, the two-scale realized volatility proxy, mean squared error, quasi-likelihood, Giacomini and White tests, ninety-five percent predictive-interval diagnostics, and the widely applicable information criterion. Model ranking is criterion dependent rather than winner take all. The skew Student t model attains the best widely applicable information criterion, the Student t model attains the best mean squared error, the asymmetric Laplace model attains the best quasi-likelihood loss, and the skew variance gamma model yields the narrowest near nominal predictive interval. The results show that non-Gaussian return laws materially change model ranking for Bitcoin across density fit, point forecasting, and interval calibration.

E1723: Deep hedging under CRRA preferences*Presenter:* **Mahoro Fukui**, Keio University, Japan

Deep reinforcement learning is leveraged to calculate option prices under the Constant Relative Risk Aversion (CRRA) utility function in heterogeneous markets. Two-Stage Optimization (TSO) is proposed to address challenges that may have prevented traditional deep hedging models from implementing hedging strategies under CRRA preferences. Since TSO does not require a performance measure to be cash-invariant or to take an exponential form, it can be generalized to broader, non-CRRA preferences as well.

E1542: A Bayesian panel data study on socioeconomic factors influencing exercise habits in Japan*Presenter:* **Tomoki Toyabe**, Kanazawa Gakuin University, Japan*Co-authors:* Makoto Nakakita, Sakae Oya, Naoki Kubota, Teruo Nakatsuma

Bayesian estimation of a panel logit model for binary longitudinal data using Markov chain Monte Carlo is examined. As an empirical application, panel data on exercise habits from the Japanese population is analyzed. Building on related existing approaches, the Ancillarity-Sufficiency Interweaving Strategy is incorporated into the MCMC algorithm to improve computational efficiency. An application of this approach to Japanese data is provided and its practical usefulness for the analysis of binary panel outcomes is illustrated.

EO192 Room 102 STATISTICAL MODELING AND INFERENCE WITH STRUCTURAL INFORMATION**Chair: Catherine Liu****E1842: Bayesian analysis of Cox regression with partly linear covariate effects via reversible jump MCMC.****Presenter:** Chun Yin Lee, Hang Seng University of Hong Kong, Hong Kong**Co-authors:** Kin Yau Wong, Hengtao Zhang, Yuanke Qu

The partially linear Cox-type regression model provides a robust and flexible framework for exploring potential nonlinear effects of continuous variables on censored outcomes in the context of complex diseases. Most contemporary research studies estimation methods for such models based on the Frequentist paradigm, necessitating the selection of bandwidth and/or the number of spline basis functions for smoothing. A Bayesian estimation approach is proposed that eliminates this requirement by employing the reversible jump Markov chain Monte Carlo (RJMCMC) algorithm. The proposed method can inherently estimate both the number and location of knots in the unknown function in a data-adaptive manner during the posterior inference process, thereby enhancing applicability and predictive accuracy. Finite-sample performance of the proposed method is evaluated through simulation studies. The effectiveness of the proposed method is illustrated through analyses of two medical datasets.

E2032: Change point detection under the heterogeneous factor strength**Presenter:** Catherine Liu, The Hong Kong Polytechnic University, Hong Kong

Detecting group structural instability is urgently needed for high-dimensional grouped vector-valued sequences in finance and many other application fields; overlooking it may incur severe risks. The group structure gives rise to mixed factor strengths under the latent factor structure containing both group and global signals. Existing methods fail to detect group structural breaks because of weak group signal strength. The global and group structural abruptness are modeled by a break-pair, which allows for a versatile change profile, including segmentation and ordering of breaks. In the coexistence of strong and weak factors, a new estimation paradigm is developed to guarantee the consistency of quasi-maximum likelihood-type estimators of the break-pair. Specifically, the strong and weak factors are separately estimated and the convergence rate of each estimator is enhanced by removing either the influence of the other or their interaction. This strategy builds upon a three-phase partition of the eigenvalues of the sample covariance matrix of observations, where weak group factors bring about one phase. Simulations demonstrate the unique ability to detect group structural change. The application is illustrated with the HKEX stock and S&P500 datasets. The proposed method remains applicable, after certain simplification, when the group structure is unknown or absent.

E2047: Certifiable perception: Generating safety maps via conformal segmentation**Presenter:** Rui Luo, City University of Hong Kong, Hong Kong

In safety-critical domains such as medical imaging and autonomous driving, standard image segmentation models often fail to provide reliable uncertainty quantification, leading to dangerous false negatives. While existing Conformal Risk control methods offer marginal coverage guarantees, they often perform inconsistently across diverse images, providing excessive coverage for simple inputs while failing to protect complex ones. Conformal Risk Adaptation (CRA) is a novel framework designed to enhance Conditional Risk control in segmentation by leveraging Adaptive score functions that adjust prediction sets based on the difficulty of individual images. Furthermore, Conditional Optimal Adaptive Thresholding (COAT) is presented as an end-to-end differentiable optimization approach that learns to predict the Optimal, image-specific threshold, effectively minimizing the Risk gap. Experiments demonstrate that these methods provide valid marginal Risk control while significantly improving consistency in Conditional Risk performance, offering a robust, principled solution for high-stakes perception tasks.

E1897: Inference and variable selection for two-phase studies with high-dimensional covariates**Presenter:** Kin Yau Wong, Hong Kong Polytechnic University, Hong Kong**Co-authors:** Haoyang Wang, Qingning Zhou

The two-Phase study design is widely used to improve estimation efficiency and reduce cost. In many two-Phase studies, the outcome and inexpensive covariates are obtained on all subjects in Phase I, while expensive covariates are measured only on a subset of subjects in Phase II. As a result, regression analysis of two-Phase studies faces a missing data problem. When two-Phase studies involve high-dimensional covariates, a more challenging high-dimensional missing data problem arises. For this problem, complete-case analyses are generally inefficient, while imputation or likelihood-based methods require a model for the missing covariates, which is almost impossible to correctly specify. To overcome these limitations, a two-step estimation method is proposed that refines a complete-data estimator by incorporating the incomplete data and auxiliary information if available to improve efficiency. This method does not require modeling the distribution of the missing covariates, and the resulting estimator is guaranteed to be at least as efficient as the complete-data estimator. Theoretical properties of the proposed method are established, including estimation consistency, inference validity, and variable selection consistency. The performance of the proposed method is evaluated via simulation studies and an application to a major cancer study is provided.

E2080: Covariate-augmented tensor factor models**Presenter:** Lingxiao Li, The Hong Kong Polytechnic University, China**Co-authors:** Catherine Liu, Elynn Chen

A covariate-augmented CP factor model for high-dimensional tensor-valued time series is studied. The proposed model extends the standard CP factor model by decomposing each mode-specific loading vector into a covariate-relevant component and a covariate-orthogonal component. The covariate effects are modeled nonparametrically through additive sieve approximations, while the residual component allows for latent loading variation not captured by the observed covariates. An iteratively projected CP procedure, IP-SO, is proposed for estimating the loading vectors and latent factors. The method first projects tensor observations onto low-dimensional sieve spaces generated by mode-specific covariates and then applies an iterative simultaneous orthogonalization step to separate the CP components. Convergence rates for the estimated loading structures and latent factors are established under weak temporal dependence. The theory shows that, when the observed covariates are informative about the loadings, the projection step reduces the effective noise dimension and leads to sharper estimation than methods that ignore auxiliary information. Monte Carlo experiments corroborate these findings, showing that IP-SO improves finite sample accuracy in weak signal and high dimensional settings.

EO156 Room 103 STATISTICAL MACHINE LEARNING FOR COMPLEX DATA**Chair: Tsung-I Lin****E1814: Robust estimation and inference for accelerated failure time models via double machine learning****Presenter:** Kun Ren, Hong Kong

Valid inference in accelerated failure time models with right-censored time-to-event data is challenging, particularly when infinite-dimensional nuisance functions are approximated via complex machine learning methods. A multi-stage estimation method leveraging double machine learning techniques is proposed to enable robust inference. The proposed approach mitigates biases derived from nonparametric nuisance estimation tasks and achieves asymptotically normal estimators under mild conditions. Furthermore, the approach generalizes prediction-powered inference to right-censored settings, providing a flexible guideline for maintaining statistical interpretability with black-box predictive models.

E1815: Deep transfer learning by model averaging for survival analysis*Presenter:* **Xingyu Lin**, The Hong Kong Polytechnic University, Hong Kong*Co-authors:* Xingqiu Zhao

In many clinical studies, data are collected from multiple sources, and these datasets may be small and different but related. A transfer learning model averaging prediction procedure for the Cox proportional hazards model with semi-parametric regression function is proposed to leverage the strength of these datasets. The proposed method incorporates model selection and model averaging techniques, and it allows for the possibility that the source datasets may not be comparable to the target dataset. The asymptotic optimality and weight convergence of the proposed method are established under some regularity conditions. Simulation studies are conducted to evaluate its performance. The results show that the proposed method can effectively improve the estimation of the target regression function by leveraging information from the source datasets.

E2063: A general framework for fair and robust regression*Presenter:* **Wenhai Cui**, The Hong Kong Polytechnic University, China*Co-authors:* Xiaoting Ji, Wen Su, Xingqiu Zhao

Fair regression methods typically rely on squared error loss, making them fragile under heavy-tailed noise. A general framework for robust regression under demographic parity (DP) is proposed that applies to a wide class of M-estimators, including Cauchy, Huber, least absolute deviation, quantile, and Tukey losses. An optimal fair transformation is introduced that guarantees DP while achieving the minimum population risk among all rank-preserving fair predictors, with convergence rates established for the resulting estimators. To balance fairness and predictive accuracy, an interpolation scheme is developed whose risk decreases while unfairness grows linearly with the interpolation parameter. The proposed framework can be further extended to conditional DP to account for legitimate covariates. Extensive simulation studies and real data applications demonstrate clear improvements over existing fair regression approaches in both robustness and predictive performance.

E2064: Optimal transport learning: Balancing value optimization and fairness in individualized treatment rules*Presenter:* **Xiaoting Ji**, The Hong Kong Polytechnic University, China*Co-authors:* Wenhai Cui, Wen Su, Xiaodong Yan, Xingqiu Zhao

Individualized treatment rules (ITRs) have gained significant attention due to their wide-ranging applications in fields such as precision medicine, ridesharing, and social welfare distribution. However, when ITRs are influenced by sensitive attributes such as race, gender, or age, they can lead to outcomes where certain groups are unfairly advantaged or disadvantaged. To address this gap, a flexible approach based on optimal transport theory is proposed, which is capable of transforming any optimal ITR into a fair ITR that ensures demographic parity. A trade-off ITR is introduced, designed to balance value optimization and fairness while accommodating varying levels of fairness through parameter adjustment. Additionally, a theoretical upper bound on the value loss for the trade-off ITR is established. The proposed method is demonstrated through extensive simulation studies and application to the Oregon Health Insurance Experiment dataset.

E2078: Deep nonparametric inference for conditional mean models with panel count data*Presenter:* **Anyin Feng**, The Hong Kong Polytechnic University Shenzhen Research Institute, China*Co-authors:* Qiang Wu, Xiangbin Hu, Xingqiu Zhao

In the statistical analysis of panel count data, semiparametric models such as the proportional mean model and the accelerated mean model are often used to investigate the occurrence rate of recurrent events. Misspecification of the prespecified model may lead to biased estimation and poor predictive performance. Therefore, it is essential to develop hypothesis testing methods to assess the adequacy of the assumed model. However, most existing diagnostic methods rely on graphical tools and lack theoretical justification. To address this issue, a hypothesis testing method to assess model adequacy with theoretical support is developed. Specifically, a deep extended mean model is proposed that integrates commonly used nonparametric and semiparametric models within a general framework. Treating the proposed model as the encompassing model, the hypothesis test is constructed based on the discrepancy between the parameter estimates obtained under the encompassing and prespecified models. Extensive simulation studies and an application to data from a skin cancer chemoprevention trial demonstrate the superior performance of the proposed method.

EO103 Room 105 SPATIAL STATISTICS**Chair: Pei-Sheng Lin****E1338: Tree-based methods for spatially dependent data: A case study of housing prices in Taipei***Presenter:* **Yi-Hung Kung**, Fu Jen Catholic University, Taiwan

With the rapid intensification of globalization and urbanization, housing prices have become a key indicator of economic and social development. Taipei, as the capital of Taiwan, exhibits complex housing price dynamics driven by multiple interacting factors, including transportation accessibility, environmental characteristics, building attributes, socioeconomic conditions, and the distribution of public facilities, resulting in substantial spatial heterogeneity. A spatial regression tree (SRT) framework is proposed to identify the key determinants of housing prices by integrating spatial dependence into nonlinear modeling structures. While conventional regression and tree-based methods effectively capture variable relationships, they often fail to adequately account for spatial autocorrelation and localized variation. To address this limitation, spatial information is incorporated through a spatial weight matrix, and a spatial lag mechanism is embedded within the regression tree framework, resulting in a geographically adaptive tree-based model. Empirical results show that the proposed SRT model effectively captures regional heterogeneity and outperforms traditional approaches in terms of predictive accuracy. The findings provide insights into the spatial mechanisms underlying urban housing markets and offer a flexible analytical framework for spatially heterogeneous data, with important implications for urban planning and housing policy.

E1370: Efficient and distribution-free variable selection in spatial models with zero inflation*Presenter:* **Chun-Shu Chen**, National Central University, Taiwan*Co-authors:* Chia-Ming Hsu

Extreme climate events, such as heavy rainfall, are often characterized by strong spatial dependence and a substantial proportion of zeros due to their rarity and localized occurrence. Modeling such data poses significant challenges, as it requires simultaneously accommodating spatial dependence, excess zeros, and complex covariate effects. A spatial modeling framework with zero inflation is utilized, with a particular emphasis on developing an efficient and distribution-free variable selection procedure for extreme climate analysis. The core contribution is a new distribution-free selection criterion that avoids reliance on a fully specified likelihood, thereby enhancing robustness against model misspecification, while remaining computationally efficient and scalable in high-dimensional spatial settings. Inspired by the regularization principle of Lasso, the proposed method enables fast and stable identification of influential covariates while preserving modeling flexibility. Extensive simulation studies demonstrate its accuracy, robustness, and computational efficiency. An application to Taiwan's 2016 daily extreme rainfall data further illustrates its practical value for environmental risk assessment and climate-related decision-making.

E1373: Threshold-boundary Poisson regression models for counting data analysis*Presenter:* **ChihHao Chang**, National Chengchi University, Taiwan

A threshold boundary Poisson regression (TBPR) model for analyzing count data with heterogeneous generating mechanisms is proposed. The proposed model allows the underlying regression structure to vary across subpopulations determined by an unknown threshold boundary, which can be either linear or nonlinear in the covariate space. To estimate the model parameters, an iterative two-stage estimation procedure is developed. In the first stage, a classification rule is constructed to partition the data into two groups based on their relative fitting performance under competing Poisson regression models. This classification rule is implemented via linear or nonlinear classifiers, enabling flexible modeling of complex threshold structures. In the second stage, separate Poisson regression models are fitted to each subgroup to update the regression parameters. Furthermore, the TBPR model is extended to accommodate zero-inflated count data, providing an alternative modeling framework to the conventional zero-inflated Poisson (ZIP) regression. Comprehensive simulation studies demonstrate that the proposed TBPR model achieves competitive or superior performance compared to the ZIP model, particularly when the data exhibit latent heterogeneity rather than excess zeros arising from a single mixture mechanism. An application to real-world data illustrates the practical effectiveness and interpretability of the proposed estimation algorithm.

E1436: Predicting income inequality: A non-stationary spatial covariance approach*Presenter:* **Ya-Mei Chang**, National Chung Cheng University, Taiwan*Co-authors:* Yi-Chen Lin, Pei-Hsin Huang, Truc Thuy Thi Le Nguyen

Inequality predictions are vital for effective cohesion policy. A spatial model for the Gini index is proposed to account for the heteroskedasticity and non-stationary spatial dependence found in European regional income inequality data. Model residuals are represented as a sum of multiple basis functions, stationary processes, and a white-noise process. To enhance computational efficiency, the least absolute shrinkage and selection operator (lasso) is employed to simultaneously select relevant processes and estimate parameters. An analysis of data from 29 European countries over the period 2009-2017 reveals that the Baltic and Nordic regions exhibit high variance and strong spatial correlation. In contrast, western and southern Europe show weak spatial dependence. By accommodating non-stationary spatial covariance, the proposed spatial model reduces the mean squared error by approximately 20% compared to standard non-spatial models that assume independent and identically distributed (i.i.d.) residuals.

E1494: Autonomous spatial statistical analysis with large language model agents*Presenter:* **Tzai-Hung Wen**, National Taiwan University, Taiwan

Recent advances in large language models (LLMs) have accelerated task-level automation, but current AI agents remain limited in their ability to conduct rigorous spatial statistical analysis. Valid spatial inference requires more than computation; it depends on appropriate problem formulation, neighborhood specification, and diagnostic checking. An autonomous AI agent is proposed that links natural-language research questions to reproducible spatial statistical workflows under curated data settings. The key methodological contribution is a structured mapping framework that translates unstructured queries into standard operating procedures for spatial analysis. The agent integrates exploratory spatial data analysis, automated spatial weight construction, Moran's I and local cluster analysis, spatial regression modeling, and internal self-verification through geometry validation, residual diagnostics, and spatial consistency checks. The system is evaluated through two case studies: regional educational inequality and epidemiological hotspot detection. Robustness is assessed against expert-constructed benchmark workflows in terms of methodological fidelity, interpretive accuracy, and execution stability. Results show that the agent can autonomously produce end-to-end spatial statistical analyses with outputs highly consistent with benchmark expert results, suggesting a practical path toward reproducible and accessible spatial statistical modeling.

EO071 Room 106 RECENT DEVELOPMENTS IN STATISTICAL COMPUTING AND APPLICATIONS**Chair: Shih-Feng Huang****E1377: Indicator-based Bayesian variable selection for ordinal probit regression model***Presenter:* **Ray-Bing Chen**, National Tsing Hua University, Taiwan*Co-authors:* Tsun-Cheng Chang, Kuo-Jung Lee

The ordinal probit regression model is appropriately utilized when the response variable is ordinal. Within this modeling framework, Bayesian variable selection is applied to accurately identify the active variables that significantly influence the response. The proposed methodology introduces a specific indicator for each variable and develops a corresponding Markov Chain Monte Carlo algorithm to generate posterior samples of these indicators, thereby facilitating robust statistical inference. This approach is evaluated using both simulated and empirical datasets to ensure the variable selection and predictive capabilities perform optimally. During the empirical application, high pairwise correlations among certain variables are observed, which introduce analytical complexities. To resolve this issue, the proposed method is expanded to include group selection, categorizing variables based on prior knowledge and assigning an indicator to each respective group. Ultimately, this group selection strategy enhances overall performance and yields superior analytical outcomes in the practical application.

E1375: A Bayesian approach for change point detection in multivariate time series*Presenter:* **Chi-Hsiang Chu**, National University of Kaohsiung, Taiwan*Co-authors:* Sangyeol Lee, Shih-Feng Huang

A Bayesian framework is proposed for detecting structural change points in multivariate time series through a structural break vector autoregressive (SBVAR) model. The proposed approach extends the univariate structural break autoregressive (SBAR) model to a higher-dimensional setting, allowing for dynamic interactions among multiple time series while accommodating time-varying parameter regimes. To identify structural breaks and estimate segment-wise coefficients, a three-stage Bayesian procedure is developed. In the first stage, change points are represented by binary indicators associated with groups of regression coefficients, and a stochastic matching pursuit (SMP) algorithm is employed to explore the model space via add-delete operations. The marginal likelihood for each group is derived in closed form by integrating out the corresponding parameters under a Gaussian prior, thereby enabling efficient computation of posterior inclusion probabilities. In the second stage, candidate change points are constructed based on posterior inclusion probabilities, followed by an information criterion-based refinement to determine the final set of structural breaks. In the final stage, segment-specific VAR coefficients are estimated accordingly. Simulation studies are conducted to demonstrate the feasibility and effectiveness of the proposed approach in identifying structural changes in multivariate time series.

E1646: Particle-filter variational Bayesian estimation for VAR models with compound Poisson jumps*Presenter:* **Wei-Ting Lai**, Feng Chia University, Taiwan*Co-authors:* Shih-Feng Huang, Ray-Bing Chen

A particle-filter-assisted variational Bayesian (PF-VB) approach for vector autoregressive models with compound Poisson jumps is proposed to jointly capture continuous dynamics and jump behavior in multivariate time series. The method integrates particle filtering for jump detection with variational Bayesian inference for parameter estimation and structure learning. Extensive simulation studies under different scenarios evaluate estimation accuracy and jump detection performance. Results demonstrate that the approach provides accurate parameter estimation and effectively detects large jumps. For empirical analysis, the method is applied to daily realized variance data of multiple international stock market indices.

constructed from high-frequency returns. The proposed approach provides a flexible and practical framework for modeling time series with jump behavior.

E1376: **Portfolio optimization via dynamic networks and vine copulas**

Presenter: **Tzu-Hsin Chien**, National Central University, Taiwan

Co-authors: Meihui Guo, Shih-Feng Huang

The application of vine copulas combined with network-based methods for portfolio optimization is explored. A DeGARCH technique is employed to preprocess each series to address inherent characteristics such as autocorrelation, conditional heteroscedasticity, and volatility clustering in financial time series. A similarity matrix is then computed from the multivariate DeGARCH data and used to construct a global minimum spanning tree (MST), which facilitates the identification of suitable stocks for portfolio construction. Subsequently, a local MST (LMST) is built from the selected stocks, and a vine copula is applied based on the LMST structure to model the joint distribution. This copula-network based distribution is then used to determine portfolio weights. An empirical analysis conducted on component stocks of the S&P 100 index over the 2019-2023 period using a rolling-window framework shows that the proposed method achieves competitive cumulative returns compared to benchmark approaches.

E1374: **Statistical inference for a contaminated zero-inflated compound Poisson-gamma process with applications**

Presenter: **Shih-Feng Huang**, National Central University, Taiwan

The estimation problem for a contaminated zero-inflated compound Poisson-gamma (ZiCPG) process is considered. The contaminated ZiCPG model provides a flexible framework for modeling the dynamics of stochastic processes with jumps by simultaneously incorporating zero inflation and random jump sizes into a Poisson process with noise contamination. Consequently, computing the maximum likelihood estimator is challenging. To address this issue, an effective procedure based on jump-state identification and particle filtering is proposed to estimate latent jumps and mitigate the computational challenges arising from noise contamination. A prediction model is further developed based on the estimated jumps. Numerical results from various simulation scenarios demonstrate that the proposed estimation scheme achieves satisfactory performance. Moreover, when applied to an electricity consumption dataset, the proposed prediction model yields superior performance in forecasting potential abnormal electricity consumption compared with competing methods.

EO223 Room 107 ADVANCES IN CAUSAL INFERENCE: FROM STRUCTURE LEARNING TO EFFECT ESTIMATION Chair: Shohei Shimizu

E1310: **Recent advances in reward modeling for reinforcement learning**

Presenter: **Masashi Sugiyama**, RIKEN/The University of Tokyo, Japan

Reinforcement learning (RL) has achieved remarkable success in robotics, games, and language model post-training, where reward signals are essential for effective agent training. Recent research on advanced reward modeling will be presented. This includes robustifying RL through transfer and weakly supervised learning, coping with interval-based rewards, and exploring diverse reward aggregation frameworks that move beyond the traditional discounted sum.

E1472: **Toward interventionally fair decision-making under partial causal graph knowledge**

Presenter: **Yoichi Chikahara**, NTT Communication Science Laboratories, Japan

Machine learning is increasingly used to support high-stakes decisions such as loan approval, hiring, and abuse detection, but its use raises serious concerns about discriminatory predictions based on sensitive attributes such as gender and race. Because discrimination is often legally assessed in terms of causal effects, recent studies have used causal inference to ensure fairness in predictions. However, most existing methods assume that the true causal graph among variables is fully known. This assumption limits practical deployment because it requires solving the highly challenging problem of perfectly identifying the causal graph from observational data. A method is proposed for learning predictors that satisfy interventional fairness using a coarse causal representation called a cluster causal graph, which describes causal relationships among user-specified groups of variables. Although cluster causal graphs are easier to estimate in high-dimensional settings, they introduce graph uncertainty because the internal causal structure within each cluster is unknown. To address this challenge, (1) a graph algorithm that enumerates adjustment sets for identifying interventional distributions and (2) a barycenter kernel MMD constraint that efficiently measures distances between interventional distributions using multiple adjustment sets are developed. Experiments on synthetic and real-world data show improved accuracy-fairness trade-offs under graph uncertainty.

E1626: **From prediction to understanding: Causal discovery for data science and AI applications**

Presenter: **Shohei Shimizu**, The University of Osaka, Japan

Causal discovery aims to go beyond prediction and infer cause-effect relationships directly from data, which is essential for reliable decision-making in science, industry, and government. Recent advances in causal discovery are presented together with real-world applications in medicine and environmental science, in a way accessible to a broad data science and AI audience. On the methodological side, extensions of existing approaches to more realistic settings are introduced, including cases with hidden common causes and datasets that mix discrete and continuous variables. These challenges are common in practice but remain difficult for standard methods. Results show how causal discovery can still provide useful insights under such conditions. On the application side, these methods are demonstrated on real problems. In medicine, causal discovery helps identify clinically meaningful relationships from observational data, supporting more interpretable analyses. In environmental science, a study on adverse outcome pathways is presented, where causal relationships among biological events are inferred from data. Overall, causal discovery complements machine learning by enabling a shift from prediction to understanding how interventions influence outcomes.

E1962: **Robust and efficient g-estimation of structural nested mean models in randomized trials with partial compliance**

Presenter: **Tomohiro Shinozaki**, The University of Tokyo, Japan

Treatment switching and concomitant therapies often make randomized trials only partially compliant with allocation, complicating causal interpretation. G-estimation of structural nested mean models (SNMMs) in such trials is revisited and an estimator is proposed that combines two sources of information: randomization-based estimating equations and equations based on no-unmeasured confounding for observed treatment histories. The equations are solved jointly for a common SNMM using generalized method of moments (GMM). Compared with conventional randomized g-estimation, the approach can include heterogeneous causal effects of non-randomized treatments and use covariates to improve precision. Compared with observational g-estimation, it retains randomization as an instrumental source of variation and may be less sensitive to unmeasured confounding. This perspective also connects to recent work on orthogonal and debiased estimating equations, although the focus is on classical g-estimation and GMM rather than machine-learning implementation. The method is evaluated in simulations covering non-compliance patterns, proportions of non-compliers, unmeasured confounding, and misspecification of auxiliary outcome and propensity-score models. An application uses data from the SELECT BC Study, a randomized trial of first-line chemotherapy for metastatic breast cancer in which concomitant therapies and treatment switching were frequent.

E1193: **Selection of mediators and dependence structure for high-dimensional mediation analysis**

Presenter: **Lijia Wang**, University of Toronto, Canada

Co-authors: Yeying Zhu, Richard Cook

Causal mediation analysis examines the potential causal pathways between an exposure variable and outcome through intermediate variables

with the goal of estimating direct and indirect effects. In practice, intermediate variables may be high-dimensional, in which case one may first aim to identify the true mediators among them. The dependence structure among mediators may then be studied with the goal of identifying a simple sufficient structure. A two-stage penalized estimation procedure is proposed to meet these goals. The first stage involves selecting mediators by identifying non-zero indirect effects via a penalized regression. The second stage aims to simplify the correlation structure among selected mediators enabling the estimation of individual, grouped or joint effects. Through transformation of variables, the correlation selection problem can be reformulated as a standard LASSO problem. The two stages can be performed jointly or sequentially and the performance of each implementation is studied through simulation studies. Finally, the proposed approach is applied to a psychiatry study in which the aim is to identify methylation loci that mediate the causal effect of childhood trauma on adult stress level.

E0229 Room 202 HIGH-DIMENSIONAL ANALYSIS FOR HEALTHCARE, FINANCE, AND ENVIRONMENT
Chair: Amanda Chu
E1428: On desparsified focused information criterion for high-dimensional linear models: with an application to PM2.5 data

Presenter: **Cy Sin**, National Tsing Hua University, Taiwan

A modification of the covariate selection criterion (CSC) is proposed to minimize the mean squared errors (MSE) of the focused parameter in causal methods such as difference-in-differences (DiD). Following a previous approach, a debiased Lasso is first applied to a high-dimensional linear model. All regularized models then proceed through the focused information criterion (FIC), and the MSE of the average treatment effect (ATE) parameter is minimized. The method is applied to PM2.5 data comprising 365 city-level observations. Apart from socioeconomic and meteorological covariates, the analysis includes 365 cross-section-specific dummies and covariates.

E1399: Periodic intraday volatility modeling and forecasting

Presenter: **Edward Meng-Hua Lin**, Tunghai University, Taiwan

An innovative conditional autoregressive range (CARR) model with periodic time-varying parameters is proposed, which is referred to as the periodic conditional autoregressive range (PCARR) model. This model is able to display time series in which the volatility distribution changes over time following a seasonal or periodic pattern. The construction of parameter estimation and volatility forecasting involves Bayesian Markov Chain Monte Carlo sampling methods. The model's ability to forecast and capture stylized facts is increased by incorporating intraday range data with the day-of-the-week effect, as shown in simulations and empirical analyses.

E1945: Model-based clustering with item response and textual data

Presenter: **Thomas Chan**, Hong Kong University of Science and Technology, Hong Kong

Co-authors: Mike So, Amanda Chu, Chin Sum Shui

A flexible model-based clustering framework is developed that integrates mixed-type data, including ordinal, nominal, and textual items, into a unified Item Response Theory (IRT) model. By assuming that latent traits follow a mixture of elliptical distributions, the framework provides a robust alternative to standard Gaussian mixtures, effectively accommodating heavy-tailed data and outliers. The model captures complex Response patterns through Item group factors and spline smoothing, while qualitative textual data are incorporated by embedding topic models directly within the latent trait space. To ensure an interpretable and parsimonious cluster structure, a repulsive normal prior is employed to mitigate the risk of redundant or overlapping clusters. Model estimation is conducted via a variational Bayes approach optimized with the Adam algorithm, facilitating efficient computation for high-dimensional data. Simulation results indicate that the proposed model outperforms classical multidimensional IRT and latent Dirichlet allocation (LDA) in both clustering accuracy and adjusted mutual information. The methodology is applied to a real-world dataset of shuttle bus service evaluation, demonstrating its ability to identify distinct respondent profiles and provide data-driven recommendations for service optimization.

E1934: Advancing psychosocial health assessment with Automatic Speech Analytics Program (ASAP)

Presenter: **Amanda Chu**, The Education University of Hong Kong, China

ASAP is developed as a machine learning-based system designed for early detection of psychosocial distress, with a focus on statistical methods for linguistic and acoustic feature extraction. ASAP records client Speech, transcribes conversations in real time, delivers immediate risk evaluations through an intuitive interface, and generates automated reports to reduce administrative workload. Linguistic features were obtained by transcribing 100 audio recordings and developing a project-specific dictionary of 53 words, organized into 14 topics and four themes. Each word was mapped to three risk indicators: the Beck Depression Inventory-II, Caregiver Burden Inventory, and Family Resilience Assessment Scale. Principal component analysis was applied to reduce dimensionality and minimize redundancy and bias. Acoustic features were extracted using openSMILE, encompassing 65 low-level descriptors related to pitch, voice quality, spectral, and energy characteristics, which were further expanded into 6,503 statistical features representing waveform variability. The integration of these linguistic and acoustic features enabled accurate psychosocial risk classification, achieving 86% accuracy in differentiating stress levels. The combination of dictionary-based linguistic analysis with advanced signal processing and statistical techniques highlights the potential of feature engineering in Speech Analytics for improving Resilience Assessment and Depression detection.

E1918: Integrating LLM sentiment, emotional word scoring, and syntactic dependency in speech for depression detection

Presenter: **Shun Hin Chan**, The Hong Kong University of Science and Technology, Hong Kong

Advancing statistical approaches for psychosocial health assessment requires integrating diverse analytic perspectives across speech and text. A unified framework combines Large Language Model (LLM) sentiment analysis, syntactic dependency features, and emotional word scoring to improve depression detection. First, a Beyond Holistic Context approach applies randomized chunking and statistical aggregation of LLM outputs, generating 1,000 severity rating iterations per participant, with summary statistics used as predictors of depression. Second, syntactic features, including mean sentence length reflecting average words per sentence and mean dependency distance derived from dependency grammar to capture grammatical relationships, were extracted from caregiver speech. Third, dimensional sentiment analysis using valence and arousal scores quantified emotional word use, with valence reflecting pleasantness and arousal indicating intensity. These predictors were integrated using stacking, an ensemble technique that statistically combines multiple learners. Results demonstrate that stacking meta-models consistently outperformed individual models, achieving stronger predictive performance. This multi-level statistical framework demonstrates the value of combining LLM sentiment analysis, syntactic dependency analysis, and emotional word scoring in speech, underscoring the potential of multidimensional measurements for robust, scalable psychosocial health evaluation.

EO142 Room 203 DISCUSSION ON EXPERIMENTAL DESIGN AND DIMENSION REDUCTION**Chair: Shao-Hsuan Wang****E1439: Optimal test planning for heterogeneous wiener processes****Presenter:** Ya-Shan Cheng, National Central University, Taiwan**Co-authors:** Chien-Yu Peng

Degradation models based on heterogeneous Wiener processes are commonly used to assess information on the lifetime of highly reliable products. An optimal test plan given limited resources is generally obtained using numerical methods for heterogeneous Wiener processes. However, numerical searches for optimal test plans have the disadvantage of being time-consuming and may provide unclear explanations for the findings. To overcome these difficulties, an explicit expression is derived for decision variables (such as the termination time, number of measurements, and sample size) of D- and V-optimal test plans with cost constraints. The theoretical results not only ensure that the optimal test plan is found, but also provide clear insights into the decision variables affected by model parameters and experimental costs. Some numerical examples are presented to support the efficiency and applicability of the optimal test plans.

E1659: On the theoretical limits and structure of optimal designs in group testing regression**Presenter:** Shih-Hao Huang, National Central University, Taiwan

D-optimal designs for group testing regression models are investigated under idealized settings where both covariates and group structures are fully controllable. Theoretical limits of estimation efficiency and the associated optimal design structure are characterized. For the single-covariate case under the complementary log-log link, analytical results show that the optimal design has a simple form and is invariant in a canonical scale. For multiple covariates under constraints, optimal designs are characterized numerically, revealing that key structural features from the single-covariate case persist in more general settings. The results provide theoretical benchmarks for evaluating design efficiency and guiding the construction of practical designs.

E1884: Bayesian sparse principal coordinates analysis with microbiome discoveries**Presenter:** Shao-Hsuan Wang, National Central University, Taiwan

Principal coordinates analysis (PCoA) is a widely used tool for exploring relationships among samples based on pairwise dissimilarities. However, classical PCoA typically produces dense loading structures, which makes interpretation difficult, particularly in ultrahigh-dimensional microbiome studies where identifying biologically relevant features is essential. A sparse PCoA framework that incorporates regularization to produce interpretable coordinate axes with explicit feature selection is introduced. To further accommodate flexible sparsity structures, a Bayesian sparse PCoA model is developed that employs global-local shrinkage priors from the three-parameter beta-normal family to achieve adaptive sparsity. Asymptotic results clarify how sample size and dimensionality influence the behavior of PCoA. Through simulation studies and analyses of the Hadza gut microbiome and pancreatic cancer tumor microbiome datasets, the proposed method substantially improves sparsity and interpretability while preserving key ecological and clinical signals.

E1969: Approximate mixture designs under the DuMouchel-Jones Bayesian D-optimality criterion**Presenter:** Hsiang-Ling Hsu, National University of Kaohsiung, Taiwan

Optimal designs for mixture experiments are typically constructed under a specified regression model, but their performance can be sensitive to model misspecification or omitted terms. A Bayesian modification of D-optimality is employed to distinguish primary inferential terms from potential terms representing possible model expansion. Approximate designs for Scheffe quadratic mixture models are investigated on a continuous simplex under this Bayesian criterion. An equivalence theorem is established to characterize optimality and to verify candidate designs. For three- and four-component mixtures, the optimal supports are shown to concentrate on geometrically interpretable boundary points, specifically simplex vertices and edge midpoints. The associated design weights vary systematically with the prior adjustment parameter, reflecting the balance between the primary model and potential expansions. Numerical investigations for five- and six-component mixtures suggest that these structural patterns persist in higher-dimensional mixture spaces. Existing discrete designs are evaluated using D-efficiency under the same criterion, and computational verification is supported by particle swarm optimization and an interactive Shiny application. The resulting framework supports robustified approximate designs for mixture experiments under model uncertainty.

E2076: Robust estimation and efficient design for group testing under error uncertainty**Presenter:** Chun-Ting Chen, National Central University, Taiwan**Co-authors:** Shih-Hao Huang, Zih-Jing Lin

Group testing is widely used in public health, but its performance can be seriously affected by test errors. Adopting simplified error models can easily lead to substantial model misspecification bias, while more flexible models tend to increase the variance of the estimator. A robust estimation method is introduced that automatically downweights potentially misclassified observations to mitigate bias. An adaptive data collection strategy based on this estimation is further proposed to reduce the variance of the estimator. Simulation studies under various error mechanisms show that the proposed methods usually have competitive or even better performance relative to conventional MLE-based approaches.

EO202 Room 205 BAYESIAN AI: FOUNDATIONS, ALGORITHMS, FAIRNESS**Chair: Rafael Weissbach****E1513: Blockwise-fairness gap penalty for subgroup-fair clustering****Presenter:** Jeong Hankyo, Seoul National University, Korea, South

AI fairness is a critical concern in automated decision-making systems. A fair clustering algorithm is proposed for settings with multiple sensitive attributes. Subgroup-blocks are introduced to represent sensitive groups at multiple scales, ranging from marginal groups to subgroups, and a fairness metric called the blockwise-fairness gap is defined over a given collection of subgroup-blocks. Because this gap is discrete and non-smooth, it is difficult to optimize directly. Therefore a smooth surrogate fairness gap and an efficient algorithm that minimizes clustering cost while penalizing the surrogate fairness gap is developed, yielding simple closed-form updates and enabling numerically stable optimization. Theoretical guarantees are provided showing that reducing the proposed surrogate fairness gap reduces the blockwise-fairness gap. Experiments on benchmark datasets demonstrate that the proposed method achieves favorable cost-fairness trade-offs, with computational efficiency and numerical stability compared to existing approaches.

E1530: Bayesian one-pass online learning for generalized linear models**Presenter:** Minwoo Chae, Pohang University of Science and Technology, Korea, South**Co-authors:** Jeyong Lee, Junhyeok Choi, Dongguen Kim

A Bayesian one-pass online learning algorithm is proposed for large-scale streaming data where observations are processed individually and exactly once. By utilizing a local quadratic approximation of the log-likelihood within a Gaussian variational framework, the method achieves computationally efficient, closed-form updates. An online analogue of the Bernstein-von Mises theorem is established, proving that the sequentially updated posterior is asymptotically equivalent to the full batch posterior and provides valid frequentist uncertainty quantification. The analysis and numerical experiments demonstrate that a critical warm-start phase ensures stability and allows the algorithm to match the efficiency of the batch MLE, consistently outperforming first-order methods like SGD.

E1531: Bayesian analysis of competing risks data with dependent left truncation via a trivariate Gaussian copula*Presenter:* **Kota Izumi**, Kitasato University, Japan*Co-authors:* Hirofumi Michimae

For competing-risks data subject to left truncation, independence between the latent failure times and the truncation time is often assumed. If this assumption is violated, ignoring the dependence may lead to biased inference. A Bayesian model for competing risks with dependent left truncation is developed by specifying the joint distribution of the two latent failure times, corresponding to competing causes, and the truncation time using a trivariate Gaussian copula. Weibull distributions are assumed for the marginal distributions of the latent failure times, and a uniform distribution is assumed for the marginal distribution of the truncation time, so that the truncation mechanism and the competing-risks process are modeled jointly. Bayesian estimation is performed in Stan via Markov chain Monte Carlo, and uncertainty is quantified through the posterior distributions of the Weibull and dependence parameters. For comparison, maximum likelihood estimation under the same joint model and an alternative model that assumes independence between the failure and truncation times are also considered. In simulation studies, data are generated under prespecified sample sizes, truncation rates, and dependence structures, and the methods are compared in terms of estimation accuracy and interval coverage. The inferential impact of ignoring dependent truncation and the practical value of explicitly modeling it are examined.

E1536: Bayesian neural networks for functional ANOVA model*Presenter:* **Choeun Kim**, Seoul National University, Korea, South

With the increasing demand for interpretability in machine learning, functional ANOVA decomposition has gained renewed attention as a principled tool for breaking down high-dimensional function into low-dimensional components that reveal the contributions of different variable groups. Recently, Tensor Product Neural Network (TPNN) has been developed and applied as basis functions in the functional ANOVA model, referred to as ANOVA-TPNN. A disadvantage of ANOVA-TPNN, however, is that the components to be estimated must be specified in advance, which makes it difficult to incorporate higher-order TPNNs into the functional ANOVA model due to computational and memory constraints. A Bayesian inference procedure for the functional ANOVA model with TPNN basis functions is proposed, termed Bayesian-TPNN, enabling the detection of higher-order components with reduced computational cost compared to ANOVA-TPNN. An efficient MCMC algorithm is developed and it is demonstrated that Bayesian-TPNN performs well by analyzing multiple benchmark datasets. Theoretically, it is proved that the posterior of Bayesian-TPNN is consistent.

E1687: Truncation-adjusted maximum likelihood for Weibull duration models*Presenter:* **Tobias Jansen**, University of Cologne, Germany*Co-authors:* Dominik Wied

Duration data subject to left truncation arise when entry into an observation window is conditioned on survival. Maximum likelihood estimation based on the unconditional density ignores this selection mechanism and yields inconsistent estimators. A truncation-adjusted maximum likelihood estimator for Weibull duration models is developed under double truncation induced by a uniform entry mechanism. A selection probability α_0 generalizes the closed-form exponential case to arbitrary Weibull shape and is evaluated by numerical quadrature. Identification is established via an injectivity argument based on the linear independence of Weibull log-density components on the observation region, dispensing with the memorylessness assumption used in the exponential case. Consistency and asymptotic normality follow from standard M-estimator arguments. Truncation contracts both information curvature and score variance without offsetting, so the sandwich form does not reduce to the inverse information. A feasible plug-in estimator based on a numerical Jacobian and outer-product score delivers valid inference. Monte Carlo simulations across truncation severities and Weibull shape regimes confirm bias reduction and near-nominal coverage relative to the naive maximum likelihood benchmark. An empirical application to German firm insolvency data quantifies the hazard shape and the magnitude of the truncation correction.

EO133 Room 208 STATISTICAL AND AI METHODS FOR SPATIAL OMICS**Chair: Can Yang****E1327: CoxFormer enables spatial omics inference with multimodal generative modeling***Presenter:* **Jin Liu**, The Chinese University of Hong Kong, Shenzhen, China

CoxFormer (co-expression pre-trained transformer) is presented. This universal gene-embedding framework uniquely integrates literature-derived gene knowledge with data-driven co-expression networks from bulk tissues and large-scale single-cell atlases to model transcriptome-wide gene-gene relationships. CoxFormer learns robust 512-dimensional embeddings for 32,016 human genes, capturing their functional and regulatory structure. It also serves as a generative prior that is deployed through a flexible framework for transcriptome-wide spatial inference under multimodal contexts. It is demonstrated that this generative framework unifies four key spatial applications beyond measured genes: histology-based imputation, gene activity prediction for chromatin accessibility, subcellular super-resolution inference for unassayed genes, and the detection of pathological tissue regions. Ultimately, CoxFormer provides a versatile foundation for multimodal spatial omics analysis, leading to whole-transcriptome insights beyond the constraints of limited gene panels.

E1773: Cross-modal denoising and integration of spatial multi-omics data with CANDIES*Presenter:* **Mingxuan Cai**, City University of Hong Kong, Hong Kong

Spatial multi-omics data offer a powerful framework for integrating diverse molecular profiles while maintaining the spatial organization of cells. However, inherent variations in data quality and noise levels across different modalities pose significant challenges to accurate integration and analyses. CANDIES is introduced as a method that leverages a conditional diffusion model and contrastive learning to effectively denoise and integrate spatial multi-omics data. With its innovative model and algorithm designs, CANDIES enhances the quality of spatial multi-omics data and yields a unified and comprehensive joint representation, thereby enabling many downstream analyses. Extensive evaluations on diverse synthetic and real datasets, including MISAR-seq data from the mouse brain, spatial CITE-seq data from human skin biopsy tissue, spatial Mux-seq and spatial ATAC-RNA-seq data from the mouse embryo, and 10x Visium data from human lymph nodes, demonstrate that CANDIES shows superior performance on various downstream tasks, including denoising, spatial domain identification, spatiotemporal trajectory reconstruction, and spatial association mapping for complex human traits. In particular, CANDIES representations can be integrated with the rich resources from genome-wide association studies (GWASs), allowing the spatial domains to be linked with complex human traits, yielding spatially resolved interpretations of complex traits in their relevant tissues.

E1811: MultiGATE: Integrative analysis and regulatory inference for spatial multi-omics*Presenter:* **Zhixiang Lin**, The Chinese University of Hong Kong, Hong Kong

New spatial multi-omics technologies that jointly profile transcriptomic and epigenome/proteome markers for the same tissue section have expanded the frontiers of spatial techniques. MultiGATE utilizes a two-level graph attention auto-encoder to integrate multi-modality and spatial information in spatial multi-omics data. The key feature of MultiGATE is that it simultaneously performs embedding of spatial pixels and infers cross-modality regulatory relationships, which allows deeper data integration and provides insights into transcriptional regulation. By effectively integrating spatial multi-omics data, MultiGATE enhances the extraction of latent embeddings of pixels and improves the inference of transcriptional regulation for cross-modality genomic features.

E1812: Automated landmark discovery method for integration of spatial omics data*Presenter:* **Xingjie Shi**, East China Normal University, China

Integrating multiple spatial omics datasets is essential for building spatially coherent atlases and performing comparative analyses, but is challenged by variability between tissue slices. Landmark-based alignment is a robust strategy, yet it relies on laborious manual annotation or inaccurate image-based methods dependent on prior knowledge. ARIEL, a framework for robust alignment and information transfer powered by a fast automated omics-driven landmark discovery method, is presented. ARIEL significantly outperforms existing methods in accuracy and efficiency, especially in complex cross-platform scenarios. The utility of ARIEL is demonstrated through construction of a reusable landmark library and spatial atlas of the human liver lobule, enabling automatic annotation of anatomical structures in new samples and leading to identification of disease-associated spatial gene programs missed by conventional analyses. ARIEL provides a versatile solution for spatial omics integration, repurposing landmarks as tools for biological discovery.

E1818: NicheScope: Identifying multicellular niches and niche-regulated cell states in spatial transcriptomics*Presenter:* **Jiashun Xiao**, Sun Yat-sen University, China

The functional state of a Cell is intricately linked to its local microenvironment, or Cell niche, composed of multiple interacting Cell types. Deciphering how these Multicellular Niches regulate Cell States is fundamental to understanding tissue biology and disease mechanisms, yet remains challenging with current computational approaches. NicheScope, a unified framework for identifying Multicellular Niches (MCNs) and their corresponding NicheRegulated Cell States (NRCSSs) from spatial transcriptomics (ST) data, is introduced. NicheScope builds on the principle that a Cell's transcriptional state is influenced by its local Multicellular composition. By jointly modeling the neighborhood cellular composition and the gene expression profile of a target Cell type, NicheScope enables the discovery of MCNs defined by specific neighboring Cell type combinations and their associated NRCSSs characterized by distinct gene programs. NicheScope's power and versatility are demonstrated across diverse tissues and ST platforms. The results establish NicheScope as a robust, scalable, and interpretable framework for dissecting the spatial and functional organization of complex tissues, providing new insights into Multicellular coordination in health and disease.

EC278 Room B103 APPLIED ECONOMETRICS**Chair: Dmitri Blueschke****E1163: Subsidy promise and anticipation effects: Evidence from electric motorcycles***Presenter:* **Fang-Chang Kuo**, National Chung Cheng University, Taiwan

The aim is to examine how the anticipation of a subsidy promise made during an election campaign influences consumer behavior in durable goods markets. Leveraging the case of Taiwan's 2022 local elections, where a mayoral candidate pledged to provide free Gogoro electric motorcycles, the Synthetic Control Method is employed to construct counterfactual sales paths for both electric and gasoline-powered motorcycles. Results reveal a substantial decline in electric motorcycle registrations immediately following the policy announcement and again after the candidate's electoral victory, while gasoline motorcycle sales remain largely unaffected. This pattern indicates that the anticipated subsidy induced forward-looking consumers to delay purchases, resulting in significant intertemporal substitution. In contrast, the absence of a response in gasoline motorcycle sales suggests limited cross-category substitution. A complementary difference-in-differences analysis conducted after the policy's formal rollout in July 2023 further shows that ignoring these anticipation effects would lead to overestimated policy impacts. Together, the findings highlight the importance of accounting for implementation lags and consumer expectations in the evaluation of subsidy programs.

E1204: The honeymoon effect in team performance: Evidence from Counter-Strike*Presenter:* **Jan Rejthar**, Prague University of Economics and Business, Czechia*Co-authors:* Petr Krautwurm

Teams are believed to perform above expectations after a change in their roster, temporarily. The presence of the honeymoon effect within teams is investigated using data from the entire history of professional Counter-Strike: Global Offensive. Semiparametric generalized additive mixed-effects models are utilized to analyze team performance trajectories following lineup changes. It is shown that teams tend to overperform immediately after a lineup change, but performance dips afterward. Furthermore, performance does not recover after the initial dip but plateaus at a lower level. Over the average lifespan, team performance declines by about 16 percent of the interquartile range. These results challenge theories predicting post-dip recovery, suggesting instead that established teams suffer from stagnation. The findings indicate that new members tend to bring fresh stimuli that temporarily boost performance, and imply that roster adjustments may revitalize stagnant teams and combat inertia in broader labor markets.

E1889: European mineral trade and technology-driven energy transition: Stringent policy vs. geopolitical risks*Presenter:* **Dmitri Blueschke**, University of Klagenfurt, Austria*Co-authors:* Monirul Islam, Oleg Mariev, Viktoria Blueschke-Nikolaeva

The aim is to investigate the relationship between European imports of aluminum and copper and clean energy technology-driven energy transition, while considering the roles of mineral prices, environmental policy stringency, inflationary pressures, and geopolitical risks from 2000 to 2023. Applying the method of moments quantile regression, the results reveal distinct responses: aluminum imports consistently rise with solar technology-driven energy generation across all market conditions, whereas copper imports respond positively to wind technology-based energy production in bear markets but negatively in bull markets. Environmental policies show a negative impact on aluminum imports but a mixed effect on copper, with negative responses in bear markets and positive responses in extreme bull markets. Geopolitical risks have no significant influence on imports of either mineral, while inflationary risks lead to increased demand for both. Notably, environmental policy stringency fails to mitigate inflationary or geopolitical risks in the aluminum import model but succeeds in the copper import model. To ensure robust findings, the Driscoll-Kraay standard errors and high-dimensional fixed effects methods are implemented to address cross-sectional dependency and endogeneity issues, respectively. These findings highlight the need for proactive, adaptive policies to secure mineral supply chains and safeguard Europe's technology-driven clean energy transition.

E1910: Economic policy uncertainty, financial development, and shadow economy: Evidence from cross-country data*Presenter:* **Chen-Sheng Lin**, Tamkang University, Taiwan

Since 2020, surging economic policy uncertainty (EPU) has disrupted global economic systems and financial intermediation. While recent studies link EPU to shadow economy (SE) expansion, they largely overlook financial development (FD) and EPU's moderating role, risking omitted variable bias. Using a 28-country panel (2000-2024) and a panel quantile autoregressive distributed lag (PQARDL) model, the long-run nexus among EPU, FD, and the SE is investigated, controlling for other variables. Applying the pooled mean group estimator across conditional SE quantiles unveils significant heterogeneous effects. First, robust long-run convergence exists across all quantiles. Second, at middle and high quantiles, both EPU and FD inherently restrain the SE. However, their interaction reveals a critical moderating effect: amid heightened uncertainty, risk shifting and informal financing severely undermine FD's ability to reduce the SE. Conversely, at lower quantiles, FD successfully fortifies formal financial channels, further suppressing SE size. Regarding policy implications, for countries with a substantial SE, merely promoting financial deepening is insufficient. Policymakers must concurrently strive to maintain a stable, low-uncertainty macroeconomic environment. Only through the synergistic combination of both can a mature financial system effectively optimize resource allocation, thereby fulfilling the long-term development goal of curtailing the SE.

E1307: The impact of Japan's negative interest rate policy on bank efficiency: A multi-dimensional approach*Presenter:* **Kuan Chieh Chen**, National Quemoy University, Taiwan

In 2016, Japan implemented a negative interest rate policy (NIRP) to stimulate economic activity amid prolonged deflation and sluggish growth. Although many scholars argue that NIRP exerts adverse effects on banks by compressing interest margins and weakening profitability, the actual impact on domestic banks is reexamined from a multidimensional perspective. Using the stochastic frontier approach, revenue, cost, and profit efficiency are assessed before and after the policy implementation. The results show that while revenue efficiency declined slightly, cost efficiency improved particularly among internet-only banks. To capture the overall operational reality, profit efficiency is further examined and found to have increased, indicating that banks have effectively restructured their business models to sustain profitability despite the challenges posed by unconventional monetary policy.

EC291 Room 104 VARIABLE SELECTION**Chair: Alexandra Soberon****E1926: Clustering high-dimensional continuous and categorical data via variable selection***Presenter:* **Somi Cha**, Chonnam National University, Korea, South*Co-authors:* Kwangmin Lee

A test-based variable selection framework is proposed for clustering high-dimensional data containing both continuous and categorical variables. When only a small subset of variables carries the cluster structure, clustering based on all variables may fail to recover the underlying groups. The main idea is to use initial cluster assignments as pseudo-labels, turning variable selection for clustering into variable-wise testing with false discovery rate control. The procedure first obtains initial cluster assignments from a dummy-expanded representation of the data, identifies informative variables through multiple testing, and then re-clusters the data using only the selected variables. A misclustering bound is established for the initial clustering step under the corresponding dummy-expanded model, showing that the initial assignments become asymptotically close to the true cluster labels under a sufficient signal condition. Simulation experiments demonstrate that the proposed framework improves clustering accuracy compared with using all variables, and an application to the 2024 Seoul Survey on Foreign Residents demonstrates that the selected variables yield interpretable cluster profiles.

E1955: Signal recoverability: A unified measure for proxy modality validation*Presenter:* **Suin Kim**, Seoul National University, Korea, South*Co-authors:* Yoonsuh Jung

A signal recoverability functional is formalized to quantify how much of the predictive signal in an expensive primary modality for an outcome is recoverable from a cheaper secondary modality, defined as the fraction of the variance of the primary-modality conditional mean of the outcome that is captured by the secondary modality. Three estimators are constructed from Neyman-orthogonal scores with cross-fitting, differing in their structural assumptions on the numerator: a fully nonparametric two-stage debiased estimator, a quadratic-form estimator imposing a finite-dimensional basis on the secondary modality, and a cross-covariance estimator with matched bases on both modalities. For each estimator, orthogonality is established, exact second-order bias identities are derived, and nuisance-rate conditions for asymptotic normality are provided. The framework is illustrated on recovering the cerebrospinal fluid, amyloid PET, and tau PET biomarker signals for cognitive impairment from an inexpensive plasma panel, where the functional ranks candidate plasma-marker subsets by recoverability and yields a debiased selection of the panel that best reconstructs each primary biomarker signal.

E1985: Joint structural break detection and variable selection in high-dimensional functional models: Sleep EEG application*Presenter:* **Mengfei Ran**, Xi'an Jiaotong-Liverpool University, China

Disentangling localized structural regime shifts from smooth time-varying covariate effects in ultra-high-dimensional functional data ($p \gg n$) remains a fundamental statistical challenge. Standard functional data analysis (FDA) methods, which predominantly prioritize smoothness, are fundamentally ill-equipped to simultaneously perform variable selection and identify abrupt structural changepoints. To bridge this gap, Adaptive Joint Learning (AJL) is introduced, a hierarchical regularization framework designed to jointly identify active functional predictors and detect shared structural changepoints in multivariate time-varying models. Distinct from conventional simultaneous estimation, a theoretically grounded two-stage screening-and-refinement procedure is proposed. Theoretically, by employing a Primal-Dual Witness construction and enforcing a formalized undersmoothing condition, the AJL estimator achieves consistent support recovery, accurate changepoint localization, and asymptotic normality for valid inference, even under high design correlation. To demonstrate its practical utility, the proposed framework is applied to the Sleep-EDF database. AJL successfully identifies a synchronized structural break marking the NREM-REM transition and reveals the nonlinear, time-varying dampening effect of age on deep sleep, offering rigorous and interpretable biomarkers for complex physiological dynamics.

E1164: Consistent variable selection for high-dimensional complex regression structures: A two-stage screening approach*Presenter:* **Wei-Cheng Hsiao**, Soochow University, Taiwan

The focus is on the challenge of variable selection in high-dimensional settings where the data generating process exhibits complex regression structures beyond standard additive assumptions. In many real-world applications such as bioinformatics or environmental monitoring predictors often influence not only the central tendency but also the variability and joint dependency patterns of the response variable. To capture these features, a consistent model selection procedure, OHT, and its extension MOHT is proposed. This framework integrates the orthogonal greedy algorithm for sequential screening with the high-dimensional information criterion to optimize the model path, followed by a trimming step to ensure parsimony. The method is designed to effectively identify significant variables governing both non-linear effects and distributional properties. Simulation results demonstrate that the proposed algorithm significantly outperforms existing methods, such as iFORT and RAMP, in terms of variable coverage and mean squared prediction error.

E2083: An empirical Bayes algorithm for variable selection with applications in genetic fine-mapping*Presenter:* **Xingbo Guan**, Hong Kong Baptist University, Hong Kong

Identifying causal variants within genome-wide association study loci is challenging due to linkage disequilibrium, multi-signal architectures, and the need for calibrated uncertainty at a large scale. A variational EM algorithm is proposed that formulates fine-mapping as Bayesian variable selection with binary inclusion indicators and learns feature-specific prior inclusion probabilities via empirical Bayes (EmpBVS). The E-step delivers closed-form updates for the variational Gaussian factor on effects and the noise precision, while the M-step updates per-SNP inclusion priors by maximizing the lower bound of marginal data likelihood. Irrelevant SNPs are shrunk out automatically as their weights contract to zero, yielding threshold-free selection. Computational sparsity results show geometric decay of null inclusion weights across iterations, and statistical consistency establishes that posterior mass concentrates on the true model and variational means consistently recover effect sizes. Simulations and comparative experiments demonstrate accurate posterior inclusion probabilities and credible sets with competitive runtime. The framework thus preserves automatic relevance learning while providing discrete selections and rigorous guarantees tailored to genetic fine-mapping.

EC252 Room 201 COMPUTATIONAL METHODS**Chair: Liang Zhong****E1890: An evolutionary approach to obtain a Nash equilibrium in a coalition game****Presenter:** **Viktoria Blueschke-Nikolaeva**, University of Klagenfurt, Austria**Co-authors:** Dmitri Blueschke, Reinhard Neck

The aim is to analyze dynamic interactions in a monetary union with three fiscal players (governments of countries) and a common central bank in the presence of exogenous shocks. A meta-heuristic solution technique based on the Differential Evolution method is employed. The proposed meta-heuristic approach for obtaining Nash equilibrium of a nonlinear dynamic game is designed with two linked procedures: a heuristic approach (the Differential Evolution algorithm) for finding individual optimal strategies and an approach for finding an approximate game equilibrium. The main advantage of using a heuristic evolutionary-based framework consists in its flexibility. Using the new methodology, Nash equilibrium can be obtained for a dynamic coalition game with various restrictions such as non-symmetric objective functions and different coalition designs.

E1991: Accelerated generalized estimating equations via inexact quasi-Newton preconditioned conjugate gradient**Presenter:** **Tom Chen**, Harvard Pilgrim Health Care and Harvard Medical School, United States

Generalized estimating equations (GEE) are a standard tool for marginal regression with clustered or longitudinal data, but the classical Fisher-scoring implementation scales poorly when within-cluster sizes are large, as each outer iteration requires multiple dense linear solves per cluster. An inexact quasi-Newton framework recasts GEE as a nonlinear root-finding problem. The outer iteration is driven by an L-BFGS update in root-finding form, while inner cluster-level solves use preconditioned conjugate gradient with adaptive forcing and structured preconditioners tailored to the working correlation. The framework accommodates flexible additive working correlations, including classical, kernel-based, and basis-expansion components, with a block-coordinate update of the regression and correlation parameters. Simulation studies indicate sizeable computational gains over standard implementations as cluster size grows, with negligible loss of statistical accuracy. The methods are implemented in an open-source R/Rcpp package.

E2001: Scaling and convergence rates of subsampling distributions for big data**Presenter:** **Melanie Zetlaoui**, Paris Nanterre University, France**Co-authors:** Patrice Bertail

Analyzing massive datasets poses significant computational challenges, often requiring substantial resources to calculate complex statistics. While subsampling, including the divide-and-conquer strategy, is commonly used to reduce complexity, its potential for improving statistical precision is often overlooked. Subsampling is demonstrated to be not merely a tool for computational efficiency but a robust framework for enhancing the convergence of statistical estimators. Theoretical derivations of precise rates of convergence for subsampling distributions and their corresponding quantiles are provided, showing how strategic sampling can refine these quantities. Furthermore, standardization techniques designed for unstandardized statistics within large-scale environments are introduced. The use of multiple subsampling scales provides critical insights into the behavior of learning procedures. Specifically, this approach allows for: Accelerating and estimating the rate of convergence of various algorithms; Quantifying the variability of complex, high-dimensional statistics; Constructing confidence intervals for out-of-sample errors and interpolating their values at larger scales. These findings are validated through simulations and applications on benchmark datasets: EMNIST (digit recognition) and VeReMi (vehicular network misbehavior), demonstrating that subsampling is a cornerstone for scalable and precise statistical learning.

E1859: PCA-based layer compression for Siamese neural networks**Presenter:** **Elizabeth Chou**, National Chengchi University, Taiwan

Siamese neural networks are widely used in similarity learning and pairwise classification, especially when paired observations provide useful supervision. However, their hidden layers may contain redundant neurons, increasing model complexity and computational cost. A principal component analysis (PCA)-based layer compression approach for Siamese neural networks is proposed. The method extracts the weight matrix of a target fully connected layer, applies PCA to its weight structure, and replaces the original layer with a lower-dimensional layer formed from the retained principal components. Thus, the compressed model is obtained through PCA-induced layer replacement rather than direct neuron pruning. The proposed method is interpreted from a low-rank approximation perspective. The retained principal components preserve dominant weight variation, while the explained-variance threshold controls the tradeoff between compression strength and information preservation. Empirical studies on both image and tabular datasets indicate that PCA-compressed models can reduce the number of neurons while maintaining, and in some cases improving, pairwise classification performance. These results suggest that PCA-based layer compression provides a simple and interpretable strategy for balancing model simplicity and predictive accuracy in Siamese representation learning.

E1371: Generative adversarial network stopping for robust surrendering**Presenter:** **Bowen Jia**, City University of Macau, China

An optimal stopping problem under model uncertainty is studied, where the underlying data-generating distribution is unknown. A distributionally robust framework is formulated by introducing an ambiguity set of probability measures around a reference model, capturing robustness against misspecification. To address the resulting high-dimensional and analytically intractable problem, a data-driven approach that combines generative adversarial networks (GAN) and deep neural networks (DNN) is proposed. The GAN approximates the worst-case distribution within the ambiguity set, while the DNN learns the optimal stopping strategy. Theoretical guarantees on the convergence of the proposed method under suitable conditions are provided. Numerical results demonstrate that the approach achieves robust performance and outperforms benchmark methods, particularly in high-dimensional settings where classical techniques are infeasible.

EC259 Room 209 BIostatistics**Chair: Masao Ueki****E1274: Semiparametric brier score framework for evaluating prediction accuracy under interval censoring****Presenter:** **Yi-Kuan Tseng**, National Central University, Taiwan**Co-authors:** Tzu-Ling Wang

A semiparametric framework is proposed to evaluate prediction accuracy for interval-censored survival data using the Brier score. Brier score estimators are first developed at fixed prediction horizons, accommodating a broad class of survival models, including Cox proportional hazards, accelerated failure time (AFT), and transformation models. To enhance clinical interpretability, the Murphy decomposition is extended to interval-censored settings, partitioning the prediction error into three distinct components: miscalibration, discrimination, and intrinsic uncertainty. Furthermore, the methodology is generalized to time-dependent Brier scores, facilitating dynamic performance assessment. This extension incorporates joint modeling and imputation strategies to address challenges such as measurement errors, sparse inspection schedules, and incomplete covariate histories. The finite-sample performance is validated through extensive simulations and applied to the Taiwan AIDS cohort data to demonstrate its practical utility. Finally, user-friendly software is developed to facilitate the adoption of these robust tools in modern survival analysis.

E1283: Integrating Bayesian generative AI with weighted logistic regression for imbalanced biomedical data classification*Presenter:* **Charlotte Wang**, National Taiwan University, Taiwan*Co-authors:* Yu-Ting Huang

Analyzing imbalanced data is a critical challenge in biomedical and public health research, where rare events are often of primary interest. Traditional data-level approaches frequently fail to capture underlying distributions in complex scenarios, leading to inconsistent classification performance. A hybrid framework is proposed that integrates a Bayesian generative AI approach with subject-weighted logistic regression. A BNNVAE model is introduced that incorporates Bayesian Neural Networks into a Variational Autoencoder to generate representative samples. These synthetic samples augment the dataset to achieve class balance. Subject-weighted logistic regression is then employed, assigning weights based on sample representativeness and proximity to decision boundaries. Simulation and real-world applications demonstrate that the BNNVAE framework significantly outperforms traditional methods such as SMOTE. It achieves superior stability and performance across F1-score, G-mean, and balanced accuracy metrics. By leveraging Bayesian neural networks, the model provides more effective estimates of data distributions, reducing reliance on large training data, a common limitation in practical applications. The proposed framework can generate more representative synthetic data and, through sample reweighting, construct interpretable classification models, thereby enhancing its practical value and applicability in biomedical and public health research.

E1464: On the inadmissibility of a well-known population genetic estimate and its improvement using shrinkage*Presenter:* **Andreas Futschik**, JKU Linz, Austria

An important goal in population genetics is to infer key parameters describing population history from DNA sequence samples, notably the scaled mutation and recombination rates. Historically, such parameters have been estimated from summary statistics, with unbiasedness given high priority, whereas more recent likelihood-based and machine-learning approaches are often computationally demanding. It is shown how shrinkage can improve common estimators. First, work is presented demonstrating that the popular Ewens-Watterson estimator is inadmissible under the Wright-Fisher model and can be uniformly improved by shrinkage, and then investigation is made into whether other estimators, such as the maximum likelihood estimator, can also be improved. An explanation is provided for why shrinkage is helpful within a framework involving Poisson processes over random intervals, and then the potential gains from shrinkage are considered. Finally, applications to next-generation sequencing data are discussed, where sequencing errors and cost-saving pooling designs must be accounted for.

E1992: KISS with the standard error: A sample size determination for a feasible sample size*Presenter:* **Takashi Omori**, Juntendo University, Japan*Co-authors:* Yumi Takagi, Sae Murakami

KISS - Keep It Simple, but Scientific. Sample size determination, the number of participants required to detect a clinically relevant treatment effect, is essential at the planning stage of clinical trials. Usually, the size is determined based on the assumed effect and the specified statistical power and type I error rate for a statistical test. However, sometimes the feasible sample size is limited. A paradigm shift suggests that for a given sample size, researchers can maximize the amount of information obtained. A calculation formula based on the standard error is developed that does not rely on a statistical test but on a similar logical approach and can be implemented using standard software. The approach determines the sample size based on the width of the treatment effect (that is, the treatment effect plus or minus its standard error) along with the probability that the effect lies within that width. The focus should be on the estimated effect with uncertainty, such as Wald statistics, and departure from the type I error rate when the sample size is limited.

E1998: On assessment of the pre-specified treatment effect in clinical trials*Presenter:* **Yumi Takagi**, Doshisha University, Japan*Co-authors:* Sae Murakami, Hiroshi Yadohisa, Takashi Omori

In clinical trials, sample size is statistically determined during the planning stage based on a pre-specified treatment effect, δ , for the primary endpoint. The effect δ is a decided value based on the results of previous clinical trials on the target population and therefore represents the value of interest to researchers conducting clinical trials. Nevertheless, researchers do not assess the magnitude of this value after the study data are observed and analyzed. Since δ is a predefined value at the planning stage, it is necessary to assess whether the obtained data and δ are in agreement for validity. Few clinical trial results have compared the obtained data with δ , and these comparisons have been limited to qualitative assessments using point estimates and 95% confidence intervals. Three methods for assessing δ are examined: the S-value, which interprets a p-value as a measure of information; the Bayes factor, which allows comparison of two hypotheses; and the Relative belief ratio, which uses a ratio of the posterior distribution to the prior distribution. These three methods are assessed and compared by applying them to clinical studies, and the resulting interpretations of δ are discussed.

EP001 Room 204 POSTER SESSION I**Chair: Louisa Kontoghiorghe****E1349: Geometric metric for identifying long-range interactions in three-dimensional chromatin structure***Presenter:* **Yunchae Jung**, Chungnam National University, Korea, South*Co-authors:* Insu Jang, Minsu Park

Three-dimensional genome reconstruction from Hi-C data provides a geometric perspective on chromatin structure that complements traditional contact intensity analysis. While interactions between genomically adjacent loci typically dominate contact matrices, identifying distal pairs that are spatially close in reconstructed space is essential for understanding large-scale chromatin organization. A geometric metric is formulated by integrating spatial distance in reconstructed space and linear genomic distance along the polymer. This metric captures the geometric properties of the data without relying on parametric decay functions and maintains stability across multiple resolutions. Simulation studies under diverse structural configurations demonstrate that the metric effectively highlights long-range interactions often masked by strong local contact patterns. Application to Hi-C datasets confirms the successful identification of structural features not detectable through contact intensities alone. By integrating reconstruction models with genomic distance information, the proposed method offers a robust tool for comparative analysis in contexts where long-range interactions play a critical role.

E1869: Persistence image-based deep Cox modeling for MCI-to-AD conversion time under right-censoring*Presenter:* **Somin Lee**, Chungnam National University, Korea, South*Co-authors:* Sunghun Kim, Eunjee Lee, Bo-yong Park

Early identification of Alzheimer's disease progression risk is important for adjusting follow-up frequency, planning interventions and medications, and allocating care resources. In clinical cohorts, conversions are often not observed within the follow-up window; right-censoring is common, and standard binary classification (0/1) may be inadequate. A pipeline that derives topological features from hippocampal surface measurements and performs censoring-aware risk modeling via Deep Cox objectives is presented. Compared with using surface maps directly, topological summaries aim to capture multiscale structural patterns and provide fixed-dimensional representations for learning. Hippocampal radial distance is arranged into an image-like scalar field, and cubical-complex persistent homology is computed to obtain birth-death intervals (Persistence Diagrams, PD). PDs are vectorized into Persistence Images (PI) via Gaussian smoothing and grid integration, yielding fixed-dimensional inputs. Experiments were conducted on baseline MCI subjects (N=322; events=188; censored=134) using 5-fold cross-validation. MLP-DeepCox on flattened PI vectors

achieved mean C-index 0.5714+0.0131; patch-token ViT-DeepCox achieved 0.5545+0.0335; RawCNN trained on raw surface maps without PI yielded 0.5400+0.0430. Results are preliminary due to limited sample size and fold-to-fold variation, but suggest the PD-PI-Deep Cox pipeline may be useful for time-to-conversion risk ranking under right-censoring.

E1936: **A framework for estimating lifetime disability trajectories and costs via survival extrapolation**

Presenter: **Wei-Ming Wang**, Fu Jen Catholic University, Taiwan

An analytical framework for estimating Lifetime Disability trajectories and related costs from administrative data via survival extrapolation is proposed. Using longitudinal Disability records, the framework combines observed Disability histories with extrapolated survival functions to estimate the expected duration that patients spend in different Functional Disability states over their remaining Lifetime. The Lifetime Distribution of Functional Disability Years (LDFY) is constructed as a summary measure of the long-term Functional burden of disease. As an illustrative application, patients with spinal cord injury identified from Taiwan Disability registry records are analyzed and their Lifetime Years spent in mild, moderate, severe, and profound Disability states are estimated. The framework can be extended to estimate Disability-state-specific Lifetime costs by incorporating subsidy, care, or other cost-related administrative data. This approach demonstrates how routinely collected administrative records can be used to quantify long-term Functional outcomes and resource needs in disease populations.

E1947: **fSKAT: A set-based association test for functional responses in genetic studies**

Presenter: **Jingyeong Jeong**, Chungnam National University, Korea, South

Co-authors: Eunjee Lee

Set-based association testing is widely used in genetic studies, but most existing methods are designed for scalar or low-dimensional outcomes. In biomedical and Neuroimaging studies, phenotypes are often functional data, such as trajectories, signals, imaging profiles, or connectivity curves indexed by time, space, or anatomical distance. Functional SKAT (fSKAT), a score-type variance-component test for associations between genetic variants and a functional response, is proposed. fSKAT captures genetic effects that are weak at individual points but accumulate across the functional domain. The method is based on function-on-scalar regression and constructs a kernel-based statistic from residual functions under the null model, with inference based on a mixture-of-chi-square approximation. Simulations show controlled type I error and improved power over scalar-summary approaches when genetic effects are distributed across the functional domain. fSKAT is applied to resting-state fMRI and genetic data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), using seed-based functional connectivity curves as a motivating phenotype. For each subject and seed region, the curve is indexed by Euclidean distance to other brain regions, and curve values represent functional connectivity. The ADNI analysis examines genetic associations with early connectivity changes before clinical diagnosis and demonstrates the utility of fSKAT for complex functional phenotypes.

E1987: **Survival trees for competing risks data with time-varying covariates**

Presenter: **Seungjoo Kim**, Sookmyung Women University, Korea, South

Co-authors: Yang-Jin Kim

In real-world clinical data, patients may experience terminal events from various causes. To accommodate this data structure, various methodologies have been developed under the competing risks framework. When time-varying covariates are present during patient follow-up, the traditional linear assumptions applied in the PH model may not be appropriate. However, studies on competing risks models incorporating time-varying covariates remain limited. A new survival tree approach is proposed wherein time-varying covariates are handled by transforming the data into an LTRC (left-truncated and right-censored) structure via the counting process formulation, and a new test statistic is introduced as the splitting criterion suitable for this data structure. The predictive performance of the proposed model is evaluated across various simulation environments using the Integrated Brier Score (IBS) and the C-index. Additionally, the practical applicability of the proposed method is demonstrated by applying it to real-world data from patients in the intensive care unit (ICU) of a university hospital, where the occurrence of hospital-acquired pneumonia (HAP) is treated as a time-varying covariate, and ICU death and discharge alive are defined as competing events.

E2002: **Explainable writer verification through dynamic handwriting feature analysis**

Presenter: **Seungjin Baek**, Pusan National University, Korea, South

Writer verification is a crucial task in biometric authentication and forensic handwriting examination, where reproducibility and reliability of decisions are required. While offline handwriting analysis mainly relies on the final written form, online handwriting data provide dynamic information, such as coordinates, pressure, velocity, and acceleration, that reflects individual writing habits and motor patterns. Existing studies have improved verification performance using complex deep learning models; however, the lack of transparency in these models makes their decisions difficult for forensic experts to trust in practical settings. Therefore, writer verification must balance predictive performance with explainability. An explainable writer verification framework is developed using multivariate online handwriting data from the AI Hub online handwriting dataset. A bidirectional LSTM-based model is employed to learn complex dynamic patterns from continuous coordinate, pressure, and kinematic features. To support interpretability, Shapley value-based feature attribution is applied to conduct feature-level interpretation. This analysis shows how each dynamic component contributes to the final verification decision, highlighting specific handwriting movements that support genuine or forged classification. Ultimately, the proposed approach goes beyond performance-oriented prediction and supports transparent forensic decision-making.

E2020: **Enhancing the reliability of deep learning-based shoeprint image analysis via explainable AI (XAI)**

Presenter: **Siwon Jeong**, Pusan National University, Korea, South

In forensics, shoeprints are vital for identification due to their Randomly Acquired Characteristics (RACs); however, their evaluation is often constrained by subjective judgment and a lack of quantitative metrics. Despite strong performance, the black-box nature of deep learning-based shoeprint analysis limits forensic interpretability and evidential reliability. A deep learning framework is proposed for shoeprint image analysis, integrating a Siamese network with explainable AI, transfer learning, and triplet-based metric learning with hard negative mining to improve discrimination between close non-matches under limited data. Two complementary XAI strategies are adopted: Grad-CAM localizes salient regions in the embedding network, whereas SINEX explains the final similarity score through a perturbation-based analysis that measures how masking segmented regions alters the Siamese output. The resulting heatmaps quantify the contribution of local visual evidence to each match decision. The framework jointly assesses predictive performance and explanation quality via ROC-AUC, EER, and Insertion/Deletion, alongside Localization Agreement (alignment with expert-identified RACs) and Pairwise Consistency (robustness of highlighted features across different image pairs). By achieving closer alignment with discriminative characteristics than existing methods, the framework provides a statistically grounded, reproducible basis for objective evidence assessment in judicial contexts.

E2095: **Predicting X-ray plateau occurrence in gamma-ray bursts using statistical and machine learning approaches**

Presenter: **Nuha Al Maashari**, Sultan Qaboos University, Oman

Co-authors: Iman Al Hasani

The occurrence of an X-ray plateau phase in gamma-ray burst (GRB) afterglows is investigated to determine whether it can be predicted from observable burst properties. Data from the Swift/XRT catalogue, including spectral, energetic, temporal, and redshift-related properties, were analysed using multiple logistic regression, Random Forest classification, dimensionality reduction, and clustering methods. Both supervised models showed limited predictive performance, with Random Forest providing improved discrimination compared with logistic regression but without achieving reliable classification. Dimensionality reduction and clustering analyses further revealed substantial overlap between plateau

and non-plateau GRBs, indicating that the observed properties do not form distinct groups associated with plateau behaviour. These findings suggest that X-ray plateau formation is governed by complex multivariate processes that are not fully represented by the available observable characteristics. The findings highlight the limitations of current GRB observables for plateau prediction and demonstrate the value of combining statistical modelling and machine learning approaches in astrophysical data analysis.

E2096: High-dimensional clustering and random forest regression for pixel-level crop classification and suitability in Oman

Presenter: **Shahad Al Siyabi**, Sultan Qaboos University, Oman

Co-authors: Iman Al Hasani

In countries that lack extensive spatial agricultural data, using satellite and climate data can be beneficial, with Oman having national self-sufficiency in agriculture as one of its goals, and the lack of a fine resolution crop map or non-fieldwork tool that can be used remotely to watch the progress of the goal and the country's targets. A high dimensional feature matrix that covers the time series 2018-2026 has been used with satellite and climate data using an unsupervised clustering approach with KMeans and dimensionality reduction using UMAP, the cropland of Oman produced 12 different agricultural systems at 100m resolution. Suitability for target crops like onion, wheat, and potato is modelled using local water, soil, and terrain data with Random Forest, and the results show the top pixels to grow the crops in with more than 90 percent accuracy, and that water availability is the most important SHAP driver, highlighting that water infrastructure is where policymakers should prioritise for these crops and Oman's agricultural goals, then an interactive dashboard was created to make the data more accessible with a dynamic map.

Sunday 09.08.2026

08:15 - 09:30

Parallel Session F – EcoSta2026

EO233 Room B101 ADVANCES IN ECONOMETRICS**Chair: Chen Qiu****E1554: Experimental design for policy choice***Presenter:* **Samuel Higbee**, University of North Carolina at Chapel Hill, United States

A method is proposed for designing experiments when the resulting data will be used to choose a policy. A planner seeks to maximize Bayes expected welfare by choosing a policy whose effects depend on an unknown finite-dimensional parameter. The planner has access to a first wave of experimental data but has the opportunity to design and run a second wave before choosing the policy. The resulting experimental design–policy choice problem involves a dynamic program with a very high-dimensional state and is generally intractable in finite samples. A tractable approximation based on the limit experiment is proposed and shown to be asymptotically optimal using a new asymptotic representation theorem for adaptive experiments with continuous treatments. The method is applied to a conditional cash transfer experiment to quantify the gains from tailoring the experiment to the policy choice.

E1713: Safe anytime-valid optimal policy recommendation*Presenter:* **Kohei Izumi**, University of Rochester, United States*Co-authors:* Keita Sunada

Treatment choice under partial identification has largely been studied in static settings. The sequential version is studied: evidence arrives over time from heterogeneous populations, the planner may stop at any data-dependent time, and the false-recommendation probability must be controlled uniformly over all stopping times. This anytime-valid safety requirement is imposed as a constraint on the policy class. An e-process threshold then defines an admissible class of policies within which the planner minimizes conditional Gamma-minimax regret over a time-consistent (or rectangular) ambiguity set for future evidence. The resulting constrained optimal stopping problem admits a dynamically consistent Bellman equation, and a finite-horizon calibration of the e-process threshold preserves size control while tightening the Ville bound. The analysis recovers, in this dynamic setting, the Type I and II regret asymmetry documented for hypothesis-testing rules in static treatment choice: the welfare cost of anytime-valid safety arises almost entirely as Type II regret—failure to recommend effective treatments—and peaks at moderate positive treatment effects. Relaxing the significance level, calibrating the threshold, and using a data-driven ambiguity set can each mitigate this cost. Two COVID-19 vaccine applications—the BNT162b2 primary series and an mRNA booster against Omicron—illustrate how the framework distinguishes overwhelming from ambiguous evidence.

E1791: Orthogonal dynamic programming for sequential treatment allocation*Presenter:* **Heejun Lee**, Brown University, United States

Sequential treatment allocation with weak treatment effects and rewards confounded by time-varying nuisance parameters is studied. In standard Bayesian frameworks, optimal planning via dynamic programming is hindered by the curse of dimensionality, as the state space grows with the history of observations. A local asymptotic framework overcomes this computational barrier by collapsing the history into low-dimensional sufficient statistics. However, nuisance variation complicates this framework: accommodating nuisance parameters increases computational complexity and requires specifying priors that may be prone to misspecification, while ignoring them introduces bias. To address this, orthogonal planning is proposed, a robust dynamic programming framework based on orthogonality.

EO075 Room B102 EMPIRICAL ANALYSIS OF ECONOMIC, FINANCIAL, AND ENERGY DATA**Chair: Xiaojing Cai****E1568: Investor attention as a driver of connectedness across FX, bond, stock, and commodity markets: Evidence from the FFR***Presenter:* **Wenting Zhang**, University of Niigata Prefecture, Japan

The aim is to examine how investor attention to the federal funds rate influences return spillovers across foreign exchange, bond, stock, and commodity markets. The framework consists of two segments. First, a time-varying parameter vector autoregressive connectedness model estimates the dynamic spillover effects across these markets from 2004 to 2025; Google search volume is employed as a proxy for investor attention and regressed on connectedness indices to evaluate the influence of attention. The findings suggest that increased attention significantly reduces the total connectedness index, indicating that attention can improve market efficiency. Second, the same influence appears on the total directional connectedness of the stock and commodity markets, helping stabilize shocks that flow into and out of both markets. However, the impact of investor attention on spillovers during different crises is heterogeneous, depending on the specific conditions at the time. These findings highlight the role of investor attention in cross-market risk contagion and suggest that policymakers monitor attention indicators and enhance investor communication to reduce information asymmetry.

E1577: Asian FX market interconnectedness and the quantile-frequency linkage with global macroeconomic factors*Presenter:* **Lei Xu**, Okinawa University, Japan

This study examines volatility transmission in Asian foreign exchange (FX) markets using a multi-method framework that combines DCC-GARCH, Diebold-Yilmaz (DY) spillovers, Barunik-Krehlik (BK) frequency decomposition, and Wavelet Quantile Correlation (WQC). Using data from nine Asian currencies and eight global variables, including interest rates, uncertainty indices, and commodity prices, the analysis captures time-varying, horizon-specific, and tail-dependent spillovers. Results reveal that long-term and upper-Quantile dynamics dominate systemic risk propagation. The Japanese yen (JPY) and Singapore dollar (SGD) are consistent net transmitters, while the Korean won (KRW), Indonesian rupiah (IDR), and Thai baht (THB) are recurrent receivers of external shocks. U.S.-centric variables (e.g., Treasury yields, VIX) exert the most persistent influence, whereas China's bond market volatility shows limited regional transmission. FX interconnectedness in Asia is far from uniform, exhibiting nonlinear and asymmetric traits that most standard models do not capture well. From a risk management perspective, these dynamics suggest a need for strategies that can adapt across time horizons and respond to changing market conditions as they unfold.

E1447: Natural gas prices and macroeconomies: Evidence from exporting and importing countries*Presenter:* **Xiaojing Cai**, Okayama University, Japan

The macroeconomic effects of natural gas price shocks across eight major countries -four exporters and four importers- over 1997-2025 are examined using a constant-parameter structural vector autoregression (SVAR) model and a time-varying parameter (TVP) VAR model with stochastic volatility (SV). The responses of real gross domestic product (GDP), consumer price inflation (CPI), and interest rates are analyzed for exporter-importer asymmetries and time variation in transmission. Clear asymmetries in output responses are found: exporters experience positive GDP effects during periods of market stress, while importers exhibit weaker or negative reactions. CPI responses are generally positive, with stronger pass-through in hub-based pricing systems. Monetary policy reactions vary across countries, reflecting institutional differences. The TVP-VAR-SV results indicate that gas price transmission is episodic and state-dependent, with significant effects concentrated in extreme episodes rather than evolving smoothly over time. Robustness checks confirm the stability of these findings. Overall, the results highlight the heterogeneous and regime-dependent nature of gas price transmission in global commodity markets.

EO155 Room B104 RANDOM MATRIX THEORY AND ITS APPLICATIONS IN HIGH DIMENSIONAL STATISTICS**Chair: Zeng Li****E1843: Mean-shift PCA by knockoff mean****Presenter:** **Zeng Li**, Southern University of Science and Technology, China

Removing noise is difficult, but adding noise is easy. The aim is to show how to eliminate mean-shift noisy components from PCA by deliberately introducing knockoff mean-shift perturbation. Standard PCA is highly sensitive to shifts in the sample mean: a small fraction of samples from a shifted distribution can cause large deviations in the leading principal components. In high-dimensional regimes, existing Robust PCA approaches cannot handle the mean-shift contamination structure inherent in the mixture models. Using tools from Random Matrix Theory, it is proven that the mean-shift spikes are spectrally separable from the stable eigenvalues of the original covariance. Furthermore, the original eigenspace remains asymptotically invariant to the contamination, independent of the mixture weight. Exploiting this spectral stability, a simple, two-stage PCA algorithm is proposed by adding knockoff mean that identifies and removes the mean-shift component using only standard PCA operations.

E2003: High-dimensional white noise tests via autocorrelations**Presenter:** **Qinwen Wang**, Fudan University, China

White noise testing is studied for high-dimensional time series. Two bias-corrected portmanteau-type statistics based on sample autocorrelation matrices are proposed, using diagonal and full covariance standardization, respectively. Under the high-dimensional regime where the dimension is proportional to the sample size, their limiting null distributions are derived and their local power is studied under a VMA(1) alternative. The analysis shows that lag aggregation has a non-monotone effect on power: adding more lags may either reduce or improve power, depending on the dimensional ratio and signal strength. Simulation results support the theoretical findings and show good finite-sample performance.

E2065: Rank-based sparse regression in principal components space under measurement error**Presenter:** **Xiaoyi Wang**, Beijing Normal University at Zhuhai, China**Co-authors:** Long Feng, Xiaoyi Wang, Le Zhou

High-dimensional regression in principal components space is studied when the predictors are observed with measurement error and the response errors may be heavy-tailed. A penalized rank-based pilot estimator in the principal component space is first proposed and its prediction error is analyzed. The prediction error consists of a term with the minimax rate plus an additional term representing the impact of measurement error which diminishes as the dimension grows. The error distribution is not required to have any finite moment in the theory, and the pilot estimator therefore enjoys the blessings-of-dimensionality property in heavy-tailed measurement error regression. To further reduce bias, an adaptively reweighted second stage rank estimator is then proposed and its strong oracle property is established. In particular, the second stage estimator is capable of improving the prediction error to the oracle rate corresponding to the active set of principal components. Simulation studies demonstrate that the proposed method remains competitive under light-tailed errors and is substantially more efficient under heavy-tailed errors, especially when predictor contamination is present. A real data example is analyzed to demonstrate the application of the method.

EO132 Room Virtual R01 RECENT DEVELOPMENTS IN FEATURE SCREENING AND FDR CONTROL**Chair: Chenlu Ke****E1267: Feature screening for high-dimensional data with measurement errors using adjusted martingale difference correlation****Presenter:** **Wenbo Wu**, University of Texas at San Antonio, United States**Co-authors:** Baoying Yang

High-dimensional data with measurement errors pose significant challenges for statistical modeling and inference due to the "curse of dimensionality" and the unavailability of direct measurements of variables. To reduce the dimensionality of data, feature screening is an effective method for identifying informative variables among a large number of observed features. While feature screening has been extensively studied in the literature, limited research has focused on feature screening with random variables affected by measurement errors. An adjusted martingale difference correlation (AMDC) is proposed to measure conditional mean dependence in the presence of measurement errors. The proposed AMDC is further extended to measure the dependence of the conditional quantile and conditional k-th central moment through specific transformations of the error-prone variables. Invariance relationships between the AMDC and the original martingale difference correlation are established and different feature screening methods are developed for the conditional mean, conditional quantile, and conditional k-th central moment of the response variable under a measurement error model. Sure screening properties are established for all the proposed procedures and extensive simulation studies are conducted to illustrate their effectiveness and advantages. Finally, microRNA expression data is analyzed using the proposed AMDC-based screening methods to demonstrate their practical usefulness.

E1491: Data-adaptive conditional quantile screening with FDR control in ultra-high-dimensional data**Presenter:** **Saidat Abidemi Sanni**, The University of Texas at San Antonio, United States**Co-authors:** Yan Yu, Zhigen Zhao

A conditional quantile feature screening procedure for ultra-high-dimensional data with false discovery rate control is proposed. Motivated by genetic association studies, the problem of screening ultra-high-dimensional predictors associated with a specific conditional quantile of the response, given a potentially multivariate set of covariates, is considered. The approach uses quantile partial correlation to assess quantile-specific conditional associations, constructs quantile mirror statistics, and applies a data-adaptive thresholding rule to control the false discovery rate. This method provides a practical tool for quantile-specific feature screening with covariate adjustment, data-adaptive thresholding, and error-rate control in ultra-high-dimensional settings. Simulation studies and a real-data analysis of genetic risk factors for abnormal blood pressure demonstrate the method's performance relative to existing approaches.

E1727: Improving predictive model false discovery rate estimation**Presenter:** **Yet Nguyen**, Old Dominion University, United States

Although predictive modeling is fundamentally driven by error minimization, interpreting the underlying relationship between predictors and the response remains a significant research priority. Recent literature regarding False Discovery Rate (FDR) estimation within predictive frameworks is examined, and a refined procedure to improve the precision of these estimations is introduced.

EO178 Room 101 QUICKEST CHANGE DETECTION ALGORITHMS FOR COMPLEX SYSTEMS (VIRTUAL) Chair: Patrick Vincent Lubenia**E1715: Multiple-experiment quickest change detection under cost constraints***Presenter:* **Patrick Vincent Lubenia**, University of Pittsburgh, United States*Co-authors:* Taposh Banerjee

Classic models of quickest change detection typically involve a single experiment used to monitor a stochastic process. This study considers the scenario where, at each observation time, an observer must select from a variety of experiments, each characterized by distinct information qualities and associated costs. The main objective is to minimize the worst-case average detection delay while satisfying constraints on false alarm rate and cost. To achieve this, the 2E-CUSUM algorithm is introduced for scenarios involving two experiments. The study then explores more complex designs involving multiple experiments, extending the 2E-CUSUM framework accordingly. The proposed algorithms are demonstrated to be asymptotically optimal.

E1765: Nonparametric multi change point detection for Markov chains via adaptive clustering*Presenter:* **Imon Banerjee**, Purdue University, United States

Offline change point detection identifies time points of distribution change in data sequences and is routinely used in signal processing, speech processing, and climatology. Despite broad applicability across economics, computer science, and planetary sciences, rigorous nonparametric techniques for change point detection with non-independent and identically distributed (non-i.i.d.) datasets have remained elusive. A nonparametric clustering algorithm is proposed that accurately obtains change points from Markovian datasets of length n by bridging Rademacher complexities of Markov chains and adaptive clustering via penalization. A DKW-type inequality for the empirical distribution of the Markov chain is derived using recent advances in Rademacher complexities of regenerating Markov chains. This inequality is then used to show that an adaptive clustering algorithm recovers the correct change points for a Markovian sequence. The tightness of the rates is established by showing that they essentially coincide with the best known rates for i.i.d. data. Computational considerations of the problem are discussed.

E1772: Asymptotically optimal change detection for unnormalized pre- and post-change distributions*Presenter:* **Arman Adibi**, Augusta University, United States

The problem of detecting changes when only unnormalized pre- and post-change distributions are accessible is addressed. This situation occurs in many scenarios in physics such as in ferromagnetism, crystallography, magneto-hydrodynamics, and thermodynamics, where the energy models are difficult to normalize. The approach is based on the estimation of the Cumulative Sum (CUSUM) statistics, which is known to produce optimal performance. First, an intuitively appealing Approximation method is presented. Unfortunately, this produces a biased estimator of the CUSUM statistics and may cause performance degradation. The Log-Partition Approximation Cumulative Sum (LPA-CUSUM) algorithm is then proposed, based on thermodynamic integration (TI) in order to estimate the log-ratio of normalizing constants of pre- and post-change distributions. It is proved that this approach gives an unbiased estimate of the Log-Partition function and the CUSUM statistics, and leads to asymptotically optimal performance. Moreover, a relationship between the required sample size for thermodynamic integration and the desired detection delay performance is derived, offering guidelines for practical parameter selection. Numerical studies demonstrate the efficacy of the approach.

EO122 Room 102 RECENT ADVANCES IN AI AND STATISTICAL METHODS FOR NEUROSCIENCE**Chair: Jocelyn Chi****E1431: Flexible inference of animal learning rules using neural networks***Presenter:* **Yuhan Helena Liu**, Princeton University, United States*Co-authors:* Victor Geadah, Jonathan Pillow

Understanding how animals learn is a central challenge in neuroscience, with growing relevance for developing aligned artificial intelligence. Existing approaches typically assume fixed parametric learning rules (e.g., Q-learning, policy gradient), which may not capture the complexity of real behavior. A framework is developed to infer learning rules directly from behavioral data during de novo task learning. The decision policy is modeled using a generalized linear model (GLM) and the learning rule, the mapping from task variables to trial-by-trial weight updates, is parameterized using a deep neural network (DNN). This approach enables flexible, data-driven inference while preserving interpretability of the policy. To capture history dependence, a recurrent neural network (RNN) variant is introduced that relaxes the Markov assumption and integrates information across trials. Simulations show accurate recovery of ground-truth learning rules. Applied to large-scale mouse behavior, the methods outperform standard RL models in predicting held-out learning trajectories and reveal reward-history-dependent updates, with larger changes following rewarded sequences. These results provide a flexible framework for inferring learning dynamics from behavior, with implications for adaptive training and data-driven behavioral models.

E1851: Sequential mixed-methods evaluation of AI-assisted digital health platforms in dementia care*Presenter:* **Qiping Fan**, Clemson University, United States

A sequential mixed-methods statistical framework was developed and applied to evaluate AI-assisted digital health platforms in dementia care. Grounded in the Build-Measure-Learn iterative design, the framework integrated qualitative and quantitative approaches across iterative development cycles. The framework was implemented across Two study phases spanning 2022 to 2025. Phase One employed qualitative thematic analysis from in-depth interviews to characterize caregiver needs across financial, legal, and functional domains, followed by instrument-based usability assessment to guide platform refinement. Phase Two applied regression-based modeling to examine technology acceptance as a function of sociodemographic and behavioral covariates among a nationally recruited sample, complemented by qualitative thematic analysis to capture user-reported experiences and attitudes toward AI integration. Quantitative and qualitative findings were integrated at the interpretation stage to assess platform usability and acceptance across successive development iterations. Implications for statistical methodology in digital health platform evaluation within neurodegenerative disease and neuroscience research contexts are discussed.

E1867: Data-driven and normative approaches to modeling neural data*Presenter:* **Jacob Yates**, University of California, Berkeley, United States

Modern computational neuroscience has developed two complementary ways of building statistical models of neural activity. Data-driven approaches start from neural recordings and seek flexible descriptions of the relationship between stimuli, behavior, latent state, and spiking responses. Their strength is empirical fidelity: they can capture structure in large neural populations without requiring that the model itself solve the animal's task. Normative approaches begin from the opposite direction. They train artificial systems to solve perceptual or inferential problems, then ask whether the resulting representations, dynamics, or errors resemble those observed in the brain. Recent work on Poisson variational autoencoders offers a bridge between these approaches. Poisson models are a natural statistical language for spiking data, but they can also define latent generative models that perform inference. This shared statistical structure makes it possible to build models that are both useful tools for describing neural responses and candidate accounts of the computations those responses implement.

EO117 Room 103 MATHEMATICAL FOUNDATION OF LANGUAGE MODELS**Chair: Junwei Lu****E1176: Preference inference for language model evaluation****Presenter:** Junwei Lu, Harvard T.H. Chan School of Public Health, United States

Human preference alignment has been shown to be effective in training large language models (LLMs). It allows LLMs to understand human feedback and preferences. Despite extensive literature on algorithms that align with human preference rankings, uncertainty quantification for ranking estimation remains underexplored and is of great practical significance. For example, overcoming hallucination in LLMs for medical applications requires inferential methods for ranking LLM outputs. A novel framework called Fisher random walk is presented to conduct semi-parametric efficient preference inference for language models, with application to medical knowledge tasks.

E1177: A theoretical framework for prompt engineering: Approximating smooth functions with transformer prompts**Presenter:** Ryumei Nakada, Harvard University, United States**Co-authors:** Wenlong Ji, Tianxi Cai, James Zou, Linjun Zhang

Prompt engineering is a powerful technique for steering large language models (LLMs) and improving performance across many tasks. As LLMs are increasingly deployed as agents that reason, decide, and adapt, a sharper theoretical account of prompting is needed. A framework is proposed in which a Transformer, given a carefully designed prompt, acts as a *configurable computational system* by emulating a “virtual” neural network during inference: the prompt specifies the network configuration, enabling dynamic computation without parameter updates. Within this view, an approximation theory for C^β functions is established, showing that Transformers can approximate β -times differentiable targets to arbitrary precision via structured prompts. A limitation for recurrent architectures is also proven: for any fixed RNN with hidden dimension $q < d^2$, varying the prompt alone cannot realize all linear maps $\mathbb{R}^d \rightarrow \mathbb{R}^d$, highlighting a structural advantage of Transformers for prompt-based computation. Finally, the framework helps rationalize common prompting practices—longer structured prompts, filtering irrelevant information, increasing token diversity, and multi-agent interactions—by linking them to the capacity and error behavior of the induced virtual network.

E1178: Algorithm-dependent learning analysis of over-parameterized neural networks**Presenter:** Yuan Cao, The University of Hong Kong, Hong Kong**Co-authors:** Junwei Lu

Modern machine learning models, such as large language models, often contain a huge number of parameters. For such over-parameterized models, there can be infinitely many minimizers of the training loss function, and different training algorithms may thus converge to different solutions. While these solutions may all yield zero training error, they can have drastically different prediction errors on new data. Therefore, understanding modern machine learning requires analyzing the impact of training algorithms on generalization performance. Recent studies on algorithm-dependent learning guarantees for over-parameterized neural networks are discussed. Theoretical analyses of the benign overfitting phenomenon are first presented, and then the test performance of gradient descent, full-batch Adam, and stochastic Adam are compared. The findings highlight the crucial role of optimization algorithms in determining the practical success of modern large-scale neural networks.

EO201 Room 104 MODERN STATISTICAL METHODS FOR CLASSIFICATION**Chair: Larry Tang****E1189: Learning subtle neurodegeneration for early Alzheimer’s disease detection****Presenter:** Zheyu Wang, Johns Hopkins University, United States

Early Alzheimer’s Disease (AD) is characterized by subtle and spatially distributed neurodegenerative changes that are difficult to detect with conventional MRI summaries. While MRI is non-invasive and widely available, standard volumetric and regional measures have limited sensitivity for identifying individuals in the earliest disease stages. While incorporating known risk factors such as age and ApoE4 status improves diagnostic performance, the added value of MRI in early detection remains limited. This leads to a crucial question: is this limitation inherent to MRI’s temporal placement within the biomarker cascade (e.g., typically changing after amyloid pathology but before major cognitive decline), or because its full diagnostic potential is currently underutilized? Two novel approaches are explored to enhance MRI’s capability for early AD detection: 1) high-dimensional distance measures that capture complex, multiregional patterns of subtle neurodegeneration, and 2) an attention-based model that learns how to integrate information across imaging representations to emphasize biologically meaningful signals. The results highlight the potential of MRI to capture early disease-related changes when coupled with modern approaches.

E1227: Exploring scientific literature using topic modeling: A practical framework for discovery and classification**Presenter:** Larry Tang, University of Central Florida, United States**Co-authors:** Amir Alipour Yengejeh

The increasing volume and diversity of scientific publications poses challenges for scalable and interpretable topic discovery and automated document categorization. An integrated framework is proposed that combines probabilistic topic modeling with supervised classification to support large-scale scientific literature analysis. Using 3689 abstracts from the Journal of Forensic Sciences, Latent Dirichlet Allocation (LDA) is applied to uncover latent thematic structures, assess topic diagnosticity across forensic disciplines, and analyze temporal research trends. The resulting document topic representations are then used for supervised abstract classification. Across multiple models and resampling scenarios, the strongest and most stable performance is achieved under a grouped category configuration.

E1848: Strategies for testing the equality of properties of two distributions of scores**Presenter:** Christopher Saunders, South Dakota State University, United States**Co-authors:** Clarissa Giefer, Janean Hanka, Semhar Michael

A common question when deciding whether to consider a type of forensic evidence is the degree to which trace objects from different sources can be distinguished; this degree is commonly referred to as the discriminating power of the comparison process. Regardless of the quality of the two trace objects being compared, if the comparison process does not give rise to an exclusionary difference between the two sources, then they are said to have indistinguishable profiles with respect to the comparison process. A new level-alpha test for the equality of scores is defined, based on all pairwise comparisons using both within-source and between-source comparisons. The applicability of this method is demonstrated using a handwriting dataset; this test compares average within-writer scores with average between-writer scores. The approximate distribution of the test statistic is discussed and the corresponding p-values from all tests of equality between all pairs of sources in a mixture model are used to make a statement about the proportion of p-values that arose from two writers with indistinguishable profiles, which in turn leads to an indirect inference of the discriminating power.

EO169 Room 105 FLEXIBLE AND INTERPRETABLE STATISTICAL LEARNING**Chair: Hoseung Song****E1207: Tweedie multivariate semi-parametric credibility with the exchangeable correlation****Presenter:** Himchan Jeong, Simon Fraser University, Canada

A framework is proposed for determining credibility premiums for multiple coverages in a compound risk model with Tweedie distribution. The framework builds upon previous results on credibility premium and provides an explicit multivariate credibility premium formula that is applicable to the tweedie family assuming that the unobserved heterogeneity for the multiple coverage have the common correlation. The practical applicability of the proposed framework is evaluated through simulation and empirical analysis using the LGPIF dataset, which includes claims and policy characteristics data for various types of coverages observed over time. The findings suggest that the proposed framework can be useful in ratemaking practice by incorporating a non-trivial dependence structure among the multiple types of claims.

E1649: Explainable machine learning by SEE-Net: Closing the gap between interpretable models and DNNs**Presenter:** Beomseok Seo, Sookmyung Womens University, Korea, South**Co-authors:** Jia Li

Deep Neural Networks (DNNs) have achieved remarkable accuracy for numerous applications, yet their complexity often renders the explanation of predictions a challenging task. This complexity contrasts with easily interpretable statistical models, which, however, often suffer from lower accuracy. The underperformance may stem more from inadequate training methods than from the inherent limitations of model structures. The Synced Explanation-Enhanced Network (SEE-Net), a novel architecture integrating a guiding DNN with a shallow Neural Network functionally equivalent to a two-layer mixture of linear models, is introduced. This shallow Network is trained under the guidance of the DNN, effectively bridging the gap between the prediction power of deep learning and the need for explainable models. Experiments on image and tabular data demonstrate that SEE-Net can leverage the advantage of DNNs while providing an interpretable prediction framework. SEE-Net embodies a new paradigm in machine learning: it achieves high-level explainability with minimal compromise on prediction accuracy by training an almost white-box model under the co-supervision of a black-box model, which can be tailored for diverse applications.

E1762: Asymptotic theory for exponentially weighted aggregated penalized splines**Presenter:** Youngseok Song, West Virginia University, United States

Theoretical properties of an exponentially weighted aggregated (EWA) penalized spline estimator are studied. Rather than selecting a single smoothing parameter for penalized splines, the EWA estimator aggregates spline estimates over a range of smoothing parameters using a data-driven, exponentially weighted pseudo-posterior distribution. Conditions on prior distributions are derived under which the estimator achieves the optimal convergence rate in terms of average mean squared error, as well as consistency and asymptotic normality. In particular, the EWA estimator is shown to be asymptotically equivalent to regression spline estimators through aggregation over smoothing parameters under such priors. Technical challenges are addressed by developing a novel analysis of eigenvalue problems associated with the B-spline basis, which differs from traditional random matrix theory. A simulation study illustrates the finite-sample performance of the proposed estimator, and a data application demonstrates its practical use.

EO188 Room 106 STATISTICAL LEARNING FOR COMPLEX STRUCTURED DATA**Chair: John Stufken****E1385: A model-free correlation coefficient for censored data****Presenter:** Linlin Dai, Southwestern University of Finance and Economics, China**Co-authors:** Tengfei Li, Kani Chen

In clinical studies, assessing statistical associations between covariates and survival outcomes is crucial. To date, there has been no formally defined model-free correlation coefficient for right-censored data that can measure the strength of associations. Traditional methods, such as the Cox proportional hazards model, often struggle with the complexities of non-monotonic or nonlinear relationships. A censored rank-based correlation coefficient (CRC) is introduced. It consistently estimates a new dependence measure taking values in $[0,1]$ and equaling 0 or 1 if and only if the variables are independent or one is a measurable function of the other. The CRC is entirely model-free without depending on the distributions of the variables. It facilitates quick computation with a complexity of $O(n \log n)$ and can effectively detect nonlinear and non-monotonic effects, even under heavy censoring. The p-values for testing independence can be obtained using a power-consistent permutation method. The CRC shows strong consistency and asymptotic normality, outperforming the Cox model and other methods in detecting nonlinear associations in both simulations and real data from the Alzheimers Disease Neuroimaging Initiative, successfully identifying proteins that existing methods fail to detect.

E2074: Learning directed latent variable networks with measurement error**Presenter:** Siliang Zhang, East China Normal University, China**Co-authors:** Yunxiao Chen, Irini Moustaki

Recovering directed dependence structure among latent constructs is challenging when they are observed only through error-prone indicators. A common two-stage strategy estimates factor scores and applies standard causal discovery algorithms to them; treating these noisy estimates as observed variables distorts the conditional independence relations among the latent constructs. A likelihood-based procedure is proposed for learning sparse directed latent graphs in linear-Gaussian structural equation models with an identified measurement model. Estimation proceeds via the marginal observed-data likelihood, propagating measurement error from the indicators into the structural model. Acyclicity is imposed through a smooth, differentiable DAG constraint, and the resulting non-convex program is solved by an augmented Lagrangian scheme. Because this step is tuned to preserve true adjacencies at the cost of redundant edges, BIC-based backward pruning is then applied to recover a sparse graph. Identifiability of the latent Markov equivalence class is established under standard assumptions. Simulation studies demonstrate that explicitly modeling measurement error yields substantial gains in structural recovery over two-stage procedures based on estimated factor scores. An application to the Policing by Consent module of European Social Survey Round 5 contrasts the inferred latent structure with the conclusions of the official topline report.

E2075: Self-balancing neural network: A novel method to estimate average treatment effect**Presenter:** Yuqi Qiu, East China Normal University, China

In observational studies, confounding variables affect both treatment and outcome. Within this causal framework, instrumental variables further influence the treatment assignment mechanism. Such dependencies distinguish the study context from standard randomized controlled trials, where assignment is random. Consequently, the estimated average treatment effect (ATE) becomes biased. To address this issue, a standard approach is to incorporate estimated propensity scores. However, these methods incur the risk of misspecification in propensity score models. To resolve this limitation, the Self-balancing neural network (Sbnet), which allows the model to endogenously obtain its pseudo propensity score from a balancing net, is proposed. The Sbnet estimates the ATE by using the balancing net as a key part of the feedforward neural network. This formulation resolves the estimation of the ATE in a single step. Moreover, a multi-pseudo propensity score framework is developed, which is estimated from diversified balancing nets and used for enhancing estimation accuracy. Finally, the proposed methods are compared with state-of-the-art methods on four simulation setups and real-world datasets. Empirical results demonstrate that the Self-balancing neural network yields performance superior to

state-of-the-art baselines.

EO042 Room 107 COMPLEX DATA: DATA INTEGRATION AND SCALABLE BAYESIAN INFERENCE
Chair: Myungjin Kim
E1543: Cross-cohort data integration for complex surveys with unobserved target outcomes

Presenter: **Danhyang Lee**, Southern Methodist University, United States

Large-scale surveys fielded at different time points often share predictors but differ in outcome availability, preventing direct inference for the more recent population. A transportability framework is developed for integrating repeated cross-sectional surveys under complex multistage designs when the target outcome is observed only in the source cohort. Under covariate shift, survey design weights are combined with density-ratio importance weights and a target-directed penalized pseudo-log-likelihood extending importance-weighted cross-validation to complex surveys is proposed, paired with a doubly robust AIPW estimator. When the outcome model itself may have shifted, bridge-calibrated estimation is introduced: shared intermediate outcomes observed in both cohorts proxy for drift in the target outcome relationship. A target-weighted penalized procedure selects informative bridges, and the resulting estimate is embedded in a sensitivity analysis yielding partial identification intervals anchored by observed bridge discrepancies. Simulations evaluate performance under varying covariate shift, outcome-model drift, and bridge informativeness. The framework is applied to Head Start FACES 2009 and 2014 cohorts, producing transportability-adjusted estimates of kindergarten math achievement.

E1641: Scalable Bayesian classifier with many relevant features

Presenter: **Jieun Lee**, Kyungpook National University, Korea, South

Co-authors: Myungjin Kim, Gyuhyeong Goh

Probabilistic classification based on the Bayes rule is often infeasible in high-dimensional settings due to the need to estimate a large covariance structure among features. While Naive Bayes classifiers circumvent this issue by imposing conditional independence, their performance can deteriorate substantially when features exhibit strong dependence, which is common in modern applications. A scalable Bayesian classification framework is developed that accommodates complex dependence among a large number of relevant features without requiring full covariance estimation. The proposed approach employs a Vecchia-type approximation to construct a sparse representation of the joint distribution. A key challenge in this setting is the absence of a natural notion of neighborhood outside spatial contexts. To address this, a data-driven neighborhood construction based on the link between variable selection and conditional independence is introduced. The resulting classifier achieves a balance between statistical accuracy and computational efficiency in high-dimensional regimes. The performance of the method is demonstrated through simulation studies and real data analysis.

E1731: Scalable data augmentation for high-dimensional Bayesian regression and classification

Presenter: **Gyuhyeong Goh**, Kyungpook National University, Korea, South

With the increasing prevalence of high-dimensional data, shrinkage priors have become standard tools in Bayesian regression modeling. As most shrinkage priors used for sparse regression belong to the class of normal-scale mixtures, Gibbs sampling has attracted considerable attention for performing exact Bayesian inference. However, when the dimension of the coefficient vector is large, the conventional Gibbs sampler under normal-scale mixture priors becomes computationally expensive. A scalable data augmentation approach enables exact Bayesian inference for high-dimensional regression and classification under a general class of normal-scale mixture priors, including spike-and-slab and horseshoe priors. A distinguishing feature of the proposed posterior sampler is that its per-iteration computational complexity can be made to scale linearly with the size of the design matrix.

EO053 Room 201 RECENT DEVELOPMENTS IN MULTIPLE TESTING AND INDUSTRIAL STATISTICS
Chair: Yuhang Xu
E1288: Multivariate control charts based on multiple testing procedures

Presenter: **Arthur Yeh**, Bowling Green State University, United States

The multiple testing procedures have garnered some degree of prominence in modern-era statistical applications, especially in genome and genes sequencing research. Multiple testing procedures can be adapted to construct multivariate control charts. The control charts thus developed can not only monitor correlated variables but also provide instantaneous diagnostics of out-of-control parameters when an out-of-control signal is detected, without much further effort. In essence, the multiple testing procedures-based multivariate control charts simultaneously tackle and provide solutions to two major challenges in multivariate control chart developments and applications, monitoring/detection and diagnostics. Furthermore, these types of charts offer potential solutions to problems that have received little or almost no attention in existing literature, e.g., multivariate control charts for monitoring correlated variables of different types. The discussions focus on some recent work, as well as potential future research directions along the same line.

E1363: Large-scale fault diagnosis for multi-group data with auxiliary information via distributed multiple testing

Presenter: **Zhihan Zhang**, East China Normal University, China

Co-authors: Wendong Li, Fugee Tsung, Dongdong Xiang

In the era of big data, efficiently diagnosing faults in high-dimensional data streams (HDS) is critical for numerous industrial applications. A novel decentralized fault diagnosis problem involving multi-group HDS with multi-sequence auxiliary information (MAI) is addressed. Traditional diagnostic methods, which are designed for a single group of single-sequence HDS, struggle with the complexity and volume of such data, often leading to suboptimal diagnostic performance. To overcome this challenge, a distributed fault diagnosis framework is proposed that leverages advanced multiple testing techniques and data fusion strategies to analyze multi-group HDS with MAI. Under this framework, a generalized multi-sequence local index of significance for the data streams in each group is introduced, based on a Cartesian hidden Markov model, to effectively fuse information from auxiliary sequences. This is then integrated into a distributed multiple testing procedure for group-wise diagnosis of the target data sequence. The proposed procedure minimizes the group-wise expected number of false positives in the target sequence while controlling the overall group-wise missed discovery rate at a specified level. Numerical studies demonstrate that the proposed method outperforms state-of-the-art diagnostic techniques, providing more reliable and effective fault diagnostics.

E1365: Screening main and interaction effects in a Plackett-Burman design

Presenter: **Yuhang Xu**, Bowling Green State University, United States

In a supersaturated design, the number of factors is greater than the number of observations. A design that is not supersaturated but has more parameters to be estimated than observations results in a supersaturated model (SSM). An SSM may occur when two-factor interactions exist in addition to main effects. The goal is to find the few active effects from the many in the SSM. Several approaches may be used to find a set of potential models and then to select the best model using some criterion. This may be done in a single stage by considering all main effects and two-factor interactions simultaneously, or by screening main effects first, then identifying the active interactions. A single-stage and multiple two-stage approaches are studied as well as different algorithms and criteria for selecting the models. For the Plackett-Burman design studied, the SCAD algorithm with the corrected akaike information criterion (cAIC) generally performs well, but the best procedure depends on the true, unknown, underlying model.

EO163 Room 202 ADVANCES IN MODERN DATA ANALYSIS**Chair: Sang-Yun Oh****E1693: Uncovering public health priorities at the World Health Organization****Presenter:** **Xiao Hui Tai**, University of California, Davis, United States**Co-authors:** Lauren Peritz, Kathryn Russ, Nick Ulle, Carl Stahmer

Government delegates convene each year in the World Health Assembly, the decision-making body of the World Health Organization, to set a global Health agenda. They allocate scarce resources for crisis response and long-term programs. With thousands of diplomatic statements on diverse topics, the systematic analysis of WHA meetings has remained a challenge for researchers. This gap is addressed using natural language processing. Thousands of delegate statements across seven major Health issues are extracted and classified using fine-tuned BERT models, demonstrating superior results over traditional word frequency-based classifiers and a chat-based large language model. A longstanding model interpretability problem is addressed by demonstrating how classification determinations are made. The analysis highlights tensions between industrialized and developing countries, particularly around infectious disease mitigation.

E1819: A Bayesian approach for producing subnational population estimates using a population base statistical register**Presenter:** **Jairo Fuquene**, UC Davis, United States

Subnational Population Estimates (SPE) in Latin America are useful for implementing new public policies in subnational areas with internal armed conflicts or difficult access. A Bayesian approach combining a Population Base Statistical Register (PBSR) and Official Population Projections (OPP) is proposed to produce SPE. The proposed procedures are useful for computing SPE of Population size or SPE as a percentage SPE (%). However, the focus is on SPE (%) due to data restrictions and to ensure data confidentiality. The PBSR is constructed using multiple administrative sources with registers from health, education, vital statistics systems, tax registration, and victim records from the current internal armed conflict in Colombia. New fast Markov Chain Monte Carlo algorithms are proposed to produce SPE (%) using data augmentation procedures to address complications caused by the resulting joint posterior containing gamma functions. The proposal is implemented to compute SPE (%) by age and sex groups in the municipality of Jamund in Colombia, which is currently affected by poverty, forced displacement, and internal armed conflict, with accuracy evaluated using a Population Census.

E1879: Lipschitz-regularized probability divergences and applications to generative modeling**Presenter:** **Ziyu Chen**, University of North Carolina at Chapel Hill, United States

Probability divergences such as the Kullback-Leibler (KL) divergence provide an information-theoretic measure of discrepancy between two probability distributions, but they require absolute continuity of one distribution with respect to the other. This assumption often fails for empirical measures or distributions supported on low-dimensional structures. In contrast, Wasserstein metrics quantify the transport cost between distributions without requiring absolute continuity, but they can break down when one of the distributions is heavy-tailed. Lipschitz-regularized divergences, introduced as a special class of (f, Gamma)-divergences, combine f-divergences and the Wasserstein-1 metric via their variational (dual) formulations and inherit the advantages of both. Key theoretical properties of Lipschitz-regularized f-divergences are presented, demonstrating how these results lead to robust generative modeling methods under minimal assumptions on the target data distribution, including cases with heavy tails, low-dimensional structure, or fractal-like support.

EO234 Room 203 DESIGN AND ANALYSIS OF RANDOMIZED EXPERIMENTS**Chair: Jizhou Liu****E1584: Inference for bipartite experiments: A superpopulation perspective****Presenter:** **Hongchang Guo**, Northwestern University, United States**Co-authors:** Yuehao Bai, Xun Huang, Jizhou Liu

The analysis studies bipartite experiments in which outcome units may be linked to many intervention units and outcomes depend on the numbers of treated and untreated neighbors. Causal comparisons between all linked intervention units treated and all linked intervention units untreated are considered, with the target parameter being a degree-weighted average of the corresponding exposure contrasts. A covariate-driven sparse random graph model is introduced in which degrees are approximately Poisson with bounded mean. Under Bernoulli assignment or complete randomization, the estimator is shown to be asymptotically normal as the numbers of intervention and outcome units grow. The asymptotic variance has two components: sampling variation from the outcome side and sampling variation induced by the intervention side.

E1619: A new design-based variance estimator for finely stratified experiments**Presenter:** **Xun Huang**, The University of Chicago, United States

Design-based inference for the average treatment effect in finely stratified experiments is considered. Design-based inference refers to settings where the only source of uncertainty stems from the randomness in treatment assignment itself; finely stratified experiments refer to settings where units are first stratified into groups of a fixed size according to baseline covariates and then, within each group, a fixed number of units are assigned uniformly at random to treatment and the remainder to control. A novel estimator of the variance of the difference-in-means based on pairing adjacent strata is presented. This estimator is well defined even in the challenging setting where there is exactly one treated or control unit per stratum. The estimator is proven to be upward-biased, and thus can be used for inference under mild restrictions on the finite population. Comparison with well-known estimators proposed previously in this setting demonstrates that, while these estimators are also upward-biased, the proposed estimator has smaller bias and therefore leads to more precise inferences whenever adjacent strata are sufficiently similar.

E1664: Regression-based and design-based causal inference in two-sided experiments**Presenter:** **Pengfei Tian**, Tsinghua University, China**Co-authors:** Peng Ding, Jizhou Liu

Regression-based and design-based causal inference in two-sided experiments under the simple multiple randomization design is studied, in which buyers and sellers are independently and completely randomized, and pair-level treatment is induced by their joint assignments. Under a local interference assumption, a broad class of linear estimands including total effects, buyer-side and seller-side spillover effects, and direct effects is considered. The natural plug-in estimator is shown to be equivalent to a saturated regression estimator, providing a unified regression representation for causal estimation in this setting. A design-based asymptotic theory is developed using a Hoeffding decomposition into buyer-side and seller-side first-order projections and a second-order interaction remainder, yielding a central limit theorem with a transparent leading variance. For inference, the conventional two-way clustered OLS variance estimator is shown to be asymptotically conservative. A weighted class of conservative variance estimators is further introduced, asymptotically containing the two-way clustered variance as a special case, with a closed-form optimal member that is sharper than any other estimator in the class. The framework also extends to regression adjustment.

EO285 Room 204 TOPICS IN TIME SERIES ANALYSIS**Chair: Kejin Wu****E1835: Tensor stochastic regression for high-dimensional time series via CP decomposition****Presenter:** Yao Zheng, University of Connecticut, United States

As tensor-valued data become increasingly common in time series analysis, there is a growing need for flexible and interpretable models that can handle high-dimensional predictors and responses across multiple modes. A unified framework for high-dimensional tensor stochastic regression based on CANDECOMP/PARAFAC (CP) decomposition is proposed, which encompasses vector, matrix, and tensor responses and predictors as special cases. Tensor autoregression naturally arises as a special case within this framework. By leveraging CP decomposition, the proposed models interpret the interactive roles of any two distinct tensor modes, enabling dynamic modeling of input-output mechanisms. Both CP low-rank and sparse CP low-rank estimators are developed, together with efficient alternating minimization algorithms, and their non-asymptotic statistical error bounds are established. Simulation studies confirm the theoretical properties and demonstrate the computational advantage. Applications to mixed-frequency macroeconomic data and spatio-temporal air pollution data reveal interpretable low-dimensional structures and meaningful dynamic dependencies.

E1902: Simultaneous inference for time series functional linear regression**Presenter:** Yan Cui, Illinois State University, United States**Co-authors:** Zhou Zhou

Joint simultaneous confidence band (JSCB) construction for regression coefficient functions of time series scalar-on-function linear regression is considered when the regression model is estimated by a roughness penalization approach with flexible choices of orthonormal basis functions. A simple and unified multiplier bootstrap methodology is proposed for JSCB construction and is shown to achieve the correct coverage probability asymptotically. Furthermore, the JSCB is asymptotically robust to inconsistently estimated standard deviations of the model. The proposed methodology is applied to a time series data set of electricity demands to visually investigate and formally test the overall regression relationship, as well as to perform model validation.

E1907: Prediction inference of time series with standard ReLU deep neural networks**Presenter:** Kejin Wu, Loyola University Chicago, United States

A methodology based on standard ReLU Deep Neural Networks (DNN) is proposed to make prediction inference with uncertainty quantification. Classically, linear, nonlinear, or nonparametric kernel methods have been relied upon to fit and predict time series. As the universal approximation ability of DNN was revealed, the application of DNN has become increasingly popular for prediction tasks in various scientific areas. However, the corresponding uncertainty quantification has not been studied thoroughly. Particularly, the uncertainty in prediction consists of two parts: (1) future variability and (2) estimation variability within training data. To capture both variabilities, a pertinent prediction interval (PPI) is built with the DNN model estimator. First, the consistency property of the DNN estimator with Beta-mixing dependent data is explored. Subsequently, it is shown that the implied forward bootstrap series is still Beta-mixing and possesses the same stationary distribution as the original time series in probability, which is a key condition for enabling the PPI. Finally, the desired PPI is constructed after imposing minimal conditions on the limiting distribution of predictive roots. Simulations and real-data analysis are deployed to evaluate the method against standard nonparametric and purely Deep-learning methods.

EO092 Room 205 ADVANCED STATISTICAL MODELLING FOR HEALTH DATA AND SURVEILLANCE**Chair: Kuan Liu****E1459: Bayesian hierarchical bootstrap framework for causal subgroup estimation with a time-to-event outcome****Presenter:** Mengyao Shi, University of Toronto, Canada**Co-authors:** Kuan Liu

Estimating treatment effects within prespecified subgroups is a central goal of causal inference. For time-to-event outcomes, subgroup causal estimands are defined via g-computation by averaging a conditional survival model over the subgroup-specific covariate distribution. Existing Bayesian approaches typically treat these distributions independently, which can be unstable when subgroups are small or imbalanced. Right censoring further reduces information, increasing uncertainty in subgroup survival contrasts. The hierarchical Bayesian bootstrap (HBB) is extended to subgroup causal inference with right-censored outcomes. The HBB places a nonparametric hierarchical prior on subgroup covariate distributions, enabling borrowing of information across related subgroups while preserving heterogeneity. This is combined with a Bayesian accelerated failure time model to perform posterior g-computation, propagating uncertainty from both the survival model and the covariate distribution. The proposed framework stabilizes subgroup causal survival estimates in sparse settings without parametric assumptions. Simulation studies evaluate performance across varying subgroup sizes and censoring levels, and compare with Bayesian additive regression trees for heterogeneous survival effects.

E1523: Assessing the reliability of longitudinal electronic dental records through agreement of periodontal markers**Presenter:** Aya Mitani, University of Toronto, Canada**Co-authors:** Hanlong Chen, Yuning Wang, Christopher McCulloch

Electronic dental records (EDR) are increasingly used to study oral diseases, yet their measurement reliability is not well assessed. one aspect of EDR reliability is evaluated by treating probing pocket depth (PPD) as an ordinal measure and comparing site-level PPD recorded at two exams within 120 days in 1,742 Canadian Armed Forces members. using a generalized linear mixed-effects model with nested random effects, kappa for agreement and weighted kappa for association are estimated. agreement was fair, and association was moderate, with lower values in males, younger individuals, and tobacco nonusers. these findings suggest EDR can support epidemiologic research, though caution is warranted.

E1632: Two-phase study design under differential unit costs: Insights from metaheuristic algorithms**Presenter:** Osvaldo Espin-Garcia, University of Western Ontario, Canada**Co-authors:** Jiaqi Yuan

Two-phase studies realize cost-efficiency by utilizing phase 1 data to select an informative subsample for expensive variable collection in phase 2. Previous work assumes equal cost of collecting expensive data across samples. This assumption is challenged and differential unit-cost two-phase designs are investigated. The proposed approach penalizes phase 2 subsamples based on budget constraint satisfaction. Three metaheuristic algorithms are compared under heterogeneous per-sample costs: Genetic Algorithm (GA), Simulated Annealing (SA), and Comprehensive Evolutionary Algorithm (CEA). The resulting designs aim to minimize the variance of the parameter of interest under the defined constraints. Simulations based on Poisson regression and inference via semiparametric maximum likelihood evaluate statistical performance across a range of design parameters. GA and CEA consistently achieve best performance while SA exhibits greater variability and occasional convergence failures.

E1690: Probabilistic reconstructions of epidemic curves using wastewater data**Presenter:** Justin Slater, University of Guelph, Canada

Determining the current and historical prevalence of an infectious disease is crucial for understanding disease burden. Wastewater data is a relatively low-resource way to measure changes in prevalence and, unlike other forms of surveillance data, is less affected by preferential testing. However,

how to best utilize wastewater data to estimate prevalence is an area of active research. A novel framework is proposed for reconstructing epidemic curves using wastewater data within a binomially thinned Poisson autoregressive framework. The effective reproduction number is modeled as the logarithmic derivative of the wastewater signal to make probabilistic inference about the true historical and current case counts. Several applications of this framework are demonstrated.

EO077 Room 208 STATISTICAL METHODS FOR THE ANALYSIS OF DATA IN BIOMEDICAL STUDIES
Chair: Yisheng Li
E1720: Statistical learning of pediatric mental health emergency department visits before, during, and after COVID-19 pandemic

Presenter: **Anqi Chen**, University of Alberta, Canada

Co-authors: Joan Hu, Rhonda J Rosychuk

A novel statistical framework for analyzing zero-truncated recurrent event data over time is presented, motivated by data from a Pediatric Mental Health Care (PMHC) program. The primary goal is to understand how patterns of emergency department (ED) visits for Mental Health concerns among children evolve across distinct phases of the COVID-19 pandemic: before, during, and after. The modeling framework is developed in a stepwise manner, guided by model fitting using a collection of Mental Health-related ED records. Stratified regression analysis is applied to investigate changes in visit frequencies and covariate effects, accounting for pre-specified period cutoffs and coarsened individual follow-up information. Real-world PMHC data serve as both the motivation and a demonstration of the practical utility of the approach.

E1970: A framework of Polya-Gamma Gibbs sampler for analyzing panel counts and terminal event time data

Presenter: **Chia-Hui Huang**, National Chengchi University, Taiwan

A joint analysis of panel counts and terminal event time models is developed under the Bayesian framework. The counts and event status are collected at discrete points, where the two types of events may exhibit either positive or negative dependence. The mean function of the counts and hazard rate of the terminal event are linked by frailty variables whose covariance matrix determines the underlying association. A Gaussian process is applied to model the logarithm of the baseline mean and hazard rate functions. Under the Bayesian framework, the Gibbs sampler with Polya-Gamma latent variables provides an efficient Markov chain Monte Carlo method to obtain posterior samples. The proposed method is applied to a clinical trial dataset consisting of panel counts and a terminal event.

E1977: High-dimensional error-in-variables survival analysis

Presenter: **Jinfeng Xu**, City University of Hong Kong, Hong Kong

High-dimensional survival data with measurement errors are prevalent in biomedical studies, where numerous clinical or genetic variables are collected for risk assessment. The presence of measurement errors in covariates substantially complicates parameter estimation and variable selection, often leading to challenging non-convex optimization problems. An error-in-variables additive hazards regression model for high-dimensional noisy survival data is proposed. By employing the nearest positive semi-definite matrix projection, a fast Lasso approach and its soft-thresholding variant are developed, both supported by strong theoretical guarantees. Under mild assumptions, model selection consistency, oracle inequalities, and asymptotic distributions are established for the proposed methods. Simulation studies and applications to two real datasets demonstrate the superior performance of these methods in handling high-dimensional data. Notably, they exhibit remarkable robustness even in scenarios with missing values, highlighting their practical utility in complex biomedical settings.

EV274 Room 209 MULTIVARIATE AND HIGH-DIMENSIONAL METHODS (VIRTUAL)
Chair: Elizabeth Juarez-Colunga
E1903: Globally aligned principal component analysis for multi-group data

Presenter: **Federico Severino**, Université Laval, Canada

Co-authors: Hedayat Fathi, Marzia Cremona

A novel principal component analysis (PCA) method for multi-group datasets is proposed, where multiple numerical variables are measured across different groups. The method combines group-specific principal components with global ones through an explicit alignment mechanism based on regularized optimization. A globally aligned covariance matrix is introduced that incorporates weighted contributions from global principal directions, balancing the preservation of within-group variance with global coherence. The alignment strength is controlled by regularization parameters that can be tuned to achieve the desired trade-off. A comprehensive simulation study demonstrates that the aligned approach achieves a favorable compromise between capturing local variation within groups and maintaining interpretability and stability across groups. The method addresses an important gap in the PCA literature, as existing approaches either ignore group structure entirely, focus exclusively on local structure through group-wise PCA, or impose restrictive assumptions of common principal components. The proposed approach respects the multi-group nature of data while improving global comparability of components. An application to 2021 Canadian Census socioeconomic data demonstrates that the proposed alignment yields more comparable and stable region-specific components than pooled or purely region-wise PCA.

E1892: The blessing of overfitting in instrumental variables: Inference with non-sparse high-dimensional signals

Presenter: **Ruei-Chi Lee**, Rutgers, United States

The Factor IV model is revisited in the context of high-dimensional IV regression from a new perspective, providing theoretical justification for recent empirical findings that the first stage signal is dense. Moreover, the implied first stage coefficients shrink toward zero as the number of instruments diverges, exhibiting local-to-zero behavior commonly used to describe settings with many weak instruments. Building on this characterization, an IV estimator is proposed based on pseudo-inverse OLS (ridgeless regression) combined with sample splitting. This procedure remains well defined when the number of first-stage regressors exceeds the sample size. The asymptotic distribution of the resulting sample-splitting IV estimator is derived under regimes with very many weak instruments, and it is shown that, when the number of first-stage regressors is sufficiently large, its asymptotic variance coincides with that of an oracle Factor IV estimator that directly observes the latent factors. The analysis also provides practical guidance on constructing many regressors from control variables and on deliberately adding noise instruments.

E2007: Change-point prediction/detection as alternative to insurer traditional hedging for equity benefit guarantees

Presenter: **Mark Zanecki**, IHA Consultants, United States

An AI-enabled daily change point prediction and detection process is introduced, including a downloadable Excel summary model for hands-on analysis. Prospective change point identification is demonstrated for selected USA indexes, ETFs, and stocks, alongside retrospective change point detection with near 100% accuracy. Theoretical foundations using discrete Fokker-Planck approximation are reviewed. Statistical support is established through t-tests, ANOVA, and linear algebra eigen-analysis via principal component methods. Prospective change points within the subsequent 3 trade days are predicted with 90% or greater accuracy and confirmed retrospectively with near 100% accuracy given sufficient run-out time. The Henriksson and Mertons sufficient statistic for investment return value, defined as the sum of condition probabilities, exceeds 1.0 and approaches 2.0. An accurate daily prospective and retrospective change point framework with sufficient accuracy provides on-demand, timely equity benefit guarantee and risk mitigation at significantly lower cost compared to traditional hedging. Time permitting, a synopsis of other applications is discussed.

Sunday 09.08.2026

10:00 - 12:05

Parallel Session G – EcoSta2026

EI005 Room 201 NEW ADVANCES IN SPORTS ANALYTICS**Chair: Anqi Chen****E1157: The problem of aging in soccer****Presenter:** Tim Swartz, Simon Fraser University, Canada

The important problem of assessing aging in soccer is considered. The problem relies on tracking data and begins with a review of average aging curves in sport. Then, a new approach is introduced which addresses personal aging curves in soccer, an essential problem of interest which has not been previously addressed. The underlying methods are taken from functional data analysis.

E1158: Understanding football receiver routes through functional data clustering**Presenter:** Tianyu Guan, York University, Canada

Running routes by receivers are central to offensive strategy in football because they shape how teams create space, advance the ball, and design passing plays. Functional principal component analysis (FPCA) is proposed to cluster these routes by treating each trajectory as a smooth functional curve. FPCA extends classical PCA to functional data and provides a low-dimensional representation of the trajectories. By projecting routes onto a small number of functional principal components (FPCs), the FPC scores that represent each route are obtained. Clustering methods such as K-means can then be applied to these FPC scores, and each cluster corresponds to a common route type run by receivers. These route clusters provide insight into the strategies teams use to advance the ball on each play. From a football analytics perspective, they are interpretable and useful for downstream tasks. For example, the FPC scores for each cluster can be used as predictors in a regression model to predict whether a route is successfully completed. Using a group lasso approach, with each group corresponding to a cluster, allows one to assess the contribution of different route types to completion outcomes.

E1606: Hierarchical Bayesian joint modeling of athlete performance trajectories and latent peak ages**Presenter:** Kangyi Peng, Simon Fraser University, Canada**Co-authors:** Tianyu Guan, Xiong Zhao

The focus is on the fundamental problem of determining when athletes reach peak performance, a key question in competitive sports. Traditional analyses often rely on simple summary statistics or per-athlete curve fitting, which limit inference and lack principled uncertainty quantification. A Bayesian joint modeling framework is developed that characterizes athlete performance trajectories together with their latent peak ages, specifying performance curves flexibly and embedding peak age in a hierarchical survival sub-model. The framework enables information borrowing across athletes and provides uncertainty quantification even with sparse or irregular observations. Simulation studies show that the method yields stable estimation and improves uncertainty assessment compared with conventional approaches. Analysis of World Athletics data across ten sport categories reveals discipline-specific peak-age patterns and covariate effects. Beyond sports, the framework is broadly applicable to problems involving joint modeling of longitudinal trajectories and function-derived latent features.

E1733: A Bayesian frailty model for the timing of corner kicks in association football**Presenter:** Riley Isaacs, Simon Fraser University, Canada

Corner kicks are becoming an increasingly important event in soccer because they are often the result of strong attacking play and can be of keen interest to sports fans and bettors. The current Premier League season (2026) is seeing goals generated by corner kicks make up the highest proportion of goals in a decade. Corner kick data are framed as right-censored event times, and the mixture feature of corner kick times caused by previous corner kicks is formulated to explore patterns of corner kicks associated with several factors. A frailty model is introduced to account for the correlation between the timing of corner kicks taken by the same team and within the same game. An analysis of the corner kicks in the 2019 season of the Chinese Super League is conducted in the Bayesian paradigm to provide an alternative view on factor effects.

EI003 Room 202 MACHINE LEARNING AND STATISTICS: BENEFITS AND PITFALLS**Chair: Ansgar Steland****E1218: How can mathematics contribute to AI?****Presenter:** Johannes Lederer, University of Hamburg, Germany

Artificial intelligence is arguably the hottest topic in science. Computer science and engineering currently set the agenda in this field, sidelining mathematics to a large extent. It will be highlighted, however, that mathematics has a lot to offer. Mathematical guarantees that provide deep insights into the inner workings of AI will be introduced, and it will be shown how statistical principles can make AI more efficient. More generally, the role of mathematics, especially mathematical statistics, in AI and data science will be discussed.

E1210: Machine learning and statistical methods for high-throughput experimentation data**Presenter:** Tatyana Krivobokova, University of Vienna, Austria

Machine learning (ML) and artificial intelligence (AI) techniques are transforming the way chemical reactions are studied today. Datasets from high-throughput experimentation (HTE) are generated to better understand the reaction conditions crucial for outcomes such as yields and selectivities. However, it is often overlooked that datasets from such designed experiments possess a specific structure, which can be captured by a statistical model. Ignoring these data structures when applying ML/AI algorithms can result in misleading conclusions. In contrast, leveraging knowledge about the data-generating process yields reliable, interpretable, and comprehensive insights into reaction mechanisms. A particularly complex dataset is available for the Buchwald-Hartwig amination. Using this dataset, a statistical model for such HTE-generated chemical data is introduced, and a parameter estimation algorithm is developed. The approach is applicable to a wide range of HTE-generated data for chemical reactions and beyond.

E1261: Personalized health monitoring through AI and sensor technology**Presenter:** Kwok Tsui, The University of Texas at Arlington, United States

Due to the advancement of computation power, sensing technologies, and data collection tools, the field of healthcare and public health monitoring and management has evolved over the past several decades under different names and application domains, such as statistical process control (SPC), process monitoring, health surveillance, prognostics and health management (PHM), personalized medicine, etc. There are tremendous opportunities in interdisciplinary research of personalized health monitoring through integration of AI, data analytics, and sensor technologies. In this talk views and experiences in the evolution of personalized health monitoring and management, its challenges and opportunities, as well as its applications in neurodegenerative diseases such as Alzheimer and Parkinson diseases will be presented.

E1695: Reinforcement learning in transportation**Presenter:** Ostap Okhrin, Technische Universität Dresden, Germany

Reinforcement learning (RL) has emerged as a powerful method for solving complex control tasks across various domains, from autonomous driving to maritime navigation. Research in RL, particularly in value-based algorithms, addresses critical issues such as overestimation bias, proposing innovative solutions like the T-Estimator (TE) and K-Estimator (KE) for bias control and algorithmic robustness. These advancements

are validated through modifications to Q-Learning and the Bootstrapped Deep Q-Network (BDQN), demonstrating superior performance and convergence. A spatial-temporal recurrent neural network architecture has been developed for autonomous ships, enhancing robustness in partial observability and compliance with maritime traffic rules. A modular framework for autonomous surface vehicles on inland waterways has been created, utilizing DRL agents for path planning and following, which significantly outperforms traditional control methods. Work on dynamic obstacle avoidance environments for mobile robots and drones emphasizes the importance of controlled training difficulty for better generalization and robustness. This approach has been successfully applied across different platforms, reducing the simulation-to-reality (Sim2Real) gap and improving performance in real-world scenarios. Through these contributions, the practical application and reliability of reinforcement learning in diverse and dynamic environments are advanced.

EO186 Room B101 ROBUST METHODS FOR CAUSAL LEARNING AND DECISION-MAKING
Chair: Abhimanyu Gupta
E1617: Hypothesis testing for detecting network interference in randomized experiments
Presenter: Christopher Harshaw, Columbia University, United States

Co-authors: Chao Gao, Fredrik Savje, Yitan Wang

Across disciplines, the randomized experiment is a widely used method for estimating causal effects. In order to meaningfully estimate causal effects, experimenters have to assume structure on the potential outcome functions. The simplest assumption is that of no-interference, although a growing body of work has provided various network interference models under which causal effects can be estimated. Still, this raises the question: to what extent can these assumptions be tested? For example, can interference be detected when it is present in the experiment? Can one test between stronger and weaker models of network interference? To this end, previous work has focused on controlling Type I error, but Type II error of proposed methods remains less well understood. Hypothesis testing frameworks for detecting interference are further investigated. The main contribution is a series of negative results. Roughly speaking, it is shown that there is no test for detecting interference which has small Type I and Type II error, even as the sample size increases. This finding has serious consequences for the epistemological nature of the no-interference and various network interference assumptions: namely, that they are, in a rigorous sense, untestable. On the positive side, it is shown that there exists a uniformly consistent test of no-interference when the alternative hypotheses are restricted to a linear-in-means Type model and the graph is sufficiently sparse.

E1840: When-if decision-making using synthetic survival control
Presenter: Jessy Xinyi Han, Broad Institute / MIT, United States

Understanding the impact of decisions on when a target event occurs, not just whether it occurs, is central to many fields, including patient Survival in healthcare, criminal recidivism in policy evaluation, and customer churning in business. A when-if decision-making framework integrates causal inference and Survival analysis to support decisions with observational data where the timing of the target event is the key quantity of interest and the data are often sparse, censored, or confounded. The motivation concerns evaluating the efficacy of different therapies for T-Cell Lymphoma across a heterogeneous patient population. Synthetic Survival Control (SSC) is a new method for estimating counterfactual hazard trajectories in panel data with censoring and unobserved confounding. The method extends the traditional panel data literature by utilizing a causal Survival panel framework with an underlying low-rank structure that naturally arises under classical parametric Survival models. Within this framework, identification of the causal estimand and finite-sample guarantees for SSC are established. Full validation of the proposed method is provided through application to T-Cell Lymphoma treatment evaluation in collaboration with clinicians at Massachusetts General Hospital.

E1896: Stable causal estimation with optimal transport
Presenter: Yixin Wang, University of Michigan, United States

In causal inference, the overlap (a.k.a. positivity) assumption is frequently violated in practice. Limited overlap causes severe instability for widely used estimators such as inverse propensity weighting (IPW), whose variance depends on density ratios that can explode as propensity scores approach zero or one. To address this challenge, the Stable Transport Estimator (STE), an optimal Transport-based approach for estimating average causal effects under limited overlap, is proposed. Unlike IPW, the variance of STE scales with the Wasserstein distance between treated and control covariate distributions, a quantity that remains finite under bounded moments, even when density ratios explode. By replacing reweighting with a geometry-aware Transport map, STE constructs Stable counterfactual comparisons without requiring density estimation or propensity score modeling. STE further extends to settings with no overlap at all: when covariate supports are disjoint, a linear interpolation Transport map enables principled extrapolation with explicit bias-variance trade-offs. Consistency, asymptotic normality, and sharp bias-variance characterizations are established under conditions substantially weaker than those required by standard methods. Across synthetic and real datasets, STE demonstrates improved robustness, reduced bias, and markedly more Stable variance compared to existing estimators.

E1911: Robust signal maximization
Presenter: Evan Munro, University of Chicago, United States

Co-authors: Peter Hull, Kirill Borusyak

The optimal design and analysis of experiments for estimating spillover effects is studied, assuming a known (e.g., linear) exposure mapping. The treatment-assignment distribution and regression-based estimator that minimize worst-case asymptotic variance against a broad class of distributions of unobservables are characterized. The design problem yields an intuitive solution in which the planner trades off spillover signal strength against diffusion of spillover variation. The analysis problem yields a simple recentered instrumental variable estimator to best leverage this variation. The framework produces natural solutions in several benchmark cases such as clustered exposure and suggests computationally tractable approximations for general networks, including bipartite settings. These tools are illustrated in semi-synthetic experiments based on two applications from development economics. The approach yields large standard error reductions in both experiments, increasing effective sample sizes by approximately 50%-100%.

EO232 Room B102 NEW DEVELOPMENTS IN ECONOMETRICS
Chair: Katsumi Shimotsu
E1931: A simple and adaptive confidence interval when nuisance parameters satisfy an inequality
Presenter: Gregory Cox, National University of Singapore, Singapore

Inequalities may appear in many models. They can be as simple as assuming a parameter is nonnegative, possibly a regression coefficient or a treatment effect. The focus is on the case that there is only one inequality and proposes a confidence interval that is particularly attractive, called the inequality-imposed confidence interval (IICI). The IICI is simple. It does not require simulations or tuning parameters. The IICI is adaptive. It reduces to the usual confidence interval (calculated by adding and subtracting the standard error times the $1 - \alpha/2$ standard normal quantile) when the inequality is sufficiently slack. When the inequality is sufficiently violated, the IICI reduces to an equality-imposed confidence interval (the usual confidence interval for the submodel where the inequality holds with equality). Also, the IICI is uniformly valid and has (weakly) shorter length than the usual confidence interval; it is never longer. The first empirical application considers a linear regression when a coefficient is known to be nonpositive. A second empirical application considers an instrumental variables regression when the endogeneity of a regressor is known to be nonnegative.

E1954: Testing the solvability of systems of linear inequalities*Presenter:* **Eric Mbakop**, OSU, United States*Co-authors:* Leonard Goff

The focus is on the problem of testing whether a system of linear equality and inequality constraints admits a solution when the coefficients must be estimated. A wide range of inferential questions in partially identified models can be formulated as hypotheses of this form. The approach exploits an alternative characterization of the hypothesis based on whether the value of a certain linear program equals zero. Building on this characterization, bootstrap-based testing procedures are developed and their uniform validity is established over large classes of data-generating processes. Simulation results demonstrate good finite-sample performance, even for moderate sample sizes. Two empirical applications illustrate the usefulness of the approach.

E1913: RDD without a discontinuity: Correcting for Berkson measurement error in the running variable*Presenter:* **Junlong Feng**, The Hong Kong University of Science and Technology, Hong Kong*Co-authors:* Leonard Goff

Measurement error is a pervasive phenomenon in empirical research. In the context of the regression discontinuity design (RDD), measurement error in the running variable can be especially problematic. Individuals who are marginal with respect to the treatment assignment rule might be spread across values of the mismeasured running variable, completely eliminating any observable discontinuity in the conditional mean of the outcome at the cutoff. When measurement error in the running variable takes the so-called Berkson form, that is when it is independent of the observed running variable, it is still possible to identify the typical sharp RDD estimand. The identification procedure requires that binary treatment status be observed without error, for the support of the measurement error to be suitably rich, and an assumption that the measurement error is not informative about potential outcomes once one conditions on the true running variable.

E1940: Semiparametric identification of the discount factor and payoff function in dynamic discrete choice models*Presenter:* **Katsumi Shimotsu**, University of Tokyo, Japan*Co-authors:* Hiroyuki Kasahara, Yu Hao

The aim is to investigate how the discount factor and payoff function can be identified in stationary infinite-horizon dynamic discrete choice models. In single-agent models, common nonparametric assumptions on per-period payoffs such as homogeneity of degree one, monotonicity, concavity, zero cross-differences, and complementarity provide identifying restrictions on the discount factor. These restrictions take the form of polynomial equalities and inequalities with degree bounded by the cardinality of the state space. Furthermore, these restrictions can identify the discount factor and the payoff function even without a normalization on the payoff at one action. In dynamic game models, firm-specific discount factors can be identified under assumptions such as irrelevance of other firms' lagged actions, exchangeability, and the independence of adjustment costs from other firms' actions. The results demonstrate that widely used nonparametric assumptions in economic analysis can provide substantial identifying power in dynamic structural models.

E1915: Initial-condition-robust inference in autoregressive models*Presenter:* **Ming Li**, National University of Singapore, Singapore*Co-authors:* Donald Andrews, Yapeng Zheng

A new confidence interval (CI) for the autoregressive (AR) parameter in an AR model with an AR parameter that may be close to or equal to one is introduced. Existing CIs rely on the assumption of a stationary or fixed initial condition to obtain correct asymptotic coverage and good finite sample coverage. When this assumption fails, their coverage can be quite poor. The proposed CI for the AR parameter has coverage probability that is completely robust to the initial condition, both asymptotically and in finite samples. This CI incurs only a small price in terms of its length when the initial condition is stationary or fixed. The CI is also robust to conditional heteroskedasticity of the errors.

EO062 Room B103 ECONOMETRICS AND STATISTICS FOR FINANCIAL AND DEMOGRAPHIC DATA**Chair: Toshiaki Watanabe****E1467: Heterogeneous autoregressive models driven by Gaussian processes with the generalized Cauchy covariance function***Presenter:* **Tsunehiro Ishihara**, Takasaki City University of Economics, Japan*Co-authors:* Toshihiro Hirano

Recent studies on volatility have highlighted the importance of path roughness measured by the fractal dimension D or the Hurst index H in modeling the dynamics of asset return volatility. Most existing approaches rely on fractional Brownian motion (fBM), which imposes $H + D = 2$. This constraint conflates long memory in levels with roughness in increments, preventing separate identification of the two properties. A Gaussian process approach with generalized Cauchy (GC) covariance functions has been recently proposed. The GC covariance function allows D and H to vary independently. Heterogeneous autoregressive (HAR) models for log realized volatility driven by Gaussian process innovations with GC covariance functions are proposed to estimate roughness, persistence, and heterogeneity separately. The proposed method is compared with several models, including AR(1), HAR, and HAR with jump components under three innovation specifications: i.i.d. Gaussian, fractional Gaussian noise (FGN), and the GC process. Bayesian estimation is conducted via MCMC with an adaptive Metropolis algorithm that reduces the number of matrix manipulations. Rolling-window forecasting performance is evaluated using MSE and QLIKE robust loss functions. In an application to daily realized volatility of the TOPIX index, prediction results and posterior quantities of $D + H$ are reported.

E1476: Tree-embedded Bayesian factor models for multidimensional categorical distributions*Presenter:* **Keisuke Sasaki**, University of Tokyo, Japan*Co-authors:* Naoki Awaya, Genya Kobayashi, Shonosuke Sugawara

Estimating common and heterogeneous structures from multiple data sources via hierarchical models is a central Bayesian inference task, where Bayesian factor models are widely used. A novel Bayesian latent factor model is proposed to parsimoniously describe many observed distributions. Social science applications often involve grouped data, such as regional age or income distributions. Standard mixture models can be inefficient here, as these distributions may lack clear clustering structures. To address this, a tree-based transformation is introduced that embeds distributions into Euclidean space, constructing the factor model there. This transformation allows straightforward extensions of the Bayesian hierarchical model. For instance, spatial dependence is incorporated using a simultaneous autoregressive prior. The model is nonparametric, imposing no strict assumptions on the observed distributions' forms. Numerical experiments with real population data show the model outperforms standard Dirichlet mixture models.

E1477: Unified mixture sampler for state-space models: Application to stochastic conditional duration models*Presenter:* **Daichi Hiraki**, University of Tokyo, Japan*Co-authors:* Yasuhiro Omori

A unified mixture sampler (UMS) is proposed that provides a universal estimation framework for nonlinear state-space models with exponential-exponential likelihood kernels. Unlike existing methods that require deriving new mixture approximations for each specific distribution, the approach dynamically adapts an existing ten-component mixture through a deterministic recentering and rescaling algorithm. Applied to the stochastic conditional duration (SCD) model, the proposed sampler efficiently handles unknown shape parameters such as those in Weibull or Gamma dis-

tributions by updating mixture components near-instantaneously during MCMC iterations. The UMS not only simplifies implementation but also ensures exact inference via a lightweight Metropolis-Hastings step. Numerical examples demonstrate that the method substantially outperforms conventional slice sampling approaches, significantly reducing autocorrelation in MCMC samples while maintaining high computational efficiency. This unified framework encompasses a wide range of applications, including logit, Poisson, and various SCD model specifications, providing a highly efficient alternative to model-specific samplers.

E1637: **Examining volatility roughness in the Japanese stock market**

Presenter: **Xuzhu Zheng**, The University of Osaka/MPX, Inc., Japan

Co-authors: Masato Ubukata, Kosuke Oya

The existence of rough volatility, which has recently attracted considerable attention and is characterized by volatility dynamics that cannot be fully captured by conventional volatility models, is examined. Specifically, the analysis investigates whether the observed roughness in volatility is merely an artifact induced by microstructure noise inherent in high-frequency price data, or whether such rough behavior persists after accounting for noise effects. The empirical analysis utilizes high-frequency data from the Nikkei 225 index and two actively traded individual stocks. Applying several representative volatility estimators, volatility series are constructed and their Hurst exponents are estimated using a nonparametric procedure. The results show that, regardless of microstructure noise, the estimated Hurst exponents consistently take low values, indicating persistent rough behavior. Furthermore, the predictability of log volatility constructed from these alternative volatility measures is evaluated using the rough fractional stochastic volatility model, the autoregressive model, and the heterogeneous autoregressive model under a unified p-ratio criterion. The analysis compares forecasting performance across models and volatility measures, while examining factors that may account for differences in predictive accuracy from the perspective of rough volatility.

E1734: **Reassessing price discovery measures in high-frequency data: Evidence from the Japanese stock market**

Presenter: **Makoto Takahashi**, Hosei University, Japan

Co-authors: Masahiro Yamada

Price discovery across investor types is studied using high-frequency data from the Tokyo Stock Exchange, showing that two widely used measures, Weighted Price Contribution (WPC) and Hasbrouck's VAR model, suffer from Structural biases that can lead to misleading conclusions about which traders incorporate information into prices. Through Monte Carlo simulations with a calibrated data generating process, it is shown that (i) WPC overestimates the Contribution of noise traders by capturing transitory Price fluctuations and volume heterogeneity as Price discovery, and (ii) VAR-based measures produce different results depending on the Cholesky ordering of variables. A seasonality-adjusted long-horizon WPC (Adj LH-WPC) is proposed that simultaneously addresses three sources of bias: transitory noise via long-horizon Price changes, volume heterogeneity via regression coefficients, and intraday seasonality via Weighted least squares. The method recovers the true permanent Price impact shares in simulations and provides stable, economically interpretable estimates of each investor type's Contribution in empirical analysis. Building on these findings, a Structural VAR framework is explored that exploits the U-shaped intraday volatility pattern as an identification condition, offering a path toward resolving the simultaneous equation bias between order flow and Price changes that remains in reduced-form methods.

EO129 Room B104 ADVANCES IN DATA SCIENCE OF LARGE-SCALE TIME SERIES

Chair: Yoshimasa Uematsu

E1448: **Unveiling large dynamic systems via simultaneous impulse response inference**

Presenter: **Mikihito Nishi**, University of Tokyo, Japan

Co-authors: Yoshimasa Uematsu, Takashi Yamagata

An inferential procedure is proposed to investigate the shock propagation mechanism in large vector autoregressions. The goal is to uncover dynamic links between shock variables and outcomes, both of which may be high-dimensional. Given the high-dimensionality, the dynamic relationships are measured via the generalized impulse response functions, which are invariant to the ordering of variables. Formulated as multiple hypotheses testing, the attempt to discover potential dynamic relationships within a large system is designed to control the false discovery rate. The use of asymptotic e-variables based on post-lasso OLS estimators is proposed. Theoretical justifications and numerical experiments are presented.

E1527: **A finite-horizon mixture cure model with application to online flea market data**

Presenter: **Yuji Komiyama**, Tohoku University, Japan

Co-authors: Yasumasa Matsuda, Masakazu Ishihara

A mixture cure model is proposed that latently divides a population based on event occurrence within a finite time horizon. Conventional mixture cure models rely on event occurrence over an infinite horizon, introducing untestable assumptions that often lead to issues with identifiability and interpretability. By shifting the estimand to a specific period of interest, the proposed approach reduces reliance on these infinite-tail assumptions and aligns interpretations more closely with finite-horizon decision-making objectives. Through simulation studies, the statistical properties of the proposed estimator are first evaluated, including estimation bias and variance. It is further shown that relying on conventional infinite-horizon models for finite-horizon decision-making can lead to erroneous judgments. Finally, the model is applied to transaction data from Mercari, a Japanese online flea market platform. The empirical results reveal that the proposed model identifies different significant variables compared to the conventional model, offering interpretations that better reflect seasonal variation in user behavior. These findings suggest that the finite-horizon framing is broadly applicable to settings where decision-makers operate within well-defined time windows, such as marketing campaigns or contract renewals, providing practitioners with a statistically rigorous yet interpretable tool for real-world decision-making.

E1673: **Data-driven forecasting of wind vectors over the Tohoku region by convolutional LSTM and copulas using NWP outputs**

Presenter: **Keigo Sato**, Tohoku University, Japan

Co-authors: Yusaku Katsuragi, Yasumasa Matsuda

A data-driven probabilistic forecasting model for station wind vectors over the Tohoku region is developed, trained on AMeDAS observations. The model post-processes wind fields predicted by numerical weather prediction (NWP) and jointly outputs prediction intervals of wind speed relevant to wind-power operation. As inputs, lead-time sequences of wind fields predicted by NWP on the Tohoku grid are used. A convolutional LSTM is trained on AMeDAS station observations and maps these wind fields to a bivariate Gaussian predictive distribution for the wind vector (u,v) at each lead time and station. The (u,v) formulation yields a predictive distribution of wind speed $s = \sqrt{u^2 + v^2}$ and prediction intervals computed from the predictive distribution via sampling, without any iterations to obtain multi-step-ahead forecasts, while retaining the physical dynamics already embedded in NWP. Furthermore, the model is extended to multivariate probabilistic forecasting using copulas to account for spatial dependencies, enabling the modeling of joint probabilities and spatial uncertainty structures across the region. The accuracy of predicted wind-speed intervals is evaluated using empirical coverage and average interval width, together with reliability diagnostics. Results are compared against quantile regression baselines that directly predict wind-speed quantiles from station-mapped NWP predictors.

E1680: **Testing conditional independence via the spectral generalized covariance measure: Beyond Euclidean data**

Presenter: **Ryunosuke Miyazaki**, Hitotsubashi University, Japan

Co-authors: Yoshimasa Uematsu

A conditional independence (CI) test is proposed based on a new measure, the spectral generalized covariance measure (SGCM). The SGCM is

constructed by expressing the squared norm of the conditional cross-covariance operator in spectral coordinates and approximating it in finite dimensions using data-dependent bases obtained from empirical covariance operators. This avoids direct estimation of conditional mean embeddings and reduces nuisance estimation to a finite collection of scalar-valued regressions. On the theoretical side, under a doubly robust product-bias condition, uniform bootstrap validity and uniform asymptotic size control are established, and nontrivial uniform power and uniform consistency over classes of projected separated alternatives are derived. The analysis also clarifies the role of spectral truncation: stronger truncation relaxes nuisance-estimation requirements, whereas weaker truncation retains more of the projected signal. To support applications beyond Euclidean data, characteristic-kernel constructions on general Polish spaces are developed via a pullback principle and non-constant completely monotone transforms of continuous negative-type semimetrics, with closure under finite tensor products. These constructions cover examples such as trajectories in metric spaces. Simulations show near-nominal size in the main settings and competitive power across a range of challenging scenarios.

E1820: Inference for sparse duration models

Presenter: **Jonas Striaukas**, Copenhagen Business School, Denmark

Co-authors: Jan Roth

Inference in sparse duration models based on a log-linearized autoregressive conditional duration (ACD) framework with high-dimensional covariates is studied through a three-step approach: first, a log-linear regression with serially correlated errors is estimated using l_1 -penalization; second, the ACD model is fitted using the first-stage estimates; and third, a decorrelated score combining both stages is constructed. The resulting statistic enables joint hypothesis testing for the sparse high-dimensional coefficient vector while accounting for serial dependence and model selection. Under suitable sparsity conditions, the test is asymptotically valid, and simulation results demonstrate good size control and power in finite samples.

EO189 Room 101 ADVANCES IN HIGH-DIMENSIONAL DATA ANALYSIS

Chair: Rahul Ghosal

E1209: Frequency selection in Bayesian spectral modeling of time series data with applications to wearable device measurements

Presenter: **Beniamino Hadj-Amar**, University of South Carolina, United States

Co-authors: Marina Vannucci, Vaishnav Krishnan

A novel Bayesian spike-and-slab framework for spectral analysis of time series data is proposed. The proposed method combines frequency selection and dimensionality reduction with a refined grid of candidate frequencies, enabling high-resolution recovery of oscillatory components while promoting sparsity through a structured spike-and-slab prior. A stochastic search algorithm efficiently explores the posterior space, yielding posterior inclusion probabilities that quantify the relevance of each frequency. The framework is extended to multivariate signals via a hierarchical prior on frequency inclusion patterns, allowing the model to capture both shared and component-specific rhythms across multiple time series. Extensive simulation studies demonstrate the method's robustness and superior performance in frequency estimation and spectral power reconstruction compared to existing approaches. Applied to actigraphy data from individuals with partial-onset seizures, the univariate model identifies clinically relevant circadian and ultradian rhythms. In a second application, for the joint analysis of physical activity and skin temperature from a healthy individual, the multivariate model reveals partially overlapping rhythmic components consistent with known physiological coupling. The approach establishes a powerful and interpretable method for spectral analysis, with broad applicability to wearable data, chronobiology, and personalized health monitoring.

E1221: A Bayesian hierarchical model for estimating cell-type-specific multi-omic networks from bulk-omics data

Presenter: **Enakshi Saha**, University of South Carolina, United States

Gene regulatory networks provide crucial insights into how genes and regulatory molecules interact to drive complex traits and diseases. These regulatory interactions exhibit significant variation across cell types. Estimating cell-type-specific networks is essential for understanding biological mechanisms of traits and diseases. While algorithms like inferelator can infer cell-type-specific networks from single-cell omics data, single cell sequencing is expensive and is typically collected from only a few individuals. In contrast, bulk omics data is abundant across large populations, but existing network inference methods for bulk data fail to account for heterogeneity in cell type compositions across samples. A hierarchical Bayesian model is proposed that addresses this limitation by first estimating cell type proportions from bulk RNA-seq data using established deconvolution techniques such as xcell and then using these proportions within a hierarchical framework to decompose co-expression networks estimated from bulk gene expression into cell-type-specific co-expression components. The method is built on a non-negative linear regression with sparsity-inducing horseshoe prior. The method is validated using simulated data and yeast bulk and single cell TF knockout data and demonstrates that the approach successfully recovers cell-type-specific regulatory relationships that are masked in traditional bulk network inference methods.

E1233: Variable selection in functional linear Cox model

Presenter: **Rahul Ghosal**, University of South Carolina, United States

Modern biomedical studies frequently collect complex, high-dimensional physiological signals using wearables and sensors along with time-to-event outcomes, making efficient variable selection methods crucial for interpretation and improving the accuracy of survival models. A novel variable selection method is proposed for a functional linear Cox model with multiple functional and scalar covariates measured at baseline. A spline-based semiparametric estimation approach is utilized for the functional coefficients and a group minimax concave type penalty (MCP), which effectively integrates smoothness and sparsity into the estimation of functional coefficients. An efficient group descent algorithm is used for optimization, and an automated procedure is provided to select optimal values of the smoothing and sparsity parameters. Through simulation studies, the method's ability to perform accurate variable selection and estimation is demonstrated. The method is applied to 2003-06 cohort of the National Health and Nutrition Examination Survey (NHANES) data, identifying the key temporally varying distributional patterns of physical activity and demographic predictors related to all-cause mortality. The analysis sheds light on the intricate association between daily distributional patterns of physical activity and all-cause mortality among older US adults.

E1232: Covariate-adjusted conditional quantile regression for functional data

Presenter: **Salil Koner**, University of California, Riverside, United States

Quantile regression is a powerful tool in functional data analysis, as it enables the characterization of heterogeneity, robustness to outliers and departures from normality, and the investigation of tail behavior. While pointwise conditional quantile functions are straightforward to model, they do not fully capture the distributional structure of functional responses. Spatial quantiles provide a principled framework for describing the depth and centrality of a function within a functional data cloud. A spline-based methodology is developed for estimating conditional spatial quantiles when both the response and covariates are functional. The proposed framework accommodates data observed on either dense or sparse domains. Estimation is carried out by solving a not-strictly convex optimization problem using a fast and efficient minorization-maximization (MM) algorithm, which yields a globally unique minimizer. Extensive simulation studies demonstrate the strong finite-sample performance of the approach, including under sparse sampling designs. The method's practical utility is further illustrated by applying it to real-world functional data.

E1269: AI-assisted STR genotyping from high-resolution melt curves for rapid forensic screening

Presenter: **Indranil Sahoo**, Virginia Commonwealth University, United States

Co-authors: Parastoo Bybordi, Rahul Ghosal, Kiersten Fultz, Tracey Dawson Green, Chenlu Ke

Rapid and reliable DNA profiling is essential for modern forensic investigations, yet conventional short tandem repeat (STR) genotyping requires time-intensive amplification and electrophoresis. High-resolution melt (HRM) analysis provides a faster alternative by generating fluorescence-based melt curves whose shapes reflect underlying genotypes; however, manual interpretation is subjective and intensity-based computational approaches are sensitive to DNA concentration and lack forensic relevance. A shape-based, AI-driven framework is proposed for early STR genotype prediction from single-source HRM curves. After interpolation to a common temperature grid, curves were modeled as functional data and functional principal component analysis (FPCA) scores were used as inputs to a random forest classifier within a nested cross-validation framework with class weighting and SMOTE to address imbalance. Grouping genotypes into three groups with similar derivative profiles further improved separability. Across five outer validation folds, the model achieved a mean accuracy of 0.623 (SD 0.071) and balanced accuracy of 0.632 (SD 0.054), with high specificity (0.760). These results show that functional representations capture discriminative melt-curve shape information and provide a rapid, concentration-invariant screening tool for prioritizing samples and generating early exclusionary information in forensic workflows.

E1844: **Bayesian variable selection for censored spatial responses with application to PFAS concentrations in California**

Presenter: **Suman Majumder**, Indian Statistical Institute, India

Per- and polyfluoroalkyl substances (PFAS) are persistent contaminants of public health concern due to their resistance to degradation, widespread occurrence, and potential health effects. Statistical analysis of PFAS concentrations in groundwater is complicated by censoring from detection limits, strong spatial dependence, and high-dimensional covariates. While PFAS levels are believed to be influenced by diverse sociodemographic, industrial, and environmental factors, their relative contributions remain unclear, motivating statistical approaches that can isolate key predictors from a large candidate set. A Bayesian hierarchical framework is developed that embeds censoring within a spatial process model via approximate Gaussian processes and employs a global-local shrinkage prior for effective high-dimensional variable selection. To refine inference, three post-selection strategies are compared based on their predictive accuracy, robustness to censoring, and stability of variable inclusion. Application of this framework to PFOS concentrations in California groundwater identifies a scientifically meaningful set of drivers including demographic factors, industrial sources, proximity to airports, traffic density, and environmental features such as herbaceous cover, elevation, and ozone concentration. The proposed framework provides interpretable inference while simultaneously offering actionable insights into the factors driving PFAS contamination in groundwater.

EO240 Room 102 RECENT DEVELOPMENTS IN THEORY OF COMPLEX DATA

Chair: Cy Sin

E1294: **Multinomial logistic regression utilizing external machine learning predictions**

Presenter: **Chi-Shian Dai**, National Cheng Kung University, Taiwan

In many modern applications, a carefully designed primary study provides high-quality individual-level data for interpretable modeling, while additional external information is available only through black-box, nonparametric machine-learning predictions. Although summary-level external information has been studied in the data integration literature, there is limited methodology for leveraging external nonparametric predictions to improve statistical inference in the primary study. A general empirical-likelihood framework is proposed that incorporates external predictions through moment constraints. A key advantage of nonparametric prediction is that it induces a rich class of valid moment restrictions that remain robust to covariate shift under a mild overlap condition without requiring explicit density-ratio modeling. The focus is on multinomial logistic regression as the primary model and addresses common data-quality issues in external sources, including coarsened outcomes, partially observed covariates, covariate shift, and heterogeneity in generating mechanisms. Large-sample properties of the resulting fused estimator are established, including consistency and asymptotic normality under regularity conditions. Moreover, sufficient conditions are provided under which incorporating external predictions delivers a strict efficiency gain relative to the primary-only estimator. Simulation studies and an application are provided.

E1302: **Variable selection for high-dimensional regression models with higher-order interactions**

Presenter: **Hsueh-Han Huang**, Academia Sinica, Taiwan

The network orthogonal greedy algorithm (network OGA) is proposed, an efficient method designed to capture higher-order (beyond second-order) interactions. By integrating the concepts of ranking and stepwise forward regression, network OGA leverages the advantages of both approaches. The algorithm is applicable to high-dimensional interaction models of arbitrary unknown orders. The sure screening property for network OGA is established and it is demonstrated that, when coupled with a high-dimensional information criterion (HDIC), the method achieves variable selection consistency. Simulation studies further validate its superior performance.

E1719: **Cross-validated estimation window averaging for multi-step forecasting**

Presenter: **Jen-Che Liao**, National Chengchi University, Taiwan

A frequentist model averaging approach is developed to address estimation window uncertainty in multi-step forecast regressions. The proposed data-driven methods for window selection and averaging are based on a leave-h-out cross-validation criterion, where h is the forecast horizon. The procedures are robust to heteroskedasticity and serial correlation of unknown forms, both of which are inherent in multi-step forecast errors. Theoretical properties of the methods are established and their performance is demonstrated through simulations.

E1501: **Double/debiased machine learning estimation of the conditional tail average treatment effect in mediation analysis**

Presenter: **Yu-Min Yen**, National Chengchi University, Taiwan

A double/debiased machine learning (DML) estimator for the conditional tail average treatment effect (CTATE) in mediation analysis is studied. The CTATE captures heterogeneity and provides aggregated local information on treatment effects across different quantile levels. It is closely related to the concepts of second-order stochastic dominance and the Lorenz curve, making it a valuable tool for policy evaluation. At a given quantile level, the CTATE is decomposed into indirect and direct effects, which facilitates analysis of how the treatment and mediator jointly contribute to the causal impact. Identification results are established and a DML estimator based on the joint influence function of the quantile and the conditional tail expectation of potential outcomes is developed. Monte Carlo simulations are conducted to evaluate the performance of the proposed approach. Its usefulness is further illustrated through an empirical application that examines the effects of a job training program using data from the National Job Corps Study.

E1622: **Non-stationary challenges in partial least squares: Applications to forecasting**

Presenter: **Shou-Yung Yin**, National Taipei University, Taiwan

The behavior of Partial Least Squares (PLS) is examined when both the target variable and predictors are non-stationary. It is shown theoretically that PLS may mechanically remove low-frequency stochastic trends in sample, so that the residual appears stationary even when regressors are structurally irrelevant. This filtering effect does not carry over out of sample: prediction errors continue to display non-stationary behavior. To clarify this contrast, an operator-based characterization of PLS is developed and connected to the Karhunen-Loeve expansion of non-stationary processes. Building on this distinction, a backward-greedy variable selection procedure is proposed, guided by a Partial-sum energy criterion, which measures the squared cumulative sum of out-of-sample residuals. Unlike standard unit-root test statistics, which exhibit a cliff-shaped landscape that defeats conventional search, this criterion provides a smooth signal for subset quality. Monte Carlo experiments confirm non-trivial

power to recover the true cointegrating variables, and an empirical application to FRED-MD data shows that out-of-sample stationarity criteria yield predictor sets that differ materially from those selected by in-sample fit.

E1745: **Robust testing of the matrix semidefiniteness hypothesis: A weighted eigenvalue sum approach**

Presenter: **Le-Yu Chen**, Institute of Economics, Academia Sinica, Taiwan

Co-authors: Jerzy Szroeter

Testing whether a given parameter matrix is positive or negative semidefinite is a challenging problem that occurs widely in econometric models of optimizing agents, particularly when matrix eigenvalues have unknown multiplicity. A robust test method is developed based on a sum of estimated eigenvalues weighted by data-dependent factors that compensate for ignorance of true multiplicities. New non-asymptotic bounds on errors incurred when approximating eigenvalue sum perturbations by linear functions of corresponding matrix perturbations are derived to support the theory. The test exhibits correct asymptotic size in the uniform sense over the null-restricted data distribution space, allowing for arbitrary eigenvalue multiplicity, and establishes test consistency with local power properties. The test is operationally simple with a fixed critical value, specifically standard normal. Monte Carlo simulations illustrate the usefulness and robustness of the proposed approach in finite samples.

EO236 Room 103 RECENT ADVANCES IN STATISTICAL LEARNING

Chair: Wenbo Jing

E1340: **Learning implicit multivariate interactions via Gibbs relaxation**

Presenter: **Tongseok Lim**, Purdue University, United States

Co-authors: Wooseok Ha, Kyeongsik Nam

Estimating joint densities in high dimensions is challenging, especially when dependence lies on lower-dimensional structures that classical non-parametric methods miss and parametric models oversimplify. A structured nonparametric framework is proposed based on an exponential-family form in which the log-density decomposes into marginal potentials and an interaction term capturing dependence beyond marginals. Interpreting maximum likelihood estimation through a statistical-mechanics lens as learning an energy landscape unifies three perspectives. First, it connects to entropic optimal transport (EOT): unlike EOT, which fixes interactions via a cost and uses empirical marginals, the approach learns both components from data. Second, jointly learning marginals and interactions separates marginal effects from intrinsic dependence, making the interaction directly interpretable. Third, imposing a one-sided constraint on the log-density yields a contact set that acts as a latent monotone structure, providing a soft (positive-temperature) relaxation of hard monotonic constraints. Theoretically, convergence rates for the MLE are derived using a pseudo self-concordance bound on the third derivative of the free energy, leading to error bounds governed by the complexity of the log-density class. Empirical results demonstrate the effectiveness of the approach.

E1419: **Comparator-adaptive Φ -regret: Improved bounds, simpler algorithms, and applications to games**

Presenter: **Mengxiao Zhang**, University of Iowa, United States

In the classic expert problem, Φ -regret measures the gap between the learner's total loss and that achieved by applying the best action transformation. A recent work introduces an adaptive algorithm whose regret against a comparator $\phi \in \Phi$ depends on a certain sparsity-based complexity measure of Φ , recovering and interpolating optimal bounds for standard regret notions such as external, internal, and swap regret. A general idea is proposed to achieve an even better comparator-adaptive Φ -regret via much simpler algorithms. Specifically, a prior distribution over all possible binary transformations is discovered and it is shown that it suffices to achieve prior-dependent regret against these transformations. Then, two efficient algorithms are proposed to achieve so, where the first one learns over multiple copies of a prior-aware variant of the kernelized MWU algorithm, and the second one learns over multiple copies of a prior-aware variant of the BM-reduction. To showcase the power of these methods and the advantages, it is also shown that the second approach can be extended to the game setting to achieve accelerated and adaptive convergence rate to Φ -equilibria for a class of general-sum games. When specified to the special case of correlated equilibria, the bound improves over existing ones.

E1944: **Statistical inference in multi-timescale stochastic approximation**

Presenter: **Zehua Lai**, Chinese Academy of Sciences, China

The multi-timescale stochastic approximation framework is introduced, with applications in composite optimization, minimax optimization, and gradient descent with momentum. Convergence and central limit theorems are discussed for both the last-iterate estimator and the averaged estimator. An estimator for the covariance matrix is constructed, enabling efficient construction of confidence intervals.

E1258: **Data-driven knowledge transfer in batch Q^* learning**

Presenter: **Wenbo Jing**, City University of Hong Kong, Hong Kong

Co-authors: Xi Chen, Elynn Chen

In data-driven decision-making across marketing, healthcare, and education, leveraging large datasets from existing ventures is crucial for navigating high-dimensional feature spaces and addressing data scarcity in new ventures. Knowledge transfer in dynamic decision-making is investigated by focusing on batch stationary environments and formally defining task discrepancies through the framework of Markov decision processes (MDPs). The transfer fitted q -iteration algorithm with general function approximation is proposed, which enables direct estimation of the optimal action-state function Q^* using both target and source data. Under sieve approximation, the relationship between statistical performance and the MDP task discrepancy is established, highlighting the influence of source and target sample sizes and task discrepancy on the effectiveness of knowledge transfer. Theoretical and empirical results demonstrate that the final learning error of the function is significantly reduced compared to the single-task learning rate.

E2023: **Large-scale linear hypothesis testing for high-dimensional renewable M-estimator**

Presenter: **Myeonghun Yu**, Ewha Womans University, Korea, South

Co-authors: Tate Jacobson, Daeyoung Ham

Renewable estimation targets streaming-data settings where historical raw observations are unavailable or cannot be repeatedly processed. Existing renewable methods are computationally efficient, but batchwise approximation errors may accumulate and invalidate high-dimensional inference. A renewable inference framework is developed that is first-order insensitive to errors from previous estimators. The method uses only the current data batch and stored summary statistics, and introduces a surrogate loss that removes the leading accumulated approximation error. For linear hypothesis testing in high-dimensional sparse models, the surrogate loss is combined with partial penalization and local linear approximation algorithms for the full and reduced models. The computed LLA solutions coincide with the corresponding renewable oracle estimators with overwhelming probability. Using this strong oracle property and the asymptotic distribution of the renewable oracle estimators, the resulting partially penalized test statistic is asymptotically chi-square. The proposed framework yields implementable online tests without repeated access to historical raw data and applies to a broad class of models, with finite-sample performance supported by numerical experiments.

EO101 Room 104 STATISTICAL METHODS FOR FUNCTIONAL AND HIGH-DIMENSIONAL DATA (VIRTUAL)**Chair: Eliana Christou****E1703: Gaussian and bootstrap approximations for functional principal component regression****Presenter:** Hyemin Yeon, Kent State University, United States

Asymptotic inference using functional principal component regression (FPCR) has long been considered difficult, largely because, upon any scalar scaling, the FPCR estimator fails to satisfy a central limit theorem, leading to the prevailing belief that it is unsuitable for direct statistical inference. A new result establishes that, upon suitable operator scaling, valid Gaussian and bootstrap approximations hold for the FPCR estimator. This finding is applied to hypothesis testing for the significance of the slope function in functional regression models, demonstrating strong numerical performance of the resulting tests. The results yield powerful inferential tools for functional regression and pave the way for new lines of inferential methodology in more complex functional regression settings.

E1706: Nonlinear sufficient dimension reduction for conditional quantiles in scalar-on-function single-index models**Presenter:** Shanshan Wang, The University of North Carolina at Charlotte, United States**Co-authors:** Eliana Christou, Eftychia Solea, Jun Song

Functional data analysis is crucial in many applications, yet its high-dimensional nature necessitates effective dimension reduction techniques. While existing approaches primarily focus on linear reductions, a nonlinear sufficient dimension reduction framework for conditional quantiles of single-index models when the predictors are random functions is introduced. The approach constructs two nested functional spaces: a Hilbert space representing the functional data and a reproducing kernel Hilbert space that captures nonlinearity. The kernel in the latter is determined by the inner product of the former, leading to a natural hierarchical structure. Dimension reduction is characterized at the general level of σ -fields and proceeds to that of classes of functions, leading to the notion of the central quantile class. The proposed estimator, called the τ th functional generalized central quantile subspace (τ -fGQS), is introduced and its convergence rate is established. Performance of the estimator is demonstrated through simulations and real-world applications to health studies, examining various health indicators, such as ADHD, Parkinson's disease, and BMI.

E1737: MediEncoder: Nonlinear factor learning via autoencoders for high-dimensional causal mediation analysis**Presenter:** Debarghya Mukherjee, Boston University, United States**Co-authors:** Shi Bo, AmirEmad Ghassami

Causal mediation analysis aims to understand the mechanisms through which a treatment affects an outcome by decomposing the total causal effect into indirect and direct pathways. Modern applications increasingly involve ultra-high-dimensional covariates and mediators, where observed variables are often driven by a smaller number of latent biological processes. Existing approaches typically rely on assumptions such as sparsity or linear factor models for dimension reduction, which can be restrictive when the relationship between latent factors and observed variables is nonlinear. To address this limitation, MediEncoder is proposed as a representation learning framework for nonlinear ultra-high-dimensional mediation analysis. The method jointly learns latent representations of covariates and mediators using an encoder-decoder architecture that captures their structural dependence, enabling flexible nonlinear factor modeling while preserving causal pathways for estimating mediation effects. Based on the learned latent representation, an estimator of the direct and indirect causal effects is developed using an efficient influence function that is multiply robust and semiparametrically efficient, without strong structural assumptions. Numerical analyses show that MediEncoder improves accuracy compared with existing dimensionality reduction approaches. Finally, its utility is demonstrated on a high-dimensional biomedical dataset.

E1768: Covariance estimation for derivatives of functional data using an additive penalty in P-splines**Presenter:** Steven Golovkine, Universita Laval, Canada**Co-authors:** Yueyun Zhu, Andrew Simpkin, Norma Bargary

P-splines provide a flexible and computationally efficient smoothing framework and are commonly used for derivative estimation in functional data. Including an additive penalty term in P-splines has been shown to improve estimates of derivatives. A method that incorporates the fast covariance estimation (FACE) algorithm with an additive penalty in P-splines is proposed. The proposed method is used to estimate derivatives of covariance for functional data, which play an important role in derivative-based functional principal component analysis (FPCA). An algorithm for estimating the eigenfunctions and their corresponding scores in derivative-based FPCA is provided. The algorithm is evaluated against existing functions in simulation. The algorithm is extended to multivariate cases, referred to as derivative-based multivariate functional principal component analysis (DMFPCA). DMFPCA is applied to joint angles in human movement data, where the derivative-based scores demonstrate strong performance in distinguishing locomotion tasks.

E1846: Generalized heterogeneous functional model with applications to large-scale mobile health data**Presenter:** Fei Xue, Purdue University, United States

Physical activity is crucial for human health. With the increasing availability of large-scale mobile health data, strong associations have been found between physical activity and various diseases. However, accurately capturing this complex relationship is challenging, possibly because it varies across different subgroups of subjects, especially in large-scale datasets. A generalized heterogeneous functional method is proposed that simultaneously estimates functional effects and identifies subgroups within the generalized functional regression framework. The proposed method captures subgroup-specific functional relationships between physical activity and diseases, providing a more nuanced understanding of these associations. Additionally, a pre-clustering method is developed that enhances computational efficiency for large-scale data through finer partitioning of subjects compared to true subgroups. A testing procedure is introduced to assess whether different subgroups exhibit distinct functional effects. In a real data application examining the impact of physical activity on the risk of dementia using the UK Biobank dataset with over 96,433 participants, the proposed method outperforms existing methods in future-day prediction accuracy, identifying three distinct subgroups with detailed scientific interpretations for each. The theoretical consistency of the methods is also demonstrated.

EO181 Room 105 ADVANCED STATISTICAL MODELING OF COMPLEX SPATIOTEMPORAL SYSTEMS**Chair: Yan Sun****E1357: Stochastic spatial stream networks for scalable inferences of riverscape processes****Presenter:** Xinyi Lu, Utah State University, United States

Spatial stream networks (SSN) models characterize correlated ecological processes in dendritic ecosystems. Conventional ssn models rely on pre-processed stream networks and point-to-point hydrologic distances. However, this data processing may be labor-intensive and time-consuming over large spatial domains. Therefore, a stochastic inference approach for the functional connectivity of stream networks is proposed. This physically-guided model utilizes the knowledge that water flows from high elevation to low elevation, and flow rate typically increases when two tributaries merge. The hierarchical branching architecture of dendritic networks is also leveraged to alleviate computing and reduce uncertainty. Spatial autoregressive models composed of inferred SSNs propagate stochasticity between network connectivity and dynamic ecological processes in a Bayesian framework. Simulated examples show that this mechanistic model facilitated learning about the functional network and enhanced predictive performance. The approach is also demonstrated in a large-scale case study using native brook trout (*Salvelinus fontinalis*) count data. A population model based on the stochastic ssn outperformed that with a conventional SSN in predicting abundance and expedited the analysis by circumventing data processing.

E1358: A random quantile approach for prior covariance estimation in Bayesian maximum entropy*Presenter:* **Kinspride Duah**, Utah State University, United States*Co-authors:* Yan Sun, Maha Moussa

Bayesian maximum entropy (BME) provides a flexible framework for spatial prediction by integrating precise measurements with uncertain information. However, conventional BME (CBME) often assumes homogeneous uncertainty, limiting its performance when data exhibit strong heterogeneity such as skewness and heavy tails. A quantile-based BME (QBME) approach is introduced that leverages quantile information to better represent asymmetric uncertainty in soft data. By incorporating multiple quantiles into the prior specification, QBME improves the characterization of uncertainty and enhances prediction accuracy, particularly in data-sparse regions. A comprehensive simulation study evaluates QBME against CBME across a range of controlled scenarios. Results show that QBME consistently outperforms CBME under diverse data conditions and error metrics. The proposed method is further demonstrated using snow water equivalent (SWE) data from Montana, United States, which include both precise and interval-valued observations. QBME more effectively captures large-scale spatial patterns while preserving local variability. These findings highlight QBME as a robust extension of BME for spatial prediction with asymmetric and uncertain data.

E1524: Extending Bayesian maximum entropy using random sets theory for interval-valued spatial data*Presenter:* **Thamali Amanda Gamlath Dissanayakalage**, Utah State University, United States

Bayesian Maximum Entropy (BME) is a powerful framework for spatial prediction when both exact measurements (hard data) and uncertain observations (soft data) are available. One of the main challenges in BME is defining the prior covariance structure when the soft data are provided as intervals. In many applications, these intervals are simplified into single-point values (e.g., midpoint) but doing so could overlook important uncertainty information. A random-set based method is introduced for building prior covariances in BME. Each interval observation is viewed as the realization of a one-dimensional random set, and its dynamics are described by the center and radius. As a result, covariance expressions are obtained that systematically capture interval uncertainty in a relatively simple format. An extended simulation study demonstrates that this approach extends the standard BME framework and provides a more flexible and informative method for spatial prediction under uncertainty.

E1728: Jointly modeling zeros, regimes, and lags in epidemic count data via penalized spatio-temporal mixture models*Presenter:* **Zudi Lu**, City University of Hong Kong, China*Co-authors:* Weiao Gan, Ngai Hang Chan

Daily epidemic count data frequently violate standard spatio-temporal Poisson assumptions, exhibiting excess zeros, multiple latent transmission regimes under similar histories, and dependence on high-dimensional temporal and spatial lags that induce collinearity and model-selection uncertainty. Existing methods typically address these issues in separate, stagewise procedures, which are computationally costly and ill-suited to mixture settings where predictor relevance may vary by regime. A unified observation-driven framework is proposed that jointly models zero inflation, latent regimes, and predictor selection within a single spatio-temporal conditional model. Excess zeros are handled via a structural-zero mechanism, latent regimes through a finite mixture with regime-specific lag structures, and complexity through dual penalization that simultaneously selects active regimes and regime-specific predictors. By conditioning on observed history, the model targets one-step-ahead prediction without introducing latent state dynamics. Estimation is achieved using a stabilized penalized EM algorithm with tractable weighted generalized linear model updates. The resulting approach is interpretable, computationally efficient, and well-suited to moderately high-dimensional epidemic surveillance data, as demonstrated in simulations and real-world data applications.

E1343: AI-guided single-index varying coefficient logistic modeling: A rain-snow partitioning case study*Presenter:* **Yan Sun**, Utah State University, United States*Co-authors:* Maha Moussa, Wei Zhang, Shandian Zhe

Varying-coefficient (VC) models make effect heterogeneity explicit, but traditional statistical estimation can be computationally demanding and unstable in high-dimensional settings. In parallel, modern artificial intelligence (AI), especially deep learning, delivers highly accurate predictions but typically lacks built-in guarantees of identifiability, shape constraints, or easily interpretable effect summaries. An AI-guided single-index VC logistic framework is developed in which a flexible "teacher" network supplies structured, data-dependent regularization for an interpretable "student" model. The student is a semiparametric single-index VC logistic regression whose parameters are modeled as smooth, shape-constrained functions of a learned contextual index, and are estimated via penalized logistic likelihood with AI-informed penalties. As a case study, the framework is applied to precipitation-phase partitioning (rain-snow classification), using approximately 16.8 million reports from 11,626 stations across the Northern Hemisphere. The resulting estimator attains AI level high test accuracy, while yielding statistically interpretable outputs. More broadly, the framework illustrates how AI-derived structure can stabilize and scale the single-index VC logistic model in complex applications, providing a template for embedding deep-learning targets into penalized semiparametric estimation.

EO239 Room 106 RECENT DEVELOPMENTS IN SPATIO-TEMPORAL MODELING AND MULTIVARIATE INFERENCE Chair: Bohai Zhang
E1882: Spatio-temporal analysis of wave power around the Korean Peninsula*Presenter:* **Jaeyun Kim**, Hanyang University, Korea, South*Co-authors:* Taeyeon Kim, Jaehong Jeong

Fixed rank kriging (FRK) is employed to predict significant wave height (SWH) and wave power in space and time around the Korean Peninsula. Using reanalysis and observational data, non-stationary and non-Gaussian characteristics are accounted for to produce high-resolution SWH fields for wave power assessment. Seasonal and daily modeling scenarios reveal that the eastern coast's offshore waters have the highest wave power potential in winter. These results demonstrate the effectiveness of FRK for large-scale prediction and support the feasibility of wave energy development in South Korea.

E1957: Dynamic network modeling for the spatiotemporal progression of high-dimensional data*Presenter:* **Yei Eun Shin**, Seoul National University, Korea, South

The modeling of high-dimensional spatiotemporal data is essential for understanding complex dynamic systems, such as tracking the large-scale spreading patterns of amyotrophic lateral sclerosis (ALS) across a human body map. Various methodological developments designed to model the progression of these network dependencies over time without relying on prespecified spatial proximity are reviewed. An autologistic network model for binary data with absorbing states is first highlighted, employing sparse regularization to effectively manage a large number of pairwise spatial associations. A spatial hidden Markov model is then introduced, which accounts for diagnostic misclassifications through high-dimensional inference of latent disease states. Expanding the discussion to broader modeling strategies, ongoing research that handles longitudinal observations at their original continuous level is also presented. In this framework, complex high-dimensional network dependencies are flexibly captured using a linear mixed-effects model by specifying the covariance structure based on inter-node distances measured by the resistance metric, a type of non-Euclidean metric. Alongside these diverse approaches, additional ongoing extensions and related follow-up studies are discussed to explore the broader scalability and potential of high-dimensional network modeling.

E1967: An RP-MMD framework for multivariate goodness-of-fit testing in location-scale families*Presenter:* **Xiaoling Peng**, Beijing Normal-Hong Kong Baptist University, China

With the increasing prevalence of high-dimensional data, developing goodness-of-fit tests that are both statistically powerful and computationally efficient remains a fundamental challenge. A scalable multivariate goodness-of-fit testing framework, termed the RP-MMD test, is proposed for a broad class of location-scale distributions. Extensive simulation studies demonstrate that the proposed method achieves competitive or superior power compared with existing tests, including the Shapiro–Wilk test, across a wide range of scenarios. Applications to real data further illustrate the practical effectiveness of the approach.

E1956: Model-free feature screening via revised Chatterjee's rank correlation for ultra-high dimensional censored data*Presenter:* **Shuya Chen**, Beijing Normal-Hong Kong Baptist University, China*Co-authors:* Min Zhou, Heng Peng

In large-scale biomedical research, ultra-high dimensional data with right-censored survival times are commonly gathered. Feature screening has emerged as a crucial statistical technique for handling such data. A straightforward and robust feature screening approach leveraging the modified Chatterjee's rank correlation is introduced, suitable for a broad range of survival models. With reasonably mild regularity assumptions, sure screening properties and ranking consistency are established. The computation involved in the proposed method is direct and simple. Simulation studies and real gene expression data analysis demonstrate the superior efficacy of the proposed approach.

E1888: Inference of spatially varying coefficient model with covariate measurement errors*Presenter:* **Bohai Zhang**, Beijing Normal-Hong Kong Baptist University, China

Spatially varying coefficient (SVC) models offer an effective way to characterize the heterogeneous effects of covariates on the response variable. However, existing SVC models assume accurately measured covariates, ignoring real-world measurement errors that may arise from many aspects. A statistically and computationally efficient framework for the inference of SVC models is developed based on profile model likelihood, with explicit consideration of covariate measurement error. Focusing on spatial clustered coefficient models, an SVC model is formulated that allows functional covariate measurement errors. Bias-corrected estimators are then derived by leveraging techniques of high-dimensional measurement error models and unbiased estimating functions, ensuring consistent parameter estimation. Additionally, a joint weighted estimation procedure is developed that accounts for unequal variances of the residuals, markedly enhancing the estimation efficiency over the unweighted alternative. Finally, the effectiveness of the proposed approaches is demonstrated through comprehensive simulation studies and an analysis of the Arctic sea ice dataset.

EO286 Room 107 MODERN ADVANCES IN BAYESIAN COMPUTATION AND INFERENCE (VIRTUAL)**Chair: Ray Bai****E1613: Generative quantile Bayes***Presenter:* **Jungeum Kim**, NCSU, United States*Co-authors:* Percy Zhai, Veronika Rockova

A multivariate posterior sampling procedure is developed through deep generative quantile learning. Simulation proceeds implicitly through a push-forward mapping that can transform i.i.d. random vector samples from the posterior. Monge-Kantorovich depth in multivariate quantiles is utilized to directly sample from Bayesian credible sets, a unique feature not offered by typical posterior sampling methods. To enhance the training of the quantile mapping, a neural network is designed that automatically performs summary statistic extraction. This additional neural network structure provides performance benefits, including support shrinkage (contraction of the posterior approximation) as the observation sample size increases. The usefulness of the approach is demonstrated on several examples where the absence of likelihood renders classical MCMC infeasible. Frequentist theoretical justifications are provided for the quantile learning framework, including consistency of the estimated vector quantile, of the recovered posterior distribution, and of the corresponding Bayesian credible sets.

E1789: Parameter exploded variational Bayes for sparse high-dimensional linear regression*Presenter:* **Alexander McLain**, University of South Carolina, United States*Co-authors:* Ray Bai, Rahul Ghosal, Enakshi Saha

Variational Bayesian (VB) inference approximates posterior densities when calculations with standard Bayesian methods are computationally intensive. For instance, in high-dimensional variable selection, VB offers a more scalable approach, obtaining results in significantly less time (hours or minutes) compared to traditional Bayesian methods which might require weeks for similar accuracy. VB aims to optimize the Kullback-Leibler divergence, commonly done using the coordinate ascent variational inference (CAVI) algorithm with the mean-field assumption. This method postulates that the posterior distributions are independent. The CAVI algorithm is implemented with extreme parameter expansion, referred to as parameter explosion, for sparse high-dimensional linear regression. The parameter exploded VB (SPXLVB) algorithm also assumes independent posterior distributions; however, the expanded parameters yield posterior means and variances that match methods that do not assume independence. Hyperparameter selection is addressed, and differences with other VB approaches are presented. The empirical properties of SPXLVB are compared with similar approaches via numerous simulation studies and four real data applications.

E1796: A scalable Bayesian framework for high-dimensional mixed-effects models via compression*Presenter:* **Sanvesh Srivastava**, The University of Iowa, United States*Co-authors:* Kshitij Khare, Sreya Sarkar

Mixed-Effects models are central to the analysis of clustered and repeated-measures data, but high-dimensional methodology in this area has concentrated mainly on penalized estimators with vector-valued covariates. Bayesian methods remain limited, with no computationally tractable sampling-based framework for Mixed-Effects modeling that accommodates Tensor-valued fixed and random-effect covariates. The Compressed Mixed-Effects Tensor (CoMET) model is a Bayesian framework for scalar response repeated-measures data with Tensor-valued covariates. CoMET compresses the random-effects covariance through structured, mode-wise random projection, producing a low-dimensional parameterization that enables simple Gaussian priors and efficient imputation of Compressed random effects. For the mean structure, it integrates low-rank Tensor decomposition with margin-structured Horseshoe priors to perform fixed-effect selection. This formulation leads to a collapsed Gibbs sampler whose computational cost grows approximately linearly with Tensor dimension, making CoMET suitable for high-dimensional settings. Theoretically, conditions are derived under which CoMET's posterior predictive risk vanishes asymptotically. Across simulations and benchmark applications, CoMET shows competitive performance relative to penalized methods.

E1807: Simulation-based inference via structured score matching*Presenter:* **Yun Yang**, University of Maryland, College Park, United States*Co-authors:* Haoyu Jiang, Yuexi Wang

Simulation-based inference (SBI) provides an effective framework for statistical analysis when the likelihood is intractable but model simulations are available. A unified SBI framework is presented that leverages structured score matching for both frequentist and Bayesian inference. On the frequentist side, a likelihood-free approach combines score matching with gradient-based optimization and bootstrap procedures for parameter estimation and uncertainty quantification. On the Bayesian side, score matching is integrated with Langevin dynamics to efficiently explore complex

posterior landscapes in moderate to high-dimensional settings. In both cases, tailored score-matching estimators and architectural regularizations are designed that embed the statistical structure of log-likelihood scores, which improves estimation accuracy and scalability.

E1777: **Batch and match: Black-box variational inference with a score-based divergence**

Presenter: **Diana Cai**, Flatiron Institute, United States

Variational inference (VI) has become a popular approach for posterior inference in latent-variable models, and recent advances in black-box VI (BBVI) methods -enabled by advances in automatic differentiation- have made VI widely accessible in practice. However, standard BBVI often converges slowly for expressive variational families due to high-variance stochastic gradients. To address this limitation, the Batch-and-Match (BaM) algorithm performs variational inference by matching the scores of the variational and target distributions over batches of samples. BaM admits closed-form updates for full-covariance Gaussians and requires significantly fewer gradient evaluations than standard BBVI. Extensions to high-dimensional settings and richer variational families are also discussed.

EO161 Room 203 STATISTICAL LEARNING: CALIBRATION, INTERPRETABILITY, AND SCALABLE INFERENCE

Chair: Lin Wang

E1555: **Axis-neighbor geometry and length-scale identifiability in Gaussian process models**

Presenter: **Chenlu Shi**, New Jersey Institute of Technology, United States

Co-authors: Lin Wang

In computer experiments, experimental designs for Gaussian process surrogates are often chosen to achieve global space-filling coverage for prediction. However, the same simulator runs must also support estimation of covariance hyperparameters, and designs that are effective for prediction alone may yield weak or unstable length-scale inference, especially for anisotropic GP models. A geometry-driven framework is developed that links experimental design directly to length-scale identifiability in GP models. For general designs, lower bounds are derived for the diagonal Fisher information that depend explicitly on local axis-neighbor geometry through the minimum coordinate-isolating spacing and the multiplicity of closest such pairs. These results show that, in the small-length-scale regime, stable inference is governed by local directional structure rather than by global spread alone. Motivated by this theory, a structured design is proposed that is a strong orthogonal array of strength $2+$, and hence combines space-filling properties with a rich collection of short coordinate-isolating pairs. Explicit Fisher-information bounds and sharp small-length-scale asymptotics are further derived for the proposed design, and a structured level-expansion is introduced that refines global resolution while preserving the local geometry responsible for length-scale identifiability. The theoretical findings are further validated through extensive numerical experiments.

E1576: **Essence codings: A data-driven approach for interpreting black-box models through factorial effects**

Presenter: **Chao-Hui Huang**, Academia Sinica, Taiwan

Co-authors: Shao-Wei Cheng

Factorial designs are a common choice for conducting screening experiments. Traditionally, the codings for factorial effects are predetermined before the experimental phase begins. While this approach is simple and widely used, these predefined codings may fail to capture a substantial portion of the variability in the response, even when certain factors clearly have strong influence. A new data-driven coding system for factorial effects, referred to as essence codings, is proposed. The proposed approach constructs coding functions directly from the observed data, with the goal of capturing the dominant patterns of variation in the response. As a result, essence codings provide a more informative and interpretable representation of factor effects compared to traditional codings. Examples are presented to demonstrate the practical usefulness of the proposed method, including applications to the interpretation of black-box models. The results show that essence codings can effectively reveal meaningful structures in the data and facilitate model interpretation.

E1614: **Influence-guided subsampling**

Presenter: **Lin Wang**, Purdue University, United States

Collecting labels for every point in a large dataset is often impractical due to measurement and budget constraints. A common remedy is to label only a carefully chosen subset of design points. Computationally efficient subsampling methods are presented that select small, informative subsets from large candidate pools. Unlike most existing approaches, which are tailored to low-dimensional settings, these methods explicitly accommodate high-dimensional regimes where the number of relevant predictors can be comparable to, or even exceed, the full sample size. Theoretical guarantees are described and performance is demonstrated through extensive numerical experiments. Across a wide range of scenarios, the methods consistently outperform existing subsampling techniques, resulting in substantial savings in labeling costs.

E1963: **A unified framework for multicalibration boosting with application to survival prediction**

Presenter: **Hanxuan Ye**, University of Pennsylvania, United States

Co-authors: Hongzhe Li

Reliable risk prediction across diverse subpopulations is a central challenge when training and target populations differ. Multicalibration addresses this by strengthening classical calibration to require predictions to be unbiased across a rich class of functions encompassing prediction slices and subpopulations. A post-processing boosting algorithm for censored survival data produces multicalibrated predictors of survival probability and restricted mean survival time. Building on pseudo-observations and a functional delta-method analysis, the method is competitive with, and often outperforms, inverse propensity score weighting in unlabeled target domains. A cardiovascular risk study using two large prospective cohorts demonstrates its value across heterogeneous patient subgroups. The mechanism and theoretical foundations of multicalibration boosting are further explored, unifying variants such as multiaccuracy, BatchGCP, and BatchMVP under weaker conditions. The boosting limit is characterized as the Bregman projection of the population-optimal predictor onto the cumulative span of the audit class, identifying the function space on which multicalibration is achieved. Convergence rates, finite-sample guarantees, and a principled early-stopping rule controlling a calibration risk trade-off are derived. Universal adaptability is further extended, clarifying when multicalibrated predictors generalize across domains.

E2058: **A multi-fidelity tensor emulator for spatiotemporal outputs: Emulation of Arctic sea ice dynamics**

Presenter: **Yawen Guan**, Colorado State University, United States

Numerical models are widely used to simulate the Earth System, but they are computationally expensive and often depend on many uncertain input parameters. Their effective use requires calibration and uncertainty quantification, which typically involve running the Model across many input configurations and therefore incur substantial computational cost. Statistical emulation provides a practical alternative for efficiently exploring Model behavior. The Arctic sea ice component of the Energy Exascale Earth System Model (MPAS-Seaice) generates large spatiotemporal outputs at multiple spatial resolutions, with high-fidelity (HF) simulations being more accurate but computationally more expensive than low-fidelity (LF) simulations. Multi-fidelity (MF) emulation integrates information across resolutions to construct efficient and accurate surrogate models. A MF emulator is developed that combines tensor decomposition for dimensionality reduction, Gaussian process priors for flexible function approximation, and an additive discrepancy Model to capture systematic differences between LF and HF data. The proposed framework enables scalable emulation while maintaining accurate predictions and well-calibrated uncertainty for complex spatiotemporal fields, and consistently achieves lower prediction error and reduced uncertainty than LF-only and HF-only models in both simulation studies and MPAS-Seaice analysis.

EO227 Room 205 RECENT ADVANCES IN MIXTURE MODELS (VIRTUAL)**Chair: Jeffrey Andrews****E1698: A mixture of logistic-normal multinomial distribution and its extension to biclustering****Presenter: Sanjeena Dang**, Carleton University, Canada

Compositional count data is becoming increasingly prevalent across a variety of fields, including bioinformatics driven by advances in next-generation sequencing technologies, and topic modelling driven by large-scale digitalization of document collections. Analyzing such compositional data presents many challenges because they are restricted to a simplex, which complicates standard statistical analysis. While a Dirichlet-multinomial (DM) model can be used for such tasks, it fails to capture complex correlation structures between the variables. In contrast, the logistic-normal multinomial (LNM) distribution provides greater flexibility in capturing correlations but is computationally intensive. A computationally efficient framework for parameter estimation in LNM-based models using variational Gaussian approximations has been developed to overcome these limitations. This approach significantly reduces the computational burden and enables application to large-scale datasets. Building on this framework, a recent development of the LMN mixture model-based biclustering approach that simultaneously clusters observations and features (i.e., variables) in a dataset is discussed. Clustering of features is simultaneously achieved by introducing a block-diagonal covariance structure. The proposed approach is demonstrated on both simulated and real datasets.

E1550: Mixtures of logistic matrix-variate normal multinomial models for efficient clustering longitudinal microbiome data**Presenter: Yuan Fang**, Old Dominion University, United States

Microbiome taxa count data, derived from next-generation sequencing, are inherently high-dimensional, over-dispersed, and reveal only relative abundance, making them compositional and constrained to a simplex. To model such data, the logistic normal multinomial (LNM) approach transforms relative abundances from a simplex to real Euclidean space using the additive log-ratio transformation. Mixtures of LNM models have been developed for clustering microbiome data, adopting an efficient framework for parameter estimation using variational approximations to reduce the computational overhead. In this talk, it will be illustrated that the LNM mixture models provide a flexible framework, which can be easily adopted by assuming different data structures and distributions at the hidden layer latent space. Specifically, a matrix-LNM model is presented that introduces a matrix variate normal distribution at the latent layer, designed for time-coursed microbiome data. This approach captures both temporal dependencies and inter-sample correlations, offering a structured approach to longitudinal microbiome analysis. In addition, a family of models is also proposed by incorporating the modified Cholesky decomposition and imposing constraints on the components of the covariance matrix. Through simulation studies and real data analysis, the model's effectiveness in identifying dynamic patterns and clustering temporal microbiome profiles is demonstrated.

E1639: Mixtures of spatial factor analyzers for tensor-variate data**Presenter: Hanzhang Lu**, The University of British Columbia, Canada

A mixture of spatial factor analyzers (MSFA) is introduced to address the challenges of clustering high-dimensional spatial data. By leveraging the underlying coordinate system, the proposed framework incorporates a flexible, spline-based spatial decay covariance structure that prevents parameter inflation as dimensionality increases. To model non-spatial dependence, matrix factorization is employed for further dimensionality reduction. Parameter estimation is conducted via a variant of the expectation-maximization algorithm combined with a generalized least squares estimator. The proposed models are explored in the context of tensor-variate data analysis, where simulation studies and applications to Raman spectroscopy and hyperspectral texture databases demonstrate their capacity to accurately infer and differentiate distinct spatial patterns.

E1780: On model-based clustering of longitudinal data with missing values**Presenter: Brian Franczak**, MacEwan University, Canada

Missing values complicate the analysis of data with varying types. Longitudinal data consists of repeated measurements collected over time and frequently arises in many real-world applications. An expectation-maximization (EM) algorithm is developed to fit finite mixture models to longitudinal data with missing values. To account for time dependence between measurements, a modified Cholesky decomposition of the covariance matrices is utilized. These models are applied to cluster analysis, which aims to identify homogeneous subpopulations within datasets with unknown group labels. Model performance is assessed using both simulated and real datasets and evaluated using traditional metrics such as the adjusted Rand index and the Bayesian information criterion.

E2044: On handling outliers**Presenter: Paul McNicholas**, McMaster University, Canada

Recent approaches for finding outliers in model-based clustering will be presented.

EO241 Room 208 STEPP AND Q-TWIST: A TRIBUTE TO RICHARD D. GELBER, PHD PROFESSOR**Chair: Ann Lazar****E1172: Evaluating trade-offs in cancer clinical trials: The Q-TWiST method****Presenter: Bernard Cole**, University of Vermont, United States

The quality of life of patients is an important component for the evaluation of therapies in clinical trials. The Quality-adjusted Time Without Symptoms and Toxicity (Q-TWiST) method evaluates treatment trade-offs in clinical trials using an endpoint based on both quantity and quality of life. The methodology has been most useful for evaluating the trade-off between increased toxic side effects of treatment and improved clinical outcome. To date, most applications of the Q-TWiST method have been retrospective; however, its prospective use in clinical trials has been increasing. Key topics include the following: an Overview of the Q-TWiST method with a summary of examples in the cancer literature ranging from early applications to modern usage, Brief descriptions of related statistical tools for regression analysis and meta-analysis, and Techniques for the incorporation of longitudinal and cross-sectional utility and quality-of-life data, including data with informative missingness.

E1159: Introduction to subpopulation treatment effect pattern plot**Presenter: Ann Lazar**, , United States

Subpopulation Treatment Effect Pattern Plot (STEPP) is introduced as a method to investigate complex interactions between Treatment effectiveness and continuously measured covariates, particularly novel biomarkers. By exploring patterns of Treatment Effect across overlapping covariate intervals, STEPP avoids the need to define subgroups based on cutoff values while enabling detection of nonlinear interaction effects.

E1976: STEPP method for basic GLM models**Presenter: Wai-Ki Yip**, Miltenyi Biomedicine, Inc., United States**Co-authors:** Sergio Venturini

The STEPP method is applied to basic GLM models with binomial, linear, and Poisson outcomes. An example of the binomial model applied to an aspirin study is presented. R and STATA software implementations of the STEPP method are available as free downloads for interested users.

E1997: Subpopulation treatment effect pattern plot (STEPP): A review and a Bayesian extension for binary outcomes*Presenter:* **Marco Bonetti**, Bocconi University, Italy*Co-authors:* Luca Benetti

An overview of the STEPP methodology is provided, along with a new Bayesian extension for the specific case of binary outcomes. The focus is on the setting in which overlapping subpopulations are constructed according to a sliding window pattern, which has proven particularly useful in STEPP applications. The Bayesian approach employs a beta-binomial model to estimate the posterior distribution of subpopulation-specific treatment effects. The model assumes that overlapping intervals represent a mixture of parameters from adjacent non-overlapping intervals, with a uniform prior over the mixture weights. Estimation is performed using Metropolis within Gibbs sampling. The novel methodology is described and implemented, with application in the binary-outcome setting, and performance and sensitivity analyses regarding subgroup structure and the assumed dependence model are presented.

E1160: STEPP and Q-TWIST*Presenter:* **Richard Gelber**, TH Chan Harvard School of Public Health and Dana Farber Cancer Center, United States

The impact of STEPP and Q-Twist and their evolution throughout my research career are discussed.

EO183 Room 209 MODERN STATISTICAL METHODS: INFERENCE, OPTIMIZATION, AND LEARNING**Chair: Yao Zhang****E1328: Autoregressive learning under joint KL analysis***Presenter:* **Yunbei Xu**, National University of Singapore, Singapore*Co-authors:* Yuzhe Yuan, Ruohan Zhan

The fundamental problem of learning long sequences in autoregressive modeling and next-token prediction under model misspecification is studied, measured by the *joint* Kullback-Leibler (KL) divergence. The goal is to quantify long-horizon *error amplification* in both approximation and estimation, where the horizon H is the sequence length. By establishing matching upper and lower bounds, the first complete characterization of long-horizon error amplification under the natural joint-KL objective is provided, with improved rates and greater flexibility relative to existing work. On the approximation side, it is shown that joint KL admits a horizon-free barrier, in sharp contrast to Hellinger-based analyses that impose an $\Omega(H)$ dependence for computationally efficient methods, thereby isolating divergence choice as the source of the gap. On the estimation side, an information-theoretic lower bound of order $\Omega(H)$ is proven that holds for both decomposable policy classes and fully shared policies, matching the $\tilde{O}(H)$ upper bounds achieved by computationally efficient algorithms. It is also argued that a dynamical-systems perspective is essential for understanding autoregressive learning, clarifying the problem formulation and divergence choice that align with practical next-token training.

E1450: Statistical analysis of interleaving designs in A/B tests*Presenter:* **Nian Si**, HKUST, Hong Kong*Co-authors:* Weitao Cheng, Ningyuan Chen, Pin Gao

A/B testing for assortment recommendation systems is studied, where each user is presented with a set of items and makes a choice according to a stochastic choice model with an outside option. The goal is to estimate the global treatment effect (GTE) between treatment and control assortments when only user behavior is observed. The standard user-level randomized experiment (UE), which is unbiased but exhibits high variance, is compared with interleaving-based designs that merge treatment and control assortments to reduce variance at the cost of bias. Focus is placed on power analysis and it is shown that both UE and interleaving-based designs control Type I error, while interleaving-based designs achieve higher statistical power. Optimal interleaving mechanisms under this tradeoff are further characterized. Moreover, the framework is extended to sequential settings, where users examine recommended items sequentially in ranked order from top to bottom. Simulation studies corroborate the theoretical findings and illustrate the conditions under which interleaving designs yield substantial efficiency gains.

E1685: Online inference for optimal assortments via minimal directional perturbation radius of a max-difference statistic*Presenter:* **Shuting Shen**, National University of Singapore, Singapore

An inferential procedure is developed for testing structural properties of the revenue-maximizing assortment under a high-dimensional contextual multinomial logit model with sequentially adaptive data collection. The problem is challenging because the map from utility parameters to the optimal assortment is non-smooth and combinatorial, while the adaptive design induces complex temporal dependence. The procedure is based on data collected under a dynamic policy that updates assortments using ℓ_1 -penalized online estimation for contextual variable selection, and then performs post-learning inference on whether the optimal assortment satisfies a deployable structural rule, such as a category-mix requirement, or feasibility under inventory or resource constraints. The method is built on a revenue max-difference statistic with local perturbations of the revenue estimates, capturing uncertainty in both direction and magnitude: random unit directions account for directional error, and the minimal perturbation radius quantifies magnitude uncertainty. New tools based on martingale coupling and anti-concentration for differences of Gaussian maxima are developed. The resulting procedure is substantially more powerful than uniform error bound approaches. Finite-sample estimation rates, exact support recovery, and a non-asymptotic error decomposition for the debiased estimator are established, and asymptotic size control and power consistency are proven.

E1771: Seeing through correlations: Disentangled feature importance*Presenter:* **Jin-Hong Du**, The University of Hong Kong, Hong Kong*Co-authors:* Kathryn Roeder, Larry Wasserman

Quantifying Feature Importance with valid statistical uncertainty is central to interpretable machine learning, yet classical model-agnostic methods often fail under Feature correlation, producing unreliable attributions and compromising statistical inference. Existing approaches, such as Shapley values and leave-one-covariate-out, are designed for Feature selection and vulnerable to correlation distortion, limiting their robustness on model interpretation. Disentangled Feature Importance (DFI), a model-agnostic framework, resolves these limitations by combining principled statistical inference with computational flexibility. DFI leverages entropic optimal transport to learn flexible disentanglement maps and provide an interpretable pathway for understanding how Importance is attributed through the data's correlation structure. The framework generalizes to flow matching and differentiable loss functions, enabling statistically valid Importance assessment for black-box predictors in both regression and classification. Statistical inference theory enables valid confidence intervals and hypothesis testing with Type I error control. Empirical results on synthetic and biomedical datasets show that DFI delivers substantially higher statistical power than removal-based and conditional permutation methods, while maintaining robust, interpretable attributions under severe Feature interdependence.

E1827: Causal partial identification via optimal transport*Presenter:* **Zijun Gao**, University of Southern California, United States

In causal inference, only one of multiple potential outcomes is observed for each unit, leaving many causal quantities only partially identified (PI). This missingness closely parallels the optimal transport (OT) problem, where marginal distributions are observed but the joint coupling between them is unknown. This connection is formalized by casting the causal partial identification problem within the OT framework, enabling PI sets to be analyzed using tools from the rapidly developing optimal transport literature. When treatment effects are heterogeneous, incorporating covariate information can further sharpen PI sets. The OT formulation is generalized to a conditional optimal transport (COT) framework and new statistical

tools for COT are developed to extend the analysis of PI sets. Real-world applications and future research directions are discussed.

EP300 Room 204 POSTER SESSION II
Chair: Louisa Kontoghiorghe
E1181: Economic determinants of unemployment and forecasting the unemployment rate in Thailand

Presenter: **Kannat Na Bangchang**, Thammasat University, Thailand

The aim is to analyze the economic determinants influencing Thailand unemployment rate and to forecast future unemployment trends. Monthly secondary data from January 2017 to December 2024 were employed. The dependent variable is the unemployment rate, while independent variables include the Consumer Price Index, Foreign Direct Investment, export and import price indices, policy interest rate, total credit volume, minimum wage, COVID-19 situation, migrant labor, and exchange rate. Multiple Linear Regression was applied to examine the relationships between economic factors and the unemployment rate. Time series forecasting techniques, namely ARIMA and ARIMAX models, were used to predict unemployment trends for the next 12 months. Model performance was evaluated using Root Mean Square Error, Theil's Inequality Coefficient, and Akaike Information Criterion, along with diagnostic checking through the Ljung-Box test. The findings indicate that key economic variables significantly relate to Thailand unemployment rate in accordance with macroeconomic theories. The developed time series models demonstrate satisfactory forecasting performance and provide useful insights into future unemployment dynamics. The results can support policymakers and relevant agencies in designing effective economic and labor market policies to address future employment challenges.

E1244: Distribution-free finite-sample inference for pre-trend testing in difference-in-differences estimation with panel data

Presenter: **Sieun Kim**, Inha University, Korea, South

Co-authors: Seonghun Cho

Difference-in-differences (DID) estimation in panel data critically relies on the validity of the parallel trends assumption. Conventional pre-trend tests, however, often exhibit low statistical power and may suffer from size distortions in small samples. To address these limitations, a martingale test for exchangeability based on conformal p-values is applied to the pre-trend testing problem, treating the control group as the reference set and the treatment group as the test set. Through simulation studies and an empirical application to minimum wage and employment data, it is demonstrated that the proposed approach achieves substantially higher power in detecting trend violations while maintaining accurate control of type-I error across a wide range of settings.

E1344: Robust spatio-temporal graph modeling for behavior monitoring under heterogeneous environments

Presenter: **Jiwon Im**, Chungnam National University, Korea, South

Co-authors: Yunchae Jung, Minsu Park

Patient distress detection in clinical facilities necessitates high operational reliability to ensure patient safety. Conventional skeleton-based methods using graph convolutional networks often encounter performance bottlenecks in low-light conditions, where poor illumination induces significant noise in pose estimation and destabilizes joint detection. A real-time detection framework is developed to maintain robustness against adverse illumination by integrating skeleton-based motion analysis with scene-level context. Joint coordinates extracted from video frames are structured as spatio-temporal graphs, encoding spatial relationships and capturing dynamic motion patterns. Integrating scene-level context features complements the skeletal representation, enhancing situational inference. The challenge of performance degradation in low-light settings is addressed through an illumination-adaptive training strategy utilizing both standard and synthetically darkened data generated via gamma correction and Gaussian noise. This strategy stabilizes joint extraction and ensures reliable classification without computationally intensive image enhancement. Experimental results demonstrate that the framework consistently distinguishes routine actions from distress indicators, maintaining performance under low-light comparable to normal illumination. The achievement of high-frequency processing speeds confirms the feasibility for real-time deployment in clinical monitoring systems.

E1433: Asymptotic expansions and approximate moments for non-linear panel data models with separable errors

Presenter: **Gubhinder Kundhi**, Memorial University of Newfoundland, Canada

Co-authors: Paul Rilstone

Third-order stochastic expansions are derived for the fixed and random effects versions of non-linear panel data models with separable errors. These are used to derive the approximate bias, mean-squared error, skewness and kurtosis of the estimators. In turn, these moments are used to derive Edgeworth, saddlepoint and related expansions for these estimators. In a Monte Carlo experiment, the performance of these expansions for these models is compared to the first order approximation and other techniques commonly used in finite samples. The simulation results indicate that the confidence intervals based on these expansions have substantially better coverage properties than those based on standard asymptotic results and somewhat better than those provided by the bootstrap.

E1857: Forecasting U.S. inflation with Bayesian state-space models: Predictive coverage and posterior diagnostics

Presenter: **Keongmuk Lim**, Chungnam National University, Korea, South

Co-authors: Sangin Lee

Forecasting U.S. inflation is important for monetary policymaking, yet conventional time-series models such as AR and VAR exhibit miscalibration at long horizons, with empirical coverage falling below nominal levels. A Bayesian state-space model (BSSM) with a latent factor structure and policy inputs is applied to the inflation forecasting task, complemented by posterior diagnostics on the role of policy variables in the latent dynamics. The observation vector contains four price indices (CPI, core CPI, PCE, PPI), and the policy variables include short- and long-term interest rates; posterior inference is performed via Gibbs FFBS. In a 132-month rolling out-of-sample evaluation on FRED-MD (2015-2025), the BSSM achieves comparable point forecast accuracy to standard time-series benchmarks such as AR and BVAR, and provides higher long-horizon coverage than a comparable Bayesian VAR, reducing the gap between empirical and nominal coverage in those benchmarks.

E1908: An adaptive empirical Bayes confidence interval for mean estimation under mean-variance dependence

Presenter: **Chaewon Song**, Sookmyung Women's University, Korea, South

Co-authors: Hoyoung Park

An adaptive empirical Bayes confidence interval is proposed for the mean parameters in a heteroscedastic hierarchical model where the location and scale parameters may exhibit dependence. The proposed interval is constructed in two stages. First, the joint mixing distribution of the location and scale parameters is estimated via a two-dimensional nonparametric maximum likelihood estimator (2D-NPMLE) on a fixed bivariate grid, and posterior summaries are stabilized through a model-based bootstrap refitting procedure that yields a unified bootstrap-averaged variance estimator. Second, a unit-specific interval is formed by adaptively combining a posterior-based width with a calibrated safeguard width through a data-driven weight that quantifies the degree of posterior sharpening relative to the likelihood. The resulting interval interpolates smoothly between the two regimes: it inherits the calibrated coverage of double-shrinkage intervals when the data provide no posterior sharpening and exploits unit-specific posterior concentration when the mean-variance dependence is informative. Simulation studies across diverse independence and dependence structures, together with real-data examples, illustrate the practical utility of the proposed approach for heteroscedastic and unbalanced multilevel data.

E1948: A deep learning random-effect modelling approach for longitudinal count data*Presenter:* **Jihoon Kim**, Pukyong National University, Korea, South*Co-authors:* Il Do Ha

The deep neural network (DNN) model, a core model of deep learning, provides high predictive power to predict and classify output variables in various fields by modeling the nonlinear functional relationship between input and output variables through hidden layers. However, since the DNN has been mainly developed for independent output variables, applying it directly when output variables are correlated results in greatly reduced predictive performance. Such correlation is typically modeled via random effects, but random effect models have been primarily developed under the assumption of linearity in the functional relationship between input and output variables. A new Poisson DNN random-effect model is proposed herein. The output variables are correlated count outcomes obtained from repeated measures over time. For estimation and learning of the proposed model, an optimization algorithm based on negative marginal likelihood is developed as a loss function. Simulation and real data analysis demonstrate the validity of the proposed method. In particular, simulation results confirm that the proposed DNN model provides higher predictive performance than existing prediction models.

EV302 Room Virtual R01 FORECASTING AND MARKET DYNAMICS**Chair: Wenbo Wu****E2051: Algorithmic trading in a Hawkes flocking LOB model***Presenter:* **Hyoeun Lee**, University of Illinois Urbana-Champaign, United States*Co-authors:* Kiseop Lee

The optimal placement of Limit Orders under the Hawkes-flocking price process is studied by modeling the best ask price and best bid price using the Hawkes-flocking model. From time 0 to T , at time points $\{0, t_1, \dots, t_n = T\}$, a trader makes decisions to place sell or buy Limit Orders and may cancel existing Limit Orders. The trader maximizes its cash flow based on estimated parameters from the previous time step, under the assumption that the price follows the Hawkes-flocking model. Multiple strategies are compared, and optimal solutions are discussed based on various conditions.

E1353: Evaluating hybrid deep learning architectures for day-ahead electricity price forecasting in the German spot market*Presenter:* **Benjamin Leidig**, University of Illinois Urbana-Champaign, United States*Co-authors:* Hyoeun Lee

Forecasting day-ahead electricity prices is a longstanding focus of econometric research and energy market participants. Recent studies indicate that contemporary approaches, including hybrid and deep learning techniques, are increasingly used. However, the complexity and resource requirements of advanced techniques make custom hybrid learning architectures challenging. For example, inconsistent adherence to temporal constraints is a recurring limitation. Additionally, inappropriate model interpretations and comparisons can be misleading, often due to the black-box nature of neural networks. To address these limitations, a comprehensive selection of feature engineering and machine learning techniques was used to forecast hourly day-ahead electricity prices on the German spot market. This selection included architectures incorporating Variational Mode Decomposition, convolutional and recurrent neural networks, multi-head attention, and transformers, all under strict temporal training and forecasting constraints. The supervised models and signal decomposition techniques were tuned using Bayesian optimization and frequency separation analyses, respectively. To compare real-time performance, these frameworks were tested on a year of hourly data. The results revealed that novel hybrid deep learning architectures demonstrate superior performance during regime changes and volatility spikes inherent to energy markets compared with traditional statistical methods.

E1241: Conformal inference for multi-horizon forecasting in financial panel data: A block-wise segmentation approach*Presenter:* **Sehee Kim**, Inha University, Korea, South*Co-authors:* Seonghun Cho

The problem of multi-horizon forecasting in financial panel data is investigated, a domain where uncertainty quantification is indispensable due to the high-stakes nature of financial decision-making. While traditional approaches often depend on the validity of underlying model assumptions, modern high-performance models frequently lack formal uncertainty guarantees. To bridge this gap, the conformal prediction framework is employed, a leading approach for distribution-free inference. In particular, a block-wise segmentation strategy that partitions long time series into shorter segments is introduced, integrated with block-wise Z-score standardization to mitigate scale heterogeneity across blocks. Through extensive simulation studies and empirical analysis using U.S. stock market data, it is demonstrated that the proposed methodology, when integrated with diverse predictive models, consistently attains stable joint coverage in multi-horizon forecasting settings.

E1971: A semi-parametric additive hybrid model: An application to SSE composite index forecasting*Presenter:* **Bolun Qu**, Monash University Malaysia, Malaysia*Co-authors:* Erniel Barrios, HowChinh Lee, Siew Eu-Gene

Financial time series forecasting confronts a trade-off between the interpretability of regime-switching econometric models and the predictive capacity of deep learning. Existing hybrid architectures typically rely on globally linear baselines and static sequential estimation. This propagates specification errors into the neural network and neglects the skewed empirical distribution of market regimes. A regime-switching semiparametric additive hybrid model is proposed that integrates a threshold autoregressive process with a long short-term memory network, jointly estimated by a decoupled iterative backfitting algorithm. The model replaces the globally linear baseline with a regime-dependent parametric process, allowing the threshold mechanism to extract state-conditioned linear memory before the residual is mapped by the recurrent network. An inverse-frequency data cloning mechanism mitigates regime imbalance while preserving the temporal topology required by the recurrent architecture. The balanced cloned sample feeds only the pointwise TAR component, while the LSTM receives chronologically ordered observations. The model is evaluated on the Shanghai Stock Exchange Composite Index for joint forecasting of closing prices and logarithmic returns. Out-of-sample results indicate improved point forecasting accuracy relative to linear benchmarks, alongside statistically significant directional predictability by the Pesaran-Timmermann test.

E1937: Temporal information extraction for cross-sectional stock return prediction*Presenter:* **Kai-Xiang Chang**, National Central University, Taiwan*Co-authors:* Hsin-Yu Chiu

The aim is to examine whether temporal information from the joint evolution of firm characteristics and market states improves cross-sectional stock return prediction. While recent empirical asset-pricing studies increasingly rely on large predictor sets, such high-dimensional settings may cause static models to miss nonlinear interactions and dynamic patterns in rolling histories. To address this issue, a two-stage framework is proposed that separates temporal representation learning from final cross-sectional prediction. In the first stage, a temporal convolutional network processes twelve-month rolling sequences of firm characteristics and market-state variables to extract a sequence-based signal for future stock ranking. In the second stage, this signal is combined with contemporaneous predictors and incorporated into a tree-based model to predict next-month returns. Using U.S. equity data from 2000 to 2023, the results show that the framework generates economically meaningful out-of-sample long-

short portfolio performance and significant Fama-French five-factor alpha. The learned temporal signal provides information beyond conventional predictors and remains relevant in interpretability analyses. Overall, the evidence suggests that temporal representations can enhance cross-sectional stock return prediction, although strong single-stage benchmarks remain highly competitive.

Sunday 09.08.2026

13:35 - 14:50

Parallel Session H – EcoSta2026

EO283 Room B101 RECENT ADVANCES IN NETWORKS AND INTERFERENCE II**Chair: Mengsi Gao****E1264: Heterogeneous effects of endogenous treatments with interference and spillovers in a large network***Presenter:* **Lin Chen**, Vanderbilt University, United States*Co-authors:* Yuya Sasaki

The identification and estimation of heterogeneous effects of an endogenous treatment under interference and spillovers in a large single-network setting is studied. Endogenous treatment selection is modeled as an equilibrium outcome that explicitly accounts for spillovers and conditions guaranteeing the existence and uniqueness of this equilibrium are derived. Heterogeneous marginal exposure effects (MEEs), which may vary with both the treatment status of neighboring nodes and unobserved heterogeneity, are then identified. Estimation strategies are developed and their large-sample properties established. Equipped with these tools, the heterogeneous effects of import competition on U.S. local labor markets in the presence of interference and spillovers are analyzed. Negative MEEs are found, consistent with the existing literature. However, these effects are amplified by spillovers in the presence of treated neighbors and among localities that tend to select into lower levels of import competition. These additional empirical findings are novel and would not be credibly obtainable without the econometric framework proposed.

E1421: Policy targeting with market equilibrium*Presenter:* **Gyungbae Park**, Brown University, United States

A framework is developed for individualized treatment allocation in the presence of market equilibrium spillovers. Motivated by selective subsidy programs that reduce the price of a specific good, such as healthy-food or clean-energy coupons, the framework highlights when universal distribution is optimal, when targeted allocation dominates, and how the planner's targeting problem can be solved tractably. It is shown that the policymaker's welfare function is supermodular in treatment assignments under broad and interpretable conditions. Supermodularity implies that treatments are complements, so the welfare gain from treating one group is larger when others are also treated. This property both clarifies the trade-off between universal and targeted distribution and ensures computational feasibility, since maximizing a supermodular function can be solved in polynomial time using off-the-shelf algorithms. Statistical guarantees are also established that account for estimation uncertainty in the demand and supply equations. Finally, the proposal is illustrated in a coupon allocation problem calibrated with households expenditure data from the Philippines.

E1422: Endogenous interference in randomized experiments*Presenter:* **Mengsi Gao**, University of Southern California, United States

Identification and inference for treatment effects in randomized trials with social interactions are studied. Two key features introduce endogeneity: (1) latent variables that affect both networks and outcomes, and (2) treatment-induced network changes that mediate treatment effects. A potential-outcomes framework with post-treatment networks is developed, and direct and network-mediated effects are defined. For estimation, a shift-share IV strategy with asymptotic theory for sparse networks is proposed, but it fails in denser settings. A denoised, eigen-decomposition-based SSIV estimator is then developed that restores consistency in denser networks and improves convergence rate. An application illustrates both direct effects and network-mediated effects.

EO106 Room B102 MODELING FOR MULTIVARIATE DATA AND CAUSAL INFERENCE METHODS**Chair: Gyuhyeong Goh****E1635: Interactive fixed effects models for macro-policy analysis***Presenter:* **Jisang Yu**, Korea University, Korea, South*Co-authors:* Gyuhyeong Goh

Interactive Fixed Effects (IFE) models have become a popular methodological approach in causal inference literature. While Two-Way Fixed Effects (TWFE) has dominated work with panel data, approaches like Synthetic Control Methods and variants thereof build on models with Interactive Fixed Effects. The potential application of Interactive Fixed Effects models to examine policy impacts in macroeconomics is explored, specifically focusing on how trade and climate policies affect economic outcomes using country-level data. Differences in estimates across TWFE and IFE are examined.

E1740: Fully Bayesian synthetic control methods with sparse convex hull restriction and Gaussian process*Presenter:* **Junwoo Jo**, Kyungpook national university, Korea, South*Co-authors:* Gyuhyeong Goh, Dipak Dey

A fully Bayesian synthetic control method is proposed that preserves the convex hull restriction of the conventional method while enabling coherent posterior inference. The approach reparameterizes the weights by introducing latent positive variables and assigning them an exact spike-and-slab prior. This construction yields a Dirichlet prior on the normalized weights, automatically respects the convex hull restriction, and allows exact zeros, thereby inducing sparsity and enhancing interpretability. A Gibbs sampling algorithm jointly updates the inclusion indicators and latent weights, resolving the dimension-changing problem via a combination of Laplace approximation and population Monte Carlo algorithm. Treatment effects are modeled with a Gaussian process prior to allow cross-time correlation. This prior flexibly captures dynamic effects and provides principled uncertainty quantification. In simulations based on a linear factor model with designed sparsity and outliers, the proposed method achieves the lowest mean squared error among competing approaches. It also delivers higher interval coverage, particularly compared with Bayesian linear models using alternative priors. Practical advantages of the approach are further illustrated in an empirical application to California's tobacco control program.

E1746: Mixture of multivariate linear models with heteroscedastic covariance matrices*Presenter:* **Keunbaik Lee**, Sungkyunkwan University, Korea, South

A critical research gap in mixture of experts models for multivariate heteroscedastic data is addressed, specifically where covariance matrices vary as a function of covariates. A systematic framework is established through a mixture model architecture integrated with the Expectation-Maximization (EM) algorithm. Hypersphere decomposition is introduced for the structured modeling of covariance matrices, strategically employed to circumvent the inherent challenges of maintaining positive definiteness in high-dimensional covariance structures, a frequent technical impediment in complex statistical modeling. The approach successfully characterizes data exhibiting heteroscedasticity across multiple latent clusters. Through extensive numerical simulations and empirical data analysis, the theoretical soundness and practical efficacy of the proposed methodology are validated, confirming its robustness. The primary contribution lies in the implementation of multivariate mixture regression models in analytical environments that were previously avoided due to their computational complexity. This methodology provides a sophisticated and pragmatic analytical tool for scenarios characterized by covariate-dependent variance and intricate correlation structures.

EO126 Room B103 DATA ANALYTICS AND DECISION MODELS IN OPERATIONS MANAGEMENT**Chair: Donghao Zhu****E1732: Loss-leader pricing on hybrid platforms: A multiproduct perspective****Presenter:** Yusuke Zenryo, Kobe University, Japan**Co-authors:** Yushi Tsunoda, Yuto Ishihara

A model of a hybrid platform that hosts multiple products and enables consumers to fill their baskets through one-stop shopping is developed. Each product is supplied by a manufacturer and sold via competing third-party sellers, while the platform also enters as a first-party seller. One-stop shopping turns otherwise independent products into effective complements, creating a cross-product demand externality (the Cournot externality). The platform may optimally price one product below cost (loss-leader pricing) to internalize this Cournot externality, particularly when manufacturers' wholesale prices are high. Ceteris paribus, loss-leader pricing benefits the platform, consumers, and manufacturers, yielding a Pareto improvement. However, this positive outcome may fail to arise -and indeed leave all parties worse off- when manufacturers strategically raise wholesale prices in anticipation of the platform's loss-leading. The analysis finds that whenever loss-leading is profitable for the platform, it also benefits consumers and manufacturers, providing a procompetitive rationale for first-party selling by hybrid platforms.

E1958: Non-stationary stochastic optimization: Adaptive algorithm and applications in operations**Presenter:** Jinzhi Bu, The Hong Kong Polytechnic University, Hong Kong**Co-authors:** Siyi Wang, Sen Yang

Online Stochastic convex optimization is studied where the distribution of the random environment evolves over time under a variation-budget constraint. The state-of-the-art restarting procedure attains a minimax-optimal dynamic regret rate but requires prior knowledge of the variation budget. This crucial assumption is relaxed by developing an Adaptive Stochastic Gradient Descent (AdaSGD) algorithm that adapts to unknown non-stationarity without using information about the variation budget. AdaSGD achieves the same dynamic regret bound as in the case where the variation budget is known. The framework is further applied to three operations management problems and AdaSGD is extended to (i) the setting with strongly convex cost functions and (ii) the setting with time-dependent cost functions.

E1697: Green order first: Matching and dispatching under sustainability-aware demand**Presenter:** Donghao Zhu, The University of Tsukuba, Japan**Co-authors:** Huan Yu, Jianming Shi

The aim is to examine dispatching and matching policies in on-demand service platforms with sustainability-aware demand. A double-sided queueing network is developed with green and non-green orders and couriers, where green-preferred orders incur penalties when matched with non-green couriers. The platform faces a trade-off between operational efficiency and sustainability compliance. Two immediate matching policies are analyzed: a Greedy policy prioritizing speed and a Green-Priority policy enforcing green compatibility. Using Markov chain analysis, stationary performance metrics such as matching rates, queue lengths, mismatch rates, and expected revenue are characterized, and it is shown that the revenue indifference curve is linear in symmetric markets but nonlinear under asymmetry. A delayed Batch-Ranking policy is further introduced that allows limited waiting and cost-aware matching. Agent-based simulations show that delayed matching can substantially reduce delivery time, emissions, and overall cost. The results provide insights into when Green-Priority dispatching is optimal and how controlled delays improve sustainability outcomes without sacrificing long-run efficiency.

EO008 Room 101 ECOSTA JOURNAL SESSION**Chair: Kento Egashira****E1242: Multivariate change-point detection via optimal transport and characteristic functions****Presenter:** Sarka Hudecova, Charles University, Prague, Czechia

Detecting structural changes in a sequence of observations is a central problem in statistical inference, particularly when the observations are multivariate. In many applications, ranging from finance and environmental monitoring to biomedical signal analysis, data are collected as vectors, and potential changes may occur not only in the marginal distributions but also in the dependence structure of the joint distribution. It is therefore desirable to have a test that can identify a change point in the multivariate distribution while operating in a fully nonparametric setting, without imposing restrictive assumptions on the underlying distribution or on the nature of the change. A test for detecting distributional changes in a sequence of multivariate observations based on optimal measure transport is proposed. The test is constructed using empirical characteristic functions of the transported observations. The resulting procedure is distribution-free, which allows for straightforward implementation of an exact test. The asymptotic distribution of the test statistic is also derived. The finite-sample performance of the proposed method is illustrated through a Monte Carlo simulation study.

E2059: Robust multiple change point detection for object data**Presenter:** Shojaeddin Chenouri, University of Waterloo, Canada

A nonparametric multiple change point detection procedure is proposed for stochastic processes taking values on manifold-valued spaces $\mathcal{M}$. The proposed method is based on rank statistics constructed from intrinsic depth functions, allowing for the detection of changes in both location and scale without requiring parametric distributional assumptions. Finite-sample properties and asymptotic consistency results are established for the estimation of both the number and locations of change points. By expressing the test statistic as a U-statistic, sharp nonasymptotic bounds are derived for the proposed procedure. Robustness properties are investigated through breakdown value analysis and finite-sample Monte Carlo studies, demonstrating strong detection performance and stability under heavy-tailed distributions and contamination by outliers. The methodology is further illustrated through applications to real-world manifold-valued data arising in pedestrian surveillance video analysis, highway traffic monitoring, and vehicle health monitoring. Theoretical and empirical results confirm the effectiveness of the proposed nonparametric multiple change-point detection framework.

E1678: Real-time monitoring of global inflation heterogeneity**Presenter:** Yiren Wang, Hunan University, China**Co-authors:** Hongqi Chen, Liangjun Su

A real-time monitoring econometric framework for detecting structural changes in global inflation heterogeneity is developed. The framework detects changes in the inflation-macroeconomic relationship across quantiles in a timely manner as new data become available. A convolution-type smoothed panel quantile regression model with individual fixed effects is employed, and a gradient-based sequential detector with an associated boundary function is constructed. The proposed procedure preserves convexity and achieves $\sqrt{NT}$ -consistent estimation through bias correction. Asymptotic size control is established under the null hypothesis of parameter stability, and non-trivial power is demonstrated under local alternatives. Simulation verifies the theoretical properties of the proposed method. Using global inflation data with the proposed framework, new empirical evidence on real-time monitoring of global inflation dynamics is provided, documenting timely detection of changes across quantiles in the inflation-macroeconomic relationship and identifying the macroeconomic drivers associated with these changes.

EO074 Room 102 ADVANCES IN GRAPH DATA ANALYSIS**Chair: Daisuke Murakami****E1368: Network propagation regression: A general and interpretable framework for network-linked data****Presenter:** Yingying Ma, Beihang University, China**Co-authors:** Chenlei Leng

Network linked data are increasingly common in the social, biological, and information sciences, yet classical regression methods often fail to account for dependencies induced by network structure. A unified and interpretable framework for regression on network linked data—network propagation regression (NPR)—is proposed that flexibly captures both direct and higher-order network effects across diverse outcome types. NPR enables simple and efficient estimation via ordinary least squares for continuous outcomes and standard generalized linear model routines for binary, categorical, and time-to-event data. Consistency and asymptotic normality of the estimators under weak conditions are established, and hypothesis tests for the effective order of network influence are developed. Simulations demonstrate that NPR outperforms established methods, particularly under model misspecification. An application to social media sentiment analysis illustrates the practical utility and robustness of NPR. The framework provides a principled and general solution for regression analysis in complex networks.

E2028: Dynamic networks with node heterogeneity and homophily**Presenter:** Binyan Jiang, The Hong Kong Polytechnic University, Hong Kong

A framework is proposed for modeling node heterogeneity and link homophily in dynamic networks. The framework provides new insights into how networks evolve over time and supplies sophisticated tools for predicting future networks with statistical guarantees. The model accounts for link homophily associated with both observed and latent traits. Joint modeling of node heterogeneity and both observed and latent homophily effects presents a significant challenge in statistical inference due to the large number of confounding parameters. A novel normalized squared loss is introduced to enable stable parameter estimation in high-dimensional settings. Rigorous theoretical analysis of the estimation method is provided, and effectiveness is demonstrated through extensive simulations and illustration with real-world network data.

E2072: Node ranking on directed graphs under outgoing-link-profile differential privacy**Presenter:** Wanjie Wang, National University of Singapore, Singapore

Node ranking algorithms on directed graphs transform link structure into node importance, influence, and authority. However, a node's outgoing link profile may reveal sensitive preferences, affiliations, or relationships. This work studies outgoing link profile differential privacy, where neighboring directed graphs differ in one node's entire outgoing link profile. The focus is on two canonical ranking algorithms, PageRank and HITS authority scores, with perturbation-guided private mechanisms designed for both. For PageRank, row-level sensitivity is driven by low out-degree nodes. A transition regularization mechanism is introduced that raises small out-degrees, truncates the spectral norm, and then adds calibrated Gaussian noise. For HITS authority scores, perturbation analysis shows that stability is governed by the eigengap of $A^T A$; this instability is addressed through an eigengap-based propose-test-release mechanism that privately certifies stability before releasing a noisy score vector. For both mechanisms, row-level privacy guarantees and accuracy bounds are proven with trade-off effects on regularization parameters. Numerical experiments show that these methods improve the recovery of top-ranked nodes and reduce relative ℓ_2 error compared with edge-DP baselines after privacy-budget conversion. This is the first work to develop differentially private algorithms for directed-graph node ranking under outgoing link profile privacy.

EO039 Room 103 STATISTICAL LEARNING: STRUCTURAL COMPLEXITY, INFERENCE, DISTRIBUTED MODELS**Chair: Chao Zheng****E1761: A coefficient space projection approach for penalized regression problems****Presenter:** Gan Yuan, City University of Hong Kong, Hong Kong**Co-authors:** Zicheng Liu, Chun Yip Yau

A new approach views a penalized least squares (PLS) problem as a projection problem in a linearly-transformed space of regression coefficients. Specifically, the least squares criterion is expressed as an ℓ_2 -distance from a point to the ordinary least squares estimator, and the feasible region of the PLS problem is characterized as a geometric object under the transformed space. The penalized least squares problem can then be solved by projecting the ordinary least squares estimator onto this geometric object. The coefficient space projection approach is applied to develop a general method for deriving solution path algorithms for solving PLS problems with convex penalty functions, extending the scope of existing path algorithms. As a particular example, the general method yields the first solution path algorithm for the elastic net problem.

E1764: Towards reliable and efficient semi-supervised inference with black-box AI models**Presenter:** Sangwoo Park, King's College London, United Kingdom

Obtaining non-asymptotic, finite-sample guarantees for statistical inference typically requires accepting larger confidence regions. Existing statistical inference frameworks have generally evolved to achieve near-maximal efficiency levels, such as the tightest possible confidence regions. To overcome such efficiency limits, semi-supervised inference that augments limited real-world data with abundant synthetic data has been extensively studied. Recent advances in artificial intelligence have motivated the use of AI to generate synthetic data for semi-supervised inference, potentially achieving enhanced efficiency while maintaining reliability under appropriately designed frameworks. However, the unknown behavior of AI models often causes AI-based semi-supervised inference to harm efficiency in practice. Methods are described to achieve provable efficiency enhancements, or at minimum no degradation, in AI-based semi-supervised inference regardless of AI model quality, followed by discussions of potential synergies with active sampling strategies. Empirical evidence from risk-controlling LLM hyperparameter tuning is presented to support the theoretical claims.

E1776: Estimation and inference for nonparametric expected shortfall regression over RKHS**Presenter:** Yue Wang, University of Science and Technology Of China, China

Expected shortfall (ES) has emerged as an important metric for characterizing the tail behavior of a random outcome, specifically associated with rarer events that entail severe consequences. In climate science, the threats of flooding and heatwaves loom large, impacting natural environments and human communities. In actuarial studies, a key observation in modeling insurance claim sizes is that features exhibit distinct effects in explaining small and large claims. Nonparametric expected shortfall regression constitutes a class of statistical methods for tail learning. These methods directly target upper and lower tail averages and empower practitioners to address complex questions that are beyond the reach of mean regression-based approaches. Using kernel ridge regression, a two-step nonparametric ES estimator is introduced that involves a plugged-in quantile function estimate without sample-splitting. Non-asymptotic estimation and Gaussian approximation error bounds are provided, depending explicitly on the effective dimension, sample size, regularization parameters, and quantile estimation error. To construct pointwise confidence bands, a fast multiplier bootstrap procedure is proposed and its validity is established. The finite-sample performance of the proposed method is demonstrated.

EO069 Room 105 CROSS SECTIONAL ECONOMETRICS: THEORY AND APPLICATIONS**Chair: Abhimanyu Gupta****E1674: A plea offer you cannot refuse****Presenter:** Luke Nicholas Taylor, Aarhus University, Denmark

The vast majority of criminal cases in the United States are resolved through plea bargaining, yet the determinants of defendants' plea decisions remain poorly understood. Empirical analysis is complicated by selective missingness: plea offers are observed only when accepted, while trial outcomes are observed only when rejected. A structural model of plea choice extends the generalized Roy framework to allow for ex-ante uncertainty and non-risk-neutral preferences. Identification exploits quasi-random assignment of judges, which generates exogenous variation in trial conviction probabilities and sentencing outcomes. Using detailed case-level data from North Carolina, the analysis recovers defendants' expected trial utility and estimates the determinants of plea acceptance. Large plea discounts and substantial baseline aversion to trial are documented, while limited evidence of departures from risk neutrality is found. Results suggest that plea bargaining outcomes are driven less by preferences over risk and more by the structure of incentives and uncertainty embedded in the trial process.

E1679: Profile GMM estimation of spatial dynamic panels with endogenous weight matrix and interactive fixed effects**Presenter:** Jiajun Zhang, Shanghai University of International Business and Economics, China**Co-authors:** Chen Yahui, Xiaoyi Han, Liangjun Su

The estimation and inference of spatial dynamic panel data (SDPD) models with interactive fixed effects (IFE) is studied, where the time-varying (TV) spatial weight matrix is constructed from multidimensional socio-economic variables such as TV trade or mobility flows across regions. The TV spatial weight matrix is modeled using a three-dimensional (3D) panel with a multi-level factor structure, and the possible endogeneity of the spatial weight matrix is captured through the correlation between the error terms in the SDPD outcome equation and the local factors in the 3D panel. Using a control function approach, a two-stage estimation procedure is proposed. First, the local factors are estimated from the 3D panel and plugged into the SDPD equation to control for the endogeneity of the spatial weight matrix. Then, a nuclear norm regularization (NNR)-based profile GMM method is considered, followed by post-NNR GMM iterations. The asymptotic properties of these estimators are rigorously established, and the most efficient GMM estimator (GMME) with the best moment conditions is derived. A test for the endogeneity of the spatial weight matrix is also considered. Monte Carlo simulations demonstrate that the estimators and test statistic perform well in finite samples. The models and estimation methods are applied to study virus transmission through the human mobility network across U.S. states.

E1766: A nonparametric test for cross-unit spillovers**Presenter:** Abhimanyu Gupta, Queen's University, Canada**Co-authors:** Margherita Comola, Camila Comunello

A novel nonparametric test for cross-unit spillovers that may operate through peers' attributes, peers' outcomes, or both is proposed. The test is straightforward to implement, as it requires only estimation under the null of no spillovers, and it is shown to have a convenient asymptotic standard normal distribution. It is also versatile, accommodating data generated by a wide range of interaction structures. Three empirical illustrations demonstrate that the test is effective at detecting cross-unit spillovers arising in a nonparametric manner that existing approaches may fail to uncover.

EO134 Room 202 DIMENSION REDUCTION METHODS AND RELATED TOPICS**Chair: Hung Hung****E1487: Distributed and personalized PCA via matrix beta-mean aggregation****Presenter:** Zhi-Yu Jou, Academia Sinica, Taiwan

Distributed PCA (DPCA) has become increasingly important for analyzing large-scale data stored across multiple machines. A fundamental challenge in DPCA lies in efficiently aggregating information from multiple machines while retaining the statistical properties of the original dataset. Existing methods that aggregate only local leading eigenvectors, while ignoring important eigenvalue information, may lead to suboptimal estimation. To address this issue, beta-DPCA is proposed, a distributed PCA method that incorporates eigenvalue information through matrix beta-mean aggregation. In many real-world applications, however, local datasets often exhibit heterogeneity, whereas many standard DPCA methods implicitly assume a common principal subspace across datasets. To address this limitation, a personalized extension of beta-DPCA is further proposed that decomposes each local covariance structure into shared and individualized components. The proposed framework enables robust extraction of common features across data sources while preserving source-specific patterns.

E1571: Product-CCA: An ordering-robust canonical correlation analysis procedure without efficiency loss**Presenter:** Kenichi Hayashi, Keio University, Japan**Co-authors:** Hung Hung

Canonical correlation analysis (CCA) is a fundamental statistical tool for extracting shared information from two sets of variables, but its performance can be highly sensitive to outliers. Although robust methods for CCA have been studied, the literature remains less developed than that on principal component analysis (PCA), a closely related and simpler method. Recently, a random-partition-based approach called product-PCA was proposed that improves outlier-robustness while preserving the estimation efficiency of conventional PCA in the absence of outliers. Motivated by this idea, Product-CCA (PCCA) is proposed, a new CCA procedure based on random partitioning of the data. PCCA improves the outlier-robustness of conventional CCA without directly downweighting observations suspected to be outliers. This property enables PCCA to share the same asymptotic efficiency as CCA, an appealing property in real applications. The construction of PCCA, its theoretical considerations, and results of numerical experiments are presented.

E1754: Generalized information criterion for rank selection in high-dimensional CCA**Presenter:** Jia-Rou Liu, National Taiwan University, Taiwan**Co-authors:** Hung Hung

Determining the appropriate number of Canonical pairs (rank) in high-dimensional Canonical Correlation Analysis (CCA) remains challenging, particularly under model misspecification where conventional criteria such as AIC and BIC often perform poorly. A Generalized Information Criterion (GIC) for robust rank selection in CCA is proposed. An explicit closed-form expression for the GIC penalty is derived using the score and influence functions of the CCA parameters. Through the eigenvalue representation underlying CCA, the resulting penalty incorporates the magnitude of eigenvalues, in contrast to conventional criteria that depend solely on the number of parameters. The asymptotic properties of GIC are established and the gap conditions required for selection consistency are identified. Extensive simulations under varying signal strengths and noise structures, including both simple and Generalized spiked covariance models, support the theoretical findings. Compared with existing methods, GIC more effectively excludes noise eigenvalues than AIC while remaining more sensitive than BIC in detecting signal eigenvalues. Overall, the proposed method provides rigorous theoretical foundations and a practically robust approach for rank determination in high-dimensional CCA.

EO056 Room 203 RECENT ADVANCES IN COVARIANCE AND INVERSE COVARIANCE ESTIMATION**Chair: Sang-Yun Oh****E1540: Tensor portfolio****Presenter:** Tianyan Tu, University of California, Riverside, United States**Co-authors:** Tae-hwy Lee

Motivated by the multidimensional nature of financial data, Tensor Portfolios is a framework exploiting the intrinsic multiway structure of stock returns to reduce the number of free parameters required for portfolio construction. Three distinct methods tailored to specific structural assumptions are developed. Tensor and vector Portfolios are systematically compared through Monte Carlo simulations and empirical studies. Simulation results show Tensor Portfolios yield significantly higher out-of-sample Sharpe ratios whenever the data exhibits a Tensor structure. Empirical analysis further corroborates the effectiveness of Tensor Portfolios; their general outperformance over vector Portfolios in real-world markets highlights the practical significance of exploiting multiway information.

E1628: Distributionally robust PCA with data-adaptive Wasserstein geometry**Presenter:** Chuang Xu, Australian National University, Australia**Co-authors:** Yanrong Yang

A distributionally robust formulation of PCA is developed that minimizes worst-case reconstruction risk over distributions lying within a Wasserstein neighborhood of the empirical measure. The Wasserstein neighborhood is adaptively calibrated through a transport matrix G to capture heterogeneous uncertainty across dimensions. The homogeneous case, in which G is a scalar multiple of the identity matrix, recovers classical PCA. Under a general transport matrix G , a dual characterization of the associated minimax optimization problem is derived and a tractable surrogate objective function is introduced consisting of the square-root empirical reconstruction error plus a geometry-dependent residual exposure penalty. The exact and surrogate estimators are shown to be consistent for the population PCA subspace and asymptotically equivalent at the projector level. The transport geometry is allowed to be data adaptive, while the Wasserstein radius is calibrated via robust Wasserstein profile inference, yielding a data-driven radius of order $n^{-1/2}$. Theoretical guarantees are established, including consistency and local Grassmannian asymptotics exhibiting an explicit Wasserstein-induced drift determined by the limiting transport geometry and calibration level. Numerical experiments and a real-data application demonstrate that the method can improve out-of-sample performance under structured covariance shifts, moderate contamination, and certain same-distribution regimes.

E1643: Relaxed sparsest-permutation formulation for causal discovery at scale**Presenter:** Sang-Yun Oh, University of California, Santa Barbara, United States**Co-authors:** Gunwoong Park, Sunmin Oh

Despite the growing availability of large datasets, causal structure learning remains computationally prohibitive at scale. Sparsest-permutation learning for linear structural equation models is revisited, revealing that exact Cholesky factorization is unnecessary for structure recovery. This observation motivates a support-level relaxation that searches for sparse triangular factors over a precision-support screening graph. The relaxed formulation can be efficiently evaluated via masked zero-fill incomplete Cholesky factorization, enabling scalable comparison of candidate orderings. At the population level, soundness for Markov equivalence class (MEC) recovery is established under no-cancellation and sparsest Markov representation assumptions, as well as robustness to ordering misspecification. Motivated by these guarantees, SCOPE, a sparse-Cholesky pipeline, provides a scalable implementation of the relaxed formulation. Experiments on synthetic and real datasets demonstrate that SCOPE matches the MEC recovery accuracy of substantially slower baselines while achieving significantly reduced runtime and scaling to 10^4 variables.

EO141 Room 205 RECENT ADVANCES IN STATISTICAL INFERENCE FOR COMPLEX DATA**Chair: Shuyuan Wu****E1311: Robust variational Bayes by min-max median aggregation****Presenter:** Jiyuan Tu, Shanghai University of Finance and Economics, China

A robust and scalable variational Bayes (VB) framework is proposed that is designed to effectively handle contamination and outliers in datasets. The approach partitions the data into m disjoint subsets and formulates a joint optimization problem based on robust aggregation principles. A key insight is that the full posterior distribution is equivalent to the minimizer of the mean Kullback-Leibler (KL) divergence from the m -powered local posterior distributions. To enhance robustness, the mean KL divergence is replaced with a min-max median formulation. A notable discrepancy is observed in the m -powered marginal log likelihood function contingent on the presence of local latent variables. To address this, these two scenarios are treated separately to guarantee the consistency of the aggregated variational posterior. Specifically, when local latent variables are present, an aggregate-and-rescale strategy is introduced. Theoretically, a refined analysis of Bernstein-von Mises (BvM) theorem is provided to accommodate a diverging number of subsets m . The findings indicate that the two-stage approach yields a smaller approximation error compared to directly aggregating the m -powered local posteriors. Furthermore, a nearly optimal statistical rate for the mean of the proposed posterior is established, advancing existing theories related to min-max median estimators. The efficacy of the method is demonstrated through extensive simulation studies.

E1537: Multi-relational network autoregression model with latent group structures**Presenter:** Yimeng Ren, The Hong Kong University of Science and Technology, China**Co-authors:** Xuening Zhu, Ganggang Xu, Yanyuan Ma

Multi-relational networks among entities are frequently observed in the era of big data. Quantifying the effects of multiple networks has attracted significant research interest recently. Multiple network effects are modeled through an autoregressive framework for tensor-valued time series. To characterize the potential heterogeneity of the networks and handle the high dimensionality of the time series data simultaneously, a separate group structure for entities in each network is assumed and all group memberships are estimated in a data-driven fashion. Specifically, a group tensor network autoregression (GTNAR) model is proposed, which assumes that within each network, entities in the same group share the same set of model parameters, and the parameters differ across networks. An iterative algorithm is developed to estimate the model parameters and the latent group memberships simultaneously. Theoretically, it is shown that the group-wise parameters and group memberships can be consistently estimated when the group numbers are correctly or possibly over-specified. An information criterion for estimating the group number for each network is also provided to consistently select the group numbers. Lastly, the GTNAR method is applied to a Yelp dataset to illustrate its usefulness.

E2021: Robust high-dimensional multivariate regression with non-Gaussian errors via global-local shrinkage priors**Presenter:** Dongu Han, Meiji University, Japan**Co-authors:** Jichan Park, Genya Kobayashi, Taeryon Choi

High-dimensional multivariate regression models that rely on Gaussian error assumptions are often sensitive to outliers and distributional skewness, which limits their practical applicability. To address this limitation, a robust multivariate Bayesian regression framework that replaces the Gaussian error distribution with flexible alternatives, including the multivariate-t, skew-normal, and skew-t distributions is proposed. To accommodate high-dimensional settings in which the number of parameters exceeds the sample size, horseshoe+ priors are imposed on both the regression coefficients and the off-diagonal elements of the precision matrix. This specification induces joint sparsity in the regression structure and the conditional dependence graph. For posterior computation, efficient Gibbs sampling algorithms are developed that exploit scale-mixture representations of

the non-Gaussian error distributions and an alternative mean-field variational Bayes approximation that substantially reduces computational cost. Posterior and variable selection consistency are established, a tighter Kullback-Leibler risk bound over the standard horseshoe prior is derived, and robustness to unbounded outliers under the multivariate-t models is demonstrated. Simulation studies, along with applications to macroeconomic and cancer genomics datasets, demonstrate that the proposed method outperforms state-of-the-art competitors in estimation accuracy and predictive performance.

EO147 Room 208 MODERN METHODS FOR BAYESIAN COMPUTATION AND SAMPLING
Chair: Jieun Lee
E1316: Convergence analysis of parametric Fisher-Rao flow

Presenter: **Xin Tong**, National University of Singapore, Singapore

Fisher-Rao flow (FRF) is the natural gradient flow in measure space. Recent studies have shown non-parametric FRF to be converging with a rate that is constant under mild conditions. However, in practical sampling problems, it is more convenient to consider parametric FRF. The conditions under which the constant convergence rate will be maintained are discussed; and if it is not, methods to improve the convergence.

E1366: Accelerating sampling using birth-death dynamics

Presenter: **Lihan Wang**, National University of Singapore, Singapore

Birth-death dynamics for sampling multimodal probability distributions, which is the Fisher-Rao gradient flow of relative entropy, is discussed. The advantage of birth-death dynamics is that, unlike local dynamics such as Langevin dynamics, it allows global movement of mass directly from one mode to another, without the difficulty of going through low probability regions. It is proven that birth-death dynamics converges to the unique invariant measure with a uniform rate under some mild conditions, showing its potential for overcoming metastability. It is also shown that on a torus, the kernelized dynamics, which is used for numerical simulation, Gamma-converges to the idealized dynamics as the kernel bandwidth shrinks to zero.

E1668: Wasserstein Fisher Rao gradient flows: Sequential Monte Carlo and operator splitting

Presenter: **Sahani Pathiraja**, University of New South Wales, Australia

Wasserstein Fisher Rao (WFR) Gradient flows have recently been proposed for sampling from a target probability distribution, a task common across machine learning, Bayesian statistics, and statistical mechanics. Such Gradient flows have been shown to accelerate convergence over pure Wasserstein Gradient flows by combining the benefits of diffusive transport (Wasserstein) and reweighting (Fisher-Rao) type Gradient flows. WFR flows for the Kullback-Leibler divergence are connected to well-known Monte Carlo algorithms. This connection reveals a natural implementation using importance sampling and motivates a sequential Monte Carlo based algorithm for solving a WFR Gradient flow. Additionally, operator splitting can further accelerate convergence to equilibrium with a judiciously chosen order of operators.

EO287 Room 209 SURVIVAL ANALYSIS METHODOLOGIES
Chair: Takeshi Emura
E2040: Covariate hypothesis tests for the cure rate in mixture cure models

Presenter: **Ana Lopez-Cheda**, University of A Coruna, Spain

Co-authors: Maria Amalia Jacome Pumar, Antia Dorado

Survival analysis studies the time to an event in the presence of censoring. In many biomedical studies, a group of individuals does not experience the event, which motivates the use of cure models. Mixture cure models (MCM) allow for the separation of the incidence (probability of being susceptible) and the latency (survival function of susceptible individuals). A key aspect is evaluating the influence of covariates on both parts of the MCM using hypothesis tests. Two approaches of nonparametric methods, implemented in the R packages npcure and MDcure, are reviewed. Furthermore, two approaches that combine both methods are proposed. The behavior of the four methods is analyzed through an extensive simulation study. Finally, the methodology is applied to a dataset of breast cancer patients.

E1594: Proportional hazards regression for interval-censored outcomes with an interval-censored covariate

Presenter: **Dongdong Li**, Harvard Medical School, United States

Identifying predictors for viral rebound trajectories after antiretroviral therapy (ART) interruption is central to HIV cure research. Motivated by the need to determine whether the time to achieve viral suppression after ART initiation can predict the time to viral rebound following ART interruption, modeling approaches that relate an interval-censored outcome (e.g., time to viral rebound) and an interval-censored covariate (e.g., time to viral suppression) are investigated under the assumption that viral load only crosses a threshold when bracketed by consecutive assessments. Estimation and inference procedures are developed for fitting a proportional hazards regression model when both the outcome and a covariate are interval-censored, without imposing parametric assumptions on the baseline hazard functions. To accommodate participants with multiple episodes of ART initiation and interruption, the proposed method is extended to account for the clustering of repeated observations within individuals. The asymptotic properties of the proposed method are derived and its finite-sample performance is evaluated through simulation studies. Applying the method to data from the Zurich Primary HIV Infection cohort, a longer time to viral suppression during ART is found to be associated with an increased hazard of viral rebound after ART interruption.

E1588: On the implications of proportional hazards assumptions for competing risks modelling

Presenter: **Ralf Wilke**, Copenhagen Business School, Denmark

Co-authors: Ming Sum Simon Lo, Takeshi Emura

The assumption of hazard rates being proportional in covariates is widely made in empirical research and extensive research has been done to develop tests of its validity. Rather than contributing to this literature, new insights are provided on the implications of proportional hazards (PH) modelling in competing risks models. It is shown that the use of a PH model for the cause-specific hazards or subdistribution hazards can strongly restrict the class of copulas and marginal hazards compatible with a competing risks model. Empirical researchers should be aware that working with these models can be so restrictive that only degenerate or independent risks models are compatible. Numerical results confirm that estimates from cause-specific hazards models are not informative about patterns in the data generating process.

EC293 Room B104 STRUCTURAL BREAKS
Chair: Kaiji Motegi
E1893: Confidence sets for the emergence, collapse, and recovery dates of a bubble

Presenter: **Eiji Kurozumi**, Hitotsubashi University, Japan

Co-authors: Anton Skrobotov

Confidence sets are constructed for the emergence, collapse, and recovery dates of a bubble separately by inverting tests for the location of the break date. Both likelihood ratio-type tests and Elliott-Muller-type tests for detecting break locations are examined. The limiting distributions of these tests are derived under the null hypothesis, and their asymptotic consistency under the alternative is established. Finite-sample properties are evaluated through Monte Carlo simulations. The results indicate that combining different types of tests effectively controls the empirical coverage rate while maintaining a reasonably small length of the confidence set.

E1362: Weighted inference from long historical samples under structural change*Presenter:* **Etsusaku Shimada**, Iwate Prefectural University, Japan

Empirical work often relies on long historical samples even when the relevance of older observations changes over time. Weighted inference from historical data under structural change is studied, focusing on how alternative weighting schemes affect current-oriented estimation. The framework treats weights as a way to control the contribution of observations generated under different structural environments, rather than simply as a device for variance reduction. Estimation and asymptotic inference are developed under weak dependence, with particular attention to the effective sample size induced by downweighting older observations. Simulation experiments compare full-sample, rolling-window, and smoothly weighted procedures, illustrating how recency emphasis changes bias, variance, and estimation accuracy when structural relationships evolve gradually. The results clarify how long historical samples can be used more transparently when empirical environments are not stable over time.

E1282: Detecting structural breaks in financial time series*Presenter:* **Urban Ulrych**, ETH Zurich, Switzerland*Co-authors:* Tommaso Venturino, Patrick Cheridito

Financial time series are subject to frequent structural instability, which complicates forecasting, risk measurement, and real-time monitoring. A unified framework is proposed that integrates adaptive state-space diagnostics with global offline segmentation and strictly causal online regime recognition. Offline, a Kalman filter with Dempster-Shafer covariance adaptation is combined with penalized exact segmentation to localize structural breaks across heterogeneous data-generating processes. Online, break detection is formulated as a filtration-respecting probabilistic classification problem and implemented using a LightGBM learner based on strictly causal features derived from both raw returns and model diagnostics. Across synthetic processes with known ground truth, the framework achieves balanced break identification offline and stable short-horizon recognition relative to variance-based and Bayesian benchmarks. Applied to a cross-asset panel of financial series, it identifies economically interpretable regime shifts while maintaining parsimonious segmentation. The results highlight the complementarity of adaptive state-space monitoring and causal machine learning for structural break analysis in financial econometrics.

EC245 Room 104 HIGH-DIMENSIONAL METHODS**Chair: Myungjin Kim****E1230: Correlation comparison after nonlinear transformations***Presenter:* **Elio Zhang**, Seattle University, United States

Testing the equality of two or more correlation matrices is a fundamental problem in many scientific fields, especially when the number of variables is large relative to the sample size. Recent work has shown empirically that coupling entrywise transformations with permutation can substantially improve power in such settings, yet the mechanism behind these gains and principled guidance for choosing transformations remain unclear. These gaps are addressed through two contributions. First, a theoretical explanation is provided of why and how entrywise transformations, when paired with a permutation (random-partition) calibration, can elevate power: the transformation reshapes and amplifies distributional discrepancies across matrix entries while the permutation procedure preserves valid type-I error control under the null. Second, and more importantly, a power-driven criterion is developed for selecting (within a broad class of smooth transformations) an optimal transformation that maximizes asymptotic power without inflating type-I error. The optimality of the proposed choice is established both theoretically and via extensive simulations, and the practical utility of the method is demonstrated through analyses of two real data examples.

E1873: Group-sparse smoothing for longitudinal models with time-varying coefficients*Presenter:* **Yu Lu**, XJTLU, China*Co-authors:* Tianni Zhang, Yuyao Wang, Mengfei Ran

Longitudinal data analysis is fundamental for understanding dynamic processes in biomedical and social sciences. Although varying coefficient models provide a flexible framework by allowing covariate effects to evolve over time, fitting all effects as time-varying may lead to overfitting, efficiency loss, and reduced interpretability when some effects are actually constant. In contrast, standard linear mixed models may suffer substantial bias when temporal heterogeneity is ignored. To address this issue, a time-varying effect selection method, TV-Select, is proposed as a unified framework for structural identification that simultaneously selects relevant variables and determines whether their effects are constant or time-varying. The method decomposes each coefficient function into a time-invariant mean component and a centered time-varying deviation, where the latter is approximated by B-splines. A doubly penalized objective function is then constructed that combines a group Lasso penalty for structural sparsity with a roughness penalty for smoothness control. An efficient block coordinate descent algorithm is developed for computation. Under regular semiparametric conditions, selection consistency and oracle-type asymptotic properties are established, including asymptotic normality for the constant-effect component after correct structure recovery.

E2027: A latent divergence index for high-dimensional streaming markets: Cross-country dynamics in music consumption*Presenter:* **Donato Ferrari**, University of Calabria, Italy*Co-authors:* Giuseppina Damiana Costanzo, Alessandro Faranda Gancio

Digital streaming platforms are often associated with Cultural homogenization through globally synchronized recommendation systems. A multivariate statistical framework is developed to evaluate whether national music markets are instead becoming differentiated over time. Using weekly Spotify chart data from 30 countries between 2020 and 2026, high-dimensional country-week musical profiles are constructed based on acoustic, rhythmic, harmonic, and mood-related descriptors extracted from Spotify and Essentia systems. Principal component analysis, similarity networks, clustering methods, and multivariate distance measures characterize the Latent structure of streaming markets. A new Latent Cultural Divergence Index (LCDI) is introduced, defined as the average dispersion of national markets around a global Latent centroid in principal component space. The first two principal components explain approximately 68 percent of total variance. Results show a significant increase in Latent Divergence over time. Cross-country similarity declines persistently, while feature space distances increase substantially. Emotional and perceptual dimensions exhibit fragmentation patterns, whereas technical production characteristics remain comparatively stable. Cluster analysis reveals persistent country-specific Latent regimes and unstable hybrid markets.

EC268 Room 106 PROBABILISTIC METHODS AND STOCHASTIC PROCESSES**Chair: Ansgar Steland****E1361: Generalization of CNNs under $\mathcal{C}$ -mixing dependence****Presenter:** Fangjia Dong, Hong Kong Baptist University, Hong Kong

The generalization performance of convolutional neural networks (CNNs) for nonparametric regression is investigated under two fundamental challenges: heavy-tailed noise and data dependence. Unlike existing works assuming sub-Gaussian noise or finite variance, much weaker moment conditions are considered: the noise only requires a finite p -th conditional moment and a q -th marginal moment for $p, q > 1$, allowing for infinite variance. The input sequence dependence is formalized via the $\mathcal{C}$ -mixing condition. To address these challenges, a robust learning framework is developed using the Huber loss function and complexity-regularized CNN hypothesis spaces. By leveraging a recent Bernstein concentration inequality for $\mathcal{C}$ -mixing sequences, high-probability convergence rates for the empirical risk minimizer are established. The results show that, even under these minimal assumptions, CNNs can achieve minimax-optimal generalization rates up to logarithmic factors. Overall, the findings demonstrate the feasibility of robust deep learning in realistic scenarios with complex dependencies and heavy-tailed, potentially infinite-variance noise.

E1858: SEM for jump-diffusion processes based on high-frequency data**Presenter:** Shogo Kusano, Kumamoto University, Japan**Co-authors:** Masayuki Uchida

Structural equation modeling (SEM) for diffusion processes with jumps based on high-frequency data is considered. In recent years, high-frequency data have become easily available due to advances in information technology. To conduct SEM based on high-frequency data, SEM for diffusion processes has been previously proposed. However, in many applications, discontinuous sample paths are often observed. Since SEM for diffusion processes assumes continuous paths, it is not suitable for analyzing such data. To address this issue, SEM for diffusion processes with jumps is proposed. By using a threshold method based on increments, a quasi-likelihood function for the SEM is constructed. The quasi-maximum likelihood estimator is shown to have consistency and asymptotic normality. Furthermore, to examine whether a specified parametric model is correctly specified, a quasi-likelihood ratio test statistic is developed and its asymptotic distribution is derived. The proposed framework enables analysis of relationships between latent processes based on high-frequency data with jumps. Finally, numerical simulations are conducted to illustrate the finite-sample performance of the estimators.

E1877: Joint distributions of maximum drawdown and maximum drawup for spectrally negative Levy processes**Presenter:** Ceren Vardar Acar, Middle East Technical University, Turkey, Dumlupinar Bulvari No:1, Turkey

Path-decomposition and scale-function methods are presented for the joint analysis of maximum drawdown and maximum drawup of spectrally negative Levy processes observed up to an independent exponential time. The methodological decomposition of the problem is stated according to the two possible orderings of the infimum and the supremum. The path decompositions are collected at the stopping times of the infimum and the supremum, including the two orderings of these extremal times. These decompositions result in explicit conditional distributions of the maximum drawdowns and maximum drawups of the independent path segments, and the conditional laws are combined with the joint distributions of the extrema to yield the joint distribution of the maximum drawdown and maximum drawup. The final section provides a summary of the contribution and discusses extensions to durations and statistical applications.

EC258 Room 107 CAUSAL INFERENCE**Chair: Zudi Lu****E1933: Dynamic causal modelling and spectral analysis****Presenter:** Jian Zhang, University of Kent, United Kingdom

Pathophysiological modelling of brain systems from microscale to macroscale remains difficult in group comparisons partly because of the infeasibility of modelling the interactions of thousands of neurons at the scales involved. To address this challenge, a novel approach is presented to construct differential causal networks directly from electroencephalogram (EEG) data. The proposed network is based on conditionally coupled neuronal circuits which describe the average behaviour of interacting neuron populations that contribute to observed EEG data. In the network, each node represents a parameterised local neural system while directed edges stand for node-wise connections with transmission parameters. The network is hierarchically structured in the sense that node and edge parameters vary across subjects but follow a mixed-effects model. Spectral power analysis is further conducted. The method is used to identify dynamic causal networks and differential power regions for epilepsy.

E1331: A design-based approach to testing and inference in quasi-experiments with spillovers**Presenter:** Yechan Park, Harvard University, United States

Economic policies often affect not only their direct recipients but also untreated units through spatial, market, and social spillovers. Credibly measuring such spillovers is therefore central to policy evaluation, yet doing so requires researchers to specify an exposure mapping that compresses a complex assignment vector into a tractable measure of treatment exposure. In practice, both the form of this mapping and its tuning parameters are often chosen using theory, diagnostics, or model selection, then treated as fixed in subsequent inference. A finite-population, design-based framework is developed for testing and estimating exposure mappings in settings with interference. Exploiting only the known experimental or quasi-experimental assignment design, moment restrictions implied by a correctly specified exposure mapping are derived and used to (i) test whether a proposed exposure class is consistent with the data and (ii) estimate its tuning parameters without imposing additional superpopulation or outcome-distribution assumptions. A design-based law of large numbers and central limit theorem are established for the resulting GMM estimators under spatial and network dependence, efficiency bounds and optimal moment constructions are characterized, and methods are shown for propagating first-step exposure uncertainty into downstream policy estimands.

E1167: Comparing performance of three propensity score weighting methods for continuous exposures in nutritional epidemiology**Presenter:** Yuriko Muramatsu, Kyoto University, Japan**Co-authors:** Tosiya Sato, Shiro Tanaka

Inverse probability of treatment weighting (IPTW) estimates causal effects in entire study population by eliminating the bias due to observed confounders. For continuous exposures, it is necessary to deal with the large weights caused by a lack of positivity, and to identify the exposure distribution to estimate generalized propensity score (GPS), the conditional probability of receiving a certain level of exposure. Generalized overlap weighting (GOW) and generalized matching weighting (GMW) mitigate the influence of a lack of positivity, so they may estimate exposure effects with less bias and higher precision. Quantile binning approach estimates GPS by dividing exposures into bins, thereby preventing model misspecification. The primary and secondary aims are to compare the performance of three weighting methods and to evaluate the favorable number of bins. Using a dataset from an epidemiologic study of patients with diabetes, the associations between continuous nutrient intake and diabetes complications risk were estimated. The performance metrics were covariate balance, weight variability, and the precision of estimates. Results showed that the performance of the three methods was favorable with 2 or 5 bins and depended on the degree of overlap in the GPS. With reduced overlap, GOW was effective for confounding adjustment for continuous exposures.

EC255 Room 201 SHORT TALKS IN STATISTICAL MODELLING AND COMPUTING**Chair: Man Ho Ling****E2005: Exact distribution of some functions of the ordered multinomial counts: Computational aspects and applications****Presenter:** Sergio Venturini, Università Cattolica del Sacro Cuore, Italy**Co-authors:** Marco Bonetti

Testing whether a multinomial distribution is consistent with an equal-probability null hypothesis arises across a broad range of statistical applications. An alternative to classical test statistics is based on the order statistics of the multinomial counts: the maximum $N_{<1}>$, minimum $N_{<m}>$, and range $R = N_{<1}> - N_{<m}>$. Computing the exact distributions of these statistics is a non-trivial computational problem. An explicit recursive derivation of the exact CDFs of $N_{<1}>$ and $N_{<m}>$ is provided based on direct conditioning on individual multinomial counts and the closure property of the multinomial distribution. The derivation proceeds through nested binomial probabilities, making the probabilistic content of each step transparent. Formal equivalence between this construction and the matrix-based approach is established. It is also shown that the exact power of tests based on $N_{<1}>$ and $N_{<m}>$ can be computed analytically under structured one-sided alternatives, enabling closed-form sample size determination. The XOMultinom R package unifies these methods in a computationally efficient implementation and allows calculation of power functions and sample sizes. Applications involving sample size determination and the signalling of miscalibration in clinical risk prediction models illustrate the use of the package.

E1909: A functional data analysis approach for income distribution and poverty analysis**Presenter:** Gianni Betti, University of Siena, Italy**Co-authors:** Federico Crescenzi

Functional data analysis is shown to be a useful tool for investigating features of income distribution. By assuming a finite population approach and a fuzzy approach to poverty measurement, where each individual has a degree of belonging to a poverty set defined by a membership function, it is demonstrated that functional data analysis can be employed to model the membership function and unlock the potential of more complex poverty indicators over the entire income distribution domain.

E2091: On the relative importance of explanatory variables for the multivariate regression model**Presenter:** Chao-Ting Yu, National Chengchi University, Taiwan**Co-authors:** Tsung-Chi Cheng

Assessing the relative importance of explanatory variables is a central issue in regression analysis. Existing methods, such as the Lindeman-Merenda-Gold (LMG) method, proportional marginal variance decomposition (PMVD), dominance analysis (DA), and relative weights analysis (RWA), were mainly developed for models with a single response variable. However, many empirical studies involve multiple correlated outcomes that are better represented by multivariate linear regression. Relative importance analysis is extended to the multivariate setting by defining overall model fit as a scalar function of the residual and explained SSCP matrices. Trace-based and determinant-based R^2 - and P^2 -type criteria are considered, and multivariate versions of LMG and PMVD are derived. A unified matrix-based framework is proposed to decompose multivariate model fit into predictor-specific contributions. The proposed methods allow explanatory power to be allocated coherently under different fit criteria while accounting for correlations among response variables. Simulation results and a real-data application illustrate the statistical behavior, interpretability, and practical usefulness of the proposed methods compared with existing approaches.

E2092: Interactive tools for reliable omics visualization**Presenter:** Kris Sankaran, University of Wisconsin, United States

Low-dimensional views of high-dimensional data are central in genomics, including differentiation trajectories in single-cell data and community profiles in metagenomics. Such reductions are powerful but sensitive to hyperparameters and prone to misinterpretation. Two risks in high-dimensional visualization are examined. First, taxa selection for microbiome composition barplots is considered, showing variation while preventing blurred color encodings. A package for constructing color palettes across taxonomic categories at multiple resolutions is introduced, with case studies demonstrating its ability to surface associations that static barplots obscure. Second, distortions introduced by nonlinear dimensionality reduction methods t-SNE and UMAP in single-cell data are examined. These methods can generate spurious clusters and fail to preserve cell density. The RMetric algorithm from manifold learning is adapted to quantify local distortions, and two visualizations are developed to expose them: cells rendered as deformed ellipses to highlight local geometric change, and an interactive interface that reveals distortion evidence without overwhelming the viewer. Case studies on simulated and real data show the tools can flag fragmented neighborhoods, guide hyperparameter tuning, and support method selection.

EC295 Room 204 EXTREME VALUES**Chair: Jingru Mu****E1411: Multiple testing for spatial extremes with application to reanalysis data evaluation****Presenter:** Sooin Yun, CUNY Baruch College, United States

Climate data products, such as reanalysis datasets, require careful evaluation to determine their reliability. While most climate data evaluations focused on the mean and dependency of climate processes, the focus is on marginal extreme behavior instead, including return levels that often have devastating impacts on ecosystems and societies. In particular, the aim is to identify where the two climate extreme fields exhibit different marginal behavior, by simultaneously evaluating the differences over all spatial locations through multiple testing techniques. The large variation inherited in extreme model fitting makes this evaluation more challenging than that for the mean and dependency structure. A new multiple testing procedure, bivariate conditional local fdr (BiCLfdr), is proposed to efficiently detect signals from highly variable but spatially correlated hypotheses. The method takes advantage of both the smoothness of large-scale spatial variability and the local spatial correlation to enhance the power of comparing the marginal extreme distribution of two spatial extremes. BiCLfdr is applied to evaluate ERA5 reanalysis in its representation of winter precipitation extremes across the United States, using CPC-CONUS observations as reference. The analysis identifies locations along the West Coast, the western Great Plains, and the Southeast where ERA5 appears to inadequately represent observed winter precipitation extremes.

E2060: Mastering extremes in time series**Presenter:** Clara Cordeiro, Center for Research and Development in Mathematics and Applications (CIDMA), Portugal**Co-authors:** M Manuela Neves, Dora Prata Gomes, Pablo Montero Manso

Extreme events in univariate time series present major challenges in applications such as river flow modelling, financial risk assessment, and water consumption analysis. Classical forecasting models often fail to adequately capture tail behaviour, leading to poor prediction of rare observations. Extreme Value Theory (EVT) provides a rigorous framework for modelling such extremes. A hybrid methodology is proposed in which a standard forecasting model is first fitted to the time series, and the residuals are then fitted using EVT-based distributions. Particular emphasis is placed on estimating tail characteristics using bootstrap methods to estimate the shape parameter, enabling a more robust representation of extreme behaviour in dependent data. Bootstrap methods are explored and compared across real-data applications, while numerical experiments demonstrate the effectiveness of the proposed framework in improving the modelling and prediction of extreme events. The benefits of combining classical time series forecasting, EVT, and bootstrap-based inference for rare-event analysis are highlighted.

E1925: Conditional EVT-GAN for extreme precipitation risk mapping

Presenter: **Kang Sora**, Chonnam National University, Korea, South

Co-authors: Na Myung Hwan, Yoon Sanghoo

As climate change and environmental variability intensify extreme events, mapping regional risks has become essential for climate and environmental risk assessment, return-period estimation, disaster prevention, and policy planning. Extreme value theory (EVT) models block maxima or high-concentration extremes using the generalized extreme value (GEV) distribution, but conventional EVT is limited in representing gridded risk patterns. Generative adversarial networks (GANs) can learn spatial fields, but standard GANs often fail to preserve extreme upper-tail characteristics. EvtGAN combines EVT and GAN by fitting GEV distributions to block maxima, transforming data into a uniform space, and restoring samples through inverse GEV transformation. However, existing EvtGAN mainly relies on $r=1$ block maxima, which limits training samples under short observation periods. Moreover, GEV parameters are used only in transformation steps, not explicitly in the generator or loss function. An improved conditional EvtGAN is proposed that uses pixel-wise r -th order statistics and incorporates GEV parameters into an EVT-based loss function. The goal is to generate spatial risk maps based on annual maxima or upper-order statistics, rather than simultaneous event fields. This approach is expected to support environmental extreme risk mapping, risk simulation under climate and environmental change, and regional risk assessment.

Sunday 09.08.2026

16:15 - 17:55

Parallel Session J – EcoSta2026

EI002 Room 202 ADVANCED METHODS FOR FINANCIAL AND COMPLEX DATA**Chair: Cathy W-S Chen****E1152: Financial risk forecasting in the data centric era***Presenter:* **Richard Gerlach**, University of Sydney, Australia*Co-authors:* Chen Liu, Minh-Ngoc Tran, Chao Wang, Robert Kohn

How data size, model size, and architectural choice affect the performance of neural networks in financial risk forecasting is investigated. Using volatility forecasts for more than 10,000 stocks, it is found that, within the specifications studied, performance gains from expanding the size and diversity of the training data outweigh those from model scaling or architectural choice. It is further shown that, when neural networks are applied to financial data, global estimation is better aligned with their data-intensive nature and enhances their operational viability.

E1153: Dynamic factor stochastic volatility-in-mean VAR for large macroeconomic panels*Presenter:* **Yasuhiro Omori**, University of Tokyo, Japan*Co-authors:* Daichi Hiraki, Siddhartha Chib

A dynamic factor stochastic volatility-in-mean (SVM) specification for vector autoregressions (VARs) is developed that embeds an SVM component within a dynamic factor stochastic volatility structure. A small number of latent volatility factors capture common movements in conditional variances, while volatility enters the conditional mean of the VAR. This specification allows time-varying uncertainty to influence macroeconomic dynamics through both second moments and expected outcomes while preserving tractability in large panels. An efficient Markov chain Monte Carlo algorithm is constructed for estimation in this high-dimensional, non-Gaussian setting. Using quarterly data on twenty variables from the FRED-QD database, predictive performance is compared with the benchmark stochastic volatility VAR model. The dynamic factor SVM specification delivers superior forecasts for more variables during major macroeconomic disruptions such as the 2008 global financial crisis. The results indicate that allowing volatility to enter the mean captures an important transmission channel in macroeconomic dynamics.

E1154: Integration of textual and quantitative responses in multidimensional item response theory*Presenter:* **Mike So**, The Hong Kong University of Science and Technology, Hong Kong*Co-authors:* Thomas Chan, Amanda Chu

The integration of rich unstructured textual data from open-ended responses with structured quantitative questionnaire data presents opportunities to uncover nuanced latent characteristics, but their disparate formats hinder unified analysis. A novel multidimensional item response model is proposed that seamlessly combines topic modeling for textual data and item response theory for quantitative data. Drawing on embedded space modeling, topics are projected into the latent trait space via a multivariate t density function, enabling proximity-based topic preferences that align with respondents intrinsic traits. For quantitative items, item group factors are introduced, represented by non-decreasing spline smoothing, to capture group similarities and non-linear relationships with latent traits. The model accommodates ordinal responses, handles missing data by modeling response probabilities, and assumes conditional independence. Parameter estimation employs variational Bayes. Simulation studies assess the impact of topic selection methods. Application to real-world psychosocial health data yields more profound insights into respondents patterns. This framework advances joint analysis of mixed data types, facilitating a more profound understanding across a wide range of research.

EO152 Room B101 POLICY LEARNING AND STATISTICAL DECISION**Chair: Chen Qiu****E1519: Robust network targeting with multiple Nash equilibria***Presenter:* **Guanyi Wang**, Washington University in St Louis, United States

Many policy problems involve designing individualized treatment allocation rules to maximize the equilibrium social welfare of interacting agents. Focusing on large-scale simultaneous decision games with strategic complementarities, an optimal treatment allocation mechanism is developed that is robust to the presence of multiple equilibria. The approach remains agnostic about changes in the equilibrium selection mechanism under counterfactual policies, and a closed-form expression for the boundary of the set-identified equilibrium outcomes is provided. To address the incompleteness that arises when an equilibrium selection mechanism is not specified, the maximin welfare criterion is used to select a policy, which is implemented using a greedy algorithm. A performance guarantee for the method is established by deriving a welfare regret bound, which accounts for sampling uncertainty and the use of the greedy algorithm. The method is demonstrated with an application to a microfinance dataset.

E1787: Multi-objective policy learning*Presenter:* **Daido Kido**, Otaru University of Commerce, Japan

When treatment effects exhibit heterogeneity, allocating treatments to individuals who benefit most is essential for constructing efficient policies. However, treatments often influence multiple outcomes rather than a single one. In such cases, a treatment allocation optimized for one specific outcome may result in undesirable assignments from the perspective of another outcome. Therefore, understanding these trade-offs is crucial. Estimation and inference methods are developed for the Pareto frontier -the boundary where it is impossible to improve one outcome without worsening another. The estimator is shown to be consistent and confidence sets are shown to asymptotically achieve nominal coverage.

E1813: Policy learning with unstructured data*Presenter:* **Yue Fang**, The Chinese University of Hong Kong, China

Unstructured data, such as text and images, are increasingly central to empirical policy problems, yet most policy learning theory assumes low-dimensional, structured features. Policy learning when the available information takes the form of high-dimensional embeddings, from which researchers construct low-dimensional features to design welfare-maximizing policies, raises a key econometric challenge: the estimated features act as generated regressors and can introduce first-order bias into welfare evaluation. The bias is characterized by deriving the pathwise derivative of the welfare functional, revealing how the estimation error propagates through nuisance functions and the induced policy. A debiased policy learning method is proposed that removes the first-order bias and restores regret guarantees with the same rate as if the low-dimensional features were directly observed. The framework is illustrated using popular supervised and unsupervised feature-extraction methods, including average-derivative-type constructions, principal component analysis, latent semantic analysis, and clustering. An empirical application studies the use of satellite-based embeddings to guide mosquito net allocation and improve health outcomes in Nigeria.

E1788: Asymptotics for treatment decisions with partial identification*Presenter:* **Chen Qiu**, Cornell University, United States*Co-authors:* Jose Luis Montiel Olea, Joerg Stoye

A common approach in the study of treatment choice problems with partial identification assumes a normal location shift model and seeks minimax regret optimal decisions by jointly accounting for point-identified and set-identified parameters. Within this framework, substantial progress has been made when the parameter space is convex and centrosymmetric. Yet, in practice, data may not be normal and the parameter space may not be convex or centrosymmetric. A new, practical and scalable asymptotic approach to treatment decisions with partial identification is proposed. This approach justifies the common approach as in fact providing local asymptotic optimality results when the parameter space is convex and

centrosymmetric, and can accommodate a large class of models even beyond convexity or centrosymmetry. The approach is also computationally cheap, offering an effective way for practitioners to find large-sample optimal decisions in many models.

EO158 Room B102 MACROECONOMIC POLICIES AND ASSET MARKETS
Chair: Etsuro Shioji
E1569: Consumption rigidity, household heterogeneity, and nonmonotonic risk compensation

Presenter: **Shoka Hayaki**, Kobe University, Japan

A growing body of evidence documents that aggregate risk compensation declines during weak, high-volatility states, a finding that is difficult to reconcile with standard habit formation models. A mechanism based on heterogeneity in the onset of consumption rigidity is proposed: when households differ in the states at which consumption commitments bind, marginal-utility reweighting generates a transition from countercyclical to procyclical aggregate effective risk aversion. Using postwar U.S. data through 2019, risk compensation peaks near median economic states and subsequently declines as conditions deteriorate, exactly as the mechanism predicts. This pattern is robust across specifications, state variable constructions, and sample periods.

E1565: Historical debt dynamics in the US and UK: A TVP-VARX analysis from the 1800s to the 2020s

Presenter: **Hiroshi Morita**, Institute of Science Tokyo, Japan

The sources of public debt fluctuations in the United States and the United Kingdom over two centuries (1800s-2020s) are examined using a time-varying parameter VAR with exogenous variables (TVP-VARX). A novel long-run historical dataset is exploited to estimate time-varying fiscal reaction functions and conduct historical decompositions that disentangle the roles of macroeconomic shocks and discretionary fiscal policy. The identification strategy relies on sign restrictions that distinguish supply, demand, and primary balance shocks. The primary balance responds positively to the debt ratio in both countries, suggesting that fiscal policy has not been systematically irresponsible, although the magnitude of this response is modest. Historical decomposition reveals that macroeconomic shocks, rather than discretionary fiscal policy, account for the bulk of debt dynamics, particularly in recent decades. These findings highlight the period-specific nature of fiscal sustainability and suggest that debt accumulation is largely driven by good luck rather than good policy.

E1495: The role of the average maturity in the transmission of quantitative easing: Evidence from the Bank of Japan

Presenter: **Qiaoxi Li**, Kobe University, Japan

The transmission of the Bank of Japan's (BOJ) unconventional monetary policy is investigated by analyzing the maturity structure of its holdings of Japanese government bonds from 2001 to 2024. First, the analysis documents the distinct shifts in the BOJ's portfolio composition across different policy regimes, highlighting the strategic variations in maturity targeting. Second, using cointegration analysis, a stable long-run equilibrium is identified in which extending the average maturity significantly reduces the 10-year yield; regression analysis further reveals that this reduction operates through a dual channel, simultaneously compressing risk premia and lowering interest rate expectations. Finally, local projections reveal that maturity extension shocks lead to persistent increases in the price level and bank lending, alongside cyclical improvements in industrial production. Counterfactual simulations highlight a distinct contrast between policy regimes: During the quantitative easing policy period, the concentration of purchases in relatively short-term bonds reduced the average maturity, thereby offsetting potential stimulative effects; conversely, the rapid maturity extension during the quantitative and qualitative easing period significantly drove price growth and credit expansion.

E1486: Is the market for the Japanese Government Bonds still insulated from external forces?

Presenter: **Etsuro Shioji**, Chuo University, Japan

How effective are monetary policies in insulating domestic markets for government bonds from foreign pressures? The question is addressed by studying the market for the Japanese government bonds (JGBs), where the Bank of Japan (BOJ) has been a dominant player since 2013. For that purpose, a new data set on the BOJ's estimated daily holdings of the JGBs, classified by both their terms and issuance dates, is utilized to construct a daily measure of the BOJ's JGB market intervention. This allows estimation of how daily changes in external market forces, represented by the US treasury bond yields, impact both the BOJ's bond purchases and the JGB yields. It is shown that the BOJ, when it chose to do so, could effectively shut out influences of foreign markets from the domestic bond market through aggressive bond purchases. On the flip side, the finding implies that, with the termination of the BOJ's unconventional monetary policies in 2024, the JGB market could now see enhanced influences of external forces.

EO212 Room Virtual R01 DIFFERENT ASPECTS OF ROBUST STATISTICS
Chair: Giovanni Saraceno
E1661: Accurate Bayesian inference for tail risk extrapolation in time series

Presenter: **Stefano Rizzelli**, University of Padova, Italy

Co-authors: David Carl, Simone Padoan

A Bayesian approach is developed for inference on extreme values, focusing on high-threshold exceedances, and explicitly addressing two common sources of model misspecification: the choice of the peaks distribution and the presence of temporal dependence not captured by the working likelihood. In practice, peaks-over-threshold analyses typically rely on modelling assumptions (e.g., a Generalised Pareto model, justified by extreme-value theory) and independence across exceedances, both of which are typically violated. The proposed methodology is designed to remain robust to misspecification of the peaks distribution and to residual temporal dependence, delivering reliable inference for key tail quantities even when these assumptions fail. Posterior consistency, asymptotic normality, and the asymptotic validity of credible regions are established, thereby providing theoretical support for the methodology. Simulation studies illustrate the impact of such misspecification on standard methods and demonstrate the improved accuracy of the proposed approach, while applications to real data highlight its practical value.

E1742: Weighted likelihood robust inference using statistical data depths

Presenter: **Giulia Bertagnolli**, Free University of Bozen-Bolzano, Italy

Co-authors: Claudio Agostinelli, Ayanendranath Basu, Arun Kuchibhotla

A Depth-based weighting scheme for likelihood inference is proposed that delivers robust parameter estimates with minimal structural assumptions. Classical maximum likelihood equations are modified by assigning weights based on the discrepancy between model-based and sample-based statistical data depths of observations, thereby automatically downweighting atypical observations. The approach builds on weighted likelihood estimation (WLE) and introduces Depth Pearson residuals, which address well-known limitations of WLE in the multivariate setting. The resulting estimator retains asymptotic efficiency under the correct model while providing strong robustness under contamination. Consistency and asymptotic normality are established under standard regularity conditions, and these properties continue to hold in high-dimensional regimes where the data dimension increases with the sample size, without requiring density existence or smoothness assumptions. The theory is complemented with a breakdown point analysis for location and scatter in elliptical models, and practical performance of the method is demonstrated through simulations and real data examples.

E1410: Asymptotic properties of multivariate depth quantiles*Presenter:* **Giacomo Francisci**, University of Trento, Italy

Statistical depth functions describe the degree of centrality of a point with respect to a multivariate distribution and are an important tool in non-parametric and robust statistics. The point of maximum depth c defines the median of the distribution, whereas the level sets of the depth function are used to define multivariate quantiles. The quantile set at level $\alpha > 0$ may be identified with the radius function that assigns to every direction u in the unit sphere the greatest scalar s such that the depth of $c + su$ is at least α . Under appropriate conditions, it is shown that the centered and rescaled radius function converges weakly in the space of bounded functions on the unit sphere to a Gaussian process and the covariance function is expressed in terms of the depth function. This result can be used to construct confidence regions and perform hypothesis testing for multivariate quantiles.

EO226 Room 101 RECENT DEVELOPMENTS IN TIME SERIES MODELLING (VIRTUAL)**Chair: Yayi Yan****E1953: Panel data estimation and inference: Homogeneity versus heterogeneity***Presenter:* **Fei Liu**, University of Bath, United Kingdom*Co-authors:* Jiti Gao, Bin Peng, Yayi Yan, Fei Liu

An underlying data generating process is defined that allows for different magnitudes of CD, along with time series autocorrelation. This is achieved via high-dimensional moving average processes of infinite order (HDMA(∞)). The setup and investigation integrate and enhance homogeneous and heterogeneous panel data estimation and testing in a unified way. To study HDMA(∞), the Beveridge-Nelson decomposition is extended to a high-dimensional time series setting, and a complete toolkit is derived. Homogeneity versus heterogeneity is examined using Gaussian approximation, a prevalent technique for establishing uniform inference. For post-testing inference, central limit theorems are derived through Edgeworth expansions for both homogeneous and heterogeneous settings. The practical relevance of the established asymptotic theories is showcased accordingly. Theoretical findings are verified via extensive numerical studies using both simulated and real datasets.

E1993: Extracting predictability: A time-varying factor-augmented approach under mixed persistence*Presenter:* **Tingting Cheng**, Nankai University, China*Co-authors:* Jiti Gao, Yayi Yan, Xuanbin Yang

A time-varying factor-augmented forecasting model is developed to extract predictable variation in the target variable when predictors exhibit mixed persistence. The framework accommodates stationary, nonstationary, or mixed observable and latent predictors, and allows latent factors to follow either static or dynamic factor structures, thereby nesting many classical and new examples as special cases. A sieve estimation procedure for the time-varying coefficients is developed, corresponding large-sample properties are established, and prediction intervals for the forecasts are constructed. Simulation studies show that the proposed methods perform well in finite samples. In an empirical application to U.S. inflation forecasting, the proposed model performs favorably relative to competing benchmarks. Further evidence indicates that latent factors remain informative even when inflation exhibits strong autoregressive dynamics, since they absorb predictable variation that would otherwise be left in the forecast error.

E2039: Factor models of matrix-valued time series: Nonstationarity and cointegration*Presenter:* **Yayi Yan**, Shanghai University of Finance and Economics, China*Co-authors:* Degui Li, Qiwei Yao

A matrix factor model is proposed for nonstationary matrix-valued time series with common stochastic trends. Unlike traditional factor analysis, which flattens matrix observations into vectors, the matrix factor model fully explores the intrinsic matrix structure in the data, allowing interaction between row and column stochastic trends and subsequently improving estimation convergence while reducing computational complexity. The main estimation methodology is built on eigenanalysis of sample row and column covariance matrices when nonstationary matrix factors are of full rank and idiosyncratic components are temporally stationary, and is further extended to tackle more flexible settings when matrix factors are cointegrated and idiosyncratic components may be nonstationary. Under mild conditions allowing the existence of weak factors, convergence theory for estimated factor loading matrices and nonstationary factor matrices is derived. The developed methodology and theory are applicable to the general case of heterogeneous strengths over weak factors. An easy-to-implement ratio criterion is adopted to consistently estimate the size of the latent factor matrix. Simulation and empirical studies examine the numerical performance of the developed model and methodology in finite samples.

E2046: Model averaging of multi-layer time-varying network vector autoregressions*Presenter:* **Boyao Wu**, University of International Business and Economics, China

A time-varying multi-layer network vector autoregression (VAR) model framework is introduced for large-scale time series, allowing agents in dynamic systems to interact through multiple channels and incorporating multiple adjacency matrices to capture network spillover effects. A penalized model averaging method is proposed to determine a time-varying optimal combination of multi-layer network VAR candidate models whose number is allowed to be divergent. Under regularity conditions, asymptotic properties such as asymptotic optimality and convergence rates of the proposed time-varying weight estimation are derived in the contexts of both in-sample fitting and out-of-sample prediction. Conformal prediction is introduced to construct interval forecasts. Monte Carlo simulation and empirical studies illustrate the reliable finite-sample estimation and predictive performance of the developed methodology.

EO066 Room 102 NOVEL METHODS FOR COMPLEX DATA**Chair: Hernando Ombao****E1323: Splitting high dimensional Markov chains with application to bootstrap***Presenter:* **Patrice Bertail**, Paris Nanterre University, France*Co-authors:* Anna Dudek, Karolina Marek-Lukasiewicz

Large-dimensional Markov chains appear in many models and many applications including analysis of electro-encephalogram data. A multidimensional approximative regenerative block bootstrap (MARBB) is introduced, a bootstrap algorithm designed for high-dimensional Markov chains. The focus is on a vector autoregressive (VAR(1)) process with a low-rank structure (but the same ideas apply for single index Markov chain models). A reduction algorithm is first used that transforms the original high-dimensional time series into a lower-dimensional Markov chain. In the cointegration (nonstationary) case, it is assumed that there are very few cointegration relationships. Once the chain is reduced, the regenerative properties of Harris recurrent Markov chains within a general state space are leveraged, using the Nummelin splitting technique to extend existing results from the one-dimensional settings to the multidimensional case. This approach enables the identification of approximate regeneration times of the lower-dimensional Markov chain, which in turn leads to the splitting of the original high-dimensional Markov chain into (almost independent) regenerative blocks. These blocks are then used to bootstrap relevant statistics based on regenerative blocks. The MARBB consistency results are obtained, and the algorithm is applied to simulated data.

E1822: Testing for sphericity of autocovariance matrix in high-dimensional time series*Presenter:* **Mikoto Kita**, Waseda University, Japan*Co-authors:* Yousei Yoshida, Yan Liu

The problem of testing the sphericity of autocovariance matrices for high-dimensional time series is studied. In the presence of temporal dependence, implementation of classical test statistics typically requires the estimation of long-run variances, which becomes increasingly challenging as the dimension grows. To address this issue, a bootstrap-based testing procedure is proposed that avoids explicit variance estimation. The method enforces the spherical structure under the null hypothesis at lag zero, while preserving the dependence structure at nonzero lags through tapered covariance estimation. This construction yields a feasible approach to inference in settings where direct variance estimation is difficult. The asymptotic validity of the proposed bootstrap is established for a class of test statistics under a general (N, p) -asymptotic framework, allowing for temporal dependence. In addition, explicit convergence rates are derived that reflect the joint effects of dimensionality and dependence. Simulation studies demonstrate that the proposed method achieves accurate size control and competitive power across a range of dependence structures.

E2011: Extremal vulnerability*Presenter:* **Paolo Victor Redondo**, King Abdullah University of Science and Technology, Saudi Arabia*Co-authors:* Raphael Huser, Hernando Ombao, Miguel de Carvalho

In many complex systems, identifying the most vulnerable components is essential for effective prevention, intervention, and risk management. Extreme events rarely occur in isolation but instead arise jointly across interconnected components. A method is developed to identify the nodes that are most vulnerable to such systemic shocks. These joint extreme occurrences provide insight into patterns of Tail Dependence and help identify the components that are systematically exposed to extreme shocks propagating through the network. The notion of Extremal Vulnerability is introduced, defined as the long-run tendency of a component to be affected by extreme events propagating from other components. A novel framework is developed that builds on the Tail Dependence Matrix (TDM), which captures directed Extremal Dependence, and leads to the Extremal Vulnerability Rank (XVRank) a PageRank-inspired algorithm for quantifying Extremal Vulnerability. Theoretical properties of the resulting estimators, including consistency and asymptotic normality, are established, and their performance is validated through Monte Carlo simulations. As an empirical application, the XVRank method is applied to the Magnificent Seven (MAG7) equities to identify assets most exposed to severe market downturns, and to 30 industry portfolios to characterize Extremal Vulnerability across sectors of the US economy.

E2042: A fractal-dimension framework for quantifying self-similarity in chromatin folding*Presenter:* **Anass El Yaagoubi Bourakna**, King Abdullah University of Science and Technology, Saudi Arabia

A graph-based framework for quantifying self-similarity in chromatin folding from Hi-C contact data is introduced. Genomic loci are represented as nodes and Hi-C interactions as weighted edges, enabling the estimation of fractal dimension using correlation and sandbox methods. Validation on synthetic networks demonstrates that the approach captures hierarchical multiscale organization while distinguishing non-fractal clustered structures. Applied to IMR90 Hi-C data, the framework reveals approximate scaling behavior consistent with fractal-like chromatin organization, with larger chromosomes tending toward more compact, space-filling structures and smaller chromosomes exhibiting lower-dimensional folding patterns. These results suggest that graph-based fractal dimension provides an interpretable quantitative summary of genome organization across scales.

EO090 Room 103 NEW DEVELOPMENTS IN TIME SERIES AND SPATIAL LEARNING**Chair: Zudi Lu****E1455: A simple and effective random forest modelling for nonlinear time series data***Presenter:* **Shihao Zhang**, University of Southampton, United Kingdom*Co-authors:* Zudi Lu, Chao Zheng

Random Forests by Random Weights (RF-RW) are proposed as an alternative forest-based method for nonlinear time series modelling, addressing the limitation that the bootstrap procedure in original random forests (RF) disrupts the serial dependence structure inherent in time series data. In particular, RF-RW avoids bootstrapping, thereby preserving the temporal dependence, while assigning independent random weights for the series to reduce tree-to-tree correlations. Non-asymptotic concentration bounds and uniform consistency results are established under both fixed- and high-dimensional feature spaces to provide theoretical guarantees for the proposed RF-RW. Extensive numerical studies are conducted, including simulation experiments and UK COVID-19 daily cases prediction, which demonstrate that RF-RW outperforms existing RF-based methods and competing benchmarks.

E1493: Semiparametric ultra-high dimensional model averaging of expected shortfall for nonlinear dynamic time series*Presenter:* **Jiangtao Wang**, Huazhong Normal University, China

A semiparametric model averaging scheme is proposed for expected shortfall (ES) in a nonlinear dynamic time series setting with a very large number of covariates, including exogenous regressors and autoregressive lags. The objective is to obtain more accurate estimates and forecasts of ES via using a large number of conditioning variables in a nonparametric way. The proposed scheme consists of two steps. The first step is to forecast ES in a nonparametric manner based on a few variables. And the second step is to obtain the final forecast of ES by a combination regression based on the obtained prediction of ES derived from the first step. Asymptotic properties for the proposed scheme are investigated under some regularity conditions. Numerical studies including both simulation and an empirical application are given to illustrate the proposed methodology.

E1611: A 3D semiparametric spatial autoregressive model for multisubject data*Presenter:* **Jinxuan Bai**, University of Southampton, United Kingdom*Co-authors:* Zudi Lu, Chao Zheng, Ian Galea, Aravinthan Varatharaj

Spatial data analysis is essential for medical imaging but poses substantial methodological challenges. Neuroscience studies of the blood-brain barrier (BBB) produce massive three-dimensional datasets, demanding spatial models that are both interpretable and computationally tractable with statistical efficiency for medical research. To address this need, a three-dimensional semiparametric spatial autoregressive (3D-S2AR) model is introduced. The framework captures local voxel-level variation through a spatial autoregressive component based on three-dimensional distance-weighted matrices, while simultaneously modeling large-scale smooth nonparametric trends. A key feature of the framework is its explicit accommodation of multi-subject neuroimaging data, allowing each participant's spatial domain to differ across subjects to reflect individual variation in brain geometry, without requiring a common spatial domain. Crucially, it achieves computational efficiency even for datasets with millions of spatial locations. To mitigate the computational burden, a profile quasi-likelihood estimation method is developed and consistency and asymptotic normality of the resulting estimators are established.

E1677: Adaptive group fused lasso for panel threshold model*Presenter:* **Lulu Wang**, Beijing Normal-Hong Kong Baptist University, China*Co-authors:* Zudi Lu

Panel threshold regression has been one of the most popular methods in nonlinear panel time series analysis. The most common method to determine the number of threshold parameters is using a bootstrap procedure to approximate the sampling distribution under the assumption of

cross-sectional independence, which may work poorly when dealing with strong dependence in most climate and finance data. A panel threshold model is considered where both regressors and residuals are allowed to be cross-sectionally dependent via adaptive group fused Lasso. It is shown that with probability approaching one, the proposed method can correctly determine the unknown number of threshold parameters and estimate regression parameters consistently. Asymptotic theories of the Lasso estimators of regression coefficients are established. Simulation studies demonstrate that the proposed estimation method works well in finite samples under cross-sectionally dependent conditions.

EO040 Room 104 RECENT ADVANCES IN PANEL DATA MODELS
Chair: Alexandra Soberon
E1381: Understanding real estate matches through a semiparametric panel model of the matching function

Presenter: **Taining Wang**, Capital University of Economics and Business, China

Co-authors: Feng Yao, Adam Nowak

A semiparametric panel model of the real estate matching function is proposed that allows matching elasticities to vary flexibly with macroeconomic conditions. Extending the conventional Cobb–Douglas framework, the elasticities of (log) supply and demand in the matching function are modeled as unknown functions of environmental variables, while accounting for unobserved search frictions through time and individual fixed effects. One challenge is that buyer flows are typically unobserved and potentially endogenous. To address this issue, a two-step semiparametric instrumental variables estimator is developed, where instruments are constructed internally from innovations in sellers entry flows. The proposed estimator eliminates fixed effects via multi-differencing without imposing normalization restrictions, and shows appealing performance in simulations. The model is applied to monthly data from 100 U.S. metropolitan housing markets over the period 2012–2018. The empirical results reveal nonlinear negative effects of mortgage rates and unemployment rates on matching elasticities. The estimated matching elasticity of buyers averages around 0.67, lower than values reported in the literature. Counterfactual analyses based on the estimated model show that changes in macroeconomic conditions alone can generate sizable fluctuations in housing matches, even when market tightness is fixed.

E1392: Gravity models with smooth time-varying coefficients and fixed effects

Presenter: **Juan Manuel Rodriguez-Poo**, Universidad de Cantabria, Spain

Co-authors: Daniel Henderson, Alexandra Soberon, Stefan Sperlich

Gravity models with exporter-time, importer-time, and pair fixed effects are now standard for analyzing bilateral flows across a range of settings (e.g., international trade, foreign direct investment, remittances, and migration) where unobserved heterogeneity across origins, destinations, pairs, and time must be controlled to draw credible causal inference. In long panels, however, two empirical challenges increasingly dominate: (i) key policy elasticities may evolve over time, and (ii) the high dimensionality of fixed effects yields an incidental-parameter problem that complicates estimation and inference. A time-varying-coefficient gravity model in levels is proposed together with a concentrated local-moment estimator that eliminates pair fixed effects, leverages smoothness in time, accommodates zeros, and avoids log-linearization. Under fixed cross-sectional dimension and large time dimension, asymptotic normality of the estimator is established and practical bootstrap inference with optional bias correction is provided. Simulation evidence confirms solid finite-sample performance when bilateral covariates exhibit sufficient temporal variation. Several applications uncover economically meaningful dynamics missed by common parametric trend interactions, and diagnose when data provide too little time variation for flexible estimation.

E1468: Estimating choice models with piecewise smooth objective functions: Application to joint retirement

Presenter: **Siqi Wei**, IE University, Spain

Co-authors: Tatiana Rosa

Choice models with piecewise smooth objective functions are studied. By introducing well-defined latent variables derived from regional components (auxiliary functions), conditions are provided under which the choice model can be transformed into a reduced form resembling tobit-like models, with the introduced latent variables as potential outcomes. The potential outcome framework not only serves as an alternative solution method for the choice model, but also facilitates the implementation of the stochastic expectation-maximization (SEM) algorithm to obtain MLE. The framework is illustrated using two simple examples: 1) responses to tax and transfers, and 2) the (S,s) model. A joint retirement decision model among European couples based on the interdependent duration model is further empirically estimated, applying the proposed solution and estimation methods. The results provide empirical evidence of complementarity between spouses in their transition out of the labor force in Europe.

E1662: Empirical Likelihood based inference for varying coefficient panel data models with cross-sectional dependence

Presenter: **Luis Antonio Arteaga Molina**, Universidad de Cantabria, Spain

Local empirical likelihood based inference for semiparametric varying coefficient panel data models with cross-sectional dependence is investigated. A common factor structure is used to characterize the cross-sectional dependence. First, it is shown that the naive empirical likelihood ratio for the individual-specific and pooled varying coefficients are asymptotically standard chi-squared distributed when undersmoothing is employed. To avoid undersmoothing, bias-corrected empirical likelihood ratios that also have standard chi-squared limit distributions are presented. As a by-product, the maximum empirical likelihood estimators for the individual-specific and pooled varying coefficients are derived and their asymptotic distributions are obtained. The main advantage of empirical likelihood is that, unlike normal approximation, the plug-in estimator of the variance matrix is not needed. The feasibility of the technique and its small sample properties are demonstrated through a Monte Carlo simulation exercise and an empirical application.

EO176 Room 105 STATISTICS FOR SPATIAL AND TEMPORAL DATA
Chair: Fabian Mies
E1573: Detecting change-points of univariate time series using the Wasserstein distance

Presenter: **Anton Imm**, RWTH Aachen University, Germany

Co-authors: Fabian Mies, Ansgar Steland

The detection of change-points in univariate nonstationary time series in a nonparametric setting is studied. A statistic based on a truncated Wasserstein distance between localized empirical distribution functions of the time series is introduced, suited to the testing problem. The one-dimensional Wasserstein distance is characterized by the sequential quantile process, and the weak convergence of this process to a Gaussian limit is established. To address the nonlinearity of the quantile process, a new Bahadur representation result is developed, allowing consideration of the asymptotic behavior of the empirical process instead of the quantile process. The proof requires further analysis of the modulus of continuity of the empirical process. For feasible inference, a Gaussian multiplier bootstrap scheme is introduced and its performance is analyzed via a simulation study.

E1652: Uniform central limit theorems for non-stationary processes via relative weak convergence

Presenter: **Nicolai Palm**, LMU Munich, Germany

Co-authors: Thomas Nagler

Traditional central limit theorems (CLTs) have limited applicability to non-stationary processes, not least because there is no fixed Gaussian limit to converge to. Relative weak convergence is introduced as a generalization of classical weak convergence that provides a natural framework for CLTs in non-stationary settings. Within this framework, multivariate, uniform, and sequential relative CLTs can be proved under assumptions that

parallel those used in stationary settings. These results bridge a critical gap in asymptotic statistics, enabling rigorous study of the asymptotic behavior of dynamic sequences that do not satisfy traditional assumptions of stationarity.

E1769: **Parameter estimation for Matern random fields from local measurements**

Presenter: **Randolf Altmeyer**, Imperial College London, United Kingdom

Gaussian random fields with Matern covariance structure are fundamental models in spatial statistics. Given discrete observations on a regular grid, parametric estimation of key parameters governing variance, range, and smoothness is studied. The approach is based on the representation of the random field as the solution to an elliptic stochastic partial differential equation (SPDE). A key insight from this representation is that the model parameters exhibit distinct local scaling behaviour, reflecting the local structure of the underlying differential operator. Building on this idea, spatially localised linear combinations of the data, referred to as local measurements, are constructed to approximate localised features of the field. By analysing how these quantities scale across resolutions, a novel class of estimators based on localised quadratic functionals is developed. This can be viewed as a multidimensional extension of quadratic variation techniques from time series analysis. By carefully designing test functions with prescribed moment cancellation properties, explicit estimators for the variance, range, and smoothness parameters are obtained. Asymptotic normality is established under infill and large domain asymptotics, and explicit expressions for the estimator variances are derived using Gaussian moment identities. The resulting methods are computationally efficient, with linear complexity in the number of observations.

E2016: **Balancing the edge effect and dimension of spectral spatial statistics**

Presenter: **Anne van Delft**, Columbia University, United States

Co-authors: Holger Dette

Distributional properties of a class of spectral spatial statistics are investigated under irregular sampling of a random field defined on $\mathbb{R}^d$, and used to obtain a novel test for isotropy via a minimum distance approach. More precisely, an explicit expression for the minimum L_2 -distance between the spectral density of the random field and its best approximation by a spectral density of an isotropic process is derived in terms of certain integrals of the spectral density, which are then estimated from the data. Within this context, edge effects are well-known to create a bias in classical estimators commonly encountered in the analysis of spatial data. This bias increases with dimension d and, for $d > 1$, can become non-negligible in the limiting distribution of such statistics to the extent that a nondegenerate distribution does not exist. A general theory is provided for a class of integrated spectral statistics that are used to estimate the minimum L_2 -distance, which enables: 1) significantly reducing this bias and 2) ensuring that asymptotically Gaussian limits can be derived for $d \leq 3$ for appropriately tapered versions of such statistics. This theory addresses crucial gaps in the literature and demonstrates that tapering with a sufficiently smooth function is necessary to achieve such results.

EO084 Room 107 NEW FRONTIERS IN CAUSAL AND STRUCTURAL INFERENCE

Chair: Sotetsu Koyamada

E1735: **Latent population size estimation in directed networks via microfounded random graph models**

Presenter: **Yuki Iwanaga**, Kobe University, Japan

Co-authors: Teruyoshi Kobayashi

Estimating the true size of a population from network data is challenging when nodes with no recorded connections are systematically absent from observation. While link formation probabilities admit various specifications, the generalized random graph model offers a particularly tractable and economically interpretable foundation. A microfounded model is proposed for directed networks, where each link probability is derived from a discrete choice framework in which nodes differ in attractiveness and link formation cost. Under Gaussian heterogeneity in node-level parameters, the probability that a node goes unobserved is characterized as a function of the distributional hyperparameters and the latent population size. This isolation probability is used to identify the latent population size via moment conditions. A moment-matching algorithm is developed to jointly estimate the population size and hyperparameters by minimizing discrepancies between observed and theoretically implied degree sequence moments, without requiring node-level parameter inference. Simulation studies confirm the accuracy of the proposed estimator across a range of network configurations, and applications to real-world directed networks demonstrate its practical utility.

E1636: **Extracting causal variables through causal representation learning**

Presenter: **Hiroshi Morioka**, Shiga University, Japan

Causal discovery aims to estimate causal relationships among variables, typically under the assumption that the relevant causal variables are directly observed. In practice, however, it is often unclear what the true causal variables are. This challenge motivates causal representation learning (CRL), which seeks to recover both latent causal variables and the causal relations among them from complex observations. The CRL perspective reveals fundamental limitations of this problem: it is generally ill-posed and cannot be solved without suitable inductive biases on the underlying generative process. A recent CRL framework demonstrates that identifiability can be achieved under assumptions based on multimodal observations. These results illustrate how structural assumptions on data generation make it possible to recover meaningful causal variables from high-dimensional data, extending causal discovery beyond the classical setting of directly observed variables.

E1408: **Denoised IPW-lasso for heterogeneous treatment effect estimation in randomized experiments**

Presenter: **Mingqian Guan**, The University of Osaka, Japan

Co-authors: Komei Fujita, Naoya Sueishi, Shota Yasui

A new method is proposed for estimating conditional average treatment effects (CATE) in randomized experiments. Inverse probability weighting (IPW) is adopted for identification; however, IPW-transformed outcomes are known to be noisy, even when true propensity scores are used. To address this issue, a noise reduction procedure is introduced and a linear CATE model is estimated using Lasso, achieving both accuracy and interpretability. It is theoretically shown that denoising reduces the prediction error of the Lasso. The method is particularly effective when treatment effects are small relative to the variability of outcomes, which is often the case in empirical applications. Applications to the get-out-the-vote dataset and the Criteo Uplift Modeling dataset demonstrate that the method outperforms fully nonparametric machine learning methods in identifying individuals with higher treatment effects. Moreover, the method uncovers informative heterogeneity patterns that are consistent with previous empirical findings.

E1757: **Beyond ICA: Group theoretic identifiability via mechanism-domain contrast**

Presenter: **Pengzhou Wu**, Kyoto University, Japan

Mechanism-domain contrast is introduced as a group-theoretic principle for identifiability in unsupervised latent-variable models. Mechanisms are branches of a piecewise mixing function f ; domains are components of a latent mixture Z , whose parameters vary across a structured family (e.g., exponential families). Contrast appears as algebraic symmetry breaking: trivial mixture symmetry groups (domain contrast), unique branch toggling at boundaries (mechanism contrast), and mechanism-domain interaction contrast. This unifies and extends domain-contrast results such as time-contrastive learning. Instantiated on the observable $f_*(Z)$ with piecewise-affine f and Gaussian mixture Z , the conditions yield hierarchy-law, map, posterior, and pointwise identifiability up to a global affine transform, without requiring continuity, injectivity, or independence of latents. Under independence and diagonal covariances, the ambiguity reduces to classical ICA form. Preliminary extensions cover other piecewise-parametric maps and latent families. Applications to causality are also discussed. Discontinuities are a resource: combined with domain contrast, they tighten symmetry groups, enabling identifiable deep generative models beyond ICA.

EO243 Room 201 ADVANCES IN DIRECTIONAL STATISTICS**Chair: Thanh Mai Pham Ngoc****E1194: Local Frechet regression with Riemannian manifold predictors****Presenter:** Chang Jun Im, Seoul National University, Korea, South**Co-authors:** Jeong Min Jeon

Recent advancements in object-oriented data analysis, particularly Frechet regression, have provided a robust framework for handling responses in general metric spaces. However, the existing literature predominantly assumes that predictors are Euclidean. A novel regression framework is proposed that accommodates responses in a general metric space and predictors residing on a Riemannian manifold. Intrinsic local constant and local linear estimators are developed that completely respect the underlying geometry of the manifold predictor space. By mapping the data to the tangent space via the logarithmic map and utilizing the volume density function, the theoretical properties of the proposed estimators are comprehensively established. Specifically, both pointwise and uniform convergence rates are derived under mild geometric regularity conditions. Notably, it is demonstrated that the estimators achieve the d -dimensional minimax optimal rate (where d is the intrinsic dimension of the manifold), exactly recovering the standard Euclidean rate when the curvature of the margin condition is quadratic.

E1590: Spherical deconvolution for an atomic distribution**Presenter:** Nicolas Conanec, Université Sorbonne Paris Nord, France

Observations of a spherical random variable $Z = \varepsilon X$, defined as the convolution of a spherical random variable X with a random rotation $\varepsilon \in SO(3)$ having known density k , are considered. The variable X follows a mixture distribution: with probability p , it is deterministic and located at the North Pole, while with probability $1 - p$, it admits a density f on the sphere. Based on an i.i.d. sample $(Z_1, \dots, Z_n)$, the goal is to estimate the couple (p, f) . Estimators of p and f are proposed that rely on Fourier inversion techniques. By characterizing the degree of ill-posedness of the associated inverse problem, convergence rates for the proposed estimators are established when f belongs to a Sobolev class.

E1708: Optimal geodesic regression for directional response**Presenter:** Gaspard Bernard, Academia Sinica, Institute of Statistical Science, Taiwan

A directional response variable Y taking values on the unit sphere in dimension 3 is considered, together with linear covariates X . The problem of estimating a regression function valued on the unit sphere has received considerable attention. However, most of the existing work focuses on nonparametric methods, for which both interpretability and optimality may be difficult to achieve. A single-index framework is considered in which the regression function is assumed to be a geodesic curve and the noise is assumed to lie in the tangent space. This model has been introduced in the context of general Riemannian manifolds, but the aim is to propose a natural analogue for directional responses of the classical linear model. Under the assumption that the tangent space noise distribution is spherically symmetric with specified radial density, this approach enables the derivation of locally and asymptotically optimal tests and confidence intervals for both the geodesic curve itself and the coefficients of the single index. The potential asymptotic cost of performing inference on only one of these two parameters when the other is treated as a nuisance is then investigated. Subsequently, the question of how to preserve the validity of the proposed tests when the radial density of the noise is unspecified while retaining their optimality properties is studied. Finally, the extension of the proposed framework to responses taking values on hyperspheres of arbitrary dimension is discussed.

EO076 Room 203 MEASURING DIFFERENT DIMENSIONS OF POVERTY: METHODS AND APPLICATIONS**Chair: Stefano Marchetti****E1580: Measuring food poverty across regions and vulnerable groups: Subnational evidence from Italy during high inflation****Presenter:** Ilaria Benedetti, University of Tuscia, Italy**Co-authors:** Tiziana Laureti, Luca Secondi

Food poverty is a key dimension of material deprivation, particularly during periods of high inflation, when rising prices disproportionately affect vulnerable households. An examination of regional and sociodemographic disparities in food poverty in Italy over 2021-2023, a period marked by a sharp increase in food prices, is conducted using microdata from the Italian Household Budget Survey (HBS). The analysis adopts a distribution-sensitive approach, showing that the inflationary shock was strongly regressive: lower-expenditure households experienced higher effective inflation due to their consumption structure, with a larger share devoted to essential goods, especially food. A central contribution concerns the measurement of food poverty at the sub-national level. Given the complex Survey design and the nonlinear nature of poverty indicators, standard variance estimators may be unreliable for small domains. This issue is addressed by implementing resampling techniques to obtain robust estimates of sampling variability and enable valid inference across regions and population subgroups. Results reveal significant territorial disparities, with higher food poverty in southern regions and smaller municipalities. The findings highlight the importance of robust sub-national measurement for policy design, as national averages may conceal substantial inequalities.

E1607: Food poverty and transportation poverty at local level in Italy**Presenter:** Stefano Marchetti, Dipartimento di Economia e Management, Università di Pisa, Italy

Food poverty and transport poverty are investigated as two dimensions of poverty, each conceptualized from multiple perspectives. Food poverty is defined as the inability of a Household to afford a basket of food items consistent with a healthy and sustainable diet. Transport poverty is defined according to the 10% threshold criterion, whereby a Household is considered to be in transport poverty if its expenditure on transport exceeds 10% of total expenditure. The analysis is conducted at the local level (provinces, NUTS 3) in Italy, where sample sizes are typically limited. Small area estimation methods are employed to obtain reliable estimates at the desired level of disaggregation. The analysis combines data from the 2022 Household Budget Survey, used to compute direct estimates, with administrative data from the Tax Agency Register, which are incorporated as auxiliary variables in the small area models. The results reveal substantial territorial heterogeneity across Italy for both dimensions of poverty. These findings differ from conventional indicators of absolute and relative monetary poverty, which generally highlight a pronounced North-South divide. By focusing on specific aspects of poverty and social exclusion at the local level, the analysis highlights challenges faced by households even in relatively affluent areas of the country.

E1816: Transport poverty and energy cost burden across municipal types: A sub-national analysis for Italy**Presenter:** Luca Secondi, University of Tuscia, Italy**Co-authors:** Tiziana Laureti, Francesco Maria Morosini, Pierdomenico Dutillo

Mobility costs are a key dimension of transport poverty, as high expenditure on transport may limit access to employment, education, healthcare and other essential activities, especially for economically vulnerable households and across territorially differentiated contexts. Transport-related deprivation may also overlap with energy poverty, since both reflect the pressure of necessary and recurrent expenditures on household budgets. An analytical framework for studying transport affordability in Italy using official microdata is developed, with particular attention to the relationship between transport and energy poverty across different municipal types. The analysis considers regional and local estimation domains to capture spatial heterogeneity in expenditure structures and mobility needs. A set of affordability indicators is introduced, focusing on recurrent mobility costs, including fuel, maintenance, insurance and public transport expenditure, and on their interaction with domestic energy spending. Attention is paid to differences across metropolitan, intermediate and less urbanised areas, where car dependence and the composition of mobility expenditure

may produce distinct patterns of disadvantage. The results provide policy makers with evidence to identify the territorial and social groups most exposed to the combined burden of transport and energy costs and to support more targeted interventions.

E1823: **Measuring digital poverty among young people: An application at regional level in Italy**

Presenter: **Caterina Giusti**, Dipartimento di Economia e Management, Università di Pisa, Italy

Co-authors: Monica Pratesi, Antonella D Agostino, Tommaso Budini

Evaluating Digital Poverty among young people is crucial to understand whether new generations have Access to the resources required to exercise their rights and to prevent marginalization within an increasingly digitized society. Analyses conducted at the local level provide the detailed evidence necessary for developing effective policy actions aiming at decreasing Digital Poverty. An analytical Framework is proposed to measure Digital Poverty across Italian regions, focusing on individuals aged 6 to 19 and adopting a data-driven approach based on survey micro-data. Drawing on the Digital Poverty Alliance model, Digital Poverty is framed as a multidimensional condition that includes several interconnected aspects. Specifically, the Framework outlines five key dimensions: Devices & Connectivity, Access, Skills, Motivation, and Support. Additionally, the European Digital Competence Framework for Citizens serves as a complementary benchmark for the Skills dimension. Within this study, three dimensions are operationalized. Devices & Connectivity concerns the availability of Digital tools and internet connections. Access refers to the extent to which individuals can reliably and directly use such resources in their daily lives. Skills capture competencies in using Digital technologies for diverse purposes.

EO136 Room 204 ESTIMATION AND INFERENCE OF EXTREME VALUES

Chair: Gilles Stupfler

E1854: **Corrected inference about the extreme expected shortfall in the general max-domain of attraction**

Presenter: **Gilles Stupfler**, University of Angers, France

Co-authors: Abdelaati Daouia, Antoine Usseglio-Carleve

The use of expected shortfall as a solution for various deficiencies of quantiles has gained substantial traction in risk assessment over the last 20 years. Existing approaches to its inference at extreme levels remain limited to distributions that are both heavy-tailed and have a finite second tail moment. This constitutes a strong restriction in areas such as finance and environmental science, where the random variable of interest may have a much heavier tail or, conversely, may be light-tailed or short-tailed. Under a wider semiparametric extreme value framework, comprehensive asymptotic theory is developed for expected shortfall estimation above extreme quantiles in the class of distributions with finite first tail moment, regardless of whether the underlying extreme value index is positive, negative, or zero. The obtained asymptotic theory is contrasted with existing knowledge in the literature.

E1785: **Extreme conditional tail moment inference**

Presenter: **Solune Denis**, Université d'Angers, France

Conditional Tail Moments (CTMs) are useful tools in risk quantification. Widely used in finance and climate science (see for instance the Expected Shortfall, a particular case of CTM), their inference is currently limited to heavy-tailed models satisfying an appropriate finite second moment condition in the literature. Semiparametric estimation procedures for CTMs are developed. Asymptotic results are proved for the resulting class of CTM estimators above extreme levels of quantiles, for a wide range of distributions and under minimal moment constraints. Asymptotic confidence intervals are constructed based on the theoretical results by relying on moment estimators of the scale and shape extreme value parameters.

E1247: **A general theory for extremal regression in heavy-tailed models**

Presenter: **Abdelaati Daouia**, Toulouse School of Economics, France

Co-authors: Yasser Abbas, Gilles Stupfler

Studying rare events at the heavy tails of conditional Pareto-type distributions, in the presence of high-dimensional covariates, is a burgeoning science with many applications in actuarial, financial, and environmental risk management. The most prominent risk measures to quantify these events utilize conditional quantiles, Expected Shortfall, and expectiles at extreme levels. The few attempts to tackle this extreme value problem involve location-scale regression models with heavy-tailed noise. A more flexible and complex model is employed which better balances model generality with estimation efficiency. A general theory is developed that relies on residual-based estimators of the three regression risk measures at both intermediate and extreme levels, and fully explores their asymptotic behavior in generic settings. Simple sufficient criteria are also provided for verifying the main high-level assumption, which facilitates the construction of weighted Gaussian approximations for the tail quantile residual process, ultimately ensuring the asymptotic normality of all produced extreme value estimators. This generic extremal regression framework is then applied to linear, nonlinear and nonparametric estimation scenarios. Simulations show the undeniable potential of the methodology for various distribution types, outperforming the best available competing estimation approaches. An application to real financial data further solidifies their dominance.

E1700: **Multivariate robust extremiles**

Presenter: **Luca Merlo**, Link Campus University, Italy

Co-authors: Valeria Bignozzi, Lea Petrella, Nicola Salvati

A novel extension of univariate extremiles to a robust multivariate framework is proposed. Among the possible multivariate generalizations, the approach based on multivariate M-quantiles with Huber's multidimensional loss function is adopted. Key theoretical properties of the proposed functional are established, including existence and uniqueness under mild conditions, symmetry relations, location and orthogonal equivariance, and positive homogeneity in limiting cases. The proposed extremile sets lie inside the convex hull of the data while allowing for different levels of robustness. From an inferential perspective, the main asymptotic properties of these robust multivariate extremiles are proven, namely consistency and asymptotic normality, using the theory of Z-estimators. Their empirical performance is assessed through a series of examples on artificial and real-world data.

EO193 Room 205 RECENT ADVANCES IN MULTIVARIATE STATISTICAL METHODS

Chair: Kwangmin Lee

E1824: **Disentangling location and concentration effects in von Mises-Fisher regression**

Presenter: **Kipoong Kim**, Changwon National University, Korea, South

Co-authors: Sungkyu Jung, Joern Schulz

In many problems the response of interest is a direction, that is, a point on the surface of a sphere. In regression with a spherical response, a covariate can affect the response in two ways, as in ordinary regression: it can move the mean direction, a location effect, and it can change how concentrated the response is around that direction, a concentration effect. Existing methods fall into two groups: spherical regression methods report a single coefficient for each covariate and so mix the two effects, whereas several testing methods provide a separate test for each effect but cannot adjust for other covariates. A generalized linear model is proposed that links the natural parameter of the von Mises-Fisher distribution linearly to the covariates. This reparameterization puts the density in canonical exponential-family form, so the maximum likelihood estimator is consistent and asymptotically normal under mild conditions. The contribution is a projection-based inference framework: projecting the estimated coefficient onto the estimated reference direction and onto its orthogonal complement yields separate test statistics for the location and concentration effects.

Simulation studies show that the location and concentration tests keep the Type I error rate at the nominal level while attaining comparable power. The method was also applied to hippocampal shape data from the ParkWest study of Parkinson's disease.

E1841: **Nonparametric intensity estimation in covariate-driven Poisson processes using deep learning**

Presenter: **Atsutomo Yara**, Graduate School of Engineering Science, Osaka University, Japan

Co-authors: Yoshikazu Terada

Nonparametric maximum likelihood estimation of the intensity function for covariate-driven Poisson processes, where the intensity depends on observed covariates, is studied. This formulation naturally captures real-world phenomena in which event occurrence rates vary with external factors. A key challenge in analyzing the nonparametric maximum likelihood estimator (NPMLE) in this setting is that the Kullback-Leibler (KL) divergence may diverge without strong assumptions on the true intensity function, rendering standard excess risk analyses inapplicable. To address this issue, a unified theoretical framework based on the Hellinger distance is developed and an oracle inequality for evaluating the estimation error of the NPMLE is established under mild conditions. As an important application, deep neural network estimators are investigated and their convergence rates are derived. The results show that, when the true intensity function has a compositional structure, deep learning-based NPMLE can mitigate the curse of dimensionality and automatically adapt to low-dimensional Riemannian manifold structures. Furthermore, the obtained convergence rates are proved to be nearly minimax optimal under such compositional-structure assumptions.

E1856: **Direct sampling from conditional distributions of discrete exponential families**

Presenter: **Shuhei Mano**, The Institute of Statistical Mathematics, Japan

Sampling contingency tables with fixed marginal totals is a classic problem to which MCMC has been applied. Seminal work showed that an irreducible Markov chain can be constructed by using computational algebra. It is demonstrated that discrete exponential families are associated with a system of hypergeometric partial differential equations and that the Groebner basis can guarantee that direct sampling is possible in principle. Moreover, sequential MLEs may be used if approximations are allowed, and the approach is exact if the model is rational.

E2031: **Conformalized outlier detection for mass spectrometry data**

Presenter: **Yangha Chung**, Seoul National University, Korea, South

Co-authors: Johan Lim, Xinlei Wang, Soohyun Ahn

Quality control procedures are crucial for ensuring the reliability of mass spectrometry (MS) data, vital in biomarker discovery and understanding complex biological systems. However, existing methods often concentrate solely on either sample or peak outlier detection, rely on subjective criteria, and employ overly uniform thresholds based on asymptotic distributions, thereby failing to adequately capture the characteristics of the data. A novel approach, CPOD (Conformal Prediction for Outlier Detection), is introduced leveraging conformal prediction for outlier detection in MS data analysis. CPOD simultaneously identifies outlier samples and peaks based on data-driven and distribution-free principles. Rigorous numerical evaluations and comparisons with existing methods demonstrate superior diagnostic performance. Application to real LC-MRM data underscores practical utility, enhancing data reliability and reproducibility in MS studies.

EO228 Room 208 STATISTICAL APPLICATIONS IN CLINICAL MEDICINE

Chair: Ruoyu Miao

E1229: **Joint modelling of longitudinal creatinine trajectories and kidney disease risk in children with autoimmune disorders**

Presenter: **Qendresa Selimi**, The University of Manchester, United Kingdom

Co-authors: Christiana Charalambous, Taban Baghfalaki

Research on paediatric kidney disease remains limited, and existing approaches do not fully exploit the information contained in longitudinal biomarkers that evolve over time. Longitudinal data, consisting of repeated measurements of a variable over time, are often used in clinical research to capture subject-specific trajectories. In parallel, survival data record the time to an event of interest -such as death or diagnosis- and are crucial in understanding clinical outcomes. As both data types are typically collected on the same subjects, they are linked; analysing them separately may lead to biased estimates. A joint modelling framework was employed to investigate the relationship between longitudinal serum creatinine measurements and the risk of adverse kidney outcomes in paediatric patients with autoimmune disorders at Great Ormond Street Hospital for Children NHS Foundation Trust, London. The framework combines creatinine trajectories with the time to death or diagnosis of acute kidney injury or chronic kidney disease. The results demonstrate a strong association between evolving creatinine profiles and the risk of the composite event. Dynamic risk predictions were generated using patients' observed creatinine trajectories to demonstrate the practical utility of the proposed framework. These predictions illustrate the potential of joint models to support personalised medicine and clinical decision making in paediatric nephrology through real-time risk assessment.

E1284: **Improving microbiome modelling and prediction by using an optimised and class-weighted classifier in imbalanced datasets**

Presenter: **Chengxin Li**, Leeds Inst of Medical Research (LIMR), United Kingdom

Co-authors: Henry Wood, Arief Gusnanto, Rishabh Bezbaruah

Motivation: Distinct gut microbiome profiles between colorectal cancer (CRC) patients and healthy individuals have enabled the application of machine learning (ML) approaches to microbiome-based colorectal cancer classification. However, the limited transferability of models across cohorts remains a major challenge, as severe class imbalance and heterogeneous analysis pipelines can lead to unstable performance and reduced clinical utility. Results: The impact of normalisation, taxonomic resolution, and class imbalance handling on microbiome-based CRC prediction was evaluated. Using a real-world screening cohort, Random Forest and XGBoost classifiers were developed and evaluated. GMPR normalisation consistently improved performance across models. At the genus level, including uncultured taxa, XGBoost with class weighting achieved the highest performance and outperformed Random Forest. Both models exceeded the performance of previously published approaches on the same cohort. External validation showed higher mean AUCs compared with prior studies, although XGBoost exhibited reduced generalisability across more dissimilar cohorts.

E1591: **Consensus Monte Carlo for mixtures of categorical distributions**

Presenter: **Julie Fendler**, University of Cambridge, United Kingdom

Co-authors: Paul Kirk

Motivated by privacy-sensitive electronic health record (EHR) data, a novel Consensus Monte Carlo (CMC) algorithm for Bayesian mixture models in a federated learning setting is presented, addressing scenarios where data cannot be fully shared or pooled across compute nodes. CMC assumes that the dataset is split into S shards, and an MCMC algorithm is run independently within each shard. For a given shard, the target posterior of the MCMC algorithm, called the local posterior, is proportional to the likelihood computed within the shard multiplied by a fractionated prior. To draw samples from the global posterior (i.e., the posterior over the entire dataset), different methods exist for combining the local posteriors. A naive approach is to average the draws from the local posteriors at each iteration. Alternatively, since the data may be heterogeneously distributed across shards, a weighted mean can be considered. Prior work proposes to infer these weights, called aggregation weights, within a variational inference framework. However, in the context of finite mixture models, existing approaches assume that the number of clusters and the mixture weights are known. Such methods are extended to more general settings and considers an application to large-scale EHR data.

E1235: Asymmetry analysis of bilateral shapes**Presenter:** **Xiangyu Wu**, University of Leeds, United Kingdom**Co-authors:** Kanti Mardia, John Kent, Balvinder Khambay, Colin Goodall

Many biological objects possess approximate bilateral symmetry about a midplane. The landmark-based methods are used to measure departures from bilateral symmetry. A vector of unsigned elementary features is constructed to describe the asymmetry of each object based on these landmarks. Two approaches are proposed to compare the level of asymmetry between two groups of objects. In the first approach the elementary features are combined into a scalar composite asymmetry score for each object; then standard univariate tests are used on the composite asymmetry score to assess asymmetry. In the second approach a univariate test statistic is constructed for each elementary feature; then the maximum of these statistics is used for a union-intersection test. The methodology is illustrated on two data sets collected to assess the success of two types of surgery: (1) cleft lip surgery, where the level of asymmetry is compared between a group of cleft lip subjects and a group of normal subjects; and (2) orthognathic surgery, where for each patient, the level of asymmetry after surgery is compared to that before surgery.

EO288 Room 209 RECENT ADVANCES IN SURVIVAL ANALYSIS AND DEEP LEARNING**Chair: Il Do Ha****E1583: Survival tree for current status data with dependent censoring****Presenter:** **Yang-Jin Kim**, Sookmyung Women University, Korea, South

In survival analysis, the independent censoring assumption is commonly used for inference. However, this assumption is often violated in observational studies, and inference ignoring dependency produces invalid and biased results. A survival tree method for current status data with dependent censoring is proposed by extending several survival forests such as cforest and randomForestSRC. Current status data arise in cross-sectional studies where each subject is examined to determine whether an event of interest occurs at a given censoring time. The copula graphic estimator is applied to construct the evaluation and splitting criterion. Several methods are compared to evaluate predictive performance.

E1624: DeepMsurv: M-spline based DNN framework for time-to-event data**Presenter:** **Vu Tuan Anh**, Pukyong National University, Korea, South**Co-authors:** Takeshi Emura, Il Do Ha

In survival analysis, deep neural networks (DNNs) have been widely used to capture complex nonlinear relationships in time-to-event data. However, standard DNN-based models such as DeepSurv primarily optimize the Cox partial likelihood and do not directly estimate the baseline hazard function during training. To address this limitation, a novel M-spline-based DNN survival framework for time-to-event data, referred to as DeepM-surv, is proposed, which is efficiently optimized via a loss function based on the full log-likelihood under a five-parameter cubic M-spline-based hazard. By integrating the M-splines into the network architecture to flexibly model the baseline hazard, the approach enables the direct construction of interpretable hazard plots. Through comprehensive simulation studies and practical real-data applications, the proposed method demonstrates competitive predictive performance compared with state-of-the-art models, while substantially improving computational efficiency, interpretability, and clinical utility. The DeepMsurv method was implemented using PyTorch.

E1691: Poisson-gamma neural network for clustered count data**Presenter:** **Hangbin Lee**, Chungnam National University, Korea, South**Co-authors:** Il Do Ha, Changha Hwang, Youngjo Lee

Large-scale biomedical datasets often involve clustered count outcomes with complex correlation structures arising from high-cardinality categorical features. Standard neural network approaches typically overlook such dependencies, limiting their ability to deliver subject-specific predictions. A Poisson-gamma neural network is introduced that incorporates gamma random effects into a neural network within a hierarchical likelihood framework. The proposed model captures both nonlinear population-level patterns and subject-specific variability in a unified manner. To further improve computational efficiency and stability, an adjustment scheme for random effects and variance components is developed. Simulation studies demonstrate that the proposed model consistently outperforms existing models in terms of mean squared Pearson error and mean deviance across a range of random effect distributions. Applications to real biomedical datasets demonstrate its practical advantages for clustered count data analysis.

E1541: Left-truncation of the exponential distribution**Presenter:** **Rafael Weissbach**, University of Rostock, Germany**Co-authors:** Fiete Sieg

In a longitudinal study (panel data), statistical units of the population can be excluded from sampling (left-truncated) due to a death event before the study. Assuming a distribution for the lifespan is mandatory if the life expectancy (or if effects on it, or if the distribution of one covariates that may affect mortality, or if association between covariates of which at least one may affect mortality) are to be reported. For illustration an Exponential distribution is used. The distribution of the birth date (if observed), or equivalently the age at the begin of the study (the first wave of the panel), as second measurement to a unit, does not need to be specified. Its distribution can be profiled-out (as is the unobserved latent sample size with the theory of point processes) with a functional-analytic projection method. The resulting so-called efficient score can then be analyzed as a usual parametric M-estimator. The approach can be extended to two additional design defects, right-censoring and right-truncation. For the latter case of a double truncated sample, a data example of 55 thousand observed German enterprise lifespans, a life expectancy of 6.2 years results, at a standard error of ten weeks. The model can be extended to copula-dependent truncation which is important if mortality changes in time.

EC265 Room B103 FORECASTING**Chair: Kaiji Motegi****E1197: Constrained machine learning and stacking for real-time survey forecast combination****Presenter:** **Adam Csapai**, The Institute of Economic Research of Slovak Academy of Sciences and University of Economics in Bratislava, Slovakia

Survey-based forecast combinations are often evaluated using simple averages, as more flexible methods tend to overfit in small real-time samples. The aim is to examine whether machine learning and stacking can improve the combination of professional survey forecasts when estimation is tightly constrained. Monthly forecasts from Slovak commercial banks for current-year GDP growth and inflation over 2007-2024 are combined using a strictly real-time rolling design that accounts for data publication lags. Linear shrinkage methods, non-linear machine learning models, and non-negative stacking based on out-of-sample base predictions are compared to the equal-weight benchmark. For inflation, regularised linear combinations deliver stable gains, reducing mean squared forecast errors by 10-20 percent, while non-linear methods systematically underperform. For GDP growth, several non-linear models and restricted stacking schemes yield large improvements, with error reductions of up to 40 percent relative to the benchmark. The results demonstrate that the usefulness of machine learning in survey forecast combination is target-specific and depends critically on real-time constraints, providing guidance on when complexity adds value in small panels of professional forecasters.

E1845: Adaptive multi-scale forecasting and gate-localized conformal prediction for multivariate nonstationary time series**Presenter:** **Ziling Ma**, King Abdullah University of Science and Technology (KAUST), Saudi Arabia**Co-authors:** Junshu Jiang, Angel Lopez Oriona, Hernando Ombao, Ying Sun

ABF-T-GLCP is a model-agnostic framework for forecasting and uncertainty quantification in nonstationary multivariate time series. The central

idea is to learn an adaptive predictive state representation for point forecasting and reuse it for Conformal calibration. The forecasting module combines horizon-specific temporal experts through a learned gate and refines predictions using sparse predictive transfer across related series. The uncertainty module, Gate-Localized Conformal Prediction (GLCP), uses the learned gate state together with temporal recency to select locally relevant calibration residuals, thereby coupling uncertainty calibration to the predictive regimes used by the forecasting model. This shared representation allows point forecasts and Prediction intervals to adapt consistently under evolving temporal dynamics while retaining the model-agnostic nature of Conformal Prediction and yielding approximate local coverage under mild stability conditions. Experiments on a large-scale high-frequency commodity forecasting benchmark demonstrate consistent gains in point forecasting accuracy and substantially narrower Prediction intervals with empirical coverage close to the nominal level. Additional results indicate that the framework extends beyond the motivating financial application.

E1986: **Evaluating forecast performance at the Bank of England**

Presenter: **Emi Mise**, University of Leicester, United Kingdom

Co-authors: Anthony Garratt

The Bank of England has historically reported predictive densities for key macroeconomic variables in the form of split-normal distributions. This practice was criticised in the Bernanke report, which evaluated the Bank's forecasting performance. A Bayesian time-varying stochastic VAR model is developed, building on the specification proposed by Clark, McCracken and Merten (CMM-SV). The approach uses nowcast forecast errors and exploits the accounting identity between prediction errors and revisions to forecasts at different horizons in order to model time-varying uncertainty. Real-time predictive densities for U.K. macroeconomic variables are constructed and evaluated against the Bank's split-normals, a simple normal error benchmark, and densities derived from the Survey of External Forecasters, using evaluation methods including coverage rates, continuous ranked probability score (CRPS) and probability events of interest. The analysis also jointly models forecast errors and forecast revisions for inflation and output growth. Results indicate that this method, which delivers sizeable gains for U.S. data relative to prediction intervals based on historical forecast-error variances, including during periods of heightened uncertainty, yields only limited gains over the Bank's split-normal distributions and other institutional benchmarks. These findings suggest that the benefits of introducing stochastic volatility for density prediction may be context-dependent.

E2004: **Bayesian SARIMA modelling with a spatial error structure for electricity demand in Japan**

Presenter: **Hanzhi Yu**, Aoyama Gakuin University, Japan

Co-authors: Haruhisa Nishino

Regional electricity demand in Japan exhibits strong seasonality and cross-regional dependence, requiring forecasting models that jointly account for temporal and spatial structures. A Bayesian seasonal autoregressive integrated moving average (SARIMA) framework is developed for regional electricity demand, and alternative spatial specifications are compared using quarterly and monthly regional data. Predictive performance is evaluated using WAIC and Pareto-smoothed importance sampling leave-one-out cross-validation. The findings indicate that combining seasonal dynamics with a spatial error structure improves predictive performance and provides a computationally efficient way to capture spatial dependence. In the quarterly data model, an electricity price proxy, defined as sales revenue divided by electricity sales volume, is also incorporated into the model and analysed. The Great East Japan Earthquake appears to have a largely temporary effect on electricity demand, whereas electricity market liberalization is interpreted as a regime shift. Overall, jointly modelling seasonal patterns and spatial error dependence can be useful for regional electricity demand forecasting in Japan.

EC262 Room B104 FINANCIAL ECONOMETRICS

Chair: Jackson Lautier

E1161: **Stationary but not profitable: A critical look at pairs trading**

Presenter: **Davide Pandini**, International Federation of Technical Analysts (IFTA), Italy

Pairs trading is a market-neutral statistical arbitrage strategy that targets mean reversion in spreads between economically related assets. Conventional implementations select pairs using distance measures or cointegration tests, yet much of the evidence is drawn from pre-2008 crisis sample periods, a restricted investment universe and optimistic assumptions about costs and parameter stability, leaving its effectiveness in modern equity and equity-commodity markets uncertain. A fully reproducible end-to-end workflow for spread construction, cointegration diagnostics and rule-based trading is developed, evaluated under a strict train/test protocol with explicit transaction-cost modeling on daily data from 2007 till 2025. Results indicate that equity and index spreads provide the most consistent net risk-adjusted performance, whereas commodity-linked spreads seldom exhibit cointegration after 2018 and typically underperform once costs are applied, in contrast with prior findings in the commodity pairs-trading literature. The evidence further shows that cointegration is informative but neither sufficient nor necessary for out-of-sample profitability: some in-sample cointegrated relations deteriorate following abrupt changes in market behavior, while certain non-cointegrated spreads still yield short-horizon convergence returns. The workflow and diagnostics delineate when spread trading adds value as a controlled relative-value and risk-management tool in recent markets.

E1880: **Quantile regression and fuzzy-probabilistic inference in financial decision-making**

Presenter: **Tomas Tichy**, VSB-TU Ostrava, Czech Republic

Co-authors: David Nedela, Michal Holcapek

Weighted quantiles are essential tools in statistical analysis and regression, particularly for time-series data and moving quantile functions. Traditionally, their computation has relied on linear programming methods, which, while effective, can be computationally intensive. An alternative approach utilizing the right derivatives of the associated piecewise linear function is introduced. The corresponding weighted quantile is derived by minimizing this function. This alternative approach retains the precision and reliability of traditional techniques while simplifying the computation. Furthermore, this approach is combined with probabilistic fuzzy rules and inverse quantile fuzzy transforms to propose a framework for estimating running quantiles and predictive modelling. Additional results on financial decision-making are provided, which strengthen the dimensionality reduction and robust regression techniques. For comparison purposes, parametric and nonparametric approximation techniques are also considered. The proposed models are applied to a real-world dataset of financial assets, demonstrating their effectiveness in outperforming traditional approaches.

E1270: **On the recovery of meaningful structure from financial transfer entropy networks**

Presenter: **Haotian Yang**, Bocconi University, Italy

Co-authors: Yifeng Li

Financial network studies frequently use transfer entropy to identify systemic hubs under high-dimensional constraints, typically at $T/N < 5$. A matched data-generating process audit evaluates the node-level estimation reliability of these methods using realistic financial dynamics, including GARCH and common factors. Monte Carlo simulations reveal that at $T/N < 5$, ordinary least squares-based estimators yield precision below 17%, rendering hub detection close to random. LASSO regularization achieves higher precision but collapses to near-empty networks on empirical S&P 500 data. Furthermore, a statistical power analysis demonstrates that estimation noise alone destroys realistic network-return channels; an embedded 10% annualized network premium yields an insignificant estimated t-statistic of 0.74 at $T/N = 5$, compared to an oracle t-statistic of 5.74. While aggregate connectivity trends partially survive estimation noise, specific node-level topological claims are fundamentally unreliable.

The findings establish a binding T/N barrier, indicating that reliable edge-level estimation requires a ratio of at least 8 to 10, necessitating sector-level aggregation or high-frequency data for valid inference.

E1356: Granger causality in expectiles: An M-vine copula test

Presenter: **Roberto Fuentes-Martinez**, Universidad de Alicante - IMT Altı Studi Lucca, Italy

Co-authors: Irene Crimaldi

A model-free measure of Granger causality in expectiles is proposed, generalizing the traditional mean-based measure to arbitrary positions of the conditional distribution. Expectiles are the only law-invariant risk measures that are both coherent and elicitable, making them particularly well-suited for studying distributional Granger causality where risk quantification and forecast evaluation are both relevant. Based on this measure, a test is developed using M-vine copula models that accounts for multivariate Granger causality with $d + 1$ series under non-linear and non-Gaussian dependence, without imposing parametric assumptions on the joint distribution. Strong consistency of the test statistic is established under some regularity conditions. In finite samples, simulations show accurate size control and power increasing with sample size. A key advantage is the joint testing capability: causal relationships invisible to pairwise tests can be detected, as demonstrated both theoretically and empirically. Two applications to international stock market indices at the global and Asian regional level illustrate the practical relevance of the proposed framework.

EC306 Room 106 SHORT TALKS IN STATISTICAL METHODS AND APPLICATIONS

Chair: Jisang Yu

E1901: Children's health shocks and maternal job separation: Evidence from Japan

Presenter: **Tomoe Naito**, Seikei University, Japan

The aim is to investigate how children's health status affects mothers' labor supply using individual-level data from the Longitudinal Survey of Newborns in the 21st Century in Japan. A lagged fixed effects model is employed to capture the dynamic effects of outpatient visits and hospitalizations on maternal employment. The results indicate that when children's outpatient visits or hospitalizations persist for more than two years, mothers' probability of job separation decreases by about 1.5 percentage points on average. However, these effects vary by the child's gender and age, as well as by household income. In lower-income households, the negative effects of hospitalization weaken, lose statistical significance, and even become positive. Among households earning below 2 million yen, which is often below the relative poverty line in Japan, prior hospitalization increases the probability of maternal job separation by about 9 percentage points. These findings suggest that, for the overall sample including high-income households, children's health shocks reduce the likelihood of maternal job separation. In contrast, among low-income households, such health shocks may increase the probability of maternal job separation, likely reflecting reservation wage. While work-family conflict reduces job separation, it does not alter the impact of children's health on employment outcomes.

E2085: A simulation study of tvwLDA parameter estimators

Presenter: **Louisa Kontoghiorghe**, Kings College London, United Kingdom

Co-authors: Ana Colubi, George Kapetanios

A new model, the time-varying weighted Latent Dirichlet Allocation (tvwLDA), has been previously proposed as an extension of the standard Latent Dirichlet Allocation (LDA). Unlike LDA, tvwLDA estimates document-topic and topic-term weights using a weighted log-likelihood over time while accommodating temporal variation in the number of topics. To evaluate the performance of the proposed estimation procedure, Monte Carlo simulations are conducted. The results are used to assess whether the means of the parameter estimators approach the corresponding population parameter values as the sample size increases.

E2090: Application of a regression model for random star-shaped sets:

Presenter: **Jose Grana Colubi**, University of Oviedo, Spain

Co-authors: Gil Gonzalez-Rodriguez, Ana Belen Ramos-Guajardo

A simple linear regression model for random elements taking values in a space of star-shaped sets is introduced. The least-squares estimators are derived, and their consistency and asymptotic behavior are examined both theoretically and by means of simulation studies. The model is subsequently applied to a real-world problem involving tree trunk growth. Specifically, a publicly available dataset of tree trunk cross-sections is analyzed. The cross-sections are modeled as star-shaped sets, and the proposed model is employed to explore potential linear relationships between the growth ring of a given year and those formed in earlier years.

E2034: On an essential extension of the G-LASSO

Presenter: **Piotr Graczyk**, Universite Angers, France

Co-authors: Bartosz Kolodziejek, Hideto Nakashima, Maciej Wilczynski

The Graphical LASSO is improved by addressing the estimation of high-dimensional precision matrices, a fundamental problem in modern statistics. The Graphical LASSO with its L1-penalty is a standard approach for recovering sparsity patterns of K , with mathematical properties presented in previous work. However, many statistical models exhibit richer patterns, such as colored Graphical models with equality constraints in K , which the Graphical LASSO cannot capture. Recovering these richer structures through high-dimensional estimation of the precision matrix with atomic norm penalties is addressed. The unit balls of these penalties are polytopes, and induced patterns correspond to the polytope's facial structure by belonging to a cone in the related normal cone partition of $\mathbb{R}^p$. Theoretical guarantees are established for recovering the true pattern of K . The methods extend the primal-dual witness methodology. The analysis provides conditions on the deviation between sample and true covariance matrices for successful pattern recovery, given a novel Irrepresentability Condition (IR) for any atomic penalty. When specialized to Graphical LASSO, the results improve it by requiring weaker deviation conditions and a less restrictive IR, leading to tighter bounds and better asymptotic performance than prior work. The proposed general IR, based on a new thresholding concept, provides a unified perspective on model selection consistency.

Monday 10.08.2026

08:15 - 09:55

Parallel Session K – EcoSta2026

EI007 Room 202 DATA SCIENCE IN TECH**Chair: Lihua Lei****E1942: Data science in tech: Driving product impact from telemetry to the AI era***Presenter:* **Meeyoung Park**, Google, United States

In the modern tech ecosystem, data science is the engine driving strategic differentiation. Data-driven paradigms have fundamentally transformed product development and operations through critical pillars including robust telemetry, rigorous experimentation, continuous product optimization, and deep customer insights. By converting massive datasets into actionable intelligence, data science has evolved from a supportive analytical function into a core driver of business strategy and product innovation. Simultaneously, the rapid ascent of Artificial Intelligence and Large Language Models has triggered a profound shift in the data science landscape, significantly expanding the traditional boundaries of data exploration and predictive modeling. Navigating this transition successfully requires additional skillsets and mindsets. Data scientists are pivoting from builders of isolated models to architects of intelligent systems, cultivating a strong sense of product ownership, adaptability, and the ability to balance the nuances of AI orchestration with an enduring need for statistical rigor. Drawing from hands-on and leadership experiences within Google Cloud, the historical impact of data science in tech, its modern evolution alongside generative AI, and the future-ready capabilities required to lead in this new era are examined.

E2066: Marketplace optimization for doordash ads*Presenter:* **Hao He**, DoorDash, United States

placeholder for now, will update this next week. The main talk will be around the whole marketplace optimization system in doordash

E2067: Causal adaptive learning for decision making at scale*Presenter:* **Maria Dimakopoulou**, Uber, United States

Sequential decision making and accurate model estimation from adaptively collected data lie at the heart of decision making at scale. Reliable decision making that adapts to new data via contextual bandit or reinforcement learning algorithms requires accurate model estimation. The focus is on how to incorporate state-of-the-art methods from the causal inference literature into model estimation for decision making systems and how to pair them with efficient exploration strategies such as Thompson Sampling. The performance gains unlocked by this approach in the presence of real-world challenges such as selection bias, covariate shift, model misspecification, and bias due to adaptive data collection are discussed.

EO230 Room B101 ADVANCES IN ECONOMETRICS (VIRTUAL)**Chair: Jose Luis Montiel Olea****E1520: Evidence aggregation with ignorance in mind: Learning what is not known for archetypes discovery***Presenter:* **Davide Viviano**, Harvard University, United States*Co-authors:* Emily Breza, Arun Chandrasekhar

When evaluating policy interventions, researchers often pursue two related goals: identifying which individuals or contexts benefit most, and determining whether patterns of treatment effect heterogeneity can be used to aggregate evidence across environments. A framework is developed that partitions observations, defined by individual and environmental characteristics, into groups within which treatment effects can be stably aggregated, while setting aside contexts in which extrapolation is unreliable and further evidence is needed. The procedure therefore learns both how to summarize heterogeneity and when researchers should admit ignorance. Finite-sample regret guarantees are derived for estimation and model selection, and inference procedures that quantify the value of follow-up data collection. The approach is illustrated by reanalyzing a multifaceted anti-poverty program implemented in six countries.

E1779: Decision theory for the archetype discovery problem*Presenter:* **Jose Luis Montiel Olea**, Cornell University, United States

In the archetype discovery problem, a researcher wants to summarize N heterogeneous policy effects of interest that vary over a discrete set of covariates. The goal is to partition the set of covariates into $K < N$ groups—the *archetype sets*—and to provide a summary of the policy effects for each group. Decision theory shows that, under a weighted mean-squared-error criterion, the archetype discovery problem can be viewed as a weighted K -means clustering problem applied to the N heterogeneous policy effects. The results imply that the procedure which minimizes average risk for a given prior can be obtained by clustering the different values of the posterior mean estimate of the policy effects of interest. Similarly, an approximately minimax procedure in large samples can be obtained by clustering a consistent estimator of the policy effects. In both of these cases, an exact solution to the weighted K -means clustering problem can be found using a simple and well-known dynamic programming algorithm. The main results extend to allow for the possibility of *abstention*; that is, the researcher can admit ignorance about the effects of the policy of interest for some archetype sets. A set of assumptions exists under which a minor modification to the baseline dynamic programming algorithm solves the archetype discovery problem with abstention.

E1792: Estimating treatment effects in finite horizon experiments with rolling enrollment*Presenter:* **Jaume Vives-i-Bastida**, Stanford GSB, United States*Co-authors:* Stefan Wager, Ramesh Johari

All experiments unfold over a finite horizon. In combination with rolling enrollment of experimental subjects, this complicates the estimation of treatment effects. Subjects arriving later in the experiment have shorter observation horizons, leading to a Censoring problem. Standard methods relying on simple difference-in-means (DiM) estimators conflate early subject arrivals with treatment effects, leading to potentially misleading long-term estimates. Formal modeling of the Censoring process inherent in rolling enrollment finite horizon experiments identifies the adoption/conversion rate curve -the Probability that a subject realizes its outcome within a fixed time- as the primary object of interest. To estimate the adoption rate curve, an Inverse Probability of Censoring Weighting estimator (IPCW) and an Augmented Inverse Probability of Censoring Weighting (AIPCW) estimator are introduced. Theoretical guarantees are provided for the IPCW and AIPCW estimators, and the efficiency gains of the AIPCW estimator are quantified theoretically. A framework for targeting long-run effects is considered, and the MSE optimal estimators among a class of adoption curve weighting estimators are derived. Extensive simulations and applications to A/B tests run at Airbnb highlight the differences between alternative estimators and demonstrate the importance of estimating the conversion rate curve directly for evaluating treatment effects.

E1810: On the asymptotic properties of debiased machine learning estimators*Presenter:* **Amilcar Velez**, Cornell University, United States

Debiased machine learning (DML) is considered when the number of cross-fitting folds, K_n , may grow with the sample size n . Existing fixed- K asymptotic theory implies that DML1 and DML2, two variants of DML estimators, are asymptotically equivalent, providing no guidance on which variant to use or how to choose K_n . It is shown that when $K_n \rightarrow \sqrt{n}$, DML1 can exhibit asymptotic bias, implying that standard inference based on DML1 may fail, whereas inference based on DML2 remains valid. Under an algorithm stability condition, the standard asymptotic theory for DML2 remains valid for any $K_n \leq n$. Finally, for scalar DML2 estimators whose first-step estimators admit a linear stochastic expansion, a valid

second-order asymptotic approximation is derived showing that larger K_n lowers second-order asymptotic bias and mean-squared error, although the marginal improvements decline with K_n . In particular, among common choices for DML2, $K_n = 10$ is preferred over $K_n = 5$.

EO055 Room B102 STATISTICS AND ECONOMETRICS IN SPORTS AND CONSUMER FINANCE
Chair: Jackson Lautier
E1398: Injury risk management in the National Basketball Association: An actuarial approach

Presenter: **Hashan Savinda Peiris Kalugama Gardige**, Simon Fraser University, Canada

Co-authors: Jackson Lautier, Himchan Jeong

Major professional sports leagues have become large-scale financial institutions, increasing the financial risk associated with athlete injuries and the need for injury risk management. Injuries in the national basketball association are analyzed from the perspective of insurable risks using a three-part actuarial compound model for injury frequency and a two-part severity structure capturing missed games per injury and financial loss per missed game. The model is fit on a novel dataset from the 2022-23 NBA regular season combining injury reports, tracking and box score metrics, salary and demographic information, travel, television market variables, attendance, and estimated single-game values. Higher expected losses are predicted for players with higher salaries, average minutes played above 32 per game, and larger markets. The top ten players account for about \$453m in at-risk injury losses per season, exceeding 25% of total estimated at-risk losses. Risk mitigation implications follow from the actuarial structure. A uniform 10% reduction in average minutes is estimated to reduce expected compound loss for the highest-risk players by 30–35% (about \$9–25m per player). Leaguewide, shortening NBA games from 48 to 40 minutes is estimated to reduce total expected claims by about \$493m.

E1608: Beyond univariate analysis: Detecting mean and variance shifts across North American sports leagues and baseball history

Presenter: **Mena Whalen**, Loyola University Chicago, United States

Co-authors: Brian Mills, Gregory Matthews

Recent methodological developments in multivariate change point analysis are applied to examine structural shifts across two distinct domains of sports economics. The method identifies changes in both level (mean) and variance simultaneously, offering a more complete picture of structural change than prior univariate approaches. In the first application, competitive balance in North American professional sports leagues is tracked by simultaneously examining nine competitive balance measures spanning game uncertainty, playoff uncertainty, and consecutive season uncertainty. This approach identifies the specific measure(s) driving each estimated change point and captures changes to the variance structure of competitive balance. Results suggest that rival leagues, labor supply increases, expansion, and changes to playoff structures are the factors most commonly concurrent with structural breaks. In the second application, the same framework identifies distinct historical eras in Major League Baseball using team-level performance statistics from the late 1800s through 2021. The method recovers well-established eras including the Dead Ball Era, Integration Era, Steroid Era, and Post-Steroid Era, while also pinpointing periods of team-level dynastic dominance and collapse.

E1397: Discrete time-to-event regression analysis under left-truncation with applications to consumer finance

Presenter: **Jackson Lautier**, Bentley University, United States

Co-authors: Jun Yan, Vladimir Pozdnyakov

Asset-backed securities (ABS) play a vital role in financing American consumer automobile debt. Recently, economic analysis into ABS has benefited from the public release of data, which provides a new, rich source of loan level consumer auto loan information. Because this loan lifetime data is discrete-time and subject to random left-truncation, however, it is nontrivial to analyze. This has attracted recent study, but there is still no suitable approach to model this ABS loan lifetime data that incorporates regression coefficients for the lifetime of interest. The conditional, bivariate distribution is thus generalized to link to covariates, while keeping the left-truncation distribution unspecified. The high-dimensional, constrained likelihood-based parameter estimation problem is solved numerically, using a block coordinate descent design. Under suitable regularity conditions, the complete large sample, multivariate normal distribution of the estimators is provided. This allows for large sample inference into variable and model selection. All results are proven and verified through simulation studies. The methods are then applied in an economic study of borrower prepayment behavior for 1,553 consumer auto loans from the 2017-3 Ally Auto Receivables Trust ABS bond. It is found that borrowers with pick-up trucks prepay slower, all else equal, among other consumer finance insights.

E1521: abslife: Estimating discrete left-truncated lifetime random variables with applications to asset-backed securities

Presenter: **Lucas da Cunha Godoy**, University of California Santa Cruz, United States

Co-authors: Jackson Lautier

The consumer auto lending market in the United States operates on a massive scale: consumer auto asset-backed securities (ABS) issuance tops 200 billion and total consumer automobile debt exceeds 1,400 billion. Despite this scale, there is a lack of ready-to-use software implementing statistical methods for drawing inference on the lifetime distribution of individual consumer loans sampled from ABS. Such tools would benefit ABS investors because the source of the ABS level cash flow is the sum of its underlying individual assets. Hence, investors require precise time-to-event distribution estimates to accurately model ABS trust level performance. Furthermore, the convenience of statistical software designed for this purpose would be valuable in the high-paced environment of fixed-income trading. The ABS data is nontrivial to analyze, however, and a review is provided of statistical methods for left-truncated, right-censored, discrete time-to-event data, including competing risks. Next, asymptotic distributions for the cumulative and probability mass functions for the time-to-event distribution estimates are introduced. The abslife package for R is then presented, designed specifically to analyze such data. The package's utility is illustrated by analyzing 275,948 consumer auto loans spanning four distinct ABS bonds, demonstrating how abslife enables financial analysts to perform robust inference for discrete time-to-event data.

EO200 Room B103 RECENT ADVANCES IN RANDOM MATRIX THEORY AND HIGH-DIMENSIONAL INFERENCE
Chair: Xiaocong Xu
E1438: On spectrum of sample correlated matrices from large fold tensor vectors

Presenter: **Wangjun Yuan**, Southern University of Science and Technology, China

The limiting spectral distribution of the sample correlation matrix is investigated, whose sample vectors are k -fold tensor products of n -dimensional vectors with i.i.d. entries. The focus is on the limiting regime $n, k \rightarrow \infty$ with $k = o(n)$, and it is shown that the limiting spectral distribution is the Marcenko-Pastur law. As a consequence, it is shown that the limiting spectral distribution of the Wishart matrix from the k -fold tensor product of independent uniformly distributed unit vectors in $\mathbb{C}^n$ is the Marcenko-Pastur law.

E1458: Algorithmic inference in proportional high dimensions

Presenter: **Xiaocong Xu**, University of Southern California, United States

Many modern learning problems are studied in a proportional high-dimensional regime, where the feature dimension is not negligible compared to the sample size. An algorithmic perspective on this regime views gradient descent and proximal gradient iterates as statistical estimators along the optimization path, rather than focusing only on the final empirical risk minimizer. Results characterizing the distribution of these iterates show how this characterization can be used to construct data-driven estimates of generalization error and debiased iterates for statistical inference, including in settings beyond linear regression. Simulations illustrate how tuning for prediction and tuning for inference may differ along the optimization trajectory.

E1556: The asymptotic properties of the extreme eigenvectors of high-dimensional spiked models*Presenter:* **Xiaozhuo Zhang**, Northeast Normal university, China*Co-authors:* Zhangni Pu

Asymptotic behaviors of the extreme eigenvectors in the general spiked covariance matrix and sample canonical correlation matrix are investigated, where the dimension and sample size increase proportionally. The restrictive assumption of the block diagonal structure and the 4th moment in the population covariance matrix is eliminated. As for the SCC matrix, the asymptotic distribution of inner product of population and sample spiked eigenvectors is calculated. Simulation results are also presented to validate the key theoretical findings.

E2082: Mesoscopic dynamics in complex networks: A random matrix perspective*Presenter:* **Zhenggang Wang**, Southeast University, China

Complex networks often display patterns that are neither reducible to node-level fluctuations nor fully captured by global or coarse-grained structure. A random-matrix perspective on such intermediate dynamics in temporal networks is proposed. The mesoscale is treated as a transient dynamical regime, where local interactions have begun to organize but the system has not yet settled into stable large-scale behavior. Random matrix theory provides spectral baselines and diagnostics for identifying when such intermediate patterns are unlikely to be explained by noise alone.

EO166 Room B104 CAUSAL INFERENCE AND STRUCTURAL LEARNING IN COMPLEX SETTINGS**Chair: Shuoyang Wang****E1170: Inference for quantile mediation effects in the presence of complex confounding***Presenter:* **Shuoyang Wang**, University of Louisville, United States

Traditional mediation analysis methods face challenges when dealing with a large number of mediators. In practice, these challenges can be compounded by outliers and the complex relationships introduced by confounders. To address these issues, a novel quantile-based partially linear mediation analysis method is proposed that can handle high-dimensional mediators and introduce deep neural network techniques to model intricate relationships in confounders. Unlike most existing works that focus on mediator selection, the emphasis is on inference on mediation effects. Theoretical analysis shows that the proposed procedure controls type I error rates for hypothesis testing on mediation effects. When the dimension of mediator is high, the proposed method consistently selects important features in the outcome model. Numerical studies show that the proposed method outperforms existing approaches under a variety of settings, demonstrating its versatility and reliability as a modeling tool for complex data. Application of the proposed method to study DNA methylation's mediation effect of childhood trauma on cortisol stress reactivity reveals previously undiscovered relationships by providing a comprehensive profile of the relationship at various quantiles.

E1205: Joint low-rank and individual sparse modeling for multi-response matrix-variate trace regression*Presenter:* **Daeyoung Ham**, The University of Texas at San Antonio, United States*Co-authors:* Brad Price

Multi-response regression with matrix-valued predictors is studied. A unified joint-structure learning approach is proposed that separates signal shared across responses from response-specific heterogeneity, enabling simultaneous borrowing of strength and individualized adaptation in high dimensions. A convex regularized estimator is introduced that combines low-rank structure for the common component with sparse structure for the individual deviations, and a scalable optimization algorithm based on alternating direction methods is developed. The same framework extends naturally from continuous outcomes to multivariate generalized linear models such as multivariate logistic regression while retaining the same decomposition and computational strategy.

E1228: Structural learning and causal interference in multiplex networks: A two-stage framework for financial distress*Presenter:* **Tianhai Zu**, University of Texas at San Antonio, United States

Analyzing systemic risk in high-dimensional corporate networks presents a dual statistical challenge: first, estimating the unobserved dependency structure among thousands of firms; and second, identifying causal effects in the presence of network interference. A novel framework is proposed that bridges these two objectives to analyze corporate bankruptcy. The structural learning problem is first addressed by developing a multi-layered latent position model for multiplex networks, integrating supply chain and co-investment data into a unified low-rank embedding. This approach effectively reduces dimensionality while capturing complex transitivity and homophily, significantly enhancing predictive performance over baseline models. Building on this estimated topology, the inquiry is then extended to network causal inference. A pilot framework is explored that utilizes the learned latent structure to model spillover effects, aiming to disentangle the causal propagation of financial distress from mere structural correlation. By synergizing latent space modeling with causal identification strategies, a robust statistical pathway is offered for understanding how risk is structured and transmitted in complex economic systems.

E1784: Impact of drug-free communities program on youth development*Presenter:* **Wei Fu**, University of Louisville, United States

The Drug-Free Communities Support Program aims to reduce youth substance use by fostering multi-sector collaboration and implementing locally tailored strategies within the community. This study is among the first to evaluate the Program's impact on youth development outcomes in the United States. The DFC Program significantly reduces drug-related juvenile crime and improves academic performance. The effects are particularly pronounced in Communities where coalitions include government agencies, highlighting the critical role of institutional coordination in mobilizing local resources. Behavioral mechanisms are explored, documenting reductions in marijuana use and opioid-related inpatient hospitalizations. A cost-benefit analysis indicates substantial welfare gains, suggesting that early-life prevention represents a cost-effective investment in public health.

EO087 Room Virtual R01 RECENT DEVELOPMENTS IN BIOSTATISTICAL METHODS AND APPLICATIONS**Chair: Yisheng Li****E1866: TxMicroSim: A web-based app for multi-stage treatment decision making***Presenter:* **Xuelin Huang**, University of Texas MD Anderson Cancer Center, United States

Multistate disease progression complicates the comparison of treatment strategies, motivating the development of a user-friendly web tool for multistate disease progression simulation and treatment decision support. Microsimulation is a natural approach for identifying optimal treatment strategy. The framework employs microsimulation using a multistate model to capture state-to-state transition hazards. TxMicroSim is a web-based interface built with R Shiny to determine the optimal strategy of sequential treatment by comparing restricted mean survival times (RMST) under alternative treatment strategies. The application consists of three main steps: defining the multistate structure, building the multistate model, and performing microsimulation. Throughout the app, interactive visualizations are provided for the multistate structure, estimated hazard ratios and baseline hazards, and RMST comparisons across treatment strategies.

E1904: A general framework for incorporating identification uncertainty in individualized treatment rules**Presenter:** Muxuan Liang, The University of Texas MD Anderson Cancer Center, United States

Estimating individualized treatment rules (ITRs) from observational data or clinical trials with non-adherence is challenging due to potential unmeasured confounding bias. Instrumental variables (IV) can offer partial identification of the possible values for conditional average treatment effects (CATEs). However, optimal treatment decisions under partially identified CATEs can be uncertain, and current literature fails to inform such uncertainty in treatment decisions. A deferred treatment option is adopted and a novel class of optimal ITRs called Enhanced IV-optimal ITRs with deferred option (EIV-ITRs) is developed to guide treatment decisions and account for identification uncertainty. The deferred option extends beyond the original treatment options, identifying patients susceptible to identification uncertainty and allowing for the collection of additional information or physicians' guidance to aid in optimal decisions. To estimate the EIV-ITR, a weighted classification framework is developed with a modified hinge loss function, where the weights are nonsmooth transformations of nuisance parameters. An augmented empirical risk minimization approach is further proposed to estimate the EIV-ITRs, achieving a fast convergence rate even if the nuisance parameters are estimated using nonparametric or machine learning methods.

E1921: Bayesian nonparametric model for heterogeneous treatment effects with zero-inflated data**Presenter:** Yisheng Li, The University of Texas MD Anderson Cancer Center, United States**Co-authors:** Chanmin Kim, Ting Xu, Zhongxing Liao

One goal of precision medicine is to develop effective treatments for patients by tailoring to their individual characteristics. To achieve this goal, statistical models must be developed that can identify and evaluate heterogeneous treatment effects in a robust manner. Existing methods for assessing treatment effect heterogeneity are often based on parametric models with interactions or conditioning on covariate values, the performance of which is sensitive to omission of covariates and/or choice of their values. A new Bayesian nonparametric (BNP) method is proposed for estimating heterogeneous causal effects in studies with zero-inflated data. The method employs enriched Dirichlet process (EDP) mixtures in a BNP approach, establishing a connection between outcome and covariate DP mixtures. This enables estimation of posterior distributions concurrently, facilitating flexible inference regarding individual causal effects. Simulation studies demonstrate that the proposed method outperforms two other BNP methods in terms of bias and mean squared error of the conditional average treatment effect estimates. The proposed model has the advantage of appropriately reflecting uncertainty in regions where the overlap condition is violated. The method is applied to a study of the relationship between heart radiation dose and the blood level of high-sensitivity cardiac troponin T (hs-cTnT) to examine whether the effect of radiation dose on hs-cTnT varies by baseline characteristics.

E1973: All-in-one toolkit for biobank-scale whole-genome sequencing data management and analysis**Presenter:** Zilin Li, Northeast Normal University, China

Biobank-scale Whole-Genome Sequencing (WGS) studies are increasingly pivotal in unraveling the genetic bases of diverse health outcomes. However, managing and analyzing the sheer volume and complexity of these datasets presents significant challenges. vcf2agds is proposed, an all-in-one toolkit that efficiently converts WGS Data from Variant Call Format (VCF) Format to the annotated Genomic Data Structure (aGDS) Format. vcf2agds significantly reduces Data size while supporting seamless Genomic and functional Data integration for comprehensive genetic analyses. STAARpipeline equipped with aGDS files enables scalable, comprehensive, and functionally informed WGS analysis, facilitating the detection of common and rare coding and noncoding phenotype-genotype associations. Application of STAARpipeline to analyze Alzheimer disease (AD) in 459,216 samples from the UK Biobank demonstrated analyses that scale well in computation time and memory, yielding several potentially new significant associations with AD. As WGS datasets continue to expand in size and complexity, these tools will be increasingly useful for unlocking the full potential of Genomic research.

EO204 Room 101 ADVANCEMENTS IN CHANGE-POINT DETECTION FOR MULTIVARIATE AND OBJECT DATA**Chair:** Lynna Chu**E1839: Distribution-free change-point detection in high dimensions: A generalized homogeneity approach****Presenter:** Xianyang Zhang, Texas A&M University, United States**Co-authors:** Runmin Wang

A distance-based method is proposed to detect and localize general structural breaks in high-dimensional data beyond the first two moments, a problem largely unexplored compared to standard mean and covariance shifts. The approach uses generalized homogeneity tests and constructs a cumulative sum process in an embedded Hilbert space, for which the limiting null distribution and asymptotic consistency are rigorously established under the high-dimensional medium sample size (HDMSS) framework. To estimate multiple changepoints, the statistic is integrated with the Narrowest-Over-Threshold (NOT) strategy and componentwise monotone transformations are introduced for enhanced robustness. Empirical studies on simulated and U.S. stock return data demonstrate the method's superiority. The methodology is implemented in the R package KDist, available at <https://github.com/zhangxianyang-tamu/KDist>.

E2026: Online change-point detection using foundation probabilistic forecasting models**Presenter:** Zhengyuan Zhu, Iowa State University, United States**Co-authors:** Alan Moore, Lynna Chu

Online change-point detection is a fundamental problem in sequential monitoring, where the goal is to detect distributional changes in a time series as quickly as possible while controlling false alarms. A general framework for online change-point detection is proposed that leverages foundation probabilistic forecasters, such as Chronos-2, which can capture complex temporal structure including trend, seasonality, heteroskedasticity, and nonlinear dependence. The approach transforms each incoming observation relative to its predictive quantile forecast into an approximately standard Gaussian monitoring statistic, which can then be used in classical sequential detection procedures such as CUSUM and MOSUM. To reduce adaptation of the forecaster to post-change observations, a buffer is introduced between the forecasting context window and the forecast target. Because this design can induce serial dependence in the monitoring statistics, a parametric bootstrap procedure is developed to calibrate critical thresholds and control false alarms in practice. Extensive simulations across data-generating processes involving non-normality, trend, seasonality, heteroskedasticity, and autoregressive dependence show that the proposed method achieves competitive and often superior detection performance relative to benchmark procedures, while requiring no explicit specification of a parametric pre-change model.

E2048: Sequential change detection with active sampling**Presenter:** Georgios Fellouris, University of Illinois, Urbana-Champaign, United States

The problem of sequential change detection with active sampling is considered, where data are collected sequentially, an action is taken at each time instant, and an abrupt change occurs at an unknown time. Each action can be selected in real time based on the previously collected data and influences the distribution of the current observation both before and after the change. The objective is to jointly design an action policy and a stopping rule in order to detect the change as quickly as possible while controlling the false alarm rate. The pre-change distribution is completely specified, but a general composite post-change regime is considered. A scheme is presented that achieves first-order asymptotic optimality under Lordens minimax criterion as the mean time to false alarm and possibly the number of controls tend to infinity.

E2050: Change-point detection for object-valued time series*Presenter:* **Changbo Zhu**, University of Notre Dame, United States

Change point detection is addressed for object-valued data residing in a metric space, a topic that has attracted recent interest. Existing methods either focus on independent data or can only detect changes in the Frechet mean or variance. A self-normalization (SN) based statistic is proposed for detecting shifts in the marginal distribution of object-valued time series. The test is universally applicable to a wide range of object-valued data, such as distributional and network data, and accommodates weak serial dependence. The proposed test statistic is nearly tuning parameter free, has a pivotal limiting null distribution, and uses only pairwise distances. When combined with the Wild Binary Segmentation (WBS) algorithm, the statistic can estimate the number and locations of multiple change points. Asymptotic results for the SN-based statistic are derived under both null and local alternatives in the single change point setting. The WBS estimation consistency is established for a broad class of object-valued time series in a nonparametric setting, requiring new non-standard theoretical arguments. Extensive numerical experiments and real data analysis demonstrate the effectiveness and broad applicability of the proposed method.

EO127 Room 102 RECENT ADVANCES IN FUNCTIONAL DATA ANALYSIS**Chair: Tianyu Guan****E1640: Functional data analysis in neuroimaging data based on semiparametric transformation models***Presenter:* **Haolun Shi**, Simon Fraser University, Canada

Alzheimer's disease (AD) is a progressive neurodegenerative disorder that leads to memory loss, cognitive decline, and behavioral changes, without a known cure. Neuroimages are often collected alongside covariates at baseline to forecast patient prognosis. Identifying regions of interest within neuroimages associated with disease progression is thus of significant clinical importance. Two major complications arise in such analysis: the domain of the brain area in neuroimages is irregular, and the time to AD is interval-censored, as the event can only be observed between two revisit time points. A semiparametric approach models imaging predictors via bivariate splines over triangulation and incorporates them in a flexible class of semiparametric transformation models. Regions of interest are identified by maximizing a penalized likelihood. A computationally efficient expectation-maximization algorithm is devised for parameter estimation. An extensive simulation study evaluates the finite-sample performance of the proposed method, and an illustration with the AD Neuroimaging Initiative dataset is provided.

E1665: Nonlinear function-on-scalar regression using continuous neural networks*Presenter:* **Sidi Wu**, Fuzhou University, China

Function-on-scalar regression models the relationship between scalar predictors and a functional response, such as a growth curve or a brain signal. Classical approaches typically assume linearity or additive effects, limiting their ability to capture complex nonlinear dynamics. A novel framework for nonlinear function-on-scalar regression using continuous neural networks is introduced. In contrast to existing discrete neural network methods, which employ fixed hidden layers to map effects between predictors and static features of the functional response, this design directly incorporates continuous neurons and functional weights, enabling dynamic mapping from scalar predictors to a functional response across the entire observation interval, beginning from the first hidden layer. Specifically, scalar predictors are encoded into the initial continuous layer via functional weights, producing functional outcomes. Subsequent layers then propagate these functional outcomes and functional weights through integral transformations, maintaining the functional representation continuously from input to output. Network training is achieved through a functional gradient-based optimization algorithm combined with basis expansion applied to the functional weight parameters. Numerical experiments validate the approach, demonstrating superior prediction accuracy and smoother estimates compared to existing methods across multiple scenarios.

E1694: Outcome-guided clustering of high-dimensional omics for aging subtypes*Presenter:* **Boyi Hu**, Columbia University, United States

Clustering is widely used to uncover population heterogeneity in complex diseases. However, most existing approaches analyze high-dimensional omics or clinical phenotypes in isolation and do not jointly integrate these data sources, potentially obscuring biologically and clinically meaningful subtypes. An outcome-guided clustering method called TPClust integrates high-dimensional omics with clinical profiles to identify Aging subtypes. TPClust integrates longitudinal clinical trajectories with high-dimensional omics data to jointly identify clusters by capturing cluster-specific time-varying covariate effects on longitudinal outcomes and performing feature selection in high-dimensional omics data. The theoretical properties of TPClust are investigated and extensive simulations are conducted to evaluate its performance. Applied to Religious Orders Study and Memory and Aging Project (ROSMAP) transcriptomic data and longitudinal cognitive trajectories, TPClust identifies four Aging subtypes, from Resilient (preserved cognition, minimal pathology) to Rapid Decline (severe Alzheimers burden), as well as two intermediate subtypes (Late-Onset Decline and Early Vulnerability) that are not captured by unimodal approaches.

E1767: Multivariate planar curves: Definition, alignment and statistical analysis*Presenter:* **Cedric Beaulac**, Universite Du Quebec a Montreal, Canada

A functional framework for the analysis of multivariate planar curves extracted from images is proposed. Each object is represented as a parametric curve, and the collection of curves is treated as a structured functional variable. Building on recent advances in functional shape analysis, methods are developed to jointly align multiple contours while preserving their spatial relationships. This allows capturing both individual shape variability and interactions between objects, such as relative size, position, and orientation. Statistical models are introduced to describe this joint variability and demonstrate how these representations can be used for tasks such as classification and anomaly detection. The proposed framework is illustrated on medical imaging data, where the joint analysis of anatomical structures provides clinically relevant information. Results highlight the importance of modeling shapes as interacting objects rather than independent entities, and show that functional representations offer a flexible and interpretable approach to multivariate shape analysis.

EO179 Room 103 RECENT ADVANCES IN STATISTICAL MACHINE LEARNING**Chair: Eric Chi****E1465: Bayesian partition-based image analysis***Presenter:* **Hengrui Luo**, Rice University, United States

Bayesian partition-based image analysis is proposed, a unified framework for scalar-on-image regression that combines Bayesian spatial partitioning with minimax tree splitting. The method recursively partitions a 2D or 3D image lattice into contiguous, data-adaptive regions, avoiding reliance on fixed atlases or prespecified segmentations. A minimax splitting rule stabilizes partition learning by controlling worst-case heterogeneity across child nodes, while group spike-and-slab priors enable exact region selection, adaptive smoothing, and posterior uncertainty quantification. An efficient collapsed Bayesian algorithm is developed that jointly updates partitions, sparsity patterns, and local scales. The resulting approach provides an interpretable multiscale representation of image effects and supports prediction, signal localization, and scientific interpretation in complex imaging settings. By combining robust recursive partitioning with Bayesian structured sparsity, the framework offers a scalable and coherent approach to high-dimensional image analysis.

E1469: Frechet regression on the Bures-Wasserstein manifold*Presenter:* **Cesar Augusto Uribe Meneses**, Rice University, United States*Co-authors:* Duc Toan Nguyen

Frechet regression, or conditional barycenters, is a flexible framework for modeling relationships between covariates (usually Euclidean) and response variables on general metric spaces, e.g., probability distributions or positive definite matrices. However, in contrast to classical barycenter problems, computing conditional counterparts in many non-Euclidean spaces remains an open challenge, as they yield non-convex optimization problems with an affine structure. The existence and computation of conditional barycenters is studied, specifically in the space of positive-definite matrices with the Bures-Wasserstein metric. A sufficient condition for the existence of a minimizer of the conditional barycenter problem that characterizes the regression range of extrapolation is provided. Moreover, the optimization landscape is further characterized, proving that under this condition, the objective is free of local maxima. Additionally, a projection-free and provably correct algorithm for the approximate computation of first-order stationary points is developed. Finally, a stochastic reformulation that enables the use of off-the-shelf stochastic Riemannian optimization methods for large-scale setups is provided. Numerical experiments validate the performance of the proposed methods on regression problems of real-world biological networks and on large-scale synthetic Diffusion Tensor Imaging problems.

E1516: Robust hybrid infinite and finite dimensional tensor factorizations*Presenter:* **Jocelyn Chi**, University of Minnesota Twin Cities, United States*Co-authors:* Eric Chi

The canonical polyadic (CP) tensor decomposition represents a multidimensional data array as a sum of rank-one outer products of latent factors. Building on CP-HiFi, the hybrid infinite- and finite-dimensional CP framework, which introduces quasitensors by modeling selected modes as smooth functional factors in a reproducing kernel Hilbert space, the standard least-squares objective is replaced with a quantile-based loss. This modification enables estimation of conditional quantile functions (rather than conditional means) along functional modes. The resulting approach yields a distribution-aware tensor factorization that is robust to outliers and captures heterogeneity beyond the mean structure.

E1517: A Monte Carlo approach for nonsmooth convex optimization via proximal splitting algorithms*Presenter:* **Eric Chi**, University of Minnesota, United States

Operator splitting algorithms are a cornerstone of modern first-order optimization, relying critically on proximal operators as their fundamental building blocks. However, explicit formulas for proximal operators are available only for limited classes of functions, restricting the applicability of these methods. Recent work introduced HJ-prox, a zeroth-order Monte Carlo approximation of the proximal operator derived from Hamilton-Jacobi PDEs, which circumvents the need for closed-form solutions. The scope of HJ-prox is extended by establishing that it can be seamlessly incorporated into operator splitting schemes while preserving convergence guarantees. In particular, it is shown that replacing exact proximal steps with HJ-prox approximations in algorithms such as proximal gradient descent, Douglas-Rachford splitting, Davis-Yin splitting, and the primal-dual hybrid gradient method still ensures convergence under mild conditions.

EO043 Room 104 ADVANCES IN THE ANALYSIS OF FUNCTIONAL DATA AND NON-EUCLIDEAN DATA**Chair: Shuo-Chieh Huang****E1898: Contrastive latent functional model***Presenter:* **Damla Senturk**, University of California Los Angeles, United States*Co-authors:* Yanlong Bai, Danh Nguyen, Donatello Telesca, Esra Kurum, Abigail Dickinson, Shafali Jeste

Functional Principal Component Analysis (FPCA) is a popular tool for representing Functional data in lower dimensions along highly interpretable components depicting directions of variation in the data across time. The Contrastive Latent Functional Model (cLFM) extends FPCA to Contrastive settings involving multiple groups with a focus on time-dynamic components that are shared across samples versus those that are unique to a particular sample. The proposed Model integrates FPCA with Contrastive learning principles to capture common directions of variation across datasets while isolating distinct features specific to each sample. The Model employs a computationally efficient Expectation-Maximization (EM) algorithm for parameter estimation, ensuring orthogonality between shared and unique Functional spaces. Simulation studies demonstrate the efficacy of the proposed method across different scenarios of shared and unique variation, varying sample size and error variance. Applied to neurodevelopmental and clinical datasets, including EEG studies and longitudinal studies of kidney function, cLFM reveals novel insights into group differences between autism versus neurotypical development and among mild to severe albuminuria subgroups of chronic kidney disease patients. By bridging Contrastive Analysis with Functional data, this framework advances the capacity to disentangle complex variation patterns in multi-group Functional studies in biomedical research.

E1946: Precision physical activity prescription via reinforcement learning for functional actions*Presenter:* **Xiaoke Zhang**, George Washington University, United States*Co-authors:* Gefei Lin, Rui Miao, Jennifer Sacheck

Physical activity (PA) plays an important role in maintaining and improving health. Daily steps have been a key measure for PA which are easily accessible with common wearable devices. However, methods are lacking that can recommend a personalized optimal distribution of daily steps for an individual to follow over a period of time for the best outcomes of certain health biomarkers. A new offline reinforcement learning (RL) algorithm is developed to learn personalized and optimal PA distributions associated with cardiometabolic risk using data from the All of Us Research Program, which includes months of step counts as well as repeated measurements of key health biomarkers. In this approach, the action is a function representing the daily step distribution over a period of time. Simulation studies demonstrate the advantage of the proposed approach over existing continuous-action RL methods. The learned optimal policies from the All of Us Research Program data generally suggest people follow more consistent patterns of PA over time while offering tailored recommendations for subgroups stratified by blood glucose level, body mass index, blood pressure, age, and sex.

E1989: A Riemannian factor model for manifold-valued time series*Presenter:* **Shuo-Chieh Huang**, Rutgers University, United States*Co-authors:* Rong Chen, Yaqing Chen

A Riemannian factor model (RFM), a novel framework for analyzing potentially high-dimensional time series data observed on Riemannian manifolds, is proposed. Such time series are encountered in various applications, including economics, finance, medical imaging, and genomics and microbiome research. The proposed model is geometry-aware and accounts for the inherent nonlinearity in the data. Under a high-dimensional asymptotic regime, where the manifold dimension is allowed to diverge with the sample size n , convergence rates for the estimated loading space are established. In particular, under short-memory and strong factor conditions, a dimension-free $n^{-1/2}$ rate is obtained, which matches the convergence rate of the high-dimensional linear factor model. Finite-sample performance of the proposed RFM is demonstrated with simulated time series on the Bures-Wasserstein manifolds and products of spheres, as well as an application to monthly realized covariances of selected U.S. stock returns modeled as time series in the Bures-Wasserstein manifold, where the RFM provides demonstrably interpretable factors and yields competitive predictive performance.

E2010: Inference for dispersion and curvature of random objects*Presenter:* **Wookyeong Song**, University of California, Davis, United States*Co-authors:* Hans-Georg Mueller

There are many open questions pertaining to the statistical analysis of random objects, which are increasingly encountered. A major challenge is the absence of linear operations in such spaces. A basic statistical task is to quantify statistical dispersion or spread. For two measures of dispersion for data objects in geodesic metric spaces, Frechet variance and metric variance, a central limit theorem is derived for their joint distribution. This analysis reveals that the Alexandrov curvature of the geodesic space determines the relationship between these two dispersion measures. This suggests a novel test for inferring the curvature of a space based on the asymptotic distribution of the dispersion measures. The test can be employed to detect the intrinsic curvature of an unknown underlying space, which emerges as a joint property of the space and the underlying probability measure that generates the random objects. The asymptotic properties of the test and its finite-sample behavior are investigated for various data types, including distributional data and point cloud data. The proposed inference for intrinsic curvature of random objects is illustrated using gait synchronization data represented as symmetric positive definite matrices and energy compositional data on the sphere.

EO237 Room 105 STRUCTURED MODELS AND PREDICTION: SPATIAL DEPENDENCE AND MEASUREMENT ERROR**Chair: Juan Du****E1257: Scanner: Simultaneously temporal trend and spatial cluster detection for spatial-temporal data***Presenter:* **Xin Wang**, San Diego State University, United States

Identifying the underlying trajectory pattern in the spatial-temporal data analysis is a fundamental but challenging task. The problem of simultaneously identifying temporal trends and spatial clusters of spatial-temporal trajectories is studied. To achieve this goal, a novel method named spatial clustered and sparse nonparametric regression (scanner) is proposed. The method leverages the B-spline model to fit the temporal data and penalty terms on spline coefficients to reveal the underlying spatial-temporal patterns. The method estimates the model by solving a doubly-penalized least square problem, in which a group sparse penalty for trend detection and a spanning tree-based fusion penalty for spatial cluster recovery are used. An algorithm based on the alternating direction method of multipliers (ADMM) algorithm is developed to efficiently minimize the penalized least square loss. The statistical consistency properties of scanner estimator are established. Thorough numerical experiments are conducted to verify the theoretical findings and validate that the method outperforms the existing competitive approaches.

E1289: Robust mixture of linear mixed modeling via multivariate Laplace distribution*Presenter:* **Xiuqin Bai**, Kansas State University, United States

The assumption of normality in random effects and regression errors is the primary cause of the lack of robustness in the maximum likelihood estimation procedure for linear mixed models. A robust method is introduced for estimating regression parameters in these models, by positing that the random effects and regression errors follow a multivariate Laplace distribution. This new methodology, implemented via an EM algorithm, is computationally more efficient compared to the existing robust t procedure in the literature. Simulation studies suggest that the performance of the proposed estimation method in finite samples either surpasses or is at least on par with the robust t procedure.

E1644: Bayesian modeling of long-term blue jay spring counts in Illinois: A county-level spatiotemporal analysis*Presenter:* **Weijia Jia**, University of Illinois Urbana-Champaign, United States*Co-authors:* Weidai He

Long-term (1975-2024) Spring Bird Count (SBC) data from Illinois were analyzed to assess population trends of Blue Jays (*Cyanocitta cristata*). The SBC is a yearly, volunteer-based census conducted on the Saturday between May 4 and May 10 across all 102 Illinois counties. To account for variation in survey effort across parties and years, a Bayesian hierarchical areal modeling framework was developed incorporating an effort-adjustment term of the form $\exp(B * (effort^p - 1)/p)$ within a Markov Chain Monte Carlo (MCMC) setting. Counts were modeled across counties and years to estimate temporal trends while explicitly accounting for spatial dependence and heterogeneity in observation processes. The model incorporates relevant environmental and observational predictors to better explain variation in counts and improve inference on population dynamics. This approach provides robust estimation of Blue Jay population trends from volunteer-collected, county-level Count data and enables investigation of potential drivers underlying long-term changes.

E1744: Prediction error assessment with contaminated spatial data*Presenter:* **Rahasya Bharaniah**, Kansas State University, United States*Co-authors:* Juan Du

Accurate prediction of the underlying signal from spatial data is critical in applications such as epidemiology, agriculture, and environmental science. In practice, observations are often contaminated by measurement error or systematic bias, which may also exhibit spatial dependence, leading to degraded prediction performance. Prediction methods that aim to recover a less noisy, or true, spatial process using contaminated data are studied. Several kriging-based strategies are developed that incorporate auxiliary information from related sources. Performance is assessed with emphasis on uniform prediction error over the spatial domain. Uniform error bounds are derived, and finite-sample performance is examined through simulation studies under varying contamination levels. The proposed approach is applied to Kansas ozone data, where ground-level measurements are limited but satellite-derived column-level data are abundant, to illustrate potential improvements in prediction accuracy and design-based control of uniform prediction error compared to traditional kriging methods.

EO015 Room 106 MODERN STATISTICAL METHODS FOR COMPLEX DATA STRUCTURES**Chair: Yuan Zhang****E1589: Priority-aware Shapley value***Presenter:* **Yuan Zhang**, The Ohio State University, United States

Shapley values are widely used for model-agnostic data valuation and feature attribution, yet they implicitly assume contributors are interchangeable. This can be problematic when contributors are dependent (e.g., reused or augmented data or causal feature orderings) or when contributions should be adjusted by factors such as trust or risk. Priority-Aware Shapley Value (PASV) incorporates both hard precedence constraints and soft, contributor-specific priority weights. PASV is applicable to general precedence structures, recovers precedence-only and weight-only Shapley variants as special cases, and is uniquely characterized by natural axioms. An efficient adjacent-swap Metropolis-Hastings sampler enables scalable Monte Carlo estimation, and limiting regimes induced by extreme priority weights are analyzed. Experiments on data valuation (MNIST and CIFAR10) and feature attribution (Census Income) demonstrate more structure-faithful allocations and a practical sensitivity analysis via priority sweeping.

E1598: Multivariate inference of network moments by subsampling*Presenter:* **Wen Zhou**, New York University, United States*Co-authors:* Tianxi Li

Network moments, rescaled counts of motifs such as stars and triangles, are fundamental summaries of network structure, widely used in goodness-of-fit testing, model selection, and network comparison. While the univariate distribution of a single network moment can be approximated by subsampling, the consistency of subsampling for their joint distribution has remained unestablished. Node subsampling is proved to provide an

asymptotically accurate approximation of the joint distribution of multiple network moments under a general sparse graphon model. The theoretical analysis requires a careful characterization of the dependence structure among network moments and the corresponding multivariate asymptotic convergence, going substantially beyond existing univariate results. Building on this foundation, a practically important open problem is addressed: two-sample testing between unmatchable networks with unequal edge densities. A novel subsampling-based procedure is proposed that combines sparsification with a sample-splitting strategy, yielding the first subsampling-based inferential procedure valid for this setting.

E1800: **A snapshot of influence: A local data attribution framework for online reinforcement learning**

Presenter: **Jiaqi Ma**, University of Illinois Urbana-Champaign, United States

Online reinforcement learning (RL) excels in complex, safety-critical domains but suffers from sample inefficiency, training instability, and limited interpretability. Data attribution provides a principled way to trace model behavior back to training samples, yet existing methods assume fixed datasets, which is violated in online RL where each experience both updates the Policy and shapes future data collection. The aim is to investigate data attribution for online RL, focusing on the widely used Proximal Policy Optimization (PPO) algorithm. A local attribution framework is established, interpreting model checkpoints with respect to records in the recent training buffer. Two target functions are designed, capturing agent action and cumulative return respectively, and each record's contribution is measured through gradient similarity between its training loss and these targets. The framework is demonstrated through three concrete applications: diagnosis of learning, temporal analysis of behavior formation, and targeted intervention during training. Leveraging this framework, an algorithm called iterative influence-based filtering (IIF) is further proposed for online RL training that iteratively performs experience filtering to refine Policy updates.

E1799: **Representation learning with blockwise missingness and signal heterogeneity**

Presenter: **Weijing Tang**, Carnegie Mellon University, United States

Unified representation learning for multi-source data integration faces two important challenges: blockwise missingness and blockwise signal heterogeneity. The former arises from sources observing different, yet potentially overlapping, feature sets, while the latter involves varying signal strengths across subject groups and feature sets. While existing methods perform well with fully observed data or uniform signal strength, their performance degenerates when these two challenges coincide, which is common in practice. To address this, Anchor Projected Principal Component Analysis (APPCA) is proposed as a general framework for representation learning with structured blockwise missingness that is robust to signal heterogeneity. APPCA first recovers robust group-specific column spaces using all observed feature sets, and then aligns them by projecting shared Anchor features onto these subspaces before performing PCA. Estimation error bounds for embedding reconstruction are established through a fine-grained perturbation Analysis. In particular, using a novel spectral slicing technique, the bound eliminates the standard dependency on the signal strength of subject embeddings, relying instead solely on the signal strength of integrated feature sets. The proposed method is validated through extensive simulation studies and an application to multimodal single-cell sequencing data.

EO104 Room 107 ADDRESSING CHALLENGES FOR CAUSAL INFERENCE IN OBSERVATIONAL STUDIES Chair: Christopher Harshaw

E1312: **Nonparametric estimation of local treatment effects with continuous instruments**

Presenter: **Luke Keele**, University of Pennsylvania, United States

Instrumental variable methods are widely used to address unmeasured confounding, yet much of the existing literature has focused on the binary instrument setting. Extensions to continuous instruments often impose strong parametric assumptions for identification and estimation, which can be difficult to justify and may limit their applicability in complex real-world settings. Theory and methods for nonparametric estimation of treatment effects with a continuous instrumental variable are developed. An estimand is introduced that, under a monotonicity assumption, quantifies the treatment effect among the maximal complier class, generalizing the local average treatment effect framework to continuous instruments. Considering this estimand and the local instrumental variable curve, connections to the dose-response function and its derivative are drawn, and doubly robust estimation methods are proposed. Convergence rates and conditions for asymptotic normality are established, providing valuable insights into the role of nuisance function estimation when the instrument is continuous. Additionally, practical procedures for bandwidth selection and variance estimation are presented. Through extensive simulations, the advantages of the proposed nonparametric estimators are demonstrated. Finally, the methods are applied to data where excess travel time is an instrument for patients likelihood of receiving care at specialized health care facilities.

E1518: **Cross-balancing for data-informed design and efficient analysis of observational studies**

Presenter: **Jose Zubizarreta**, Harvard University, United States

Co-authors: Ying Jin

Causal inference starts with a simple idea: compare groups that differ by treatment, not much else. Traditionally, comparable groups are constructed using only observed covariates; however, incorporating outcome data into the study design while preserving valid inference remains a long-standing challenge. The general problem of covariate adjustment, effect estimation, and statistical inference is studied when balancing features are constructed or selected with the aid of outcome information. Cross-balancing is proposed, a method that uses sample splitting to separate the error in feature construction from the error in weight estimation. This framework addresses two cases: one where features are learned functions and one where they are selected from a potentially high-dimensional dictionary. In both cases, mild and general conditions are established under which cross-balancing yields consistent, asymptotically normal, and efficient estimators. In the learned-function case, cross-balancing achieves finite-sample bias reduction relative to plug-in estimators and is multiply robust when learned features converge at slow rates. In the variable-selection case, cross-balancing requires only a product condition on the approximation quality of the selected variables. Cross-balancing is illustrated in simulations and an observational study, demonstrating that careful use of outcome information can substantially improve estimation and inference while maintaining interpretability.

E1809: **Inference in high-dimensional linear mediation models under proportional asymptotics**

Presenter: **Rajarshi Mukherjee**, Harvard T.H. Chan School of Public Health, United States

Mediation analysis in a linear structural equation model is considered under a proportional asymptotic regime where both the number of mediators and confounders are allowed to diverge proportionally to the sample size. In this regime, consistent and asymptotically normal estimators of direct, indirect, and suitable path-specific effects are provided through a carefully devised scheme without imposing any sparsity structure on the problem. Typical assumptions used in debiasing methodology under proportional asymptotic regimes do not apply to the mediation analysis problem, and a partial ridge-based debiasing scheme is employed to address the relevant subtleties. The effect of sample splitting and cross-fitting is additionally considered, and detailed numerical experiments are performed to validate the results in finite samples.

E2073: **Where best to intervene**

Presenter: **Caleb Miles**, Columbia University, United States

One common motivation for mediation analysis is to help identify promising avenues for future interventions in order to improve the efficacy of an exposure or to mitigate a harmful effect or disparity. That is, mediation analysis is thought to help address the question “Where best to intervene?”. Mediation analysis formalizes this notion, which naturally leads to causal estimands that can inform about the effects of certain future (generally stochastic) interventions on unobserved variables along causal pathways. Some of these interventions correspond to so-called

randomized interventional analogs of natural direct and indirect effects and path-specific effects. Unlike so-called *natural* mediation effects, these effects are manipulable in the sense that they comport with the target trial principle; that is, they are estimable in a hypothetical future randomized trial. Furthermore, their identification does not rely on any cross-world counterfactual independence assumptions. Lastly, these effects contain so-called *separable effects* as a special case.

EO121 Room 201 ADVANCES IN COMPLEX DATA ANALYSIS FOR BIOMEDICAL AND ECONOMIC APPLICATIONS Chair: Yunpeng Zhao

E1306: Adaptive frequency band learning of nonstationary functional time series: An application to high-dimensional EEG signals

Presenter: **Pramita Bagchi**, George Washington University, United States

Frequency-domain properties of high-dimensional EEG (HD-EEG) signals encode rich information about brain function, varying over time through their time-varying spectral density. Instead of analyzing this density directly, researchers often summarize power within pre-defined frequency bands. However, how frequencies are partitioned crucially influences how well these summaries capture underlying dynamics, an aspect often overlooked. Treating HD-EEGs as nonstationary functional time series, a method is proposed to learn frequency-band structures that optimally summarize time-varying spectral behavior. Scan statistics and a scalable search algorithm are introduced to detect frequency-domain changes for adaptive band estimation. Theoretical results establish key properties, and simulations confirm the accuracy of band recovery. Applied to HD-EEG data from alternating eyes-open and eyes-closed conditions, the method uncovers spatially localized nonstationary patterns and cross-frequency synchronization consistent with known physiological phenomena.

E1355: CalCS: Calibrated cost-sensitive classification under strict error constraints

Presenter: **John Park**, Hong Kong University, Hong Kong

While fundamentally distinct in their objectives, the Neyman-Pearson (NP) and cost-sensitive (CS) paradigms both provide essential frameworks for classification. Standard CS methods often rely on empirical error estimates to satisfy user constraints. However, it is demonstrated that these plug-in estimators systematically underestimate true population risk. To resolve this, calibrated cost-sensitive classification (CalCS) is introduced, an algorithm that formally bridges the NP and CS paradigms. Given a strict NP error constraint, CalCS leverages exact finite-sample probability bounds to determine associated CS costs that provide the desired control. It is shown that the algorithm provides finite-sample guarantees while converging to the strict theoretical thresholds. Evaluations on synthetic and real-world datasets demonstrate that CalCS controls targeted error rates, successfully translating mathematical guarantees into robust, practical classifiers.

E1696: Covariance-based clustering and biclustering via the heterogeneous block covariance model and variants

Presenter: **Yunpeng Zhao**, Colorado State University, United States

Co-authors: Xiang Li, Ning Hao, Qing Pan, Mayson Zhang

Clustering methods are traditionally designed to group data points in a (possibly high-dimensional) Euclidean space. A different but equally important task involves clustering at the feature level, which arises naturally in gene expression analysis but has received relatively limited attention in the statistics literature. In bioinformatics, it is often desirable to identify sets of highly correlated genes whose expression levels are jointly regulated and act synergistically to carry out the same biological functions. The heterogeneous block covariance model (HBCM) addresses this problem by characterizing community structure directly within the covariance matrix while accounting for heterogeneity in how features connect within communities. A novel variational expectation-maximization algorithm enables efficient estimation of group memberships. Theoretical analysis establishes the consistency of membership recovery, and simulation studies demonstrate the superior performance of HBCM compared with existing methods. An extension of HBCM to a joint clustering and biclustering framework simultaneously partitions both features and samples. This generalization introduces new computational and theoretical challenges, for which principled solutions are proposed. Applications to real gene expression data highlight the model's ability to uncover biologically meaningful clusters and biclusters.

E1710: The population lotto game: How strategic resource allocation structures non-transitive outcomes in pairwise competitions

Presenter: **Joel Nishimura**, Arizona State University, United States

Co-authors: Giovanni Artiglio

To understand if and how strategic resource allocation can constrain the structure of pairwise competition outcomes in human competitions, a new multiplayer resource allocation Game, the Population Lotto Game, is introduced. This Game allows agents to allocate their resources across a continuum of possible specializations. While the Game allows non-transitive cycles between players, its Nash equilibrium forms a hierarchical structure between discrete leagues based on different resource budgets, with potential sub-league structure and/or non-transitive cycles inside individual leagues. An algorithm that can find a particular Nash equilibrium for any finite set of discrete sub-Population sizes and budgets is provided. Furthermore, the algorithm finds the unique Nash equilibrium that remains stable for the subset of players with budgets below any threshold.

EO208 Room 203 ADVANCES IN EXPERIMENTAL DESIGN (VIRTUAL)

Chair: John Stufken

E1609: A two-phase A/B test for binary outcomes on network data

Presenter: **Nicholas Rios**, George Mason University, United States

A/B testing is a powerful and popular tool for conducting online experiments for comparing a treatment with a control. Many applications for A/B testing involve comparing binary outcomes at the user level, such as monitoring whether or not a user clicked a button. Additionally, many of these individual users belong to a network, such as users of Facebook and LinkedIn. Unfortunately, when individuals belong to a network, the Stable Unit Treatment Value Assumption (SUTVA) can be violated because connected individuals tend to influence each other's decisions and responses. Additionally, in practice, both the mechanism of network interference and the true model for the response are unknown. An online sampling framework is proposed that dynamically minimizes a design optimality criterion based on the worst-case treatment variance under a set of plausible models and network interference mechanisms. Simulations and numerical studies on synthetic data based on real networks from Facebook users show that the proposed framework is competitive with existing approaches in terms of power and Type-I error rates.

E1604: Robust joint modeling for data with continuous and binary responses

Presenter: **Lulu Kang**, University of Massachusetts Amherst, United States

In many supervised learning applications, the response consists of both continuous and binary outcomes. Studies have shown that jointly modeling such mixed-type responses can substantially improve predictive performance compared to separate analyses. However, outliers pose a new challenge to existing likelihood-based modeling approaches. A new robust joint modeling framework for data with both continuous and binary responses is proposed based on the density power divergence (DPD) loss function with l_1 regularization. The proposed framework leads to a sparse estimator that simultaneously predicts continuous and binary responses in high-dimensional input settings while down-weighting influential outliers and mislabeled samples. An efficient proximal gradient algorithm with Barzilai-Borwein spectral step size and a robust information criterion (RIC) for data-driven selection of penalty parameters are developed. Extensive simulation studies under a variety of contamination schemes demonstrate that the proposed method achieves lower prediction error and more accurate parameter estimation than several competing approaches. A real case

study on wafer lapping in semiconductor manufacturing further illustrates the practical gains in predictive accuracy, robustness, and interpretability of the proposed framework.

E1834: **Efficient Bayesian estimation and inference for Shapley value via experimental design**

Presenter: **Wei Zheng**, University of Tennessee, United States

Co-authors: Zheng Zhou, Robert Mee, Yongdao Zhou

The Shapley value fairly allocates cooperative gains among players but requires evaluating all 2^d coalitions, computationally infeasible for large d , especially when single coalition evaluations are costly (as in AI, data science, and genomics). A Bayesian framework using Gaussian processes is proposed to infer unobserved coalition values, yielding posterior distributions over Shapley values that support both point estimation and uncertainty quantification. The posterior evaluation complexity reduces from exponential to polynomial order, and experimental design is integrated to select coalitions minimizing posterior variance. The method offers three advantages over existing approaches: (i) full statistical inference, (ii) accurate Shapley value estimation using as few as $d^2 - d + 1$ coalition evaluations, and (iii) robustness across diverse cooperative games. Simulations and case studies confirm superior accuracy and efficiency under comparable evaluation budgets.

E1964: **Exact designs for order-of-addition experiments under a transition-effect model**

Presenter: **Jiayi Zheng**, George Mason University, United States

Co-authors: Nicholas Rios

In the chemical, pharmaceutical, and food industries, the order of adding a set of components sometimes impacts the final product. These instances of the Order-of-Addition (OofA) problem seek to find the optimal sequence of components. Extensive research on this topic has been conducted, but almost all designs are found by optimizing the D-optimality criterion. However, when prediction of the response is important, I-optimal designs remain necessary. Furthermore, designs are needed for experiments where some orders are infeasible due to constraints. A new model for OofA experiments is presented that uses transition effects to model the effect of order on the response. Three algorithms are proposed to find D- and I-efficient exact designs under this new model: Simulated Annealing, a metaheuristic algorithm; Bubble Sorting, a Greedy local optimization algorithm; and the Greedy Randomized Adaptive Search Procedure (GRASP), another metaheuristic algorithm. These three algorithms are generalized to handle block constraints, where components are grouped into blocks with a fixed order. Two examples illustrate the effectiveness of the proposed designs and models, even under block constraints.

EO112 Room 204 STATISTICAL MODELING FOR ENVIRONMENTAL AND PUBLIC HEALTH SURVEILLANCE

Chair: Kangyi Peng

E1621: **Copula-based joint modeling of emergency department visits with time-varying dependence**

Presenter: **Cindy Feng**, Dalhousie University, Canada

Co-authors: Guanjie Lyu

Jointly modeling multiple correlated count time series is essential in health services research, where outcomes like emergency visits for mental health and substance use often evolve together. Ignoring these dependencies can obscure meaningful trends and limit the effectiveness of policy evaluation. A copula-based framework is proposed that combines negative binomial regression with penalized splines for flexible marginal trends and a smoothly time-varying copula to capture evolving dependence. Simulations show improved accuracy and uncertainty quantification over models assuming constant or no dependence. Applied to monthly emergency visits in Nova Scotia (2017-2023), the model reveals dynamic associations: strong positive correlation pre-COVID 19, a temporary reversal during lockdowns, and partial recovery post-reopening. This approach offers a powerful tool for analyzing complex, time-varying relationships in health care utilization data to inform public health planning.

E1702: **Monte Carlo integration of differential equation systems with epidemiological applications**

Presenter: **Willard Braun**, UBC, Canada

Co-authors: Kyeongah Nah

Monte Carlo (MC) integration is a well-known method for evaluating integrals for which deterministic numerical methods struggle, particularly in higher dimensions. MC integration is a straightforward application of the Law of Large Numbers applied to simulated data. Integrating a differential equation requires different tactics. Two new methods are presented. The first method is based on the Mean Value Theorem and performs well in comparison to other recently proposed MC approaches, even on stiff systems. Its performance is studied on the Susceptible-Infected-Removed (SIR) model, which is commonly used in epidemic modelling and forecasting. The second method is based on an approximation to continuous-time Markov chains and can be applied to delay-differential equations. An application to a delayed infection and recovery version of the SIR model is considered.

E1782: **Application of hidden Markov models to interpret the wastewater viral signal of rarely detected viruses**

Presenter: **Johanna de Haan-Ward**, University of Ottawa, Canada

Co-authors: Chandler Wong, Elizabeth Renouf, Shen Wan, Xin Tian, Robert Delatolla, Elizabeth Mercier

Since the onset of the COVID-19 pandemic, wastewater-based surveillance has emerged as a critical tool for monitoring disease spread, including early detection and anticipating periods of increased disease burden. Statistical methodology has been essential in understanding the viral level of endemic viruses such as SARS-CoV-2 and influenza, with the goal of linking the wastewater viral signal to key outcomes for public health authorities, such as the number of hospitalizations. Rare viruses present a challenge for statistical modelling due to irregular sampling of wastewater, zero-heavy distributions, and in some instances missing information about clinical cases. Hidden Markov models have been widely applied in fields such as ecology, finance, and medicine, where time-series data are assumed to result from an underlying, unobservable process. Hidden Markov models were applied to two years of wastewater samples from Ottawa, Canada analyzed for the presence of Mpox, a zoonotic virus first detected in Canada in early 2022 and spread through close human contact. Hidden Markov models specified with two and three underlying states were compared, where the time-series data were assumed to derive from a hurdle model to account for excess zeros in the time series, including during times of high disease burden. The three-state model provides inference best aligned with discipline-specific beliefs about the disease burden of Mpox.

E1808: **Structural time series for hybrid clustering/classification of mutations in wastewater to detect new disease variants**

Presenter: **Devan Becker**, Wilfrid Laurier University, Denmark

Wastewater-based epidemiology (WBE) for COVID-19 variants of concern uses genetic sequencing of wastewater to determine frequencies of individual mutations, which are compared to known variants to estimate their abundance. For WBE, variants are defined as lists of mutations. In a single sample, these models are linear, with covariates indicating the inclusion of a mutation in a variant's definition (1 if included, 0 otherwise; each mutation can belong to multiple clusters or classes) and coefficients corresponding to the abundance of that variant. By leveraging samples taken over space and time in a Bayesian Non-negative Matrix Factorization framework with structural time series, variant definitions and abundances can be simultaneously estimated. In the NMF model, abundances are constrained to the simplex and mutations are assigned to overlapping clusters. Through careful choice of prior, information about existing variants can be imposed and residual information can be used to estimate previously undiscovered variants. This hybrid cluster-classification approach reflects the nature of known variants, where variant definitions consist of small sets of mutations with many mutations of unknown presence in definitions. Variant definitions group all mutations into overlapping classes,

constituting a classification task for known variants with unknown mutations and a clustering task for novel variants.

EO184 Room 205 INFERENCE USING RESAMPLING METHODS IN MODERN STATISTICS AND MACHINE LEARNING Chair: Hyemin Yeon

E1425: Empirical likelihood for unbalanced ranked set sampling

Presenter: **Seonghun Cho**, Inha University, Korea, South

Co-authors: Soohyun Ahn, Johan Lim

Unbalanced ranked set sampling (URSS) commonly arises in practice, but incorporating it into the empirical likelihood (EL) framework requires careful handling of unequal stratum sizes. Two EL formulations for inference on the population mean under URSS are investigated. The first approach, termed deterministic balancing, combines the stratum-specific probability constraints into a single weighted normalization constraint. A theoretically justified choice of weights that is aligned with the conventional RSS mean estimator is proposed and the asymptotic distribution of the resulting empirical likelihood ratio statistic is established. The second approach, termed probabilistic balancing, retains the stratum-wise probability constraints and leads to a standard chi-square limiting distribution. The asymptotic properties of both methods are derived and their finite-sample performance is compared through simulation studies. The results show that, although both approaches provide satisfactory control of size, deterministic balancing often yields higher power across a wide range of unbalanced designs. Finally, the proposed methods are applied to test the symmetry of dental size measurements in patients with normal occlusion.

E1592: Bootstrapping network statistics using overlapping partitions

Presenter: **Sayan Chakrabarty**, University of Michigan, United States

Co-authors: Liza Levina

Bootstrapping network data efficiently is a challenging task. Existing methods tend to make strong assumptions on both the network structure and the statistics being bootstrapped, and are computationally costly. A general algorithm, OPBoot, is introduced for network bootstrap that partitions the network into multiple overlapping subnetworks and then aggregates results from bootstrapping these subnetworks to generate a bootstrap sample of the network statistic of interest. This approach is usually much faster than competing methods as most of the computations are done on smaller subnetworks. OPBoot is shown to be consistent in distribution for a large class of network statistics under minimal assumptions on the network structure, and extensive numerical examples demonstrate that the bootstrap confidence intervals produced by OPBoot attain good coverage without substantially increasing interval lengths in a fraction of the time needed for running competing methods.

E1721: Gaussian and bootstrap approximation of suprema of empirical processes

Presenter: **Alexander Giessing**, National University of Singapore, Singapore

Non-asymptotic Gaussian approximation results are presented for the sampling distribution of suprema of empirical processes when the indexing function class $\mathcal{F}_n$ varies with the sample size n and may not be Donsker. Prior approximations of this type required upper bounds on the metric entropy of $\mathcal{F}_n$ and uniform lower bounds on the variance of $f \in \mathcal{F}_n$, which limited their applicability to high-dimensional inference problems. In contrast, the results presented hold under simpler conditions on boundedness, continuity, and the strong variance of the approximating Gaussian process. The results are broadly applicable and yield a novel procedure for bootstrapping the distribution of empirical process suprema based on the truncated Karhunen-Loeve decomposition of the approximating Gaussian process. The flexibility of this new bootstrap procedure is demonstrated by applying it to three fundamental problems in high-dimensional statistics: simultaneous inference on parameter vectors, inference on the spectral norm of covariance matrices, and construction of simultaneous confidence bands for functions in reproducing kernel Hilbert spaces.

E1301: High-dimensional bootstrap and asymptotic expansion

Presenter: **Yuta Koike**, University of Tokyo, Japan

Recent work has shown that bootstrap approximation for The maximum of a sum of independent random vectors is justified even when The dimension is much larger than The sample size. In this context, numerical experiments suggest that third-moment matching bootstrap approximations outperform normal approximation even without studentization, but existing theoretical results cannot explain this phenomenon. An asymptotic expansion formula for The bootstrap coverage probability is developed and shown to explain The phenomenon. In particular, a blessing of dimensionality phenomenon is identified: The third-moment matching wild bootstrap is second-order accurate in high dimensions even without studentization if The covariance matrix has identical diagonal entries and bounded eigenvalues. The validity of these results is established under The assumption that The underlying distributions admit Stein kernels.

EO137 Room 208 SURVIVAL AND RECURRENT EVENT MODELING IN BIOMEDICAL RESEARCH Chair: Elizabeth Juarez-Colunga

E1360: Additive shared mixed frailty model for bivariate gap time data

Presenter: **Ivo Sousa-Ferreira**, FCIencias.ID, Portugal

Co-authors: Ana Maria Abreu, Cristina Rocha

Recurrent event data frequently arise in biomedical studies, where within-subject dependence and the presence of zero-recurrence subjects challenge classical survival models. A new additive shared frailty model for gap times between recurrent events is introduced, assuming that the recurrence process follows a non-homogeneous Poisson process with a Weibull rate function. The frailty acts additively on the rate function and has a non-central chi-squared distribution with zero degrees of freedom, a mixed distribution with a probability mass at zero and a continuous positive component. This formulation simultaneously accommodates within-subject correlation and zero-recurrence subjects. The resulting multivariate marginal survival function admits a competing risks interpretation with two independent causes. Since the model is fully specified, the maximum likelihood method is applied for parameter estimation using a marginal likelihood function, with particular attention to the bivariate recurrent event setting. A brief simulation study illustrates the asymptotic properties of the estimators under different censoring percentages. An application to a well-known hospital readmission data set is also presented to elucidate the practical contribution of the proposed model.

E1388: Bounded hazard ratio Cox model for the effect of time to treatment on mortality

Presenter: **Krithika Suresh**, University of Michigan, United States

Co-authors: John Rice

In resource-limited settings, there is often interest in assessing the effect of time to treatment (TTT) on subsequent mortality. It is demonstrated that the traditional Cox proportional hazards model specifying the effect of TTT as an unrestricted term in the log hazard ratio results in a mathematical anomaly that violates the expected monotonic TTT effect on survival (i.e., as TTT increases, survival probability should decrease). A class of bounded hazard ratio (BHR) Cox models is proposed that attenuate the hazard ratio for TTT towards the null with increasing treatment time, such that hazard for death after treatment does not exceed the hazard without treatment. Estimation is performed using direct optimization of the partial log-likelihood, and a linearized approximation is proposed to fit the model in standard software for large sample sizes. From BHR models, the estimated hazard ratio curve describes how the treatment effect diminishes with delays in treatment. Additionally, the marginal survival probability difference comparing immediate treatment to a treatment time in the future is presented. The performance of model estimation is evaluated in a simulation study and the use of the approach is demonstrated in an application to treatment for colon cancer using NCDB data.

E1390: A joint model for longitudinal change point analysis and recurrent events with application to cystic fibrosis outcomes*Presenter:* **Jiayuan Shi**, University of Colorado Anschutz, United States*Co-authors:* Giovanni Silva, Brandie Wagner, Elizabeth Juarez-Colunga

Many biomedical studies collect repeated biomarker measurements together with recurrent clinical event outcomes, where changes in biomarker trajectories may provide important information about underlying disease progression and recurrent event risk. A Bayesian joint model for longitudinal biomarkers with subject-specific trajectory changes and recurrent clinical events is proposed. The model links individualized biomarker trajectories to recurrent event risk through shared random effects, allowing event risk to be associated with biomarker levels, rates of change, and latent change-point timing. The longitudinal submodel captures sharp subject-specific shifts in biomarker trajectories, while the recurrent event submodel uses a counting-process formulation that accounts for left truncation due to variable entry ages. The model quantifies temporal associations between biomarker progression and recurrent outcomes. Application to data from the Early Pseudomonas Infection Control study of children with cystic fibrosis demonstrates the method's utility. The preferred two-random-change-point model estimated mean subject-specific change points at 6.7 and 15.6 years. The odds of Pseudomonas aeruginosa Infection increased modestly between the two change points, followed by a sharp increase after the second change point. Faster increases in Infection probability between change points were associated with elevated recurrent pulmonary exacerbation risk.

E1878: Characterizing the survivor and hazard functions with quantile-parameterized Meta-B distributions*Presenter:* **Bryan McNair**, University of Colorado Anschutz Medical Campus, United States*Co-authors:* Elizabeth Juarez-Colunga

A quantile-parameterized meta-B distribution (QPM_m-B) is a flexible distribution based on any basis distribution and parameterized by quantiles. It achieves arbitrary flexibility by adding additional quantiles to the basis parameter vector. These distributions are placed on a continuum, with the basis distribution on one end and the true distribution on the other. The QPM_m-B methodology proposes a nonparametric estimator of the survival function that also allows estimation of the hazard function. A log-logistic distribution is used as the basis function. The QPM_m-B methodology can handle complex features of the data in a straightforward manner, including all types of censoring and truncation. Estimation is carried out using maximum likelihood methods, and consistency of estimators is demonstrated. Simulation studies assess the performance of the proposed method for small and large sample sizes. An application of the QPM_m-B methodology is illustrated through analysis of a rheumatoid arthritis study involving left-truncation and interval censoring on a reversed time scale by modeling time of first biomarker positivity prior to rheumatoid arthritis diagnosis.

EO170 Room 209 STATISTICAL METHODS FOR IMPERFECT DATA (VIRTUAL)**Chair: Farouk Nathoo****E1201: A Bayesian semiparametric model for survival analysis with a right censored covariate***Presenter:* **Farouk Nathoo**, University of Victoria, Canada*Co-authors:* Puneet Velidi, Maryclare Griffin, Tanya Garcia, Roland Matsouaka, Sujit Ghosh, Juxin Liu, Jing Qian

Survival models incorporating censored covariates are useful for understanding Huntington's disease progression because subject observation often ends before the initial presentation of symptoms. Modeling both a censored response and censored covariate is challenging due to ordering constraints (e.g., ordering of disease stages) leading to non-independent censoring mechanisms. We develop SPARTACCUS, a Bayesian joint model with semiparametric Cox proportional hazards models for both the right-censored response and censored covariate, including uncensored covariates. A computationally efficient MCMC sampler is implemented that uses data augmentation to impute censored values using posterior predictive distribution. In simulations, SPARTACCUS outperforms complete case analysis in high-censoring settings across a range of sample sizes and hazard shapes. Applying our method to the PREDICT-HD clinical trial shows that increased baseline parietal lobe grey matter volume is associated with increased hazard of Stage II Huntington's, accounting for time to Stage I symptoms. This finding may inform the development of non-invasive brain stimulation therapies for Huntington's.

E1336: Attenuated, exaggerated, or flipped: On the ramifications of measurement error in concentration indices*Presenter:* **Sarah Lotspeich**, Wake Forest University, United States*Co-authors:* Cassandra Hung

The concentration index is a common method to quantify whether a health outcome is proportionally distributed across people or neighborhoods relative to some exposure of interest. Otherwise, health disparities may be present. However, when applying these methods to real-world data sources, like electronic health records (EHR), the exposure can be inaccurately measured. One area where this issue arises is when investigating potential disparities in hospitalization intensity relative to patients' whole-person health. The factor by which measurement error in the exposure biases the concentration index is derived. Despite the connection to linear regression, and contrary to other classical measurement error problems, it is shown that measurement error in the exposure can actually attenuate, exaggerate, or even flip the direction of a disparity. Based on this derivation, a moment-based estimator that incorporates partial validation data to correct for bias is proposed. Using simulations studies, the impact of ignoring measurement error on the concentration index in various finite-sample settings is illustrated and how the proposed method corrects for it is demonstrated. Finally, potential disparities in hospitalizations relative to a composite whole-person health score across an extensive learning health system are examined, using the moment-based estimator and partial validation data (obtained via expert chart reviews) to correct for EHR data quality issues.

E1546: SPARCC: Semi-parametric robust estimation in a right-censored covariate model*Presenter:* **Brian Richardson**, University of North Carolina at Chapel Hill, United States*Co-authors:* Seong-ho Lee, Tanya Garcia, Karen Marder, Yanyuan Ma

In Huntington disease research, a current goal is to understand how symptoms change prior to a clinical diagnosis. Statistically, achieving this goal entails modeling symptom severity as a function of the Covariate time of diagnosis, which is often heavily right-censored in observational studies. Existing estimators that handle right-censored covariates, such as the complete case estimator and maximum likelihood estimator, vary in their statistical efficiency and robustness to misspecifications of nuisance parameters (i.e., densities for the censored Covariate and censoring variable). The SPARCC estimator (SemiParametric Robust estimation in a right-Censored Covariate model) is proposed as a Robust and efficient alternative. When the nuisance parameters are modeled parametrically, the SPARCC estimator is doubly Robust, that is, consistent if at least one nuisance parameter is correctly specified, and SemiParametric efficient if both are correctly specified. When the nuisance parameters are estimated via nonparametric or machine learning methods that converge sufficiently fast, the SPARCC estimator is consistent and SemiParametric efficient. Empirical results demonstrate that the proposed estimator, implemented in the R package SPARCC, has its claimed properties. Application to Huntington disease symptom trajectories using data from the Enroll-HD study is presented.

Monday 10.08.2026

10:25 - 12:05

Parallel Session L – EcoSta2026

EI006 Room 202 TIMELESS TOPICS IN STATISTICS: NEW FORMULATIONS AND RESULTS**Chair: Marianthi Markatou****E1831: Subdata selection or sampling methods***Presenter:* **John Stufken**, George Mason University, United States

Selecting or sampling subdata from a larger dataset has received increasing attention over the past decade and many methods have been proposed. Two of these methods are examined: one model-based and one model-free. The first is based on efficient estimation of model parameters, while the second focuses on prediction. Selected strengths and weaknesses are discussed.

E1928: Ancillarity-guided goodness-of-fit testing under symmetry: A normality-testing illustration*Presenter:* **Albert Vexler**, The State University of New York at Buffalo, United States

Classical goodness-of-fit tests detect deviations from a target distribution. In many problems, symmetry is a natural structural feature under both the null model and the alternatives of interest. For example, paired data often lead to symmetric distributions after forming within-pair differences. Symmetry structures arise in financial applications, where stock-market returns are often modeled using distributions that are approximately symmetric around zero after centering. An ancillarity-guided approach to improving GOF tests in symmetry-preserving settings is presented, using normality testing as the illustration. The idea is not to delete or condition on ancillary information. Rather, the goal is to construct a test statistic whose rejection decision is approximately independent of a relevant set of ancillary statistics. Theoretical results show that reducing this ancillary dependence improves power and identify the type of ancillary structure with respect to which independence can lead to a most powerful benchmark. In the normality-testing illustration, this principle motivates simple modifications of classical procedures. An asymptotic relative efficiency analysis confirms the advantage of this construction. For symmetric alternatives, the modified tests are shown to be twice as efficient as their classical counterparts. The analysis illustrates how ancillarity-guided independence can provide a simple and practically useful route to improving GOF tests.

E1929: A framework for multivariate goodness-of-fit testing based on matrix distances*Presenter:* **Marianthi Markatou**, University at Buffalo, United States

Measures of discrepancy between two probability distributions are used in scientific literature to develop goodness-of-fit methods. A unified framework for the study of two-sample and k-sample goodness-of-fit testing based on the concept of matrix distance is presented. The matrix distance is first defined, and its elements are then used to construct test statistics for testing equality in distribution of k-samples. The two-sample test statistic is shown to be a special case of the k-sample test statistic. The asymptotic distributions of the tests under the null hypothesis are derived, and the connection of the MMD statistic with the proposed tests is illustrated. Computational considerations are presented, and the implementation of these methods is demonstrated via the QuadratiK software. Simulation results exemplify the performance of the new methods and compare it with that of state-of-the-art procedures.

EO235 Room B101 RECENT ADVANCES IN NETWORKS AND INTERFERENCE I**Chair: Liang Zhong****E1180: Unobserved heterogeneous spillover effects in instrumental variable models***Presenter:* **Huan Wu**, University of North Carolina at Chapel Hill, United States

A general framework is developed for identifying causal effects in settings with spillovers, where both outcomes and endogenous treatment decisions are influenced by peers within a known group. The generalized local average controlled spillover and direct effects (LACSEs and LACDEs) are introduced, which extend the local average treatment effect framework to settings with spillovers and establish sufficient conditions for their point identification without restricting the cardinality of the support of instrumental variables. These conditions clarify the necessity of commonly imposed restrictions to achieve point identification with binary instruments in related studies. The marginal controlled spillover and direct effects (MCSEs and MCDEs) are then defined, which naturally extend the marginal treatment effect framework to settings with spillovers and are nonparametrically point identified from continuous variation in instruments. These marginal effects serve as building blocks for a broad class of policy-relevant treatment effects, including some causal spillover parameters in the related literature. Semiparametric and parametric estimators are developed, and an application using Add Health data reveals heterogeneity in education spillovers within best-friend networks.

E1434: Randomization tests in two-stage experiments*Presenter:* **Liang Zhong**, Faculty of Business and Economics, The University of Hong Kong., Hong Kong*Co-authors:* Jizhou Liu, Azeem Shaikh

Two-stage randomized experiments are widely used to study spillover effects in clustered settings. In such designs, clusters are first randomly assigned to different treatment saturation levels, and units within each cluster are then randomly assigned to treatment, with the number treated determined by the assigned saturation level. Randomization tests are developed for general two-stage experiments with multiple saturation levels and varying cluster sizes. For sharp null hypotheses, conditional randomization tests are proposed that reduce to simple cluster-level permutation procedures; the resulting tests are exactly valid in finite samples. Weak null hypotheses on average spillover effects are then studied and it is shown that studentized conditional randomization tests based on Neymanian variance estimation are asymptotically valid. The framework is further extended to bounded spillover nulls and monotone spillover nulls across ordered saturation levels. For the latter, a novel unconditional randomization test based on pairwise imputation is developed and finite-sample validity is established. This approach also extends to more general ordered-exposure settings, including network experiments. Taken together, these results provide a unified randomization-based approach to testing sharp, weak, bounded, and monotone hypotheses in two-stage experiments.

E1623: Randomization tests in bipartite experiments*Presenter:* **Jizhou Liu**, Peking University, China*Co-authors:* Sizhu Lu, Peng Ding

Randomized experiments with bipartite structures are increasingly used to study interventions where treatments and outcomes are defined on distinct but interdependent populations. A framework for Fisherian-style randomization tests tailored for bipartite experiments is developed that formalizes randomization tests on a subset of units over a restricted assignment space defined by the bipartite structure. The proposed procedures are finite-sample valid under commonly used experimental designs, including complete randomization and Bernoulli trials, and extend naturally to more general designs through conditional sampling. General network experiments can be represented within the bipartite framework, enabling the analysis of total and spillover effects on networks. Simulation and empirical studies illustrate the validity and practical relevance of this approach.

E1849: Networks as proxy controls: Nonparametric identification and estimation of partial effects*Presenter:* **Gaoqian Xu**, University of Washington, United States

A nonparametric model is studied where a latent variable creates endogeneity by affecting both network formation and an outcome of interest. The existing network control function approach is generalized to nonparametric outcome models, using individuals' link functions to account for unobserved heterogeneity. Identification is achieved through a form of matching on unobservables: individuals are conceptually matched based on

their latent link functions. Implementation involves two steps: first, the distances or dissimilarities between the latent link functions are estimated using network data; second, functional kernel smoothing is applied over these distances to estimate the structural parameter. Asymptotic analysis reveals a fundamental trade-off: the robustness gained from this approach comes at the unavoidable cost of a slow convergence rate, driven by the difficulty of matching on latent objects. This statistical cost is characterized by deriving a minimax lower bound.

EO173 Room B103 RECENT ADVANCES IN ECONOMETRIC METHODOLOGY
Chair: Jiatong Li
E1226: Adaptive multi-task learning for multi-sector portfolio optimization

Presenter: **Ruike Wu**, Shanghai University of Finance and Economics, China

Accurate transfer of information across multiple sectors to enhance model estimation is both significant and challenging in multi-sector portfolio optimization involving a large number of assets in different classes. Within the framework of factor modeling, a novel data-adaptive multi-task learning methodology is proposed that quantifies and learns the relatedness among the principal temporal subspaces (spanned by factors) across multiple sectors under study. This approach not only improves the simultaneous estimation of multiple factor models but also enhances multi-sector portfolio optimization, which heavily depends on the accurate recovery of these factor models. Additionally, a novel and easy-to-implement algorithm, termed projection-penalized principal component analysis, is developed to accomplish the multi-task learning procedure. Diverse simulation designs and practical application on daily return data from Russell 3000 index demonstrate the advantages of multi-task learning methodology.

E1563: One-step estimation for general Gaussian processes

Presenter: **weilin Xiao**, Zhejiang University, China

The fractional Gaussian process has been widely applied in many fields, and statistical inference for it poses significant challenges due to the lack of the Markov property. A method for estimating unknown parameters in general Gaussian processes based on one-step estimation is considered, which corrects an initial guess estimator to an asymptotically efficient one. Under certain regularity conditions, the one-step estimator of general Gaussian processes is shown to be strongly consistent and asymptotically normal with the same asymptotic variance as the maximum likelihood estimator. Monte Carlo simulations show that the one-step estimator reduces computational time compared with the exact maximum likelihood estimator and outperforms method of moments estimators in terms of asymptotic variance. The estimation methods are applied to fit fractional Gaussian processes to logarithmic daily realized volatility series, where the series display the roughness phenomenon and the stationary property.

E1547: Factor-adjusted knockoff inference for high-dimensional time series

Presenter: **Jie Wei**, Huazhong Univ. of ST, China

A new knockoff inference framework for high-dimensional time series data, termed FAR-knockoff, is proposed, which incorporates factor adjustment and accounts for serial dependence. Unlike most knockoff methods that require temporal independence on covariates or errors, FAR-knockoff leverages a weighted principal component analysis to estimate latent factors from possibly serially and cross-sectionally correlated covariates, followed by Fixed-X knockoff construction for the idiosyncratic components. This design improves power by reducing collinearity while preserving theoretical validity. Rigorous finite-sample and asymptotic guarantees for false discovery rate (FDR) control are established, showing robustness against estimation error in covariance structures. Simulation studies demonstrate substantial gains in power and FDR stability compared to existing approaches. Applying FAR-knockoff to U.S. Treasury bond risk premia prediction, improved out-of-sample performance and meaningful insights into maturity-dependent predictive structures are found. The method advances knockoff inference for dependent data environments common in many areas.

E1593: Transformed estimation for panel VAR models with interactive fixed effects

Presenter: **Bin Peng**, School of Economics, Huazhong University of Sci.&Tech., China

Co-authors: Xuan Ding, Qiankun Zhou

Estimation and inference challenges in Panel Vector Autoregressive (PVAR) models with unobservable interactive effects are addressed through a transformed estimation method that does not rely on prior knowledge of factor loadings and factors. To further improve convergence speed, an average estimator based on the transformed method is introduced. Both estimators are shown to be consistent as $T \rightarrow \infty$. Impulse response functions for economic-specific shocks are derived and inference theorems for the estimators are established. Monte Carlo simulations are conducted to examine the finite sample properties of the estimators. An application exploring the relationship between sovereign debt and economic growth finds that the observed negative correlation is mainly attributed to common shocks.

EO051 Room 101 ADVANCES IN CHANGE-POINT DETECTION AND NONPARAMETRIC INFERENCE Chair: Carlos Misael Madrid Padilla
E1457: Detecting changepoints in globally-indexed functional time series

Presenter: **Drew Yarger**, Purdue University, United States

Co-authors: Derek Tucker

In environmental and climate data, there is often an interest in determining if and when changes occur in a system. Such changes may result from localized sources in space and time like a volcanic eruption or climate geoengineering events. Detecting such events and their subsequent influence on climate has important policy implications. However, the climate system is complex, and such changes can be challenging to detect. One statistical perspective for changepoint detection is functional time series, where one observes an entire function at each time point. The context considered is where each time point is a year, and a function of temperature indexed by day of the year is observed. Furthermore, such data is measured at many spatial locations on Earth, which motivates accommodating sets of functional time series that are spatially-indexed on a sphere. Simultaneously inferring changes that can occur at different times for different locations is challenging. Test statistics are proposed for detecting these changepoints, and performance is evaluated using varying levels of data complexity, including a simulation study, simplified climate model simulations, and climate reanalysis data. Changes in stratospheric temperature globally over 1984-1998 are evaluated. Such changes may be associated with the eruption of Mt. Pinatubo in 1991.

E1605: High-dimensional change point detection with missing values

Presenter: **Abolfazl Safikhani**, George Mason University, United States

Assuming fixed model parameters over a relatively large time interval is unrealistic due to possible external shocks to the dynamics of the data generating process. A more realistic assumption is to let parameters behave in a piecewise constant manner, where time points with jumps in model parameters are called change points. A change point detection problem in a high-dimensional mean shift model with missing values is considered. The presence of missing values among observations further complicates the detection problem due to possible loss of information before and after a change point. A four-step algorithm is designed to tackle this problem. First, observations are split into blocks of a certain size and a regularized estimator is defined to perform parameter estimation while missing values are imputed properly. This step is followed by thresholding and re-imputation to fine-tune the estimation results due to the occurrence of misspecification in the first step. Finally, an exhaustive search is performed to locate the change points. Theoretical properties of the proposed algorithm are established, including the consistency of estimating both the

unknown number and the location of change points under mild conditions. Furthermore, the effectiveness of the developed detection algorithm is confirmed empirically through comparison with competing methods as well as through various synthetic data and two real data examples.

E1724: **Graph-based change-point detection for regression changes under covariate shift**

Presenter: **Lynna Chu**, Iowa State University, United States

Co-authors: Seho Kim

Change-point detection in regression aims to determine whether and when the conditional relationship between covariates and response changes over time. Classical structural break tests typically rely on strong global parametric assumptions that permit extrapolation across covariate regimes, but they can be sensitive to model misspecification. Modern nonparametric methods are more flexible, yet many target changes in the joint distribution of the data and may therefore fail to distinguish conditional distributional change from covariate shift. A graph-based, covariate-aware framework for detecting conditional changes in regression is proposed that accommodates time-varying covariate distributions through local comparability. The resulting test is designed to isolate changes in the conditional mechanism from changes in the covariate distribution itself. Its asymptotic null distribution is derived and simulations demonstrate that the method performs well across a wide range of settings.

E1748: **Estimation and inference for change points in functional regression time series**

Presenter: **Shivam Kumar**, University of Chicago, United States

Co-authors: Haotian Xu, Daren Wang

Change point estimation and inference under a Functional linear Regression model with changes in the slope function is studied. A novel Functional Regression Binary Segmentation (FRBS) algorithm is presented, which is computationally efficient and achieves consistency in multiple change point detection. The algorithm utilizes the predictive power of piecewise constant Functional linear Regression models in the reproducing kernel Hilbert space framework. A refinement step is proposed that improves the localization rate of the initial estimator output by FRBS, and asymptotic distributions of the refined estimators are derived for two different regimes determined by the magnitude of a change. To facilitate the construction of confidence intervals for underlying change points based on the limiting distribution, a consistent block-type long-run variance estimator is proposed. Theoretical justifications for the proposed approach accommodate temporal dependence and heavy-tailedness in both the Functional covariates and the measurement errors. Empirical effectiveness of the methodology is demonstrated through extensive simulation studies and an application to the Standard and Poors 500 index dataset.

EO222 Room 102 DATA SCIENCE, STATISTICS, AND AI FOR DIGITAL FINANCE

Chair: Jeffrey Chu

E1707: **Understanding the evolution of Kyles lambda on digital blockchain assets**

Presenter: **Lovely Jamil**, American University of Sharjah, United Arab Emirates

While Kyle's lambda, a measure of the price impact of trading activity, is extensively studied in traditional financial markets, its behaviour in digital asset markets remains underexplored. An examination of the evolution of Kyle's lambda across major blockchain-based assets using transaction-level data from the Kraken Exchange reveals heterogeneity in Kyle's lambda across digital asset classes, with cryptocurrencies and large-cap DeFi tokens exhibiting lower liquidity sensitivity than NFTs and meme coins. The dataset contains high-frequency trade data, including timestamps, prices, and volumes, for multiple digital asset classes across USD and EUR trading pairs over 2025. Kyle's lambda is estimated using regression-based approaches, including returns on signed trading volume and dollar-volume specifications, as well as models based on trade activity and microstructure dynamics. Results show that Kyle's lambda is negatively related to dollar trading volume ($p < 0.01$) and positively associated with return volatility and spikes during market stress. Moreover, lambda dynamically precedes increases in volatility and declines in trading activity, suggesting it serves as an early signal of liquidity risk and varying market efficiency. These findings highlight time-varying liquidity and cross-sectional differences in price impact with implications for traders, researchers, and regulators concerned with liquidity risk and the stability of decentralized financial systems.

E1974: **Extreme value-informed multi-modal transfer learning for blockchain fraud detection**

Presenter: **Jeffrey Chu**, Renmin University of China, China

Blockchain-based financial systems face increasingly sophisticated fraudulent schemes, yet detecting anomalies in large-scale transaction graphs remains challenging due to high dimensionality, non-stationary behaviour, and Multimodal structure. An Extreme-Value-Informed Multimodal Transfer Learning (EVI-MTL) framework is proposed that integrates structural transaction features, participant-level temporal sequences, textual and video metadata, and extreme value-based risk measures. A Peaks-over-Threshold model is used to construct tail-risk embeddings and a temporally-aware Maximum Mean Discrepancy (TMMD) regularization to address distributional shifts across chains and over time. Empirical studies on Ethereum and Solana show that EVI-MTL substantially improves fraudulent-node classification, outperforming state-of-the-art baselines by 5-15 percentage points, with the largest gains during periods of elevated tail risk. Results indicate that incorporating extreme-value characteristics meaningfully enhances Transfer Learning and yields more reliable early-warning indicators for phishing, rug pulls, and related illicit behaviours, providing a statistically grounded approach for monitoring risk-sensitive blockchain environments.

E2055: **Does governance credibility matter: Evidence from binding and non-binding fan token polls**

Presenter: **Yuanyuan Zhang**, American University of Sharjah, United Arab Emirates

Co-authors: Stephen Chan

The aim is to examine whether governance credibility affects fan token markets by comparing binding and non-binding fan polls issued by major football clubs. Using daily price, trading volume, market capitalization, and poll data for PSG, GAL, JUV, ATM, and CITY fan tokens, a stacked difference-in-differences event-study framework is applied around governance poll events. The analysis separates immediate and short-term market reactions and tests whether binding polls generate different effects from non-binding polls. Results suggest that governance polls are associated with short-run changes in token returns, with evidence of pre-event anticipation and post-event adjustment. However, the market response varies across tokens and outcome variables, with stronger evidence for price effects than for persistent changes in trading volume, market capitalization, or liquidity. Overall, the findings indicate that fan token governance events contain market-relevant information, but the credibility premium attached to binding polls is limited and heterogeneous across clubs.

E2062: **Multi-modal enhanced graph transfer learning for digital finance fraud detection**

Presenter: **Chenguang Yang**, University of California, Riverside, United States

Fraudulent activities on blockchain networks pose serious threats to the integrity and reliability of decentralized finance ecosystems. Detecting malicious nodes, such as phishing or ransomware addresses, in large-scale blockchain transaction graphs remains challenging because these graphs are dynamic, sparse, and continuously evolving. Transfer Learning is promising for fraud detection, as different fraud types are often conducted by overlapping actor groups and may share behavioral and structural patterns across networks. By transferring knowledge from known fraud types to emerging ones, models can improve detection under limited supervision. However, digital financial systems are inherently Multi-mOdal, involving Graph structures, transaction sequences, temporal price dynamics, and textual metadata. In addition, distribution shifts across time and platforms make it difficult for existing Graph Transfer Learning methods to align source and target domains effectively. To address these challenges, the Multi-mOdal Enhanced Graph Transfer Learning framework, or MOE-GTL, integrates Graph, temporal, and textual modalities for fraudulent

node detection. The framework further introduces Temporal-aware Maximum Mean Discrepancy, or TMMD, a regularization mechanism that aligns Multi-mOdal feature distributions between source and target graphs over time. Experiments on Ethereum and Solana transaction graphs demonstrate that MOE-GTL improves fraudulent node classification accuracy.

EO165 Room 103 MODERN METHODS FOR MODELING AND LEARNING FROM COMPLEX DATA (VIRTUAL)
Chair: Pei Wang
E1191: Genetic algorithms in regression

Presenter: **Mo Li**, UL at Lafayette, United States

Co-authors: Qiqi Lu, Robert Lund, Xueheng Shi

Many statistical problems require optimization over a discrete parameter space with unknown and potentially varying dimensions. In these settings, gradient based methods are often ineffective due to non-differentiability and the combinatorial growth of candidate models. GAReg is introduced, a unified genetic-algorithm package for discrete optimization in regression problems where standard algorithms are impractical or unjustified. GAReg uses a compact chromosome representation that supports a range of tasks, including optimal knot placement for regression splines, best-subset variable selection, multiple changepoint detection, and related discrete model search problems. The package implements uniform initialization, constraint preserving crossover and mutation operators, steady state replacement, and optional island model parallelization. By enabling efficient exploration of high-dimensional model spaces, GAReg delivers near optimal solutions in regimes where exhaustive enumeration, integer programming, or dynamic programming approaches are computationally infeasible.

E1231: Penalized linked component analysis for spatial-temporal burst detection in water distribution systems

Presenter: **Shenghao Xia**, Bowling Green State University, United States

Detecting bursts from spatial-temporal (ST) hydraulic data is critical for water distribution system management. Conventional anomaly detection methods, such as statistical process control and basis expansion, are inefficient and inaccurate in detecting and localizing bursts from data, which are continuously collected from multiple potential locations. These limitations can delay response and lead to substantial economic losses from water loss and subsequent infrastructure repair. A new method based on penalized linked component analysis is proposed, which extracts ST anomaly features by differentiating the commonly shared normal features and individual anomaly features. Penalization is incorporated in the algorithm to increase the sensitivity of location detection and reduce the rate of false alarms. The effectiveness of the proposed method is demonstrated with a simulated case study.

E1440: Bovine pinkeye detection and prediction

Presenter: **Hossein Moradi Rekabdararkolaee**, Bowling Green State University, United States

Bovine pinkeye is a contagious ocular disease that significantly impacts cattle health and agricultural productivity. Conventional diagnostic methods rely on clinical observation, which can be subjective, time-consuming, and impractical for large herds or remote locations. To address these limitations, a computer vision approach for eye detection and deep learning algorithm for predicting pinkeye in cows was employed.

E1443: Finite Markov chains with absorbing states and misspecified random effects with an application to cognitive data

Presenter: **Pei Wang**, Bowling Green State University, United States

Finite Markov chains with absorbing states are valuable tools for analyzing longitudinal data with categorical responses. However, defining the one-step transition probabilities in terms of fixed and random effects presents challenges due to the large number of unknown parameters involved. To address this, a marginal model is employed to estimate the fixed effects across various choices of the distribution governing the random effects. Subsequently, an h-likelihood method is utilized to estimate the random effects based on these fixed effect estimates. The estimation approach is applied to analyze longitudinal cognitive data from the Nun Study. The findings highlight that the fixed effects remain relatively robust across a wide range of assumptions. However, the analysis of random effects utilizing tools such as AIC, Q-Q plots, and gradient plots appears to be sensitive to misspecifications in the distribution of the random effects. The proposed approach allows researchers to verify the assumptions of random effects and provides more accurate estimation of these effects.

EO217 Room 104 STATISTICAL INFERENCE FOR COMPLEX AND DEPENDENT DATA
Chair: Wenbo Wu
E1322: Spherically embedded time series with unknown trend and periodic components

Presenter: **Jiazhen Xu**, Australian National University, Australia

Co-authors: Han Lin Shang

Spherically embedded time series are time series with values naturally residing on or can be equivalently mapped to the unit sphere. Despite their ubiquity, these data often exhibit complex non-stationarity driven by latent deterministic trend and periodic components. Traditional Euclidean time series methods fail to account for the intrinsic non-Euclidean geometry of the sphere, leaving a critical gap in rigorous methodologies for modeling and forecasting. To address this methodological gap, a unified geometric framework for the analysis of non-stationary spherically embedded time series is proposed. Central to this approach is a nonparametric spherical trend-periodicity decomposition model, which utilizes an optimal-transport-based removal operator to systematically decouple the deterministic components while preserving the spherical topology. The estimation procedure operates sequentially where the global smooth trend is first estimated and extracted, followed by the identification of the unknown period and the extraction of the periodic structure. The resulting de-trended and de-seasonalized stationary residuals can be further modeled via a spherical autoregressive process. The practical utility of this methodology is validated through applications to U.S. electricity generation compositions and bike trip volume profiles in New York City, yielding significantly enhanced forecasting accuracy while providing interpretable insights into the underlying structural dynamics.

E1634: Time series Gaussian chain graph models

Presenter: **Qin Fang**, The University of Sydney, Australia

Co-authors: Xinghao Qiao, Zihan Wang

Time series graphical models have recently received considerable attention for characterizing conditional dependence structures in multivariate time series. In many applications, the multivariate series exhibit variable-partitioned blockwise dependence, with distinct patterns within and across blocks. A new class of time series Gaussian chain graph models is introduced that represent contemporaneous and lagged causal relations via directed edges across blocks, while capturing within-block conditional dependencies through undirected edges. In the frequency domain, this formulation induces a cross-frequency shared group sparse plus group low-rank decomposition of the inverse spectral density matrices, which is exploited to establish identifiability of the time series chain graph structure. Building on this, a three-stage learning procedure is proposed for estimating the undirected and directed edge sets, which involves optimizing a regularized Whittle likelihood with a group lasso penalty to encourage group sparsity and a novel tensor-unfolding nuclear norm penalty to enforce group low-rank structure. The asymptotic properties of the proposed method are investigated, ensuring its consistency for exact recovery of the chain graph structure. Superior empirical performance is demonstrated through both extensive simulation studies and an application to U.S. macroeconomic data that highlights key monetary policy transmission mechanisms.

E1747: F-type tests for functional linear models*Presenter:* **Shu-Chin Lin**, National Taiwan University, Taiwan

Hypothesis testing for functional linear regression models with a scalar response is considered under the assumption that the coefficient function resides in a reproducing kernel Hilbert space. The goal is to test whether the coefficient function belongs to a closed subspace of the reproducing kernel Hilbert space, providing a unified platform to test global nullity, partial nullity, and parametric forms for the coefficient function. A simple and efficient testing procedure is developed based on a statistic that resembles the classical F-test. The minimax rate for the testing problems is studied, and the proposed testing procedure is shown to be asymptotically close to the optimal setting under mild conditions.

E1829: Neural Wasserstein two-sample tests*Presenter:* **Zhenhua Lin**, National University of Singapore, Singapore*Co-authors:* Xiaoyu Hu

The two-sample homogeneity testing problem is fundamental in statistics and becomes particularly challenging in high dimensions, where classical tests can suffer substantial power loss. A learning-assisted procedure based on the projection Wasserstein distance is developed, motivated by the observation that there often exists a low-dimensional projection under which the two high-dimensional distributions differ. In practice, the projection directions are learned via manifold optimization and a witness function using deep neural networks. To adapt to unknown projection dimensions and sparsity levels, a collection of candidate statistics is aggregated through a max-type construction, avoiding explicit tuning while potentially improving power. The validity and consistency of the proposed test are established, and a Berry-Esseen type bound for the Gaussian approximation is proved. In particular, under the null hypothesis, the aggregated statistic converges to the absolute maximum of a standard Gaussian vector, yielding an asymptotically pivotal (distribution-free) calibration that bypasses resampling. Simulation studies and a real-data example demonstrate the strong finite-sample performance of the proposed method.

EO041 Room 105 RECENT ADVANCES IN NEUROIMAGING DATA ANALYSIS**Chair: Panpan Zhang****E1387: A multivariate harmonization framework for multi-site neuroimaging studies***Presenter:* **Qiong Wu**, University of Pittsburgh - Pittsburgh, PA, United States*Co-authors:* Jingsong Zhou, Shuo Chen

Multi-site large consortium studies involving neuroimaging and omics data have become common and essential for advancing biomedical research. Statistical analyses of such datasets rely on harmonization methods to remove technical variance and batch effects while preserving biological signals. However, most existing harmonization methods align the means and variances of neuroimaging features separately, without accounting for dependencies among imaging variables. This can result in incoherent covariance structures across study sites and suboptimal performance in downstream cross-site analyses. To address this gap, graph-guided ComBat (GG-ComBat) is proposed, a new harmonization framework that corrects batch effects in multivariate neuroimaging variables while allowing them to borrow strength from each other based on their dependencies. Specifically, GG-ComBat learns a graph structure from shared covariance patterns across study sites and leverages this structure to model parameters within the ComBat empirical-Bayes framework. Computationally efficient algorithms for empirical-Bayes estimation that explicitly account for cross-feature covariance are developed. Simulation studies and application to the adolescent brain cognitive development (ABCD) study demonstrate that GG-ComBat outperforms existing harmonization methods, including ComBat and CovBat, in recovering site-specific effects, aligning covariance across sites, and improving downstream prediction performance.

E1781: Improving the measurement of test-retest reliability for high-dimensional connectome and neuroimaging biomarkers*Presenter:* **Xin Ma**, Columbia University, United States*Co-authors:* Seonjoo Lee

Psychiatric disorders affect more than 20% of the population in the United States. Advances in neuroscience research have shown the potential clinical uses of neuroimaging markers for diagnosis, treatment selection, and disease monitoring. However, current clinical practice depends solely on clinical assessments without using biomarkers. The main obstacles are low test-retest reliability (TRR), the absence of rigorous TRR methods, methodological inconsistencies, and inadequate replication. A Generalized Imaging Intraclass Correlation Coefficients (GI2C2) reliability framework designed for high-dimensional connectome and neuroimaging data is proposed to address this issue. The goal is to create statistically rigorous, clinically relevant, and scalable tools to help researchers and clinicians identify reproducible connectome biomarkers for psychiatric illnesses.

E1786: Characterizing longitudinal sMRI biomarker trajectories in Alzheimer's disease using the DSMM*Presenter:* **Kaidi Kang**, Wake Forest University School of Medicine, United States

Characterizing longitudinal trajectories of sMRI-derived biomarkers across Alzheimer's Disease (AD) progression is critical for understanding Disease dynamics, yet remains challenging due to the fragmentary nature of available data: existing datasets capture only brief windows of an individual's decades-long Disease course. This is compounded by limited availability of repeated sMRI measurements. Together, these restrict standard analyses to participants with sufficient imaging follow-up and observed clinical events, and introduce selection bias and reduced statistical power. To address these limitations, the double anchoring events-based sigmoidal mixed model (DSMM) was extended to characterize longitudinal sMRI-derived biomarkers in AD. The DSMM employs Disease-relative time scales anchored to incident AD diagnosis and enables inclusion of participants without an observed conversion event, broadening the analyzable study population. Using harmonized multi-cohort data from the Alzheimer's Disease Sequencing Project Phenotype Harmonization Consortium (ADSP-PHC), imaging trajectories are aligned relative to AD onset across participants regardless of conversion status. This strategy mitigates selection bias, enhances statistical power, and yields clinically interpretable biomarker estimates along the Disease continuum, enabling formal comparison of the temporal ordering of multimodal biomarker changes in AD.

E1801: SIGNET: A signed network spectral clustering method for proteomic module discovery in Alzheimer's disease*Presenter:* **Panpan Zhang**, Vanderbilt University Medical Center, United States

Signed networks arise naturally in many biomedical applications, where both positive and negative relationships contain distinct and important biological information. However, most existing community detection methods either ignore edge signs or rely on transformations that do not adequately preserve negative correlation interpretation, potentially obscuring meaningful structure. SIGNET, a spectral clustering framework for signed, weighted networks, is proposed. SIGNET directly incorporates both positive and negative edge information through the signed Laplacian. Silhouette score-based strategies for selecting the number of clusters are further developed, including a data-driven silhouette-gap criterion. Extensive simulations demonstrate that SIGNET achieves robust and accurate clustering performance across a range of network configurations, including varying signal strengths, cluster sizes, and structural heterogeneity. Compared with existing approaches, SIGNET exhibits improved stability in cluster number selection and more reliable recovery of the underlying modular structure. Application to a protein-protein co-expression network from the Vanderbilt Memory and Aging Project reveals biologically coherent modules that capture biological processes relevant to Alzheimer's disease, highlighting the practical utility of SIGNET in high-dimensional biomedical studies. An R package, SIGNET, is publicly available to support implementation.

EO207 Room 106 NEW DEVELOPMENTS IN STOCHASTIC MODELING**Chair: Takashi Owada****E1342: Optimal portfolio choice with ESG considerations and asymmetric information****Presenter:** Haibo Liu, Purdue University, United States

Portfolio optimization problems incorporating environmental, social, and governance (ESG) factors are studied for three types of investors, namely brown investors who care only about their wealth, green investors who prefer assets with high ESG ratings, and mixed investors who take a position between the two. Given that some ESG ratings are not publicly available, a distinction is made between different levels of ESG information accessibility for these investors. The impact of ESG preferences and ESG information accessibility on investment strategies is explored, addressing issues such as the components of optimal portfolios of different investors and the utility indifference value of nonpublic ESG information.

E1582: Stable and Frechet limit theorem for subgraph functionals in the hyperbolic random geometric graph**Presenter:** Takashi Owada, Purdue University, United States**Co-authors:** Christian Hirsch, Ruiting Tong

Fluctuations of subgraph counts in hyperbolic random geometric graphs on the d -dimensional Poincare ball are studied in the heterogeneous, heavy-tailed degree regime. In a hyperbolic random geometric graph whose vertices are given by a Poisson point process on a growing hyperbolic ball, two basic families of subgraphs are considered: star shape counts and clique counts, with their global counts and maxima over the vertex set analyzed. Working in the parameter regime where a small number of vertices close to the center of the Poincare ball carry very large degrees and act as hubs, joint functional limit theorems are established for suitably normalized star shape and clique count processes together with the associated maxima processes. The limits are given by a two-dimensional dependent process whose components are a stable Levy process and an extremal Frechet process, reflecting the fact that a small number of hubs dominates both the total number of local subgraphs and their extremes. As an application, fluctuation results for the global clustering coefficient are derived, showing that its asymptotic behavior is described by the ratio of the components of a bivariate Levy process with perfectly dependent stable jumps.

E1676: Pooled likelihood inference for secondary phenotype models in multiple case-control studies**Presenter:** Zifu Wei, Sichuan University, China

Secondary outcomes from case-control studies are scientifically useful but statistically awkward because retrospective ascertainment hides the source-population case fraction. In a single study, intercept-like secondary regression parameters are generically confounded with that hidden prevalence and are therefore not identified without external prevalence information or an equivalent normalization. Pooling several case-control studies under a common source-population covariate law changes the problem: cross-study distortion contrasts identify the hidden prevalence vector through a linear system, and the conditioning of that system yields a natural weak-identification index. On that foundation, a pooled profile-likelihood framework is developed that profiles over the common covariate distribution and treats the prevalence vector as part of the inferential target. The framework is analyzed in three model cells: a baseline linear-mean model with known error density, an extension with unknown common error density, and an extension with unknown mean surface and known error density. Several extensions to this framework are also discussed, including dimension reduction and extreme regions.

E1716: Connectivity transitions in large-scale simplicial complexes via local limits**Presenter:** Taegyu Kang, Georgia Institute of Technology, United States**Co-authors:** Souvik Dhara

Higher-dimensional connected components in random simplicial complexes are investigated. In a d -dimensional simplicial complex, d -dimensional connectivity is defined by incidence between $(d-1)$ - and d -dimensional simplices. The phase transition of the largest d -dimensional connected component is identified in terms of the parameter that governs the number of d -dimensional simplices incident to a typical $(d-1)$ -dimensional simplex. When the parameter is larger than the threshold $1/d$, a large d -dimensional component, called the giant component, emerges. Below the threshold, all d -dimensional components are small. Additionally, when the parameter is larger than $1/d$, the proportion of 0 -dimensional simplices in the d -dimensional giant component converges to 1, meaning the giant occupies almost all vertices. The model considered is the multi-parameter random simplicial complex and its special case, the Linial-Meshulam random simplicial complex. The primary machinery is the local-weak convergence of random simplicial complexes. Local-weak convergence in probability of the multi-parameter random simplicial complex is established and the local information is leveraged for the formation of the giant component and its concentration.

EO146 Room 107 CAUSAL INFERENCE: CONFOUNDING, SELECTION BIAS, AND COVARIATE ADJUSTMENT**Chair: Fan Li****E1423: Covariate-adjusted win statistics in randomized clinical trials with ordinal outcomes****Presenter:** Zhiqiang Cao, Shenzhen Technology University, China**Co-authors:** Scott Zuo, Mary Ryan Baumann, Kendra Plourde, Patrick Heagerty, Guangyu Tong, Fan Li

Ordinal outcomes are common in clinical settings where they often represent increasing levels of disease progression or different levels of functional impairment. To compare different intervention strategies in clinical trials, the direct use of ordinal logistic regression models may not be ideal for analyzing ranked outcomes. The focus is on representing the average treatment effect for ordinal outcomes via intrinsic pairwise outcome comparisons captured through win estimates. Recognizing the value of baseline covariate adjustment toward enhanced precision, propensity score weighting estimators are first developed, including both inverse probability weighting (IPW) and overlap weighting (OW), tailored to estimating win parameters. Furthermore, augmented weighting estimators are developed that leverage an additional ordinal outcome regression to potentially improve efficiency over weighting alone. Leveraging the theory of U-statistics, the asymptotic theory for all estimators is established, and closed-form variance estimators are derived. Through extensive simulations the enhanced efficiency of the weighted estimators over the unadjusted estimator is demonstrated, with the augmented weighting estimators showing a further improvement in efficiency except for extreme cases. Finally, the proposed methods are illustrated with the ORCHID trial, and the covariate adjustment methods are implemented in an R package winPSW to facilitate the practical implementation.

E1504: Average treatment effect estimation under poor overlap via weighted estimands**Presenter:** Shunichiro Orihara, Tokyo Medical University, Japan**Co-authors:** Sho Komukai

The average treatment effect (ATE) is commonly considered as the causal effect of interest. In situations with poor overlap, where the covariate support between the treatment and control groups is limited, the inverse probability weighting (IPW) estimator may have large variance. To address this issue, weighted ATE (WATE) estimands, such as the ATE for the overlap population (ATO), are sometimes considered. However, these estimands may differ from the original target of interest such as the ATE. A novel estimation procedure is proposed that uses a class of WATEs based on the beta weight family, including ATO, and estimates the ATE via extrapolation. Specifically, the following steps are considered: 1) estimating IPW estimators for several WATE estimands; 2) regressing the IPW estimates on the hyperparameter of the WATEs; and 3) setting the hyperparameter to 0 (extrapolation). Theoretical justifications for the procedure and its properties, such as asymptotic normality, are discussed. Additionally, since the proposed procedure is easy to implement, an explanation of how to conduct the analysis in R with a simple programming is provided.

E1620: On the graphical rules for recovering the average treatment effect under selection bias*Presenter:* **Haidong Lu**, Yale University, United States*Co-authors:* Yichi Zhang

Selection bias is a major obstacle to valid causal inference in epidemiology. Over the past decade, several graphical rules based on causal diagrams have been proposed as sufficient identification conditions for addressing selection bias and recovering causal effects. However, these rules are typically tied to specific identification strategies and estimators. Two important cases of selection bias cannot be addressed by these existing rules and their associated estimators: one in which selection is a descendant of a collider between the treatment and the outcome, and another in which selection is affected by the mediator. To address selection bias and recover the average treatment effect in these settings, alternative graphical rules are proposed and identification formulas are derived using g-computation and inverse probability weighting (IPW) based on single-world intervention graphs (SWIGs). Simulation studies evaluate the performance of the proposed estimators, particularly in settings where traditional crude selected-sample analyses (i.e., complete-case analyses) yield conclusions that contradict the truth.

EO203 Room 201 EMERGING TOPICS IN DATA SCIENCE FOUNDATIONS**Chair: Li-Pang Chen****E1254: Rating of players by Laplace approximation and dynamic modeling***Presenter:* **Robert Hsuan-Fu Hua**, Florida State University, United States

The Elo rating system is a simple and widely used method for calculating players' skills from paired comparison data. Many have extended it in various ways. Yet the question of updating players variances remains to be further explored. The issue of variance update is addressed by using the laplace approximation for posterior distribution, together with a random walk model for the dynamics of players strengths, and a lower bound on players variances. The random walk model is motivated by the Glicko system, but nonidentically distributed increments are assumed to take care of player heterogeneity. Experiments on men's professional matches showed that the prediction accuracy slightly improves when the variance update is performed. It also showed that new players strengths may be better captured with the variance update.

E1256: Semiparametric efficient estimation of causal effects defined by statistical functionals*Presenter:* **Kuan-Hsun Wu**, Pennsylvania State University, United States

The problem of estimating a general class of causal effect is explored. In particular, the causal effects can be represented as a Hadamard differentiable statistical functional of the underlying distribution. The semiparametric lower bounds are found and attained by a nonparametric estimation approach. Commonly used cases, including but not limited to average, quantile, and distributional causal effects, are shown to be special cases in the general class. Moreover, a new causal effect defined on expectiles is investigated, which is also fully covered by the theory.

E1395: A knot selection algorithm for regression spline*Presenter:* **Cheng-Kuan Lin**, National Chengchi University, Taiwan*Co-authors:* Tzee-Ming Huang

In nonparametric regression, spline regression is a commonly used approach for approximating an unknown function. However, to use splines for function approximation, it is necessary to specify knots and the order, among which the selection of knot locations plays a crucial role in determining the quality of the approximation. The proposed algorithm is based on a test statistic designed to detect differences in regression coefficients, and utilizes the outcome of this test to determine the locations of the knots. Both theoretical analysis and numerical experiments demonstrate that this method outperforms the conventional approach based on equally spaced knots.

E1509: Shift-invariant attribute scoring for Kolmogorov-Arnold networks via Shapley value*Presenter:* **Ching Wang**, National University of Singapore, Taiwan*Co-authors:* Siqi Li

For many real-world applications, understanding feature-outcome relationships is as crucial as achieving high predictive accuracy. While traditional neural networks excel at prediction, their black-box nature obscures underlying functional relationships. Kolmogorov-Arnold Networks (KANs) address this by employing learnable spline-based activation functions on edges, enabling recovery of symbolic representations while maintaining competitive performance. However, KAN's architecture presents unique challenges for network pruning. Conventional magnitude-based methods become unreliable due to sensitivity to input coordinate shifts. A pruning framework using Shapley value attribution is proposed to assess node importance in a shift-invariant manner. Unlike magnitude-based approaches, ShapKAN quantifies each node's actual contribution, ensuring consistent importance rankings regardless of input parameterization. Extensive experiments on synthetic and real-world datasets demonstrate that ShapKAN preserves true node importance while enabling effective network compression. The approach improves KAN's interpretability advantages, facilitating deployment in resource-constrained environments.

EO081 Room 203 THEORIES AND METHODOLOGIES FOR HIGH-DIMENSIONAL AND COMPLEX DATA ANALYSIS Chair: Kento Egashira**E1629: Notations for dendrograms and asymptotic properties of single-linkage clustering in high-dimensional settings***Presenter:* **Yota Takao**, Tokyo University of Science, Japan*Co-authors:* Kento Egashira

Agglomerative hierarchical clustering using single linkage is analyzed for high-dimensional data randomly obtained from two or more independent populations. The aim is to clarify the relationship between dendrogram shapes formed by high-dimensional data and the parameter space of the populations under less stringent assumptions. Agglomerative hierarchical clustering is a method that groups data for which prior information such as the true number of clusters is not provided according to their similarity. Users of agglomerative hierarchical clustering can understand characteristics of the data under investigation by analyzing the resulting dendrograms. Notations representing various dendrogram shapes are introduced to facilitate discussion of dendrogram structures. Using these notations, asymptotic properties of agglomerative hierarchical clustering with single linkage are established as the data dimension diverges.

E1618: Asymptotic behaviors of kernel hierarchical clustering under high-dimensional settings*Presenter:* **Kento Egashira**, Tokyo University of Science, Japan*Co-authors:* Yota Takao

In high-dimensional settings with multiple underlying populations, the asymptotic behavior of hierarchical clustering methods, including Ward's method, has been extensively studied. Under suitable conditions, the resulting dendrogram converges to one of several distinct topological forms, which enables a precise characterization of when the true population structure is correctly recovered. Despite the widespread use and empirical success of kernel methods in high-dimensional data analysis, the theoretical properties of hierarchical clustering combined with kernel techniques remain largely unexplored. The asymptotic behavior of kernel-based hierarchical clustering is investigated to relax existing conditions required for consistent population separation. Focusing on Ward's method with a Gaussian kernel, sufficient conditions are derived under which the desired dendrogram structure is asymptotically obtained. The theoretical analysis provides an explanation for the improved clustering performance observed in high-dimensional settings and broadens the applicability of hierarchical clustering methods.

E1670: Asymptotics of empirical risk minimization in High dimension*Presenter:* **Cosme Jean Leon Louart**, Chinese university of Hong Kong Shenzhen, China

Recent results on the asymptotic behavior of regularized empirical risk minimization are presented. A complete theoretical characterization of its performance is derived under weak smoothness and convexity assumptions on the loss functions, together with general concentration assumptions on the data. More specifically, a central limit theorem for the minimizer yields an approximation of the test score in a Gaussian-convoluted form involving two deterministic parameters, which are characterized as solutions of a fixed-point equation depending on the population statistics and the loss functions. This approximation makes it possible to fully characterize the breakdown of Gaussian universality for the score, a phenomenon that has attracted significant attention in recent years. The main insights provided by these formulas for several loss functions and models commonly used in practice are discussed.

E1627: Sequential monitoring for distributional change-points in time series*Presenter:* **B Cooper Boniece**, Drexel University, United States*Co-authors:* Lajos Horvath, Lorenzo Trapani

A nonparametric sequential monitoring procedure for detecting distributional breaks in time series is developed based on a class of degenerate U-statistics associated with possibly indefinite kernels, encompassing energy distances and kernel-based maximum mean discrepancy (MMD). The framework accommodates a general class of monitoring schemes and is developed in a Hilbert space setting. The asymptotic null behavior is characterized under weak dependence, and tractable approximation of the limit distribution is developed via spectral methods. Simulations across a range of monitoring schemes, along with real data examples, illustrate the method's performance.

EO195 Room 204 STATISTICAL INFERENCE UNDER COMPLEX DEPENDENCE AND NONREGULARITY**Chair: Karl Oskar Ekvall****E1603: Scalable estimation of crossed random effects models via multi-way discretization***Presenter:* **Shota Takeishi**, Washington University in St. Louis, United States*Co-authors:* Shonosuke Sugawara

Cross-classified data frequently arise in scientific fields such as education, healthcare, and social sciences. A common modeling strategy is to introduce crossed random effects within a regression framework. However, this approach often encounters serious computational bottlenecks, particularly for non-Gaussian outcomes. A scalable and flexible method is proposed that approximates the distribution of each random effect by a discrete distribution, effectively partitioning the random effects into a finite number of representative values. This approximation allows the model to be expressed as a multi-way discrete structure, which can be efficiently estimated using a simple and fast iterative algorithm. The proposed method accommodates a wide range of outcome models and remains applicable even in settings with more than two-way cross-classification. Consistency and asymptotic normality of the estimator are theoretically established under general settings of classification levels. Simulation studies and real data applications demonstrate the practical performance of the proposed method.

E1759: When parameters are on the boundary: Likelihood ratio tests beyond Wilks' theorem*Presenter:* **Clara Bertinelli Salucci**, University of Oslo, Norway*Co-authors:* Anders Kvellestad, Riccardo De Bin

Likelihood ratio tests (LRTs) are a central tool in statistics and data science for hypothesis testing, typically justified by Wilks' theorem, which guarantees an asymptotic chi-square distribution of the test statistic under regular conditions. However, these conditions can fail in practice. In particular, they fail in a broad and practically relevant class of problems where parameters lie on the boundary of the parameter space: from a practical point of view, the classical chi-square approximation is still often used under such settings in different applications, but it can be severely misleading. Recent advances in understanding the asymptotic behaviour of LRTs with parameters on the boundary are discussed, with a focus on the geometric and probabilistic mechanisms underlying the asymptotic distribution of the test statistic.

E1601: Likelihood ratio tests of boundary points with singular information and unidentifiable nuisance parameters*Presenter:* **Karl Oskar Ekvall**, University of Florida, United States

The asymptotic distribution of likelihood ratio tests is derived when nuisance parameters are unidentifiable under the null, parameters of interest lie on the boundary of the parameter space, and the information matrix may be singular -three complications that arise simultaneously in mixture models and genetic linkage analysis. The limiting distribution is the supremum of a chi-bar-square-process indexed by the nuisance parameter. Notably, when singularity stems from the nuisance parameter rather than the parameter of interest, the form of the limit is unchanged relative to the nonsingular case. The results unify existing theory for boundary inference and recover several application-specific results as special cases.

E1826: On the inverse of covariance matrices for unbalanced crossed designs*Presenter:* **Ziyang Lyu**, University of New South Wales, Australia*Co-authors:* Scott Sisson, Alan Welsh

A long-standing open problem in the analysis of linear mixed models with crossed random effects under unbalanced designs concerns finding an analytic expression for the inverse of the covariance matrix of the observed response. The inverse matrix is required for likelihood-based estimation and inference. However, for unbalanced crossed designs, the covariance matrix is dense and the lack of a closed-form representation for the inverse has made using likelihood-based methods computationally challenging and difficult to analyze mathematically. The Khatri–Rao product is used to represent the covariance matrix and to construct a modified covariance matrix whose inverse admits an exact spectral decomposition. Building on this construction, an elegant and simple approximation to the inverse of the covariance matrix for asymptotic unbalanced designs is obtained. For non-asymptotic settings, an accurate and interpretable approximation is derived under mildly unbalanced data and an exact inverse representation is established as a low-rank correction to this approximation, applicable to arbitrary degrees of unbalance. Simulation studies demonstrate the accuracy, stability, and computational tractability of the proposed framework.

EO224 Room 205 HIGH DIMENSIONAL INFERENCE AND MULTIPLE TESTING**Chair: Johan Lim****E1333: Double truncation method for controlling local false discovery rate in case of spiky null***Presenter:* **Jaesik Jeong**, Chonnam National University, Korea, South*Co-authors:* Shinjune Kim, Johan Lim, DoHwan Park

Many multiple test procedures, which control the false discovery rate, have been developed to identify some cases (e.g. genes) showing statistically significant difference between two different groups. However, a common issue encountered in some practical data sets is the presence of highly spiky null distributions. Existing methods struggle to control type I error in such cases due to the inflated false positives, but this problem has not been addressed in previous literature. This issue was recently encountered while analyzing Set4 gene deletion data and a solution was proposed involving modeling the null distribution using a scale mixture normal distribution. However, the use of this approach is limited due to strong assumptions on the spiky peak. A novel multiple test procedure is presented that can be applied to any type of spiky peak data, including situations with no spiky peak or with one or two spiky peaks. The approach involves truncating the central statistics around 0, which primarily contribute to the null spike, as well as the two tails that may be contaminated by alternative distributions. This method is referred to as the double truncation

method. It is demonstrated numerically that the proposed method effectively controls the false discovery rate at the desired level using simulated data. Furthermore, the method is applied to two real data sets, namely the set protein data and peony data.

E1545: **High-dimensional multiple testing under latent confounding**

Presenter: **Shota Katayama**, Faculty of Economics, Keio University, Japan

Detecting group differences based on treatments or disease status is a critical challenge in high-dimensional genetic data analysis. In observational studies where random assignment is infeasible, confounding variables can affect both the variables of interest (such as treatment variables) and the outcome variables. While observable confounders can be addressed by incorporating them into the model, latent factors such as ancestry, analytical environment, and equipment may also act as confounders. Consequently, statistical inference performed without accounting for such latent confounding variables will lead to biased conclusions. A multiple testing procedure tailored for high-dimensional settings with latent confounding is proposed. Specifically, high-dimensional treatment parameters and latent variables are first estimated using L1-norm and nuclear norm regularizations, respectively, and then test statistics are constructed based on these estimators. Finally, a multiple testing procedure that can asymptotically control the false discovery rate (FDR) is proposed.

E1560: **Testing independence in high-dimensional data based on the absolute cross-covariance sum**

Presenter: **Johan Lim**, Seoul National University, Korea, South

Co-authors: Seonghun Cho, Seongoh Park

To test the independence of a variable of interest and all other variables, the absolute cross-covariance sum (ACS) statistic is proposed, defined as the sum of absolute correlations. Asymptotic normality of this sum is established as the number of variables increases under regularity conditions and finite-dependence assumptions. Consistent estimators of its mean and variance are proposed, such that the standardized ACS statistic achieves asymptotic normality. Numerical investigations of the size and power of the proposed test statistic are conducted and compared with existing methods. The ACS statistic is applied to identifying island variables, which are independent of others, in gene expression data.

E1686: **FDR control via neural networks under covariate-dependent symmetric nulls**

Presenter: **Junyong Park**, Seoul National University, Korea, South

In modern multiple hypothesis testing, the availability of additional covariate information alongside the primary test statistics has motivated the development of more powerful and adaptive inference methods. However, most existing approaches rely on p-values that are precomputed under the assumption that their null distributions are independent of the covariates. A novel framework is proposed that first derives covariate-adaptive p-values from the assumption of symmetric null distribution of the primary variable given the covariates, without imposing any parametric assumptions. Building on these data-driven p-values, a neural network model is employed to learn a covariate-adaptive rejection threshold via the mirror estimation principle, optimizing the number of discoveries while maintaining valid false discovery rate (FDR) control. Furthermore, estimation of the conditional null distribution enables the computation of p-values directly from the raw data. Simulation studies and a real data example are presented.

EO289 Room 208 EVENT AND LONGITUDINAL MODELS FOR MEDICAL RESEARCH AND FORESTRY Chair: Elizabeth Juarez-Colunga

E1352: **Dynamic mediation and latent change modeling of the dual process model of grief in Parkinson's caregiving**

Presenter: **James Haws**, University of Colorado Anschutz, United States

Co-authors: Kathryn Colborn, Lane Drew, Elizabeth Juarez-Colunga, Yizhou Fei

Caregivers of individuals with Parkinson's disease experience chronic stress, anticipatory grief, and ambiguous loss. The dual process model of grief posits coping through oscillation between loss-oriented (grief processing, yearning, emotional engagement) and restoration-oriented (role adjustment, future planning, identity reconstruction) processes, yet empirical tests of oscillation dynamics remain limited. An evaluation of the mechanistic pathways of caregiver and patient outcomes using mediation and latent change score is conducted, using data from a stepped-wedge cluster randomized trial of an outpatient palliative care intervention (N = 359 patients; N = 300 caregivers). Assessments occurred at baseline, 3-, 6-, and 12-month follow-ups. Outcomes include patient quality of life and caregiver burden and grief, with mediators including burden, mood, and spiritual wellbeing. Oscillation consistent with the DPM will be evaluated through coupled change processes between loss- and restoration-oriented indicators. The novel approach integrates longitudinal modeling with psychological theory to test dynamic mediation and regulatory processes underlying grief adaptation. Results may inform mechanism-focused palliative care interventions targeting both patient symptom management and caregiver grief adaptation.

E1795: **ldmppr: Location dependent marked point processes in R**

Presenter: **Lane Drew**, University of Colorado Anschutz, United States

Co-authors: Andee Kaplan

The ldmppr R package is presented for estimating, evaluating, simulating from, and visualizing location-dependent marked spatial point processes. To date, it has commonly been assumed that the marks associated with a point process are independent of the locations. However, when dealing with many point processes, such as those arising in forestry applications, the independence assumption proves unreasonable. A practical framework for generating marked point processes with dependence between the marks and locations is introduced. A brief discussion of the theory underpinning the modeling approach is provided along with an outline of the package use in a typical scenario involving real data. The functionality of the package for both generating from and assessing the goodness-of-fit of a given model is highlighted, enabling users to generate realistic point patterns given a reference pattern or parameter values of interest.

E1825: **Exact Poisson two-stage designs for landmark survival endpoints**

Presenter: **John Rice**, University of Michigan, United States

Landmark analyses of time-to-event endpoints are increasingly used in phase II clinical trials to enable timely efficacy assessment under fixed follow-up. Common two-stage designs rely on large-sample normal approximations or simulation-based calibration, which can lead to distorted type I error and inefficient early stopping in small to moderate samples. A class of exact two-stage designs for landmark survival endpoints based on a Poisson event-count framework is proposed. Modeling the number of events within a fixed window as a Poisson process with known exposure, exact interim and final decision boundaries are derived that control the one-sided type I error without asymptotic approximations. The designs allow early stopping for futility while preserving exact error control. Closed-form expressions are obtained for operating characteristics, including rejection probabilities, probability of early termination, and expected sample size under null and alternative hypotheses. This enables efficient design evaluation without extensive simulation. The approach is illustrated using landmark progression-free survival as a motivating example. These designs provide a transparent and computationally efficient alternative to approximate methods for landmark survival endpoints in early-phase trials.

E1714: **Estimation of treatment effects with data coarsened over informative observation intervals**

Presenter: **Paula Langner**, Veterans Health Administration, United States

Co-authors: Elizabeth Juarez-Colunga

When timing of observations in a longitudinal study is informative or related to the outcome, estimation may be affected if the observation process

is ignored. When an outcome is coarsened over each observation interval, the subsequent information loss further impacts estimation. The effect of informative variation in observation intervals on treatment effect estimation is explored when longitudinal count data are coarsened. Whether accounting for the observation process within the framework of a joint model can retain efficiency is investigated. A joint model of a recurrent event outcome and observation process, linked through a shared random effect, is proposed. Two forms of a coarsened outcome are considered, where either full counts for each interval or a dichotomized outcome indicating the presence or absence of any event in the interval are observed, with focus on the relative efficiencies of the treatment effect estimates. Simulation studies demonstrate that when the baseline event rate is low, the treatment effect estimate based on binary data is relatively efficient compared to count data, even while considering a range of other population parameters. In these cases, the joint model does not substantially improve over the outcome-only model, which ignores the informative observation times. However, with a higher baseline event rate, bias and variance of treatment effect estimates based on binary data are significant.

EO049 Room 209 STATISTICAL METHODS FOR BIOMEDICINE AND HEALTH RESEARCH
Chair: Masao Ueki
E1473: Comparing calibration fairness across ensemble learning methods: Bagging, boosting, and stacking

Presenter: **Hiroe Seto**, Kyoto Women's University, Japan

Co-authors: Michio Yamamoto, Hikari Komai, Kuniyoshi Hayashi

Fairness in prediction models has become a critical ethical concern in society. In particular, attention has focused on calibration fairness that is, whether predicted probabilities are equally accurate across subgroups defined by sensitive attributes such as gender and race. However, methods for evaluating calibration fairness have not been sufficiently established. To address this issue, several approaches for this purpose have been recently proposed, including the calibration error difference (CED) and the variable-based calibration plot (VC-plot). In this context, the question of which prediction algorithms can achieve fairer calibration remains unresolved. Identifying algorithms that are prone to achieving fair calibration is a key step toward constructing fair probabilistic prediction models. Previous work has suggested that ensemble learning methods may yield fairer calibration than non-ensemble approaches. The investigation focuses on which types of ensemble learning methods can achieve calibration fairness in probabilistic prediction models. Specifically, calibration fairness is systematically compared among a single decision tree (DT) and three ensemble learning methods bagging (random forest; RF), boosting (gradient boosting decision tree; GBDT), and stacking (DT+GBDT, DT+GBDT+RF). Using three real-world datasets requiring fairness consideration, probabilistic prediction models are constructed and evaluated with CED and VC-plot via 100 bootstrap iterations.

E1475: Effects of population stratification on gene-gene interaction detection

Presenter: **Neelotpal Das**, Nagasaki University, Japan

Co-authors: Masao Ueki

Genome-wide association studies (GWASs) have helped identify several causal variants underlying a wide range of complex diseases. To further understand the genetic architecture of these diseases, researchers often study epistasis or gene-gene interaction, which is defined as the departure from a purely additive model between two or more loci and is typically detected by testing an interaction term in a regression model. However, a key challenge in this analysis is population structure, which refers to the presence of distinct ancestral subpopulations within a sample and is known to inflate false positive rates in GWAS. How population stratification affects epistasis detection under three modelling frameworks is investigated: (i) standard linear regression, (ii) EIGENSTRAT, a widely used method for correcting false positives due to population structure in GWAS and (iii) a novel regression-based method for this setting. For each of the three models, the bias of the interaction term for a quantitative phenotype is analytically derived, where individuals are sampled from two distinct populations. Building on analytical results, simulation studies are conducted across a wide range of population structure scenarios, comparing the false positive rates produced by each model. Finally, the analytical and simulation-based findings are validated using real Alzheimer's disease data.

E1522: A suite of Spearman-like correlation measures for bivariate interval-censored data

Presenter: **Eric Kawaguchi**, University of Southern California, United States

Nonparametric tests of independence and their associated measures of association are important tools for studying bivariate failure time data. While pairwise concordance measures, such as Kendall's tau, have been studied extensively for bivariate interval-censored outcomes, comparatively less attention has been given to rank-based correlation measures. An extension of Spearman's rho for bivariate interval-censored data is proposed. The estimator relies only on marginal distributions and avoids direct estimation of the bivariate failure time surface. The method is further extended to allow for covariate adjustment, enhancing its practical relevance. Through simulation studies, it is demonstrated that the proposed estimator performs well across a range of scenarios. Finally, the utility of the approach is illustrated through applications to several real-world datasets.

E2017: Use of out-of-cluster prediction to improve model selection in regression with unsupervised clustering

Presenter: **Masao Ueki**, Nagasaki University, Japan

In regression with unsupervised clustering, the explanatory variables are first clustered, and separate regression models are then built for each cluster. The resulting models are often evaluated using in-cluster prediction criteria, such as the AIC and the BIC. Out-of-cluster prediction is explored for evaluating regression models, particularly in selecting the number of clusters. A model exclusion procedure is developed that makes use of the reduced accuracy of out-of-cluster prediction compared with in-cluster prediction, under the assumption that regression models differ between clusters, to exclude redundant models before applying model selection. The model exclusion procedure is considered within standard regression frameworks, including generalized linear and Cox regression models. For Cox regression models, a normalized partial log-likelihood is proposed to avoid divergence issues that arise when the standard partial log-likelihood is used for model selection. Selecting the number of clusters using AIC combined with the proposed model exclusion procedure achieves model selection consistency. The improved performance of the proposed exclusion procedure is confirmed through extensive simulation studies involving Gaussian linear, logistic, and Cox regression models combined with K-means clustering. An analysis of real-world data illustrates the application of Cox regression models.

EC254 Room B102 ECONOMETRIC INFERENCE
Chair: Jingru Mu
E1325: Quantile local projections: Identification, smooth estimation, and inference

Presenter: **Josef Ruzicka**, Nazarbayev University, Kazakhstan

Standard impulse response functions measure the average effect of a shock on a response variable. However, different parts of the distribution of the response variable may react to the shock differently. A popular method to capture this heterogeneity are quantile regression local projections. Structural identification is achieved by short-run restrictions or instruments and asymptotics are established. To overcome their excessive volatility, two novel smoothing estimators are introduced and information criteria for optimal smoothing are proposed. In an empirical application, it is shown that financial conditions affect the entire distribution of GDP growth and not just its lower part. Thus, financial conditions matter not only for recessions, but also in normal times and even for recoveries.

E1990: An improved inference for IV regressions*Presenter:* **Wenjie Wang**, Nanyang Technological University, Singapore*Co-authors:* Liyu Dou, Pengjin Min, Yichong Zhang

Empirical instrumental variables (IV) studies often report separate results based on low-dimensional instruments and many base instruments. A combination test is proposed that integrates these commonly reported statistics. The test linearly combines a cluster-robust Wald statistic based on low-dimensional IVs with leave-one-cluster-out Lagrangian multiplier (LM) and Anderson-Rubin (AR) statistics constructed from many IVs. Joint asymptotic normality and asymptotic optimality of the proposed test are established. The procedure yields costless efficiency improvements, automatically adapts to weak identification of many instruments, and is accompanied by a practical rule of thumb for assessing efficiency gains.

E2043: Bias-corrected method of moments for short-T dynamic panels with latent groups*Presenter:* **Muhammad Afnan Arif**, Asia-Europe Institute, Universiti Malaya, Kuala Lumpur, Malaysia, Malaysia

Dynamic panel data models with short time dimension face the incidental parameters bias, rendering within-groups estimators inconsistent. Existing bias-corrected method of moments estimators address this problem but impose homogeneous slope coefficients, an assumption violated when units belong to latent groups with heterogeneous dynamics. A bias-corrected method of moments estimator is proposed for short-T dynamic panels where slopes are common within groups but heterogeneous across groups, with group membership unobserved. Within-group bias-corrected moment conditions are derived by evaluating the bias function at the group-specific autoregressive parameter, and pooling across groups yields misspecified conditions. A classification rule assigns units to the group whose bias-corrected moment function is closest to zero in a quadratic norm, and group-specific parameters are estimated by minimizing the within-group objective. Classification consistency, post-classification consistency, and asymptotic normality are established under fixed time dimension and large cross-section. A consistent information criterion for selecting the number of groups is developed. Simulation evidence shows that the proposed estimator recovers the latent group structure and outperforms pooled bias-corrected estimators.

E1526: Regional innovation policy evaluation with endogenous regressors and fixed effects*Presenter:* **Alexandra Soberon**, Universidad de Cantabria, Spain*Co-authors:* Antonio Musolesi, Juan Manuel Rodriguez-Poo, Sylvie Charlot

This paper proposes a nonparametric estimator for the gradient of an unknown structural function in panel data models with endogenous regressors and two-way unobserved effects. Individual fixed effects and latent common factors are jointly eliminated through a cross-sectionally demeaned first-differencing transformation. Endogeneity of the regressors is addressed via a control function approach within a triangular simultaneous equations system, where reduced-form residuals enter the differenced regression additively. The gradient is recovered by locally weighted linear regression using product kernel weights that jointly localise pairs of consecutive observations around each evaluation point. Asymptotic properties of the estimator are established and finite-sample performance is assessed through Monte Carlo experiments under structural functions of varying nonlinearity and endogeneity strength. The proposed methodology is applied to the evaluation of regional innovation policies in Europe, allowing for a flexible assessment of the marginal effects of policy interventions on knowledge production while accounting for simultaneity, endogenous input allocation, and unobserved heterogeneity across regions.

EC290 Room B104 FINANCIAL TIME SERIES**Chair: Toshiaki Watanabe****E1272: A Bayesian GARCH EWMA control chart for monitoring Bitcoin***Presenter:* **Majika Jean-Claude Malela**, University of Pretoria, South Africa

Statistical process control (SPC) techniques have been applied across various fields and industries, including medicine, education, and finance. SPC is useful for monitoring the stability of a process by identifying abnormalities and investigating the sources of variability. However, traditional control charts struggle to effectively capture the dynamic nature of financial data, as such data is often characterised by volatility and autocorrelation. This situation violates the assumption of independent and identically distributed observations. Therefore, it is essential to identify an appropriate time-series model for statistical control, often explored through the Box-Jenkins approach. A new Bayesian exponentially weighted moving average (EWMA) control chart based on a generalized autoregressive conditional heteroscedasticity (GARCH) model is introduced. The newly developed control chart along with the three other GARCH EWMA-type control charts are used in a case study to monitor Bitcoin prices. The results show that the properties of the new charts are attractive and more effective in detecting abrupt shifts in the financial time-series data.

E2030: Exogenous-price adaptive routing for financial return forecasting*Presenter:* **Jiahao Zhou**, University of Macau, China

Short-horizon stock return forecasting remains difficult because prices respond to both market dynamics and external information shocks. An exogenous-price adaptive routing framework is proposed for daily stock return forecasting. The model combines adjusted OHLCV data with external signals, including FinBERT-based news sentiment and macro-policy uncertainty indicators. To reduce look-ahead bias, daily external variables are lagged by one trading day, while monthly variables are aligned with a one-month availability lag. Instead of directly adding these variables to the prediction features, the model uses external signals and price-state features to generate routing weights over multiple patch scales in a Pathformer-style structure. The hidden sequence is decoded by a GRU readout to predict next-day log returns. Across Pathformer-style benchmarks and routing ablations, models are selected using validation performance before test evaluation. The results show that the proposed framework maintains competitive log-return forecasting accuracy while improving risk-sensitive performance, with higher cumulative equity and lower maximum drawdown than Pathformer-style baselines. The findings suggest that exogenous-price adaptive routing is useful when forecasting accuracy and downside risk are considered together.

E2024: Exploiting ubiquitous local optima for forecasting emerging market credit spreads*Presenter:* **Gary Anderson**, CEMAR LLC, United States*Co-authors:* Alena Audzeyeva

A novel approach for determining support vector regression (SVR) kernel parameters in the presence of multiple local optima is proposed. In contrast to existing approaches focusing on identifying a single best tuning parameter setting, an impractical goal in many financial market applications, the framework employs a global optimization algorithm to produce a collection of competitive SVR kernel parameter candidates, and applies the model confidence set test to select the most accurate subset from the collection of promising candidates. The approach is used to predict credit spreads for four mature emerging market sovereign borrowers. Combining forecasts into simple forecast combinations and contrasting them against random forests, standard SVR, and conventional benchmark model forecasts, substantial gains in forecast accuracy are found when forecast combinations from the most accurate model sets are used. Furthermore, analysis of the forecasting results provides useful insights into credit risk pricing by international investors.

E1182: Heterogeneous predictability on mutual fund alphas: A sparse clustering GMM approach*Presenter:* **Liyuan Cui**, City University of Hong Kong, Hong Kong*Co-authors:* Guan hao Feng, Jiangshan Yang

Mutual fund managers' skills, measured by risk-adjusted alphas, are predictable using fund characteristics identified via machine learning. How-

ever, alphas predictive power varies across funds and periods, with most funds exhibiting negligible alphas. To model the heterogeneous predictability, Sparse Clustering GMM (SCGMM), a nonparametric approach to uncover latent fund group structures, is introduced. SCGMM clusters funds using estimated alphas and identifies group-specific parameters tied to market predictors. The method accounts for heterogeneity in cross-sectional grouping and time variation mechanisms across predictors, without requiring prior cluster information or time variation specifications. The proposed estimator is theoretically grounded, ensuring consistency in both the grouping and estimation of time-varying parameters. Empirical analysis of U.S. data shows that only a small group of mutual funds exhibit alphas predictably driven by market predictors.

EV304 Room Virtual R01 SURVIVAL AND BIOSTATISTICAL MODELLING
Chair: Jiayuan Shi
E2057: A copula-based defective Gompertz cure rate model for dependent censoring

Presenter: **Silvana Schneider**, Federal University of Rio Grande do Sul, Brazil

Traditional survival models are often inadequate for analyzing datasets that include a cure fraction, especially in the case of competing risks. One of the challenges in this analysis is the presence of dependent censoring, which occurs when the censoring times are not independent of lifetimes. Such censoring can occur when patients leave follow-up due to either clinical improvement or disease progression. In such cases, the assumption of independent censoring, commonly adopted in traditional models, may be violated, impacting survival estimates and statistical inference. To address this limitation, a copula-based regression model for dependent censoring is proposed, incorporating the defective Gompertz distribution to model the cure rate. The proposed model employs the Clayton copula function to capture possible dependence structures between lifetimes and censoring times. The proposed methodology is validated through simulation studies and an application to a dataset that has not yet been explored in the literature. Specifically, the model is applied to breast Cancer data obtained from the Hospital Cancer Registry of the state of Paraiba, Brazil, showcasing its ability to capture complex dependencies in survival outcomes and to estimate the cure fraction within this population.

E1551: Survival analysis with time-varying covariate: A R shiny tool for visualization

Presenter: **Yimei Li**, University of Pennsylvania, United States

Survival analysis in clinical research often needs to evaluate the association between time-varying covariates (TVCs) and time-to-event endpoints. Existing methods to analyze TVCs include naive approach (Kaplan-Meier [KM] method and Cox model), extended KM method, landmark analysis, and time-dependent (TD) Cox model. However, pros and cons of various methods are underappreciated and visualization of survival curves with TVCs is not always straightforward. An innovative R Shiny tool, TVCurve, is developed to provide a comprehensive suite for survival analysis with TVCs. The tool incorporates the above methods and generates corresponding survival curves. It also includes a parametric model assuming Weibull distributions. The tool builds in a Simulation panel to allow thorough evaluation of bias in the estimates under different methods. The use of the tool is demonstrated via applications to several clinical trials and extensive simulations are conducted. A summary table of the observed patterns in Simulation results is also provided, as a quick reference for understanding the direction of bias and implication for study conclusions in clinical applications. The findings underscore the importance of employing appropriate statistical methods for analyzing TVC in survival analysis. The easy-to-use R Shiny tool will promote adoption of the correct method and enhance the accuracy and reproducibility of survival analysis in clinical research.

E2069: Statistical modeling challenges in large-scale population database: United States renal data system modeling

Presenter: **Danh Nguyen**, University of California, Irvine, United States

The United States Renal Data System (USRDS), funded by the National Institute of Diabetes and Digestive and Kidney Diseases, is a National Data System that collects, analyzes, and disseminates information on chronic Kidney disease (CKD) and end-stage Kidney disease (ESKD) in the US (USRDS.org). It includes Data on nearly all patients on dialysis in the US. Several challenges in modeling CKD and ESKD patient outcomes are addressed: (1) profiling healthcare providers; (2) joint modeling including multivariate joint modeling of longitudinal, recurrent, and terminal outcomes and spatiotemporal modeling of patient outcomes, including longitudinal hospitalization and mortality. Several frequentist and Bayesian approaches to addressing large Data size and high-dimensional parameters associated with modeling spatial effects and parametrization of time-varying dynamic effects of risk factors on patient outcomes are presented. The discussion highlights opportunities and open challenges in modeling patient outcomes using the USRDS database.

Monday 10.08.2026

14:35 - 16:15

Parallel Session N – EcoSta2026

EO231 Room B101 ADVANCES IN ECONOMETRIC THEORY AND METHODS**Chair: Junfan Tao****E1631: Score test for order of finite normal mixtures***Presenter:* **Junfan Tao**, Kyoto University, Japan*Co-authors:* Jiaying Gu, Pujee Tuvaandorj, Stanislav Volgushev

Finite mixture models are widely used in empirical work, but determining the number of components remains a challenging problem. A score test for inference on the number of components is developed, drawing on Neymans $C(\alpha)$ framework. The test generalizes previous work, can be interpreted as a score test for neglected heterogeneity of multiple dimensions in finite Gaussian mixtures, and is closely related to the information matrix test for the same model. For a broad class of alternatives, the local expansion of the likelihood yields the same test statistic as in related likelihood-ratio-based inference on the order of finite Gaussian mixtures, thereby benefiting from the computational simplicity of a score test. A bootstrap-based implementation of the test is proposed and its consistency is established. The local asymptotic power envelope is characterized, which provides insight into its optimality. An empirical application illustrates the use of the proposed test.

E1853: Semiparametric estimation and inference for single-index models with high-dimensional covariates*Presenter:* **Ruixuan Liu**, Chinese University of Hong Kong, Hong Kong

A new estimation and inference method for high-dimensional single index models is developed. A simple two-stage estimation method based on the average derivative estimator (ADE) is proposed. This ADE comprises weighted score functions of covariates that can easily be estimated under a semiparametric Gaussian copula structure. In the first stage, standard nonparametric estimates for marginal features and a regularized estimator for the precision matrix of the Gaussian copula are used to obtain high-dimensional score estimands. In the second stage, LASSO type thresholding is conducted to obtain sparse estimates of the index vector in single index models. Both steps involve only convex minimization problems. The rate of convergence of the estimator is derived. Moreover, for inference, asymptotic normality of a de-biased estimator using the one-step Newton-Raphson update is proved.

E1863: Generic covariate adjustment for regression discontinuity designs*Presenter:* **Zhengfei Yu**, University of Osaka, Japan*Co-authors:* Jun Ma, Yuya Sasaki

An empirical entropy balancing estimator is introduced for regression discontinuity (RD) designs that incorporates covariate balance into estimation. The proposed method reweights the standard local polynomial RD estimator using entropy balancing weights, which are obtained by minimizing a divergence from uniform weights subject to covariate balance conditions. The estimator admits an empirical likelihood representation that efficiently exploits covariate balance conditions as correctly specified over-identifying moment restrictions, yielding an asymptotic variance no greater than that of the standard estimator that ignores covariates. Relative to linear regression adjustment methods, the proposed approach offers two key advantages: it guarantees estimation consistency for nonlinear causal estimands, such as quantile treatment effects in RD designs, and it improves estimation efficiency for causal parameters involving derivatives, such as those arising in regression kink designs, by flexibly incorporating balance conditions on both levels and derivatives of the covariates.

E1883: Poisson geometry of selective inference in proportional dimensions*Presenter:* **Hirofumi Ota**, University of Tokyo, Japan

Conditional selective inference for the randomized Lasso yields exact finite-sample pivots, but exact validity does not guarantee informative confidence intervals in proportional dimensions. A Poisson-geometric theory is developed that explains the loss of post-selection precision caused by inactive KKT constraints. Under Gaussian design, an exact finite-sample factorization holds: conditional on the active residual, inactive scores are independent Gaussian variables. Consequently, near-binding inactive faces converge, after rescaling, to a marked Poisson point process, and the nearest inactive faces imply that the inactive truncation width is typically of order $\sqrt{n}$. The inactive selection event contributes an exact scalar post-selection tilt, reducing the analysis to a one-dimensional residual-norm state. In a downstream-small regime this state converges to a deterministic limit; in a proportional-active regime it is confined to a deterministic compact set. A nonasymptotic lower bound for truncated-normal confidence intervals converts the Poisson width scaling into $\sqrt{n}$ scale worst-case interval inflation.

EO080 Room B102 RECENT ADVANCES IN ECONOMETRICS AND STATISTICS**Chair: Seok Young Hong****E1961: Finite-sample properties of model specification tests for multivariate dynamic regression models***Presenter:* **Koichiro Moriya**, Keio University, Japan*Co-authors:* Akihiko Noda

A new model specification test is proposed for multiple-equation systems with cross-equation error and dynamic regressor-error dependences. Conventional tests often rely on exogeneity conditions strong enough to ensure consistency of the OLS estimator. These exogeneity conditions are violated when regressors and errors are dynamically dependent, rendering conventional model specification tests invalid. To address these limitations, the relationship among alternative exogeneity conditions is clarified, the consistency of competing multiple-equation estimators is characterized, and a generalized Durbin estimator is proposed for multiple-equation systems with an intercept, cross-equation error and regressor-error dependences. The estimator remains consistent under the weakest exogeneity condition. Its asymptotic distribution is derived and Wald tests are constructed. Monte Carlo experiments confirm that the bootstrap-based Wald test substantially improves finite-sample size control. An application of the bootstrap-based Wald test to the Fama–French multifactor models leaves the null hypothesis unrejected in cases where competing FGLS-based tests reject it.

E1978: Unified inference for predictive mean and quantile regressions via empirical likelihood*Presenter:* **Seok Young Hong**, Nanyang Technological University, Singapore

An empirical likelihood framework is developed for testing return predictability in the conditional mean and conditional quantiles. A unified chi-square limit theory is established across a broad spectrum of predictor persistence, including stationary, mildly integrated, nearly integrated, unit-root, and mildly explosive cases. Two complementary approaches are provided to handle the unknown intercept: (i) a sample-splitting approach under relaxed regularity conditions and (ii) a new two-stage method that improves efficiency and accommodates quantile inference, where sample-splitting is infeasible. The finite-sample bias of the two-stage method is examined, and a bias-correction scheme and gradually saturated weights are proposed that improve performance under high persistence. Simulation evidence demonstrates that the tests exhibit competitive size and power across persistence classes, with notable gains in quantile predictability. An empirical application to the U.S. stock market shows modest evidence of mean predictability, whereas quantile-based inference reveals stronger and economically relevant predictability in the tails of the return distribution.

E2029: Recovering no-trade regions: Pontryagin-guided policy projection for transaction-cost control*Presenter:* **Jeonggyu Huh**, Sungkyunkwan University, Korea, South*Co-authors:* Hojin Ko

Transaction-cost portfolio choice is a canonical high-dimensional nonsmooth control problem: proportional costs induce no-trade regions and boundary interventions that neural policies can fail to preserve. Regularized PG-DPO, a Pontryagin-guided policy-projection method, learns this structure without solving a global value function or variational inequality. The method warm-starts a policy by direct Monte Carlo optimization, estimates fixed-policy marginal values by BPTT through continuation rollouts, and applies a closed-form dead-zone projection derived from a quadratically Regularized transaction-cost Hamiltonian. The projection exactly preserves the no-action wedge and converts marginal-value ratios into buy, hold, or sell decisions. Local guarantees are provided for fixed-policy marginal-value recovery, projection stability, and margin-based buy/hold/sell regime recovery. Experiments on multi-asset portfolio benchmarks show recovery of no-trade regions, correlation-dependent intervention geometry, mixed trading regimes, and dimension-robust local optimality diagnostics up to 50 assets.

E2081: Business cycle co-movement and stock market risk premiums over 145 years*Presenter:* **Ryuta Sakemoto**, Hokkaido University, Japan

The aim is to investigate whether individual and global output gaps are associated with expected stock market returns using long-run data covering more than 145 years. The results show that the output gap common factor predicts expected returns better than individual output gaps in both the pre- and post-World War II periods. Moreover, a positive output gap common factor shock causes negative stock market returns, while a negative stock market shock does not lead to an increase in the output gap common factor, suggesting that the output gap common factor acts as an exogenous shock. Finally, the output gap common factor predicts stock market excess returns in both periods within an out-of-sample context.

EO197 Room B103 EMPIRICAL STUDIES ON CONTEMPORARY ECONOMIC ISSUES**Chair: Shin Kinoshita****E1579: State-dependent inflation dynamics in a small open economy: A CVAR approach***Presenter:* **Takamitsu Kurita**, Kyoto Sangyo University, Japan*Co-authors:* Pal Boug, Havard Hungnes

The aim is to investigate how domestic prices in a small open economy respond to external cost shocks and assesses whether inflation dynamics have shifted during the recent surge in inflation. Using more than three decades of Norwegian data, a cointegrated vector autoregressive (CVAR) model grounded in a cost-based markup framework is estimated, linking consumer prices to unit labour and import costs. The empirical results point to a stable long-run price-cost relationship but also to evolving short-run dynamics, with pronounced pass-through from foreign prices during the recent high-inflation period and limited exchange rate pass-through. Structural CVAR-based simulation analysis delivers results consistent with the empirical findings and sheds further light on the underlying data-generating structure that may account for the observed inflation dynamics.

E1581: On proper indices*Presenter:* **Hideyuki Mizobuchi**, Doshisha University, Japan*Co-authors:* Valentin Zelenyuk

Recently, attention to the so-called circularity test for index numbers has re-emerged, and indices satisfying this property (along with certain other properties) have even been promoted as proper indices, implicitly framing alternative indices as improper and thereby undermining their perceived validity. The main purpose of the investigation is to clarify and discuss the problems associated with imposing circularity on index numbers, both theoretically and empirically. The theoretical investigation relies on both axiomatic and economic approaches to index number theory. An empirical comparison of various indices used to measure changes in outputs, inputs, and productivity is provided in order to illustrate the problems associated with proper indices, using industry-level production account data for the U.S. economy over the period 1987-2016.

E1684: Estimating the probability of Federal Reserve policy rate changes using large language models*Presenter:* **Takashi Matsuki**, Ryukoku University, Japan

While monetary policy decisions made by the Federal Reserve Board (FRB) at FOMC meetings exert a profound influence on both the U.S. and global economies, their occasional divergence from professional economic forecasts can intensify global uncertainty. The uncertainty surrounding FRB monetary policy changes is mitigated by leveraging Large Language Models (LLMs). First, key textual data reflecting the FRB's policy stance (including Statements, Minutes, Press Conference transcripts, the Beige Book, and speeches by the FRB Chair) are quantified using three LLMs (OpenAI-GPT 5.4, Google-Gemini 3.1, and Claude-Sonnet 4.6). From these quantifications, a policy stance indicator is constructed from Statements, Minutes, and Press Conference, a policy forecast indicator from speeches, and an economic assessment indicator from the Beige Book. Next, these indicators are employed to estimate a model of the probability of policy rate changes. Furthermore, by regressing the estimated probabilities of policy rate changes on key macroeconomic variables (such as economic growth rate, inflation rate, and unemployment rate), the macroeconomic factors that most significantly impact policy decisions are identified. The estimated probabilities are also compared with those derived from the FedWatch tool. Finally, the study examines which Language Model offers the most robust performance in forecasting monetary policy changes.

E1452: Analysis of energy preference from economic security perspective by a choice experiment*Presenter:* **Shin Kinoshita**, Ryukoku University, Japan

To protect the country and its people from the destabilization of the international situation, the idea of economic security in the field of energy has spread in Japan. Japan depends on imports of oil and natural gas as its main energy sources. The Japanese people face the risk of soaring electricity prices, which threatens their daily lives. Therefore, to break away from dependence on imported energy sources, it is expected that nuclear power plants, which have been shut down since the Great East Japan Earthquake in 2011, will be restarted or expanded, and that renewable energy sources such as solar power will be popularized. With these issues, the preferences of Japanese citizens regarding energy sources and import dependence from the perspective of energy-related security are measured. A choice experiment was used. It was found that respondents do not value increasing import dependence, while they hope for a reduction in CO2 emissions and stability of electricity supply. How personal attributes and attitudes affect energy preferences was also analyzed. The results showed that older people, people with politically liberal beliefs, and people who regularly engage in energy-saving behavior approve of high energy import dependence.

EO130 Room 104 HIGH DIMENSIONAL STATISTICAL LEARNING AND LOW RANK STRUCTURES**Chair: Long Yu****E1923: High-dimensional factor and clustering analysis via projected and truncated sample covariance matrix***Presenter:* **Long Yu**, Shanghai University of Finance and Economics, China

Spike identification in high-dimensional mixture factor models is addressed, where spiked eigenvalues arising from cluster-mean differences may be comparable in size to those from latent common factors. This overlap poses a significant challenge to existing methods. A novel projected and truncated sample covariance matrix is introduced, which, with suitable thresholding, consistently recovers the factor loading space regardless of the relative strength of the factor-driven eigenvalues. Importantly, this consistency holds even when the subspaces induced by the factor structure and the cluster means are partially confounded. Given the estimated factor space, a projected spectral clustering procedure is proposed that removes the factor component to reveal the underlying cluster structure. Furthermore, an iterative refinement algorithm is developed that builds upon these steps to enhance estimation accuracy. All proposed methods are supported by rigorous theoretical guarantees and validated through extensive simulations and real data analysis.

E1939: The sequential estimation of quantile factor models*Presenter:* **Manyao Shi**, Shanghai University of Finance and Economics, China*Co-authors:* Long Yu

Quantile factor models are widely used to capture heterogeneous factor structures across different distributional quantiles. However, the original iterative quantile regression (IQR) algorithm estimates all factors simultaneously, which does not guarantee nestedness across different factor numbers. A sequential estimation framework for quantile factor analysis (SQFA) is proposed. Algorithmic convergence of the proposed procedure is established, showing that the objective function is monotonically non-increasing. The estimated factors and loadings converge in probability at the same rate as the simultaneous estimation approach. A rank-minimization criterion is embedded into the procedure to consistently determine the number of factors at each quantile. Monte Carlo simulations and an empirical application demonstrate that SQFA performs well in finite samples, offering both computational efficiency and reliable statistical performance.

E1941: Robust imputation for high-dimensional panel with a latent factor structure*Presenter:* **Yiming Wang**, Shanghai University of Finance and Economics, China*Co-authors:* Long Yu

Factor models are widely used in economics and finance to capture common variations in high-dimensional panel data. However, real-world data often suffer from heavy-tailed errors and missing observations, which can severely compromise the performance of conventional Principal Component Analysis (PCA). To address these challenges, a robust estimation framework for high-dimensional factor models with random missing data is proposed. Motivated by the equivalence between PCA and constrained least squares in the complete-data factor model, an element-wise Huber loss function defined only on the observed entries (i.e., where $W_{it} = 1$) is minimized, which is solved by an iterative Huber regression algorithm. Theoretical convergence rates for the minimizer of the objective function are established and their asymptotic normality is derived under mild moment conditions. Furthermore, three criteria for robustly determining the number of factors are proposed. Extensive simulations and a real macroeconomic dataset (FRED-QD) demonstrate that the proposed method significantly outperforms existing approaches in imputation and prediction under various random missing mechanisms and heavy-tailed settings.

E1949: A sparsity-exploiting inexact interior-point method for quadratically regularized optimal transport*Presenter:* **Shunhua Yang**, Shanghai University of Finance and Economics, China

Optimal transport is a fundamental tool in modern statistical learning for comparing probability distributions and capturing geometric structure in high-dimensional data. This study examines large-scale quadratically regularized optimal transport from the perspective of structured second-order optimization. Quadratic regularization can induce sparse transport plans, but the associated dual Hessian may be unstable near the boundary, which makes classical Newton-type methods difficult to apply directly. A primal-dual interior-point formulation is instead employed, deriving a positive definite Schur complement system along the central path. The main observation is that although interior-point iterates are dense, their limit is sparse. Under strict complementarity, the zero pattern of the optimal plan can be locally identified. This motivates a sparsity-exploiting approximation of the Schur complement. The sparsified matrix preserves positive definiteness and has controlled spectral properties, which ensure convergence of the inner linear iteration. By imposing a suitable accuracy condition on the inner solve, the outer iterates remain in a central neighborhood. The resulting method requires no additional variables and does not rely on a strictly feasible initial point. It provides a structured and theoretically justified approach for large-scale optimal transport problems in high-dimensional statistical learning.

EO061 Room 105 SPATIO-TEMPORAL MODELING AND APPLICATIONS**Chair: Daisuke Murakami****E1669: Synthetic control method with block kriging for point-referenced data***Presenter:* **Naoki Hiramoto**, The Graduate University for Advanced Studies (SOKENDAI), Japan*Co-authors:* Daisuke Murakami

A change-of-support-aware synthetic control method (SCM) for settings where outcomes are observed at spatially irregular, time-varying point locations is developed. Standard SCM requires a fixed spatial panel, which is unavailable when data such as housing transactions are collected at locations varying across periods. Moreover, coarse spatial aggregation can mask heterogeneous treatment effects that differ in sign across subregions. The approach combines block kriging (BK) with SCM. BK aggregates point-level outcomes into areal means for user-specified spatial units, producing a panel compatible with SCM estimation. By explicitly modeling the latent spatial process, BK corrects for imbalanced sampling and provides kriging standard errors for uncertainty quantification. Areal averages satisfy exact linear identities across nested regions, enabling algebraic reconstruction of subregion effects from a limited number of SCM fits. Monte Carlo experiments confirm that the method recovers spatially heterogeneous and sign-reversing effects hidden under coarse aggregation. An application to the Toei Oedo Line opening in Tokyo reveals positive effects near western inner-city stations and negative effects in eastern and waterfront areas, consistent with spatial reallocation of residential demand rather than a uniform metropolitan-wide price increase.

E1638: Structural estimation of spatial equilibrium models: A nonlinear extension of spatial autoregressive models*Presenter:* **Shunsuke Segi**, Kobe University, Japan

A Full Information Maximum Likelihood (FIML) approach is proposed for the structural estimation of logit-type spatial equilibrium models with endogenous accessibility. Recent quantitative spatial models typically rely on natural experiments or instrumental variables to address endogeneity, but such identification strategies are often constrained by data availability. The logit-based spatial equilibrium structure can be interpreted as a nonlinear generalization of the spatial autoregressive (SAR) model, subject to a sum-to-one constraint on population shares. Analogous to the ML estimator for SAR models, the proposed estimator exploits the Jacobian term of the Likelihood function to correct for endogeneity bias without requiring external instruments. Using Monte Carlo simulations, FIML is compared with ordinary least squares and aggregate logit estimators, the latter being mathematically equivalent to the Poisson pseudo-Maximum Likelihood. The results show that FIML yields unbiased parameter estimates and recovers parameter values that correspond to dynamically stable equilibria under logit dynamics. These findings suggest that nonlinear extensions of SAR estimation offer an instrument-free route to structural identification in spatio-temporal equilibrium models.

E1461: Short-term forecast of repeating earthquakes using a spatiotemporal nonstationary renewal process model*Presenter:* **Shunichi Nomura**, Waseda University, Japan*Co-authors:* Naoki Uchida, Yoshihiko Ogata

Small repeating earthquakes observed along plate boundaries occur more frequently than large repeating earthquakes and have much shorter recurrence intervals. Hence, short-term forecasts for small repeating earthquakes yield relatively high occurrence probabilities, allowing for validation of forecasting methods by comparing predictions with observations over short time spans. On the other hand, forecasting small repeating earthquakes is challenging because their recurrence intervals can vary due to external influences such as afterslip following nearby large earthquakes and slow slip events. Under such conditions, stationary models assuming constant recurrence intervals can easily fail. To address this issue, a forecasting method for small repeating earthquakes based on a nonstationary renewal process model is proposed. In the proposed model, spatiotemporal variations in interplate slip rates in the surrounding region are estimated from the activity of small repeating earthquake sequences distributed along the plate boundary. These spatiotemporal variations are represented using tensor-product spline functions and estimated by maximizing a penalized likelihood with smoothing constraints. Moreover, by extrapolating the slip rates beyond the observation period, the occurrence probabilities of small repeating earthquakes over a short-term future horizon can be evaluated. The proposed model is applied to a catalog of small repeating earthquakes in northeastern Japan.

E1533: Coarse-to-fine spatial GLMM for scalable prediction and multiscale analysis*Presenter:* **Daisuke Murakami**, The Institute of Statistical Mathematics, Japan*Co-authors:* Alexis Comber, Takahiro Yoshida, Narumasa Tsutsumida, Chris Brunsdon, Tomoki Nakaya

Although a recent study suggested that coarse-to-fine learning provides a fast and flexible framework for large-scale spatial process modeling, the method was originally developed for Gaussian responses, limiting its applicability. To address this limitation, the coarse-to-fine spatial modeling framework was extended to accommodate spatial generalized linear mixed models (GLMMs), with a particular focus on count and binary data. The resulting spatial GLMMs efficiently addresses the degeneracy problem often encountered in conventional spatial GLMMs. The performance of the proposed spatial GLMMs was evaluated in terms of spatial prediction and multiscale feature extraction via Monte Carlo experiments and an empirical application. The developed method is implemented in an R package spCF.

EO063 Room 106 MODERN THEORETICAL STATISTICS FOR COMPLEX DATA**Chair: Teppei Ogihara****E1212: Regression discontinuity designs for random objects***Presenter:* **Daisuke Kurisu**, The University of Tokyo, Japan*Co-authors:* Yidong Zhou, Taisuke Otsu, Hans-Georg Mueller

A generalization of regression discontinuity designs is proposed to handle random objects residing in geodesic metric spaces. A key challenge in this setting is the absence of algebraic operations, which makes it difficult to define treatment effects using simple differences. To address this, the causal effect at the cutoff is defined as a geodesic between the local Frechet means of untreated and treated outcomes. Estimation is carried out using local Frechet regression. The proposed geodesic regression discontinuity design method is supported by theory, including convergence rate guarantees, and is demonstrated in applications where causal inference is of interest in complex outcome spaces.

E1369: On lead-lag estimation of non-synchronously observed point processes*Presenter:* **Takaaki Shiotani**, Graduate School of Mathematical Sciences, The University of Tokyo, Japan*Co-authors:* Takaki Hayashi, Yuta Koike

A new theoretical framework for analyzing lead-lag relationships between high-frequency order arrivals is introduced. The problem is formulated in terms of the cross-pair correlation function of point processes, with the estimand defined as its peak location, providing a statistical interpretation of the seminal method. A more natural estimator based on kernel smoothing with an adaptive bandwidth selection procedure is proposed. The estimator achieves a near-optimal convergence rate for the peak estimation problem. The proposed method is validated through simulations and real-data analysis, demonstrating its empirical advantages.

E1462: Asymptotically uniformly most powerful tests for diffusion processes with nonsynchronous observations*Presenter:* **Teppei Ogihara**, University of Tokyo, Japan

A quasi-likelihood ratio testing procedure is introduced for diffusion processes observed under nonsynchronous sampling schemes. High-frequency data, particularly in financial econometrics, are often recorded at irregular time points, challenging conventional synchronous methods for parameter estimation and hypothesis testing. To address these challenges, a quasi-likelihood framework is developed that accommodates irregular sampling while integrating adaptive estimation techniques for both drift and diffusion coefficients, thereby enhancing optimization stability and reducing computational burden. The asymptotic properties of the proposed test statistic are rigorously derived, showing that it converges to a chi-squared distribution under the null hypothesis and exhibits consistency under alternatives. Moreover, it is established that the resulting tests are asymptotically uniformly most powerful. Extensive numerical experiments corroborate the theoretical findings and demonstrate that the method outperforms existing nonparametric approaches.

E1666: Hidden Markov torus graph modeling for dynamic neural phase interactions*Presenter:* **Takeru Matsuda**, University of Tokyo & RIKEN Center for Brain Science, Japan

Oscillatory phase dynamics play a central role in brain information processing. Neural phases are represented as multivariate circular data on a high-dimensional torus. A method for extracting dynamic neural phase interactions is developed based on a hidden Markov torus graph model. This framework incorporates temporal dynamics into the torus graph, a graphical model for multivariate circular data. To address the computational intractability of the normalization constant, score matching and pseudo-likelihood methods are employed for parameter estimation and posterior computation. The proposed method is applied to unsupervised clustering of spatiotemporal patterns in calcium imaging data from the mouse cortex.

EO045 Room 107 RECENT ADVANCES IN CAUSAL INFERENCE AND SURVIVAL ANALYSIS**Chair: Liangyuan Hu****E1435: A fully Bayesian approach to structural nested failure time models for longitudinal causal inference***Presenter:* **Liangyuan Hu**, Rutgers University, United States

Structural nested failure time models (SNFTMs) offer a robust alternative to marginal structural models for causal inference with time-varying treatments, notably avoiding the strict positivity assumption. However, their widespread adoption is hindered by the computational intractability of frequentist g-estimation, where rank-based objective functions create a non-differentiable staircase topology that precludes standard gradient-based optimization. A Bayesian framework is proposed that fundamentally resolves this instability. By specifying a parametric baseline counterfactual distribution, the discontinuous objective is transformed into a smooth posterior surface, enabling efficient estimation via standard MCMC algorithms without the need for artificial smoothing parameters. The posterior consistency and asymptotic normality of the resulting estimators are theoretically established. Simulation studies demonstrate that this Bayesian approach yields superior stability and coverage compared to frequentist g-estimation, particularly in small-sample and high-censoring regimes. Finally, the method is applied to the CARRA Registry to evaluate the

impact of adalimumab withdrawal strategies on the timing of disease flare in juvenile idiopathic arthritis. By avoiding the data wastage associated with frequentist artificial recensoring, this framework efficiently leverages the full follow-up period to recover precise estimates of the protective effects of maintenance therapies.

E1427: **Principal stratification with U-statistics under principal ignorability**

Presenter: **Fan Li**, Yale University, United States

Principal stratification is a popular framework for causal inference in the presence of an intermediate outcome. While the principal average treatment effects are the standard target of inference, they may be insufficient when interest lies in the relative ordering of potential outcomes within a principal stratum. The principal generalized causal effect estimands are introduced to accommodate nonlinear contrast functions, providing robust, probability-scale summaries suitable for ordinal outcomes and win-loss comparisons with composite endpoints. Under principal ignorability, the theoretical results are expanded to a broader class of causal estimands in the presence of a binary intermediate variable. Nonparametric identification results are developed and efficient influence functions are derived for the generalized causal estimands in principal stratification analyses. These efficient influence functions motivate multiply robust estimators and lay the ground for obtaining efficient debiased machine learning estimators via cross-fitting based on U-statistics. The proposed methods are illustrated through simulations and the analysis of a data example.

E1430: **Weighted double score matching for causal inference in survey observational data**

Presenter: **Guangyu Tong**, Yale University, United States

Double score matching is an emerging method for causal inference that combines propensity and prognostic scores to achieve dimension reduction and model robustness. However, when the observational data are drawn from a complex survey, the existing literature does not provide theoretical guidance on the proper use of survey weights in matching for population-level causal inference. Weighted double score matching (WDSM) is developed that provides a unified framework for incorporating survey weights throughout the score estimation, matching, and outcome analysis under both retrospective (informative) and prospective (non-informative) sampling designs. Asymptotic normality is established under both known and estimated scores and a linearization-based wild bootstrap procedure is developed for variance estimation that preserves the survey-weighted matching structure. Through an extensive series of simulation studies, the performance of the estimators is examined and the results are compared to those of alternative methods. Two case studies are carried out to illustrate the application of the proposed methods with complex survey data. A discussion of the findings is provided along with practical guidelines for weighted double score matching analysis of observational data from complex surveys.

E1552: **Debiasing differentially private time-to-event data**

Presenter: **Yi Xiong**, University at Buffalo, United States

Sharing time-to-event data is essential for enabling collaborative research, designing effective interventions, and advancing patient care. However, sharing exact survival curves poses significant privacy risks. To protect individual privacy and mitigate the risk of membership inference attacks, various privacy-preserving solutions have been proposed. The differential privacy (DP) framework, in particular, offers strong and rigorous protection for data sharing. However, the noise injection required by DP can distort the probability density of the data, resulting in low utility and invalid statistical inference. Methods are proposed to mitigate these biases in differentially private, right-censored time-to-event data using a deconvolution framework via kernel density estimation. A bias-corrected nonparametric estimator for the marginal distribution of event times is provided and a corrected score approach for regression analysis is developed to ensure valid inference under privacy-preserving noise.

EO225 Room 201 RECENT ADVANCES IN DATA SCIENCE AND APPLICATIONS

Chair: Wai Leong Ng

E1755: **Evaluation of parameter efficient fine-tuning and retrieval-augmented generation for early cyberbullying detection**

Presenter: **Chun Fai Carlin Chu**, The Hang Seng University of Hong Kong, Hong Kong

Co-authors: Calvin Chun Ho Tong, Chun Hung Chiu

In the domain of cyberbullying detection, many prior studies primarily concentrate on classifying whether a piece of content is harmful or not, without addressing victim and perpetrator identities explicitly. The classification result is insufficient for developing early detection models which require not only recognizing harmful content but also understanding the roles of participants. Recent advances in generative artificial intelligence have opened up new opportunities for sophisticated content analysis. In particular, large language models with extended context windows demonstrate the capacity to comprehend complex dialogues involving multiple participants, implicit role dynamics, and evolving conversational slangs, enabling the extraction of fine-grained contextualized properties. Effective ways to model content type, victim and perpetrator identities for early cyberbullying detection are discussed. The implementation strategies of Parameter Efficient Fine-Tuning (PEFT) and Retrieval-Augmented Generation (RAG) are addressed, and their prediction performance is compared.

E1417: **A constrained least-squares ghost sample point method for unstructured high-dimensional data**

Presenter: **Kwun Lun Chu**, The Hang Seng University of Hong Kong, Hong Kong

The constrained least-squares ghost sample points (CLS-GSP) method is introduced, a novel modelling technique for uncovering underlying functions from unstructured, high-dimensional data like point clouds. The method begins with a local approach, fitting the data using a linear combination of radial basis functions. The centres of these basis functions, termed "ghost points," are strategically placed and are independent of the original data points. Furthermore, instead of using a standard least-squares fitting process, a regularization constraint is introduced that forces the local model to precisely interpolate the value at the centre of the neighbourhood. Numerical results demonstrate that this method is effective for solving complex problems, particularly those related to partial differential equations (PDEs).

E1330: **A unified two-step estimation approach for semiparametric models under two-phase sampling**

Presenter: **Hoi Min Ng**, The Hang Seng University of Hong Kong, Hong Kong

Co-authors: Qingning Zhou, Kin Yau Wong

The two-phase sampling design provides a cost-effective strategy for studying associations between disease outcomes and relevant risk factors. In the first phase, inexpensive variables are collected on a large cohort, and in the second phase, a subsample is selected for more detailed and costly measurements. This design retains a broad population view while requiring only a representative subset for expensive covariate acquisition. Existing update estimation methods construct a consistent estimator using complete cases and then refine it using information from a working model built on auxiliary variables observed in the full cohort. Although this approach can substantially improve efficiency, most existing implementations focus on updating Euclidean parameters. A more general setting is considered in which the working model involves functional parameters. An update estimation framework is developed that accommodates both Euclidean and functional parameters, thereby improving efficiency across a broader class of targets. The proposed method supports estimation and inference for Euclidean parameters, functional parameters, and their combinations. This is an important advantage because many quantities of interest in biomedical studies are inherently functional. The proposed methodology is illustrated using semiparametric models, including the transformation model and partially linear model.

E1364: Consumption and portfolio choice under information frictions and trading costs*Presenter:* **Chi Chung Siu**, The Hang Seng University of Hong Kong, Hong Kong*Co-authors:* Siu Kai Choy, Yawen Zheng

Optimal consumption and portfolio decisions are analyzed incorporating both informational and trading frictions. A utility maximizing agent divides resources between consumption and investment to hedge uninsurable labor risk while contending with transaction costs and limited information processing capacity. It is shown that trading frictions reduce consumption and investment and lower welfare, whereas informational frictions increase precautionary saving. Greater information processing ability amplifies wealth effects, raising target consumption and investment. When information processing is endogenized, higher information costs weaken processing capacity, leading to lower consumption and more conservative portfolios. Notably, the effects of informational frictions persist even after conditional variance stabilizes, underscoring enduring complexities in consumption and investment choices.

E1346: Asymptotic inference for change-points in time series*Presenter:* **Xinyi Tang**, The Hang Seng University of Hong Kong, Hong Kong

The asymptotic distribution of a change-point estimator for piecewise stationary time series across different magnitudes of break sizes is investigated. Specifically, break sizes of order $O(1/n^\alpha)$ for $0 < \alpha < 1/2$, $\alpha = 1/2$, and $\alpha > 1/2$, corresponding to large, moderate, and small break sizes, respectively, where n denotes the sample size, are examined. The results reveal that the asymptotic distributions in these three regimes differ but are all linked to the maximizer of certain functions of a two-sided drifted Brownian motion. To address the practical challenge of unknown break sizes, an asymptotically pivotal statistic that is robust across the whole range of break size regimes on $\alpha \in [0, \infty)$ is introduced. This statistic provides a unified approach for constructing confidence intervals for the change-point without requiring prior knowledge of the break size. Simulation studies show that the asymptotic inference performs well under different break sizes, while the pivotal statistic demonstrates better performance in most scenarios. Applications to financial time series further highlight the practical relevance of the proposed inference methods.

EO177 Room 202 MATHEMATICAL FOUNDATIONS OF DATA SCIENCE AND AI**Chair: Johannes Lederer****E1239: Empirical error estimates for graph sparsification***Presenter:* **Miles Lopes**, UC Davis, United States*Co-authors:* Siyao Wang

Graph sparsification is a well-established technique for accelerating graph-based learning algorithms, which uses edge sampling to approximate dense graphs with sparse ones. Because the sparsification error is random and unknown, users must contend with uncertainty about the reliability of downstream computations. Although it is possible for users to obtain conceptual guidance from theoretical error bounds in the literature, such results are typically impractical at a numerical level. Taking an alternative approach, these issues are addressed from a data-driven perspective by computing empirical error estimates. The proposed error estimates are highly versatile, and this is demonstrated in four use cases: Laplacian matrix approximation, graph cut queries, graph-structured regression, and spectral clustering. Moreover, two theoretical guarantees for the error estimates are provided, and explanation is given for why the cost of computing them is manageable in comparison to the overall cost of a typical graph sparsification workflow.

E1277: Revisiting privacy amplification by subsampling in selective release DPSGD*Presenter:* **Fang Xie**, Beijing Normal-Hong Kong Baptist University, China

Machine learning's reliance on sensitive data necessitates privacy-preserving techniques like Differentially Private Stochastic Gradient Descent (DPSGD). However, DPSGD suffers from substantial utility degradation and slow convergence due to Gradient clipping and noise injection. Prior works have attempted to improve DPSGD from various perspectives; notably, the Differentially Private Selective Update and Release (DPSUR) algorithm has achieved remarkable model utility. However, the privacy accounting in DPSUR overlooks the variation in sampling probability introduced by the Selective Release mechanism, which compromises the rigor of its privacy guarantees. To address these limitations, the privacy analysis of the Selective Release mechanism is re-evaluated and a novel algorithm is proposed: Differentially Private Selective Release based on Clipped Gradients (DPSR-CG). Through a rigorous, newly derived privacy analysis and extensive experiments on multiple datasets (MNIST, CIFAR-10, IMDB, and FMNIST), the DPSR-CG mechanism is demonstrated to maintain strict privacy guarantees while achieving exceptional model performance.

E1409: Optimal score function estimation via empirical risk minimization*Presenter:* **Thanh Mai Pham Ngoc**, Universite Sorbonne Paris Nord, France*Co-authors:* Thomas Bonis, Viet Chi Tran

Score-based generative models (SGMs) rely on estimating score functions via empirical risk minimization, yet their statistical properties under practical implementations remain poorly understood. Score estimation from i.i.d. data is studied and it is shown that minimax optimal rates over Sobolev classes can be achieved using empirical risk minimization with suitable smoothness constraints and derivative penalization. The diffusion framework of SGMs is then analyzed, where score functions must be estimated across noise levels. Despite singular behavior near zero noise (especially for data supported on low-dimensional manifolds) it is proved that properly constrained estimators achieve optimal rates for density estimation in Wasserstein distance. The results rely on the smoothing properties of the Ornstein-Uhlenbeck semigroup and provide theoretical support for the good generalization performance of neural network-based SGMs through implicit regularization.

E2053: Overview of methods for characterizing brain functional connectivity*Presenter:* **Hernando Ombao**, KAUST, Saudi Arabia

Modeling dependence between nodes in a brain network is central to understanding underlying neural mechanisms. A broad range of statistical methods for characterizing dependence in a brain network is presented. An overview of some of the most prominent frequency domain measures, such as coherence, partial coherence, and dual-frequency coherence is provided. This is followed by recent developments. First, wavelet partial canonical coherence (WPCanCoh) derived under multivariate locally stationary wavelet processes is discussed. Second, current work on information-theoretic measures of connectivity including the spectral transfer entropy (STE) and the non-linear vector coherence (NVC), which quantify the magnitude and direction of information flow from a certain frequency-band oscillation of a channel to an oscillation of another channel, is briefly described. In the final part, the use of deep learning techniques to estimate high-dimensional nonparametric models that explain how activity in one channel can be explained by potential non-linear past activity from other channels in the network is explored. These methods are illustrated through numerical experiments and provide interesting and novel findings on the analysis of EEG recordings from different cognitive tasks.

EO094 Room 203 RECENT DEVELOPMENTS IN RELIABILITY ANALYSIS**Chair: Man Ho Ling****E1237: Machine learning-based mixture model****Presenter:** Suvra Pal, University of Texas at Arlington, United States

A two-component mixture cure model (MCM) is considered. The first component of the MCM that describes the cure rate, or the incidence, is modeled using a machine learning (ML) based approach. The second component of the MCM that describes the survival distribution of the uncured, or the latency, is modeled using the Cox's proportional hazards structure. For the estimation of model parameters, an expectation maximization algorithm is developed in conjunction with the Platt scaling method which is used to convert the ML outputs to posterior probabilities of cure. When the classification boundary, with respect to classifying the cured and uncured units, is non-linear it is shown that the ML-based modeling results in more accurate and precise estimates of the cured probability when compared to the logistic regression-based modeling. This further results in improved predictive accuracy for cure. Interestingly, it is found that improving the estimation results related to the incidence also improves the latency estimation results. Finally, an application of the ML-based MCM is illustrated using a data on the lifetime of Kevlar 49 wrapped pressure vessels subject to different stress levels.

E1802: Optimal design of Type-I censoring experiments via Shannon information gain**Presenter:** Hon Yiu So, Oakland University, United States**Co-authors:** Sukanya Das

Type-I censoring is widely used in life-testing and reliability studies to control experimental duration and cost, but the choice of censoring time and sample size critically affects the information obtained from the data. A Bayesian framework for optimal design of Type-I censoring experiments based on Shannon information gain, defined as the expected Kullback-Leibler divergence from prior to posterior, is presented. A computational approach that combines augmented probability simulation with a Metropolis-Hastings algorithm is developed to explore the design space of censoring time and sample size. A monotone spline smoothing technique is further employed to estimate a stable and interpretable information surface. Theoretical results establish that the expected information gain is non-decreasing in both censoring time and sample size, with diminishing marginal returns, providing practical guidance for design selection. Simulation studies and real data analysis demonstrate how the proposed method identifies efficient censoring schemes under budget constraints. While motivated by reliability applications, the framework applies broadly to censored lifetime experiments, including settings where only limited or indirect failure information is observed.

E1485: Optimal progressively censored reliability sampling plans for log-location-scale distributions**Presenter:** Chien-Tai Lin, Tamkang University, Taiwan**Co-authors:** Shao-Lun Jian, Narayanaswamy Balakrishnan

A variable neighborhood search algorithm-based approach is introduced to determine the minimum sample sizes required for progressively censored reliability sampling plans within the flexible family of log-location-scale distributions, which includes Weibull and log-logistic distributions. The proposed method significantly reduces sample sizes compared to previously developed approaches, demonstrating its feasibility, especially for small sample sizes, in contrast to complete search methods. Optimal censoring plans are identified using A- and D-optimality criteria, a variance measure criterion, and a cost function-based criterion. The proposed approach consistently outperforms methods discussed in the literature for the case of Weibull distribution. Furthermore, an additional optimization problem with a cost constraint is addressed to illustrate the practicality and efficiency of the proposed method, and a sensitivity analysis is also carried out.

E1171: Optimal design for step-stress accelerated life testing with hybrid group censoring for non-destructive one-shot devices**Presenter:** Man Ho Ling, The Education University of Hong Kong, Hong Kong

Step-stress accelerated life tests (SSALTs) have attracted increasing attention from both industry and academia as effective methods for inducing rapid product failures and collecting extensive failure data for reliability analysis. A new framework for optimizing the design of SSALTs with hybrid group censoring is introduced, specifically for non-destructive one-shot devices, all while adhering to budget and time constraints. In this talk, an optimal design of SSALTs using a hybrid group censoring scheme will be discussed. An approach is presented to determine the optimal stopping threshold for an SSALT experiment, providing practitioners with a decision rule for terminating the experiment based on the number of failures observed. The findings demonstrate that this framework is a valuable tool for designing effective SSALTs, enabling the evaluation of mean lifetimes in a cost-effective manner for non-destructive one-shot devices.

EO151 Room 204 VARIABLE SELECTION AND RELATED TOPICS**Chair: Jia-Han Shih****E1647: Adaptive variable selection on high-dimensional non-parametric regression models****Presenter:** ChienTong Lin, National Sun Yat-sen University, Taiwan**Co-authors:** Suneel Babu Chatla, Ching-Kang Ing

Sparse linear models that incorporate non-parametric elements have gained significant prominence because of their flexibility in high-dimensional applications. While standard estimation frameworks often rely on finite basis expansions and assume an underlying sparse structure to achieve parsimony, the practical implementation of these models is hindered by the fact that both the degree of sparsity and the functional smoothness are generally unknown. Conventional approaches often circumvent this by pre-specifying the number of basis functions, focusing solely on variable selection. The convergence properties of the Orthogonal Greedy Algorithm (OGA) across diverse regimes of sparsity and smoothness are investigated. A unified framework is proposed that utilizes a high-dimensional information criterion (HDIC) to jointly determine the optimal number of OGA iterations and the basis dimension. The resulting procedure, OGA combined with HDIC (OGA+HDIC), is adaptive in the sense that it automatically achieves the optimal trade-off between variance and squared bias without prior knowledge of the sparsity or the smoothness levels. Furthermore, OGA+HDIC attains the optimal convergence rate in two fundamental high-dimensional contexts: the linear additive model and the varying coefficient model.

E1675: Regularized variable selection with missing data via stacked multiple imputation**Presenter:** Kyoji Furukawa, Kurume University, Japan**Co-authors:** Natsumi Kumano

Missing data are ubiquitous in statistical analyses and pose challenges for estimation and variable selection. While multiple imputation (MI) is widely used, its direct application to variable selection is problematic, as different models may be selected across imputed datasets, complicating interpretation and inference. Stacking approaches combine multiply imputed datasets into a single weighted dataset, enabling standard model selection without relying on Rubin's rules. A stacked MI approach that incorporates the analysis model through model-based weighting has been proposed, thereby avoiding complex procedures to ensure compatibility between imputation and analysis models. However, appropriate methods for variable selection within this framework remain unclear, and their performance has not been well examined. The proposed method integrates this approach with regularized variable selection techniques, such as LASSO and elastic net. It uses model-based weights to account for imputation uncertainty, enabling efficient penalized estimation within a single framework while remaining effective when the number of covariates is large. Through simulations under various missing data scenarios, the method improves model selection accuracy, reduces bias, and maintains predictive

performance compared with existing methods. An application to a clinical dataset further demonstrates its practical utility and computational efficiency.

E1215: **Importance inference of optimal test planning for degradation analysis**

Presenter: **YiShian Dong**, National Chengchi University, Taiwan

Co-authors: Chien-Yu Peng

Determination of the decision variables such as the inspection period, number of measurements, and sample size is crucial for planning an efficient degradation test. For widely used stochastic processes, the necessary and sufficient conditions for the explicit expression of optimal decision variables can be derived by minimizing the approximate variance of an estimator of interest under a limited budget. The importance of the decision variable is proposed to study the rate at which the objective function improves with the decision variable. The necessary and sufficient conditions for determining the importance of the optimal decision variables are theoretically investigated to elucidate the effect of the experimental costs and model parameters. Furthermore, the relative rankings of the importance of the optimal decision variables are illustrated through numerical examples.

E1496: **Variable selection based on multivariate regression association**

Presenter: **Jia-Han Shih**, National Sun Yat-sen University, Taiwan

Co-authors: Yi-Hau Chen

A multivariate regression association measure based on spatial signs is applied for variable selection. This measure quantifies the predictability of a multivariate outcome from a multivariate covariate and is therefore particularly suitable for variable selection. Since the measure is estimated nonparametrically, the corresponding variable selection procedure is also nonparametric. Thus, it can capture complicated dependence structure of response on covariate. Simulation studies are conducted to compare the performance of the proposed method with several existing approaches in the literature under various models. The results show that the proposed method performs competitively with existing methods. Finally, a real dataset is analyzed for illustration.

EO114 Room 209 RECENT ADVANCES IN METHODOLOGY FOR CATEGORICAL DATA AND APPLICATIONS Chair: Tomoyuki Nakagawa

E1751: **Quantile regression for panel brand choice data with individual random effects**

Presenter: **Masaaki Okabe**, Kansai University, Japan

Co-authors: Jun Tsuchida

In marketing research, understanding how price sensitivity and brand loyalty vary across the distribution of consumer preferences is crucial for designing targeted strategies. Standard multinomial choice models estimate only average covariate effects on relative utilities and may therefore obscure important heterogeneity. A Bayesian quantile regression framework for panel multinomial choice data is proposed that jointly estimates the conditional quantiles of latent relative utilities while accounting for individual-specific effects. Specifically, multinomial choice quantile regression is extended by introducing correlated multivariate normal random effects into the latent utility equations. The error term is assumed to follow a multivariate asymmetric Laplace distribution, whose scale-mixture representation enables efficient Gibbs sampling for all model parameters, including the random effects and their covariance matrix. By separating out persistent individual-level preferences, the proposed approach allows quantile-specific covariate effects to reveal how the impacts of price, promotion, and advertising differ between low-utility and high-utility consumer segments. An application to scanner panel data on brand choice demonstrates that the method uncovers distributional patterns in price sensitivity that remain hidden in mean-based models.

E1663: **Theoretical foundations and asymptotic confidence intervals for multiclass Matthews correlation coefficients**

Presenter: **Jun Tamura**, Yokohama City University, Japan

Co-authors: Kouji Yamamoto, Kenichi Hayashi

Classification problems arise in many fields and require appropriate performance measures. In binary classification, the Matthews correlation coefficient (MCC) has attracted considerable attention because it incorporates all cells of the confusion matrix and provides a more comprehensive summary than some commonly used alternatives. In many practical applications, however, classification problems involve more than two categories, and several extensions of MCC have been proposed for multiclass settings. Although these measures are increasingly used, their theoretical relationships have not yet been fully clarified, and methods for statistical inference remain limited. The theoretical structure of MCC-based measures for multiclass classification is clarified and asymptotic confidence intervals for these measures are derived. Paired designs, in which multiple classifiers are applied to the same set of subjects, are also considered, and asymptotic confidence intervals for differences between MCC-based measures are constructed. Simulation studies are conducted to evaluate the finite-sample performance of the proposed procedures and related methods.

E1667: **Power-divergence-based measures approximating latent correlation in contingency tables under bivariate normality**

Presenter: **Wataru Urasaki**, Tokyo University of Science, Japan

Association measures for two-way contingency tables play an important role in the analysis of ordinal categorical data, where one primary goal is to quantify the strength of association beyond testing independence. Although many divergence-based measures have been proposed, their values are not always easy to interpret in relation to familiar correlation coefficients. In many applications, observed categories may also be regarded as discretizations of underlying continuous variables, making latent correlation a natural target of interpretation. Building on a previously established closed-form approximation, the power-divergence from independence under bivariate normality is expressed as a function of the latent correlation coefficient. By inverting this relationship, a family of power-divergence-based association measures is obtained that reflects latent correlation while retaining the computational simplicity of divergence-based methods. The resulting framework includes classical measures, such as Linfoot's informational measure and Pearson's contingency coefficient, as special cases. Asymptotic distributions and confidence intervals are also derived through the delta method. The methodological background, theoretical properties, and practical advantages of the proposed measures for contingency table analysis are presented, together with a brief report on numerical experiments.

E1407: **Interpretation of marginal homogeneity based on decomposition and visualization in collapsed ordinal tables**

Presenter: **Satoru Shinoda**, Yokohama City University, Japan

Co-authors: Kouji Yamamoto, Sadao Tomizawa

Marginal homogeneity (MH) is a fundamental concept for analyzing paired ordinal data in square contingency tables. When ordinal categories are collapsed to simplify interpretation, multiple collapsing patterns arise, making it difficult to assess MH within a unified framework. A framework for interpreting MH based on marginal cumulative distributions is introduced. First, a new model, termed collapsed marginal mean equality (CoME) for collapsed tables is proposed. Together with extended marginal homogeneity (EMH), it is shown that the MH model holds if and only if both the CoME and EMH models hold. Second, to enable comparisons across tables, a Kullback Leibler information type measure of departure from MH that incorporates ordinal structure is introduced. It is further shown that this measure can be decomposed into components corresponding to CoME and EMH, allowing identification of whether deviations arise from proportional scaling or differences in distributional shape. Finally, to facilitate intuitive interpretation, a visualization approach that displays the CoME and EMH sub-measures for each collapsing pattern is presented. Real data

examples demonstrate the practical utility of the proposed framework.

EC264 Room B104 ASSET PRICING
Chair: Jackson Lautier
E1499: Pricing of long-term care: The case of elderly home

Presenter: **Chi Ho Tang**, Hong Kong Shue Yan University, Hong Kong

The existing literature on long-term care is complemented through the study of private elderly home market in Hong Kong. The contributions are three-folded. First, it studies the price dispersion of the ancillary long-term services, and explains how spatial difference or searching cost affects its price dispersion. Then, it applies hedonic pricing model to examine three levels of factors (i.e. package level, elderly home level and district level) that significantly explain the monthly rental of elderly home. Finally, it formulates the measure for healthcare premium. Directions for future research will be discussed.

E1995: Macro strikes back: Term structure of risk premia

Presenter: **Svetlana Bryzgalova**, London Business School, United Kingdom

Co-authors: Christian Julliard, Jiantao Huang

A novel priced Wold representation theorem is provided that sharply identifies shocks common to financial markets and the macroeconomy, their propagation, and the term structure of macro risk premia. The identification leverages the fact that, in equilibrium models, asset prices are jump variables and reveal priced risks in large cross-sections of returns (N tends to infinity). Macro factors' risk premia are found to be strongly time-varying and countercyclical, with sharply increasing unconditional term structures. This pattern is driven by the slow propagation of a priced common shock, distinct from TFP, that captures most of the persistence in macroeconomic variables. Macro risk premia are negligible in the short run, yet grow to match equity risk premia at the business cycle horizon. This finding is not a mechanical byproduct of persistence: while GDP, consumption, industrial production, and employment exhibit upward-sloping term structures, other similarly persistent factors, e.g., VIX or intermediary-based variables, display downward-sloping or flat ones.

E1198: Partial moment momentum and contrarian strategy in the Taiwan stock market

Presenter: **Po-Keng Cheng**, National Taipei University, Taiwan

Co-authors: Yu-Ting Chang, Ping-Chen Tsai

Volatility-adjusted momentum and contrarian strategies in the Taiwan stock market are examined using the partial moment momentum (PMM) framework. Using data from 1999-2024, traditional momentum strategies are shown to be weak and highly exposed to downside risk, while PMM significantly improves risk-adjusted performance and reduces drawdowns, especially in bear markets. Applying the same adjustment to contrarian strategies does not enhance returns, indicating that partial-moment scaling is effective for momentum but not reversals.

E1326: Valuation of carbon emission allowance-linked derivatives under regime switching

Presenter: **Yu-Min Lian**, Fu Jen Catholic University, Taiwan

The underlying spot European Union carbon emission allowance price dynamics and the forward interest rate processes are respectively driven by a Markov-modulated jump-diffusion model with stochastic volatility, jump intensity switching, and Markov-modulated jump size, and a Markov-modulated Heath-Jarrow-Morton model with the full term structure. The cost of carry model is applied to derive carbon emission allowance futures prices, and the regime-switching Esscher transform is employed to identify a risk-neutral martingale measure, which is used to derive integral expressions for prices of European-style European Union carbon emission allowance futures options. The analysis addresses regime uncertainty, jump risk, and volatility term structure impacts on carbon derivatives prices.

EC247 Room 101 TIME SERIES ANALYSIS
Chair: Ansgar Steland
E1290: The Whittle likelihood for mixed models in hydrology

Presenter: **Jakub Pypkowski**, Imperial College London, United Kingdom

Co-authors: Adam Sykulski, James Martin, Ben Marchant

Mixed models, which combine fixed and random effects, are commonly used for time series analysis in hydrology. In particular, they are used for inference, forecasting and filling in missing values in groundwater level time series. Estimating parameters of mixed models using maximum likelihood has computational complexity of order $O(n^3)$ in the general case. This is prohibitive for large datasets, leading to simplifying assumptions such as fixing certain free parameters in practical implementations. A frequency-domain method based on the Whittle likelihood is proposed to jointly estimate all parameters of mixed models. Using the fast Fourier transform, the method reduces the computational complexity to $O(n \log n)$. It is robust to non-Gaussian random errors and accommodates incomplete datasets and nonlinear means expressed through impulse response functions. In a simulation study, it is found that the method performs similarly to maximum likelihood in terms of estimation accuracy when random errors are Gaussian, and outperforms maximum likelihood when random errors are non-Gaussian. It is shown that the method also outperforms an alternative two-stage approach that estimates mean and random error covariance parameters separately. The method is applied to groundwater level time series from observation boreholes in Southern England, obtaining better-fitting models in less time than using maximum likelihood.

E1868: Reduced rank time variation for vector-valued models

Presenter: **Alessandro Celani**, Orebro University, Sweden

Co-authors: Luca Pedini, Alessandro Celani

A reduced-rank framework for time-varying parameter VAR models is proposed that tackles the curse of dimensionality in standard specifications. Conventional TVP VARs involve a number of latent states that grows quadratically with system size, making estimation infeasible in moderate dimensions. The approach preserves the matrix structure of time-varying coefficients and introduces a bilinear decomposition driven by row- and column-specific latent processes. This representation captures rich and heterogeneous dynamics across equations and variables while ensuring that the number of states grows only linearly. As a result, it delivers substantial computational gains without sacrificing flexibility. A Bayesian estimation strategy based on efficient precision sampling is developed and identification challenges arising from the multiplicative structure are addressed. Compared to existing factor-based approaches, the proposed model provides a more flexible allocation of time variation, allowing both row and column effects to evolve over time. Monte Carlo and empirical evidence show that the framework achieves accurate inference and strong forecasting performance in high-dimensional macroeconomic applications, making it well suited for settings where standard TVP-VARs are computationally prohibitive.

E1951: Robust prediction intervals for time series forecasting: A bootstrap and Bayesian approach

Presenter: **Betty X-Y Chu**, National Chung Hsing University, Taiwan

Co-authors: Cathy W-S Chen

A new bootstrap strategy, the standardized skewed Student-t block multiplier bootstrap (skew-tBMM), is proposed to capture heteroskedasticity and heavy-tailed behavior in time series forecasting with machine learning models. Prediction intervals are constructed by integrating bootstrap

resampling with machine learning methods, including support vector regression, random forest, XGBoost, and long short-term memory networks, while Bayesian structural time series (BSTS) and autoregressive integrated moving average models serve as probabilistic benchmarks. To enable fair comparisons across models and data-generating mechanisms, a scale-adjusted relative interval score is introduced for evaluating interval forecasts. Simulation studies under nonlinear dynamics, deterministic seasonality, transfer-function effects, and conditional heteroskedasticity show that skew-tBMM provides well-calibrated intervals with competitive sharpness. An empirical application to monthly suicide counts in Japan illustrates the practical applicability of the framework. Results indicate that machine learning models combined with skew-tBMM, particularly long short-term memory networks, produce reliable forecast distributions in complex and heavy-tailed settings, while BSTS performs well for probabilistic forecasting.

E1378: Mean independent component analysis for multivariate time series

Presenter: **Chung Eun Lee**, Baruch College, United States

Co-authors: Zeda Li

Mean independent component analysis for multivariate time series is introduced to reduce the parameter space. In particular, a contemporaneous linear transformation is sought that detects univariate mean independent components so that each component can be modeled separately. The mean independent component analysis is flexible in the sense that no parametric model or distributional assumptions are made. A unified framework is proposed to estimate the mean independent components from data with fixed dimension or diverging dimension. The mean independent components are estimated by martingale difference divergence so that mean dependence across components and across time is minimized. The approach is extended to group mean independent component analysis by imposing a group structure on the mean independent components. A method is further introduced to identify the group structure when it is unknown. The consistency of both proposed methods is established. Extensive simulations and a real data illustration for community mobility are provided to demonstrate the efficacy of the method.

EC246 Room 102 HYPOTHESIS TESTING

Chair: Chao Zheng

E1291: A covariate-adaptive test for replicability across multiple studies with false discovery rate control

Presenter: **Dennis Leung**, University of Melbourne, Australia

Co-authors: Ninh Tran

Partial conjunction (PC) p-values and side information provided by covariates can be used to detect signals that replicate across multiple studies investigating the same set of features, all while controlling the false discovery rate (FDR). However, when many features are present, the extent of multiplicity correction required for FDR control, along with the inherently limited power of PC p-values – especially when replication across all studies is demanded – often inhibits the number of discoveries made. To address this problem, a p-value-based covariate-adaptive methodology is developed that revolves around partitioning studies into smaller groups and borrowing information between them to filter out unpromising features. This filtering strategy: 1) reduces the multiplicity correction required for FDR control, and 2) allows independent hypothesis weights to be trained on data from filtered-out features to enhance the power of the PC p-values in the rejection rule. The methodology has finite-sample FDR control under minimal distributional assumptions, and its competitive performance is demonstrated through simulation studies and a real-world case study on gene expression and the immune system.

E1981: A doubly projected test for conditional independence with high-dimensional dependent data

Presenter: **Songyen Chen**, National Chengchi University, Taiwan

Co-authors: Tzee-Ming Huang

The problem of testing conditional independence is crucial in statistics, econometrics, and modern machine learning, yet it remains challenging in the presence of high-dimensional dependencies. A conditional independence testing framework is addressed under high-dimensional dependent data. A self-normalized type test statistic is proposed that integrates time series block smoothing, sample splitting, and flexible machine learning techniques satisfying suitable convergence rates. The key insight of the framework is that the double projection strategy maps the conditional independence problem into a family of unconditional covariate weighted orthogonality conditions, bypassing the additive noise exogeneity condition commonly required by conventional residual based conditional independence tests. In particular, the proposed test is capable of detecting both linear and nonlinear conditional dependence structures. Under mild conditions, and allowing both the dimension of the covariates and the sample size to diverge, the test statistic is shown to be asymptotically pivotal under the null, and asymptotic local power properties are established against alternatives. Simulation studies demonstrate satisfactory finite sample performance. An empirical application illustrates the usefulness of the proposed approach in testing for high-dimensional Granger non causality.

E2019: Permutation-invariant methods in sequential testing: The case of election auditing

Presenter: **Dania Freidheim**, The University of Melbourne, Australia

Co-authors: Damjan Vukcevic, Dennis Leung

Sequential testing for bounded data on $[0,1]$ is a fundamental problem in modern statistics. A canonical formulation of this problem involves testing the null hypothesis that the true population mean is at most 0.5. Modern approaches often rely on supermartingale statistics to maintain strict error control. However, many of these methods lack permutation invariance: their conclusions depend on the arbitrary order in which the data are observed, potentially introducing sampling order bias and reducing efficiency. A solution motivated by the betting intuition of investment theory is presented: F -weighted portfolio algorithms. These algorithms, which have a natural Bayesian interpretation, yield betting strategies that are permutation invariant. The framework is applied to risk limiting election audits, where verifying a reported outcome relies on sequentially sampling ballots. Across a variety of testing scenarios, carefully tuned F -weighted algorithms are shown to outperform existing sequential tests while remaining computationally tractable.

E1401: An adaptive weighted component test for phase I monitoring in high-dimensional means

Presenter: **Qianwei Lan**, University of Macau, China

Co-authors: Lianjie Shu, Xinyi Ren

In high-dimensional Phase I process monitoring, variables frequently exhibit distinct heterogeneity in both sparsity and shift magnitude, often compounded by unequal component variances. Existing methods designed under rigid assumptions, such as extreme sparsity or pure density, suffer severe loss of power when the actual shift pattern deviates from these patterns. To bridge this gap, a nonparametric adaptive weighted component test (AWCT) control chart is proposed. The approach first normalizes component-wise statistics to mitigate variance heterogeneity. Recognizing that sparsity is essentially a special case of magnitude heterogeneity where unaffected components have shift sizes of zero, the AWCT then employs data-driven adaptive weights to simultaneously accommodate unknown sparsity and varying shift magnitudes. Extensive simulation studies demonstrate that the AWCT consistently achieves higher detection power than competing methods across diverse shift patterns. A real-world example further validates the practical value of adaptive weighting in high-dimensional monitoring.

EC260 Room 103 MACHINE LEARNING**Chair: Zudi Lu****E1318: Controlling for omitted variable bias in deep neural networks****Presenter:** Manuel Pfeuffer, Humboldt-Universität zu Berlin, Germany**Co-authors:** Roshan Prakash Rane, Kerstin Ritter, Sonja Greven

Control variables are widely used in statistics and econometrics to account for omitted variable bias. However, they have seen almost no use in deep learning. This is surprising, given that deep learning models tend to learn unwanted spurious associations stemming from known sensitive or confounding covariates, such as gender or age, which can be considered a form of omitted variable bias. While many existing confound-control and fairness methods in deep learning try to restrict the correlation of such covariates with model predictions, it is shown that this does not correct for omitted variable bias. A control variable approach for deep learning models based on generalized additive modelling of the effects of model inputs and additional covariates is therefore proposed. As flexible additive models can suffer from concavity, a simple estimation procedure is introduced that refits the final layer of a pre-trained network to include covariate effects, using back-fitting with ridge penalisation. These effects can optionally be orthogonalised to include or exclude mediated effects. This yields unbiased, interpretable estimates and offers the flexibility to model the desired effects depending on the scientific or fairness objective. The approach is verified using simulated images and real world neuroimaging data with synthetic confounding.

E1538: Understanding bicycle route choice through explainable AI**Presenter:** Iryna Okhrin, Dresden University of Technology, Germany**Co-authors:** Katrin Lubashevsky, Stefan Huber, Sven Lissner

Bicycle route choice is analyzed with a focus on methodological comparison and explainable artificial intelligence (XAI). Using data from multiple German cities, detailed route-level attributes (e.g., infrastructure quality, traffic conditions, elevation) are combined with city-level characteristics (e.g., urban structure, cycling culture, environmental factors) to capture both local and contextual influences. A classical discrete-choice model (conditional logistic regression) is compared with machine learning methods (e.g., random forests, XGBoost, support vector machines, deep learning models). The emphasis is on the trade-off between predictive performance and interpretability. To address this, XAI techniques are applied to systematically evaluate feature importance and model behaviour across methods. Results indicate that machine learning models generally achieve higher predictive accuracy, while XAI enables meaningful interpretation despite their complexity. In contrast, conditional logistic regression provides transparent, theory-consistent insights but may be less predictive. Overall, the analysis demonstrates how combining statistical models, machine learning, and XAI yields a comprehensive understanding of cyclist preferences, supporting data-driven urban planning and the design of sustainable cycling infrastructure.

E1950: Machine learning estimation of Armington elasticities and their implications for trade policy simulation**Presenter:** Jong-Hwan Ko, Pukyong National University, Korea, South**Co-authors:** Il Do Ha

Computable General Equilibrium (CGE) models are widely used in trade policy analysis, but key behavioral parameters such as Armington elasticities are usually calibrated from a single benchmark Social Accounting Matrix rather than estimated from data. This approach produces fixed values without uncertainty and may distort policy results when parameters are mis-specified. A machine learning framework is proposed to estimate sector- and region-specific Armington elasticities and examine how using these estimates changes trade policy simulation outcomes. Using panel data from GTAP versions 6-12 for 1997-2023, together with bilateral trade data from UN Comtrade, XGBoost, Gaussian Process Regression, and Bayesian Ridge Regression are applied to estimate heterogeneous elasticities across 16 regions and 22 sectors. The results show substantial cross-sector and cross-region variation that is masked by conventional calibration practices. These estimated elasticities are incorporated into a 16-region, 22-sector global CGE model to simulate U.S. tariff increases against major trading partners in 2025-2026. Compared with standard calibration, the ML-based estimates produce meaningfully different policy outcomes, including larger welfare losses in import-competing sectors and weaker export gains in capital-intensive sectors. The findings suggest that data-driven elasticity estimation can improve the realism and policy relevance of CGE-based trade analysis.

E1350: Physical robot-object interaction recognition using wavelet scattering-based features for transformer models**Presenter:** Haeun Ryu, Chungnam National University, Korea, South**Co-authors:** Yunchae Jung, Jiwon Im, Minsu Park

Effective input representation design is essential for sensor-based action classification due to complex time-frequency structures and inherent stochastic noise in robotic signals. Conventional transformer models relying on raw signal patch-wise tokenization often exhibit vulnerability to high-frequency noise and fail to capture multi-scale temporal dependencies. To address these challenges, a robust classification framework is proposed that utilizes the wavelet scattering transform (WST) as the input representation for transformer encoders. WST provides a contraction mapping that is Lipschitz continuous to deformations, generating energy-preserving and shift-invariant features through multi-scale wavelet filtering and nonlinear operations. These mathematical properties facilitate the identification of distinct physical interaction patterns, including precision placement, kinetic release, and continuous surface contact. Evaluations using the DROID (dataset for robot instruction and demonstration) dataset demonstrate that WST-based representations achieve superior classification accuracy and enhanced noise robustness compared to traditional raw patch-based methods. The results indicate that integrating WST with transformer architectures offers a statistically sound feature engineering strategy, effectively supporting automated action classification and post-hoc process validation in autonomous robotic systems within industrial environments.

EC253 Room 205 INFERENCE FOR COMPLEX MODELS**Chair: Kento Egashira****E1847: Scoring-based model choice for conditional transformation models****Presenter:** Alexander Ritz, Clausthal University of Technology, Germany**Co-authors:** Benjamin Saeften

Conditional transformation models (CTMs) offer a flexible and interpretable way to obtain conditional density estimates, allowing for a more complete understanding of the response distribution given the explanatory variables than models focused exclusively on the conditional mean. At the same time, estimates of full conditional distributions are harder to assess in their ability to capture the true data generating process. In order to establish a rigorous approach to model choice for CTMs, it is possible to rely on scoring rules, which are not restricted to evaluating model performance via an implied point estimate. However, in scenarios where data is not abundant, this approach bears the risk of overestimating the model's out-of-sample performance if the same data is used to estimate as well as evaluate model fit. Therefore, it is necessary to correct for this bias in an analogous manner to established covariance penalties for prediction error estimates. A bias correction for proper scoring rules is derived and a scoring-based model choice criterion is constructed, facilitating the rigorous choice of parsimonious distributional regression models. Models chosen by this approach are further assessed in comparison to those selected via established model choice criteria like WAIC.

E1975: On prediction in linear mixed models under misspecified stochastic linear restrictions*Presenter:* **Tatjana von Rosen**, Stockholm University, Sweden

In many statistical settings, additional (or prior) information about the model parameters is available. Such information can be formulated either as linear exact restrictions, used when a specific linear hypothesis about the parameters is of interest, or as linear stochastic restrictions, used when prior information arises from previous investigations and is subject to uncertainty. A general mixed linear model subject to two competing sets of stochastic linear restrictions, M_0 and M_1 , is considered. The restrictions M_0 are assumed to be correct, whereas M_1 may be misspecified. Focusing on the best linear unbiased predictors (BLUPs) under mixed models with stochastic restrictions, necessary and sufficient conditions are derived under which the BLUPs obtained under the two competing models coincide. Further conditions are provided under which the BLUP obtained under the model with misspecified restrictions M_1 is the BLUP associated with the correct restrictions M_0 . The approach is based on solving an appropriate system of matrix equations. The same methodology can also be used to study equality of the best linear unbiased estimators of the fixed effects in this context.

E1914: Inference for constrained extremum estimators*Presenter:* **Jessie Li**, University of California, Santa Cruz, United States

Inference for constrained M-estimators and constrained GMM estimators with possibly nonsmooth or nonconvex objectives and possibly nonconvex constraint sets is studied. Test-inversion is used to construct a uniformly valid simultaneous confidence set which covers parameters either in the interior or on the boundary of the constraint set as well as parameters drifting towards the boundary at arbitrary rates. The method works for both $\sqrt{n}$ -consistent estimators and estimators with a slower-than- $\sqrt{n}$ rate of convergence to non-standard asymptotic distributions. Monte Carlo simulations illustrate the uniformly correct coverage of the method in a boundary constrained maximum likelihood model and a nonsmooth GMM model. An empirical application conducts uniformly valid inference in a nested logit model of cereal demand which imposes the constraint that the correlation parameter among choices within the same nest must lie between 0 and 1.

E1380: Support vector machine analysis under right censored data*Presenter:* **Jin-Jian Hsieh**, Department of Mathematics, National Chung Cheng University, Taiwan

Predicting patient survival time is crucial in medical treatment, providing valuable insights for doctors and patients while guiding future treatment planning. A novel approach to applying support vector machines to right-censored data, a common challenge in survival analysis, is evaluated. The goal is to enhance support vector machines' prediction accuracy for right-censored data. To achieve this, partial rank estimation and generalized additive models are integrated to impute censored values, followed by support vector regression for model training and survival time prediction. Through simulations, the method is compared with the widely used "survivalsvm" and the Buckley-James imputation approach. Finally, the approach is validated using three real-world datasets.

EC248 Room 208 BAYESIAN MODELLING**Chair: Yasuhiro Omori****E1470: Identification verification for structural VARs with sparse heterogeneous Markov switching heteroskedasticity***Presenter:* **Tomasz Wozniak**, University of Melbourne, Australia*Co-authors:* Fei Shang

A structural vector autoregressive model is proposed with a new and flexible specification of the volatility process which is called sparse heterogeneous Markov-switching heteroskedasticity. In this model, the conditional variance of each structural shock changes in time according to its own Markov process. Additionally, it features a sparse representation of Markov processes, in which the number of regimes is set to exceed that of the data-generating process, with some regimes allowed to have zero occurrences throughout the sample. These developments are complemented with a definition of a new distribution for normalised conditional variances that facilitates Gibbs sampling and identification verification. In effect, the model: (i) normalises the system and estimates the structural parameters more precisely than popular alternatives; (ii) can be used to verify homoskedasticity reliably and, thus, inform identification through heteroskedasticity; and (iii) features excellent forecasting performance comparable with stochastic volatility. Finally, revisiting a prominent macro-financial structural system, evidence is provided for the identification of the US monetary policy shock via heteroskedasticity, with estimates consistent with those reported in the literature.

E1817: A Bayesian Lasso based sparse learning model*Presenter:* **Yushu Li**, University of Bergen, Norway*Co-authors:* Ingvild Margrethe Helgoy

The Bayesian Lasso is constructed in the linear regression framework and applies Gibbs sampling to estimate regression parameters. A new Sparse learning model, named the Bayesian Lasso Sparse (BLS) model, is developed that employs the hierarchical model formulation of the Bayesian Lasso. The main difference from the original Bayesian Lasso lies in the estimation procedure; the BLS uses a learning algorithm based on the type-II maximum likelihood procedure. In contrast to the Bayesian Lasso, the BLS provides Sparse estimates of regression parameters. The BLS is also derived for nonlinear supervised learning problems by introducing kernel functions. Comparisons of the BLS model to the Relevance Vector Machine, Fast Laplace, Bayesian Lasso, and Lasso are conducted on both simulated and real data. Numerical results demonstrate that the BLS is Sparse and precise, especially when dealing with noisy and irregular datasets.

E1982: Information criterion for Bayesian trend filtering in generalized linear models*Presenter:* **Yuko Kakikawa**, Institute of Science Tokyo, Japan*Co-authors:* Yoshiyuki Ninomiya

Data observed across adjacent locations or time points often exhibit smooth structures with some abrupt changes. To analyze such data, Bayesian sparse regularization based on differences of regression coefficients has been studied. In particular, methods based on first-order differences such as fused lasso have received attention. However, it is often insufficient for representing complex variation in data, for which regularization based on higher-order differences such as trend filtering is regarded as more suitable. To select models based on such regularization, the widely applicable information criterion (WAIC) can be a useful approach. However, two points require careful consideration. First, WAIC is derived under asymptotic settings in which the influence of priors becomes negligible. Thus, WAIC might not appropriately reflect the properties of estimators. Second, since WAIC does not include a penalty term reflecting the complexity of the prior, it inevitably selects the prior class with the highest complexity when comparing models that have different classes of priors. To address this, the prior intensified information criterion (PIIC) was proposed and subsequently extended to Bayesian sparse regularization frameworks. However, existing methods are not directly applicable to Bayesian trend filtering, a typical regularization based on higher-order differences. To overcome this limitation, PIIC for Bayesian trend filtering under generalized linear models is proposed.

E1292: Bayesian presence-only data modeling validation*Presenter:* **Giovanna Jona Lasinio**, Sapienza University of Rome, Italy*Co-authors:* Gian Mario Sangiovanni, Gianluca Mastrantonio

Motivated by the need for a reliable assessment of predictive performance in spatial point process models, a revisiting of cross-validation strategies for spatially dependent observations is presented. Standard random-assignment or multinomial thinning schemes are computationally convenient,

but they ignore spatial autocorrelation and often produce overly optimistic error estimates because the training and validation sets remain strongly dependent. The presentation, therefore, argues for spatial block cross-validation, whose goal is to evaluate genuinely out-of-sample, longer-range prediction rather than short-range interpolation. It draws on leave-group-out cross-validation ideas developed for latent Gaussian models, where groups are constructed to remove the observations most informative for predicting a held-out point, using correlations in the linear predictor. Building on this motivation, the proposed direction for point processes is to define spatially meaningful blocks using a graph constructed from a reference set, and then apply clustering to obtain folds that better reflect the underlying dependence structure. This creates a principled compromise between preserving spatial structure and maintaining enough separation for honest validation. The talk also highlights open methodological questions.

EV305 Room Virtual R01 APPLIED AND DISTRIBUTIONAL METHODS
Chair: Pavel Krupskiy
E1924: Integration of time series models with multicriteria methods for dynamic competitiveness assessment

Presenter: **Gabriella Epifani**, University of Salento, Italy

Co-authors: Sabrina Maggio, Pier Paolo Miglietta

The assessment of competitiveness in the olive oil market in southern Europe is often based on static representations, overlooking the temporal dynamics of performance. Traditional multicriteria decision analysis approaches rely on static or aggregated data, neglecting the stochastic structure of time series. A two-step methodological extension of the tensor-based TOPSIS approach is proposed by integrating time series models into the evaluation of alternatives. First, temporal dependence is analyzed by using time series tools, allowing persistence and variability patterns to be identified. Second, criteria and weights are determined through a data-driven approach based on the eigenvectors of the correlation matrix. This procedure enables the synthesis of the dependence structure among variables, avoiding subjective assignments. The proposed approach, which integrates multicriteria analysis and advanced statistical methods, is applied to southern European countries active in the olive oil market. Results show that incorporating temporal dynamics and data-driven weights significantly affects the ranking of alternatives compared to static approaches, providing a robust measure of long-term competitiveness.

E1200: Structural asymmetry in transatlantic gas markets: A regime-switching analysis of imported volatility

Presenter: **Hamidreza Abshenasan**, Universidad Autonoma de Madrid, Spain

Co-authors: Prosper Lamothe Fernandez, Fernando Gallardo Olmedo

Transatlantic natural gas integration has created a 'structural asymmetry'. While US liquefied natural gas (LNG) exports physically bridge the markets, volatility spills backward, disproportionately impacting American storage valuations. A dual-identification strategy is utilized, employing generalized and orthogonalized forecast error variance decomposition (FEVD) to decouple financial from physical spillovers. Integrating these into a regime-switching real options valuation (ROV) model (2016-2025) confirms high financial connectedness (mean total connectedness index = 14.78%) but identifies a 'crisis inversion'. EU volatility drives US injections ($\beta = -0.1098$, $p < .001$), yet US flows do not set EU prices (spillover below 2%). This asymmetry creates divergent asset economics: US storage gains an 11.07% flexibility premium, while EU mandate-driven assets gain only a 1.52% premium. The findings challenge convergence theories, suggesting US policy should prioritize high-cyclability infrastructure for arbitrage, while EU regulators must explicitly price storage rigidity.

E1994: Distributional intersectional fairness in AI-supported job matching

Presenter: **Sabrina Muehlbauer**, Institute for Employment Research, Germany

Co-authors: Enzo Weber, Paula Ziethmann

Most algorithmic fairness evaluations remain limited to single-attribute group comparisons or individual-level prediction errors. A distributional approach to intersectional fairness is introduced that evaluates how machine learning models reshape the allocation of outcomes across intersecting social groups. Using large-scale administrative labour-market data from public employment services, an AI-supported job matching system is analysed. Intersectional disparities are measured via Jensen-Shannon divergence and observed occupational distributions are compared to model-implied allocations. Results show that unconstrained models do not merely reproduce existing labour-market structures but systematically amplify intersectional disparities across occupations. Fairness-aware regularization reduces aggregate disparities but introduces a pronounced trade-off with predictive performance and reallocates distortions across occupations rather than eliminating them. To identify localized fairness risks, an occupation-level audit mechanism is proposed that flags cases with unusually large distributional shifts. The findings demonstrate that fairness in large-scale decision-support systems must be evaluated at the level of outcome distributions. They highlight the limits of purely technical mitigation and the need to combine fairness-aware modelling with explainability, monitoring, and human oversight in real-world deployment.

Monday 10.08.2026

16:45 - 18:25

Parallel Session O – EcoSta2026

EO059 Room B101 ADVANCES IN ECONOMETRICS**Chair: Yuya Sasaki****E1562: Detection of multiple structural breaks with latent homogeneity for large panel data***Presenter:* **Jiatong Li**, Huazhong University of Science and Technology, China*Co-authors:* Degui Li, Hongqiang Yan

This paper studies the estimation of multiple structural breaks in large panel data, allowing for heterogeneous break structures across units. A latent homogeneity framework is introduced in which cross-sectional units are partitioned into unobserved groups sharing common break locations and numbers, while heterogeneity is preserved across groups. A three-step procedure is proposed: unit-level break estimation, clustering based on Hausdorff distances between estimated break sets, and a post-clustering refinement that pools information within groups via a localized CUSUM criterion. The proposed estimator achieves exact consistency for break locations and consistently recovers the latent group structure under general conditions with both dimensions diverging. Simulation results demonstrate that the proposed method consistently recovers break locations and latent group structures when group structure is present.

E1382: Counterfactual variance analysis with local projections*Presenter:* **Jooyoung Cha**, University of Notre Dame, United States

Counterfactual changes in unconditional volatility across regimes are studied. The counterfactual logic combining the shock environment of one regime with the propagation mechanism of another follows earlier counterfactual work in autoregressive and VAR-based settings. The contribution here is to recast that exercise in a local projections framework. A counterfactual long-run variance is defined, where propagation is represented by the regime-specific sequence of impulse responses. Estimating this propagation sequence with local projections and combining it with regime-specific estimates of shock variances in a plug-in estimator of the counterfactual variance is then proposed. Under standard structural assumptions, point identification of the population counterfactual is shown and conditions under which the plug-in estimator should be consistent and asymptotically normal are outlined. A Shapley-style averaging provides a possible additive attribution of volatility changes to shocks and propagation. The goal is to provide a flexible alternative to parametric VAR-based counterfactual analysis that avoids imposing cross-horizon restrictions on the impulse-response path.

E1386: Semiparametric inference in panel data with interactive and high-dimensional confounding*Presenter:* **Yukun Ma**, University of Rochester, United States

Factor-augmented double machine learning (FA-DML) is developed as a semiparametric framework for inference on low-dimensional causal parameters in panel data settings with interactive and high-dimensional confounding. Existing double/debiased machine learning methods for panel data typically rely on additive fixed effects and transformations such as demeaning or differencing, which fail when unobserved heterogeneity follows an interactive fixed effects structure. To address this limitation, orthogonalized machine learning is combined with principal-components-based factor estimation in a cross-fitted iterative procedure. In each fold, flexible learners first estimate the conditional expectations of the outcome and treatment given observed covariates; residualized outcomes are then used to recover latent factors and loadings, and these nuisance and factor estimates are iteratively updated until convergence. The resulting estimator delivers valid root-NT inference under joint large-N, large-T asymptotics, provided both the nuisance estimation error and factor estimation error are asymptotically negligible under stated rate conditions. Consistency of the variance estimator is also established. Monte Carlo simulations show that FA-DML attains near-nominal coverage in the presence of interactive confounding, whereas standard DML procedures that ignore latent factors exhibit severe undercoverage.

E1169: Estimating nonseparable selection models: A functional contraction approach*Presenter:* **Yi Xin**, California Institute of Technology, United States*Co-authors:* Fan Wu

A novel method is proposed for estimating nonseparable selection models. It is shown that, for a given selection function, the potential outcome distributions are nonparametrically identified from the selected outcome distributions and can be recovered using a simple iterative algorithm based on a contraction mapping. This result enables a full-information approach to estimating selection models without imposing parametric or separability assumptions on the outcome equation. A two-step estimation strategy is proposed for the potential outcome distributions and the parameters of the selection function and the consistency and asymptotic normality of the resulting estimators are established. Monte Carlo simulations demonstrate that this approach performs well in finite samples. The method is applicable to a wide range of empirical settings, including consumer demand models with only transaction prices, auctions with incomplete bid data, and Roy models with data on accepted wages.

EO125 Room B102 ESTIMATION OF SOCIOECONOMIC INDICATORS WITH LIMITED DATA**Chair: Gianni Betti****E1279: Sufficient dimension reduction meets two-sample regression estimation***Presenter:* **Masayuki Hirukawa**, Ryukoku University, Japan*Co-authors:* Di Liu, Artem Prokhorov

When conducting regression analysis using a dataset of anonymous surveys, researchers often face the situations in which important regressors are unavailable. Suppose that there is yet another dataset that contains the missing regressors as well as other variables that overlap across the two datasets. In this environment, regression coefficients may still be estimated consistently by combining data. The plug-in least squares (PILS) has been proposed to do that and its attractive properties in finite samples when compared to other available methods have been demonstrated. However, this estimator attains the parametric convergence rate only if the number of overlapping variables is three or less. To mitigate the curse of dimensionality, the scope of PILS is extended by assuming that the conditional expectation of each missing regressor given the overlapping variables takes the form of a single-index model with an unknown link function. Because the index coefficients are estimated through sufficient dimension reduction (SDR), the new estimator is referred to as PILS-SDR. PILS-SDR is robust to heterogeneity in the underlying population and allows for both continuous and binary missing regressors. It is asymptotically normal with the parametric convergence rate. Monte Carlo simulations confirm desirable finite-sample properties of PILS-SDR, and a real data example from the wage modelling illustrates empirical relevance of PILS-SDR.

E1285: Survey-to-survey imputation: From bracketed income to inequality measurement*Presenter:* **Lorenzo Mori**, University of Bologna, Italy*Co-authors:* Gianni Betti, Laura Neri

Many surveys record income in brackets, which leads to downward-biased inequality measures. A survey-to-survey imputation technique based on generalized additive models for location, scale and shape (SSIT-GAMLSS) is developed to reconstruct a continuous income variable from classed data and estimate inequality indices. A jackknife repeated replications technique is proposed to estimate the variability. The proposed method is evaluated through simulations. As an empirical application, inequality is estimated at the provincial level in Tuscany for 2021, 2022, and 2024, revealing spatial heterogeneity and temporal variation and providing timely estimates that would otherwise be unavailable.

E1806: Measuring multidimensional deprivation at the local level through survey-to-survey imputation*Presenter:* **Chiara Gigliarano**, LIUC University C.Cattaneo, Italy*Co-authors:* Annamaria Bianchi, Antonella D Agostino, Laura Neri

A comprehensive set of indicators of deprivation and well-being is developed to identify the most disadvantaged territories and population groups in terms of poverty and inequality. The analysis is based on data from the MYPEOPLE survey, conducted in July 2024 on approximately 10,000 households using a mixed CAWI-CATI methodology across three Italian regions: Lombardy, Tuscany, and Campania. Since total household disposable income is collected in brackets in the survey, a survey-to-survey imputation approach is implemented to recover a continuous income distribution. Specifically, income bracket information from MYPEOPLE is combined with detailed income data from EU-SILC. Fuzzy measures of deprivation based on income are constructed using the Integrated Fuzzy and Relative approach. This framework allows for a nuanced assessment of poverty intensity and facilitates comparisons across population groups. An in-depth focus on Lombardy examines patterns of socio-economic vulnerability in Inner Areas as defined by the National Strategy for Inner Areas. The analysis adopts a multidimensional perspective, encompassing both economic and material conditions as well as access to essential services. It further integrates objective indicators with subjective evaluations, capturing individuals' self-assessed socio-economic conditions.

E1367: Quantile time dummy product model for estimating cheapflation*Presenter:* **Federico Crescenzi**, University of Tuscia, Italy

A novel quantile time product dummy approach is introduced to measure cheapflation – the disproportionate rise in the prices of cheaper products relative to more expensive ones – using high-frequency web-scraped food price data from Italy between 2020 and 2024. Unlike existing studies that rely on fixed price categories or quality ladders, this method directly estimates price inflation for arbitrary quantiles of the price distribution without requiring prior classification. The results reveal substantial inflation gradients: lower-priced items experienced significantly faster price growth than higher-end counterparts. These findings align with recent evidence on level pass-through pricing, asymmetric markup adjustments, and demand shifts toward low-cost goods.

EO105 Room B103 APPLICATIONS OF ADVANCED ECONOMETRICS IN PRODUCTION ECONOMICS**Chair: Jisang Yu****E1174: Crop insurance and farm-level productivity: Evidence from Kansas***Presenter:* **Youngjune Kim**, Kangwon National University, Korea, South*Co-authors:* Hanbin Lee, Dustin Pendell

The influence of crop insurance participation on agricultural productivity at the farm level is investigated. Utilizing a panel dataset from 1993 to 2018 that includes production data, farm characteristics, and financial information for farms in Kansas, a control function approach is employed to estimate a production function and derive farm specific, time varying measures of total factor productivity. Based on these productivity estimates, an instrumental variable approach is used to assess the impact of crop insurance expenditures per acre on farm productivity, accounting for the potential endogeneity of insurance spending decisions. The findings indicate that increases in per acre crop insurance expenditures have a positive effect on farm level agricultural productivity.

E1175: Climate variability and input interactions in coffee productivity: Evidence from Sub-Saharan Africa*Presenter:* **Chia-Lin Chang**, Department of Applied Economics Department of Finance National Chung Hsing University, Taiwan*Co-authors:* Wondim Awoke, Jisang Yu

An examination is conducted of how climate change affects coffee productivity through its interaction with production inputs across 24 Sub-Saharan African countries over the period 1969 to 2019. Using climate-augmented and climate-dependent cobb-douglas models together with a climate-integrated translog specification, an analysis is performed of how temperature and precipitation jointly influence output and input elasticities. Panel-corrected standard errors are employed to address serial correlation and cross-sectional dependence in long-run panel data. The results show that rising temperatures significantly reduce coffee productivity, while precipitation effects are nonlinear and depend on interactions with production inputs. Importantly, climate stress alters factor returns: labor productivity declines sharply under hotter and wetter conditions, whereas capital becomes more productive, indicating a climate-driven factor substitution mechanism. Estimates from the translog model confirm strong substitutability between labor and capital, while climatic variables act as complements rather than substitutes to conventional inputs. These findings suggest that climate impacts operate primarily through input productivity channels, highlighting the need for climate-adaptive investment, input reallocation, and institutional support to strengthen resilience in climate-vulnerable agricultural systems.

E1179: Estimating impacts of large-scale agricultural intervention: The case of Ethiopian agricultural growth program*Presenter:* **Eunsun Hong**, Korea University, Korea, South*Co-authors:* Jisang Yu

This study evaluates the productivity impacts of Ethiopia's agricultural growth program (AGP I) by explicitly addressing measurement error, input endogeneity, and aggregation bias. Evaluating large-scale interventions in smallholder settings is often confounded by severe measurement error and heterogeneity in self-reported data. Using harmonized panel data from the Ethiopia LSMS-ISA, a two-stage structural empirical strategy is implemented. First, Cobb-Douglas production functions are estimated at both plot and household levels using instrumental variables (IV) to recover total factor productivity (TFP) purged of simultaneity and attenuation bias. Second, these IV-based TFP measures are employed in a difference-in-differences framework to identify the causal effect of AGP I. Results show that IV estimates diverge significantly from OLS, and that aggregating from plots to households sharply reduces unexplained productivity dispersion. Crucially, once productivity is measured at the household level using IV-based methods and appropriate aggregation, statistically significant evidence is found that AGP I increased TFP. These findings suggest that productivity detected in conventional analyses may reflect granular mismeasurement rather than the absence of genuine policy impacts. The necessity of integrating structural estimation techniques into impact evaluations to avoid misleading conclusions driven by noisy survey data is therefore underscored.

E1278: Heatwave-induced asymmetry in price transmission across marketing stages*Presenter:* **Hyoeun Sun**, Korea University, Korea, South*Co-authors:* Dong Hee Suh

The aim is to examine how heatwaves shape asymmetric price transmission along vegetable marketing stages. Using daily farm-gate, wholesale, and retail prices for Korean cabbage, machine learning techniques are applied to assess the relative importance of climate variables. These models capture nonlinearities and complex interactions in temperature dynamics. Building on this evidence, a heatwave intensity index is constructed and incorporated into a nonlinear autoregressive distributed lag (NARDL) framework to model asymmetric price transmission. Price changes are decomposed into positive and negative components and further separated into heatwave-induced and non-climate-driven movements, allowing extreme heat to influence both the magnitude and speed of price pass-through. The findings are expected to show that heatwaves amplify farm-level price increases and accelerate transmission to wholesale markets, while retail pass-through remains incomplete. The findings further suggest that intensifying future heatwaves may strengthen asymmetric transmission, thereby redistributing climate-related price risk along the marketing chain.

EO119 Room B104 RECENT ADVANCES IN MACHINE LEARNING FOR ECONOMIC INFERENCE**Chair: Eoghan O'Neill****E1324: Nonlinear autoregressive models for functional time series with bayesian additive regression trees****Presenter:** Eoghan O'Neill, University College Dublin, Ireland**Co-authors:** Jiahao Cao, Maria Grith, Anastasija Tetereva, Guanyu Hu

Bayesian additive regression tree models for function-on-function regression are introduced. The outcome function is modelled as a linear combination of data-adaptive basis functions. The coefficients of basis functions are determined by sums of trees that can split on both scalar and functional variables, including the lag of the dependent variable. Splitting rules for functions are defined by inner products between functional covariates and linear combinations of fixed basis functions, distinct from the aforementioned data-adaptive basis. A prior on functional splits is considered that allows Markov chain Monte Carlo tree samples to adapt to the data by placing higher probability on selecting a subset of relevant basis functions for a splitting rule. The forecasting performance of the method is evaluated in an application to option pricing implied volatility surface data.

E1648: Machine learning and difference-in-differences: A practitioners dilemma**Presenter:** Paolo Libenzio Brignoli, ETH Zurich, Switzerland**Co-authors:** Andrea Naghi, Alessandro Varacca

Recent econometric advances in difference-in-differences (DiD) have developed robust tools for staggered treatment adoption and expanded machine learning (ML) methods for selection on observables. Yet, an explicit methodological bridge connecting these advances is missing. A double/debiased machine learning staggered DiD estimator is introduced, showing that the statistical properties enabling ML in standard DiD naturally extend to staggered policy adoption. By formally connecting orthogonalized estimation in two-period settings with modern staggered designs, this approach enables valid, flexible estimation in complex, high-dimensional contexts. Simulation evidence demonstrates substantial improvements in empirically relevant settings with nonlinear relationships and high-dimensional sparse covariates. An application to evaluate the European Union Agri-Environmental Schemes (2014-2020) accounts for both enrollment timing and complex farm characteristics, finding that most farmers maintain previous practices after receiving environmental payments. This suggests significant windfall effects in current policy implementation, offering direct implications for future agricultural policy design.

E1864: Tail-aware density forecasting of locally explosive time series: A neural network approach**Presenter:** Julien Peignon, Paris Dauphine University, France**Co-authors:** Arthur Thomas, Elena Dumitrescu

A Mixture Density Network specifically designed for forecasting time series that exhibit locally explosive behavior is proposed. By incorporating skewed t-distributions as Mixture components, the approach offers enhanced flexibility in capturing the skewed, heavy-tailed, and potentially multimodal nature of predictive densities associated with bubble dynamics modeled by mixed causal-noncausal ARMA processes. An adaptive weighting scheme is implemented that emphasizes tail observations during training and leads to accurate Density estimation in the extreme regions most relevant for financial applications. Once trained, the MDN produces near-instantaneous Density forecasts. Through extensive Monte Carlo simulations and two empirical applications on natural gas prices and inflation, the proposed MDN-based framework delivers superior forecasting performance relative to existing approaches.

EO017 Room 101 STATISTICAL ANALYSIS OF COMPLEX DYNAMIC DATA**Chair: Weichi Wu****E1760: Online TV regression with dependent noise****Presenter:** Fabian Mies, Delft University of Technology, Netherlands**Co-authors:** Ivan Krylov

Via nonlinear regression procedures, signals of bounded variation can be recovered from noisy observations at the same asymptotic rate $n^{1/3}$, which is identical to the rate for Lipschitz signals, while allowing for spatially inhomogeneous smoothness. The problem is considered in a sequential setting and in the presence of temporally dependent error terms. Adaptivity is achieved by leveraging an online learning algorithm to choose the optimal restart times for running averages. The method achieves the optimal rate of convergence, up to log factors, for the online nonparametric regression problem with martingale difference errors, and the procedure is shown to be robust against small deviations from the martingale assumptions. To handle stronger autocorrelations, a thinning procedure is devised and shown to recover the optimal rate of convergence for signal denoising under total variation constraints.

E1798: Optimal tuning parameter selection in heteroscedasticity and autocorrelation robust inference**Presenter:** Zhou Zhou, Zhejiang University, China

The selection of a block size, window size, or bandwidth parameter is universal and important in various Heteroscedasticity and Autocorrelation Robust (HAR) inference problems. Most current proposals for choosing this parameter are ad hoc without theoretical justification of their performance or optimality. An investigation into the asymptotic bias and variance of a fundamental HAR procedure for a general class of piecewise locally stationary time series is presented. As a result, an easy to implement and asymptotically optimal tuning parameter selection algorithm is proposed. The investigation also reveals that analogous tuning parameter selection algorithms for stationary time series are biased when applied to nonstationary time series.

E1804: Simultaneous inference for nonlinear time series, a sieve M-regression approach**Presenter:** Tianpai Luo, Department of Statistics and Data Sciences, Tsinghua University, China

Simultaneous inference of conditional distributions in nonlinear time series is studied from a sieve M-regression perspective. Existing literature on sieve M-regression has primarily focused on pointwise asymptotics, leaving the development of uncertainty quantification over the entire predictor space unexplored. This gap is addressed by establishing a uniform Bahadur representation for the sieve M-estimator, accommodating dependent data and a growing number of sieve basis functions. A novel high-dimensional empirical process theory is developed for temporally dependent data, and a specifically designed M-decomposition method is utilized to control high-dimensional complexities. Building on this representation, a convex Gaussian approximation is developed to characterize the asymptotic behavior of the estimator and construct valid simultaneous confidence regions (SCRs). To facilitate practical implementation, a self-convolved bootstrap algorithm is introduced that accurately approximates the distribution of the maximal deviation. The inferential framework is supported by rigorous error bounds and validated through numerical simulations and real data applications.

E1838: Inference for structural changes in nonstationary functional time series with partial measurement error**Presenter:** Weichi Wu, Tsinghua University, China, China**Co-authors:** Lujia Bai, Qirui Hu

Change point detection and localization for a general class of locally stationary functional time series is studied. To accommodate nonstationarity and other possible complex features, such as discontinuous trajectories and heterogeneous partial measurement error of contemporary functional data, methods are proposed that do not rely on pre-processing techniques of pre-smoothing and dimension reduction, which would be less accurate

without the assumptions of stationarity and continuous trajectories. For detecting changes, a bootstrap-assisted test for structural breaks among all mean trajectories is proposed, which is asymptotically correct and can detect local alternatives of $n^{-1/2}$. For localizing changes, practical and consistent algorithms are developed for estimating single and multiple change points, which further enable the estimation of mean trajectories. To establish the theoretical properties of the proposed approaches, a new backward martingale difference inequality and a functional Burkholder inequality for nonstationary functional time series are developed, which can be of independent interest. The effectiveness of the approach is demonstrated through extensive simulation studies and real data analyses. The proposed algorithms are available in the R package *fcpsed*.

EO099 Room 102 RECENT DEVELOPMENTS IN TIME SERIES AND RELATED TOPICS
Chair: Sangyeol Lee
E1444: On the theoretical foundations of neural network-based time series: Stationarity, ergodicity, and generalization

Presenter: **Dongwon Kim**, Korea Institute for Advanced Study (KIAS), Korea, South

Stationarity and ergodicity play a central role in time series analysis, forming the theoretical basis for consistent estimation, asymptotic normality, and the validity of statistical inference. While these properties have been well studied in classical models such as ARMA and GARCH, their theoretical understanding in neural network-based time series models remains limited. Sufficient conditions for the stationarity and ergodicity of multivariate neural network autoregressive moving average (NN-ARMA) processes are developed. The analysis builds upon the iterated random function (IRF) approach, providing a general theoretical foundation for the stability of nonlinear stochastic systems driven by neural architectures. Furthermore, the generalization capability of stationary ergodic NN-ARMA models through Rademacher complexity bounds is investigated. By adapting results, generalization error bounds under weak dependence are derived, bridging statistical learning theory and time-series ergodic theory. Together, these results aim to establish a unified framework that links the theoretical stability of neural network time series models with their learning capacity, offering insights into both the probabilistic and algorithmic behavior of modern stochastic neural models.

E1502: Robust estimation for bounded bivariate time series models of counts based on density power divergence

Presenter: **Minyoung Jo**, Dankook University, Korea, South

Co-authors: Sangyeol Lee

Two types of bounded bivariate time series models are investigated, suitable for analyzing time series of counts with values within a finite range. They are a modified version of bounded bivariate integer-valued ARCH models and bounded bivariate INAR models. All these models are constructed based on the bivariate binomial distribution of type II. A main focus is on providing a robust estimation method for these models. For this, the minimum density power divergence estimator (MDPDE) is employed as a robust estimator. To assess the performance of MDPDE and validate its effectiveness, Monte Carlo simulations and real data analysis using monthly earthquake data in the United States and monthly crime data in Pittsburgh are conducted. The findings collectively confirm the efficacy and suitability of the methods.

E1544: Multivariate nonparametric Erlang mixtures

Presenter: **Kyeong-A Yang**, Sungkyunkwan University, Korea, South

Co-authors: Byungtae Seo

Multivariate positive-valued data arise in various applications, including insurance losses, survival outcomes, and waiting-time data, where skewness, heavy tails, and multimodality are commonly present. Multivariate Erlang mixtures provide a flexible and analytically tractable framework for modeling such data, with appealing properties including closure under marginalization and denseness in the class of positive continuous multivariate distributions. However, existing research has primarily focused on their structural properties and expectation-maximization (EM) based estimation under a fixed number of components. A nonparametric maximum likelihood estimator (NPMLE) for multivariate Erlang mixtures is considered, without pre-specifying the number of support points of the mixing distribution. The proposed approach uses a gradient function as a diagnostic tool to assess lack of fit and to identify promising support points. Numerical studies demonstrate that this method progressively refines the support and improves model fit for multivariate Erlang mixtures. These findings suggest that multivariate nonparametric Erlang mixtures serve as an effective and flexible framework for multivariate density estimation.

E1567: Robust change point detection in VARX models via recursive CUSUM tests

Presenter: **Sangjo Lee**, Inha University, Korea, South

A robust change point detection procedure for Vector Autoregression with Exogenous variables (VARX) models is developed in the presence of bounded non-stationarity in Exogenous policy variables, such as level shifts or regime changes. The proposed test is based on Recursive residuals from the Recursive Least Squares (RLS) algorithm. Since these residuals are constructed as one-step-ahead prediction errors, they preserve the Martingale Difference Sequence (MDS) property with respect to the natural filtration, regardless of the persistence properties of the Exogenous variables. Under the null hypothesis, the Recursive LSCUSUM statistic converges to the supremum of a Brownian bridge. Simulation results demonstrate that the method performs reliably under policy shifts and retains strong power for detecting genuine structural changes. Its practical value is illustrated through an application to U.S. financial market data.

E1682: The density function of spectral measure of skewed elliptical distributions

Presenter: **Moosup Kim**, Keimyung University, Korea, South

Joint heavy-tail modeling is a fundamental problem in financial risk management, as asset returns often exhibit heavy-tailed behavior and pronounced extremal dependence. These features can be effectively characterized within the framework of multivariate regular variation. Elliptical distributions provide a tractable parametric class for modeling such behavior; however, their inherent symmetry limits their applicability in practice, where empirical data frequently exhibit asymmetry. A parametric model based on skew-elliptical distributions enables the modeling of asymmetric multivariate tail behavior.

EO196 Room 103 MODERN STATISTICAL METHODS FOR LEARNING AND DECISION-MAKING
Chair: Tao Ma
E1195: Sampling as bandits: Evaluation-efficient design for black-box densities

Presenter: **Takuo Matsubara**, The University of Sydney Business School, Australia

Bandit Importance Sampling (BIS) is proposed. BIS is a powerful importance sampling framework tailored for settings in which evaluating the target density is computationally expensive. BIS facilitates accurate sampling while minimizing the required number of target-density evaluations. In contrast to adaptive importance sampling, which optimizes a proposal distribution, BIS directly optimizes the set of samples through a sequential selection process driven by multi-armed bandits. BIS serves as a general framework that accommodates user-defined bandit strategies. Theoretically, the weak convergence of the weighted samples, and thus the consistency of the Monte Carlo estimator, is established regardless of the specific strategy employed. A practical strategy that leverages Gaussian process surrogates to guide sample selection is presented, adapting the principles of Bayesian optimization for sampling. Comprehensive numerical studies demonstrate the superior performance of BIS across multimodal, heavy-tailed distributions, and real-world Bayesian inference tasks involving Markov random fields.

E1704: Balanced policy switching in reinforcement learning*Presenter:* **Tao Ma**, University College London, United Kingdom

Reinforcement learning (RL) aims to learn optimal policies that maximize long-term cumulative reward and has achieved widespread success across domains. In many real-world applications, however, updating or switching policies incurs non-negligible costs, making frequent changes undesirable. This challenge arises both in offline settings, where decisions rely solely on historical data, and in online scenarios where policy updates must balance improvement against switching costs. A unified framework for policy switching in both offline and online RL is introduced, leveraging ideas from optimal transport to propose a principled formulation that captures the trade-off between performance gains and switching costs. Key theoretical properties are established and a Net Actor-Critic algorithm is developed for the resulting problem. Experiments on robot control tasks and traffic signal management demonstrate the effectiveness of the approach.

E1770: U-processes of growing order and their application in mean estimation*Presenter:* **Shunan Yao**, Hong Kong Baptist University, Hong Kong*Co-authors:* Stanislav Minsker

New results on robust multivariate location estimation built on U-statistical methodology are presented. First, a Bousquet-type inequality for U-processes with near-optimal constants is derived. Second, using the derived technical results, a new location estimator based on a U-statistic version of Tukey's median is introduced. The proposed estimator is shown to be asymptotically normal and achieves efficient asymptotic covariance, in contrast to the original Tukey's median.

E1633: Kernel integrated R^2 : A measure of dependence*Presenter:* **Shakeel Gavioli-Akilagun**, London School of Economics, United Kingdom*Co-authors:* Mona Azadkia, Florian Kalinke, Zoltan Szabo, Seyedpouya Mirrezaeioudaki

The kernel integrated R^2 is a new measure of statistical dependence that combines the local normalization principle of the recently introduced integrated R^2 with the flexibility of reproducing kernel Hilbert spaces (RKHSs). The proposed measure extends integrated R^2 from scalar responses to responses taking values on general spaces equipped with a characteristic kernel, allowing measurement of dependence for multivariate, functional, and structured data while remaining sensitive to tail behaviour and oscillatory dependence structures. This measure takes values in $[0, 1]$, equals zero if and only if independence holds, and equals one if and only if the response is almost surely a measurable function of the covariates. Two estimators are proposed: a graph-based method using K -nearest neighbours and an RKHS-based method built on conditional mean embeddings. Consistency is established and convergence rates are derived for the graph-based estimator, showing its adaptation to intrinsic dimensionality. Numerical experiments on simulated data and a real data experiment in the context of dependency testing for media annotations demonstrate competitive power against state-of-the-art dependence measures, particularly in settings involving nonlinear and structured relationships.

EO058 Room 104 ROBUST INFERENCE IN HIGH DIMENSIONS (VIRTUAL)**Chair: Miles Lopes****E1415: Perturbed double machine learning: nonstandard inference beyond the parametric length***Presenter:* **Zijian Guo**, Zhejiang University, China

Inference on a low-dimensional functional β is studied in the presence of infinite-dimensional nuisance parameters. Classical inferential methods are typically based on Wald intervals, whose large-sample validity rests on asymptotic negligibility of nuisance error; for example, influence-curve based estimators (Double/Debiased Machine Learning, DML) are asymptotically Gaussian when nuisance estimators converge faster than $n^{-1/4}$. Although such negligibility can hold even in nonparametric classes, it can be restrictive. To relax this requirement, Perturbed Double Machine Learning is proposed, which ensures valid inference even when nuisance estimators converge slower than $n^{-1/4}$. The proposal is to (i) inject randomness into the nuisance estimation step to generate perturbed nuisance models, each yielding an estimate of β and a Wald interval, and (ii) filter out perturbations whose deviations from the original DML estimate exceed a threshold. For Lasso nuisance learners, it is shown that, with high probability, at least one perturbation yields nuisance estimates sufficiently close to the truth, so the associated estimator of β is close to an oracle with known nuisances. The union of retained intervals delivers valid coverage even when the DML estimator converges slower than $n^{-1/2}$. The framework extends to general machine-learning nuisance learners, and simulations show coverage when state-of-the-art methods fail.

E1418: High-dimensional Gaussian and bootstrap approximations for robust means*Presenter:* **Anders Kock**, University of Oxford, United Kingdom

Recent years have witnessed much progress on Gaussian and bootstrap approximations to the distribution of sums of independent random vectors with dimension d large relative to the sample size n . However, for any number of moments $m > 2$ that the summands may possess, there exist distributions such that these approximations break down if d grows faster than the polynomial barrier $n^{\frac{m}{2}-1}$. Gaussian and bootstrap approximations to the distributions of winsorized and trimmed means are established that allow d to grow at an exponential rate in n as long as $m > 2$ moments exist. The approximations remain valid under some amount of adversarial contamination. The implementations of the winsorized and trimmed means do not require knowledge of m . As a consequence, the performance of the approximation guarantees "adapts" to m .

E1557: Detecting non-uniform patterns on high-dimensional hyperspheres*Presenter:* **Tuan Pham**, University of Texas, Austin, United States

A new probabilistic characterization of the uniform distribution on the hypersphere in terms of the distribution of pairwise inner products is proposed. This characterization leads naturally to an Ingster-type distance for quantifying deviations from uniformity, whose asymptotic behavior can be analyzed systematically via Edgeworth-type expansions. Surprisingly, it is shown that this distance captures the minimax rates for testing uniformity simultaneously across several high-dimensional parametric models, even in the models where densities with respect to the uniform law do not exist. A simple test for spherical uniformity based on this distance is proposed. The detection rates and consistency under various classes of alternatives, both local and non-local, are studied. The proposed test is universally consistent in fixed dimensions, minimax-optimal in a variety of high-dimensional parametric models, and consistent against non-local high-dimensional alternatives. This is different from previously studied high-dimensional Sobolev tests and extreme-value-based tests, which are rate-suboptimal or inconsistent against one or more classes of alternatives.

E1597: Inference for multivariate and infinite-dimensional parameters: Beyond CLT and beyond independence*Presenter:* **Arun Kuchibhotla**, Carnegie Mellon University, United States

Construction of confidence intervals and hypothesis tests has long been based on central limit theorems for estimators and variance estimation. Such methods require knowledge of the rate of convergence and estimation of the asymptotic variance, and even when these are satisfied, CLT-based methods can fail to retain validity uniformly over large classes of data-generating processes and when the dimension increases too fast with the sample size. Furthermore, dependence between observations only worsens the issues. A framework is introduced that builds on HulC and significantly expands its scope. In addition to general results of uniform validity under independence, dependence, and diverging dimensions, specific examples are presented for constructing confidence bands for empirical processes and high-dimensional mean parameters.

EO124 Room 105 RECENT ADVANCES ON STATISTICAL ANALYSIS FOR DEPENDENT DATA**Chair: Gan Yuan****E1739: Adaptive Bayesian optimization with consistent smoothness estimation****Presenter:** Saifei Sun, City University of Hong Kong, Hong Kong

A novel algorithm, Adaptive Matern Kernel Bayesian Optimization (AMKBO), is presented to address the challenges of hyperparameter uncertainty in Gaussian Process (GP)-based Bayesian Optimization. AMKBO accelerates convergence by adaptively estimating the smoothness parameter of the Matern Kernel, improving the hyperparameter adjustment strategy, and incorporating an opposition-based learning mechanism into acquisition function Optimization. Specifically, AMKBO generates initial sampling points using random curves and constructs an estimator for the smoothness parameter of the covariance Kernel, allowing the Kernel to model functions with varying degrees of smoothness. During Optimization, a refined length-scale adjustment strategy is employed to achieve a more balanced trade-off between exploration and exploitation while preventing excessively aggressive exploration. In parallel, the opposition-based learning mechanism enhances search coverage and computational efficiency in acquisition function Optimization. Experimental results on synthetic benchmark functions and real-world problems demonstrate that AMKBO outperforms existing methods by achieving faster convergence and avoiding local optima.

E1749: Inference for change-points in time series under various break sizes**Presenter:** Chun Yip Yau, Chinese University of Hong Kong, Hong Kong

The asymptotic distribution of a change-point estimator for piecewise stationary time series is investigated across different magnitudes of break sizes. Specifically, break sizes of order $O(1/n^\alpha)$ for $0 < \alpha < 1/2$, $= 1/2$, and $> 1/2$ are examined, corresponding to large, moderate, and small break sizes, respectively, where n denotes the sample size. Results reveal that the asymptotic distributions in these three regimes differ but are all linked to the maximizer of certain functions of a two-sided drifted Brownian motion. To address the practical challenge of unknown break sizes, an asymptotically pivotal statistic is introduced that is robust across the whole range of break size regimes on $[0, \infty)$. This statistic provides a unified approach for constructing confidence intervals for the change-point without requiring prior knowledge of the break size. Simulation studies show that the asymptotic inference performs well under different break sizes, while the pivotal statistic demonstrates better performance in most scenarios. Applications to financial time series further highlight the practical relevance of the proposed inference methods.

E1774: Deep neural networks estimation for dependent data**Presenter:** Chao Zheng, University of Southampton, United Kingdom**Co-authors:** Gan Yuan, Zudi Lu, Shubin Wu

Recent years have seen substantial progress in the theoretical analysis of deep neural networks, though the majority of existing results assume independent observations. In contrast, the statistical properties of deep ReLU neural networks for modelling nonlinear, non-i.i.d. mixing sequences are investigated, encompassing a broad class of time series models. Sharp non-asymptotic error bounds for the DNN estimator are established, demonstrating that these bounds depend explicitly on the underlying dependence structure of the data, the architectural characteristics of the network, and the dependence structure under different mixing scenarios. Systematic simulation studies demonstrate that the empirical results align closely with the theoretical findings.

E1790: Opioid crisis and firm downside tail risks: Evidence from the option market**Presenter:** Weiming Zhang, IE Business School, Spain

The aim is to explore how opioid crisis exposure affects firm downside tail risks implied from equity options. Using a large sample of public firms from 1999 to 2020, firms headquartered in regions with higher opioid death rates face higher downside tail risks, as evidenced by higher costs of protection against left tail risks. The effects are reversed following exogenous anti-opioid legislation, supporting a causal interpretation. Further analysis shows that the opioid crisis heightens firm risk by lowering labor productivity. Greater impact is documented among firms with greater reliance on labor, limited local labor supply, and lower workplace safety.

EO065 Room 106 STATISTICAL ANALYSIS FOR STOCHASTIC PROCESSES**Chair: Teppei Ogihara****E1413: Change detection for the volatility in linear parabolic SPDEs****Presenter:** Yozo Tonaki, The University of Osaka, Japan**Co-authors:** Yusuke Kaino, Masayuki Uchida

Volatility change point detection is considered for second order linear parabolic stochastic partial differential equations (SPDEs) based on high frequency spatio-temporal data. Ornstein-Uhlenbeck processes can be obtained from the inner product of the solution of the SPDE and an orthonormal system, and these processes can be approximated through statistical inference for SPDEs based on high frequency spatio-temporal data. A test statistic is thus proposed to detect changes of the volatility function over time based on change point analysis for diffusion processes, and the asymptotic null distribution of the proposed test statistic is derived. The proposed test is then shown to be consistent. Additionally, some examples are provided and numerical simulations of the proposed test statistic are performed.

E1507: Estimation error and hypothesis testing for non-identifiable models, with applications to machine learning**Presenter:** Junichiro Yoshida, University of Tokyo, Graduate School of Mathematical Sciences, Japan

To bridge machine learning and classical statistical methods, it is important to analyze estimation error in non-identifiable models such as neural networks. However, in complex non-identifiable models, estimation error is difficult to evaluate explicitly. To address this problem, an estimator of the estimation error is proposed together with its confidence interval, thereby enabling classical statistical methods, such as hypothesis testing, for non-identifiable models. The key idea is to apply the resolution of singularities to singular models following Watanabe. While his theory is primarily developed in a Bayesian framework, the estimator is feasible within a frequentist asymptotic framework and is computationally tractable.

E1511: Robustified Gaussian quasi-BIC for volatility**Presenter:** Shoichi Eguchi, Osaka Institute of Technology, Japan**Co-authors:** Hiroki Masuda

There have been several studies on model selection for stochastic differential equations and robust model selection; for example, the BIC-type information criterion for locally asymptotically quadratic models and AIC- and BIC-type information criteria based on density-power divergence. In this talk, robust model comparison is considered in a class of non-ergodic continuous volatility regression models contaminated by finite-activity jumps. Using density-power weighting and Holder-inequality-based normalization of the conventional Gaussian quasi-likelihood function, two Schwarz-type statistics are proposed.

E1642: Regularized estimation for locally Cauchy Levy regression**Presenter:** Yuma Uehara, Kansai University, Japan

A continuous-time regression model driven by a locally Cauchy Levy process with high-frequency samples is considered. For this model, Cauchy quasi-likelihood estimation enables estimation of the mean and scale parameters over a fixed time period. However, the estimating function is

not convex, and thus estimation accuracy becomes unstable, especially when the dimension of covariate processes is high. As a solution to this problem, a regularized estimation scheme is proposed, and its theoretical properties are derived.

EO079 Room 107 RECENT DEVELOPMENTS IN STATISTICAL METHODS FOR CAUSAL INFERENCE
Chair: Shuichi Kawano
E1503: Causal inference with multiple versions of treatment via mixture-of-experts

Presenter: **Kohei Yoshikawa**, Kyushu University, Japan

Co-authors: Shuichi Kawano

SUTVA (Stable Unit treatment Value Assumption) includes the requirement that there are no multiple versions of treatment. In observational studies, although the implementation of treatment is not always controlled, multiple versions of treatment may exist in the treatment. Ignoring such latent versions violates the no-multiple-versions assumption and may obscure causal interpretation. Existing approaches clarify comparisons between treatments, however, they do not directly identify causal contrasts between latent versions within the treatment. A causal inference framework for multiple versions of treatment based on a mixture-of-experts model is proposed. This framework allows identification and estimation of pairwise average causal effects for treatment-version pairs even when the versions are unobserved. A maximum likelihood estimator is developed by using the EM algorithm and combined with inverse probability weighting to estimate the expectation of potential outcome for each version. Identification and large-sample properties under standard regularity conditions are also studied. Simulation studies show that the proposed method recovers the latent version structure and provides reliable estimates of causal contrasts.

E1506: Semiparametric efficiency and flexible estimation of weighted average treatment effects under two-phase sampling

Presenter: **Kazuharu Harada**, Tokyo Medical University, Japan

Co-authors: Masataka Taguri

Two-phase sampling is widely used in observational studies when important covariates are expensive to collect, but valid causal inference requires careful adjustment for the sampling design. Weighted average treatment effects under two-phase sampling are studied, including the average treatment effect, the average treatment effect on the treated or untreated, and the average treatment effect in the overlap population. The semiparametric efficiency bound for this broad class of target parameters is first derived and it is shown how it decomposes into the full-data efficiency component and an additional term induced by phase-2 subsampling. Several consistent estimators are then considered with focus on enriched estimators that incorporate information available for all phase-1 individuals. In particular, the enriched doubly robust estimator is shown to attain the semiparametric efficiency bound under suitable conditions, clarifying when phase-1 information can substantially improve efficiency. Design implications are also discussed, including how the benefit of enrichment depends on the phase-1 variables and the sampling mechanism. Finally, a data-adaptive implementation based on cross-fitting and double machine learning is outlined, which enables flexible nuisance estimation while preserving the main efficiency and robustness properties. The results provide practical guidance for both the design and analysis of observational studies with two-phase sampling.

E1558: Estimation of causal effects via conditional instrumental variables in functional linear structural equation models

Presenter: **Michio Yamamoto**, The University of Osaka / RIKEN AIP / Shiga University, Japan

Functional data analysis provides a flexible framework for modeling data observed over a continuum, such as longitudinal trajectories, temporal signals, and spatial profiles. In many applications, researchers are interested not only in associational relationships among functional variables but also in causal effects. However, causal effect estimation becomes substantially more difficult when unobserved confounding is present. A functional linear structural equation model is considered in which relations among functional variables are represented by linear operators, and identification and estimation of average causal effects using conditional instrumental variables are studied. A central challenge is that the identification equation implied by the model takes the form of an ill-posed inverse problem in an infinite-dimensional space, so standard finite-dimensional instrumental variable arguments do not apply directly. To address this issue, a Tikhonov-regularized estimator for the causal effect is proposed. Theoretical properties of the proposed method are established by deriving a convergence rate for its mean squared error under suitable regularity conditions. Finite-sample performance is also investigated through simulation studies based on synthetic functional data.

E1750: Conditional post-selection inference in propensity score analysis

Presenter: **Yoshiyuki Ninomiya**, The Institute of Statistical Mathematics, Japan

Selective inference (conditional post-selection inference) is a methodology that has attracted much attention in recent years in the fields of statistics and machine learning. Naive inference based on data that are also used for model selection tends to show overestimation, and selective inference conditions on the event that the model was selected. Selective inference in propensity score analysis is developed with a semiparametric approach, which has become a standard tool in causal inference. Specifically, for the most basic causal inference model in which the causal effect can be written as a linear sum of confounding variables, Lasso-type variable selection is conducted by adding an L_1 penalty term to the loss function that yields a semiparametric estimator. Confidence intervals are then constructed for the coefficients of the selected confounding variables, conditional on the event of variable selection, with asymptotic guarantees. Recent developments in selective inference are also incorporated.

EO149 Room 202 RECENT ADVANCES AT THE INTERFACE OF STEIN'S METHOD AND STATISTICS
Chair: Yuta Koike
E1238: Stein discrepancy for unsupervised domain adaptation

Presenter: **Anneke von Seeger**, University of Minnesota - Twin Cities, United States

Co-authors: Gilad Lerman, Dongmian Zou

Unsupervised domain adaptation (UDA) aims to improve model performance on an unlabeled target domain using a related, labeled source domain. A common approach aligns source and target feature distributions by minimizing a distance between them, often using symmetric measures such as maximum mean discrepancy (MMD). However, these methods struggle when target data is scarce. A novel UDA framework is proposed that leverages Stein discrepancy, an asymmetric measure that depends on the target distribution only through its score function, making it particularly suitable for low-data target regimes. The proposed method has kernelized and adversarial forms and supports flexible modeling of the target distribution via Gaussian, GMM, or VAE models. A generalization bound on the target error and a convergence rate for the empirical Stein discrepancy in the two-sample setting are derived. Empirically, the method consistently outperforms prior UDA approaches under limited target data across multiple benchmarks.

E1456: Incomplete U-statistics of equireplicate designs: Berry-Esseen bound and efficient construction

Presenter: **Cesare Miglioli**, University of Pittsburgh, United States

Co-authors: Jordan Awan

U-statistics are a fundamental class of estimators that generalize the sample mean and underpin much of nonparametric statistics. Although extensively studied in both statistics and probability, key challenges remain: their high computational cost, addressed partly through incomplete U-statistics, and their non-standard asymptotic behavior in the degenerate case, which typically requires resampling methods for hypothesis testing. A novel perspective on U-statistics is introduced, grounded in hypergraph theory and combinatorial designs. The approach bypasses the traditional Hoeffding decomposition, the main analytical tool in this literature but one highly sensitive to degeneracy. By characterizing the dependence

structure of a U-statistic, a Berry-Esseen bound valid for incomplete U-statistics of deterministic designs is derived via Stein's method, yielding conditions under which Gaussian limiting distributions can be established even in degenerate cases and when the order diverges. Efficient algorithms are also introduced to construct incomplete U-statistics of equireplicate designs, a subclass of deterministic designs that, in certain cases, achieve minimum variance. Beyond its theoretical contributions, the framework provides a systematic way to construct permutation-free counterparts to tests based on degenerate U-statistics, as demonstrated in experiments with kernel tests that use the maximum mean discrepancy and the Hilbert-Schmidt independence criterion.

E1481: Non-asymptotic bounds for group sequential quasi-MLE, misspecified models, and dependence

Presenter: **Jay Bartroff**, University of Texas at Austin, United States

Results based on Stein's method are presented for the limiting multivariate normality of joint parameter estimates under group sequential sampling in a setting that allows dependence within groups. In a general parametric regression setting in which the n -th observation is the n -th subject's response regressed on their covariates, the quasi-MLE (QMLE) setup is considered in which the specified quasi-likelihood may differ from the true data generating process. Existing asymptotic normality results for the fixed-sample QMLE are extended to the group sequential setting, and non-asymptotic bounds to this limit are obtained via existing Stein's method tools. The results quantify convergence in QMLE, misspecified models, and under dependence, when the quasi-likelihood assumes independence but the true likelihood does not. As an example, these results are applied to the Poisson generalized linear mixed model.

E1938: Convex distance bounds for the stable central limit theorem via Steins method

Presenter: **Robert Gaunt**, The University of Manchester, United Kingdom

The stable central limit theorem is a generalisation of the classical central limit theorem that relaxes the finite variance assumption. Quantifying the quality of the distributional approximation of the stable central limit theorem is a fundamental problem in probability and mathematical statistics, which has seen numerous contributions over the years. In particular, there have been recent advances on this problem using the Stein method in which the stable central limit theorem is quantified with respect to Wasserstein type distances. Rates of convergence in the multivariate stable central limit theorem are derived in terms of the convex distance, which is a natural multivariate generalisation of the Kolmogorov distance, using Stein's method. Lower bounds are also presented to demonstrate optimality of the bounds in certain regimes. The bounds are derived by combining recent advances on the Stein method for alpha-stable distributions together with a powerful smoothing technique.

EO148 Room 204 MODELING AND INFERENCE FOR NONSTATIONARY TIME SERIES

Chair: Manon Felix

E1203: Self-normalization for CUSUM-based change detection in locally stationary time series

Presenter: **Florian Heinrichs**, FH Aachen - University of Applied Sciences, Germany

A novel self-normalization procedure for CUSUM-based change detection in the mean of a locally stationary time series is introduced. Classical self-normalization relies on the factorization of a constant long-run variance and a stochastic factor. In this case, the CUSUM statistic can be divided by another statistic proportional to the long-run variance, so that the latter cancels. Thereby, a tedious estimation of the long-run variance can be avoided. Under local stationarity, the partial sum process converges to $\int_0^1 \sigma(x) dB_x$ and no such factorization is possible. To overcome this obstacle, a self-normalized test statistic is constructed from a carefully designed bivariate partial-sum process. Weak convergence of the process is proven, and it is shown that the resulting self-normalized test attains asymptotic level α under the null hypothesis of no change, while being consistent against a broad class of alternatives. Extensive simulations demonstrate better finite-sample properties compared to existing methods. Applications to real data illustrate the method's practical effectiveness.

E1202: Self-driving neural networks for yield curve modeling

Presenter: **Sicco Kooiker**, Vrije Universiteit, Netherlands

Co-authors: Julia Schaumburg, Marcin Zamojski, Janneke van Brummelen

A factor model with time-varying loadings for yield curve modeling and forecasting is proposed. While maintaining the interpretation of the factors as level, slope, and curvature through explicit identification restrictions, the loadings are allowed to take flexible shapes by specifying them as neural networks that evolve over time using a self-driving updating scheme based on past forecast errors, with gradient scaling to improve robustness. Using an empirically calibrated simulation study and an application to U.S. Treasury yields across 24 maturities, it is shown that flexible and dynamic factor loadings improve forecasting performance relative to standard benchmarks, including Nelson Siegel models and the random walk. The gains are strongest at medium maturities and shorter forecast horizons, highlighting the importance of capturing curvature dynamics. In-sample results further illustrate how time-varying loadings provide insight into changes in yield curve shape beyond traditional parametric specifications.

E1400: Multiple change-point inference for locally stationary time series with applications to financial data

Presenter: **Wai Leong Ng**, The Hang Seng University of Hong Kong, Hong Kong

Change-point detection and locally stationary time series modelling are two major approaches for the analysis of non-stationary data. The former aims to identify stationary phases by detecting abrupt changes in the dynamics of a time series model, while the latter employs (locally) time-varying models to describe smooth changes in dependence structure of a time series. However, in some applications, abrupt and smooth changes can co-exist, and neither of the two approaches alone can model the data adequately. A novel likelihood-based procedure is proposed for the inference of multiple change-points in locally stationary time series. In contrast to traditional change-point analysis where an abrupt change occurs in a real-valued parameter, a change in locally stationary time series occurs in a parameter curve and can be classified as a jump or a kink depending on whether the curve is discontinuous or not. It is shown that the proposed method can consistently estimate the number, locations, and the types of change-points. Two different asymptotic distributions corresponding respectively to jump and kink estimators are also established. Extensive simulation studies and a real data application to financial time series are provided.

EO244 Room 205 IDENTIFICATION AND INFERENCE IN DYNAMIC AND HETEROGENEOUS MODELS (VIRTUAL)

Chair: Yi Zhang

E1805: Semiparametric models for dynamic treatment effects and mediation analyses with observational data

Presenter: **Shenshen Yang**, Tianjin University, China

Co-authors: Sukjin Han, Sungwon Lee

A semiparametric model is proposed that captures how a sequence of interventions interacts with a sequence of outcomes. In such settings, the outcome at a given period is affected by the history of treatments and outcomes, directly or indirectly through mediators. The main challenge in understanding various channels of dynamic effects is that, in observational settings, individuals make dynamically endogenous decisions regarding treatment selection. A model with an AR(1) process on the latent variable across stages is constructed. Using instrumental variables methods, it is shown how average and quantile dynamic treatment effects and mediation effects can be point identified and efficiently estimated in a class of semiparametric models under treatment endogeneity and flexible heterogeneity. The proposed procedure requires only binary instruments. As a byproduct of the semiparametric specification, parameters reflecting the degree of endogenous selection and time invariant heterogeneity are also identified and estimated.

E1794: The projection solution to the incidental parameter problem*Presenter:* **Yuanqi Zhang**, University College London, United Kingdom*Co-authors:* Andrew Chesher, Adam Rosen

An econometric approach to panel data models is introduced for settings where many observational units each deliver a small number of observations. In such models, the presence of variables that are constant within units while varying across units results in an incidental parameter problem. The approach removes these incidental parameters via projection, which delivers a correspondence specifying all combinations of observable variables and within-unit-varying unobservable heterogeneity that are achievable by choice of some value of the unit-specific incidental parameters. With unit-specific variables removed, there is no need for assumptions concerning their joint distribution with other variables. The resulting incomplete model is typically partially identifying. Identified sets are characterized via moment inequalities using tools of random set theory. Examples of application to static and dynamic models with discrete or continuous outcomes using a variety of exogeneity restrictions, including forms of weak exogeneity and distribution-free restrictions on within-unit-varying unobservable heterogeneity, are presented.

E1959: Parameter estimation under high-dimensional dynamic effects with an application to multilayer network connectedness*Presenter:* **Wei Shi**, Jinan University, China*Co-authors:* Shiyun Cao

The estimation of regression coefficients in high-dimensional time series with complex dependencies from lagged outcome variables is examined. The outcome variable is structured as a multidimensional array and forms a tensor-valued time series. The interaction matrix is inferred from the data without using a predetermined one. Complex dynamic effects are modeled by assuming that the autoregression coefficients exhibit a low-rank structure. The low-dimensional parameters are estimated using orthogonal moments and sample splitting, with estimates shown to be $\sqrt{NT}^{-1}$ consistent and asymptotically normal. Simulations indicate that the estimators perform well in finite samples. As a practical application, the proposed method quantifies connectedness among interconnected multilayer networks by application to the study of the impact of economic development on biodiversity in China, accounting for the spatio-temporal dependence among species.

E1891: Universal inference for incomplete models*Presenter:* **Yi Zhang**, Jinan University, China*Co-authors:* Hiroaki Kaido

A growing number of empirical models exhibit set-valued predictions. Such models are attractive because they avoid ad-hoc assumptions on model components that are not understood by the analyst. A tractable inference method with finite-sample validity is developed for such settings. The proposed approach provides robust tests and confidence intervals for counterfactual objects and other functionals of the underlying parameter, and accommodates discrete and continuous covariates, and nuisance parameters. The procedure builds on a robust version of the universal inference framework of Wasserman et al. (2020). Relative to existing inference approaches based on moment restrictions, it does not rely on resampling or simulation and can be implemented without additional functional form assumptions such as linearity.

EO198 Room 208 AI-POWERED ANALYSIS OF COMPLEX DEPENDENT DATA**Chair: Ansgar Steland****E1309: Federated learning of quantile inference under local differential privacy***Presenter:* **Shuyuan Wu**, Shanghai University of Finance and Economics, China*Co-authors:* Leheng Cai, Qirui Hu

Federated learning for quantile inference under local differential privacy (LDP) is investigated. An estimator based on local stochastic gradient descent (SGD) is proposed, whose local gradients are perturbed via a randomized mechanism with global parameters, making the procedure tolerant of communication and storage constraints without compromising statistical efficiency. Although the quantile loss and its corresponding gradient do not satisfy standard smoothness conditions typically assumed in existing literature, asymptotic normality for the estimator as well as a functional central limit theorem are established. The proposed method accommodates data heterogeneity and allows each server to operate with an individual privacy budget. Furthermore, confidence intervals for the target value are constructed through a self-normalization approach, thereby circumventing the need to estimate additional nuisance parameters. Extensive numerical experiments and real data application validate the theoretical guarantees of the proposed methodology.

E1529: Pairwise maximum likelihood for multi-class logistic regression model with multiple rare classes*Presenter:* **Xuetong Li**, Central University of Finance and Economics, China*Co-authors:* Xuetong Li, Danyang Huang

The problem of multi-class logistic regression with one major class and multiple rare classes is studied, which is motivated by a real application in TikTok live stream data. The model is inspired by the two-class logistic regression model but with surprising theoretical findings, which in turn motivate new estimation methods with excellent statistical and computational efficiency. Specifically, since rigorous theoretical analysis suggests that the resulting maximum likelihood estimators of different rare classes should be asymptotically independent, multiple pairwise two-class logistic regression problems are considered instead of optimizing the joint log-likelihood function with computational challenge in multi-class problem, which are computationally much easier and can be conducted in a fully parallel way. To further reduce the computation cost, a subsample-based pairwise likelihood estimator is developed by downsampling the major class. It is shown rigorously that the resulting estimators could be as asymptotically efficient as the global maximum likelihood estimator under appropriate regularity conditions. Extensive simulation studies are presented to support the theoretical findings and a TikTok live stream dataset is analyzed for illustration purpose.

E1298: Conditional local independence testing for Ito processes with applications to dynamic causal discovery*Presenter:* **Xinwei Sun**, Fudan University, China

Inferring causal relationships from dynamical systems is central to many scientific inquiries. Conditional local independence, which describes whether the evolution of one process is influenced by another process given additional processes, is important for causal learning in such systems. A hypothesis test is proposed to examine conditional local independence between stochastic random processes. The test is grounded in the semi-martingale decomposition of the Ito process, with which a stochastic integral process is introduced that is a martingale under the null hypothesis. A test for the martingale property is then applied, quantifying potential deviation from local independence. The test statistic is estimated using the optimal filtering equation. Consistency of the estimation is established, thereby determining the level and power of the test. Numerical verification and a real-world application to causal discovery in brain resting-state fMRI are conducted.

EO064 Room 209 SURVIVAL ANALYSIS, COPULAS, AND FRAILITY**Chair: Takeshi Emura****E1752: Variable selection in distributional time-to-event models with dependent and administrative censoring****Presenter:** Annika Stroemer, Department of Medical Biometry and Statistics, University of Marburg, Germany**Co-authors:** Nadja Klein, Ingrid Van Keilegom, Andreas Mayr

A standard assumption in survival analysis is that the censoring time C and the survival time T are conditionally independent given the covariates. This assumption holds for administrative censoring but is often violated in medical studies, where censoring may depend on the event time, for example when patients withdraw from a trial due to deteriorating health, causing sicker patients to be censored earlier. A model-based boosting approach via distributional copula regression is proposed that accounts for dependent censoring and the heterogeneity of the censoring mechanism. Administrative censoring is treated as independent of the event time, while dependent censoring is jointly modeled with the survival time through a parametric copula. All distribution parameters, including the covariate-dependent copula parameter, are estimated via component-wise gradient boosting. Boosting remains feasible even for high-dimensional data with $p > n$ and enables data-driven variable selection across all distributional components. Performance is investigated in a simulation study covering scenarios with varying proportions of administrative versus dependent censoring, different censoring rates, and varying levels of dependence between T and C , with illustration using a biomedical example.

E1616: Dependent censoring with simultaneous death times**Presenter:** Mikael Escobar-Bach, University of Angers, France**Co-authors:** Salima Helali

The problem of dependent censoring models with positive probability that failure times are equal is considered. A Marshall-Olkin type model is proposed and some properties of the associated survival copula in its application to censored data are studied. Estimators for the marginal distributions and joint survival probabilities under different schemes are introduced and their asymptotic normality is established under appropriate conditions. The finite-sample performance of the approach is evaluated through a simulation study on synthetic data and an application to real data.

E1653: On the robustness of maximum h-likelihood estimators under cause-specific competing risks frailty models**Presenter:** Il Do Ha, Pukyong National University, Korea, South**Co-authors:** Hangbin Lee, Youngjo Lee, Chew Chee

In clustered competing risks data, both the events of interest and competing events are often correlated within the same cluster. This dependency can be modeled using cluster-specific frailties and the parameters can be estimated by maximum h-likelihood estimators (MHLEs). The robustness of MHLEs under the common log-normal frailty assumption is theoretically investigated when the true frailty distribution deviates from log-normality. A robust non-parametric method for cause-specific competing risks frailty models is proposed, which does not assume a specific parametric form for the frailty distribution. The proposal includes a hybrid iterative algorithm for computing non-parametric maximum likelihood estimators (NPMLEs). Simulation studies under various frailty distributions indicate that the proposed NPMLEs demonstrate superior performance, particularly for the association parameter. MHLEs of the regression coefficients yield comparable performance in terms of average relative bias and root mean squared error, which aligns with theoretical findings. Both methods are applied to real-world datasets from multi-center studies on bladder cancer and bone marrow transplantation to demonstrate their practical utility.

E1585: Inferring median survival under dependent censoring using the copula graphic estimator**Presenter:** Takeshi Emura, Hiroshima University, Japan**Co-authors:** Dennis Dobler

The median is a popular nonparametric measure to calibrate patient survival and represents clinical benefit of treatments. A key difficulty in estimation of median survival is handling right censoring. Existing inference methods have been developed under the fundamental assumption of independent censoring, which is often too strong for clinical applications. Novel statistical methods are proposed for inferring median survival when data are subject to dependent censoring. To accommodate dependent censoring schemes, the copula graphic estimator is utilized leading to the estimation and significance test for medians in the one sample and two sample settings. A Wald test for a general factorial design is also developed. Simulation studies are performed to investigate the performance of the proposed methods and to check the reliability of the R package. A real dataset on gastric cancer patients is analyzed to illustrate the proposed methods for median survival.

EC279 Room 201 APPLIED STATISTICS**Chair: Daisuke Murakami****E1449: Assessing urban water loss patterns: A comparative time series study between Brazil and Portugal****Presenter:** Elisa Henning, Universidade do Estado de Santa Catarina, Brazil**Co-authors:** Andreza Kalbusch, Amilcar Oliveira, Teresa Oliveira

Water losses in urban supply systems constitute a significant challenge in the context of climate change and increasing pressure on water resources. Understanding their temporal dynamics, key drivers, and cross-country differences is essential to support effective mitigation strategies. Statistical modeling provides a useful framework for this analysis. A comparative assessment of water losses in urban supply systems in Brazil and Portugal is conducted based on time series methodologies. Time series regression models were employed to characterize trends, identify structural differences in water loss behavior, and examine potential associations with per capita water consumption. Additionally, the Grey Model was applied for short-term forecasting, given its suitability for limited or incomplete data series. The results indicate divergent trajectories between the two countries, with a decreasing trend in water losses observed in Portugal and an increasing trend in Brazil. In contrast, per capita water consumption exhibits an upward trend in Portugal and relative stability in Brazil. These findings suggest that variations in water loss patterns are not directly attributable to per capita consumption levels, but are more likely driven by context-specific factors, such as infrastructure conditions, management practices, and regulatory frameworks. Furthermore, the Grey Model demonstrated high predictive accuracy, with forecast errors below 2%.

E2015: Advances in comparative evaluations of bovine females and hermaphrodites: A multivariate longitudinal permutation test**Presenter:** Stefano Bonnini, University of Ferrara, Italy**Co-authors:** Livio Corain, Antonella Peruffo

Intersexuality in bovines can manifest itself in many different ways. The freemartin syndrome is one of the most common. Bovine blood hormonal concentrations and morphometric measurements from postnatal to adulthood are analyzed. Specifically, the trajectories of hormonal and morphometric variables over time in female and freemartin bovines are compared. The application of a two-sample combined permutation test provides a powerful and effective inferential method to make such comparative evaluations, by identifying the variables with respect to which the two populations differ and the time of the change point.

E1988: Modeling race dynamics in circuit motorsport: A state-space approach for lap time decomposition**Presenter:** Karolina Jindrova, Prague University of Economics and Business, Czechia**Co-authors:** Petra Tomanova

Lap-level racing data combine precisely recorded outcomes with sources of variation that are only partly observed. A structural state-space approach is developed to decompose lap times in circuit motorsport into measured race effects and latent time-varying performance dynamics. The

observation equation relates lap time to exogenous regressors representing measurable influences, while the state equation captures unobserved processes such as tire behavior, track evolution, and other persistent performance shifts. Rather than correcting lap times through separate static regressions, the approach models observed and latent effects jointly and treats decomposition as a filtering and smoothing problem. Parameters are estimated by maximum likelihood using the Kalman filter, and smoothed states provide retrospective signal extraction. Standardized One-step-ahead prediction errors are used to check serial correlation, normality, and conditional heteroskedasticity. This structure allows measured race events to be analyzed together with latent dynamics that are not directly recorded in the data. Empirical results from Formula One lap data show interpretable estimates of measured effects while preserving a flexible latent component for evolving performance dynamics.

E1514: **Comparing regularization strategies for CO_2 determinants: Classical, Bayesian and maximum entropy approaches**

Presenter: **Maria Costa**, University of Aveiro, Portugal

Co-authors: Ana Helena Tavares, Jorge Cabral

Mozambique provides an empirical setting for comparing information-theoretic and Bayesian approaches in a small-sample regression problem. An annual macroeconomic dataset with a carbon-emissions response and three economic covariates is used to examine 18 static specifications, namely OLS, ridge, lasso, four Bayesian models with weakly informative, ridge, lasso and horseshoe priors, and eleven GME or GCE variants with alternative support widths, updating schemes and prior mass structures. Unit root tests indicate strong persistence in most series, and bounds tests find no robust evidence of cointegration, shifting the analysis from long-run interpretation to a short-run specification in first differences. The central question is not which static estimator recovers a long-run relation, but which family remains closer to the short-run effects identified by the dynamic model when temporal dependence is ignored. The results suggest both a stable component, with consistent sign patterns across specifications, and a method-sensitive component, with substantial variation in magnitude and relative proximity to the dynamic short-run effects. Classical and Bayesian specifications tend to remain closer to the dynamic short-run structure, whereas many entropy-based variants produce more compressed coefficient patterns. The comparison highlights regularization as an inferential choice rather than a purely technical adjustment.

EC269 Room 203 COMPLEX AND HETEROGENEOUS DATA

Chair: Sotetsu Koyamada

E1403: **Metric skewness for object data**

Presenter: **Vida Zamanifarizhandi**, University of Turku, Finland

Co-authors: Joni Virta, Janne Kujala, Oona Rainio

As datasets become increasingly complex, more advanced methods for extracting insight from data are required. Recently, analyzing data in metric spaces has gained attention and several descriptive statistics have been developed. However, important distributional characteristics, such as skewness, which provides valuable information about its structure, remain largely unexplored for object data in metric spaces. A novel method is introduced to compute the metric skewness. Moreover, its use in testing for the presence of skewness exhibits promising outcome both in level and power, surpassing its multivariate skewness counterparts in Euclidean space. Finally, the proposed metric skewness is applied as an inferential statistic to positron emission tomography (PET) data, illustrating its practical applicability.

E2006: **Identifying structural types in techno-capital complex systems**

Presenter: **Tsung-Han Ke**, National Chi Nan University, Taiwan

Co-authors: Hung-Chun Huang, Hsin-Yu Shih, Yun-Xun Hsu

A Lyapunov-based decomposition framework is developed for analyzing nonlinear dependence structures in techno-capital complex systems. Semiconductor patent activity (TT) and the NASDAQ Composite (NAQ) are transformed into Lyapunov dynamic series to measure evolutionary instability and structural variation under nonlinear conditions. The framework decomposes system dynamics into kinetic energy and potential energy components to distinguish short-term fluctuations from long-run structural accumulation. Three resonance structures are identified from the decomposition: kinetic-to-kinetic resonance reflects synchronized dynamics, kinetic-to-potential resonance captures adaptive feedback, and potential-to-potential resonance represents long-run structural coupling and self-organization. Empirical evidence reveals weak short-term kinetic synchronization but significant long-term potential resonance between TT and NAQ. The results further reveal a shared evolutionary trajectory in which semiconductor innovation dynamics lead financial market adjustment over time. These findings suggest that techno-capital systems evolve primarily through institutional co-evolution and potential-based self-organization rather than short-term market synchronization. The framework provides a statistical approach for identifying self-organizing structures and nonlinear dependence patterns in complex adaptive systems.

E2045: **Imputation under cellwise contamination**

Presenter: **Matthias Templ**, University of Applied Sciences and Arts Northwestern Switzerland, Switzerland

Cellwise contamination, in which outliers strike individual cells rather than whole observations, is pervasive in survey microdata, medical records and administrative registers. Even at a modest per-cell rate, most rows of a wide dataset contain at least one corrupted entry, so casewise-robust methods and listwise deletion fail; and when values are also missing, standard imputation propagates the contamination. Integrated procedures that jointly flag deviating cells and estimate model parameters exist but assume fully continuous, complete data; two-step detect-then-impute pipelines discard information and threshold hard at the boundary. A family of integrated cellwise-robust imputation methods is developed for mixed continuous and categorical data with missing values. Cell weights from robust conditional residuals enter an iteratively reweighted least-squares fit at the cell level. One variant performs per-variable M-regression and is provably consistent for the clean parameters under the independent cellwise model; another extends the cellwise minimum covariance determinant estimator, combining multivariate cell detection with cellwise-robust regression. Multivariate detection outperforms per-variable residuals when contamination is correlated within blocks of variables. Because cellwise breakdown points are intrinsically low, the methods complement casewise approaches; all are available in the R package VIM.

E1968: **Bayesian integration of survey data sources: Costs and quality**

Presenter: **Joseph Sakshaug**, Germany

Non-probability web surveys (NPS) have become increasingly popular as they are convenient and inexpensive. At the same time, traditional probability-based surveys (PS) are experiencing steadily declining response rates that drive up costs. Combining the two approaches to offset their respective weaknesses has become one of the central challenges in survey research. A new methodology is presented that fuses probability and non-probability samples to produce more accurate analytic inferences. The method uses a Bayesian framework that optimally weights the two data sources according to estimated measures of bias. The approach is evaluated through extensive simulations and a real-data application that pairs a large probability-based survey with several parallel non-probability web surveys fielded by different vendors, each representing distinct selection mechanisms. Results show that the combined estimator consistently achieves lower mean squared error (MSE) than either source used alone, and in the worst case produces no increase in MSE. A cost analysis further demonstrates that the approach can substantially reduce overall survey expenditure, making it especially attractive for studies seeking to lower data-collection costs while maintaining or improving data quality.

E1870: An innovative hybrid approach for forecasting spatio-temporal variables*Presenter:* **Antonella Congedi**, University of Salento, Italy*Co-authors:* Sandra De Iaco

Climate change represents one of the main ecosystem problems, and accurate predictions of meteorological factors are essential to address extreme events, rising sea levels, loss of biodiversity, and to integrate energy-saving strategies with renewables. Forecasting methodologies include statistical and machine learning (ML) models, which capture the complex spatio-temporal dynamics of climate variables. An innovative hybrid approach combining spatio-temporal geostatistical techniques and artificial neural networks is proposed to analyze two meteorological datasets. The combination of kriging with ConvLSTM2D network is applied in the first case study to model solar radiation daily recorded in 2020 over a regular grid covering the Lombardy Region. In the second application, kriging combined with SparseConvLSTM network is used to model temperature weekly observed at meteorological stations of the Veneto Region from 2019 to 2022. A comparison with alternative methods is presented, and results highlight the advantages of using hybrid approaches instead of pure traditional or ML approaches. Future studies could extend the application of the proposed methodology to further fields or combine spatio-temporal kriging with other ML approaches.

E2054: Two-stage deep learning framework for spatiotemporal interpolation and probabilistic forecasting of air temperature*Presenter:* **Monica Palma**, University of Salento, Italy*Co-authors:* Nouman Iqbal

Accurate spatiotemporal interpolation and forecasting of environmental variables are important for climate analysis and environmental monitoring. Daily air temperature exhibits nonlinear dynamics, seasonal variability, and spatial nonstationarity that are difficult to capture using traditional geostatistical methods. Classical kriging approaches rely on stationarity assumptions and predefined covariance structures, which may reduce flexibility and increase computational costs for large spatiotemporal datasets. A two-stage deep learning framework is proposed for spatiotemporal interpolation and probabilistic forecasting of daily mean air temperature. In the first stage, a spatiotemporal deepkriging model with multi-resolution basis function embeddings captures spatial and seasonal dependencies without explicit covariance estimation. In the second stage, QMLP, QLSTM, and QConvLSTM models are evaluated for multi-step probabilistic forecasting. Model performance is assessed using average MAE, RMSE, and R^2 across spatial locations. The framework is applied to NASA POWER MERRA-2 daily meteorological data from 1982-2023 at 30 locations in the Apulia region of Italy. Results indicate improved forecasting accuracy and computational efficiency compared with covariance-based geostatistical approaches.

E1872: Assessing a spatio-temporal biodiversity score through multi criteria analysis of blind source separation components*Presenter:* **Rebecca Distratis**, University of Salento, Italy*Co-authors:* Sandra De Iaco, Abdollah Jalilian, Iman Masoumi

Biodiversity is a complex, multidimensional phenomenon structured across space and time. Traditional statistical approaches often struggle with the high dimensionality, multicollinearity and spatial dependence typical of ecological data. A framework combining spatio-temporal blind source separation with the ELECTRE score procedure is proposed. First, the algorithm decomposes a multivariate random field into statistically independent latent components, separating distinct climatic, environmental and biological signals while accounting for spatial and temporal dependence. Then, these components are integrated into a biodiversity suitability index through a structured synthesis that preserves a non-compensatory logic. The methodology is applied to a spatio-temporal dataset covering Italy from 2020 to 2024 and integrating biodiversity proxies with environmental and ecological variables. Results reveal the ability to capture coherent seasonal dynamics as well as spatial gradients, such as peaks in spring and summer, greater variability in transitional periods and phenological contrasts between the Tyrrhenian and Adriatic areas. Sensitivity analysis confirms the robustness of the spatial rankings and suitability classes under moderate threshold perturbations. A transparent and reproducible workflow is provided for monitoring complex spatio-temporal ecological systems, offering a robust alternative to conventional biodiversity indices.

E1871: A spatial non-compensatory composite indicator for socio-economic disparities*Presenter:* **Lucia Simmini**, University of Salento, Italy*Co-authors:* Sandra De Iaco, Rebecca Distratis, Claudia Cappello

Measuring socio-economic disparities across territories requires composite indicators capable of synthesising multiple heterogeneous dimensions. An integrated framework is proposed for constructing a spatial non-compensatory composite indicator, the socio-economic gap indicator (SEGI), that addresses three limitations commonly found in the literature: redundancy among correlated indicators, the treatment of spatial dependence as an ex-post diagnostic step, and the use of compensatory aggregation. The framework combines spatial blind source separation, which extracts independent latent components each capturing a distinct spatial structure, with the ELECTRE score, a non-compensatory outranking procedure preventing strong performance in one dimension from masking weaknesses in others. Spatial context is embedded into the score formula via indicator kriging, which estimates a location-specific parameter reflecting the socio-economic conditions of neighbouring provinces. The methodology is applied to 107 Italian provinces using indicators of education, labour market conditions, income, and health. The resulting indicator highlights spatial patterns consistent with the north-south divide and shows stronger spatial autocorrelation than a benchmark model, offering a more robust and spatially grounded measure of territorial disparities.

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